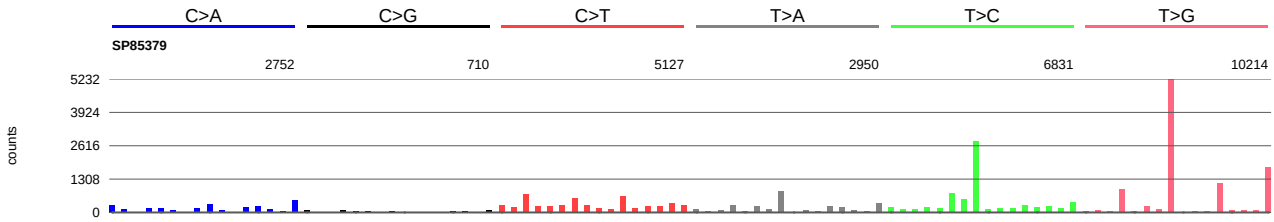
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SP85379
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710
5127
2950
6831
10214
counts
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1308
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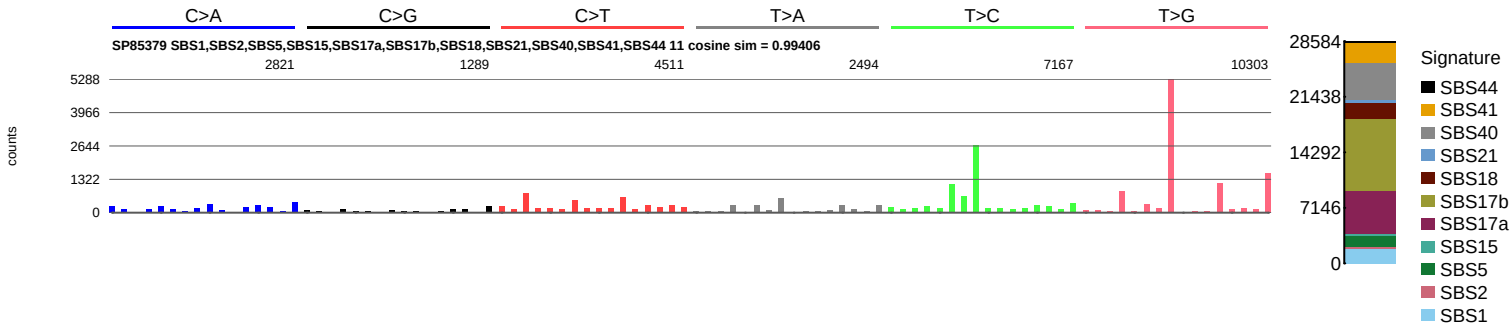
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SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99406
2821
1289
4511
2494
7167
10303
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
0
Signature
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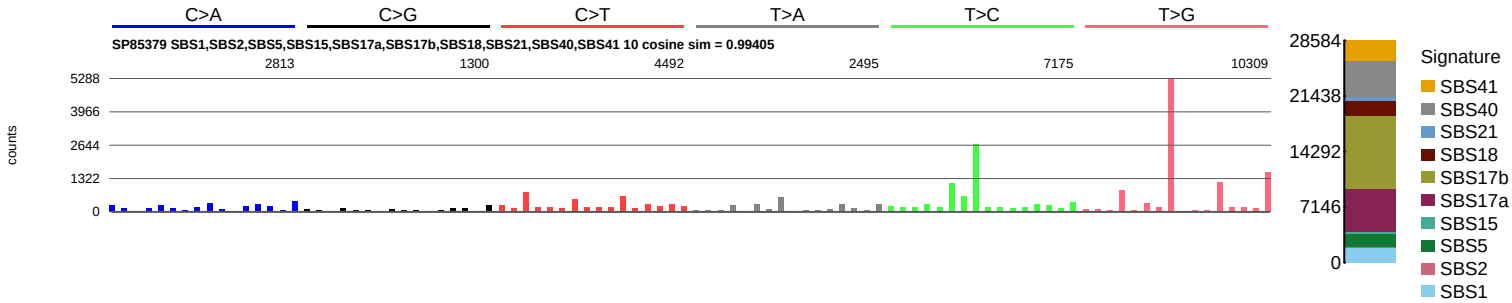
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99405
2813
1300
4492
2495
7175
10309
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
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SBS21
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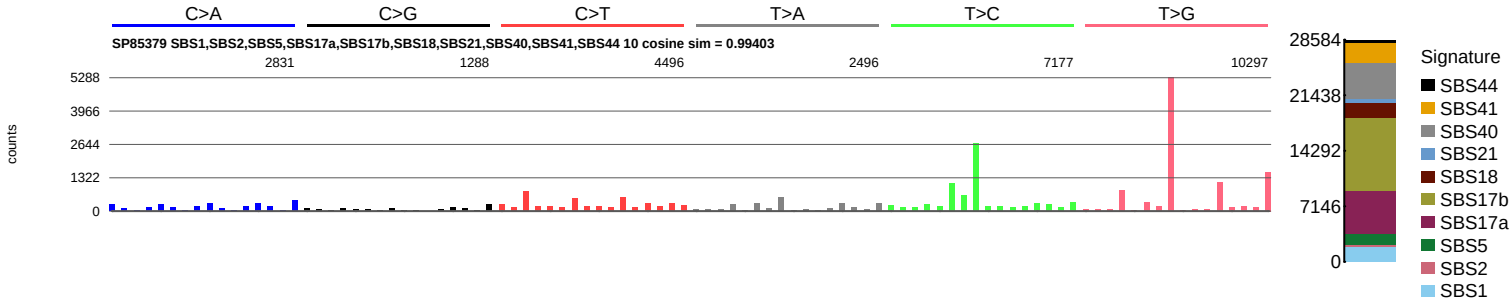
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99403
2831
1288
4496
2496
7177
10297
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
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Signature
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SBS41
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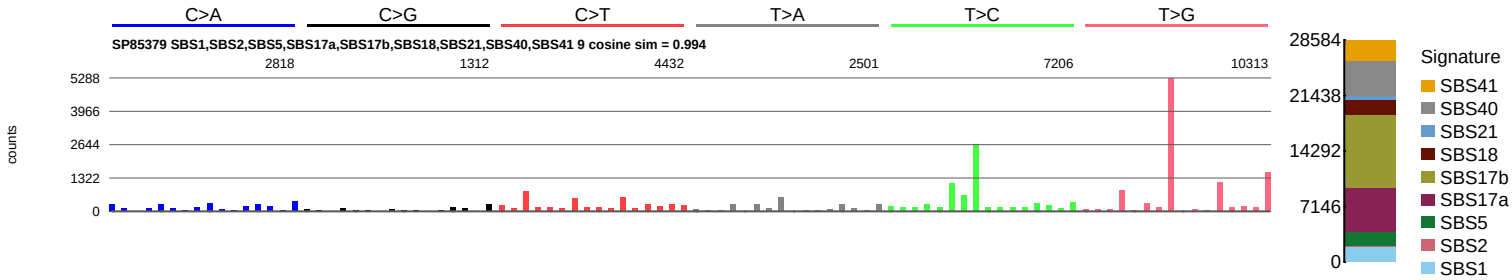
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>T
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T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.994
2818
1312
4432
2501
7206
10313
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
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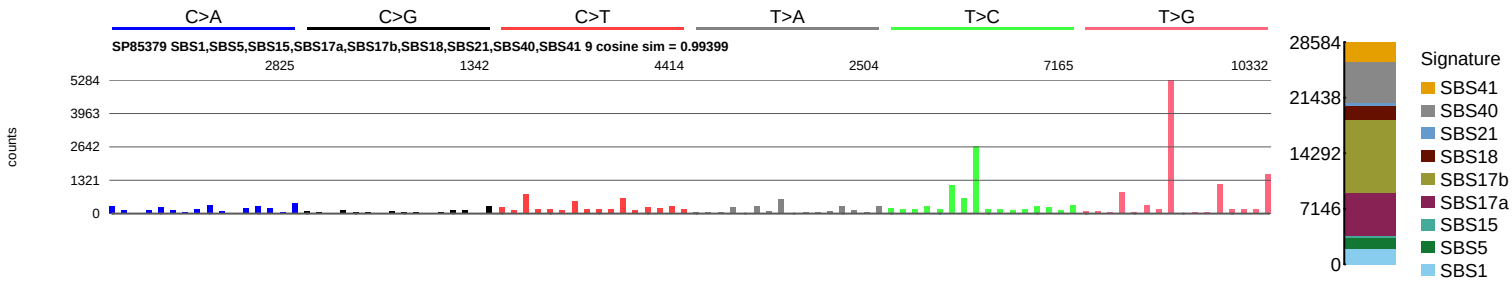
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99399
2825
1342
4414
2504
7165
10332
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
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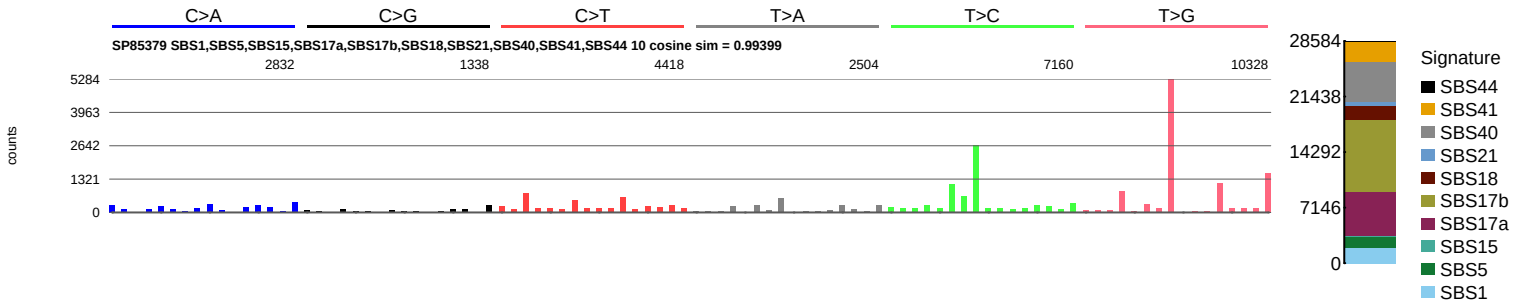
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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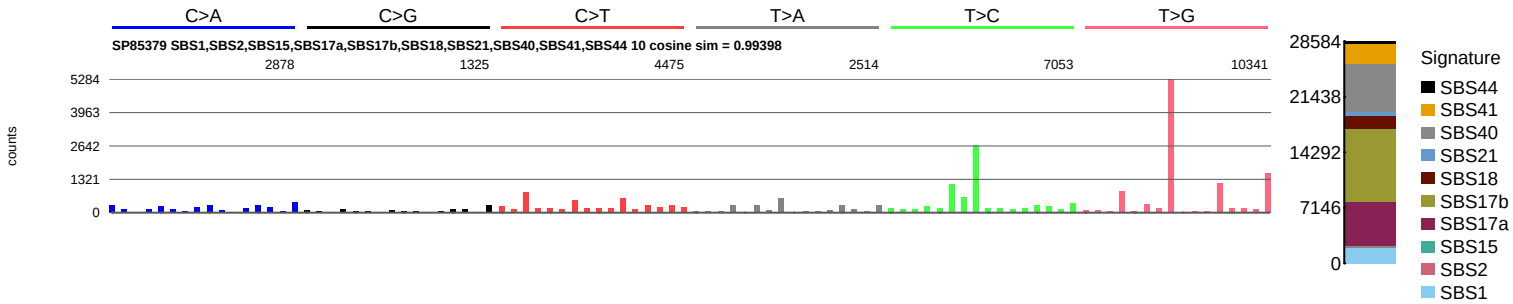
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99399
2832
1338
4418
2504
7160
10328
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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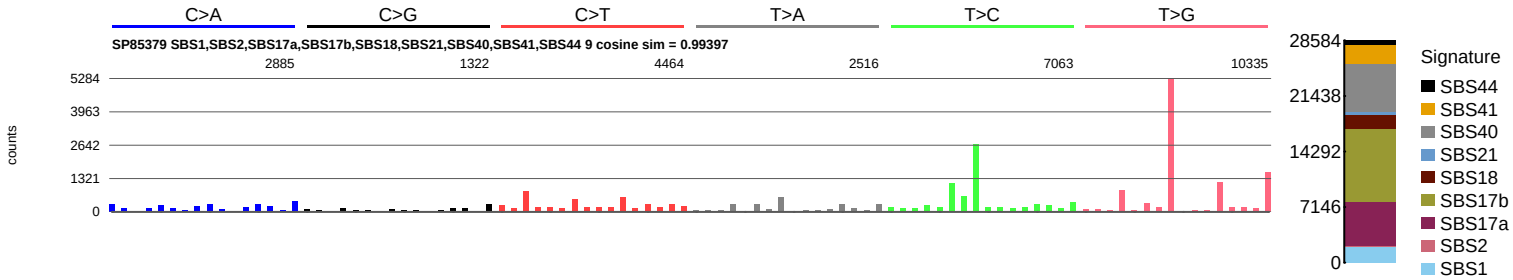
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99398
2878
1325
4475
2514
7053
10341
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99397
2885
1322
4464
2516
7063
10335
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
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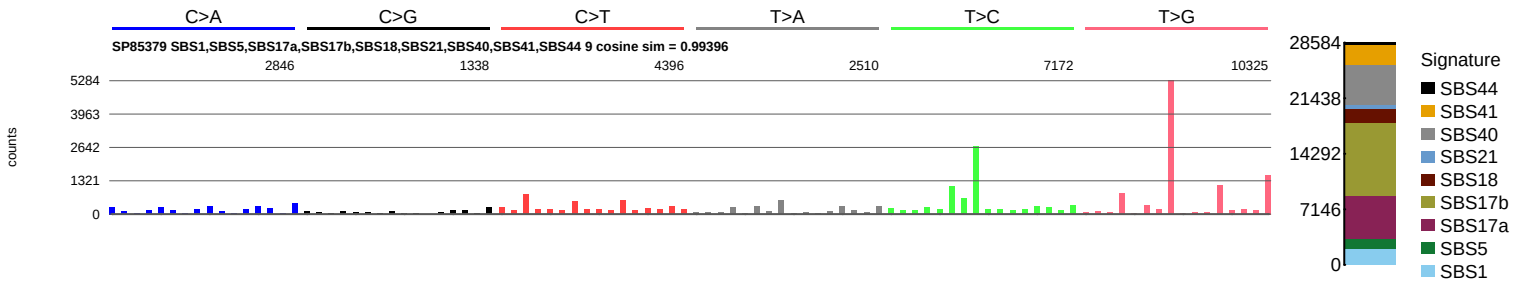
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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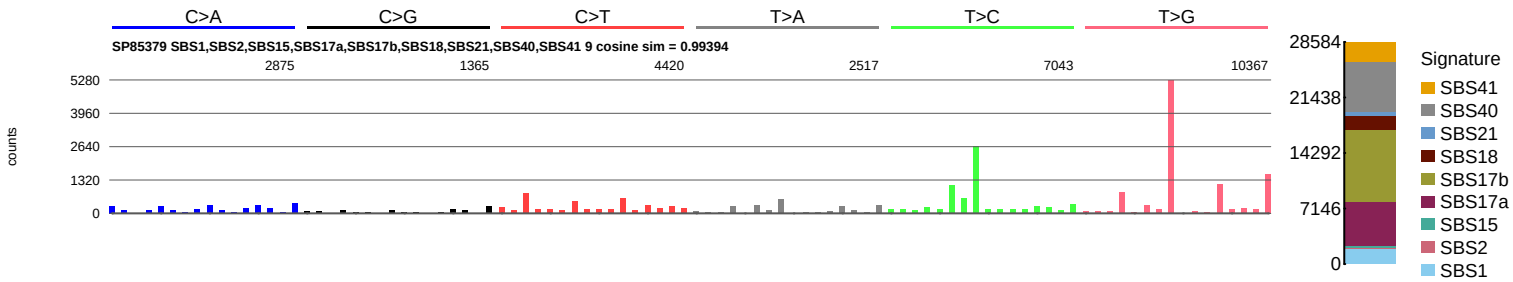
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99396
2846
1338
4396
2510
7172
10325
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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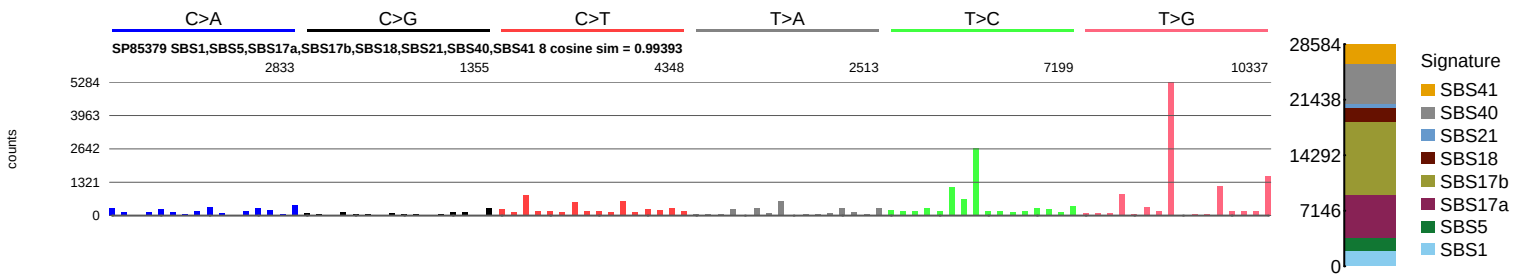
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99394
2875
1365
4420
2517
7043
10367
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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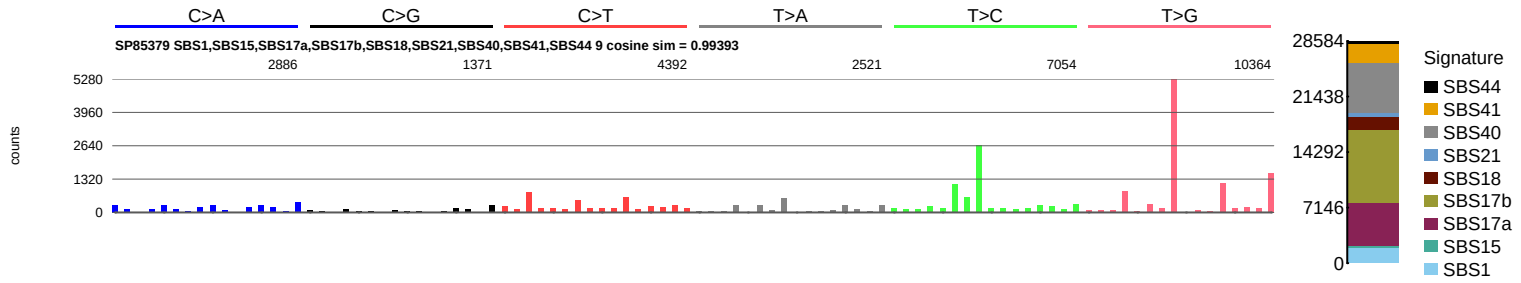
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99393
2833
1355
4348
2513
7199
10337
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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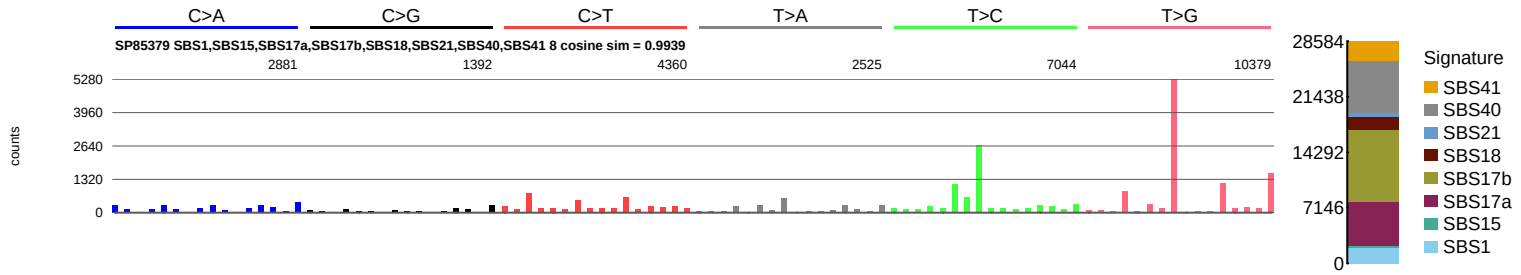
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99393
2886
1371
4392
2521
7054
10364
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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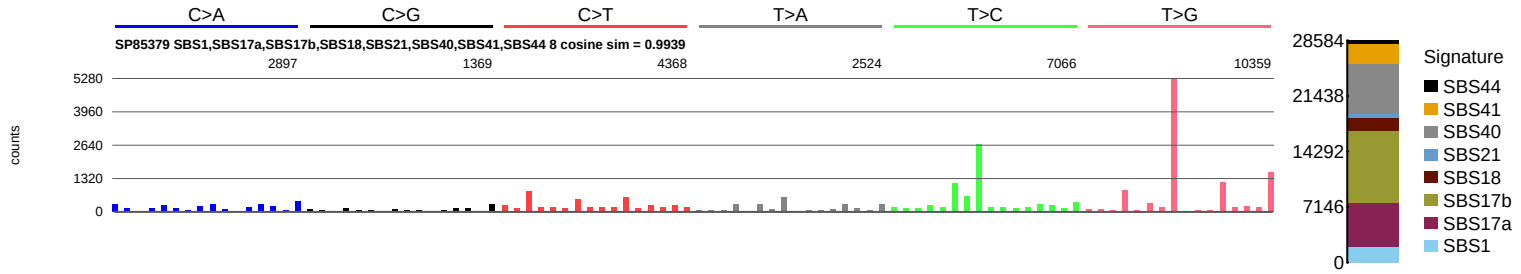
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9939
2881
1392
4360
2525
7044
10379
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9939
2897
1369
4368
2524
7066
10359
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
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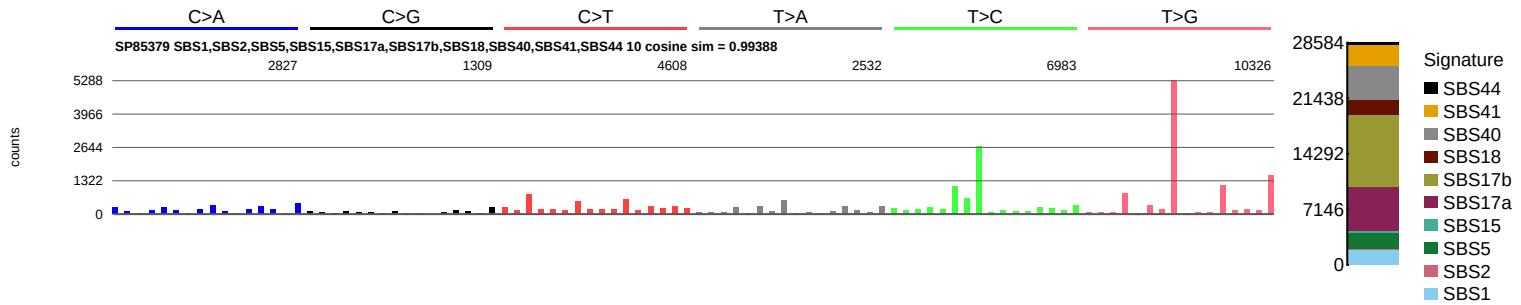
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99388
2827
1309
4608
2532
6983
10326
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
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Signature
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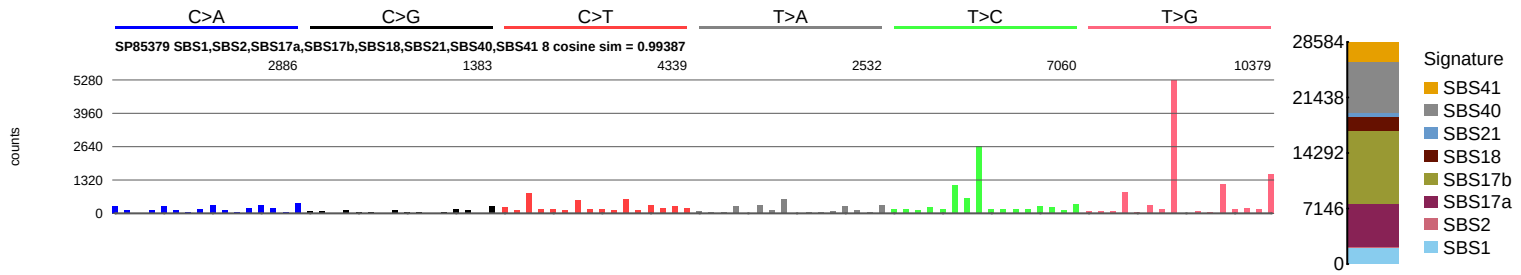
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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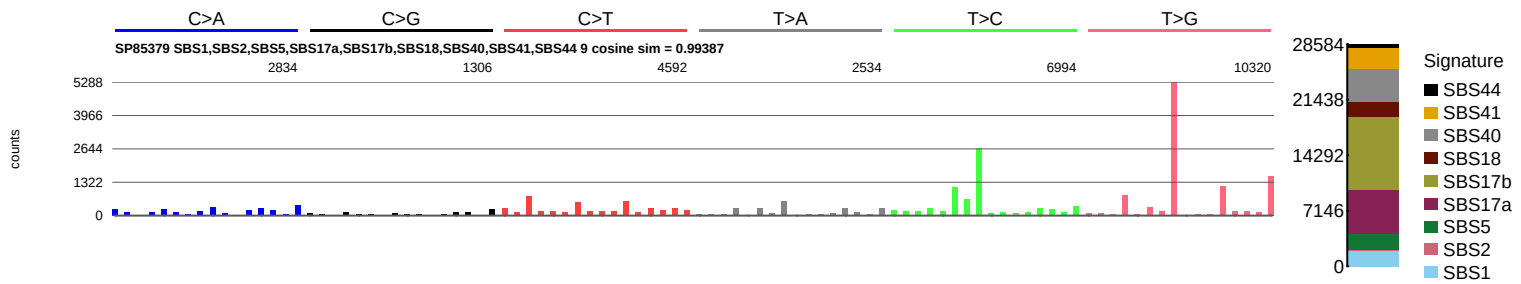
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99387
2886
1383
4339
2532
7060
10379
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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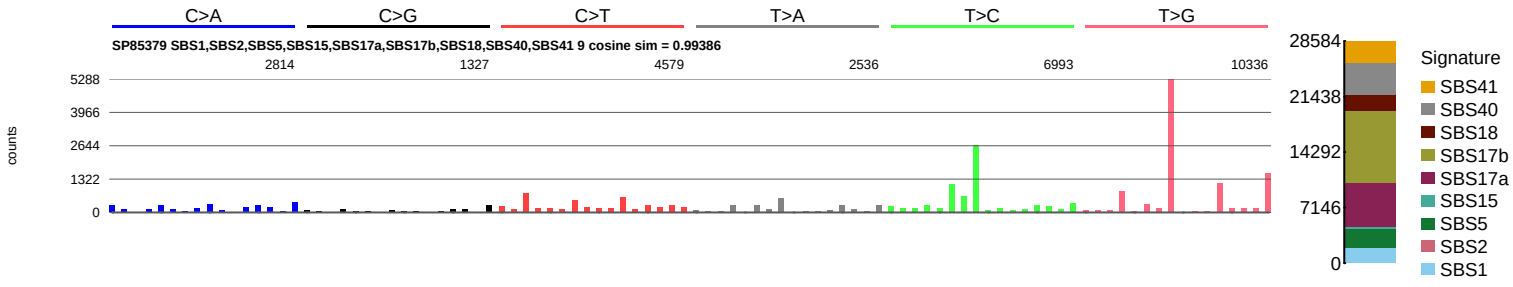
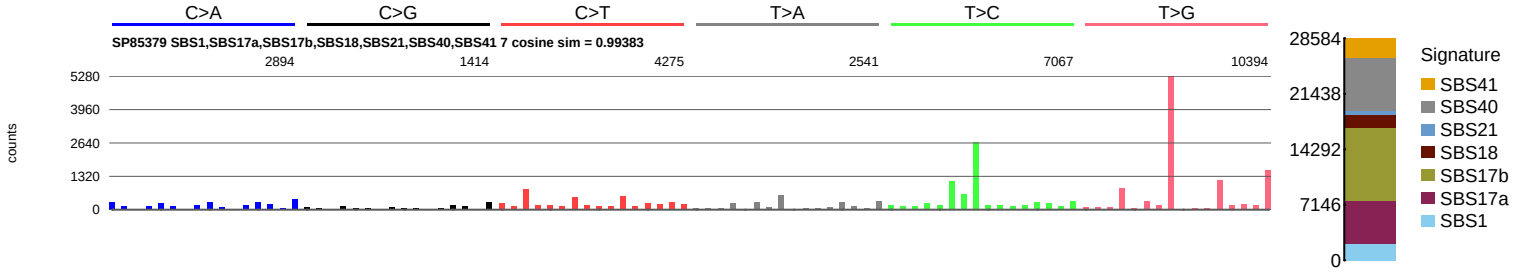
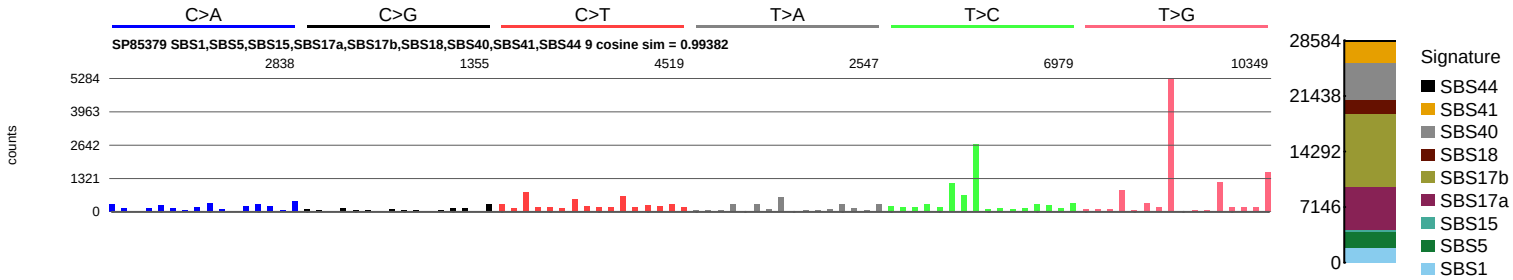
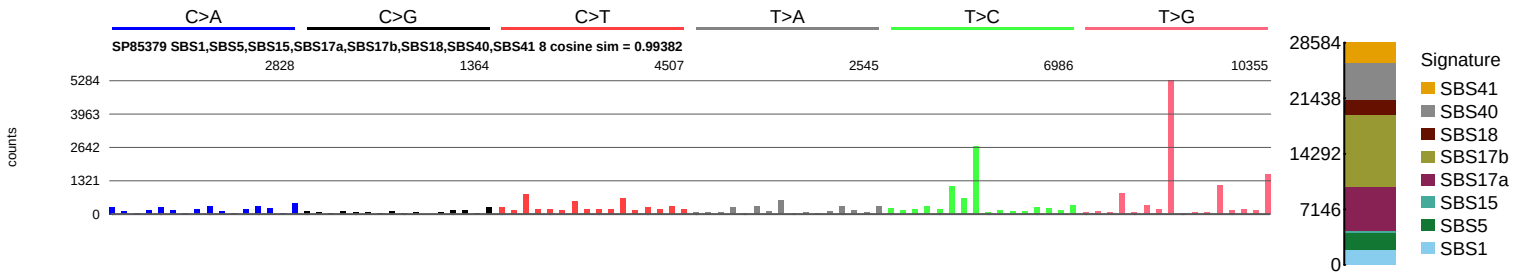
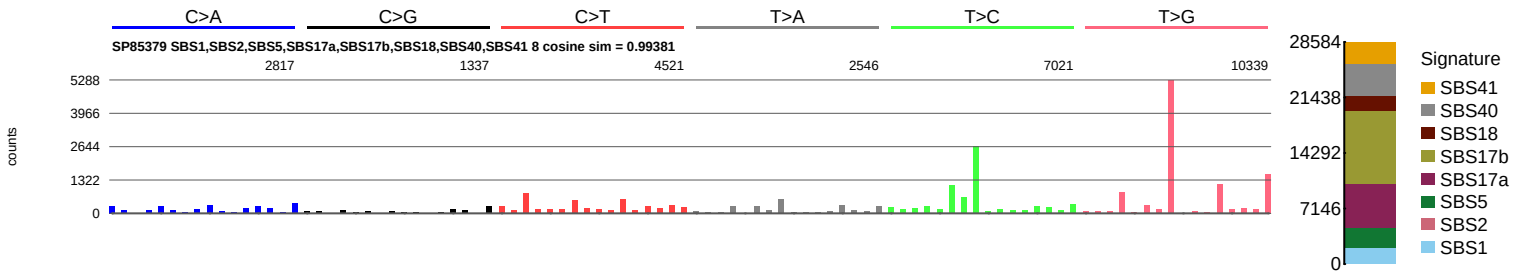
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99387
2834
1306
4592
2534
6994
10320
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

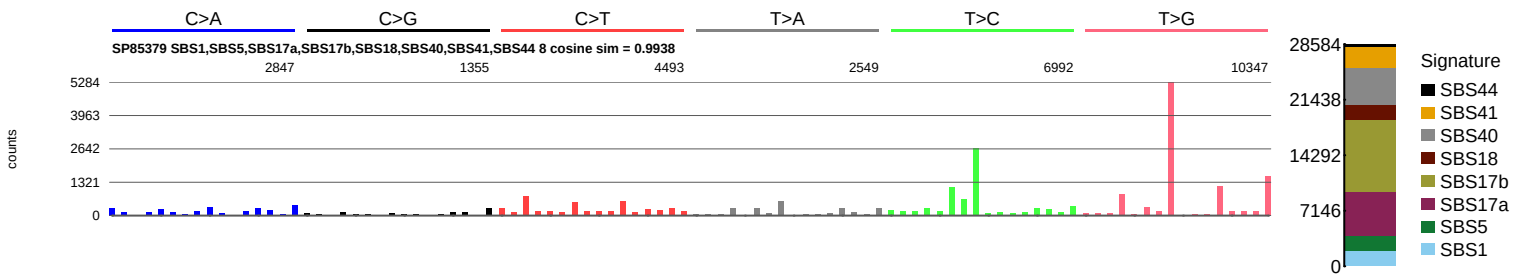

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99386
2814
1327
4579
2536
6993
10336
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

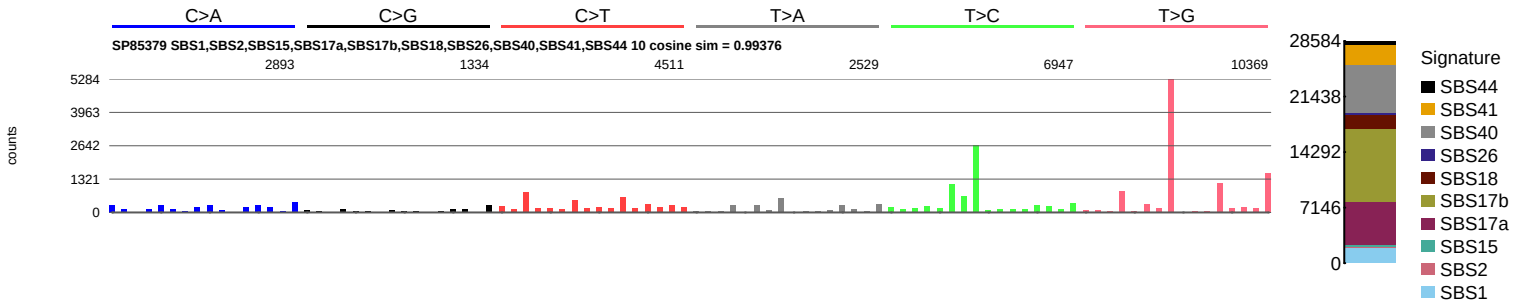

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.99383
2894
1414
4275
2541
7067
10394
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99382
2838
1355
4519
2547
6979
10349
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99382
2828
1364
4507
2545
6986
10355
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99381
2817
1337
4521
2546
7021
10339
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9938
2847
1355
4493
2549
6992
10347
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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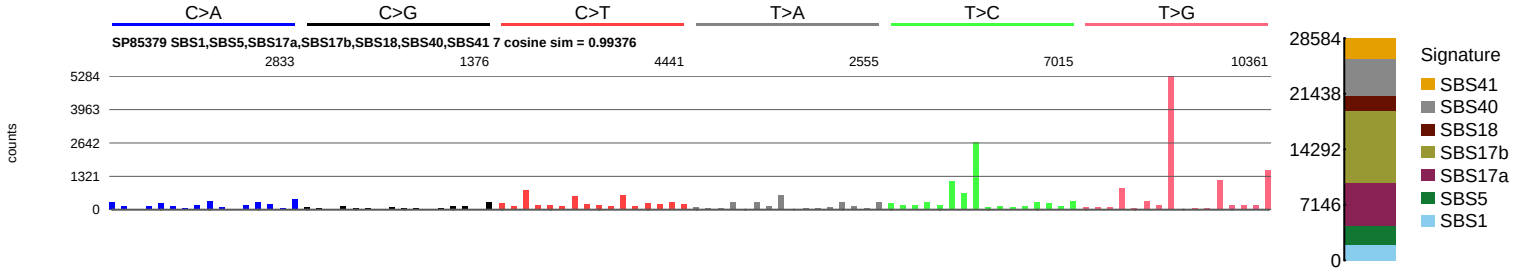
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99376
2893
1334
4511
2529
6947
10369
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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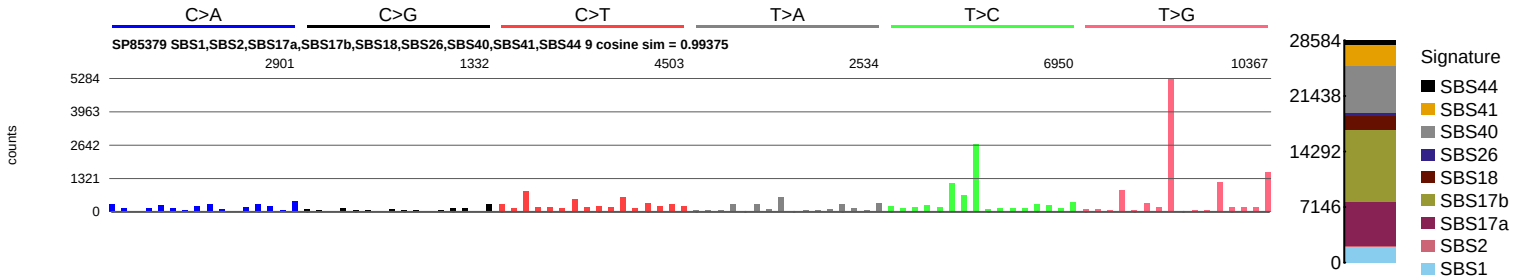
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99376
2833
1376
4441
2555
7015
10361
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99375
2901
1332
4503
2534
6950
10367
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
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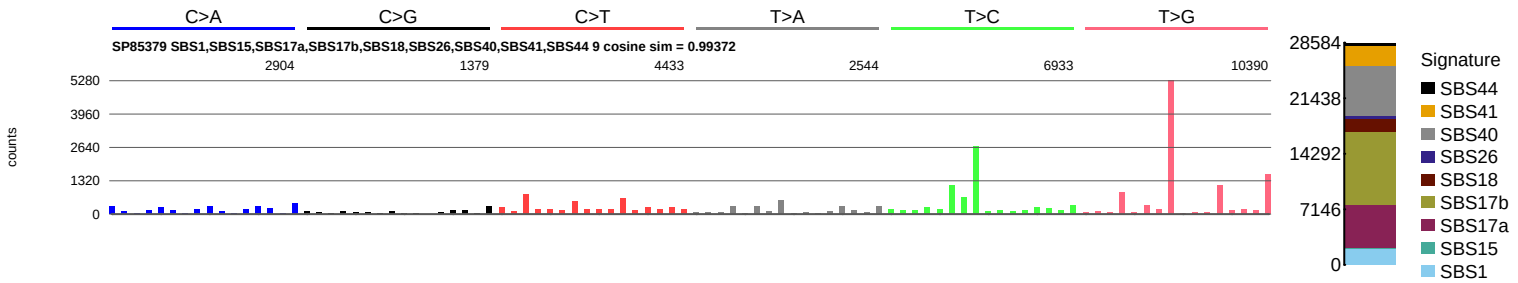
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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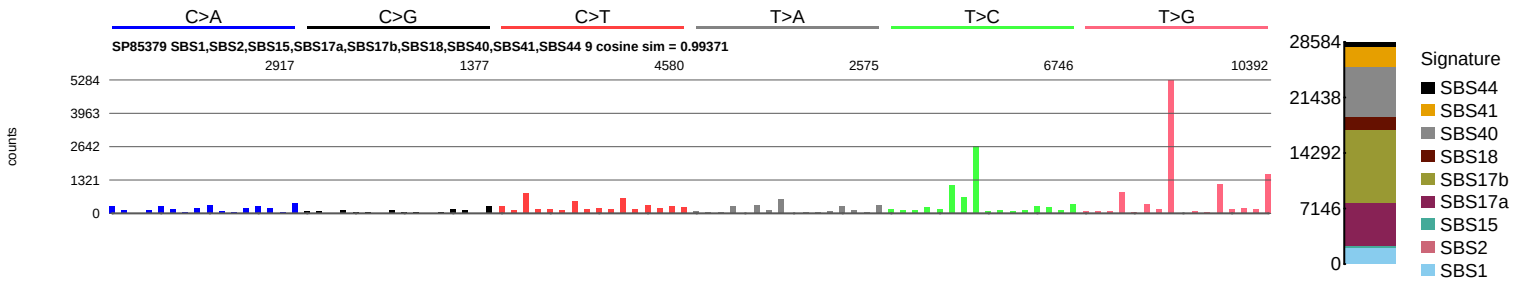
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99372
2904
1379
4433
2544
6933
10390
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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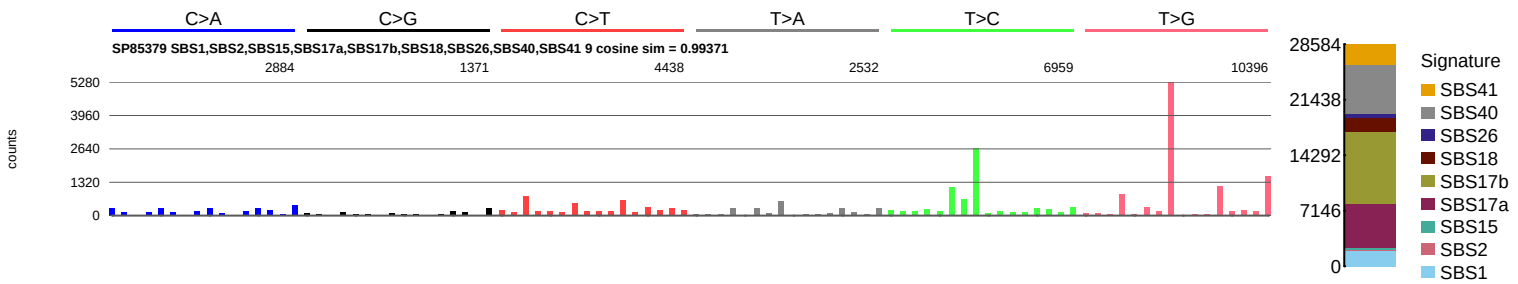
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99371
2917
1377
4580
2575
6746
10392
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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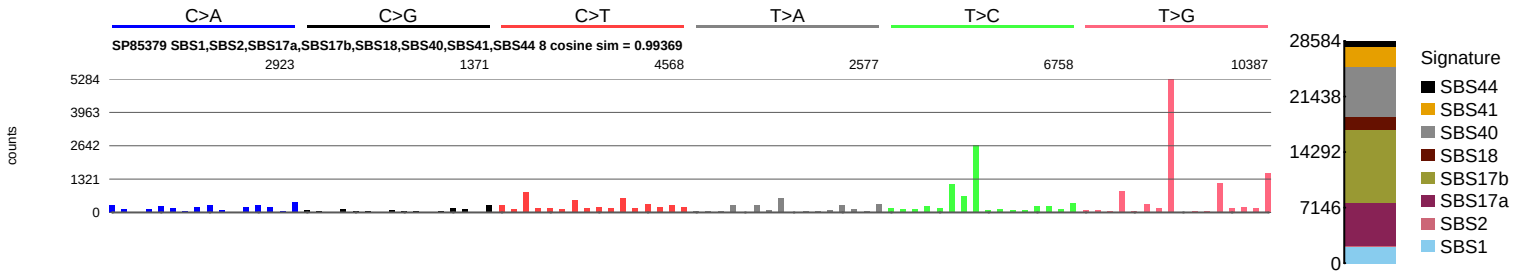
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99371
2884
1371
4438
2532
6959
10396
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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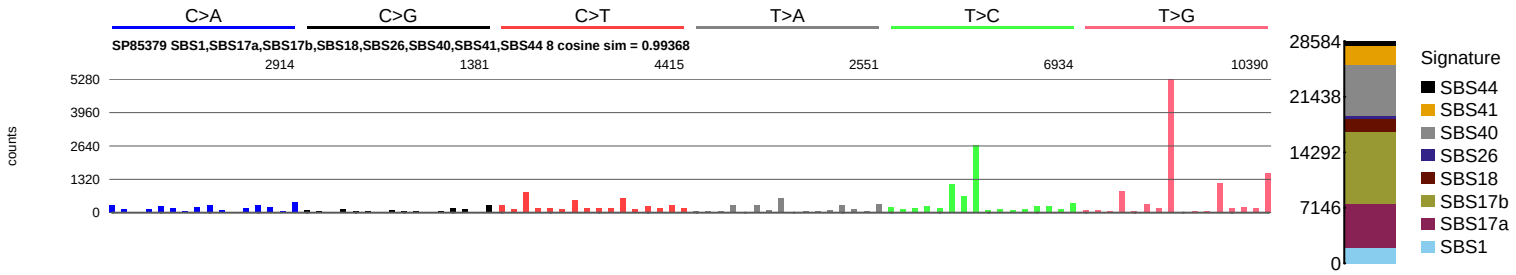
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99369
2923
1371
4568
2577
6758
10387
counts
5284
3963
2642
1321
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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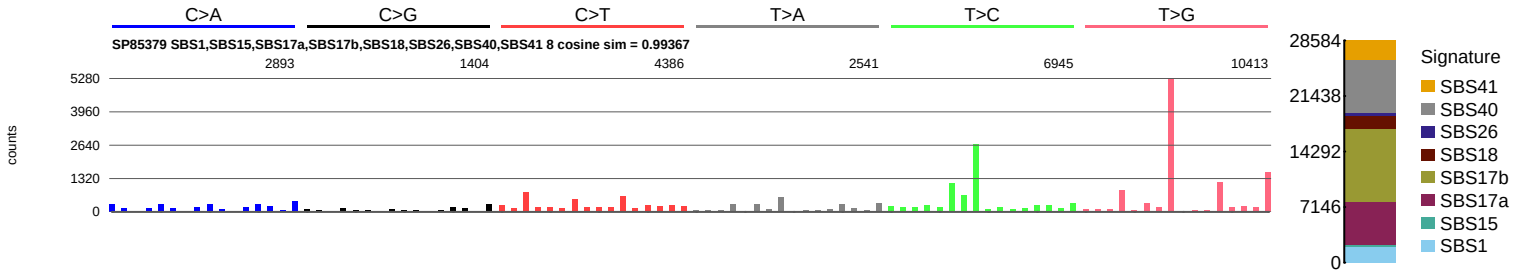
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99368
2914
1381
4415
2551
6934
10390
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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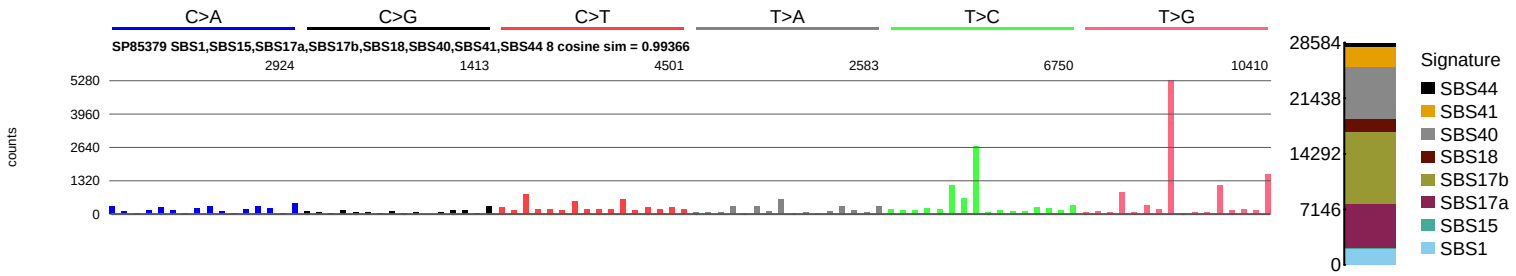
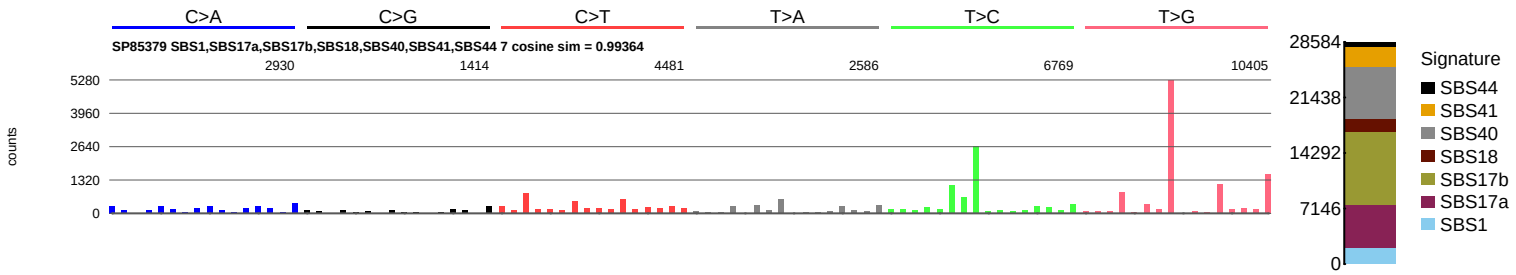
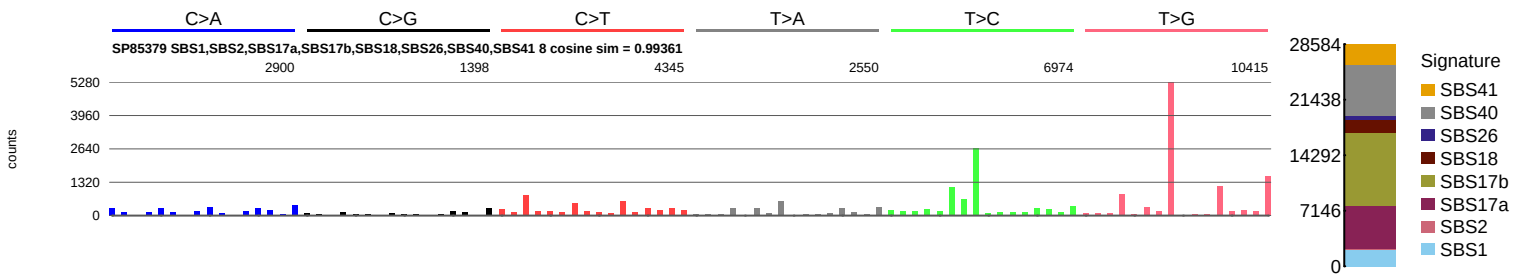
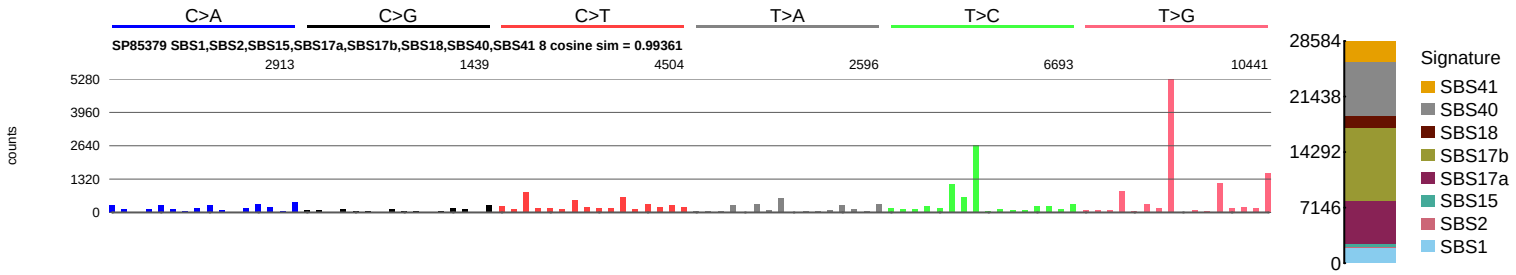
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99367
2893
1404
4386
2541
6945
10413
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99366
2924
1413
4501
2583
6750
10410
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.99364
2930
1414
4481
2586
6769
10405
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99361
2900
1398
4345
2550
6974
10415
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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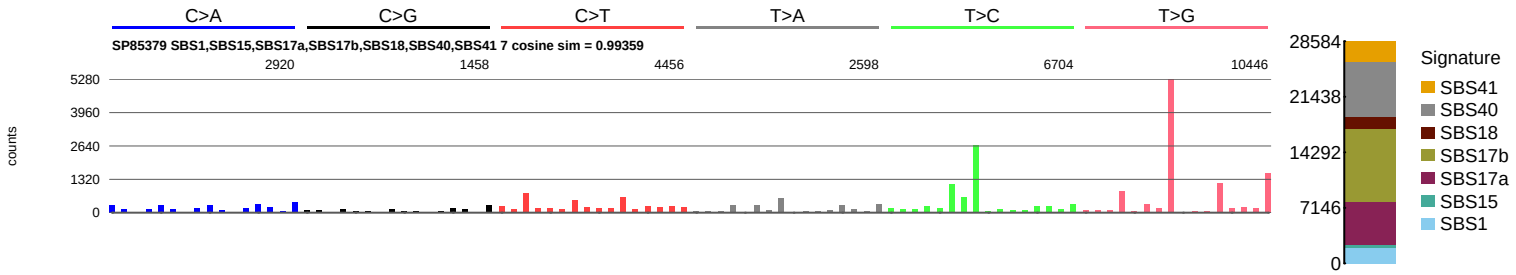
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99361
2913
1439
4504
2596
6693
10441
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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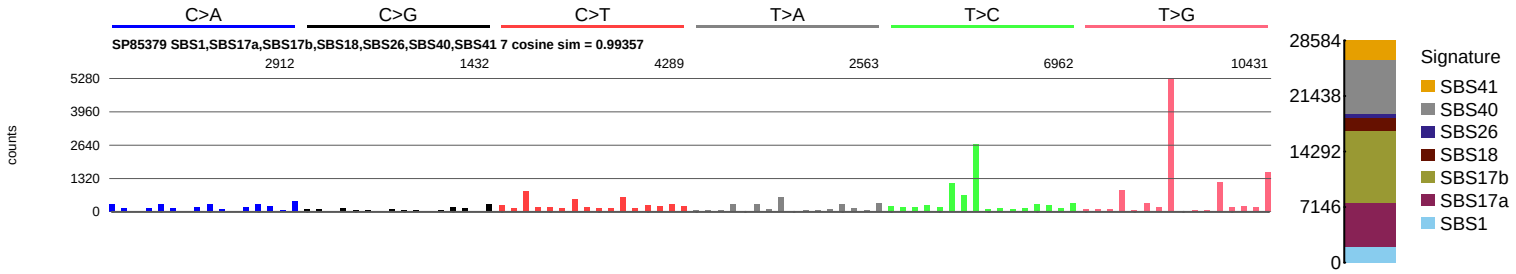
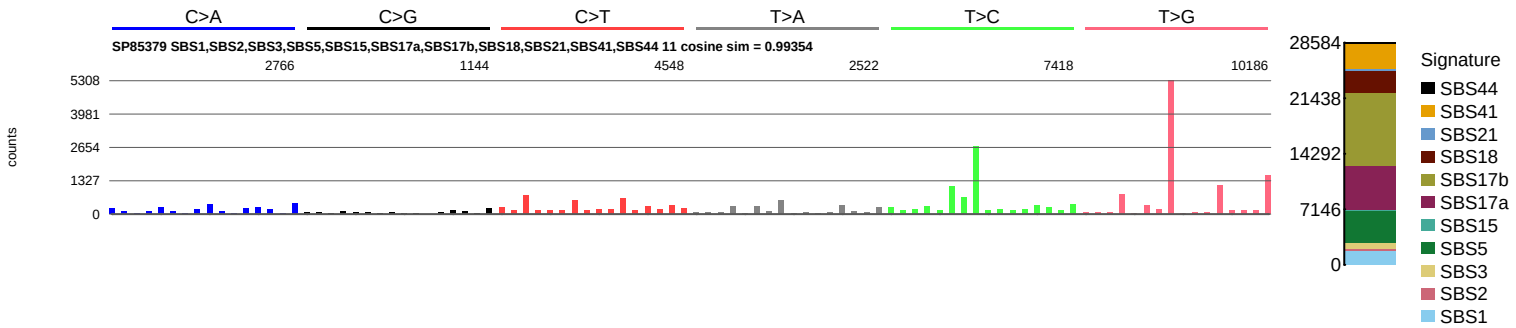
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99359
2920
1458
4456
2598
6704
10446
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.99357
2912
1432
4289
2563
6962
10431
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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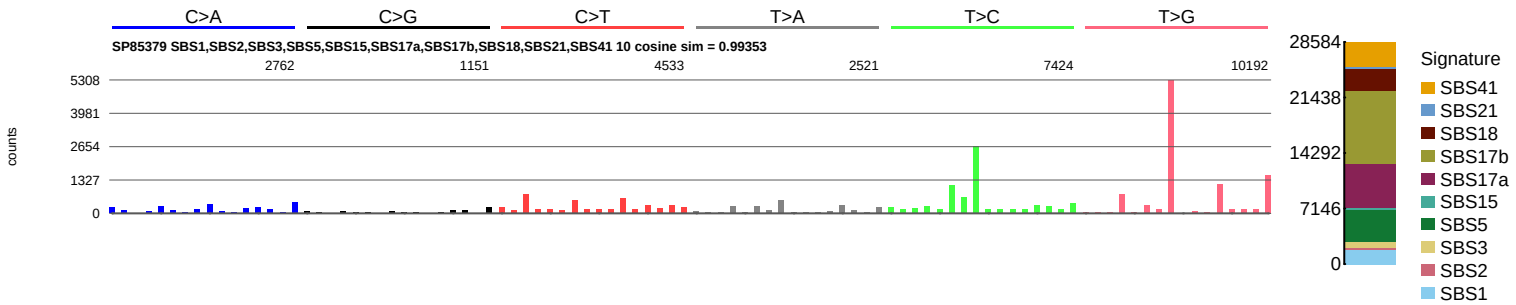
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.99354
2766
1144
4548
2522
7418
10186
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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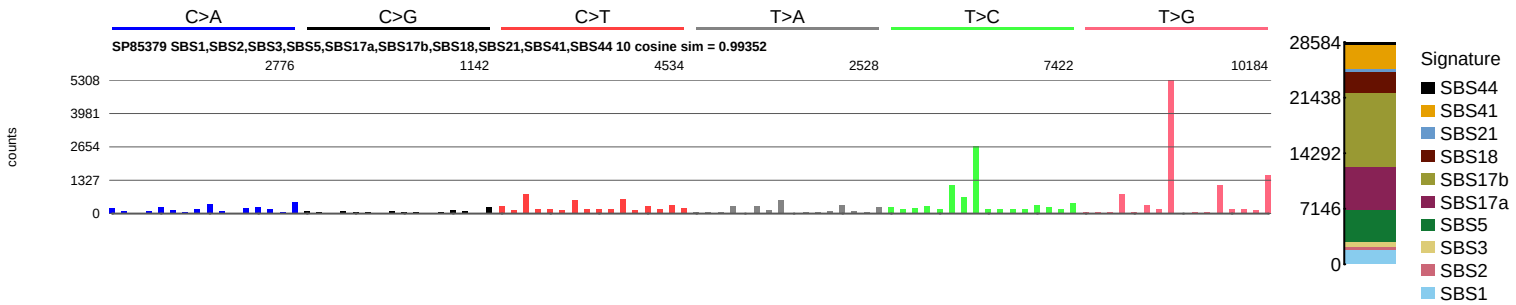
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99353
2762
1151
4533
2521
7424
10192
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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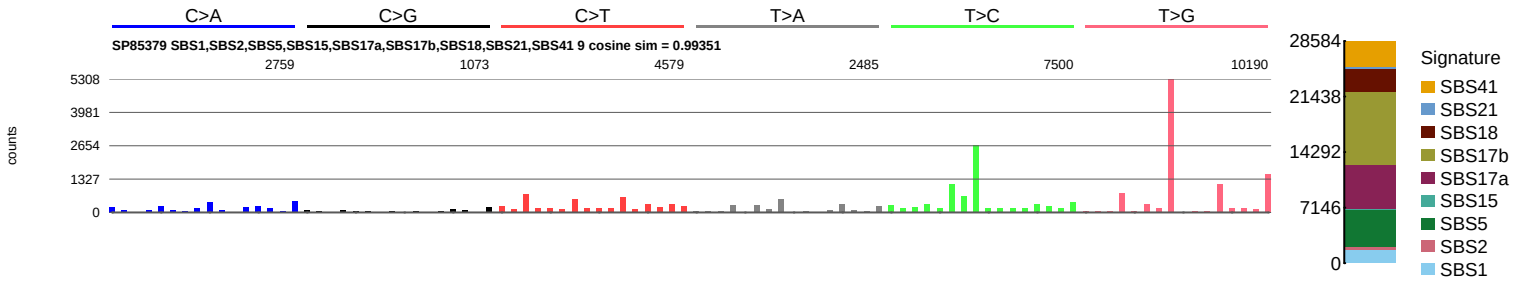
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99352
2776
1142
4534
2528
7422
10184
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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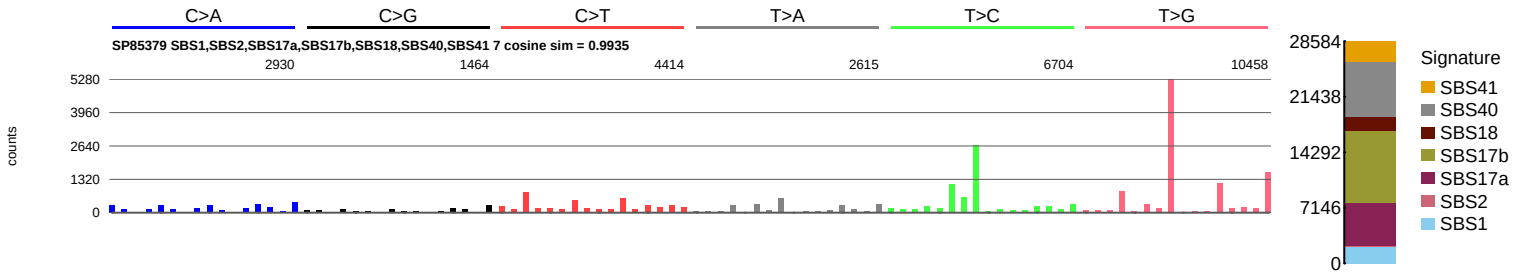
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99351
2759
1073
4579
2485
7500
10190
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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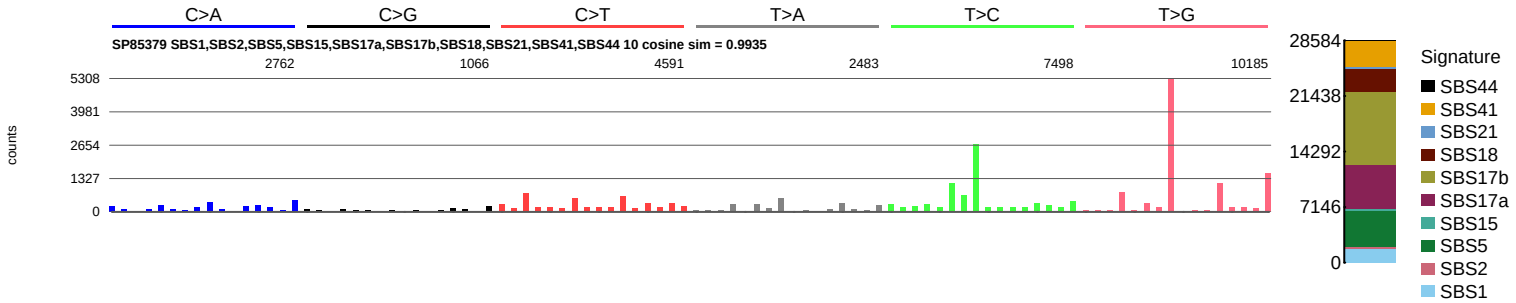
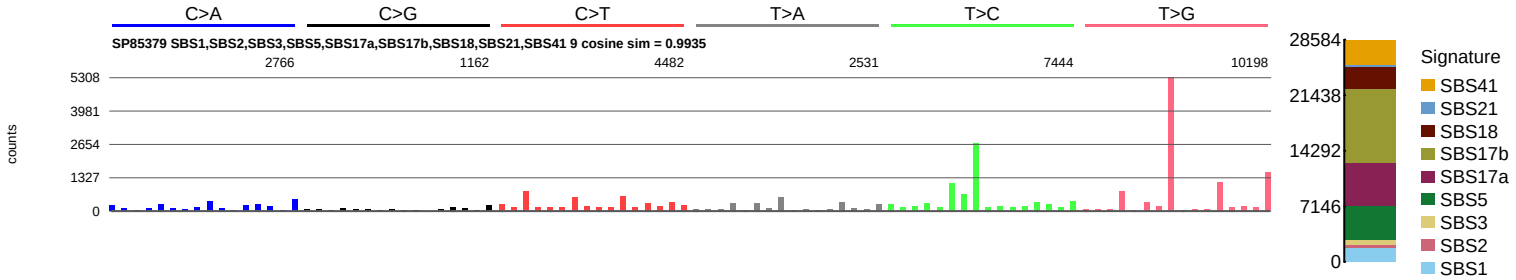
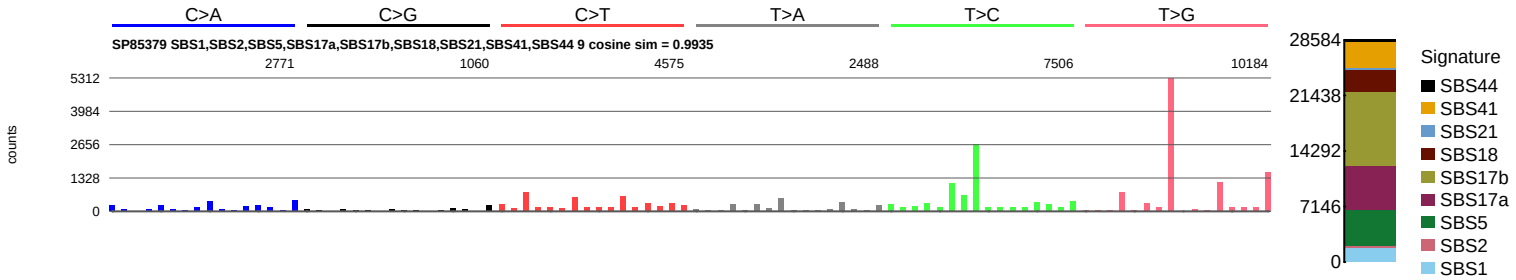
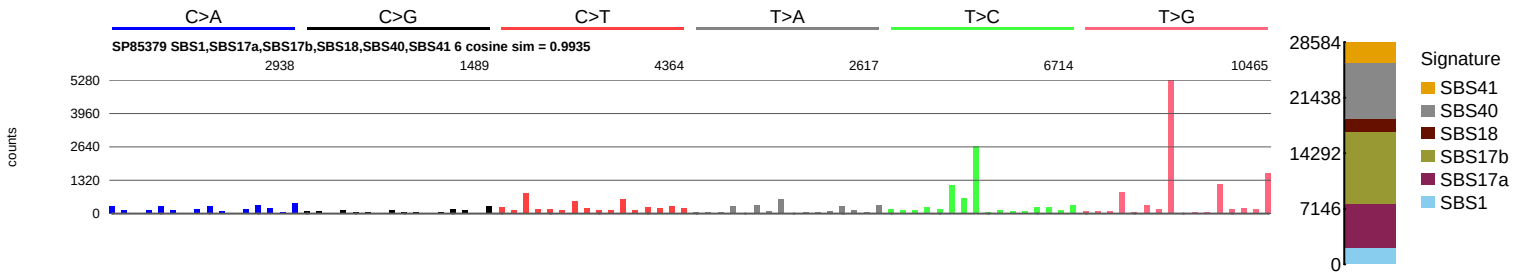
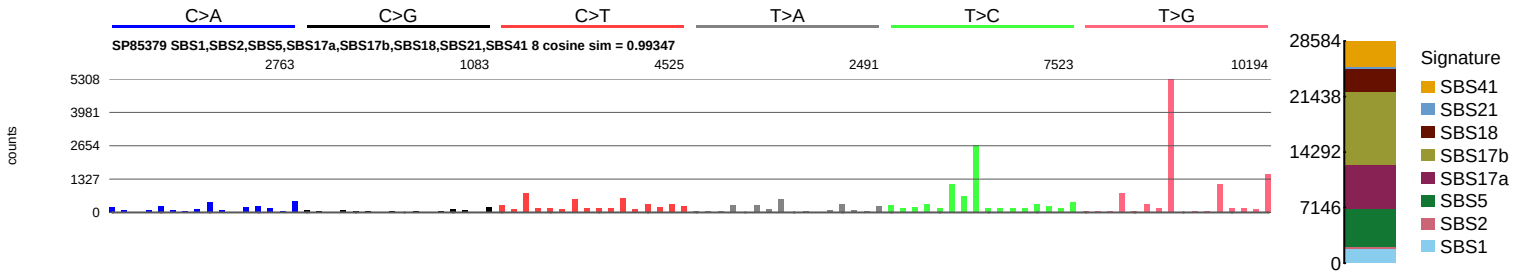
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.9935
2930
1464
4414
2615
6704
10458
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9935
2762
1066
4591
2483
7498
10185
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9935
2766
1162
4482
2531
7444
10198
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9935
2771
1060
4575
2488
7506
10184
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.9935
2938
1489
4364
2617
6714
10465
counts
5280
3960
2640
1320
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1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C>T
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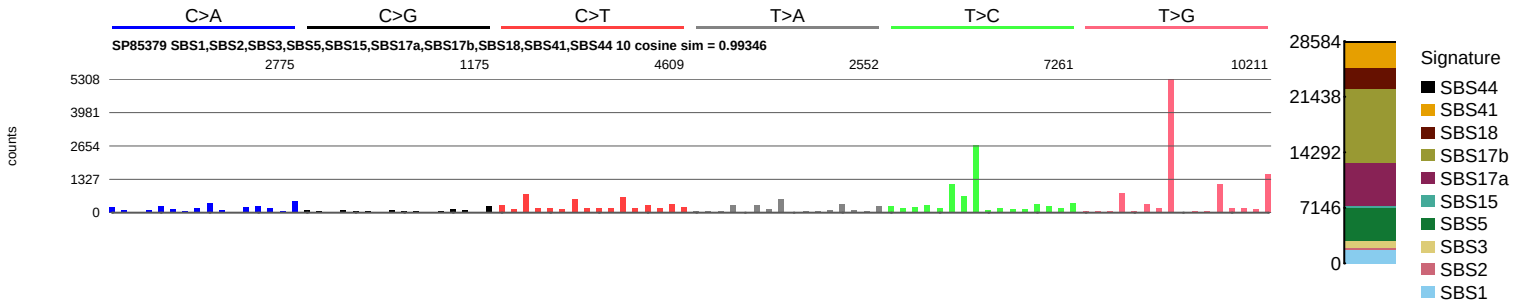
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99347
2763
1083
4525
2491
7523
10194
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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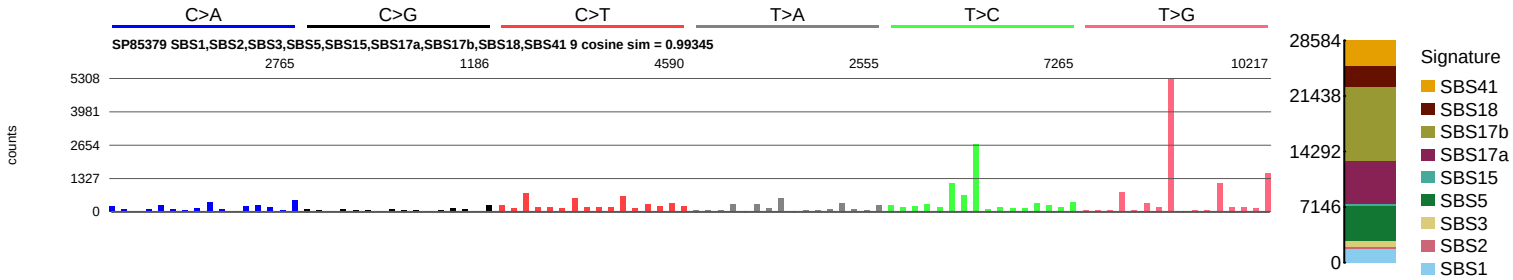
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.99346
2775
1175
4609
2552
7261
10211
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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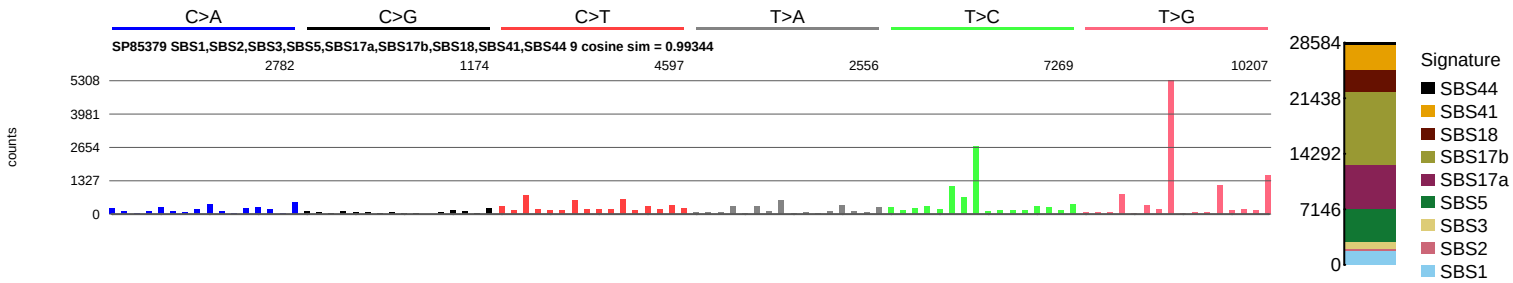
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99345
2765
1186
4590
2555
7265
10217
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
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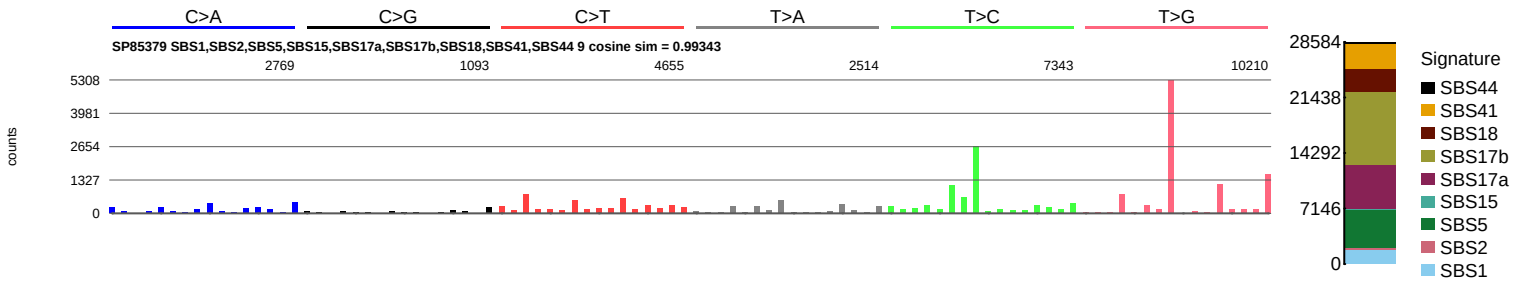
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99344
2782
1174
4597
2556
7269
10207
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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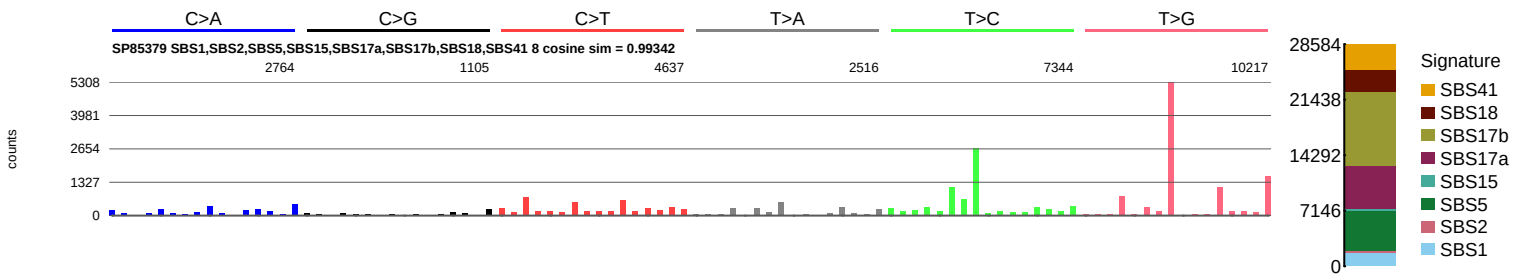
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99343
2769
1093
4655
2514
7343
10210
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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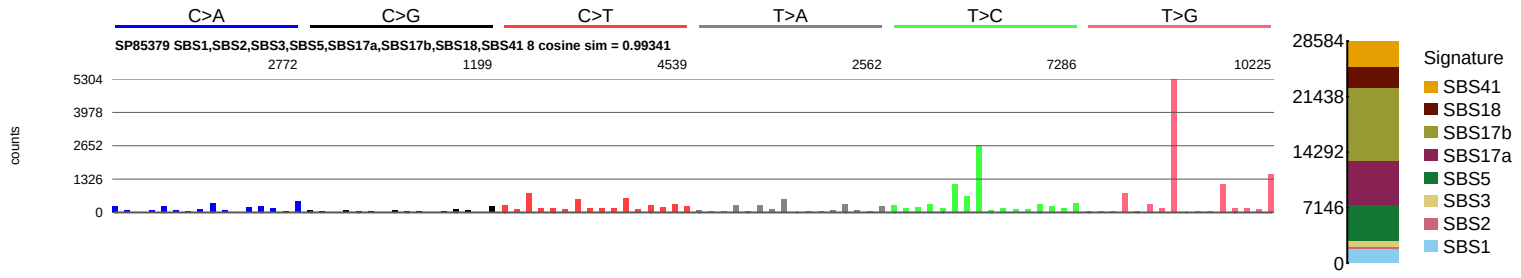
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99342
2764
1105
4637
2516
7344
10217
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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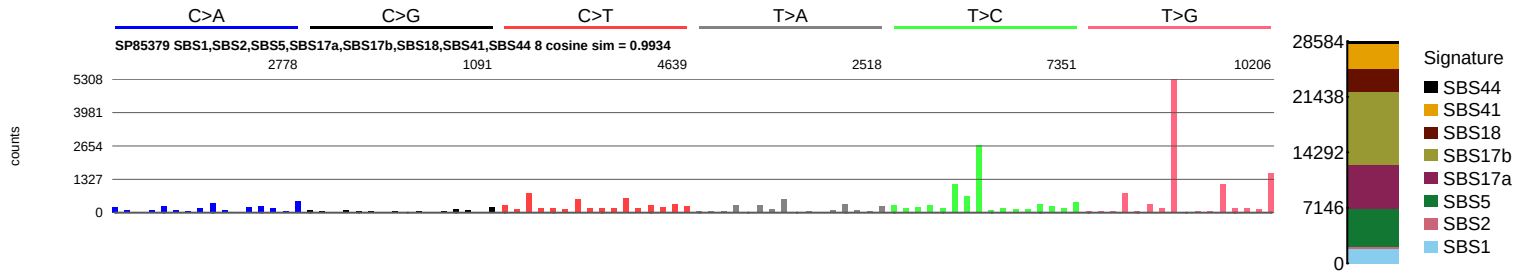
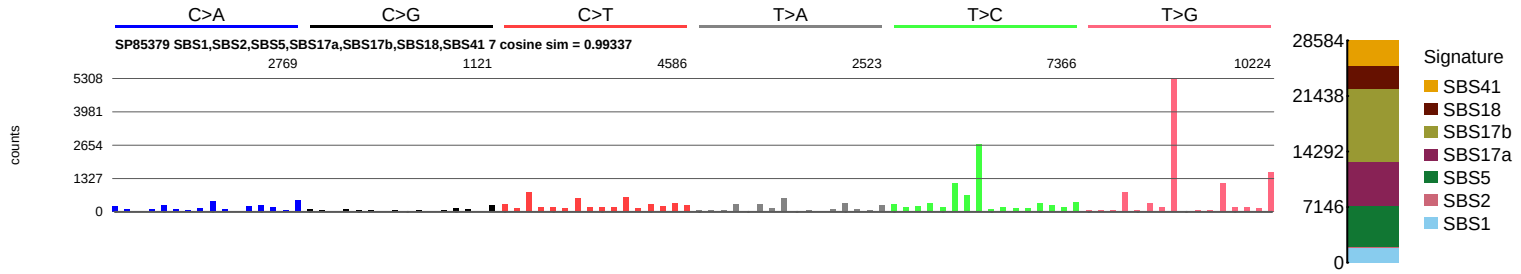
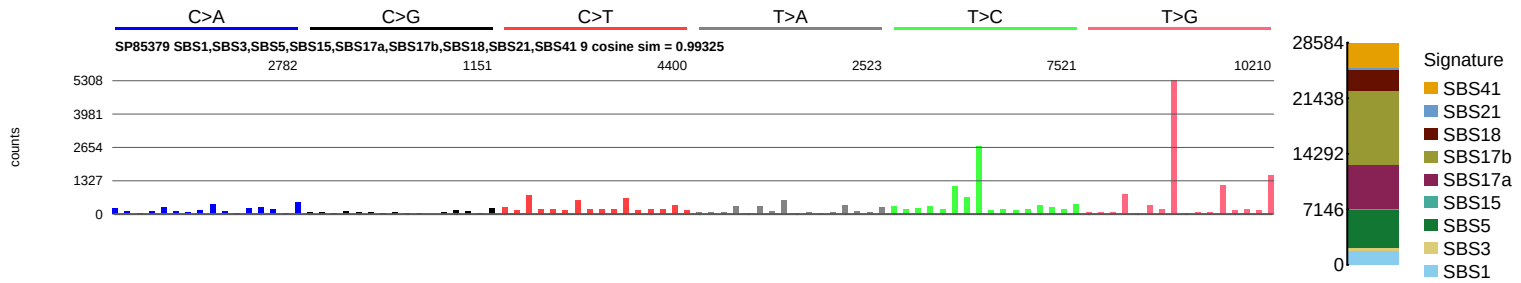
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99341
2772
1199
4539
2562
7286
10225
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9934
2778
1091
4639
2518
7351
10206
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99337
2769
1121
4586
2523
7366
10224
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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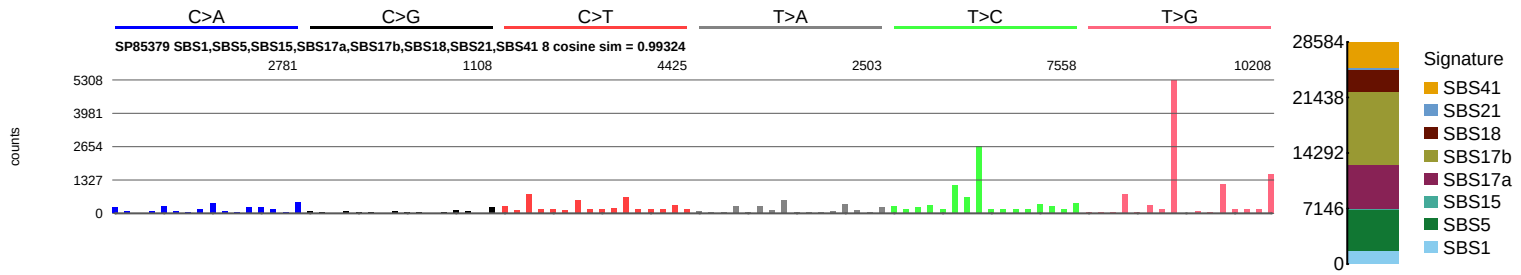
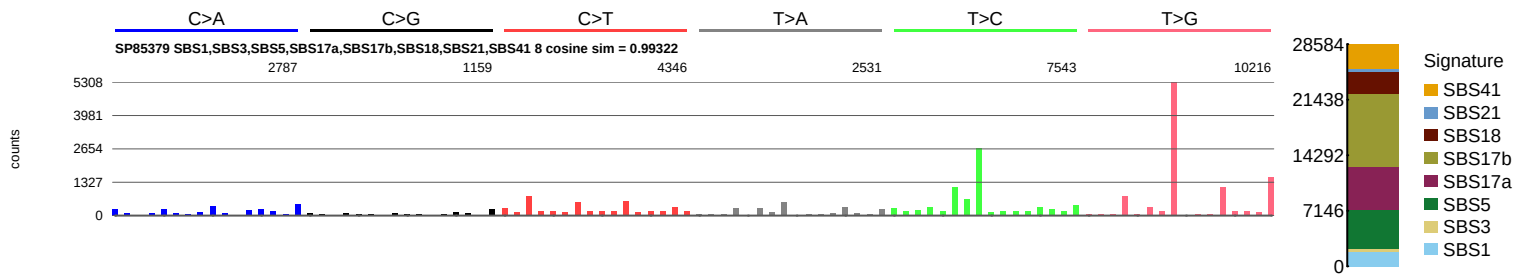
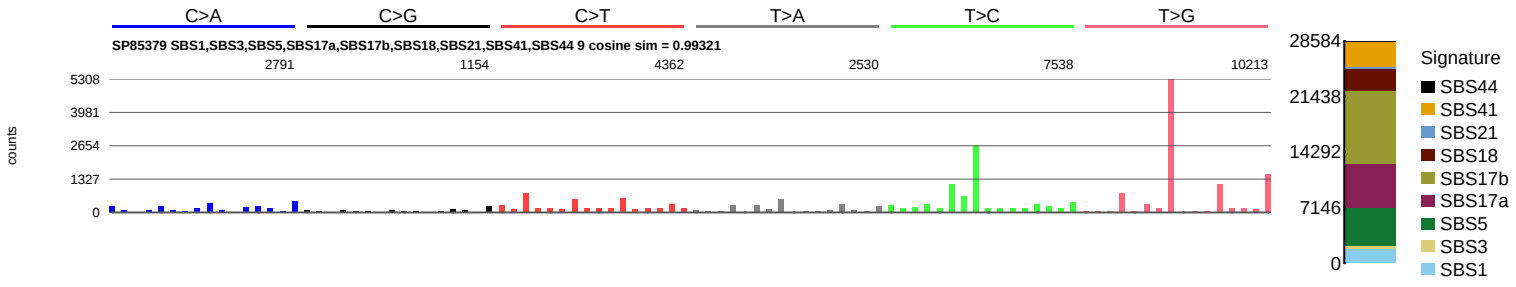
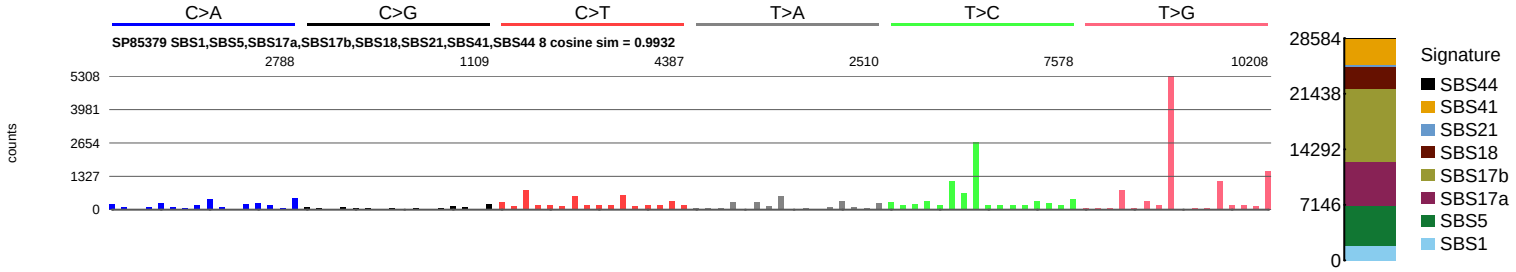
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99325
2782
1151
4400
2523
7521
10210
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99324
2781
1108
4425
2503
7558
10208
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99322
2787
1159
4346
2531
7543
10216
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99321
2791
1154
4362
2530
7538
10213
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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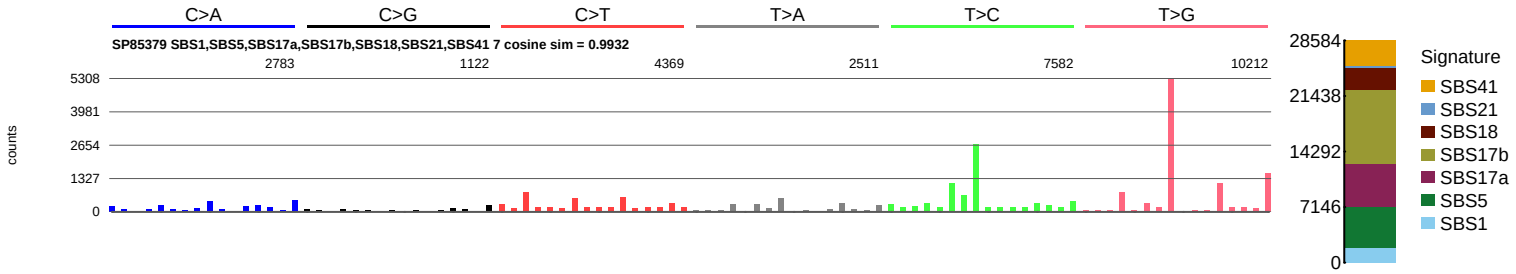
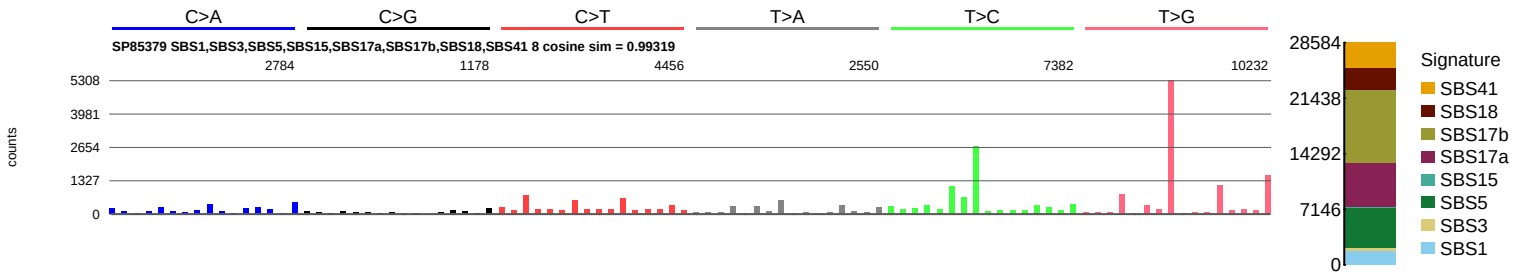
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9932
2788
1109
4387
2510
7578
10208
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9932
2783
1122
4369
2511
7582
10212
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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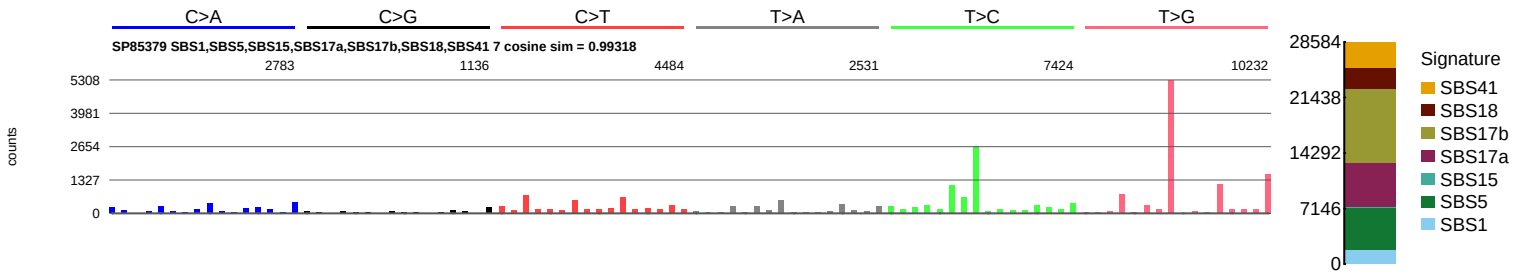
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99319
2784
1178
4456
2550
7382
10232
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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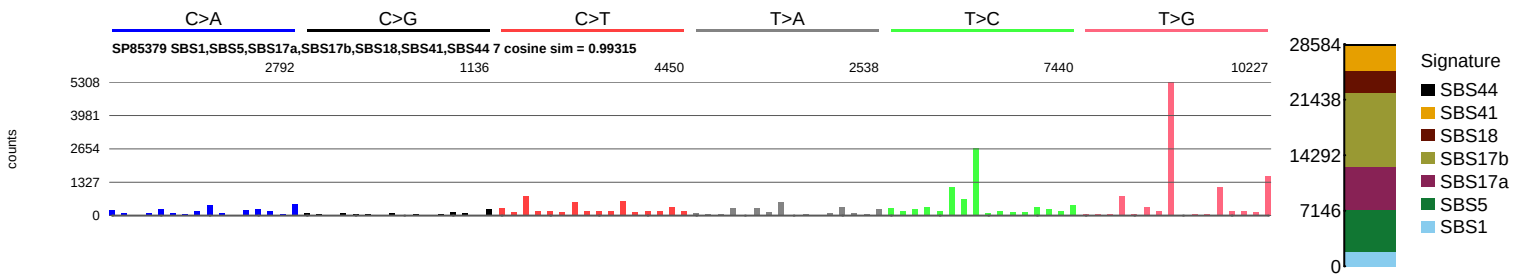
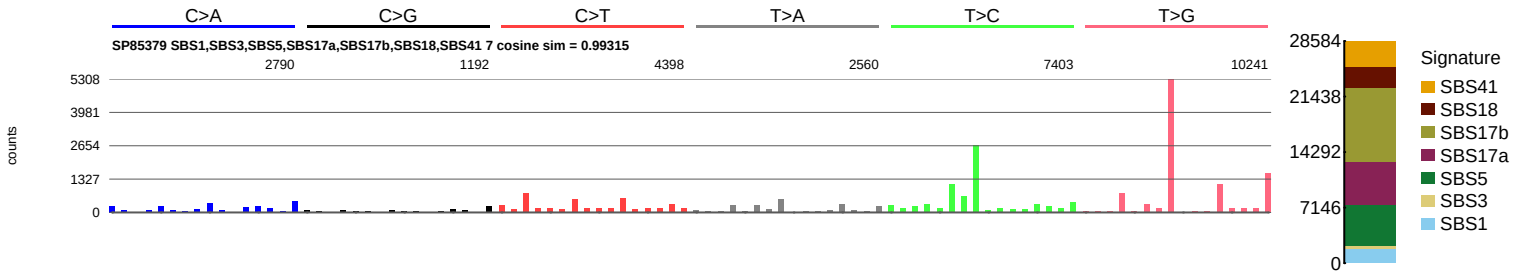
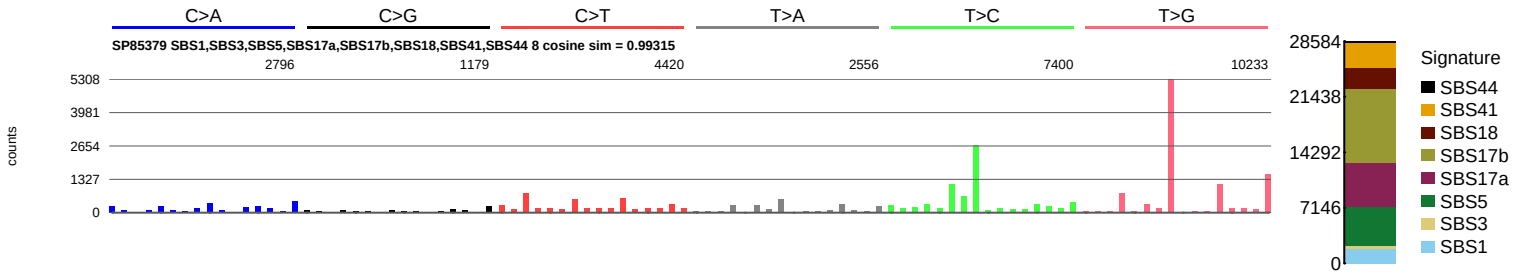
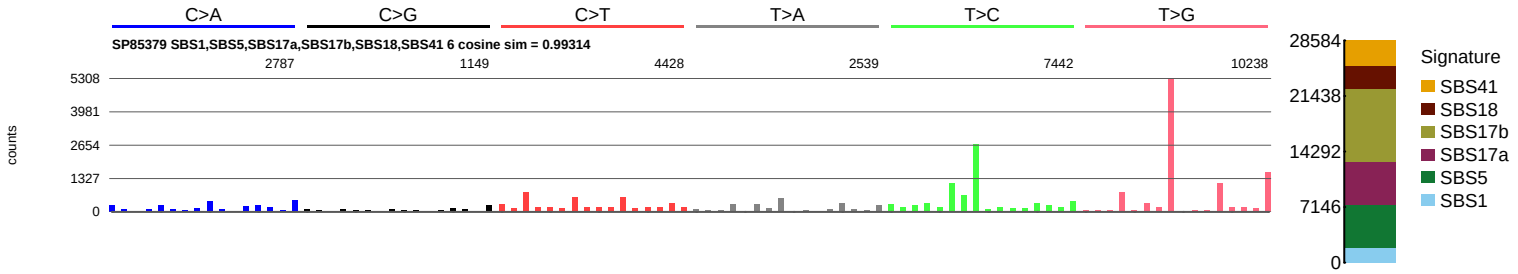
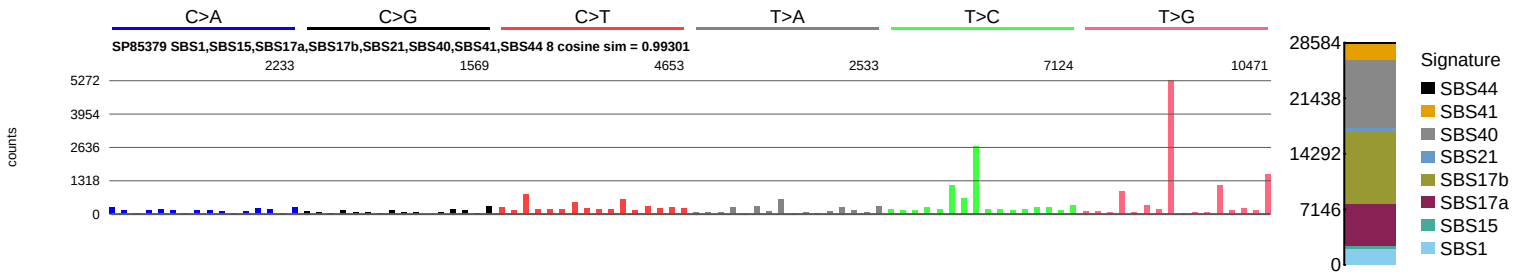
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99318
2783
1136
4484
2531
7424
10232
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

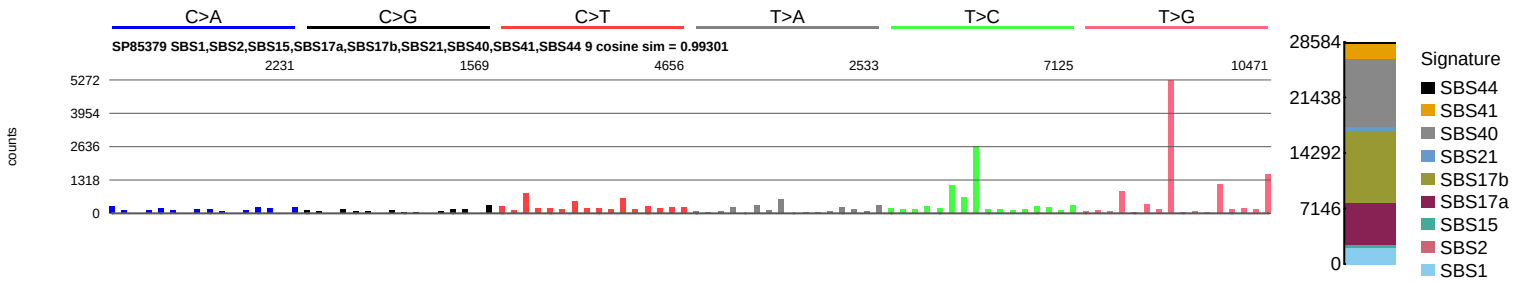

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.99315
2792
1136
4450
2538
7440
10227
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1

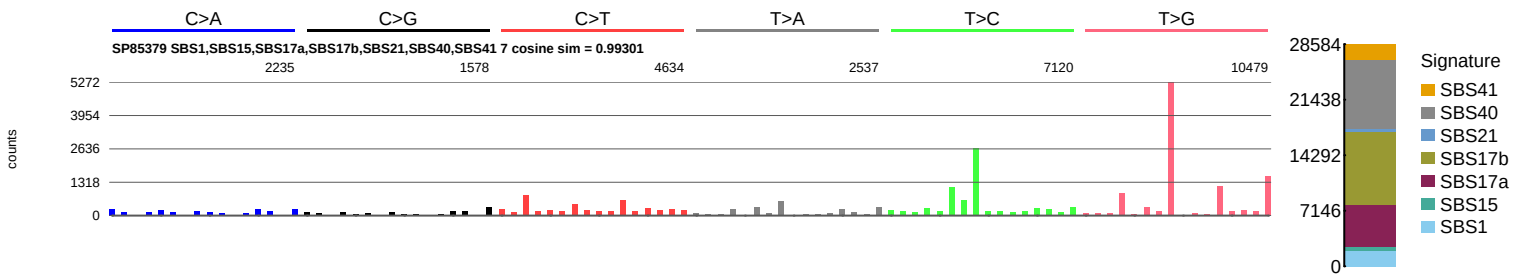

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99315
2790
1192
4398
2560
7403
10241
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

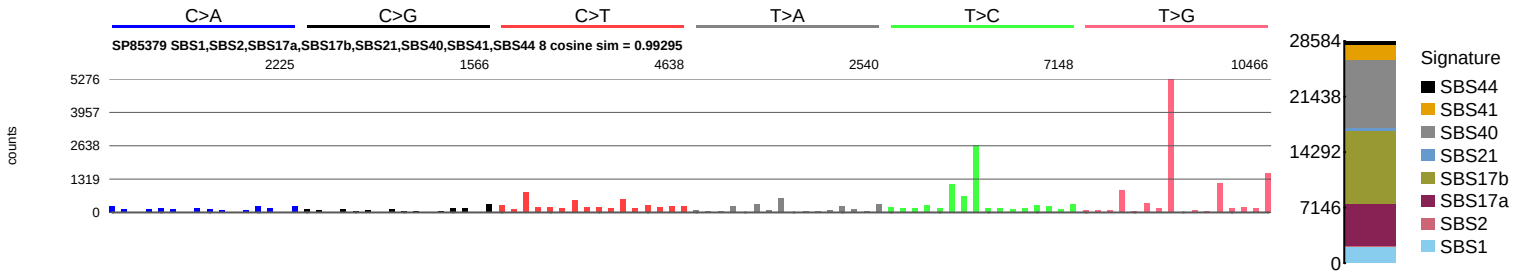

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99315
2796
1179
4420
2556
7400
10233
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

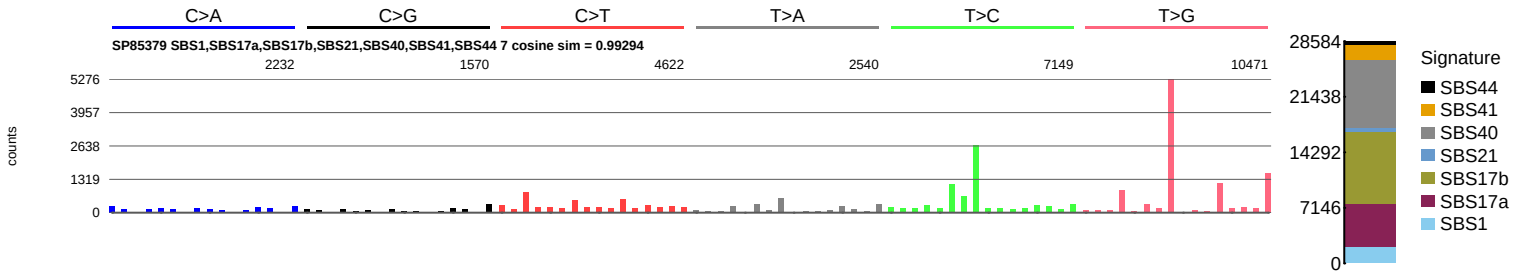

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99314
2787
1149
4428
2539
7442
10238
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1

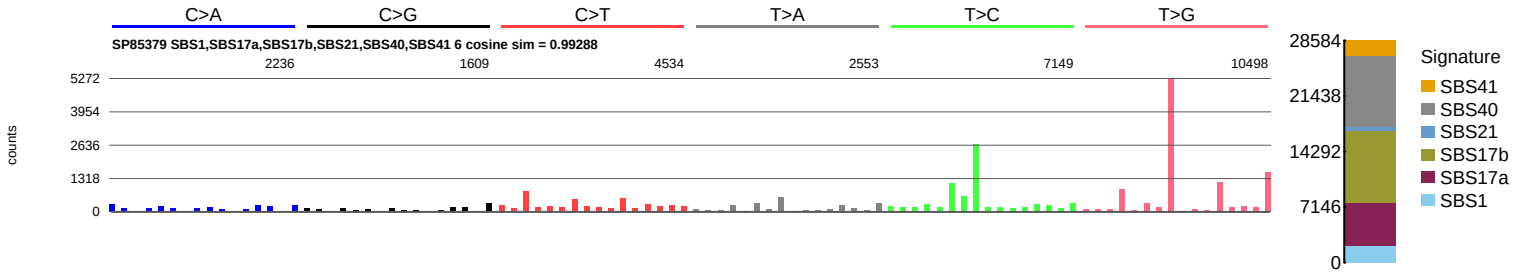

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99301
2233
1569
4653
2533
7124
10471
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99301
2231
1569
4656
2533
7125
10471
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99301
2235
1578
4634
2537
7120
10479
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99295
2225
1566
4638
2540
7148
10466
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.99294
2232
1570
4622
2540
7149
10471
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
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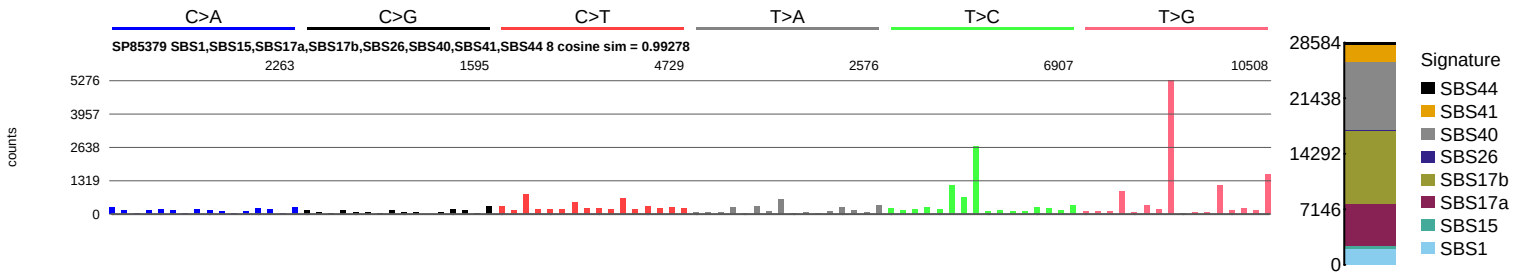
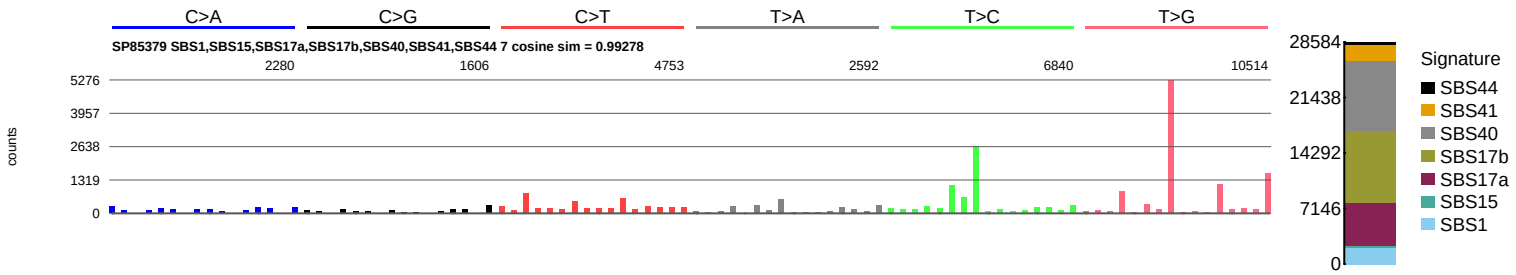
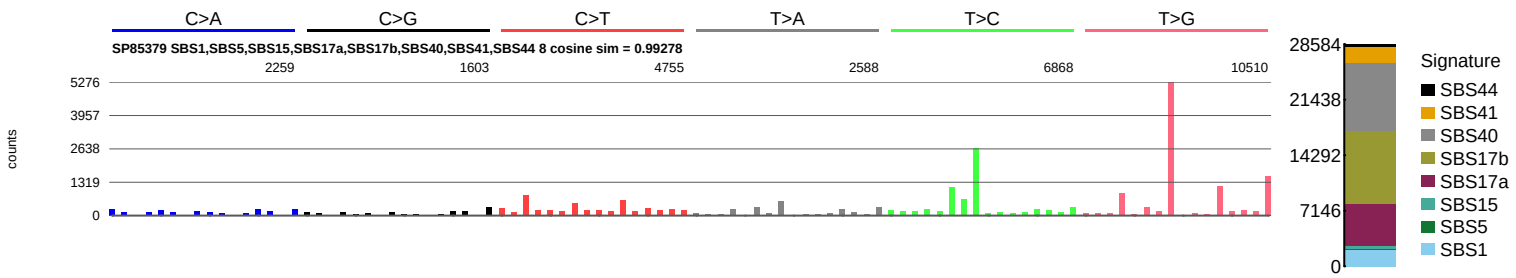
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99288
2236
1609
4534
2553
7149
10498
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99278
2263
1595
4729
2576
6907
10508
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99278
2280
1606
4753
2592
6840
10514
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1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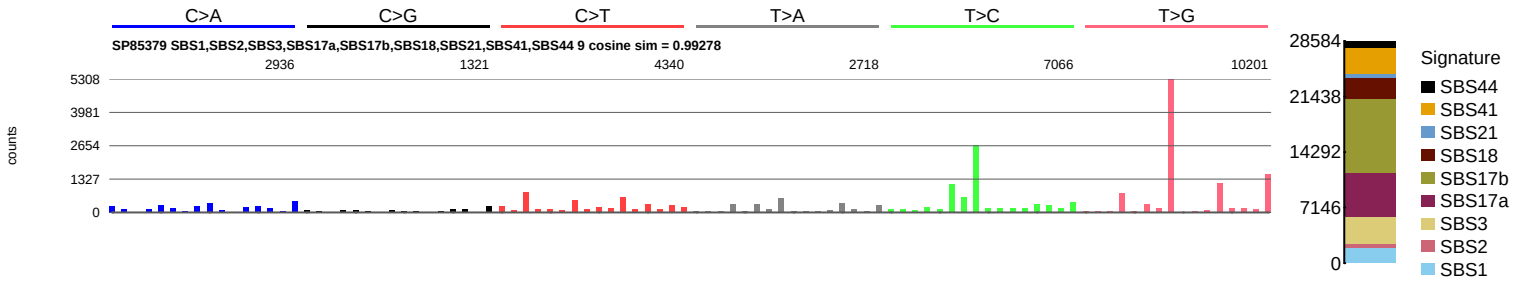
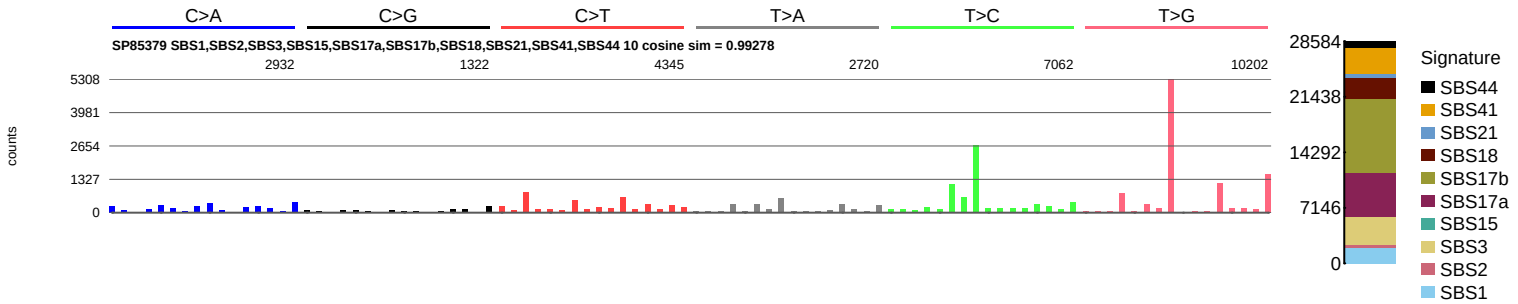
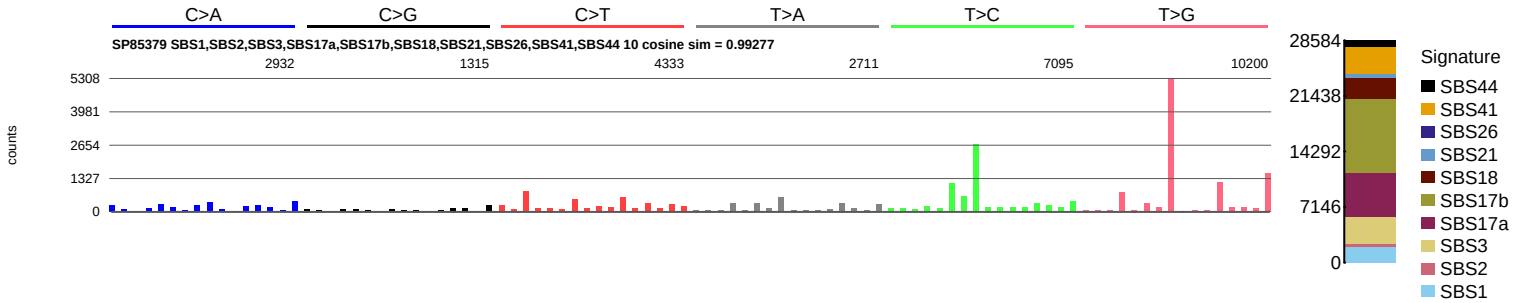
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99278
2259
1603
4755
2588
6868
10510
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99278
2936
1321
4340
2718
7066
10201
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99278
2932
1322
4345
2720
7062
10202
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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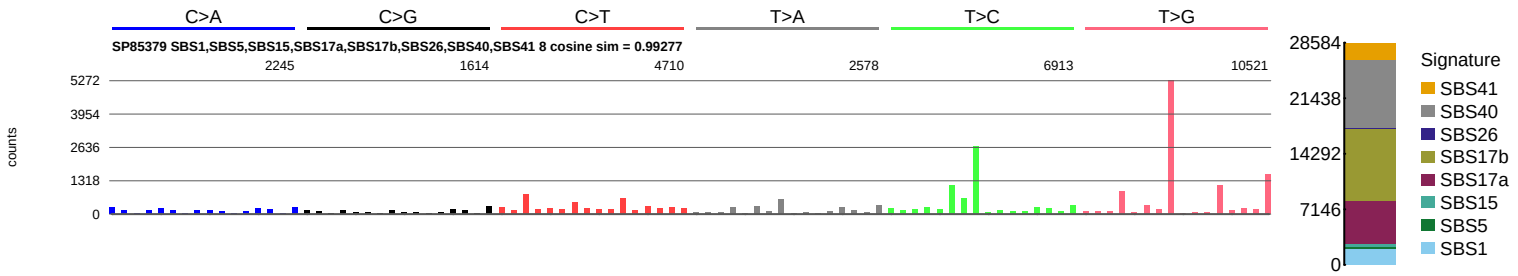
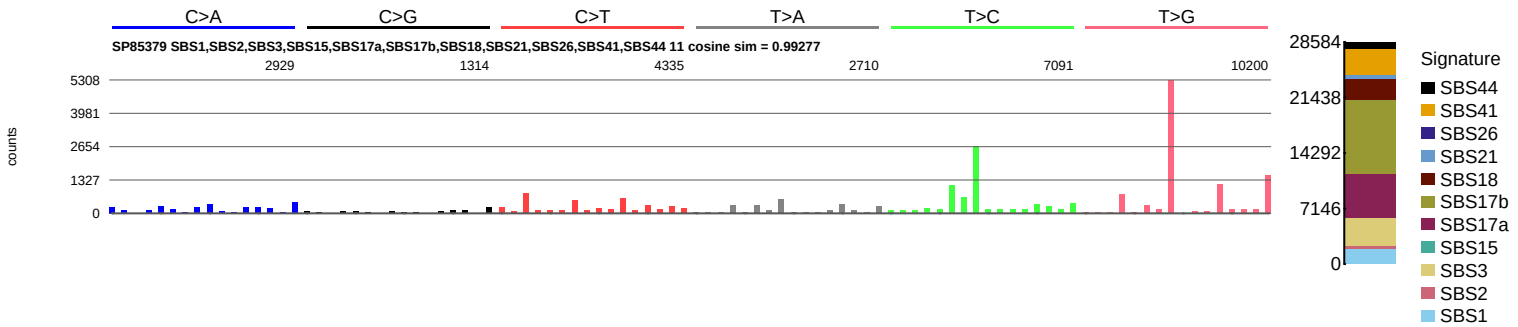
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99277
2932
1315
4333
2711
7095
10200
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99277
2245
1614
4710
2578
6913
10521
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
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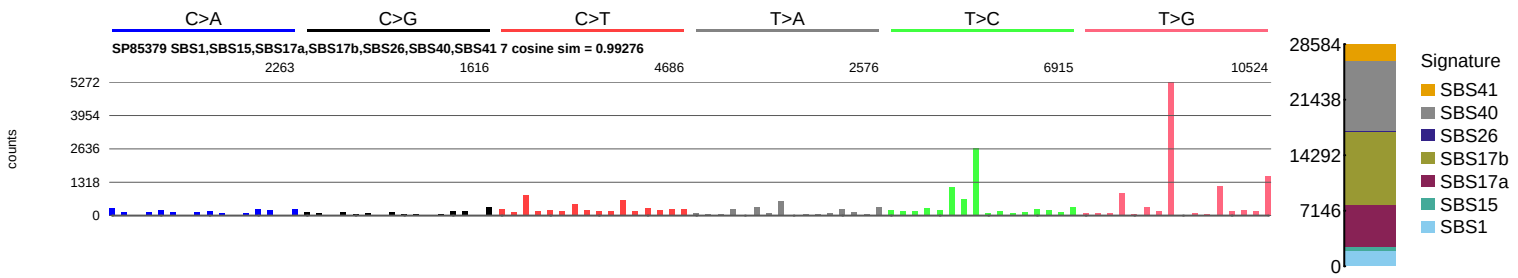
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99277
2929
1314
4335
2710
7091
10200
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99276
2263
1616
4686
2576
6915
10524
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
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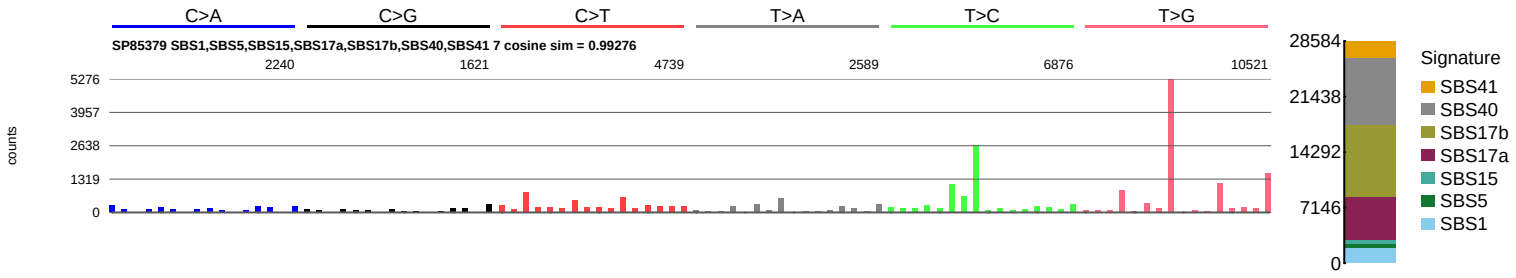
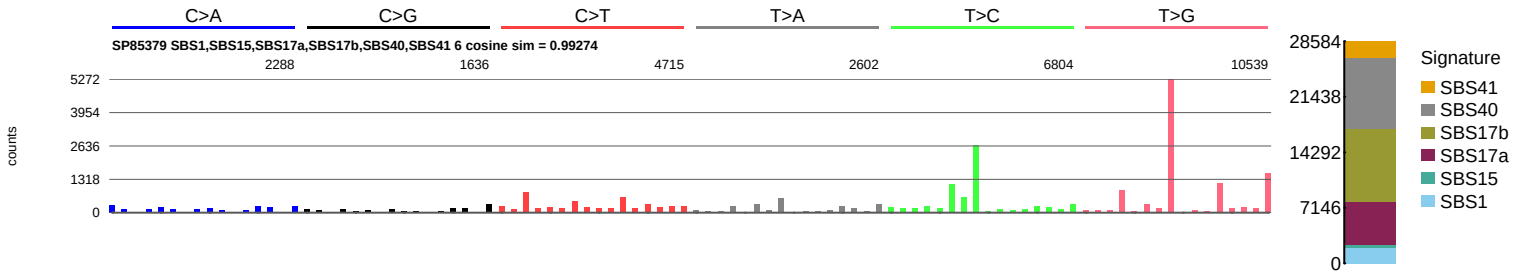
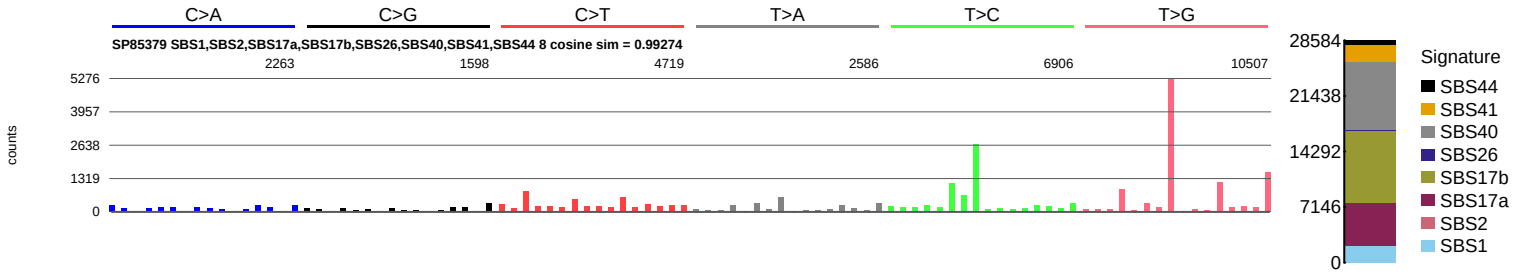
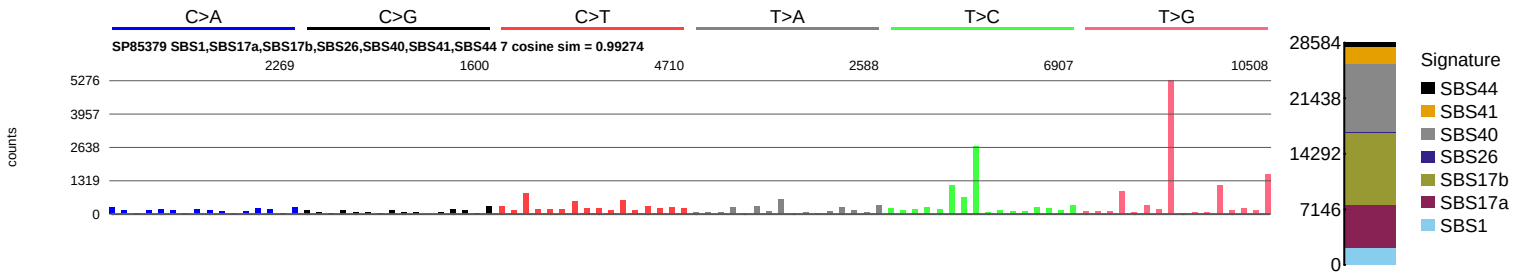
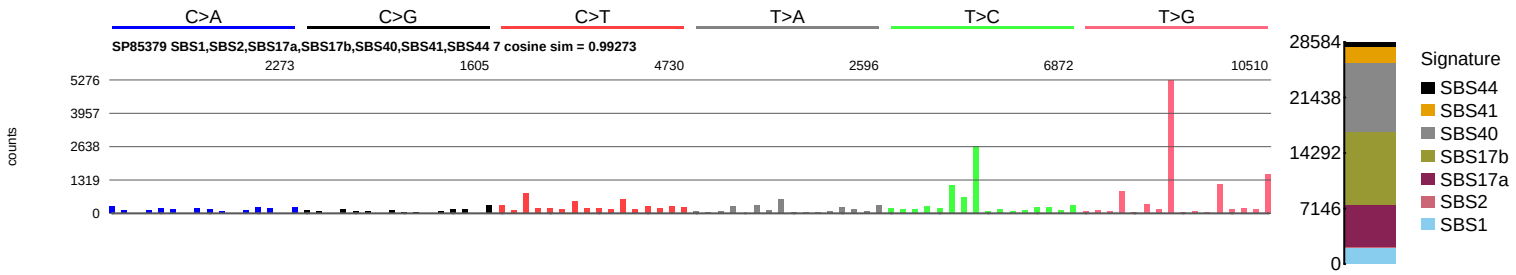
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

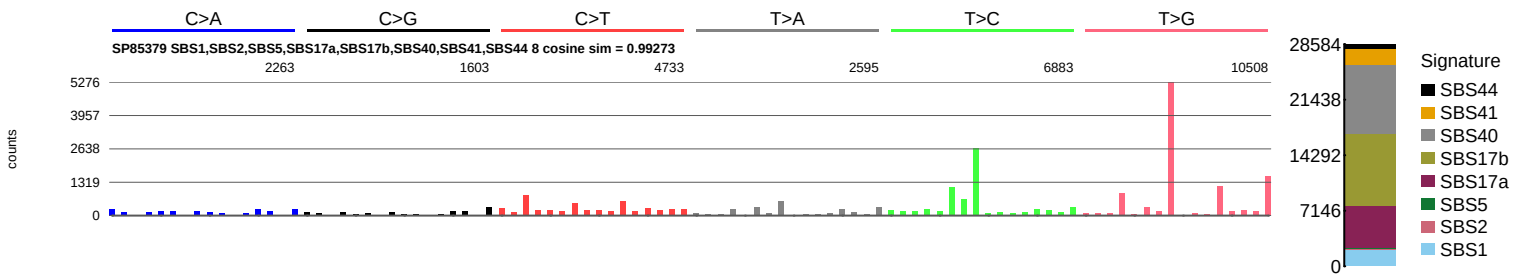

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99276
2240
1621
4739
2589
6876
10521
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1

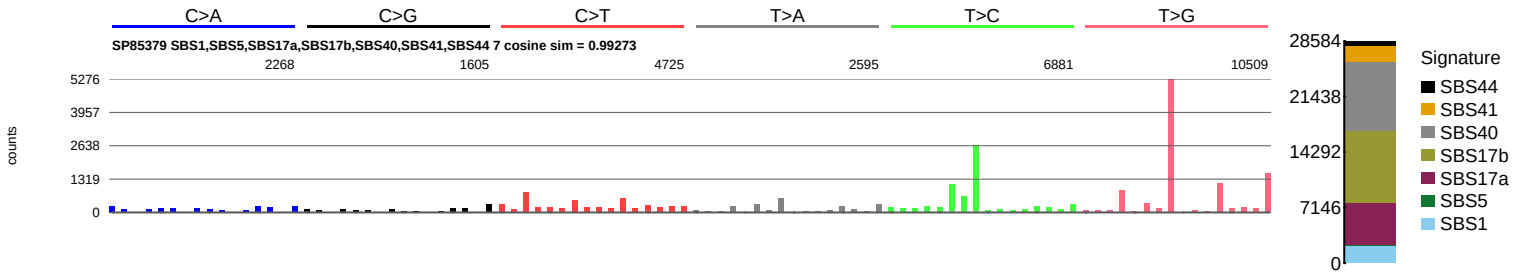

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99274
2288
1636
4715
2602
6804
10539
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

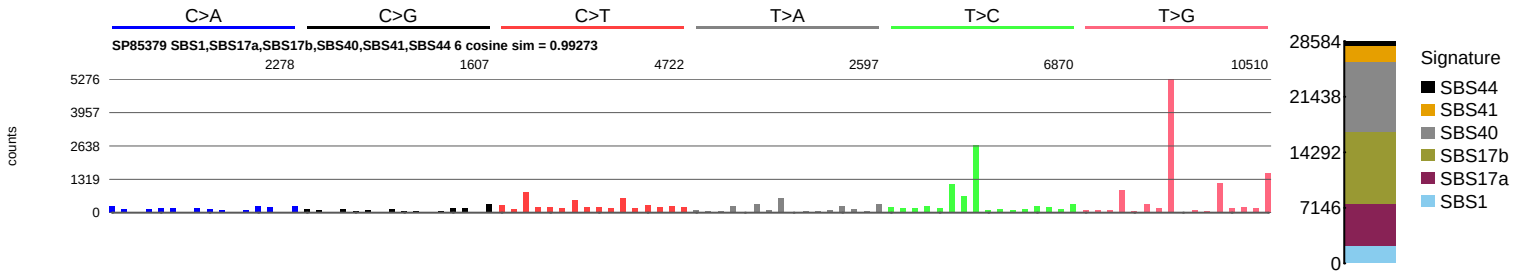

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99274
2263
1598
4719
2586
6906
10507
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

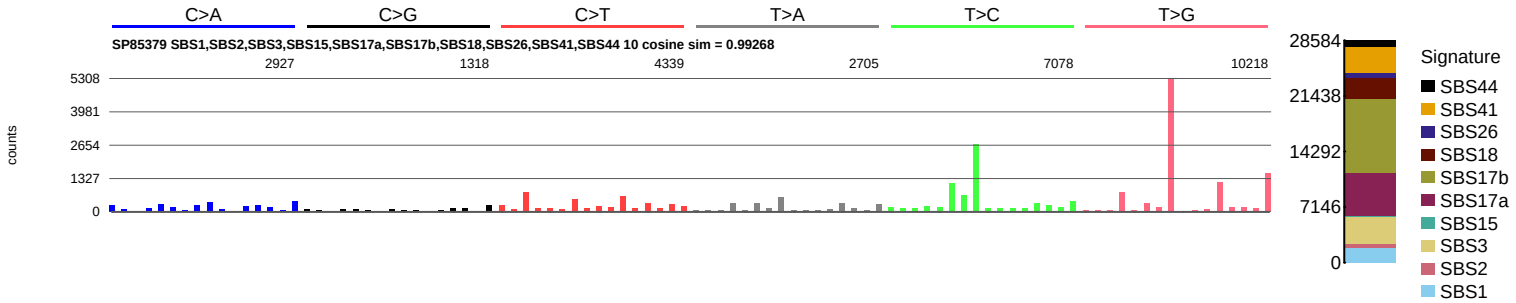

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.99274
2269
1600
4710
2588
6907
10508
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1

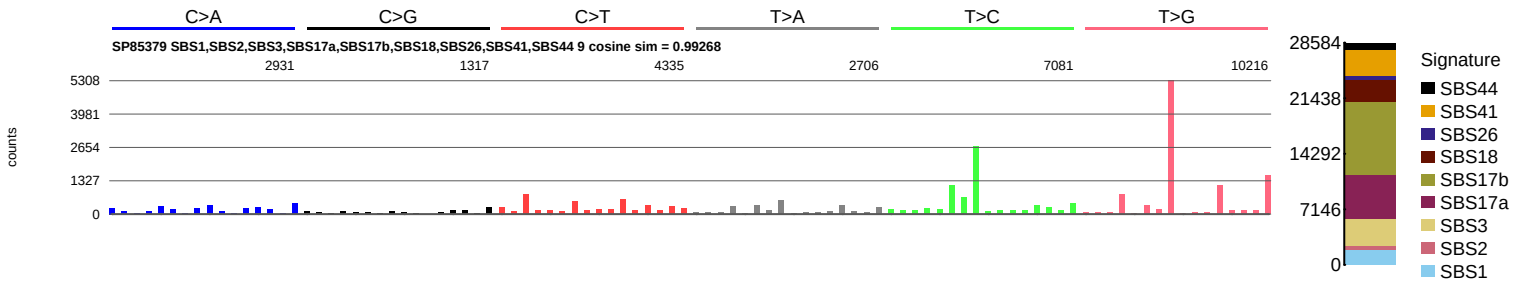

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99273
2273
1605
4730
2596
6872
10510
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1

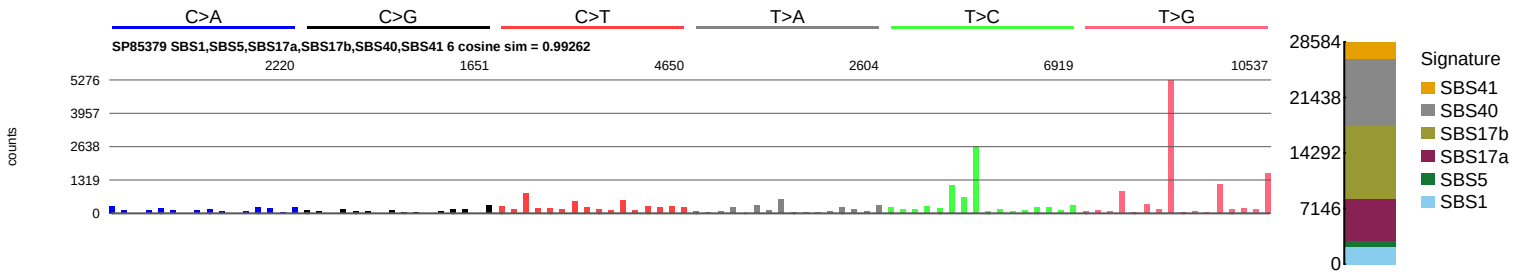

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99273
2263
1603
4733
2595
6883
10508
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1

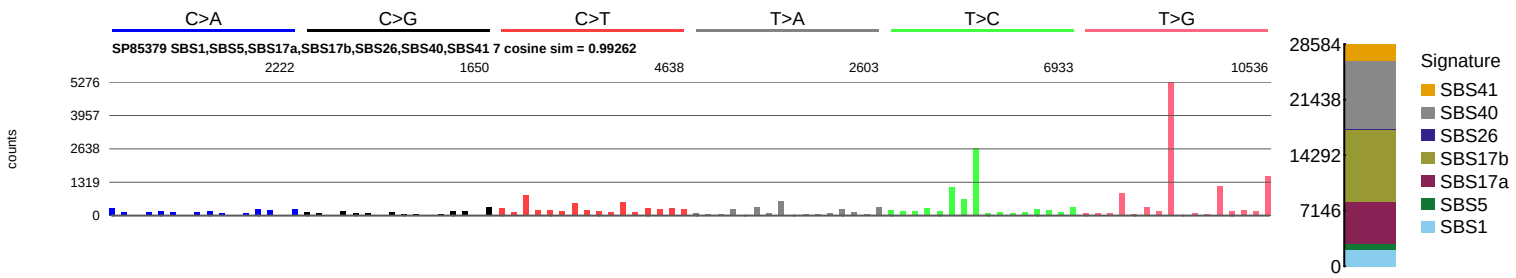

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99273
2268
1605
4725
2595
6881
10509
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1

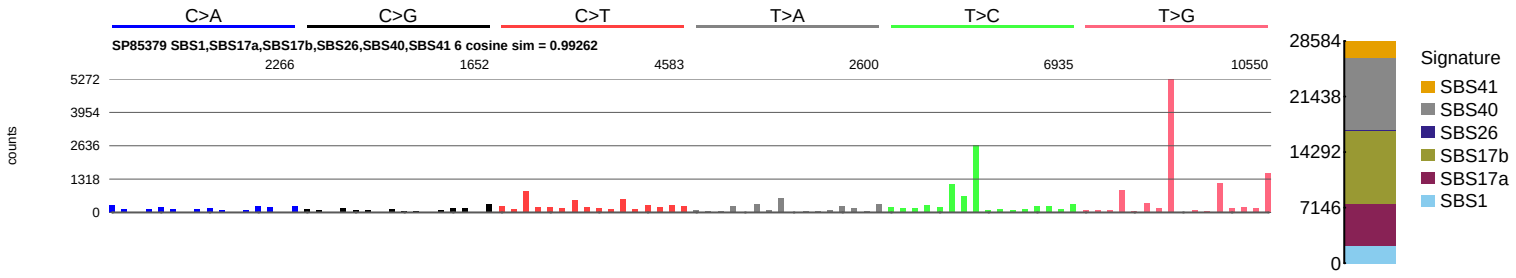

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.99273
2278
1607
4722
2597
6870
10510
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1

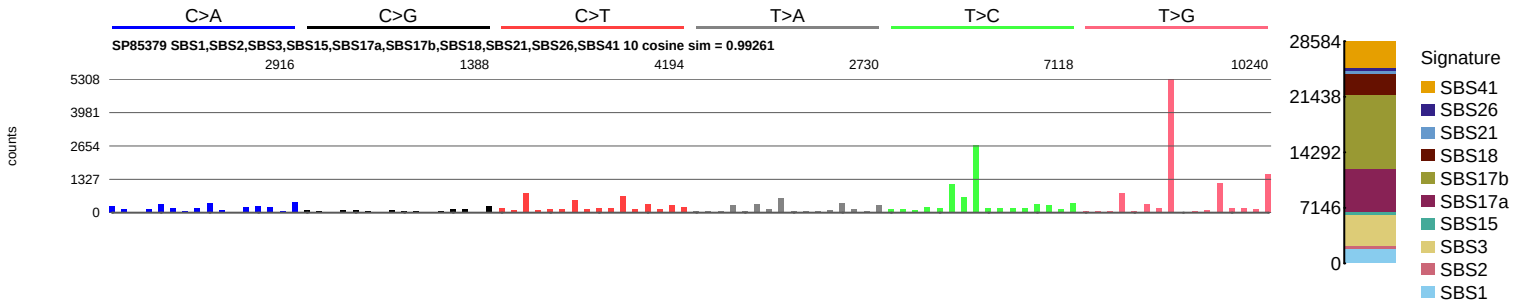

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99268
2927
1318
4339
2705
7078
10218
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99268
2931
1317
4335
2706
7081
10216
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99262
2220
1651
4650
2604
6919
10537
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99262
2222
1650
4638
2603
6933
10536
counts
5276
3957
2638
1319
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99262
2266
1652
4583
2600
6935
10550
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1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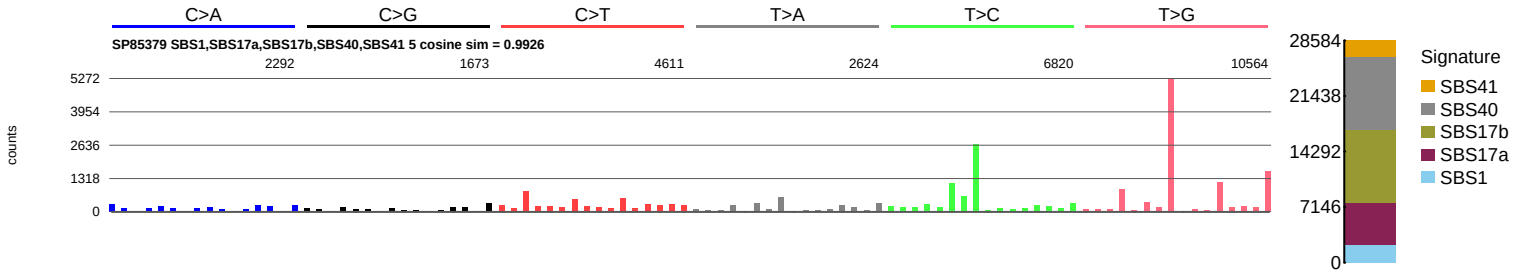
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99261
2916
1388
4194
2730
7118
10240
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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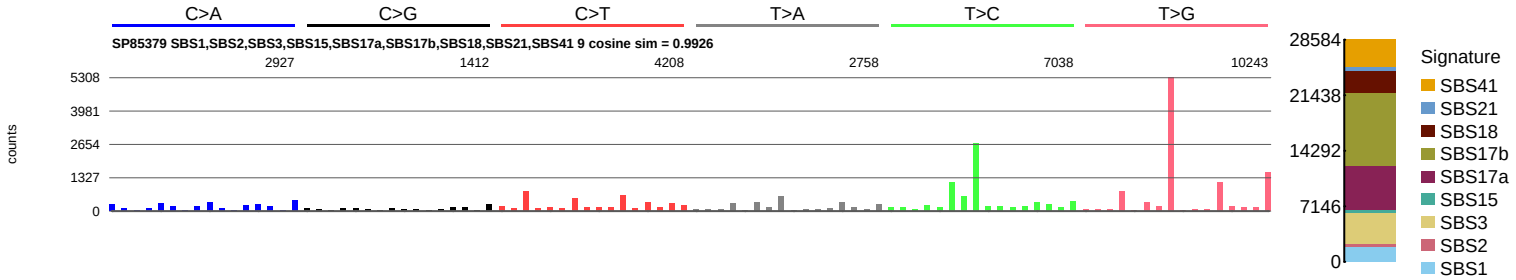
SP85379 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.9926
2292
1673
4611
2624
6820
10564
counts
5272
3954
2636
1318
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1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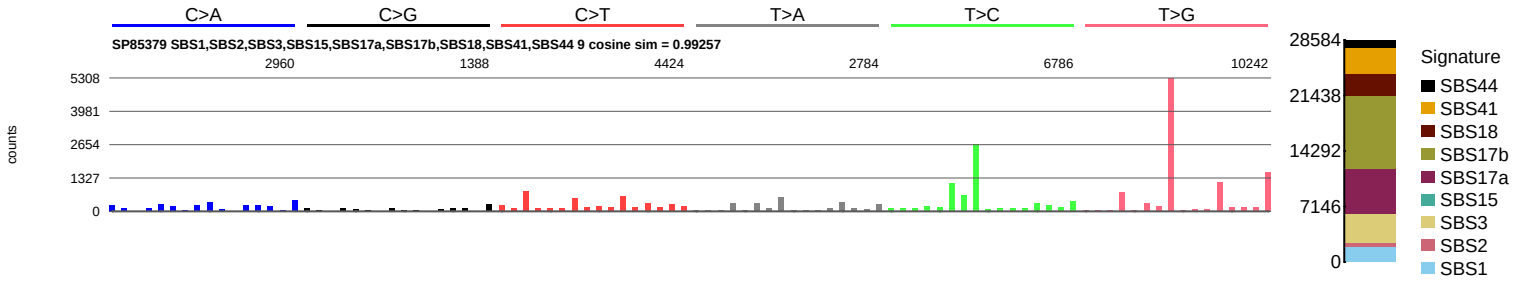
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9926
2927
1412
4208
2758
7038
10243
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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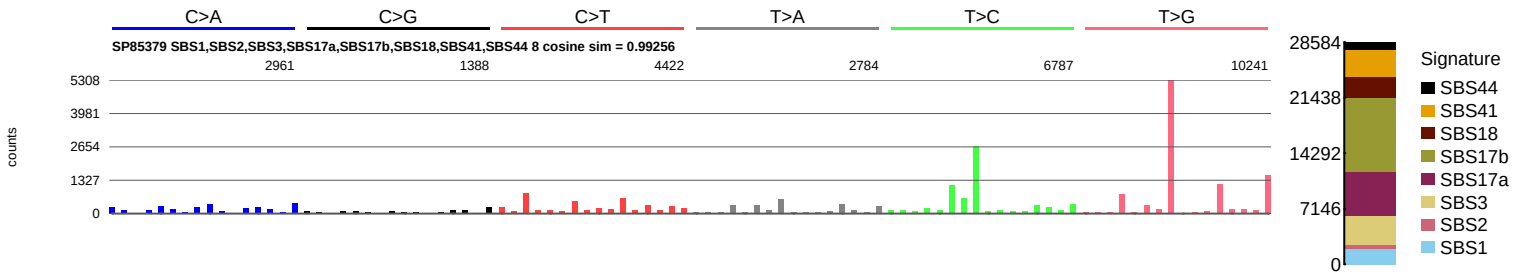
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99257
2960
1388
4424
2784
6786
10242
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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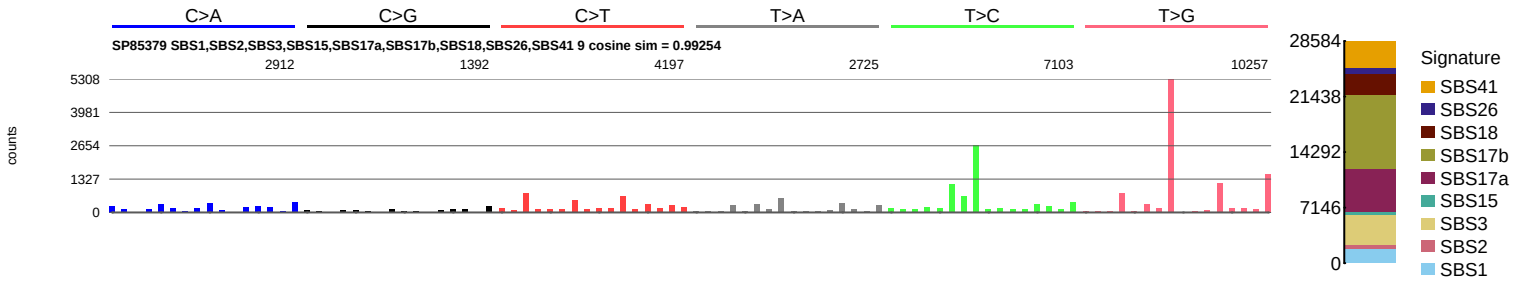
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99256
2961
1388
4422
2784
6787
10241
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>T
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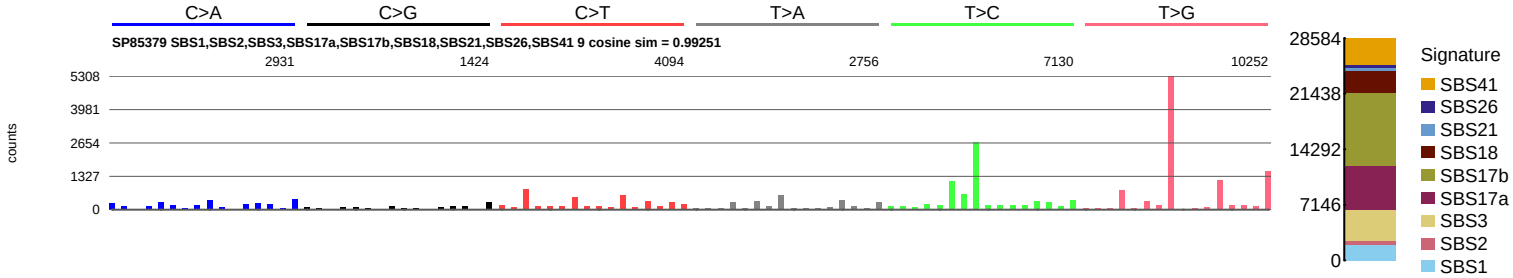
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99254
2912
1392
4197
2725
7103
10257
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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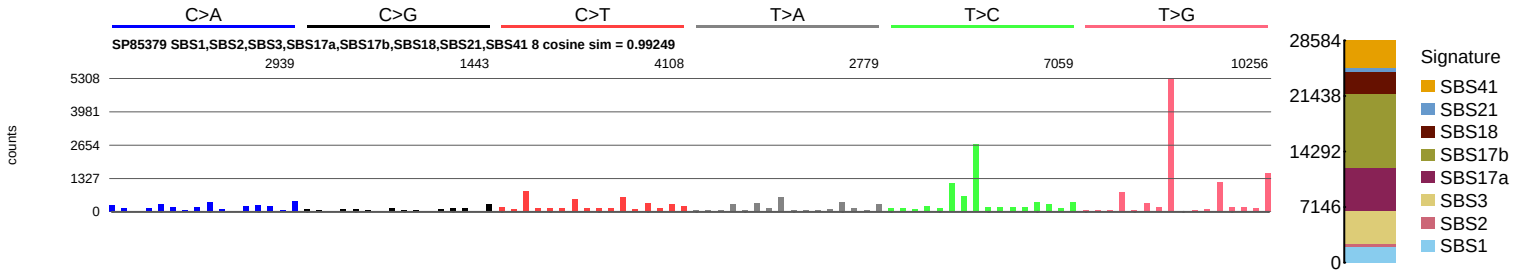
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99251
2931
1424
4094
2756
7130
10252
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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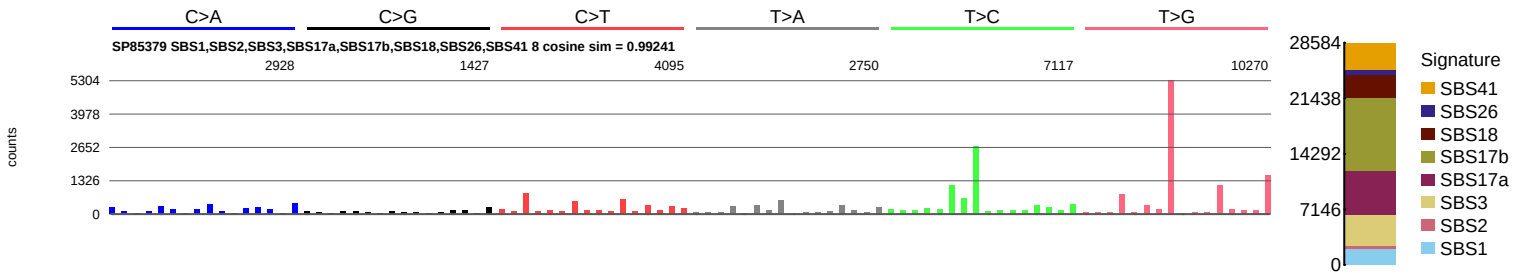
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99249
2939
1443
4108
2779
7059
10256
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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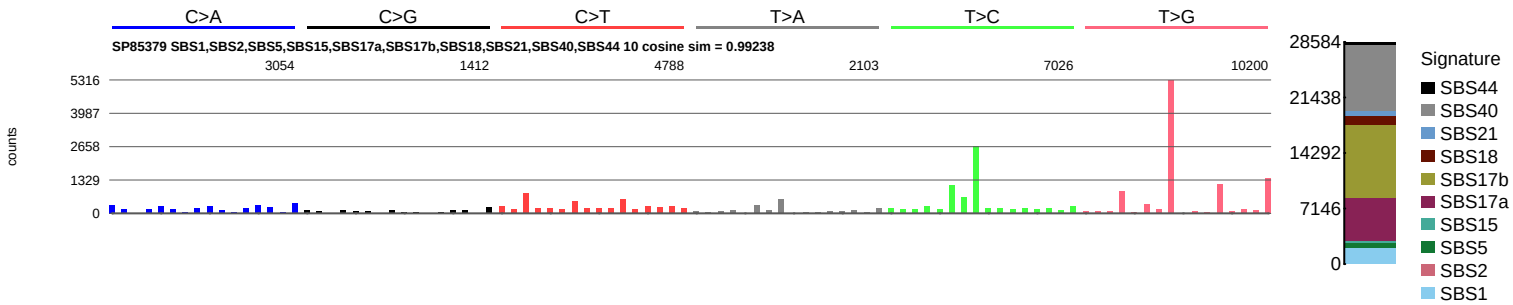
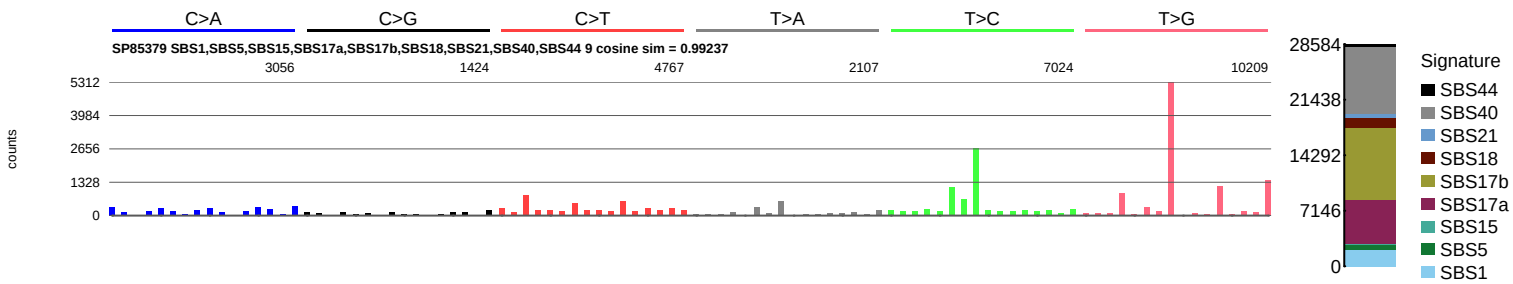
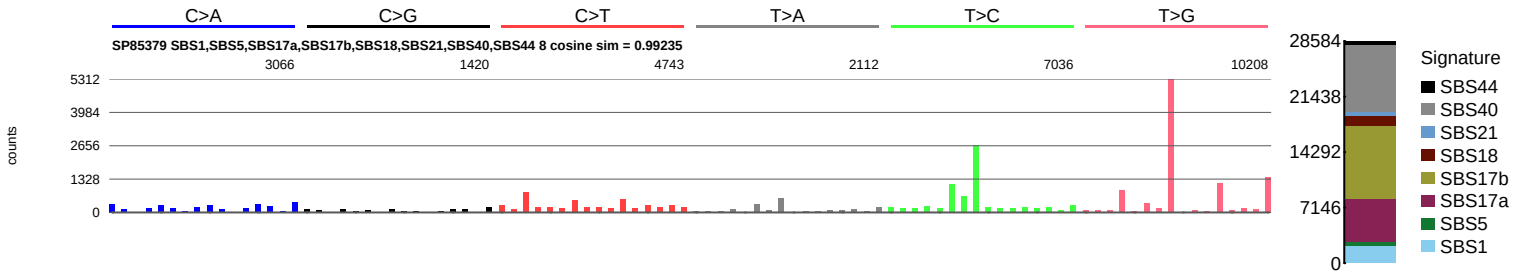
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99241
2928
1427
4095
2750
7117
10270
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.99238
3054
1412
4788
2103
7026
10200
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99237
3056
1424
4767
2107
7024
10209
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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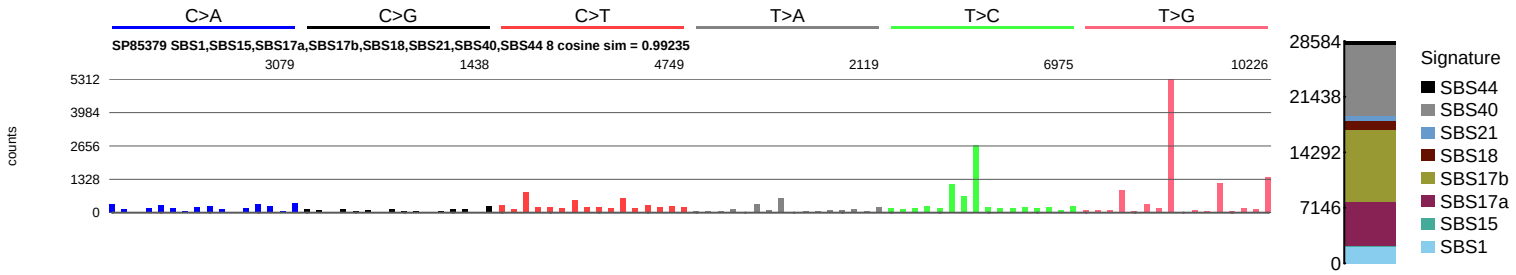
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.99235
3066
1420
4743
2112
7036
10208
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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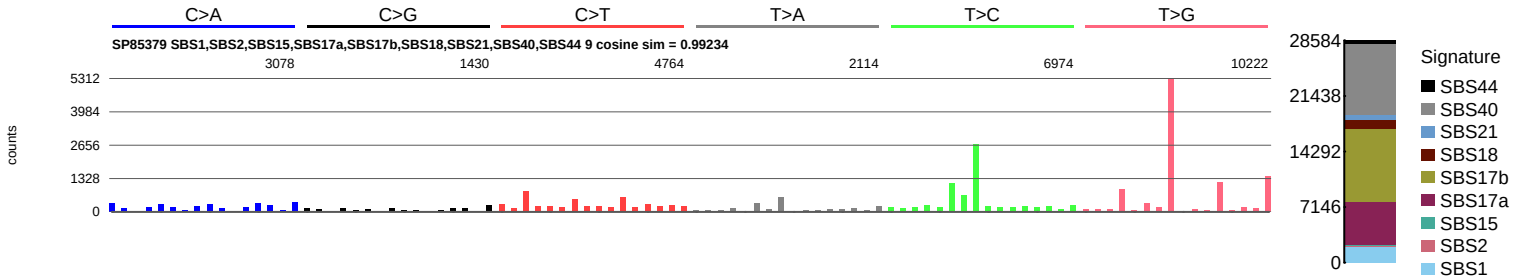
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.99235
3079
1438
4749
2119
6975
10226
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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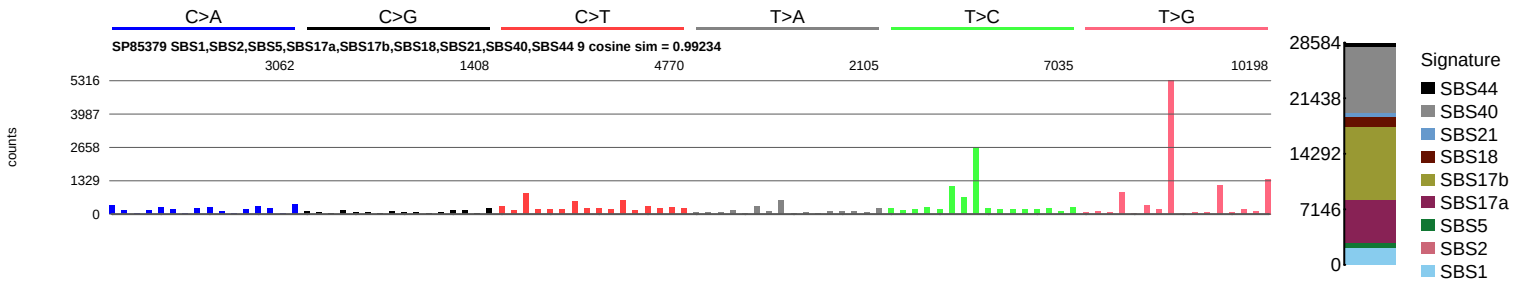
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99234
3078
1430
4764
2114
6974
10222
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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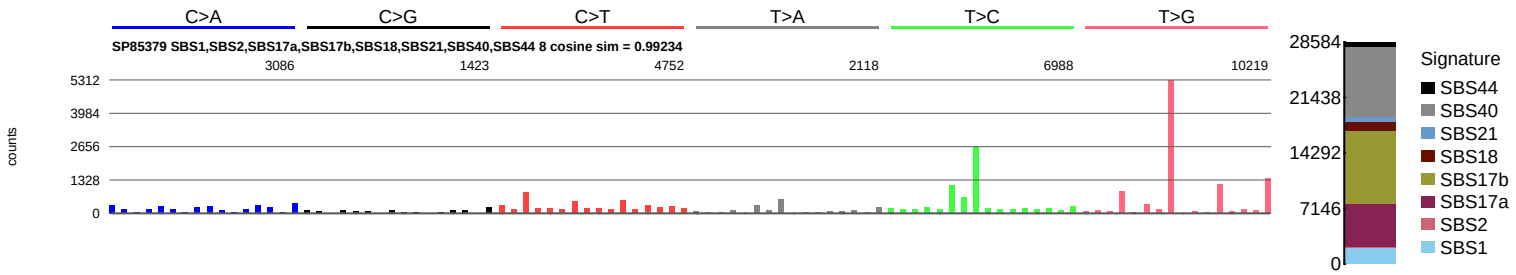
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99234
3062
1408
4770
2105
7035
10198
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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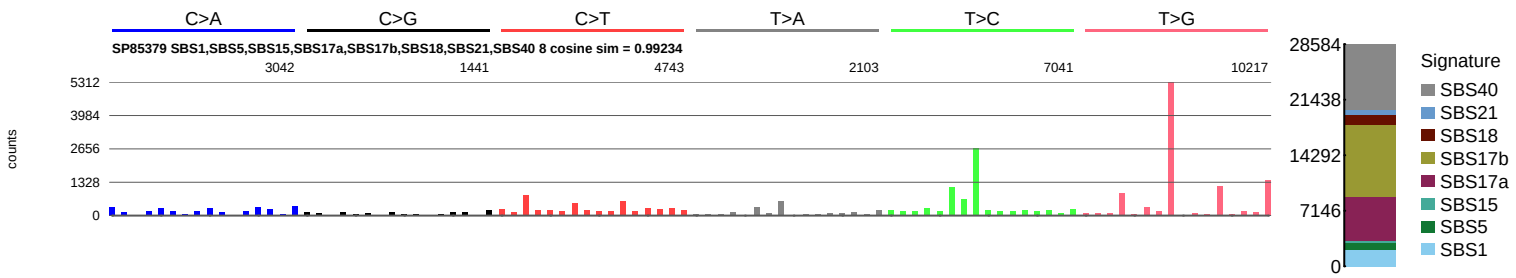
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.99234
3086
1423
4752
2118
6988
10219
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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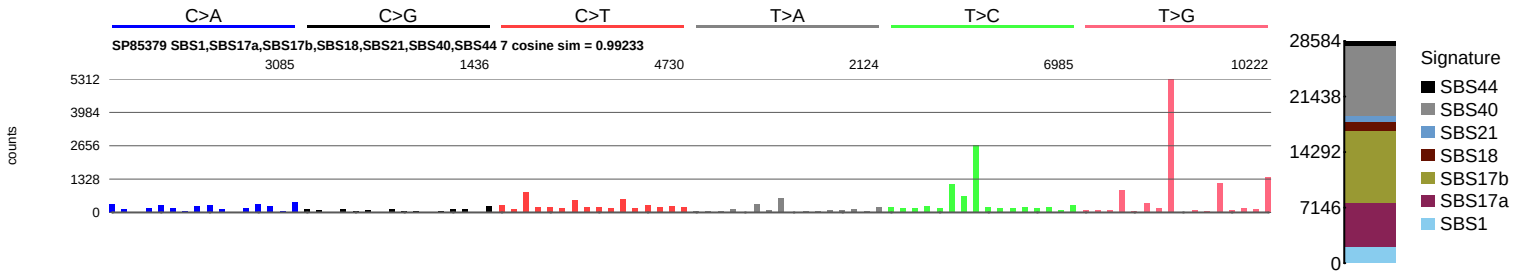
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99234
3042
1441
4743
2103
7041
10217
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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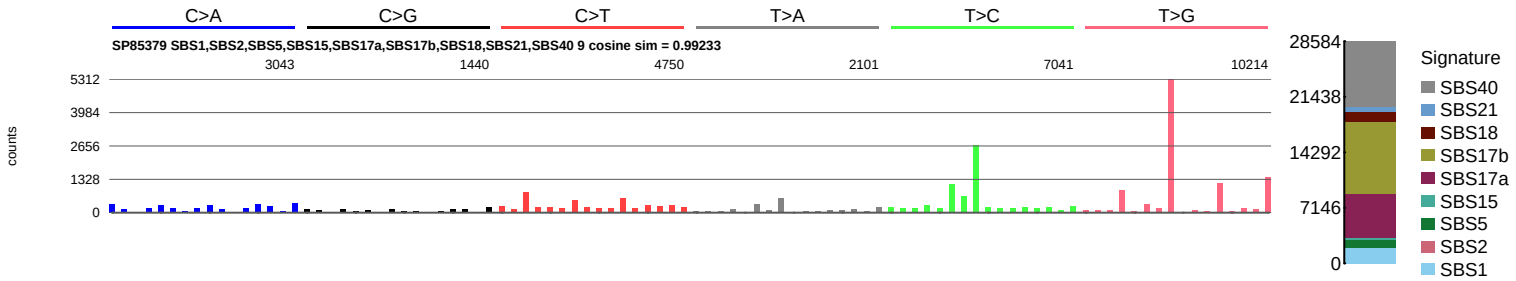
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.99233
3085
1436
4730
2124
6985
10222
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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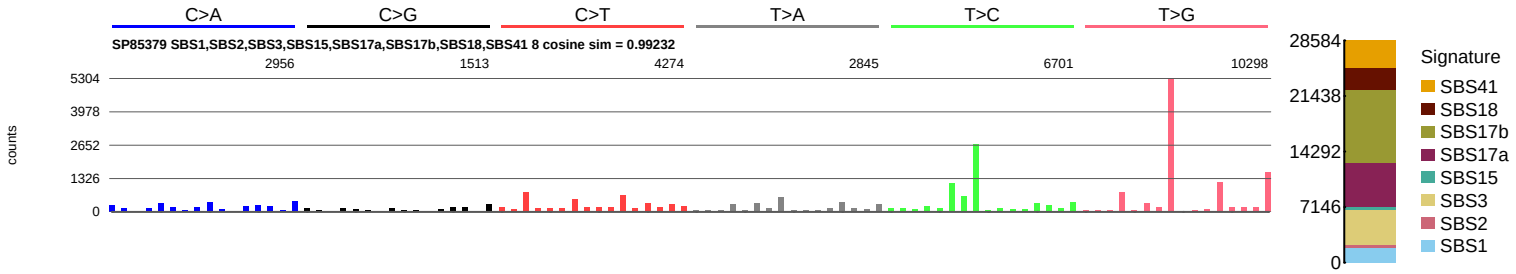
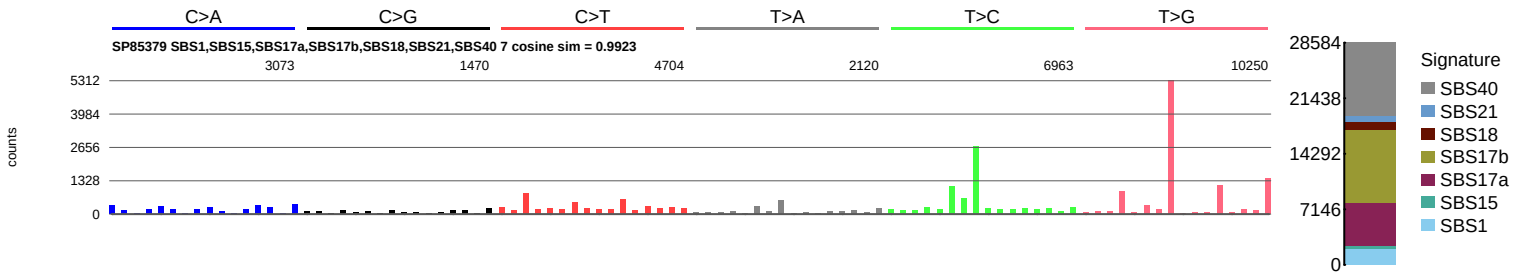
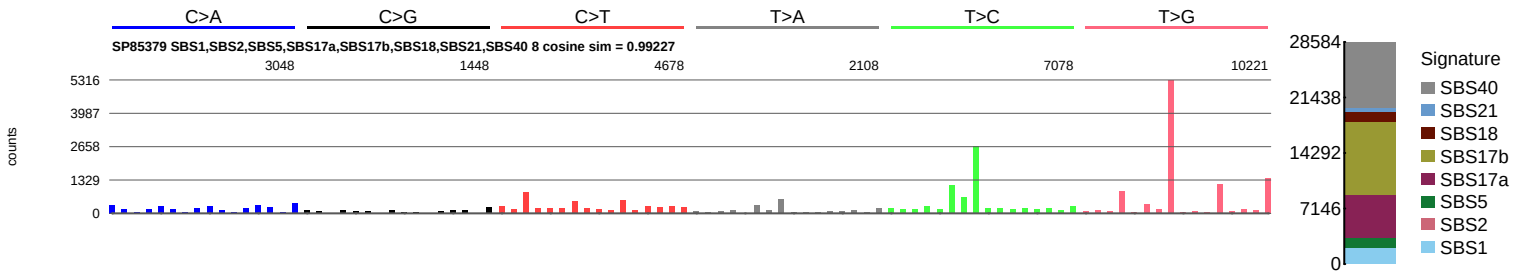
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99233
3043
1440
4750
2101
7041
10214
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99232
2956
1513
4274
2845
6701
10298
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.9923
3073
1470
4704
2120
6963
10250
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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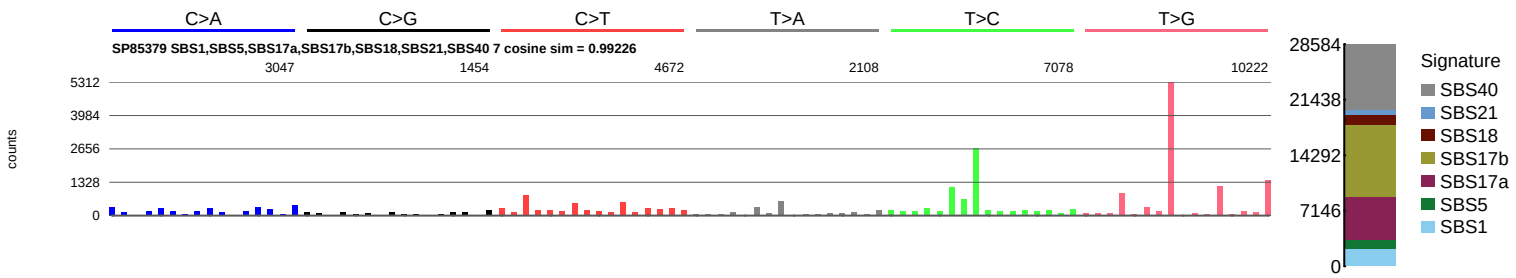
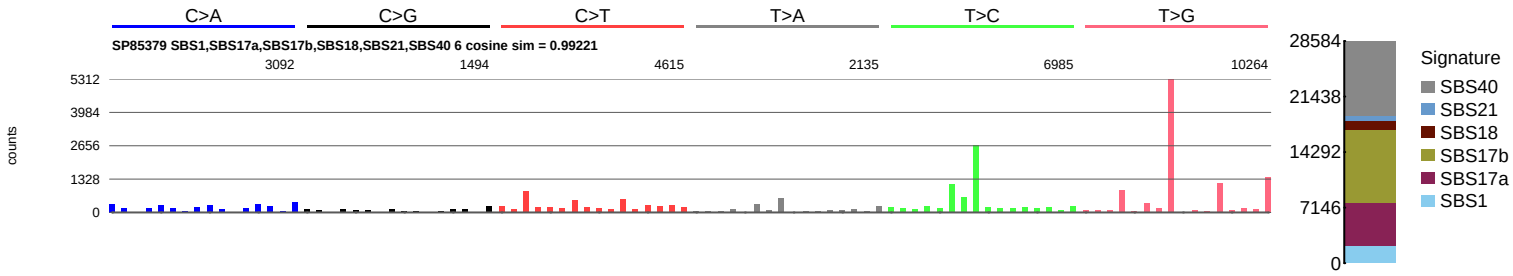
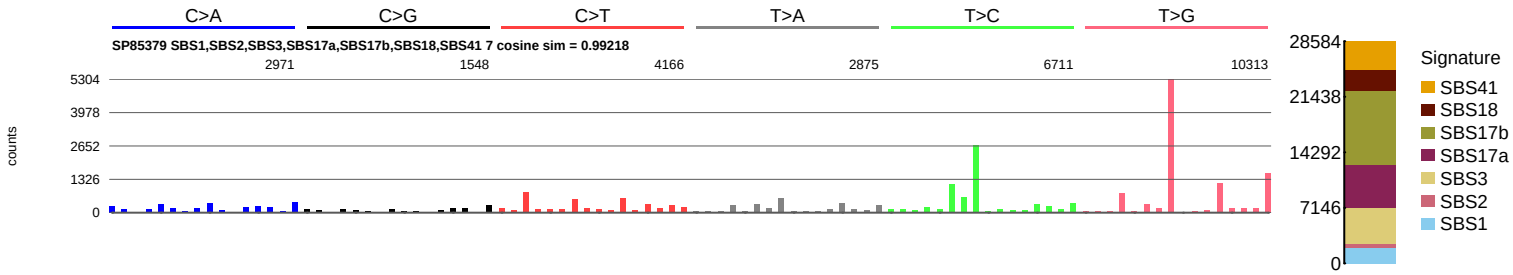
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99227
3048
1448
4678
2108
7078
10221
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.99226
3047
1454
4672
2108
7078
10222
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.99221
3092
1494
4615
2135
6985
10264
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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C>G
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T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99218
2971
1548
4166
2875
6711
10313
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
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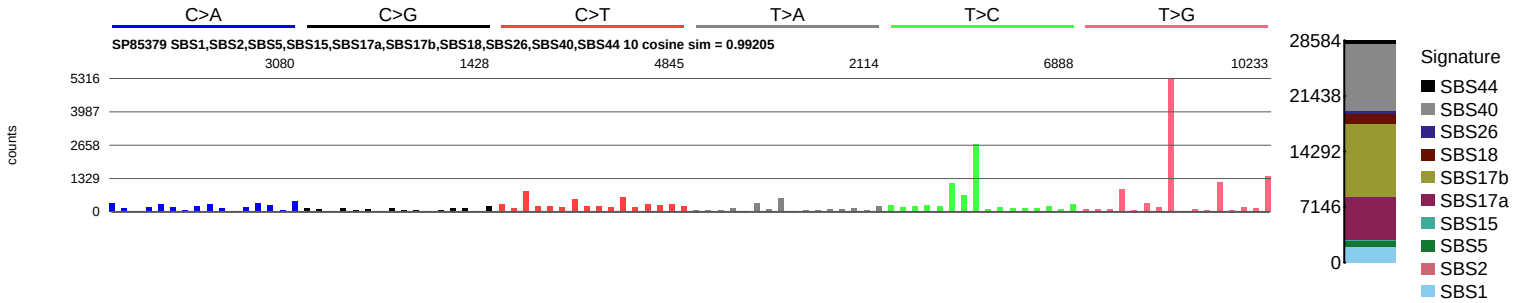
SBS17a
SBS3
SBS2
SBS1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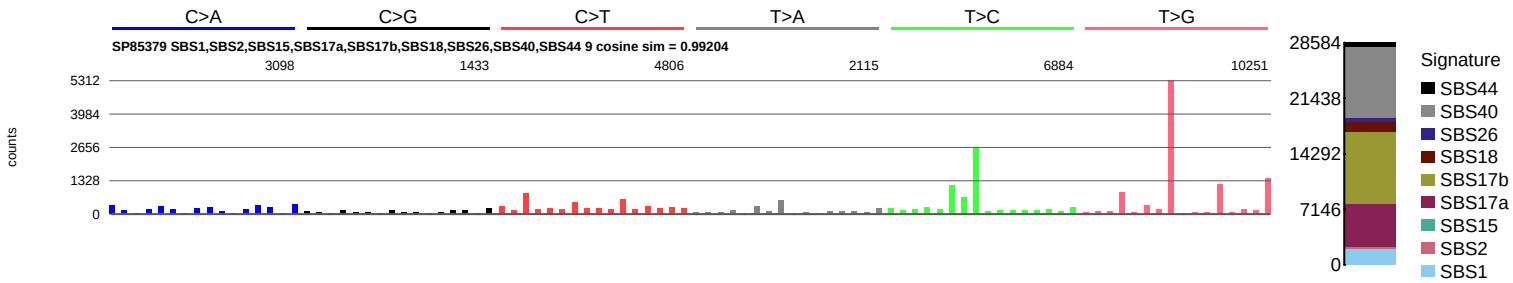
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.99205
3080
1428
4845
2114
6888
10233
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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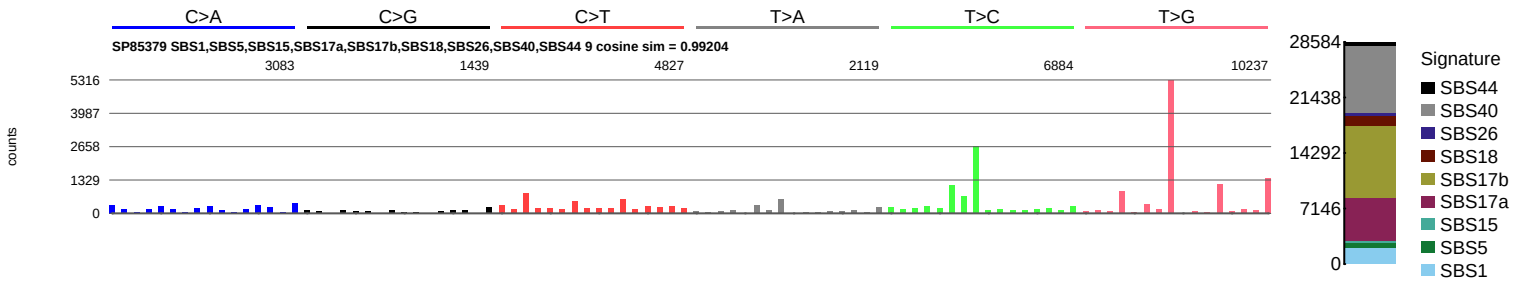
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99204
3098
1433
4806
2115
6884
10251
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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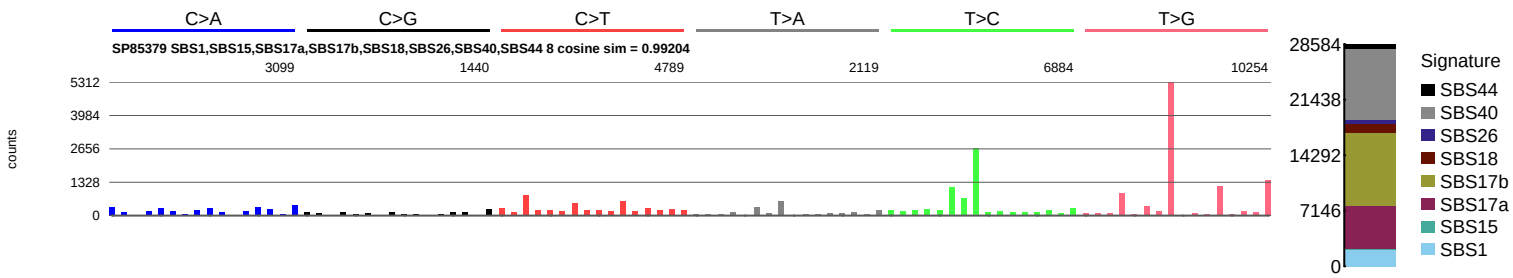
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99204
3083
1439
4827
2119
6884
10237
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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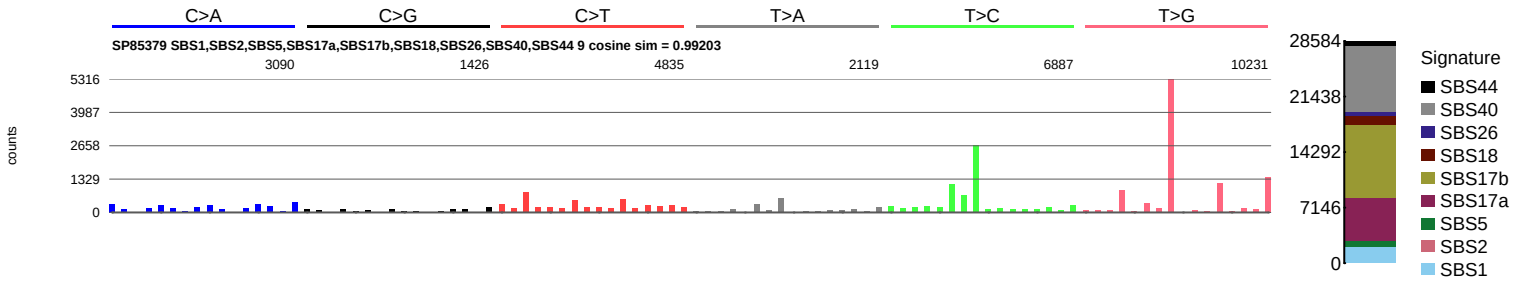
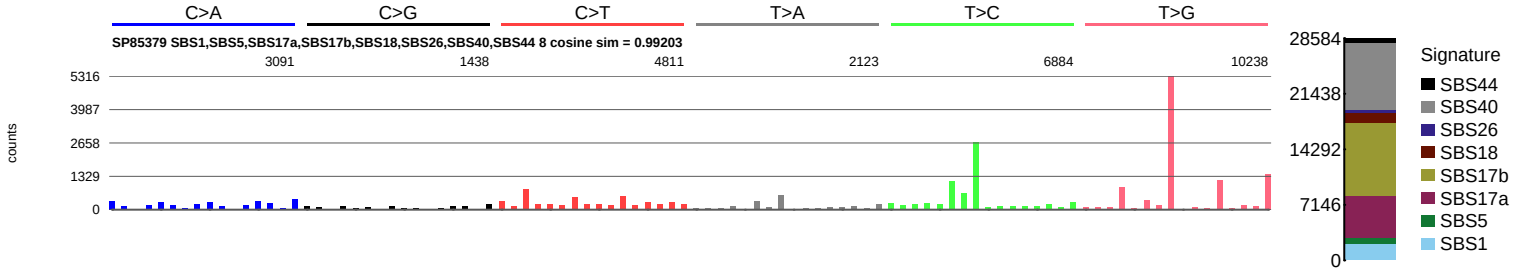
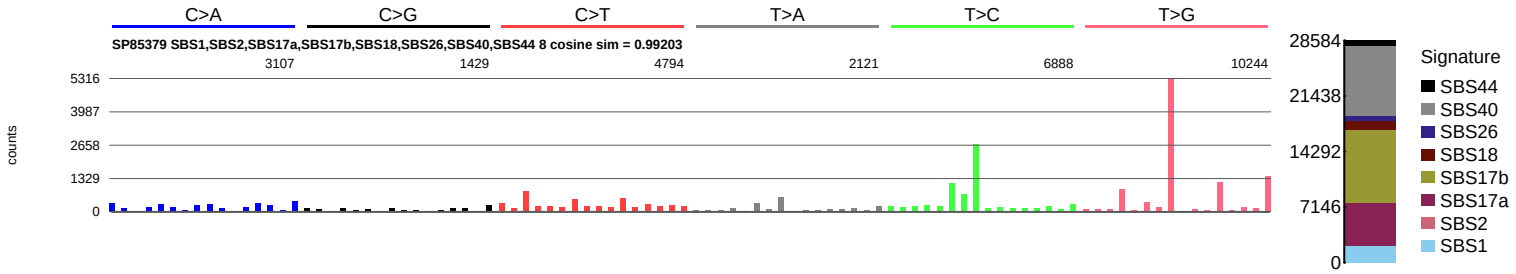
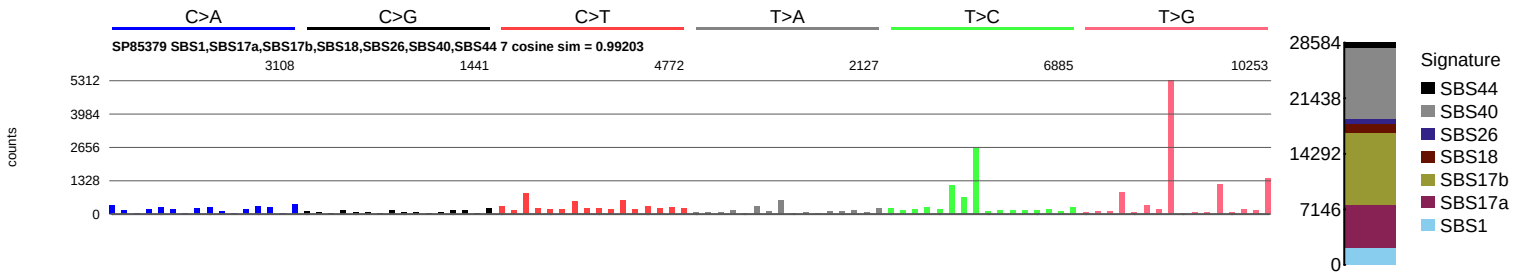
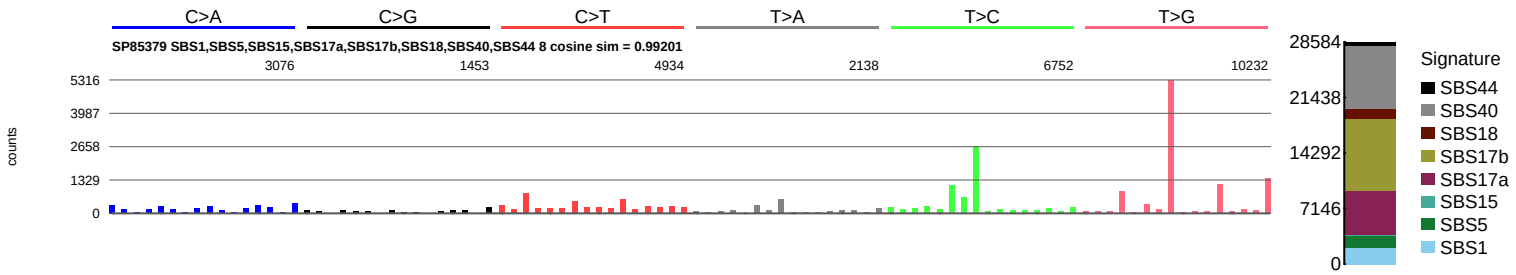
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.99204
3099
1440
4789
2119
6884
10254
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

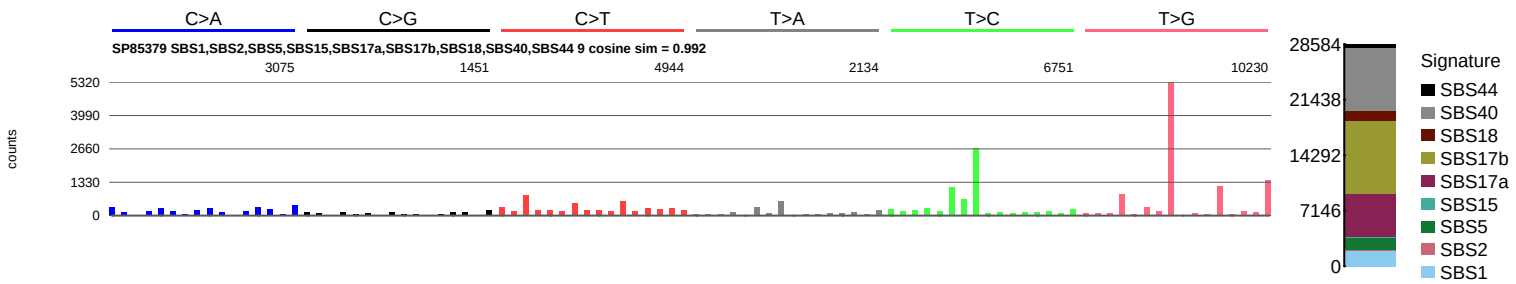

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99203
3090
1426
4835
2119
6887
10231
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

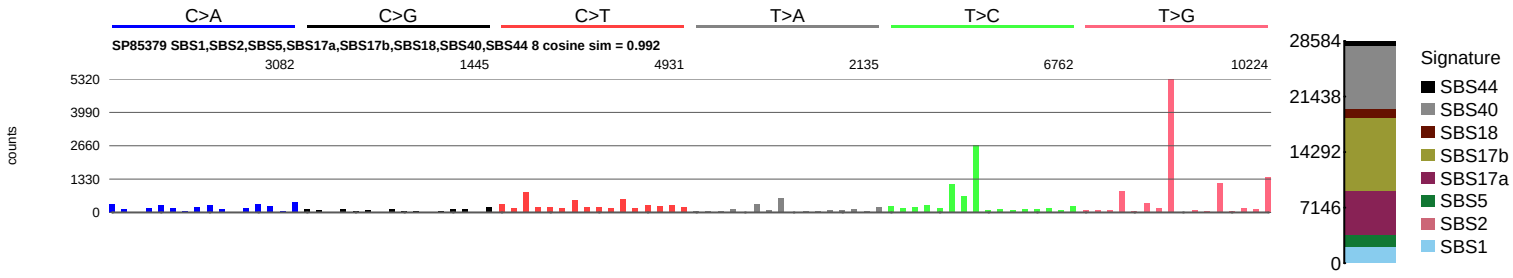

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.99203
3091
1438
4811
2123
6884
10238
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1

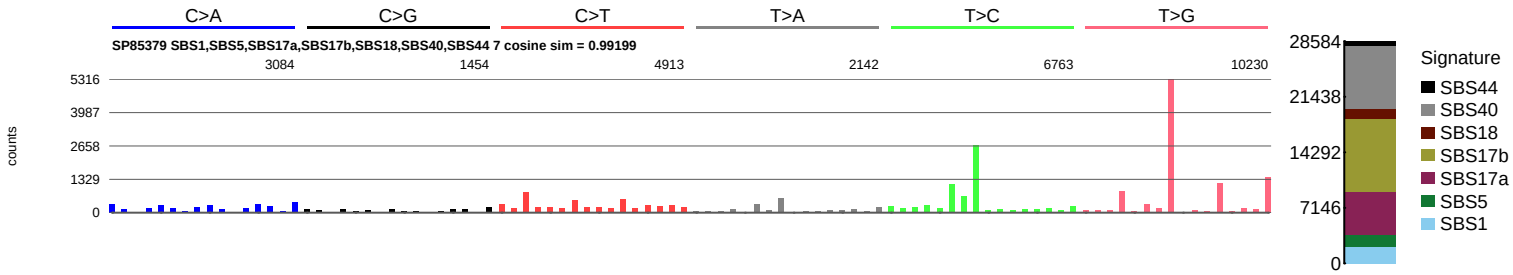

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.99203
3107
1429
4794
2121
6888
10244
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

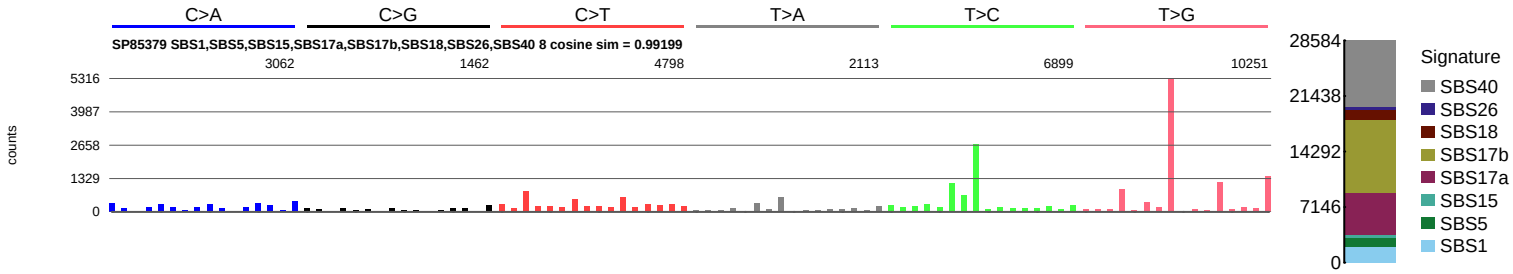

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.99203
3108
1441
4772
2127
6885
10253
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

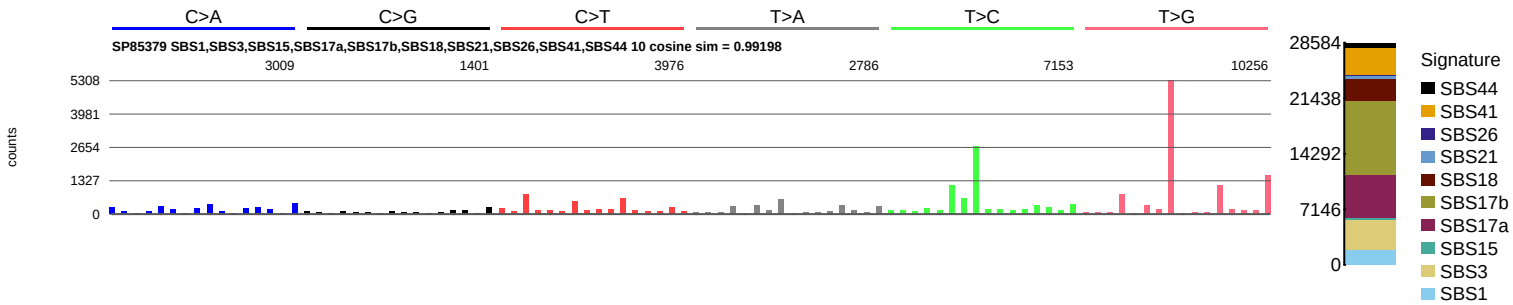

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99201
3076
1453
4934
2138
6752
10232
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

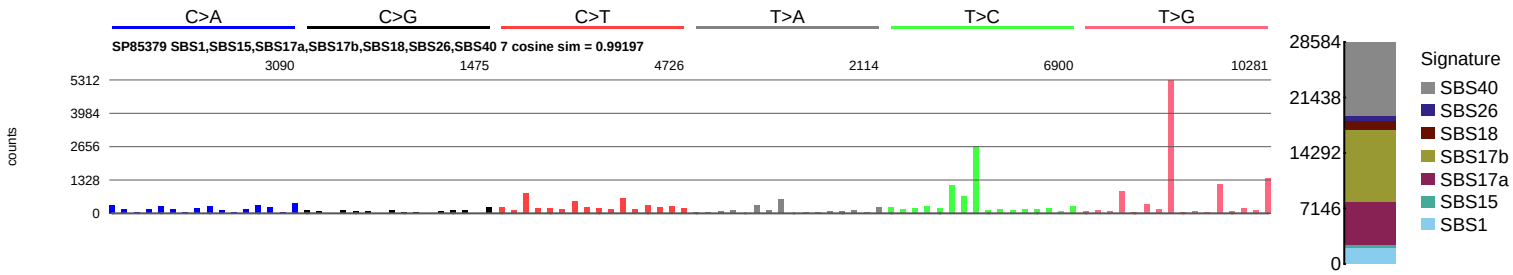

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.992
3075
1451
4944
2134
6751
10230
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.992
3082
1445
4931
2135
6762
10224
5320
3990
2660
1330
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.99199
3084
1454
4913
2142
6763
10230
5316
3987
2658
1329
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99199
3062
1462
4798
2113
6899
10251
5316
3987
2658
1329
0
counts
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99198
3009
1401
3976
2786
7153
10256
5308
3981
2654
1327
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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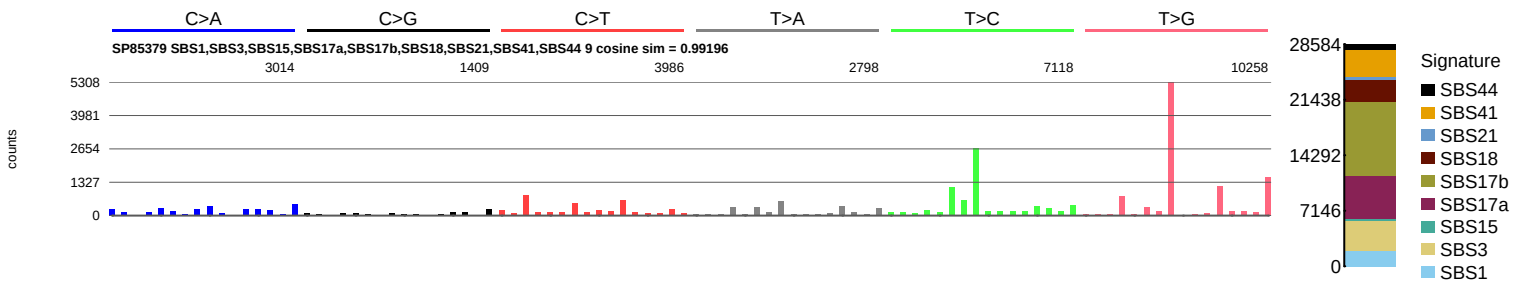
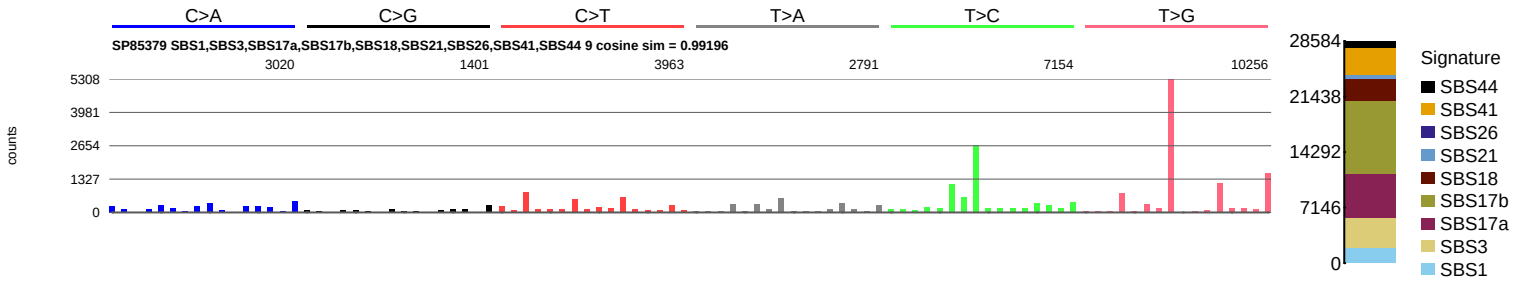
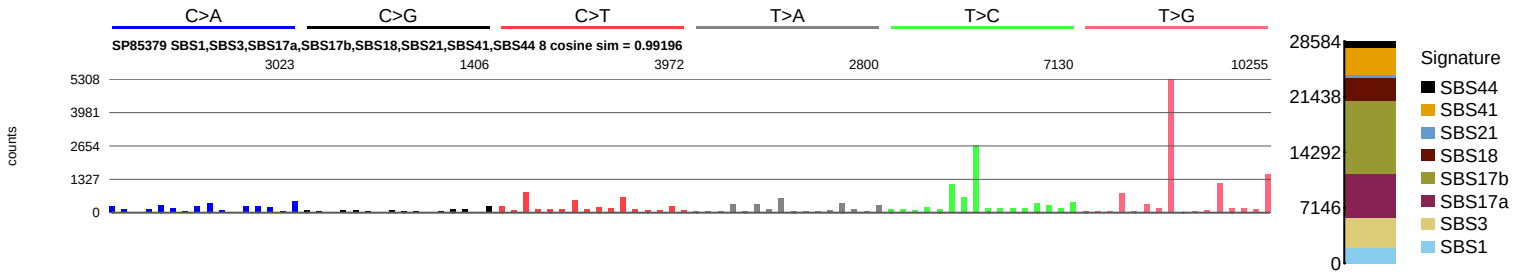
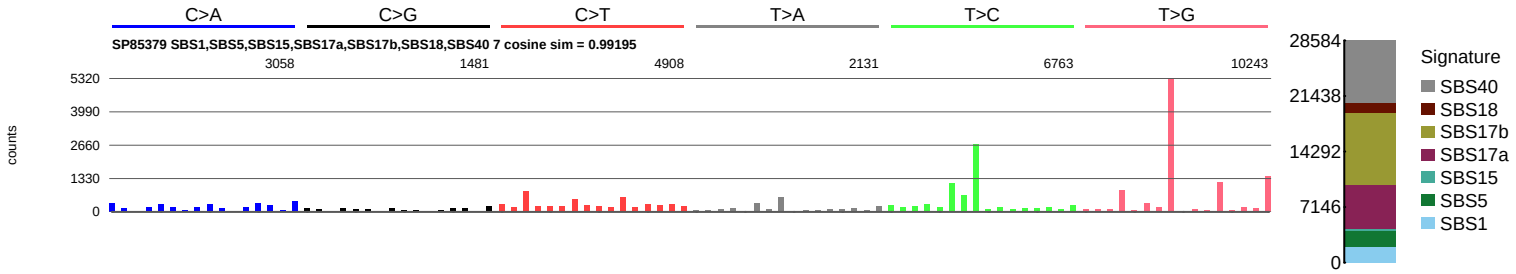
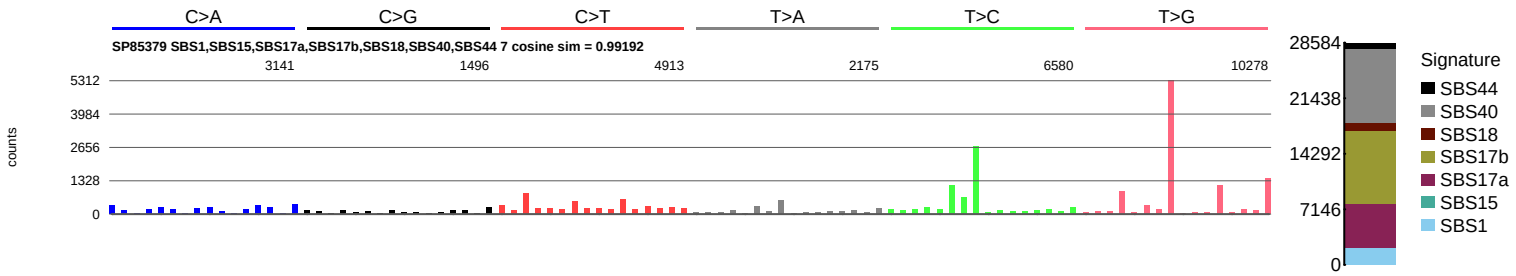
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.99197
3090
1475
4726
2114
6900
10281
5312
3984
2656
1328
0
counts
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

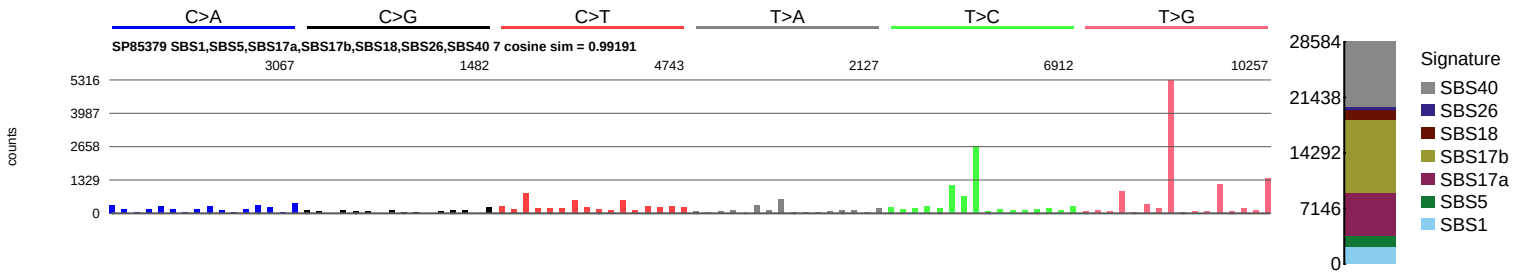

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99196
3014
1409
3986
2798
7118
10258
5308
3981
2654
1327
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

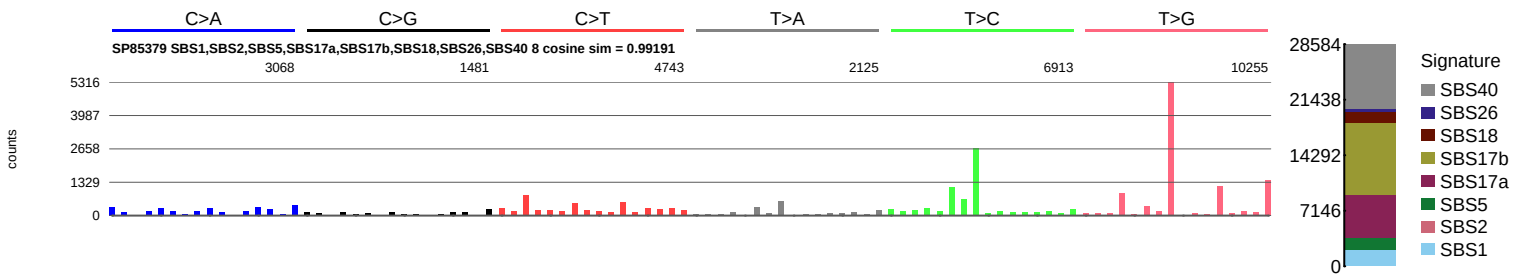

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99196
3020
1401
3963
2791
7154
10256
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

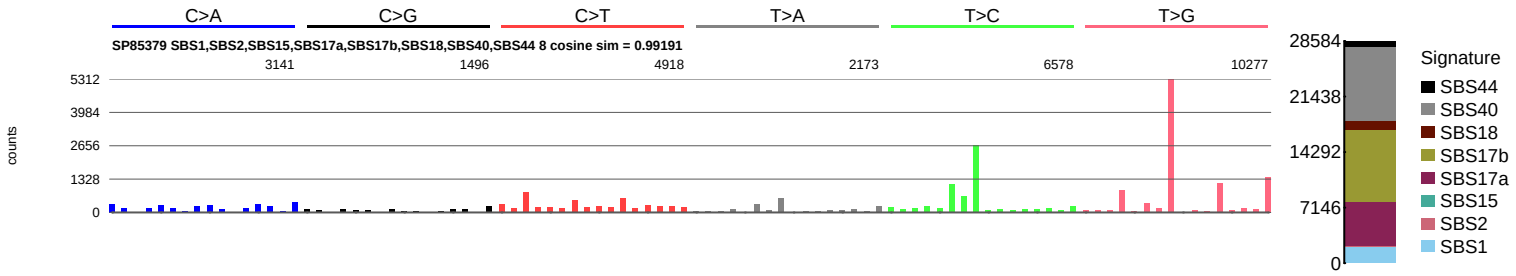

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.99196
3023
1406
3972
2800
7130
10255
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

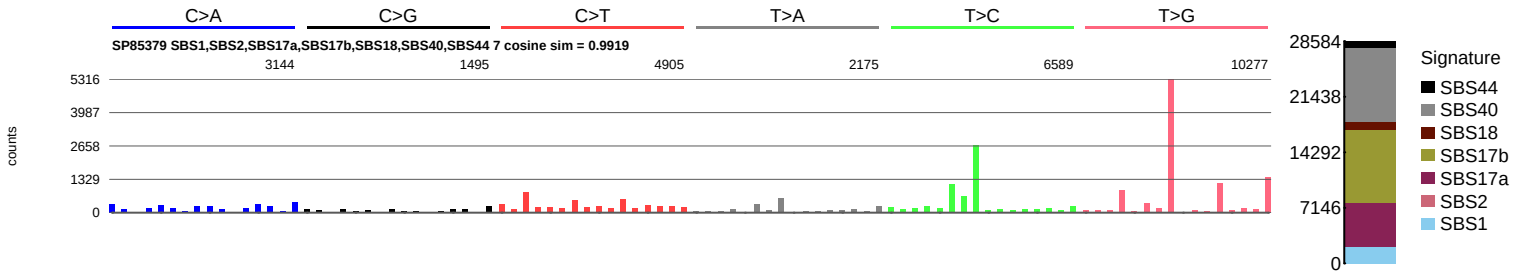

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99195
3058
1481
4908
2131
6763
10243
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

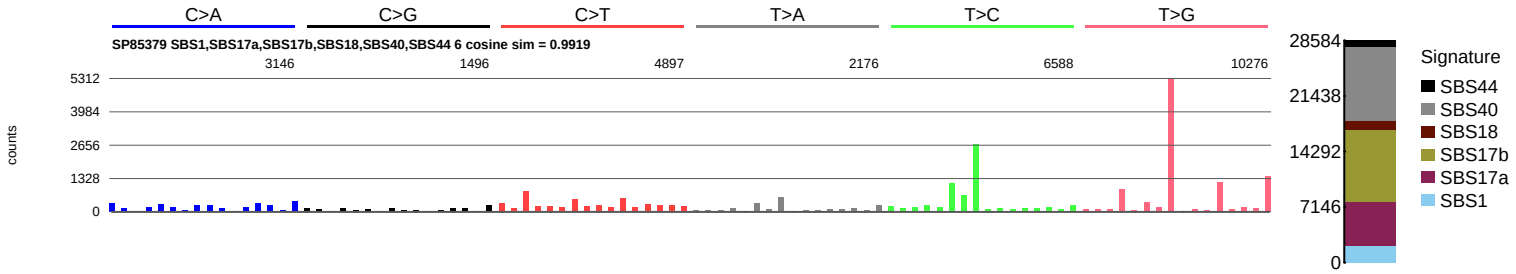

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.99192
3141
1496
4913
2175
6580
10278
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.99191
3067
1482
4743
2127
6912
10257
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99191
3068
1481
4743
2125
6913
10255
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99191
3141
1496
4918
2173
6578
10277
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9919
3144
1495
4905
2175
6589
10277
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
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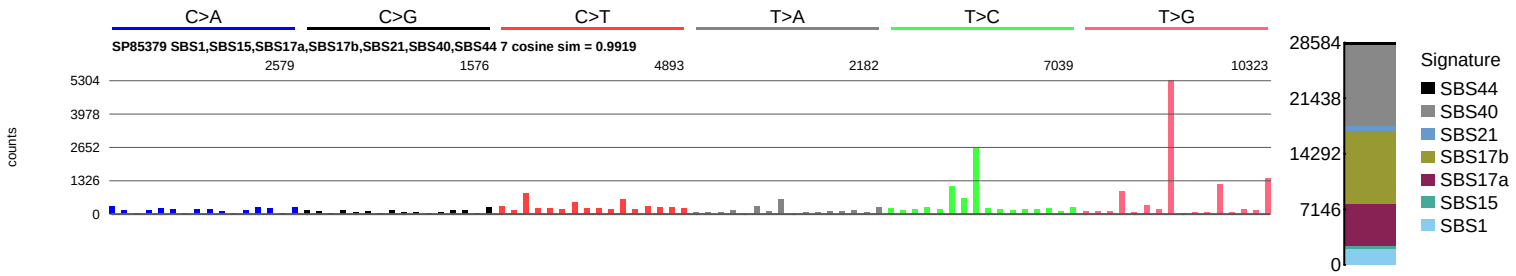
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.9919
3146
1496
4897
2176
6588
10276
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1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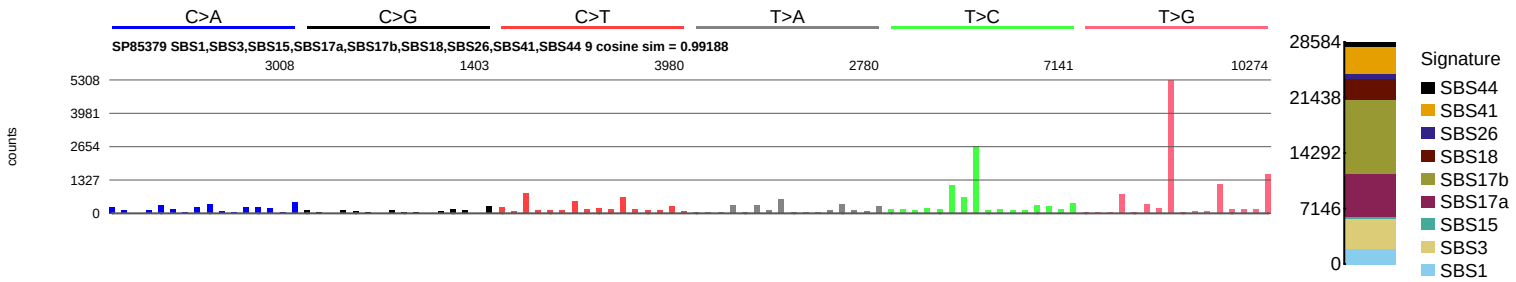
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.9919
2579
1576
4893
2182
7039
10323
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1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C>T
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T>C
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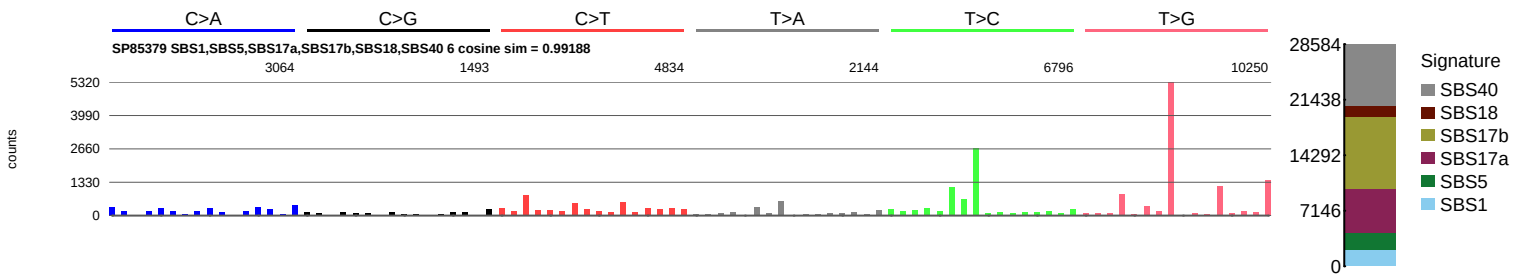
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99188
3008
1403
3980
2780
7141
10274
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
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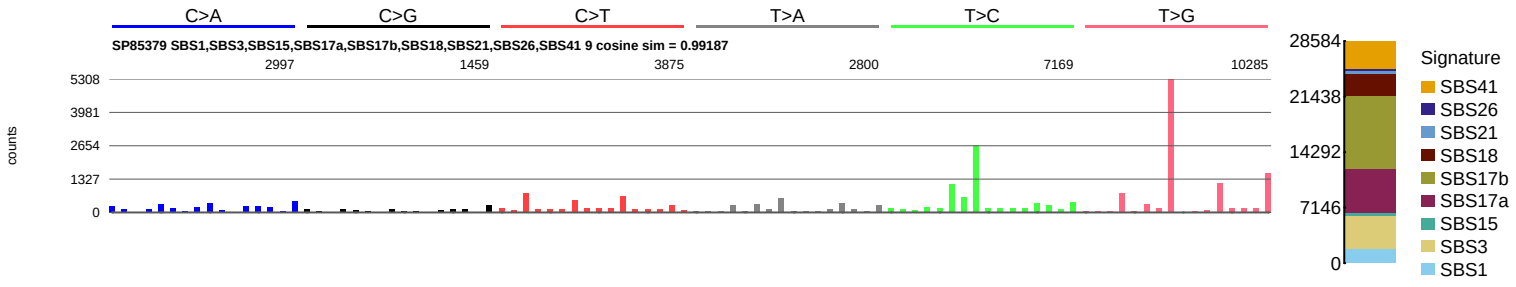
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99188
3064
1493
4834
2144
6796
10250
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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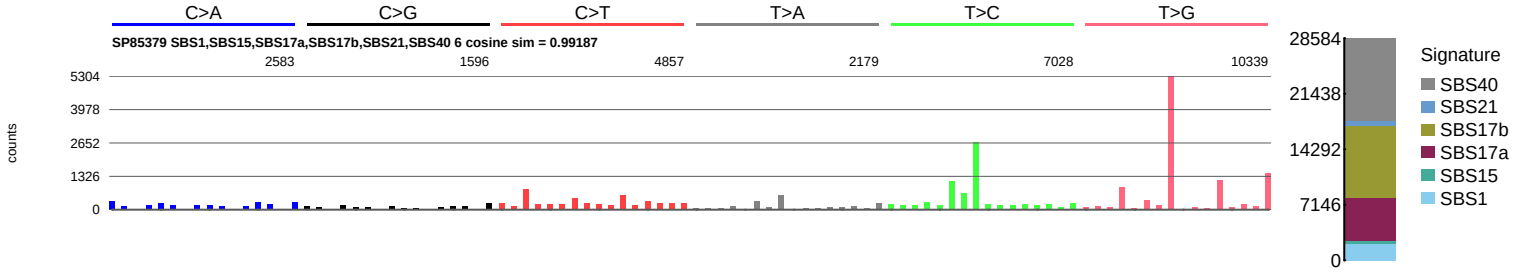
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99187
2997
1459
3875
2800
7169
10285
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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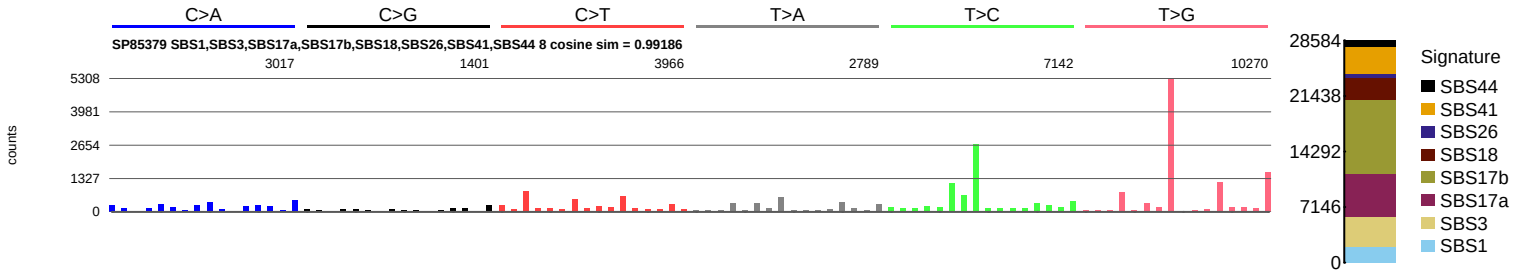
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.99187
2583
1596
4857
2179
7028
10339
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1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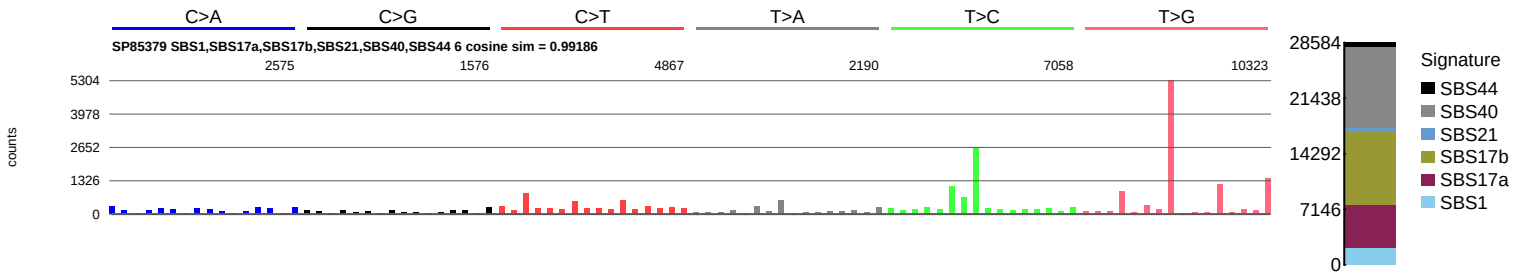
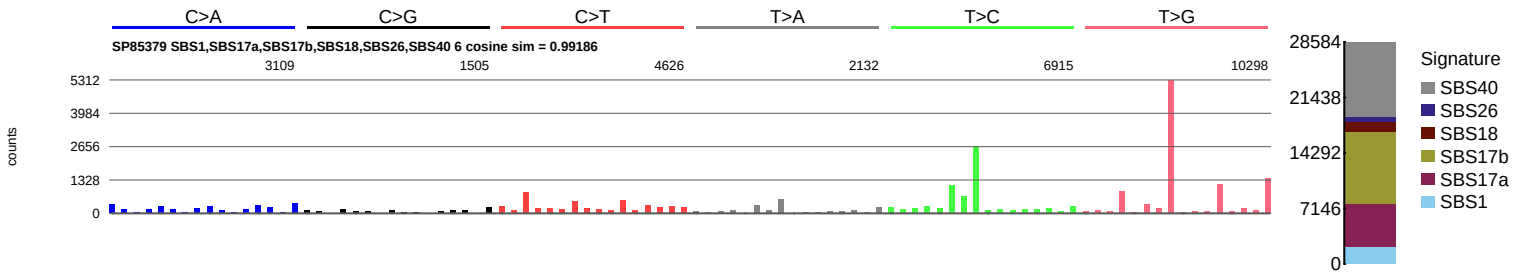
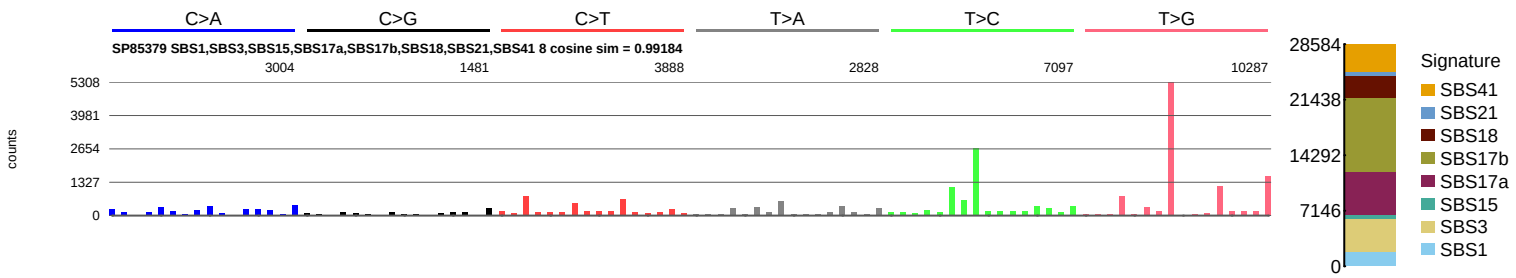
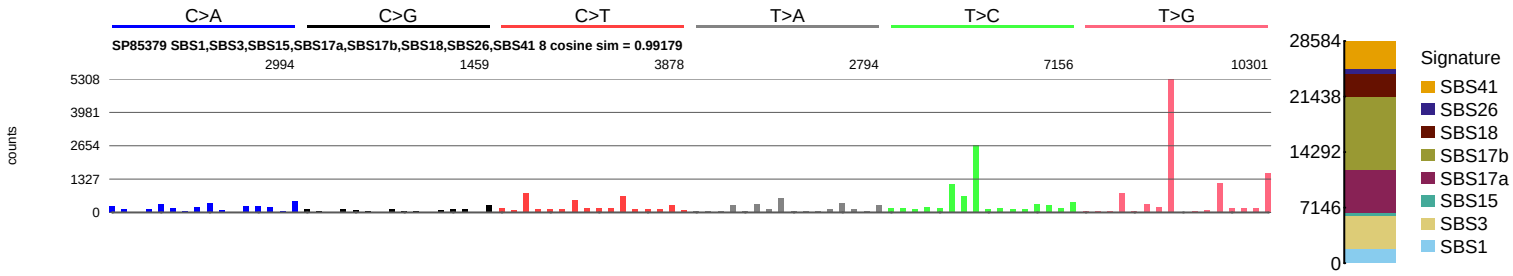
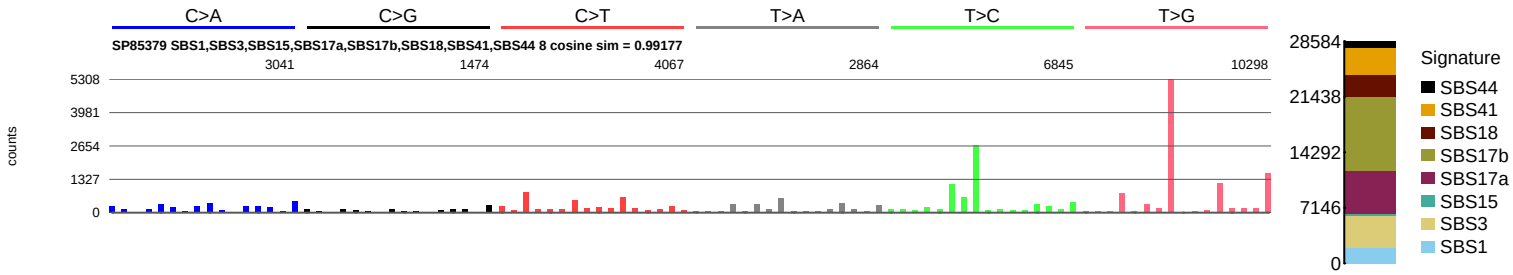
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99186
3017
1401
3966
2789
7142
10270
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

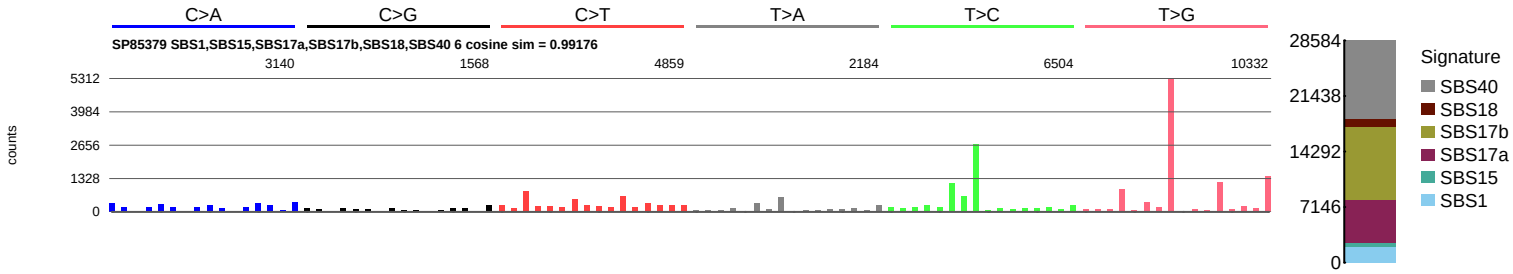

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.99186
2575
1576
4867
2190
7058
10323
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS1

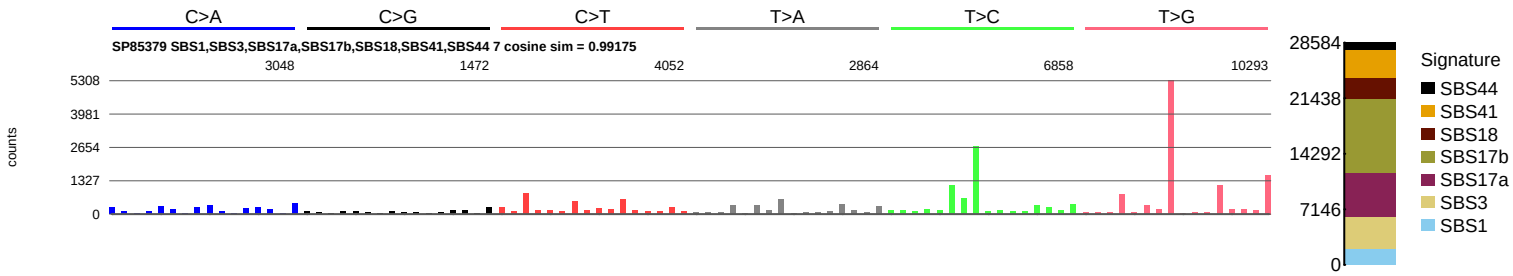

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.99186
3109
1505
4626
2132
6915
10298
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

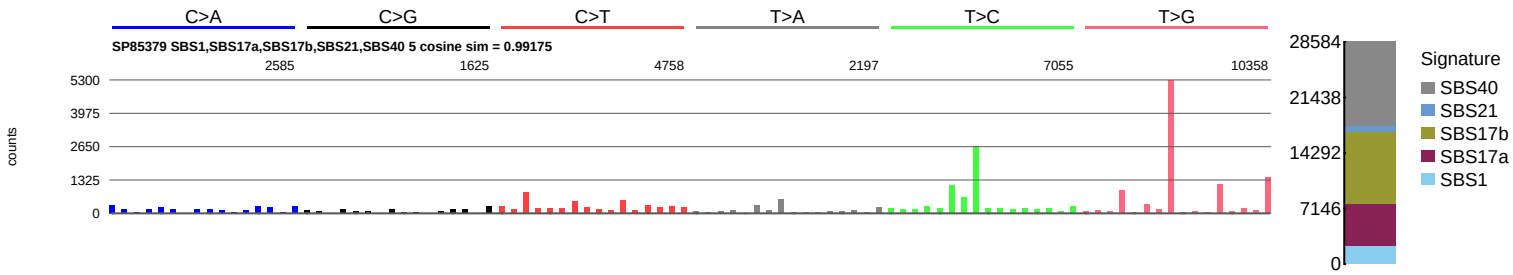

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99184
3004
1481
3888
2828
7097
10287
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

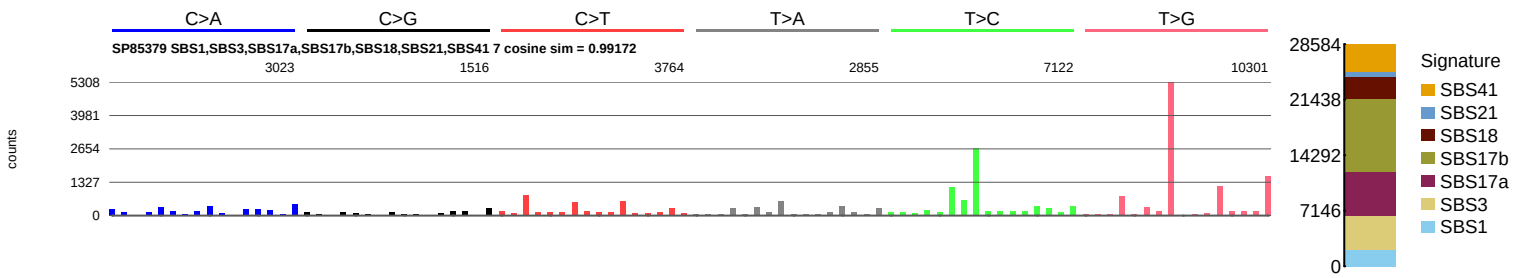

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99179
2994
1459
3878
2794
7156
10301
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99177
3041
1474
4067
2864
6845
10298
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99176
3140
1568
4859
2184
6504
10332
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.99175
3048
1472
4052
2864
6858
10293
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.99175
2585
1625
4758
2197
7055
10358
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS1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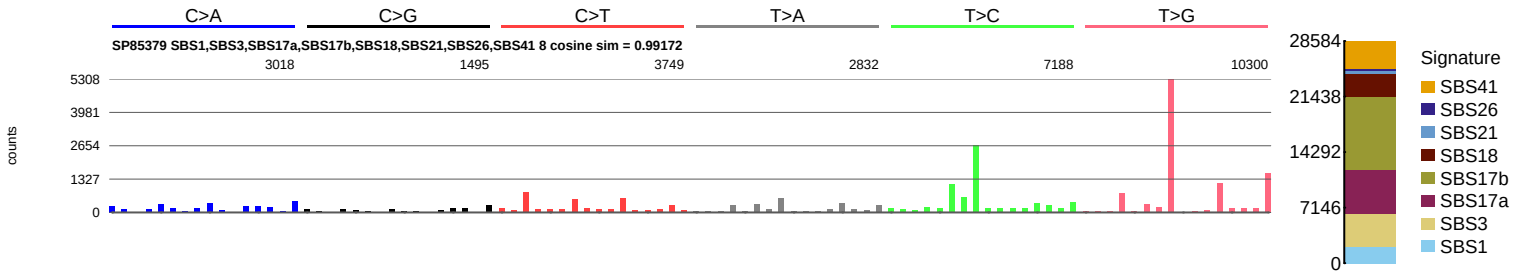
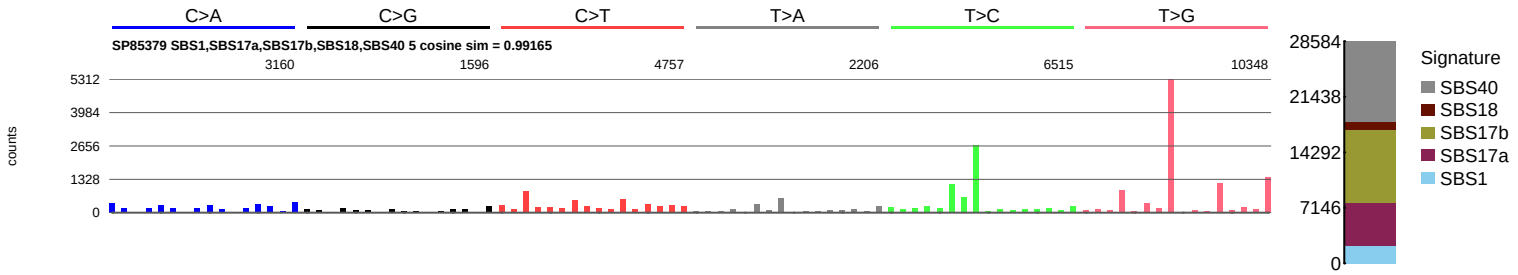
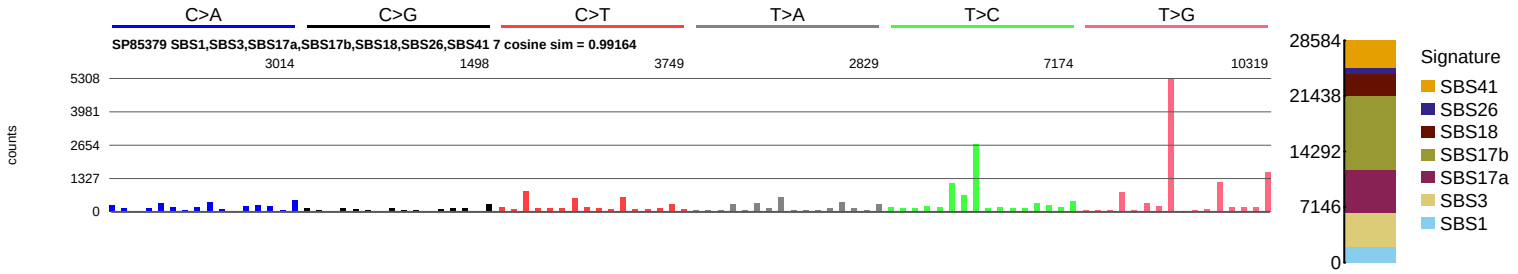
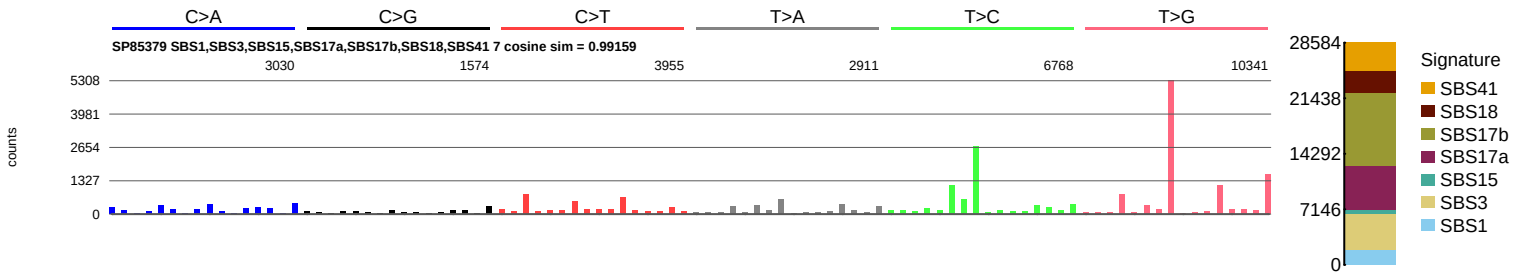
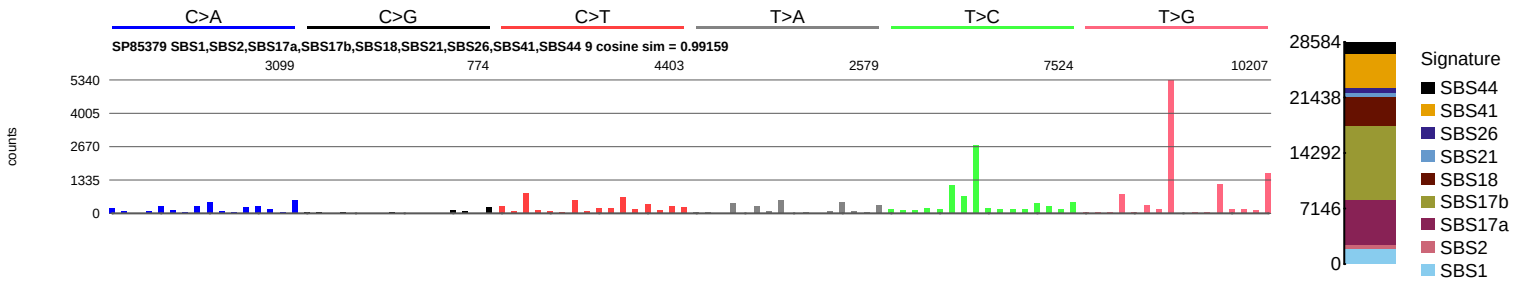
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99172
3023
1516
3764
2855
7122
10301
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

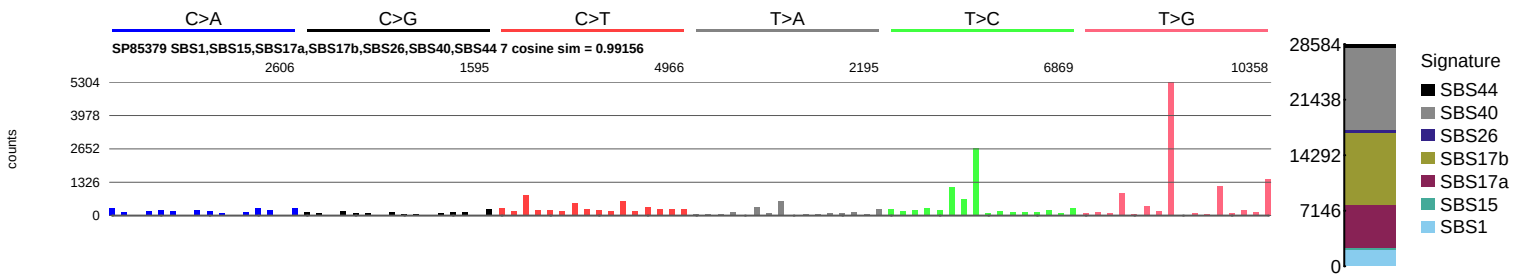

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.99172
3018
1495
3749
2832
7188
10300
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

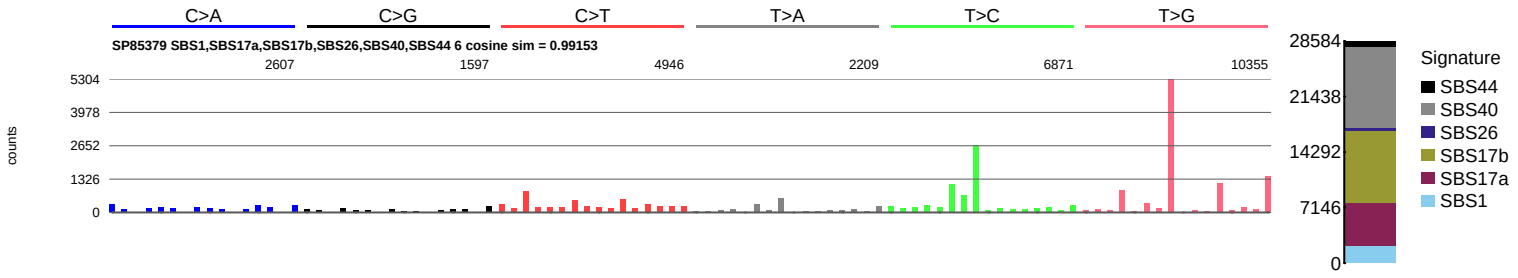

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.99165
3160
1596
4757
2206
6515
10348
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1

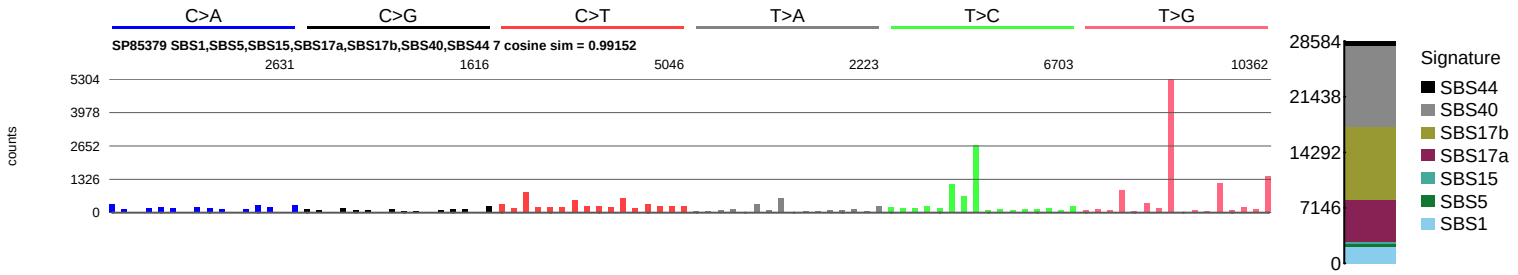

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99164
3014
1498
3749
2829
7174
10319
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

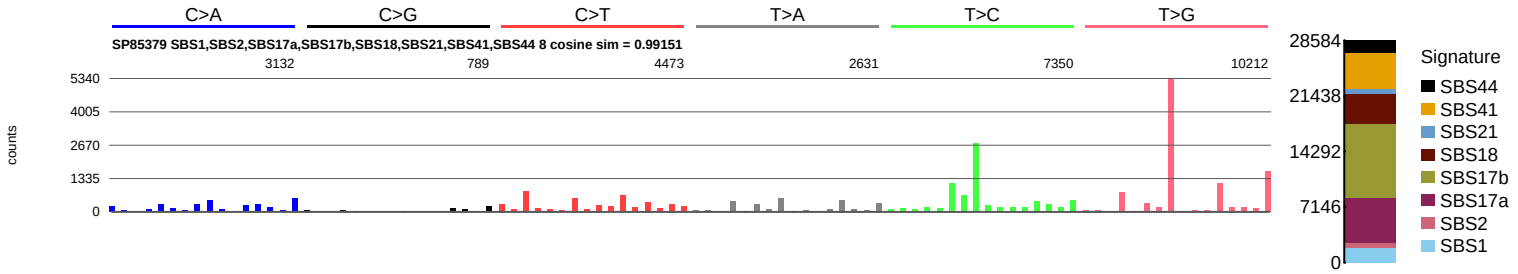

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99159
3030
1574
3955
2911
6768
10341
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

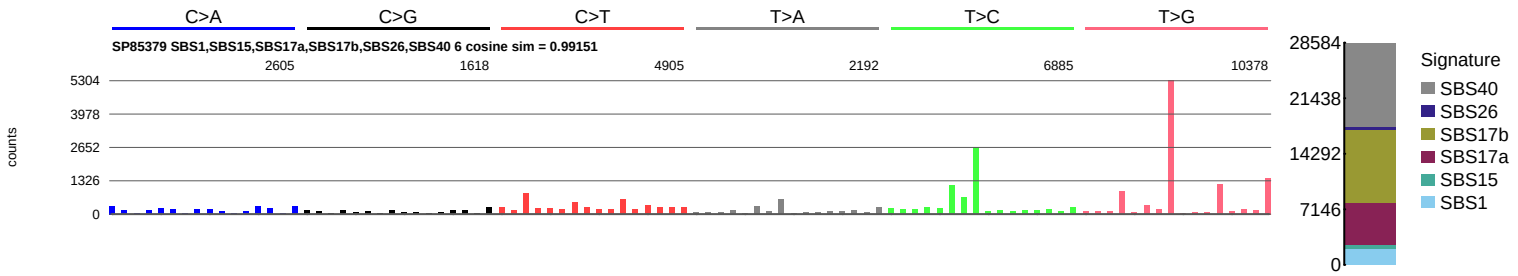

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99159
3099
774
4403
2579
7524
10207
counts
5340
4005
2670
1335
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.99156
2606
1595
4966
2195
6869
10358
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.99153
2607
1597
4946
2209
6871
10355
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.99152
2631
1616
5046
2223
6703
10362
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.99151
3132
789
4473
2631
7350
10212
counts
5340
4005
2670
1335
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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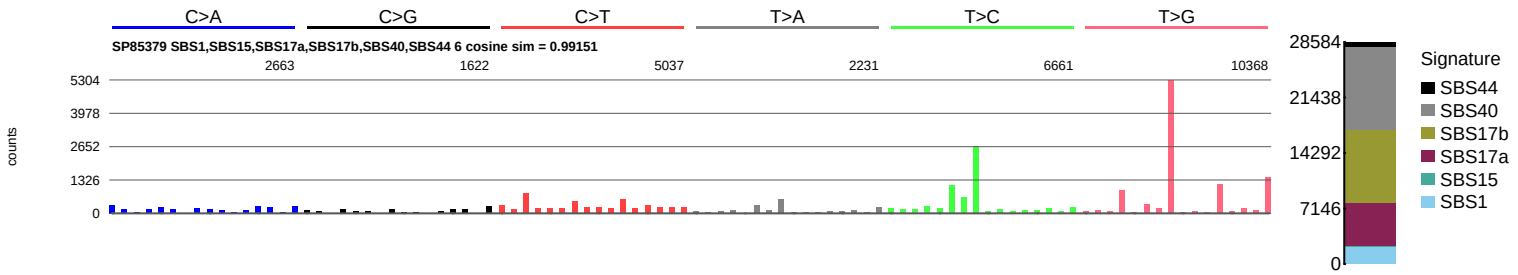
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.99151
2605
1618
4905
2192
6885
10378
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
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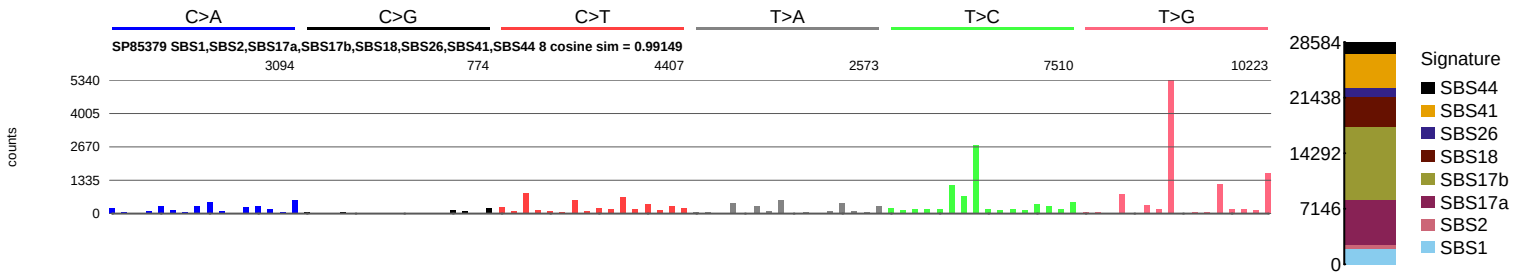
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.99151
2663
1622
5037
2231
6661
10368
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS1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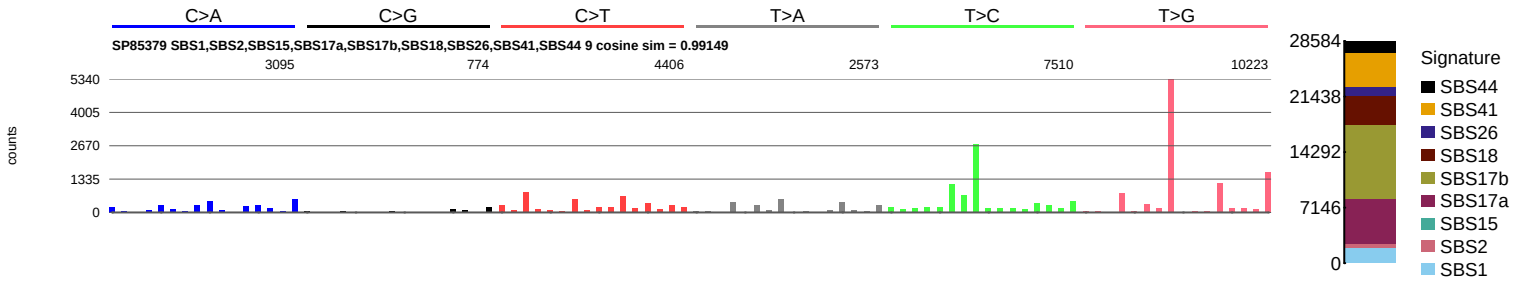
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99149
3094
774
4407
2573
7510
10223
counts
5340
4005
2670
1335
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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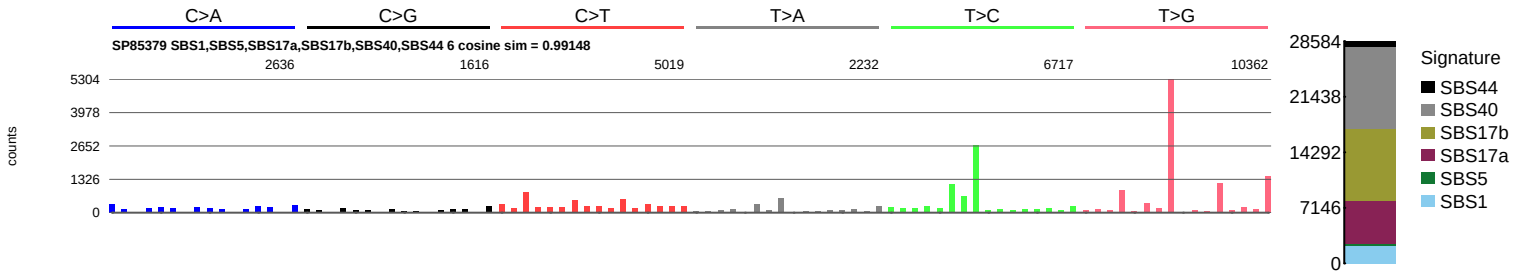
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99149
3095
774
4406
2573
7510
10223
5340
4005
2670
1335
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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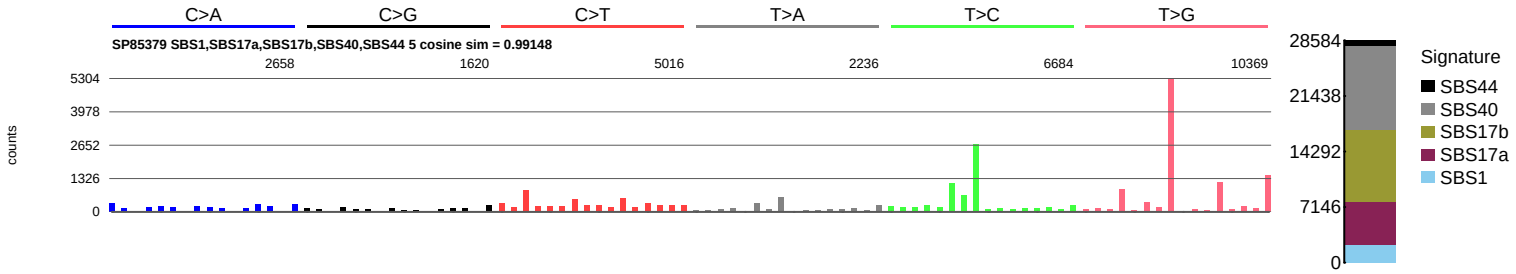
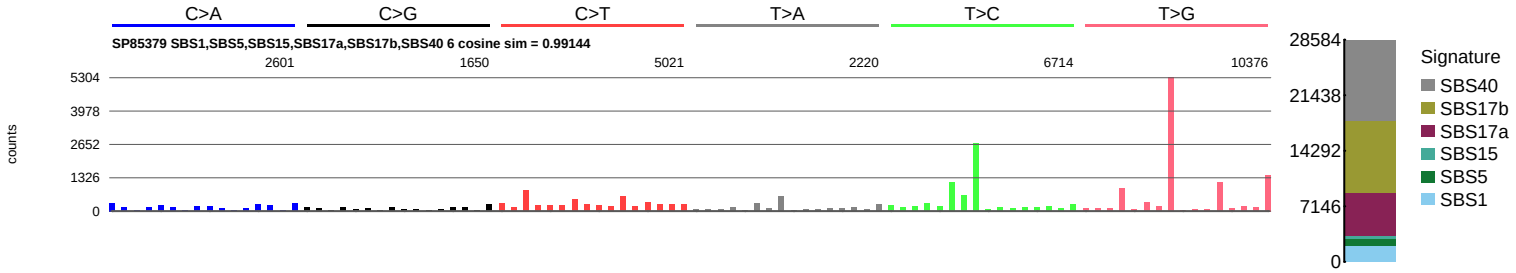
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.99148
2636
1616
5019
2232
6717
10362
5304
3978
2652
1326
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.99148
2658
1620
5016
2236
6684
10369
5304
3978
2652
1326
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1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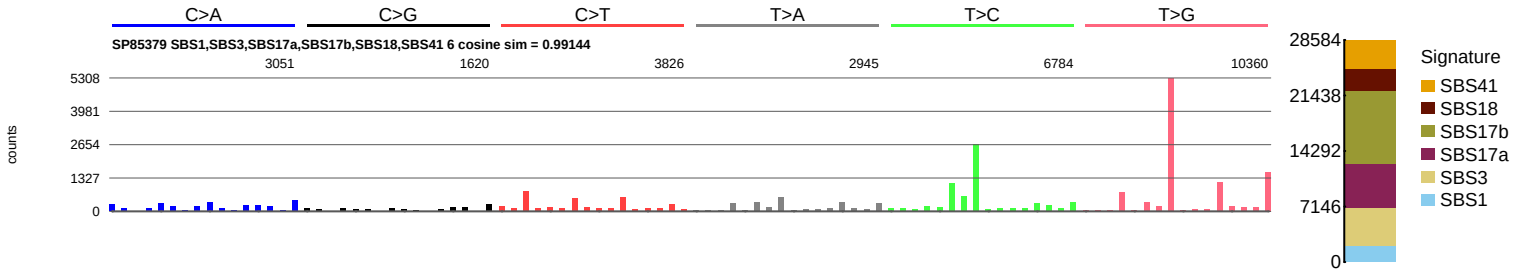
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.99144
2601
1650
5021
2220
6714
10376
5304
3978
2652
1326
0
counts
28584
21438
14292
7146
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1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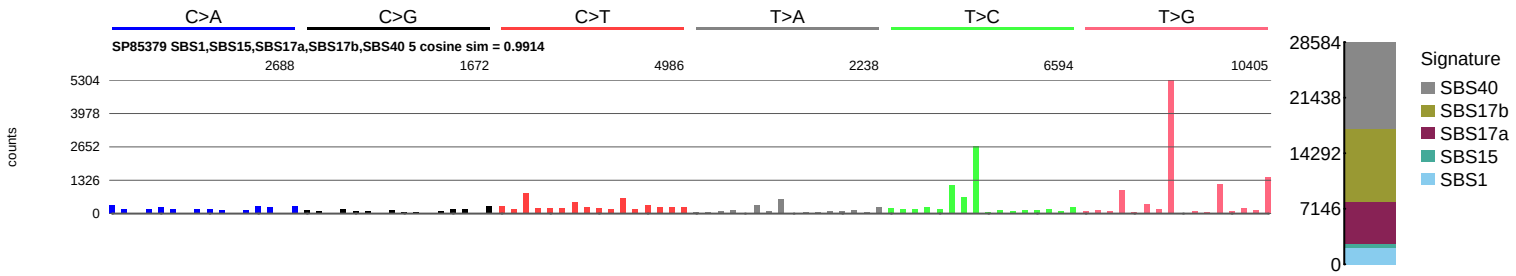
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99144
3051
1620
3826
2945
6784
10360
5308
3981
2654
1327
0
counts
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1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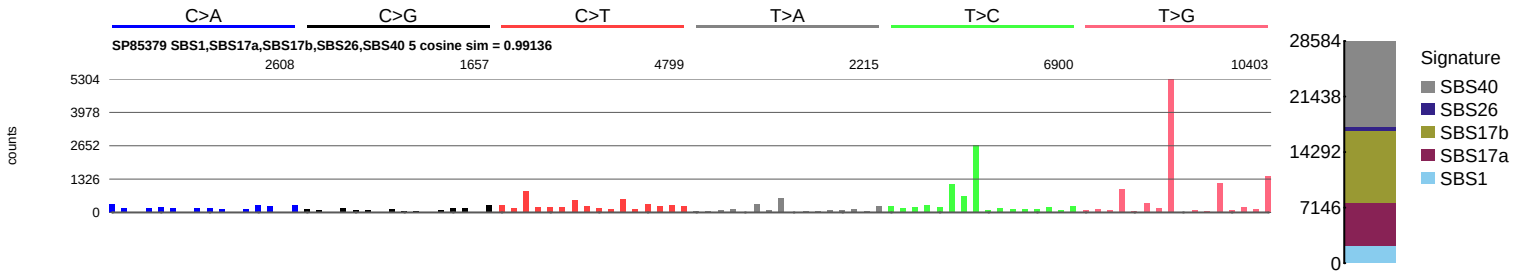
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.9914
2688
1672
4986
2238
6594
10405
5304
3978
2652
1326
0
counts
28584
21438
14292
7146
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS1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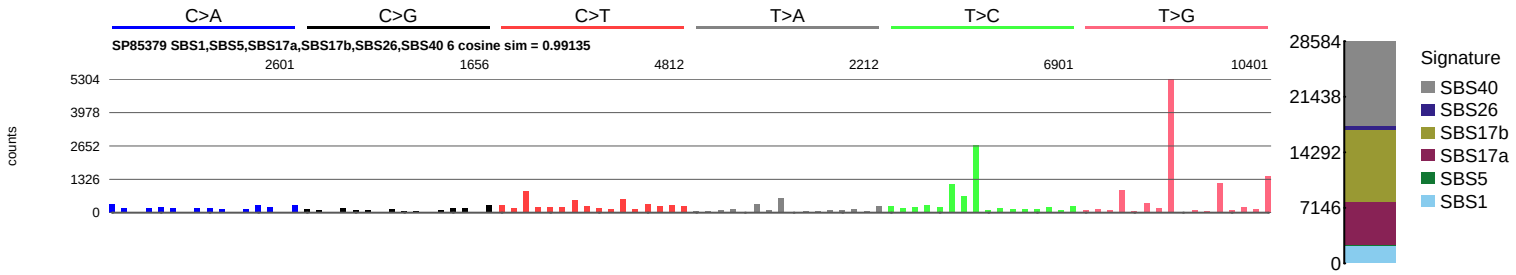
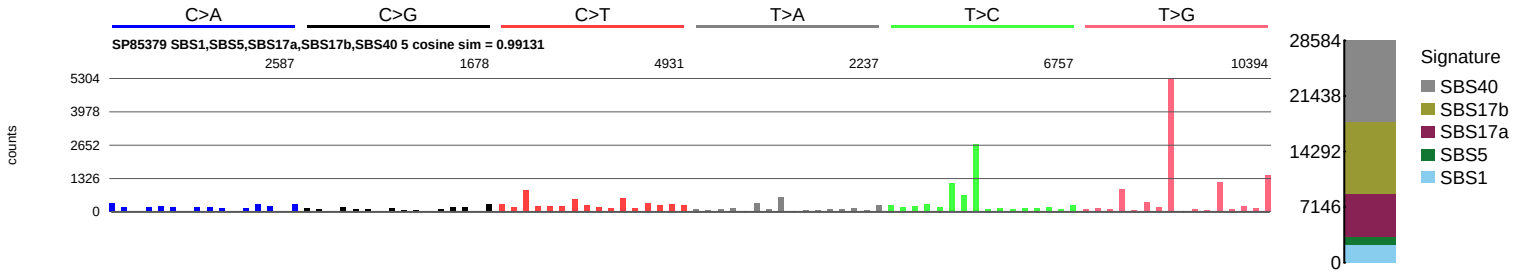
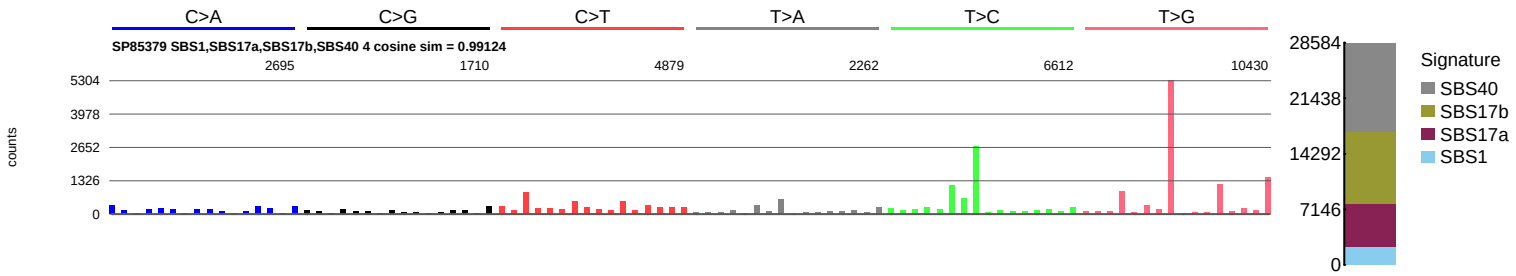
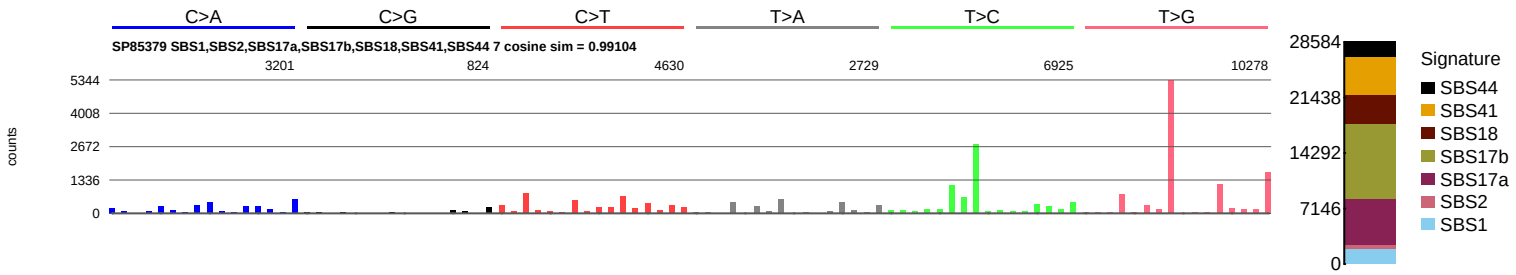
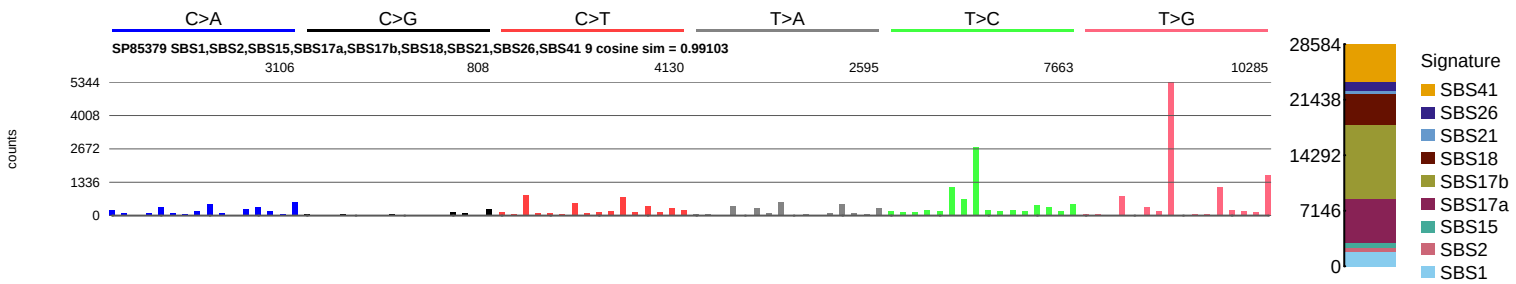
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.99136
2608
1657
4799
2215
6900
10403
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1

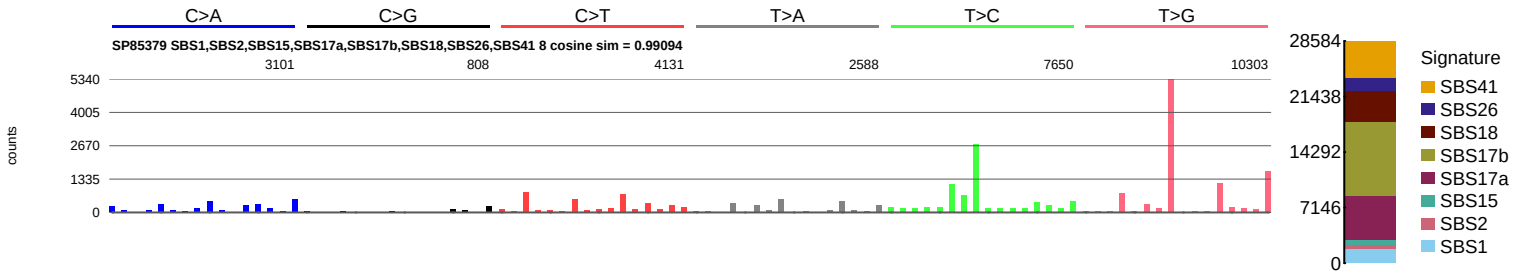

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.99135
2601
1656
4812
2212
6901
10401
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1

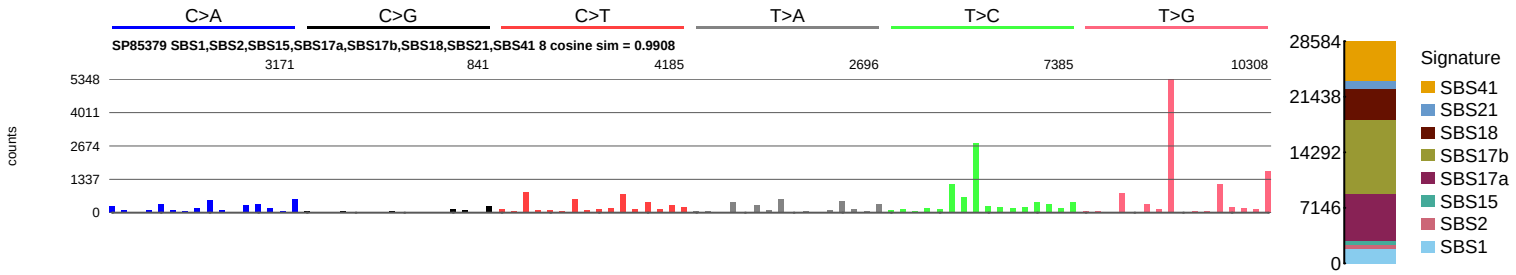

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.99131
2587
1678
4931
2237
6757
10394
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS1

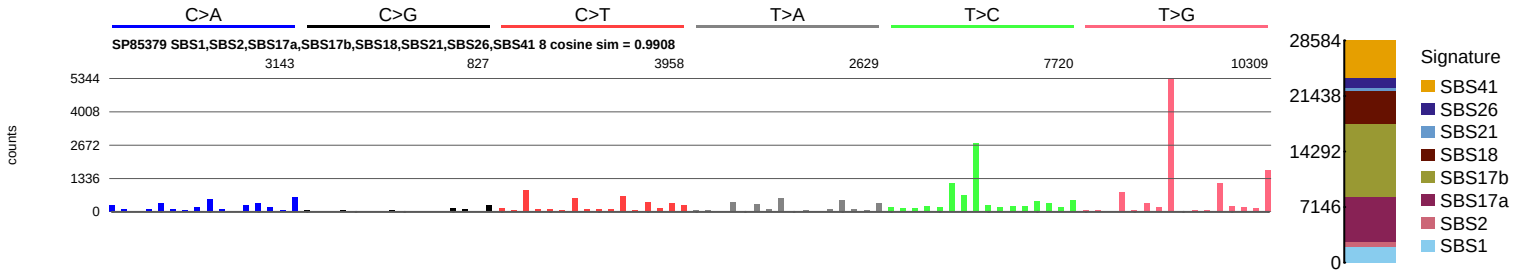

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.99124
2695
1710
4879
2262
6612
10430
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS40
SBS17b
SBS17a
SBS1

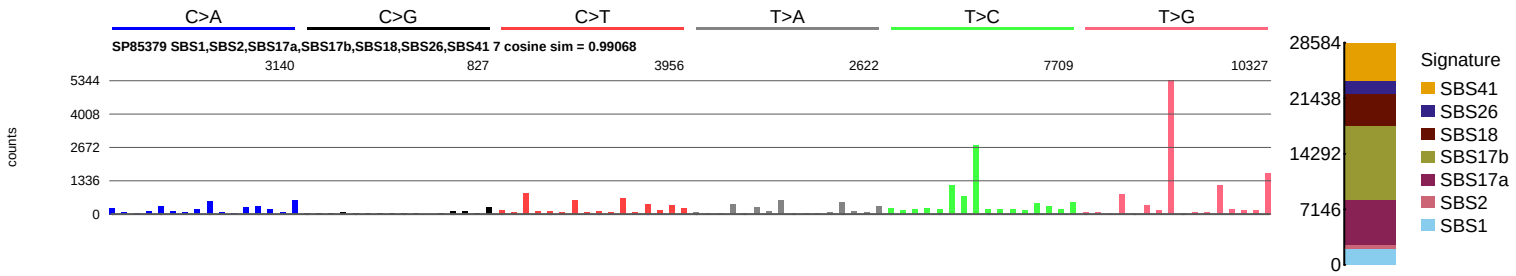

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.99104
3201
824
4630
2729
6925
10278
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1

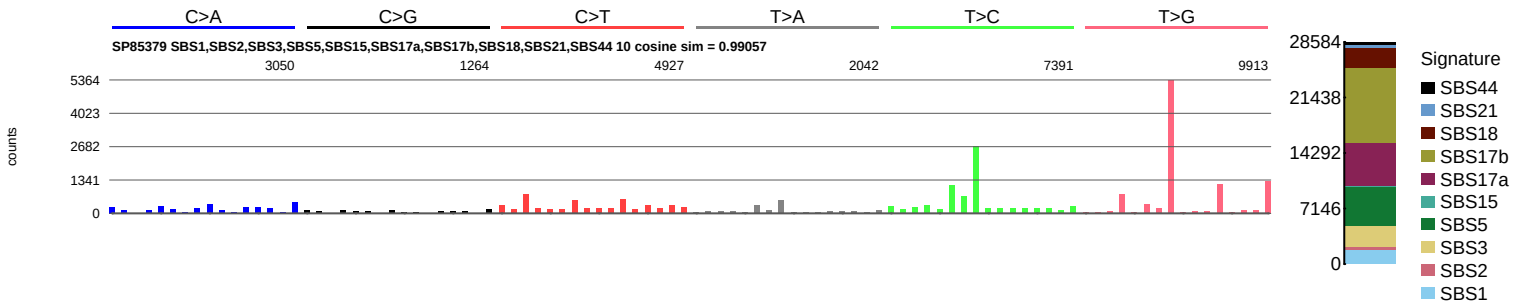

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99103
3106
808
4130
2595
7663
10285
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

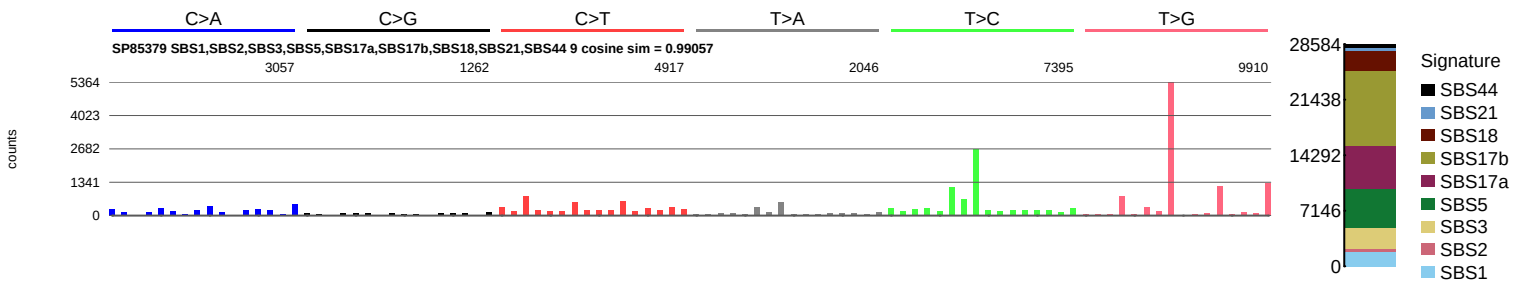

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99094
3101
808
4131
2588
7650
10303
counts
5340
4005
2670
1335
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

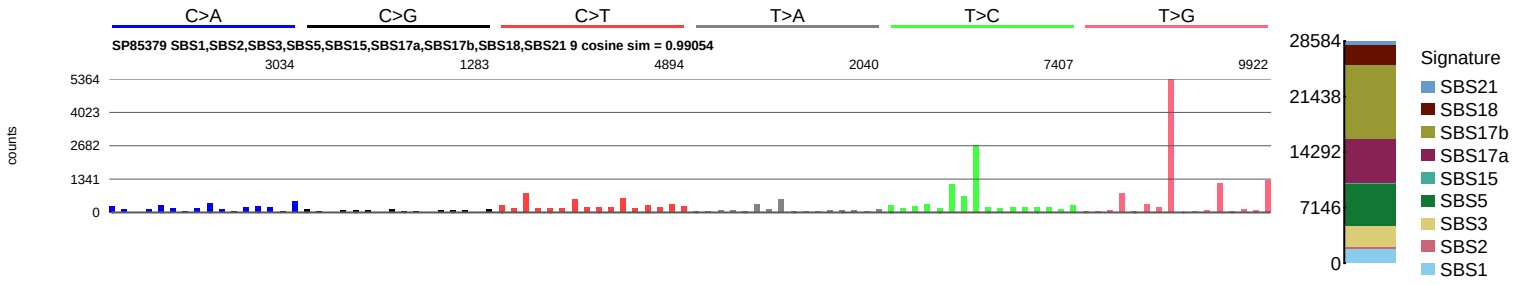

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9908
3171
841
4185
2696
7385
10308
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

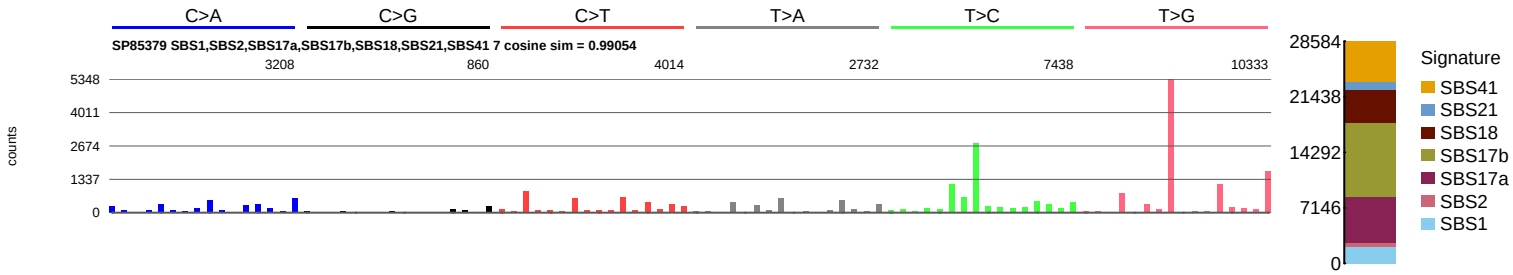

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.9908
3143
827
3958
2629
7720
10309
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

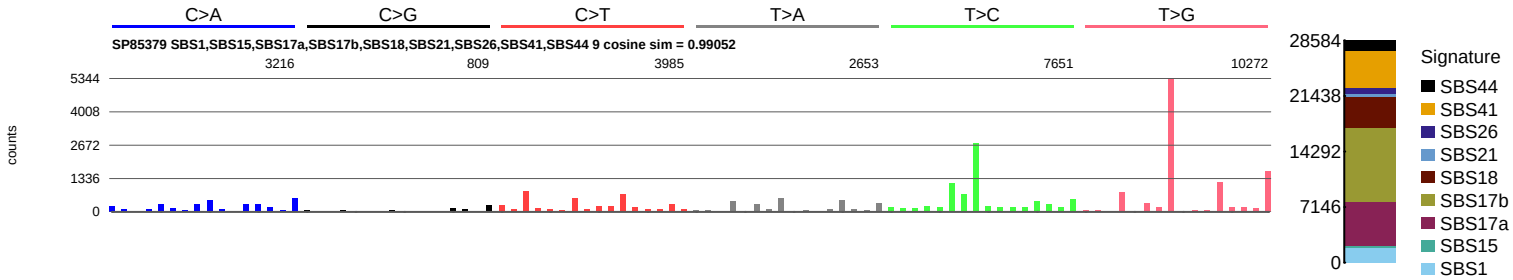

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99068
3140
827
3956
2622
7709
10327
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

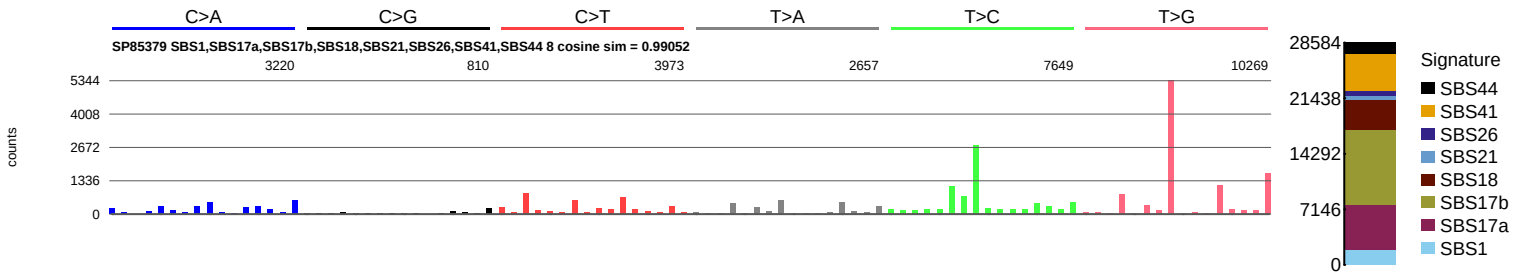

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.99057
3050
1264
4927
2042
7391
9913
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

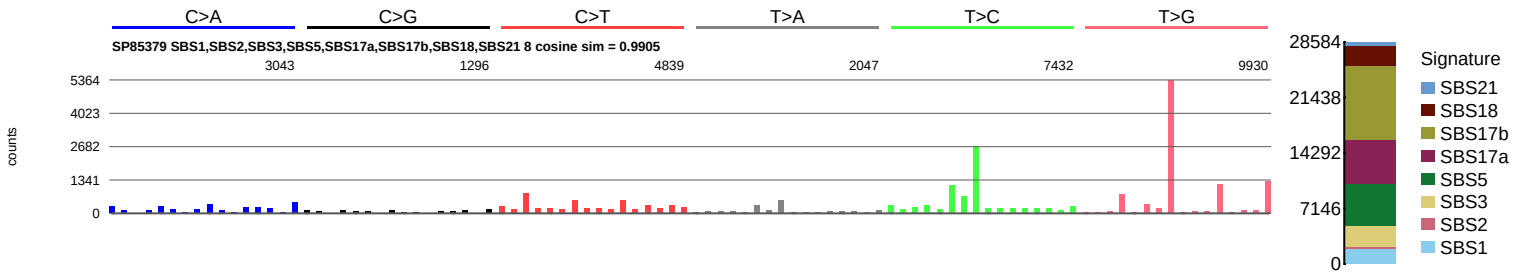

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99057
3057
1262
4917
2046
7395
9910
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99054
3034
1283
4894
2040
7407
9922
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99054
3208
860
4014
2732
7438
10333
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99052
3216
809
3985
2653
7651
10272
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.99052
3220
810
3973
2657
7649
10269
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1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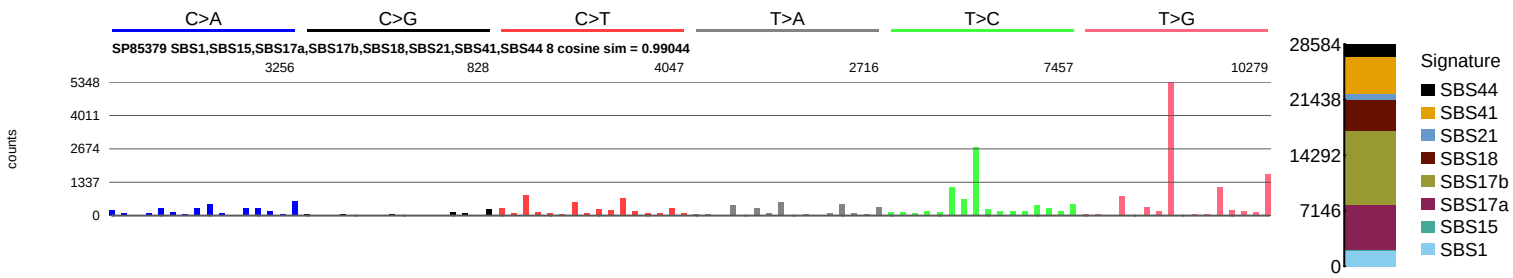
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.9905
3043
1296
4839
2047
7432
9930
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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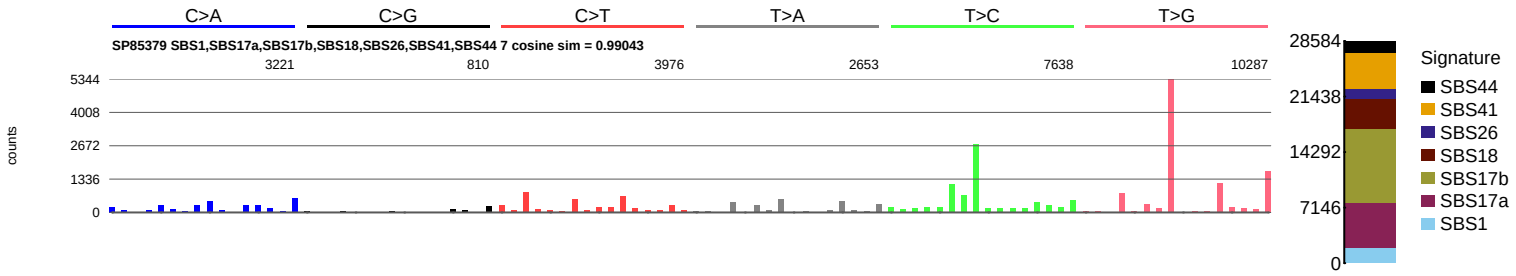
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.99044
3256
828
4047
2716
7457
10279
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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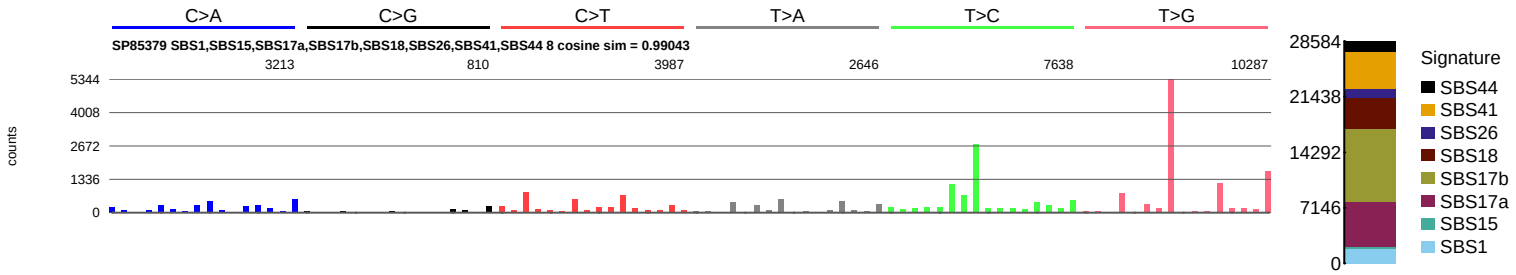
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.99043
3221
810
3976
2653
7638
10287
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
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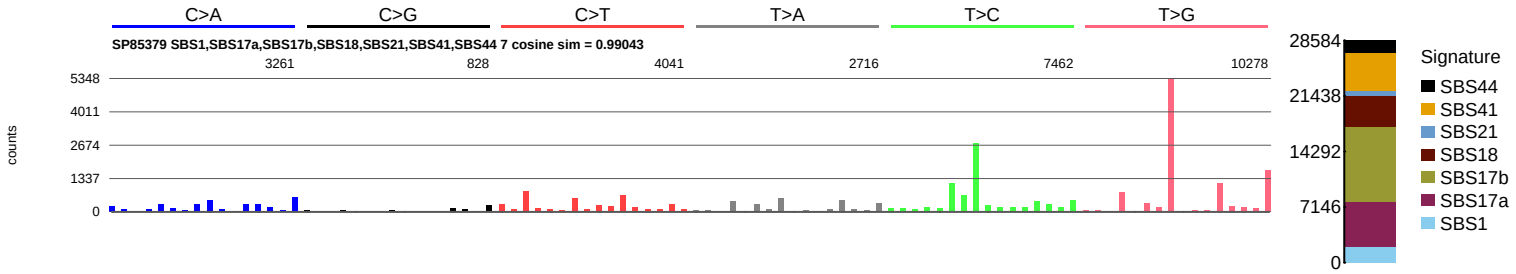
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99043
3213
810
3987
2646
7638
10287
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.99043
3261
828
4041
2716
7462
10278
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
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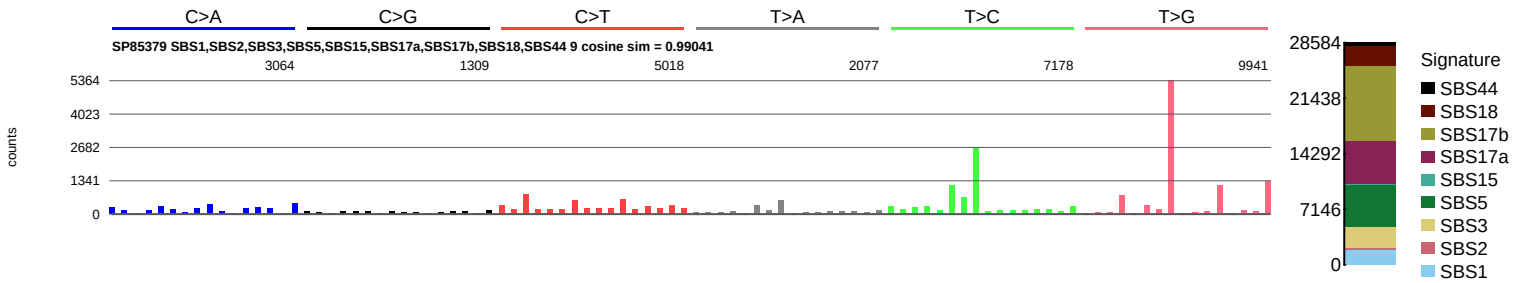
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1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C>T
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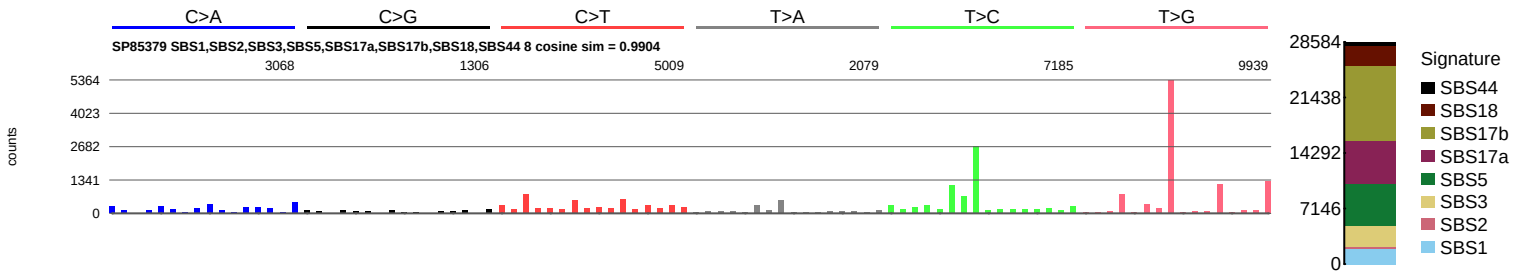
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.99041
3064
1309
5018
2077
7178
9941
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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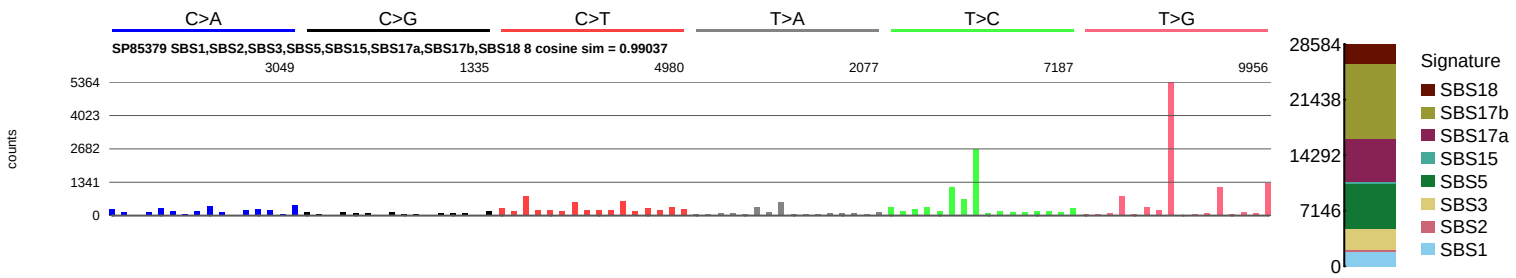
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.9904
3068
1306
5009
2079
7185
9939
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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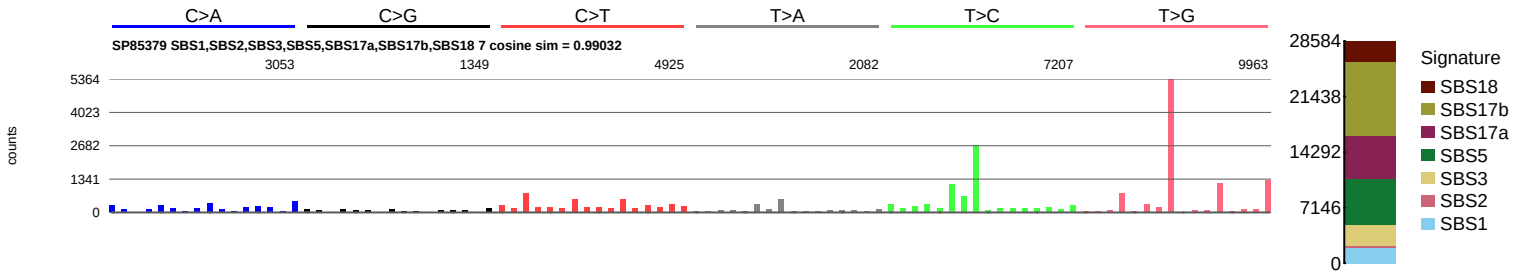
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99037
3049
1335
4980
2077
7187
9956
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
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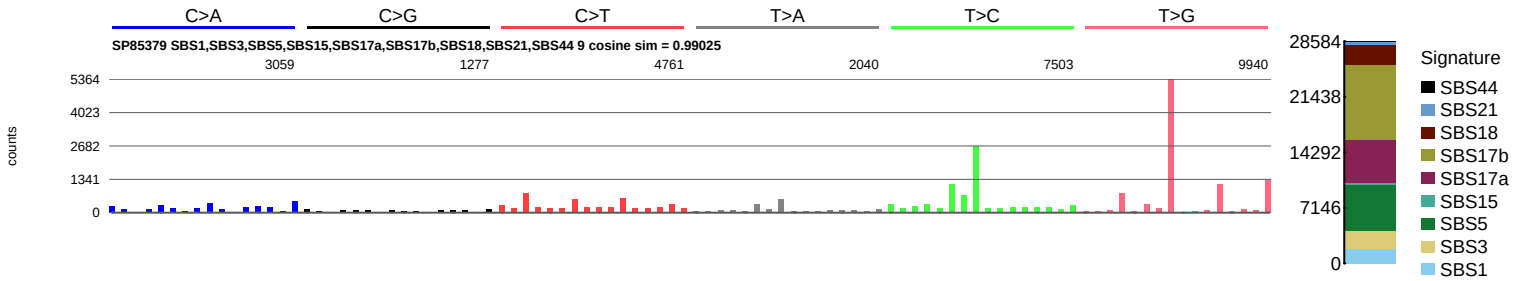
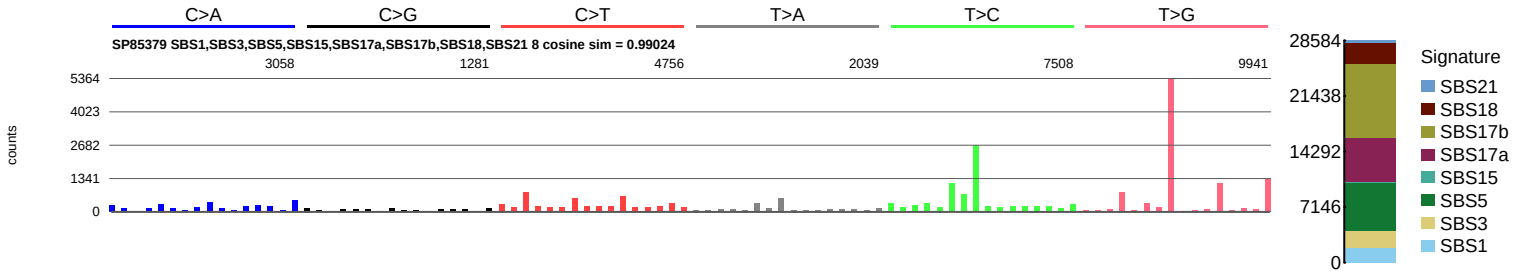
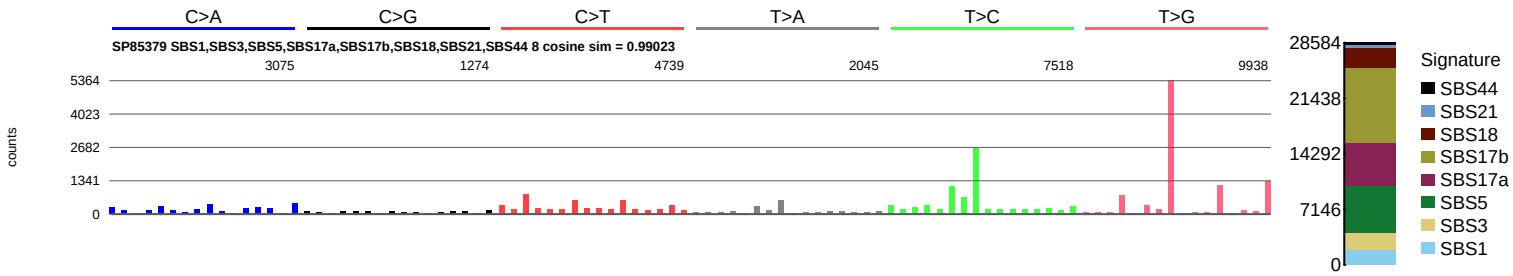
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99032
3053
1349
4925
2082
7207
9963
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99025
3059
1277
4761
2040
7503
9940
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99024
3058
1281
4756
2039
7508
9941
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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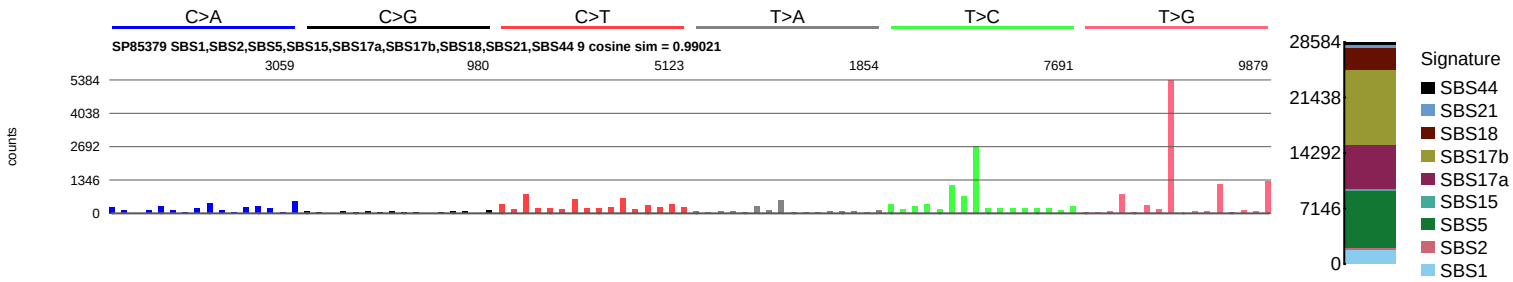
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.99023
3075
1274
4739
2045
7518
9938
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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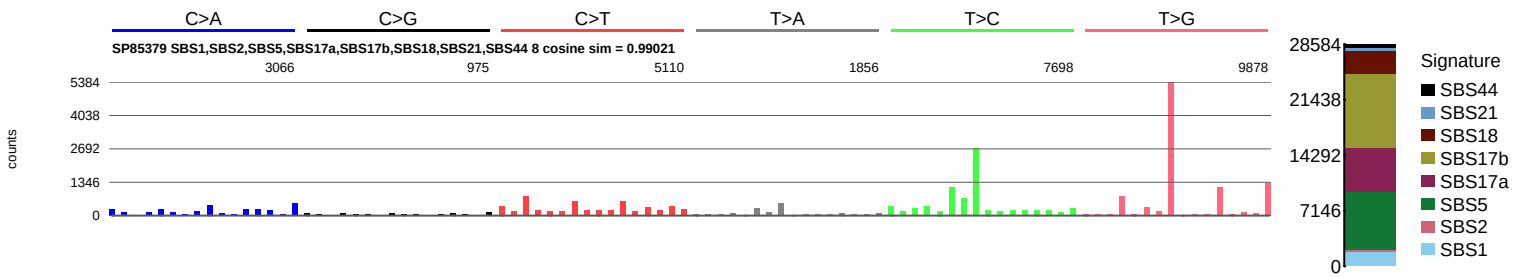
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99021
3059
980
5123
1854
7691
9879
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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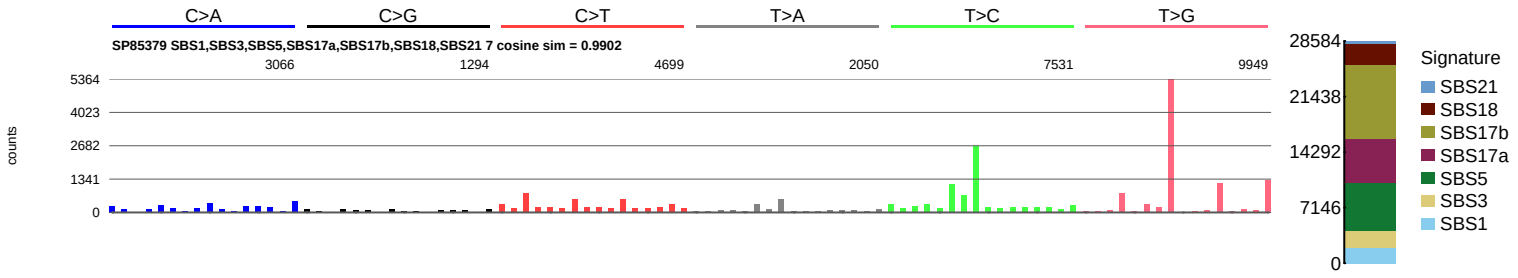
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.99021
3066
975
5110
1856
7698
9878
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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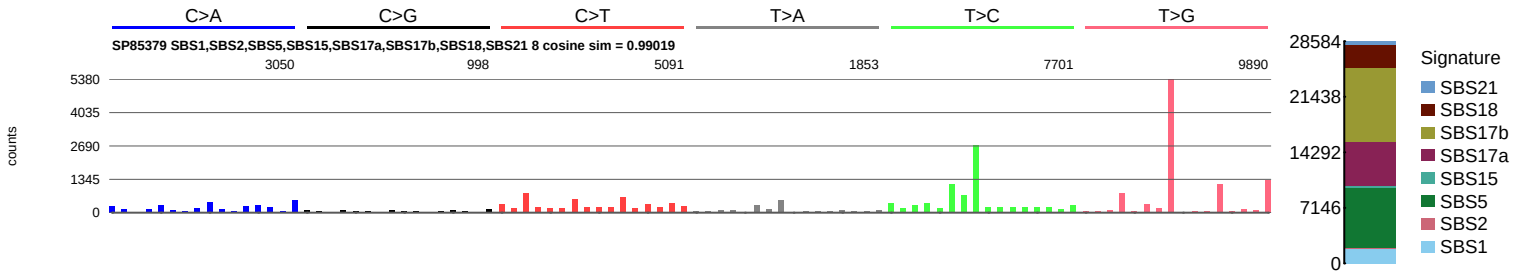
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.9902
3066
1294
4699
2050
7531
9949
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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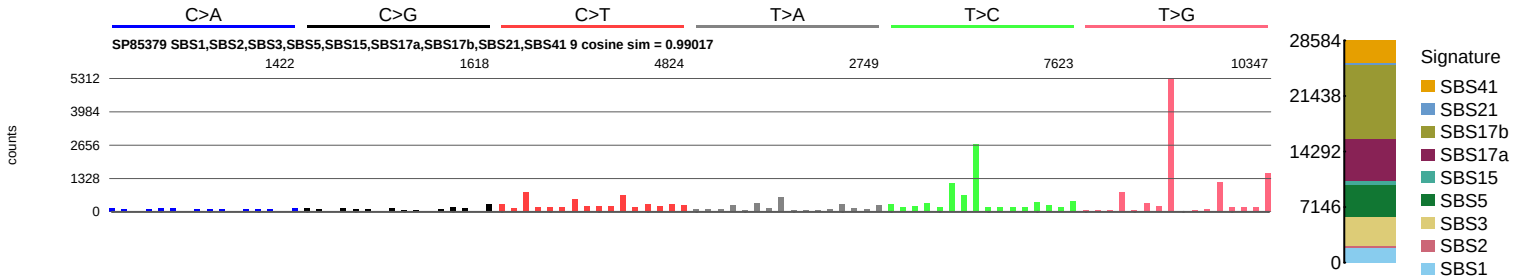
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99019
3050
998
5091
1853
7701
9890
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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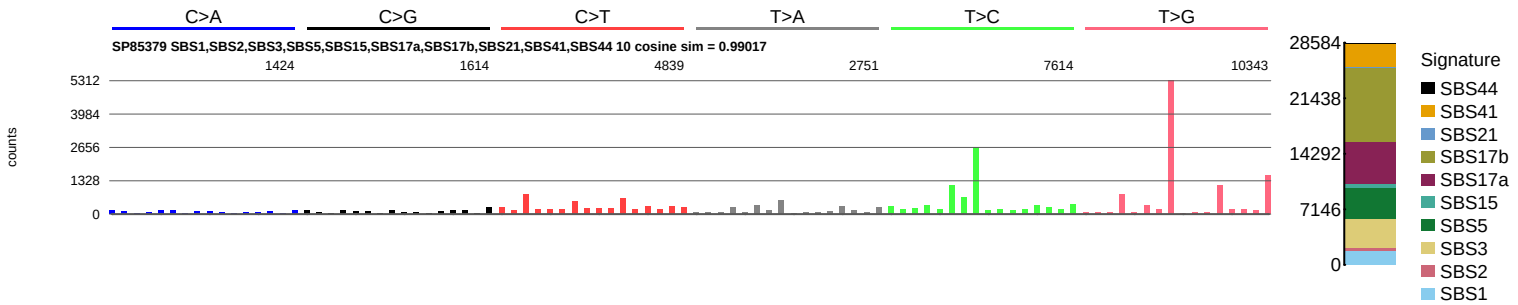
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99017
1422
1618
4824
2749
7623
10347
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
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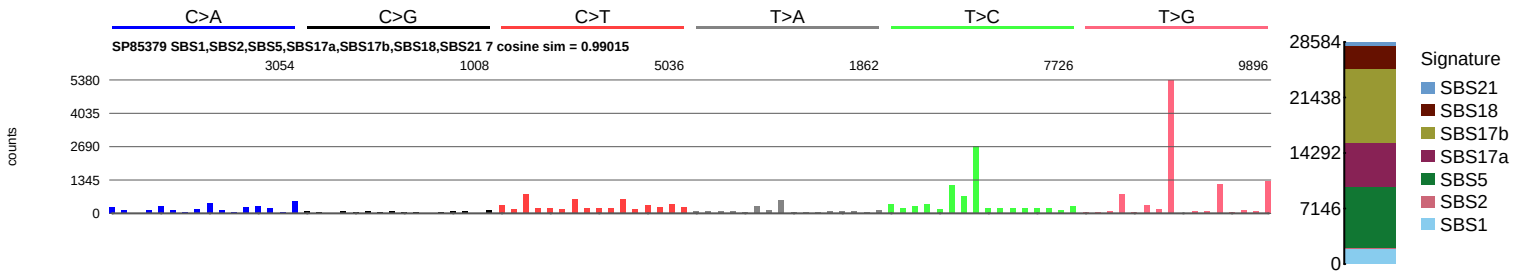
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.99017
1424
1614
4839
2751
7614
10343
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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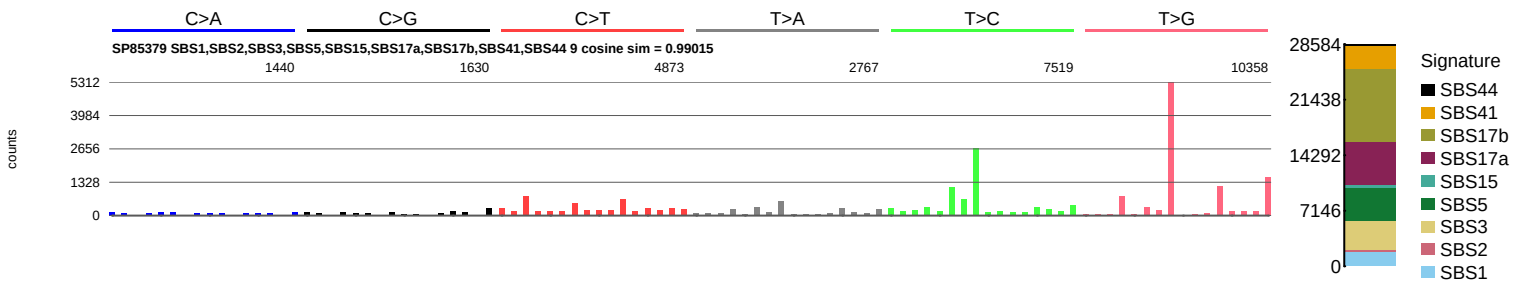
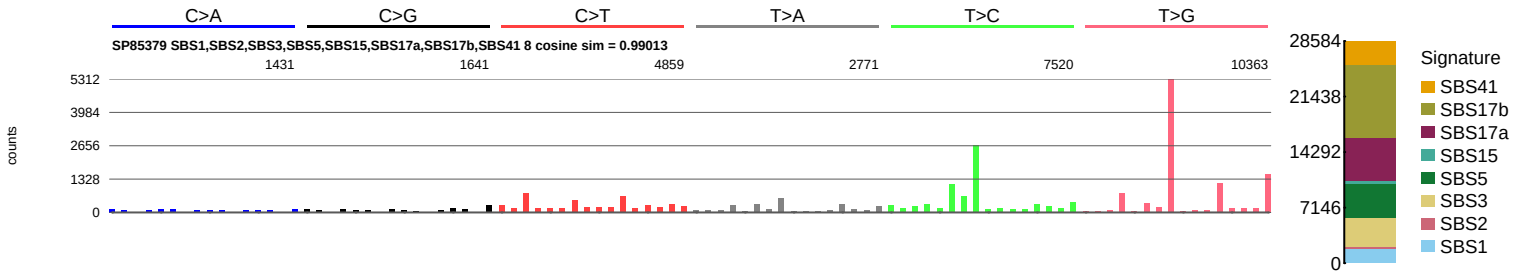
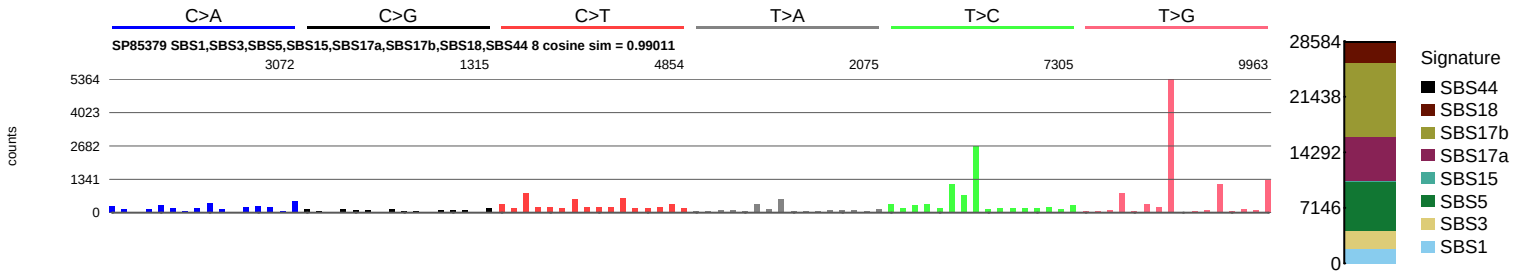
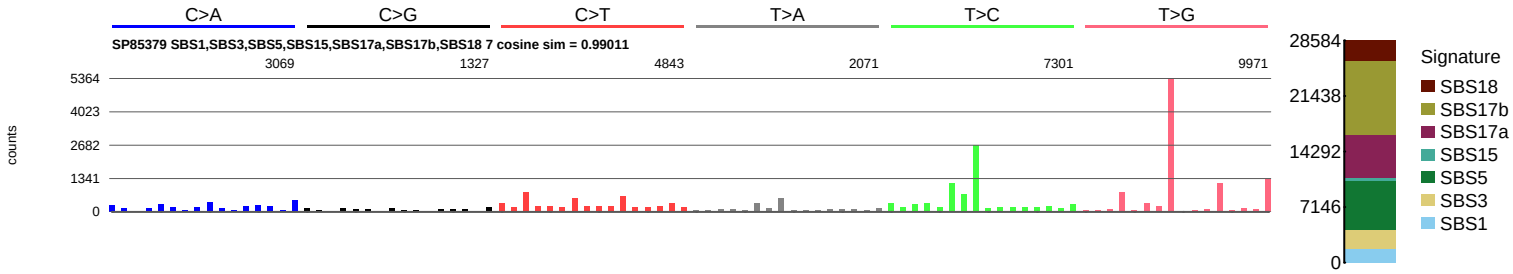
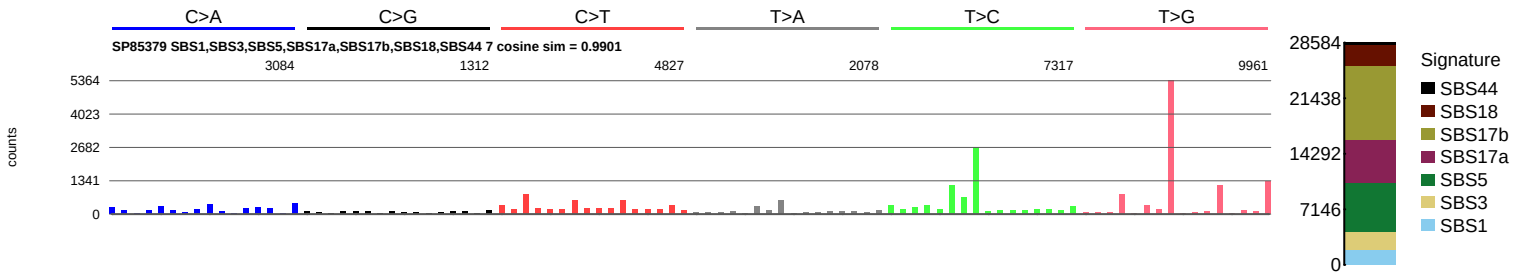
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.99015
3054
1008
5036
1862
7726
9896
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

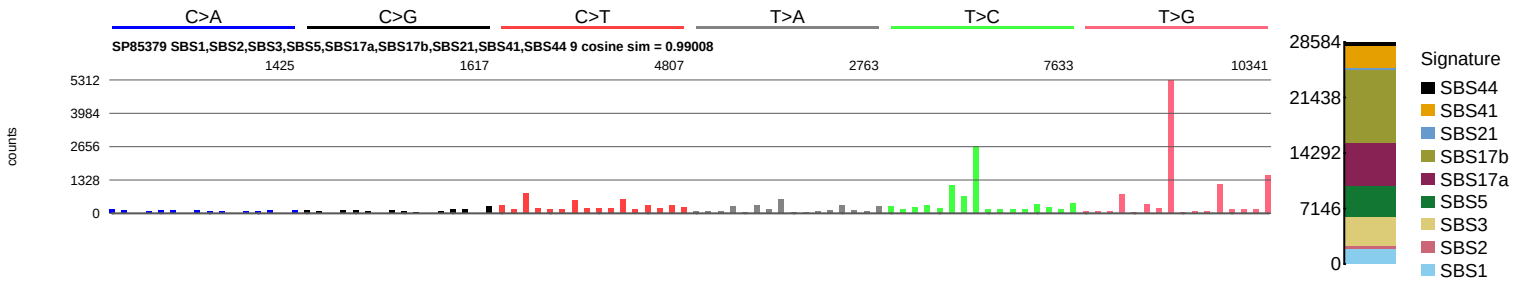

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.99015
1440
1630
4873
2767
7519
10358
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

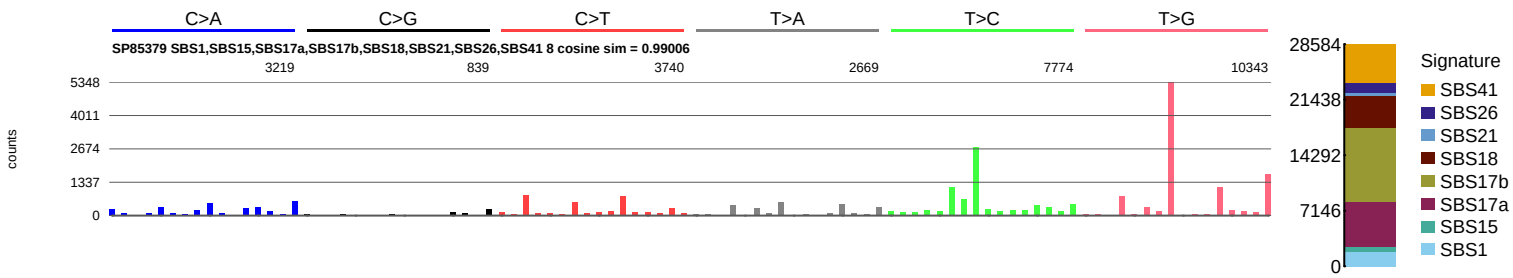

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99013
1431
1641
4859
2771
7520
10363
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99011
3072
1315
4854
2075
7305
9963
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99011
3069
1327
4843
2071
7301
9971
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9901
3084
1312
4827
2078
7317
9961
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.99008
1425
1617
4807
2763
7633
10341
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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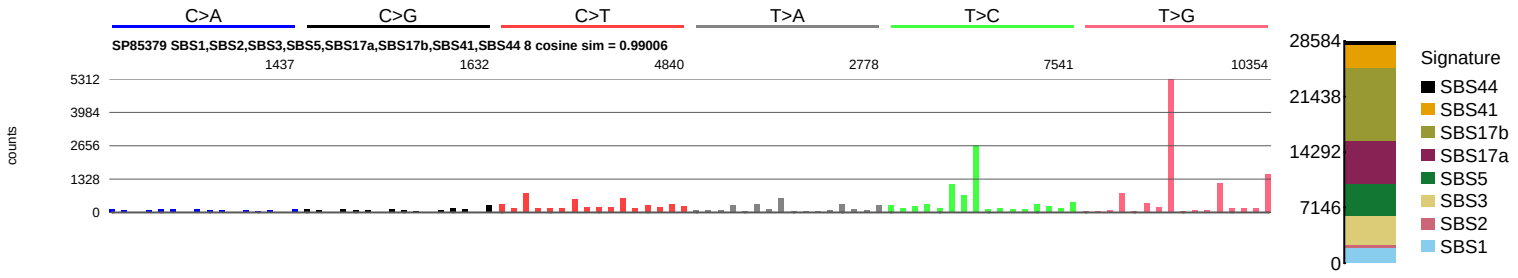
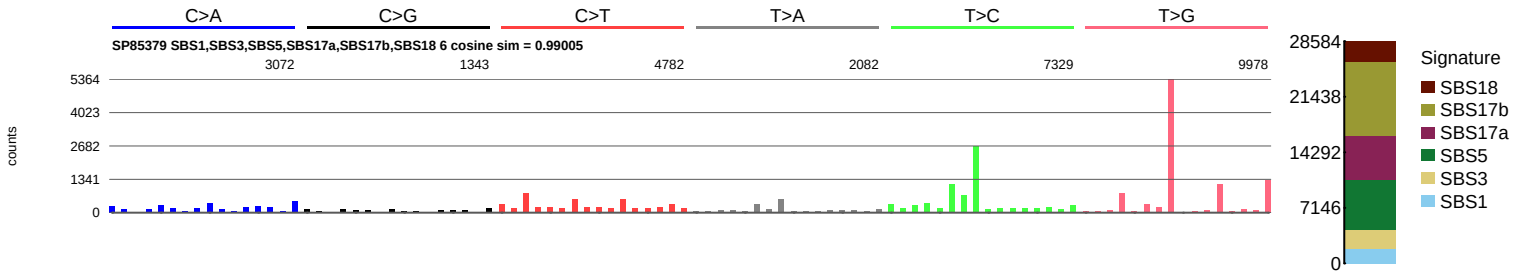
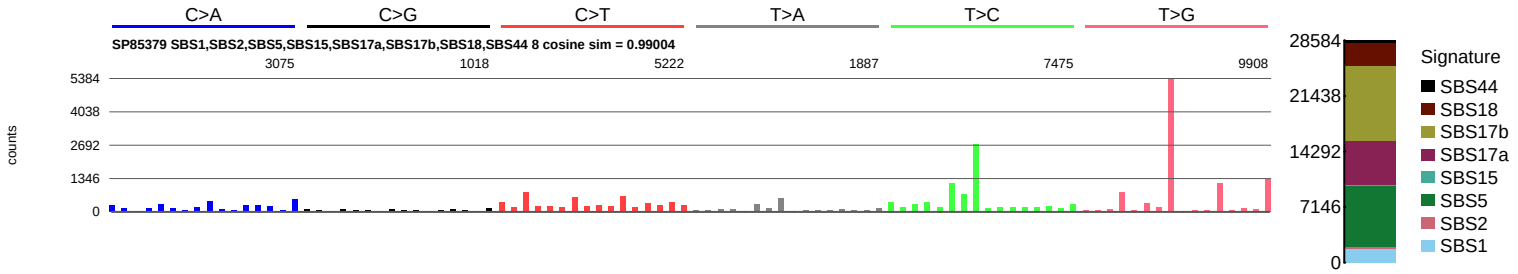
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.99006
3219
839
3740
2669
7774
10343
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.99006
1437
1632
4840
2778
7541
10354
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.99005
3072
1343
4782
2082
7329
9978
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99004
3075
1018
5222
1887
7475
9908
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
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Signature
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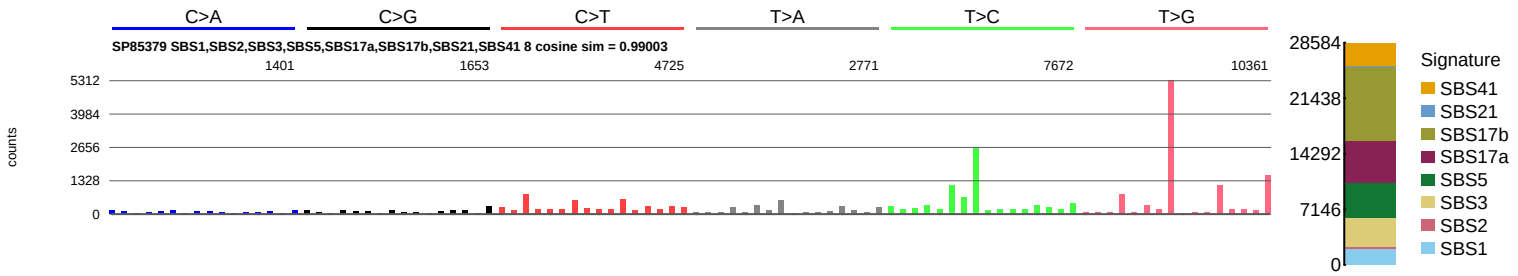
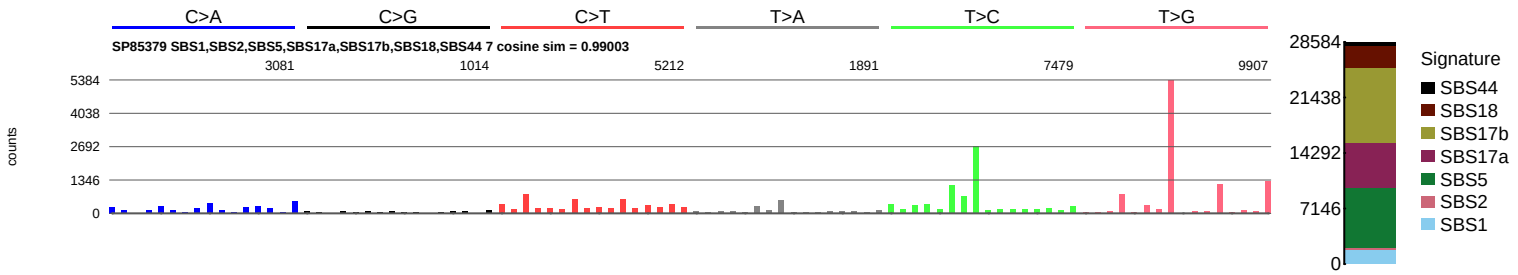
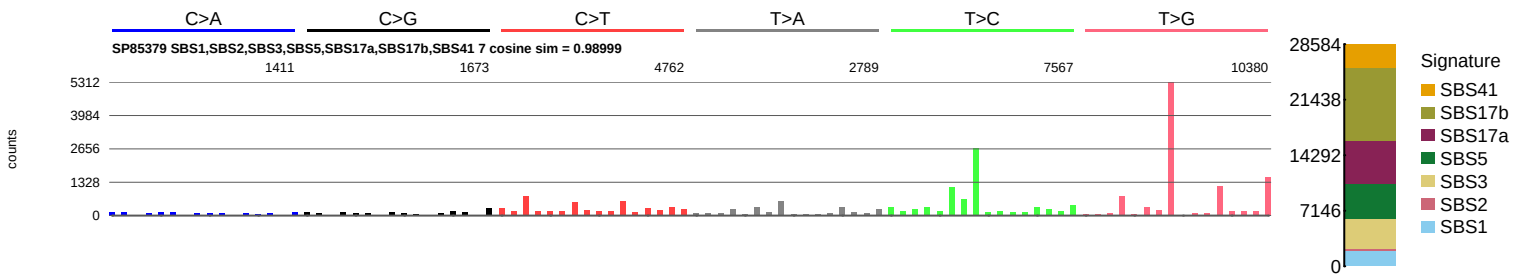
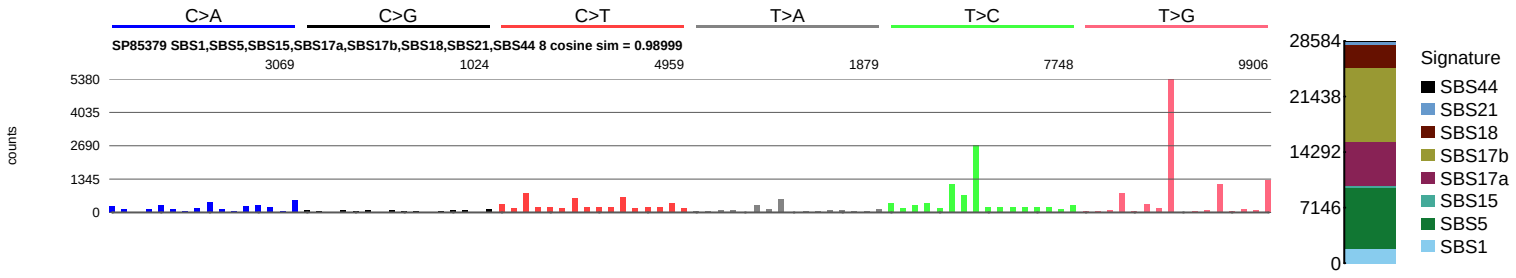
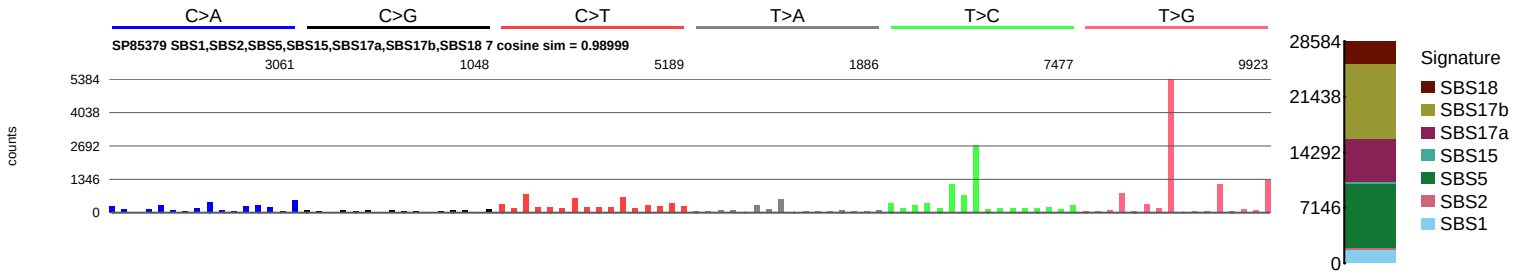
SBS2
SBS1

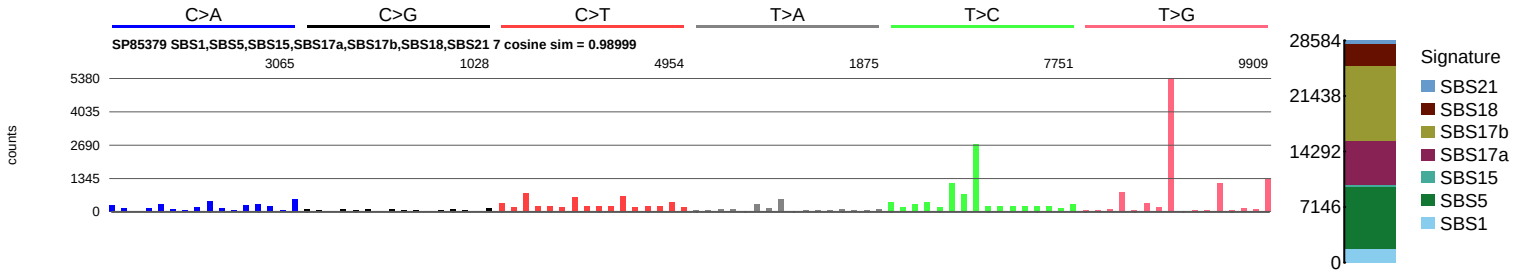

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99003
1401
1653
4725
2771
7672
10361
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

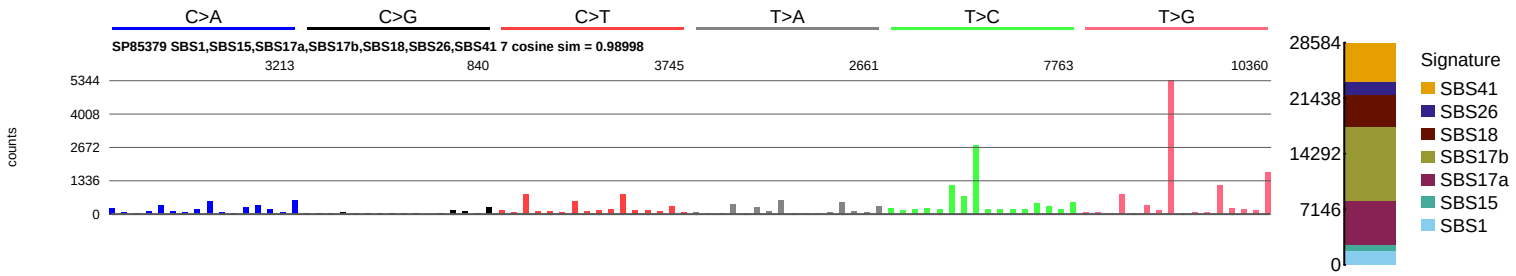

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.99003
3081
1014
5212
1891
7479
9907
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

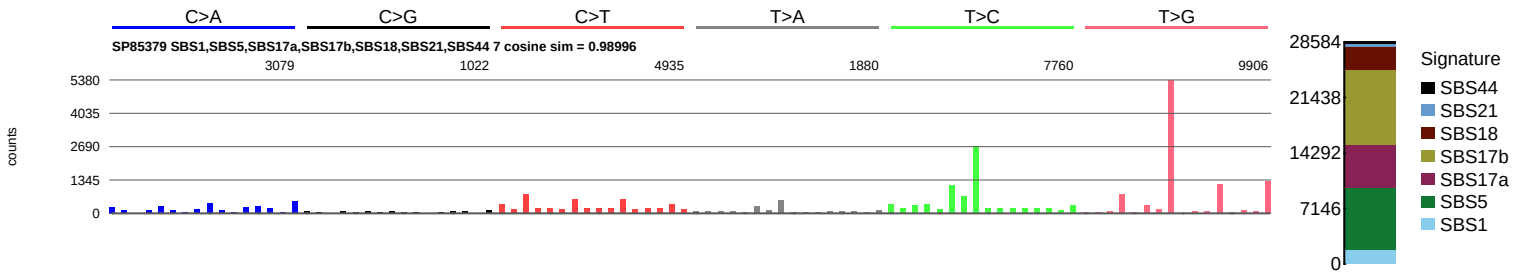

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98999
1411
1673
4762
2789
7567
10380
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

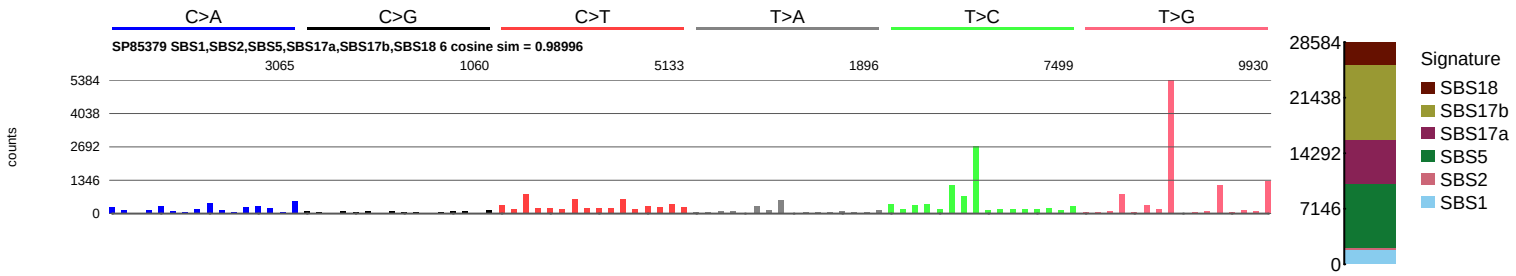

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98999
3069
1024
4959
1879
7748
9906
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

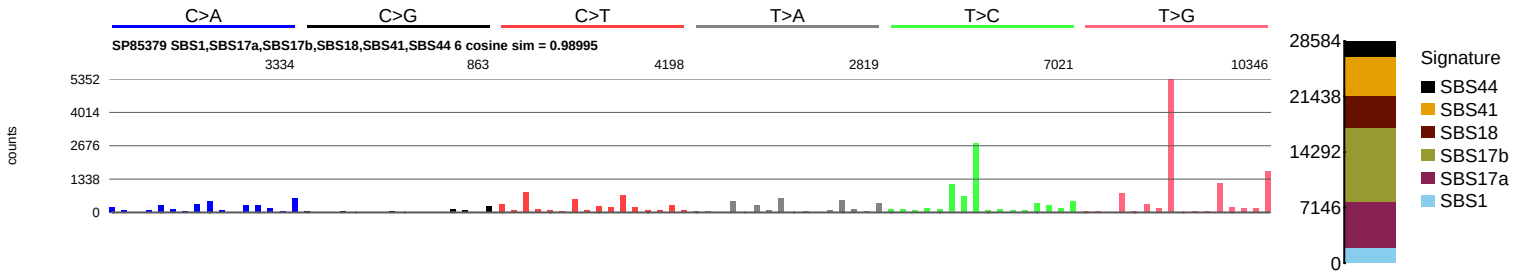

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98999
3061
1048
5189
1886
7477
9923
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98999
3065
1028
4954
1875
7751
9909
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98998
3213
840
3745
2661
7763
10360
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98996
3079
1022
4935
1880
7760
9906
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98996
3065
1060
5133
1896
7499
9930
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98995
3334
863
4198
2819
7021
10346
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
0
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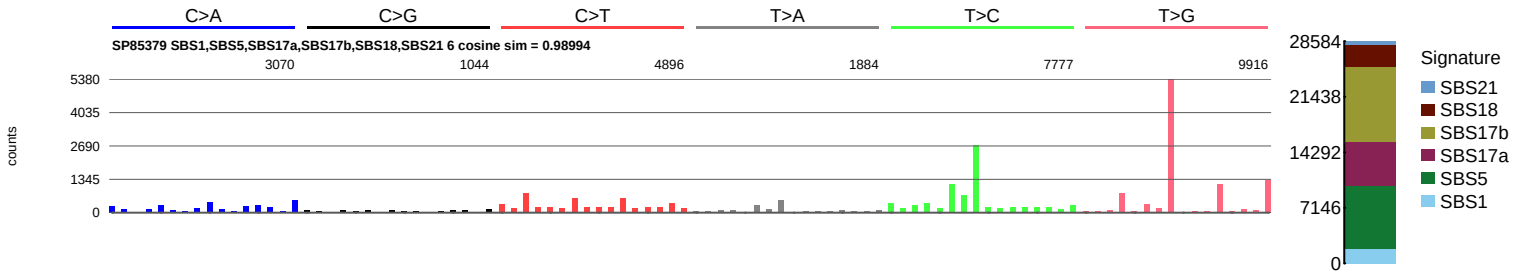
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98994
3070
1044
4896
1884
7777
9916
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
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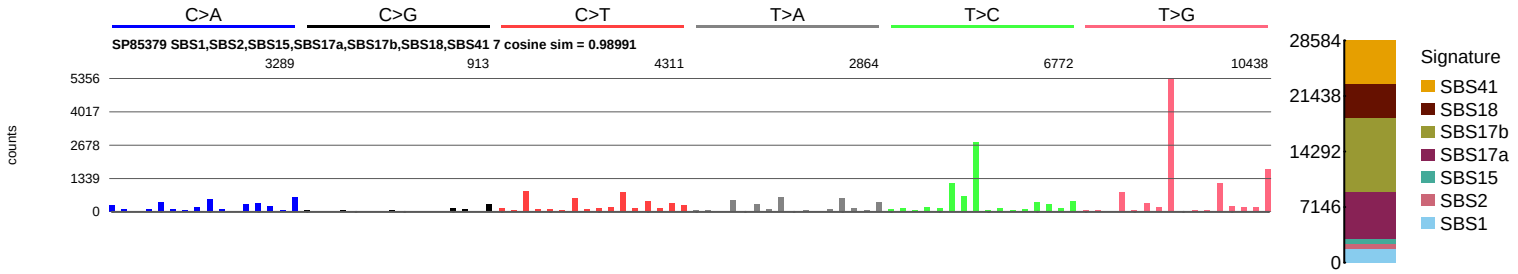
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98991
3289
913
4311
2864
6772
10438
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
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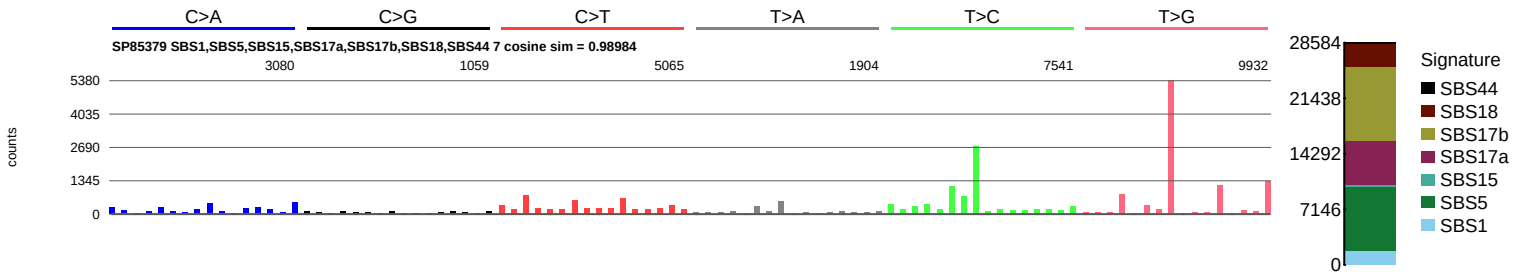
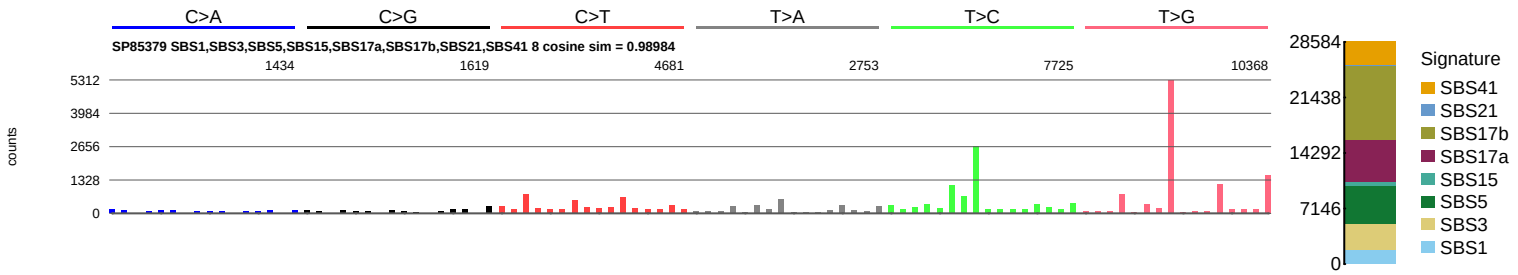
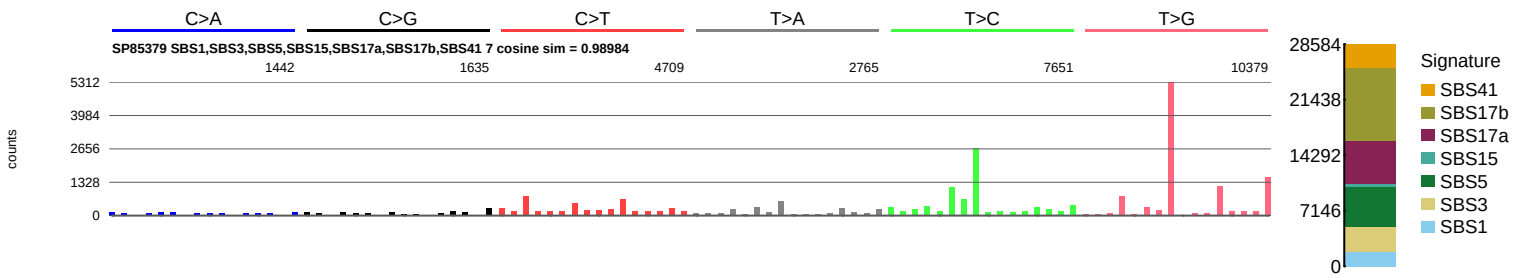
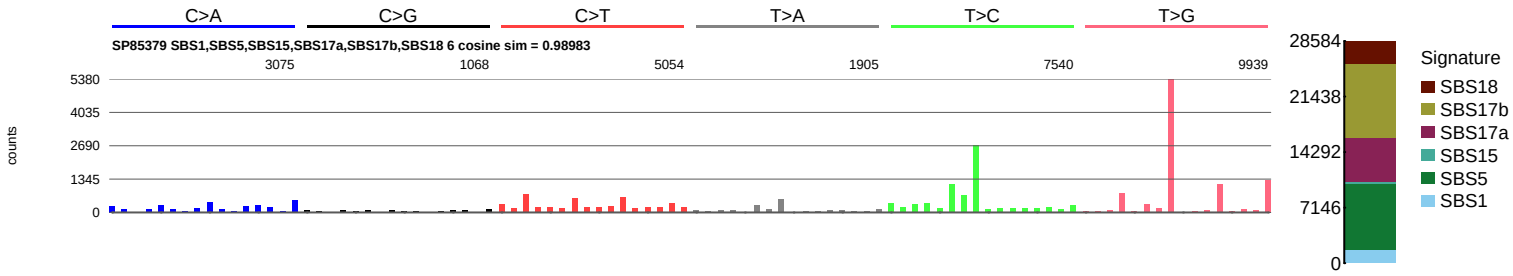
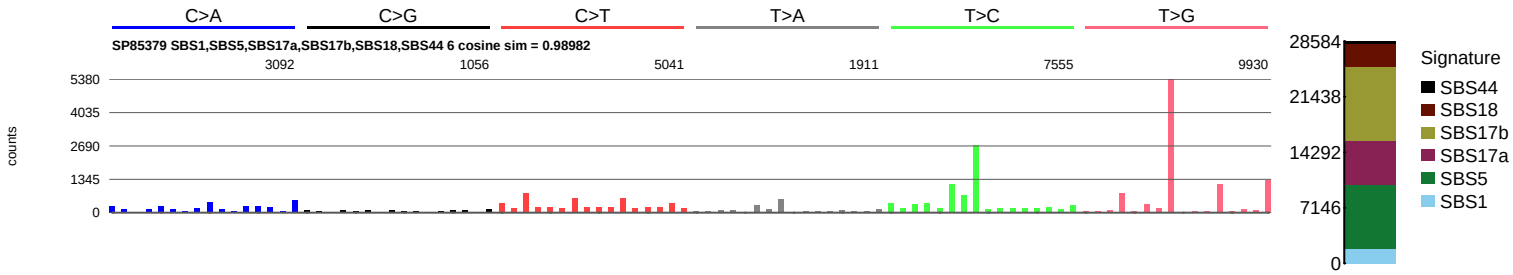
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

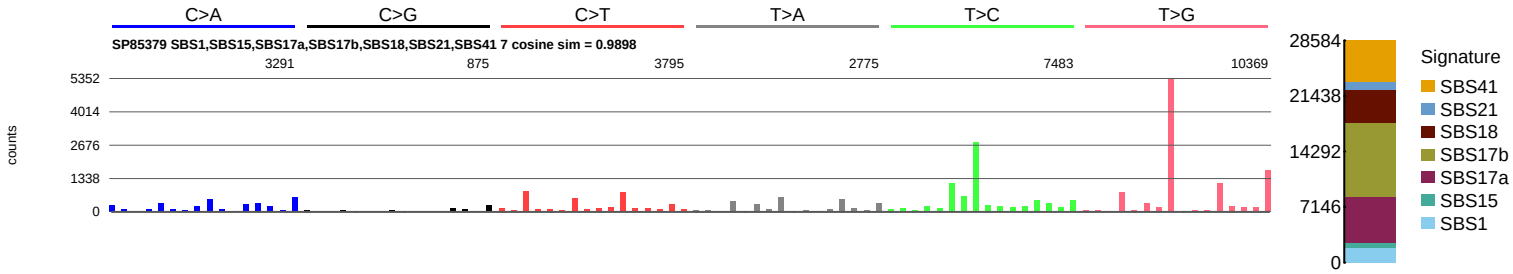

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98984
3080
1059
5065
1904
7541
9932
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

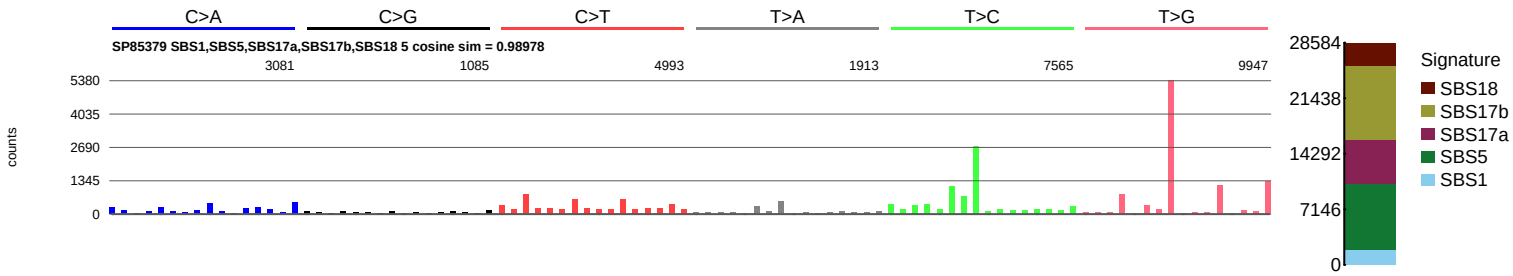

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98984
1434
1619
4681
2753
7725
10368
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

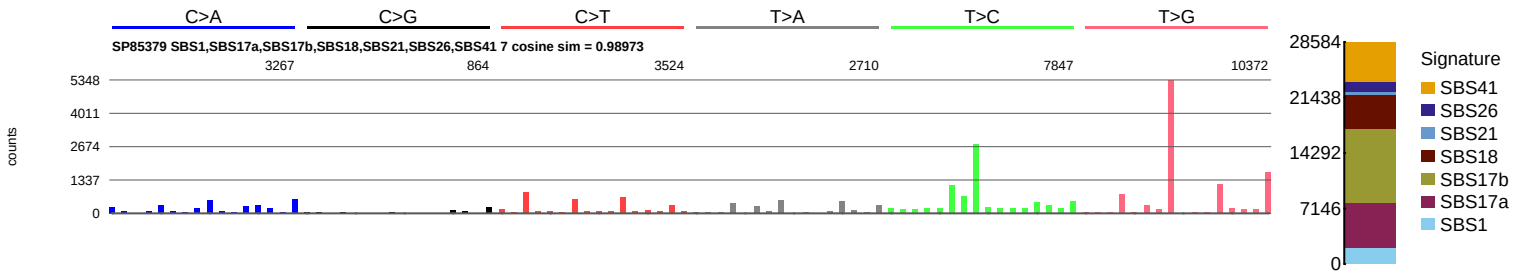

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98984
1442
1635
4709
2765
7651
10379
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

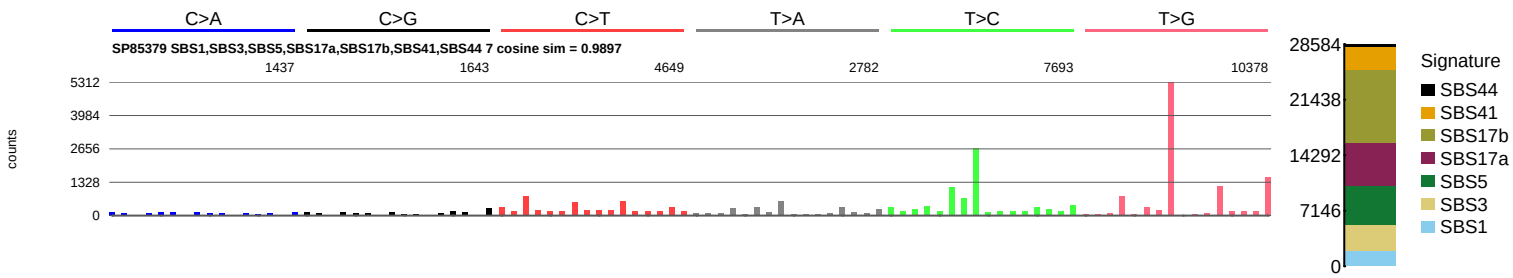

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98983
3075
1068
5054
1905
7540
9939
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

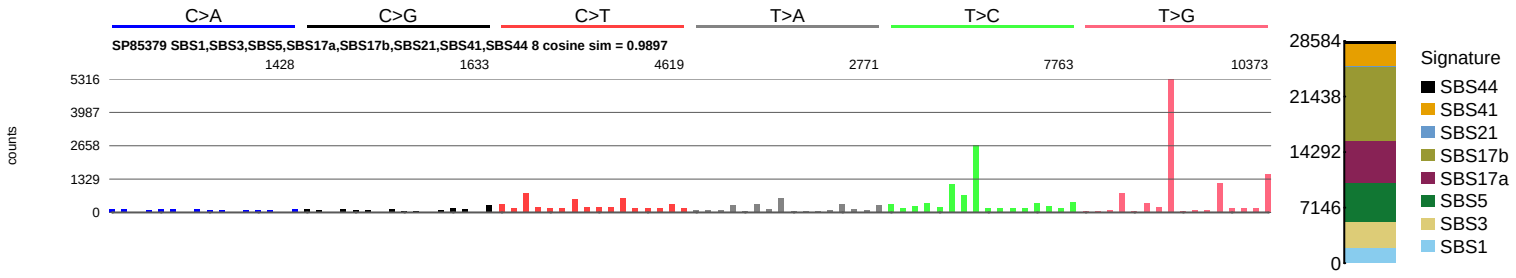

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98982
3092
1056
5041
1911
7555
9930
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9898
3291
875
3795
2775
7483
10369
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98978
3081
1085
4993
1913
7565
9947
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98973
3267
864
3524
2710
7847
10372
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.9897
1437
1643
4649
2782
7693
10378
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1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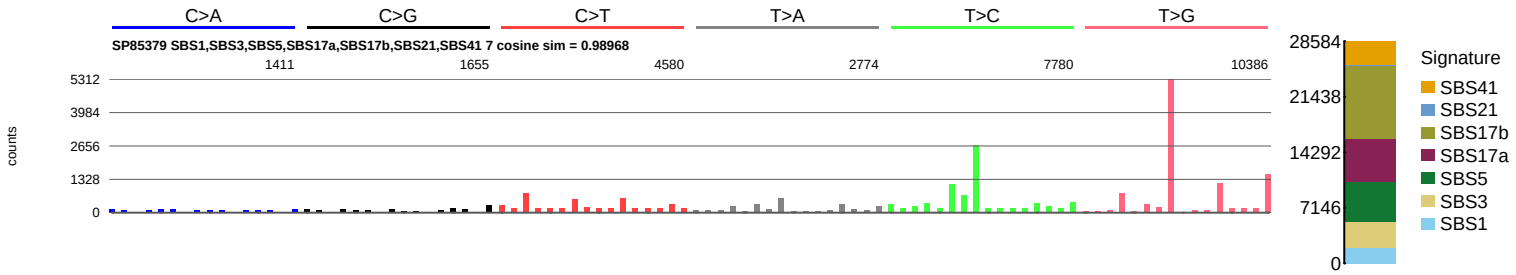
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.9897
1428
1633
4619
2771
7763
10373
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1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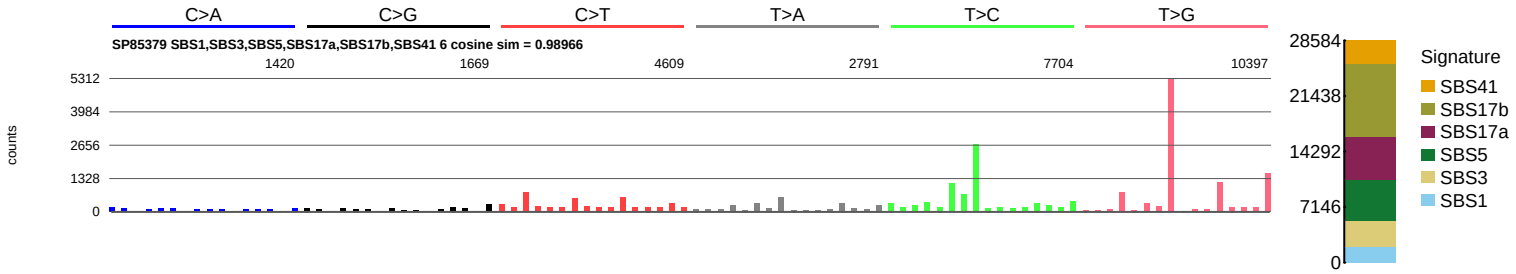
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98968
1411
1655
4580
2774
7780
10386
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
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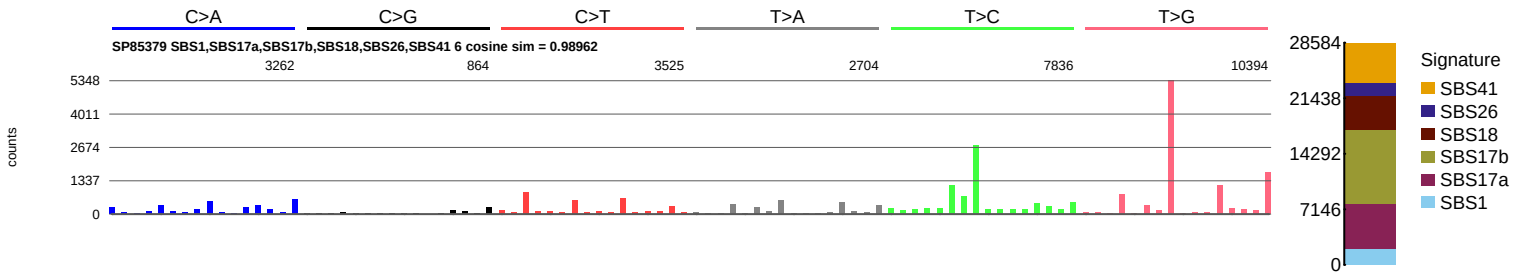
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98966
1420
1669
4609
2791
7704
10397
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
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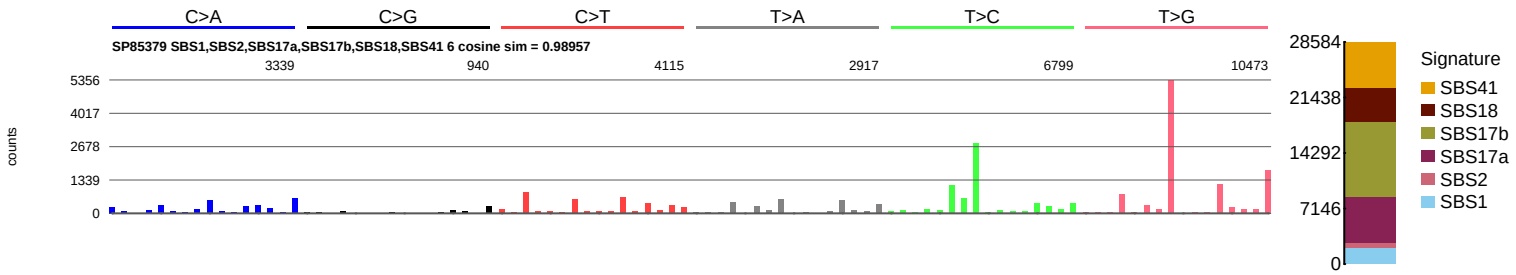
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98962
3262
864
3525
2704
7836
10394
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
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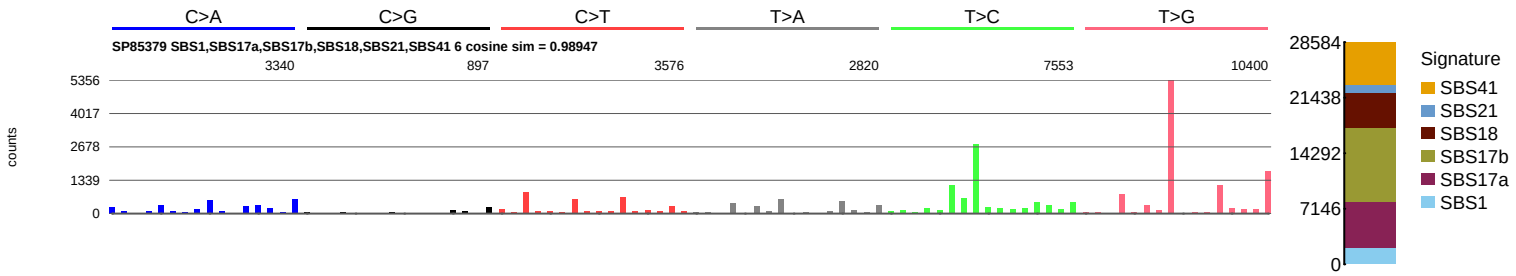
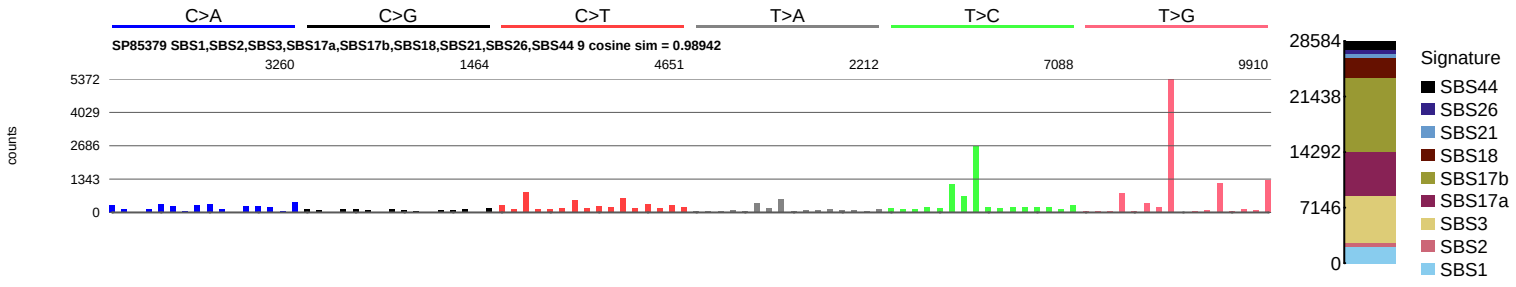
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98957
3339
940
4115
2917
6799
10473
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98947
3340
897
3576
2820
7553
10400
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1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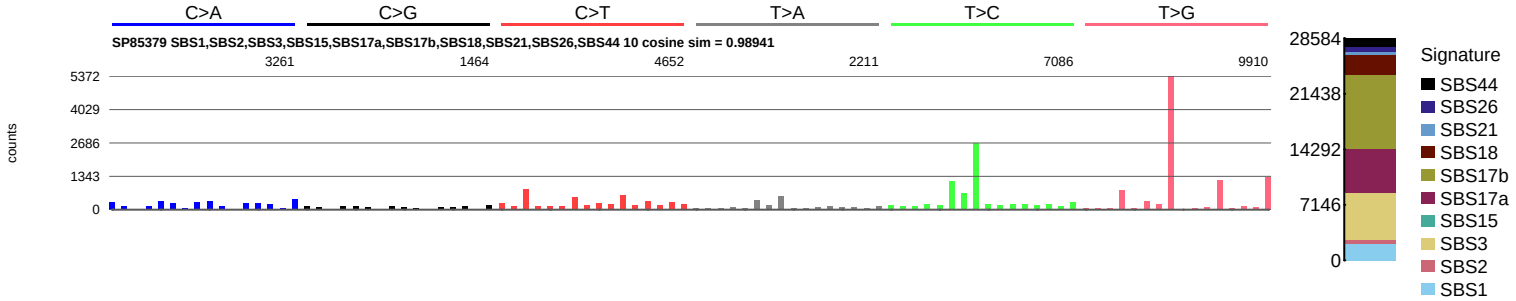
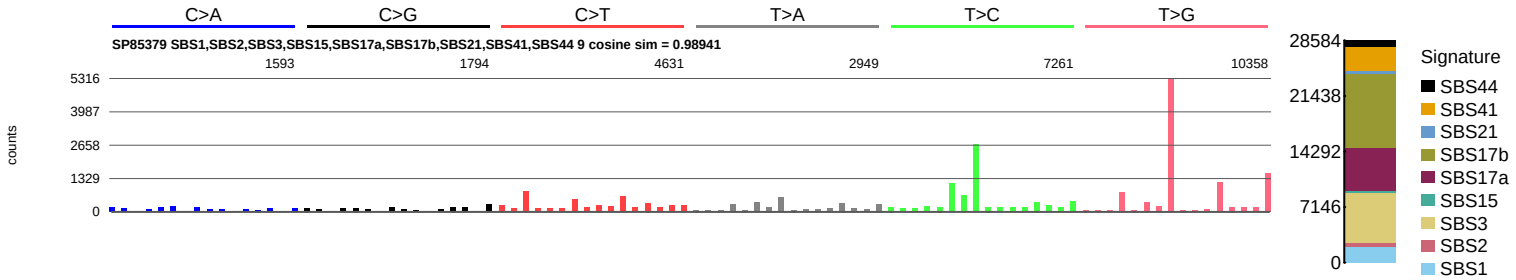
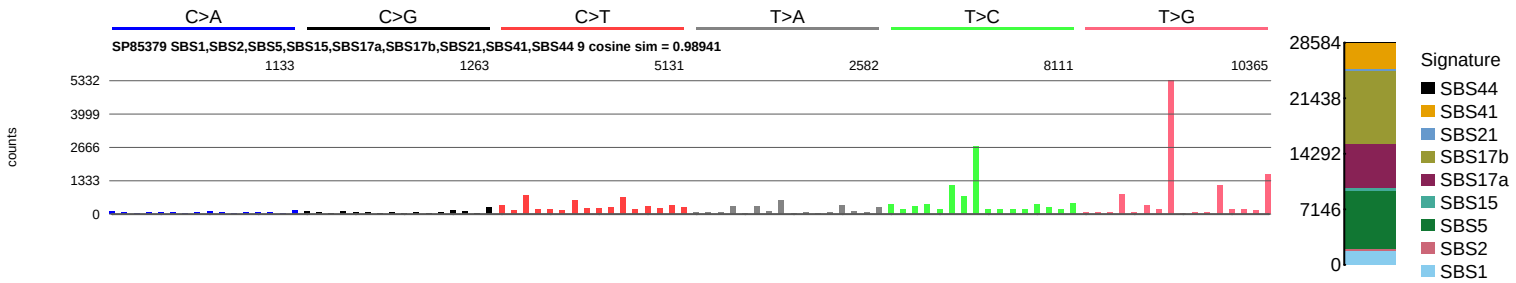
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98942
3260
1464
4651
2212
7088
9910
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98941
3261
1464
4652
2211
7086
9910
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98941
1593
1794
4631
2949
7261
10358
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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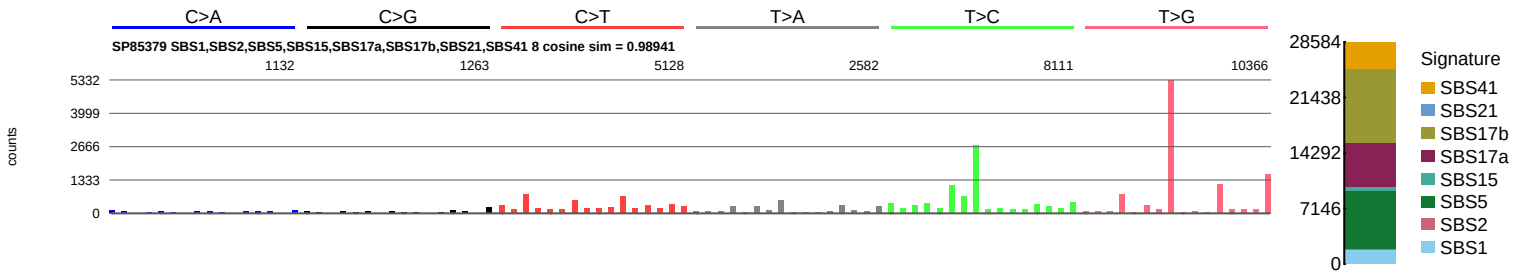
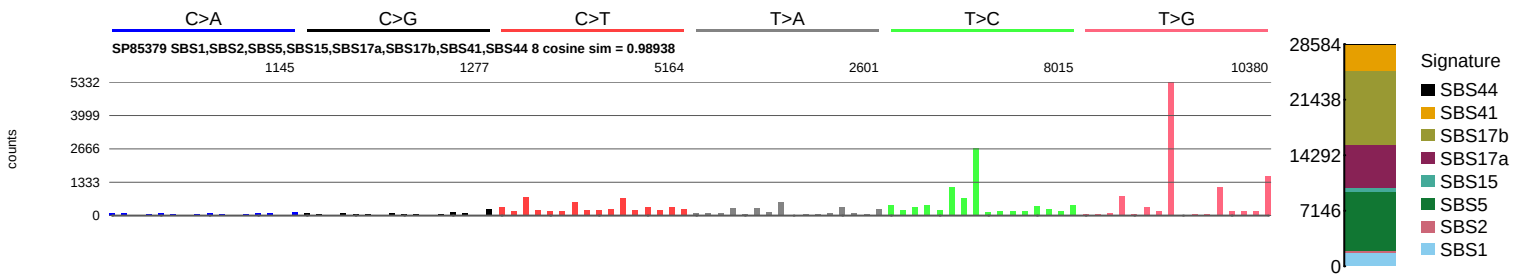
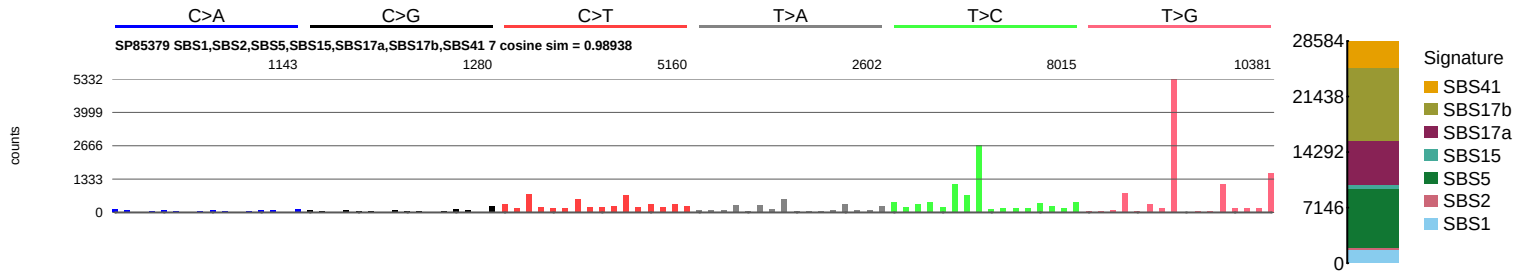
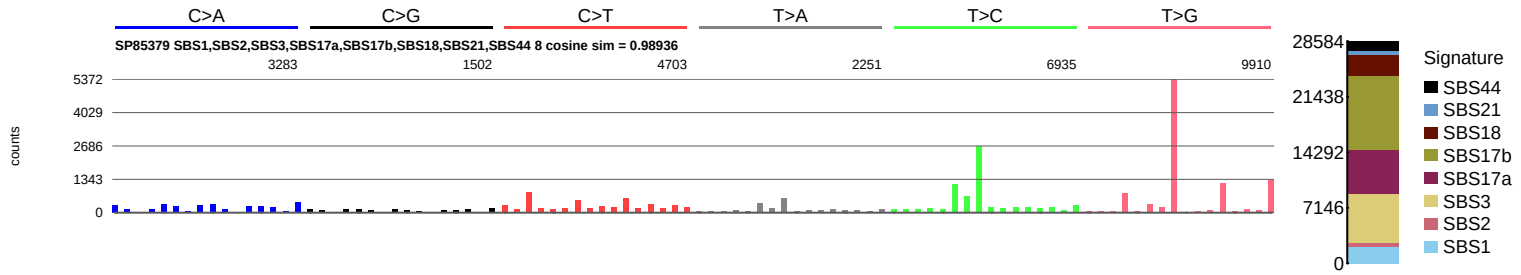
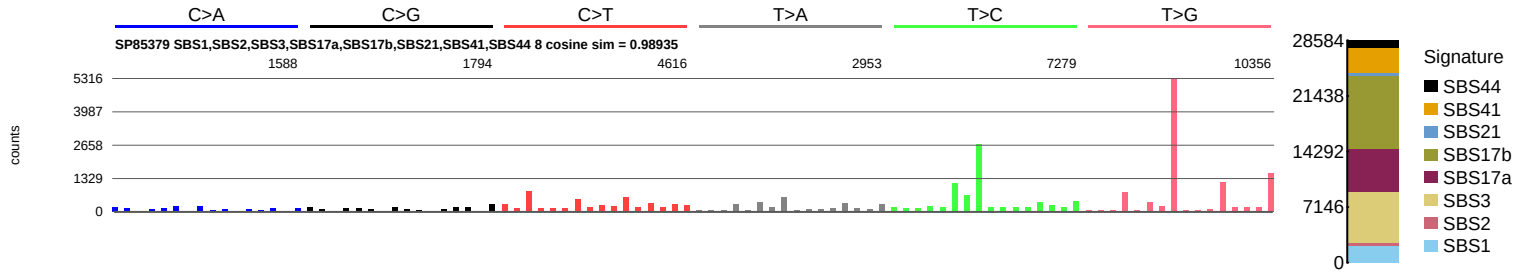
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98941
1133
1263
5131
2582
8111
10365
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98941
1132
1263
5128
2582
8111
10366
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98938
1145
1277
5164
2601
8015
10380
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98938
1143
1280
5160
2602
8015
10381
5332
3999
2666
1333
0
counts
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98936
3283
1502
4703
2251
6935
9910
5372
4029
2686
1343
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98935
1588
1794
4616
2953
7279
10356
5316
3987
2658
1329
0
counts
28584
21438
14292
7146
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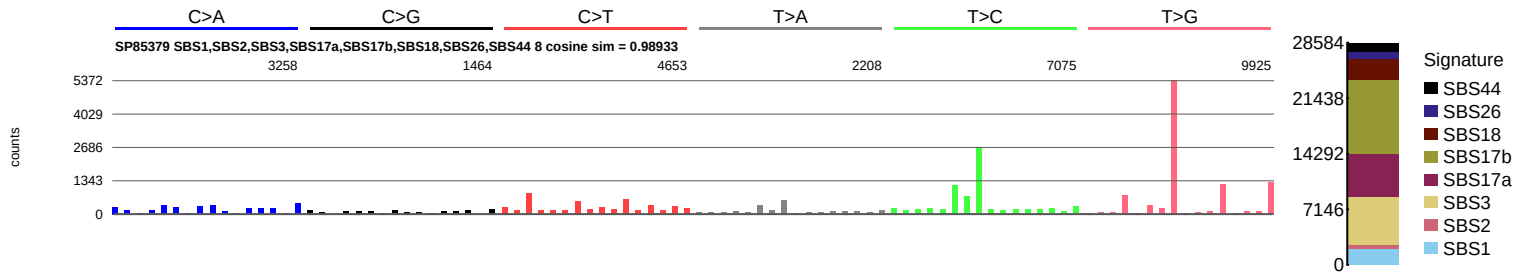
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98933
3258
1464
4653
2208
7075
9925
5372
4029
2686
1343
0
counts
28584
21438
14292
7146
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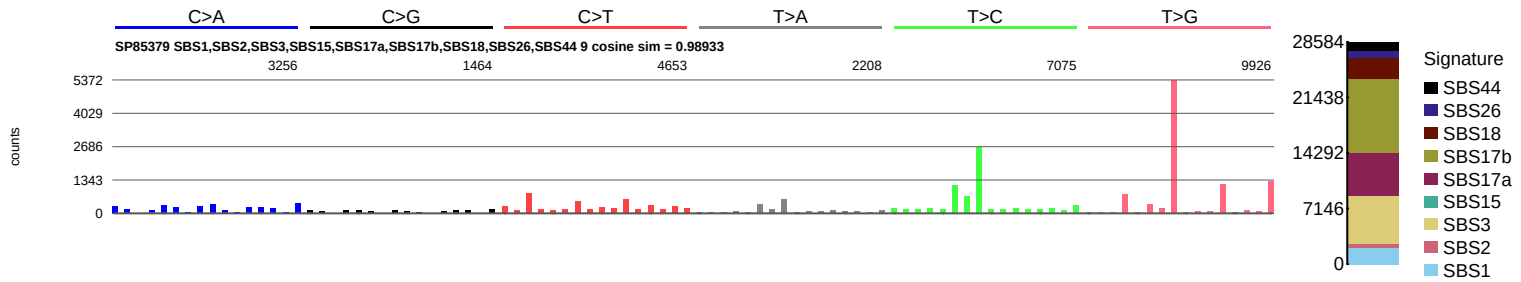
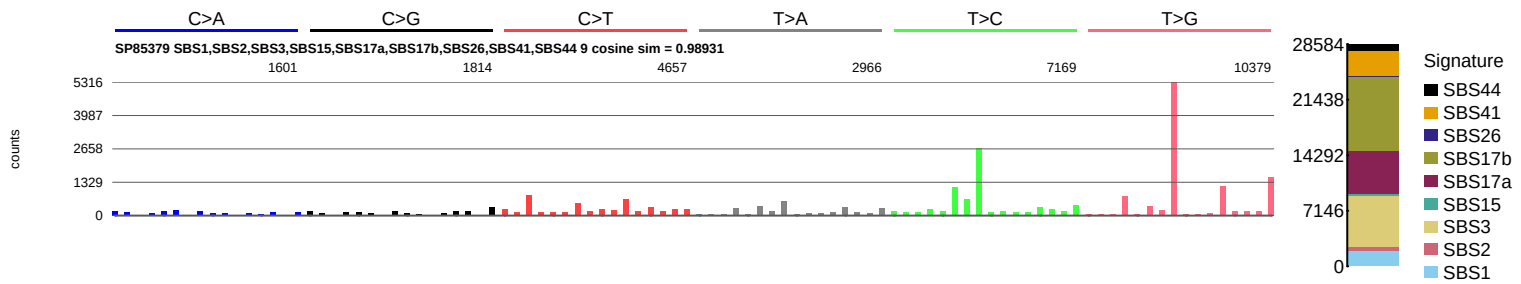
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98933
3256
1464
4653
2208
7075
9926
5372
4029
2686
1343
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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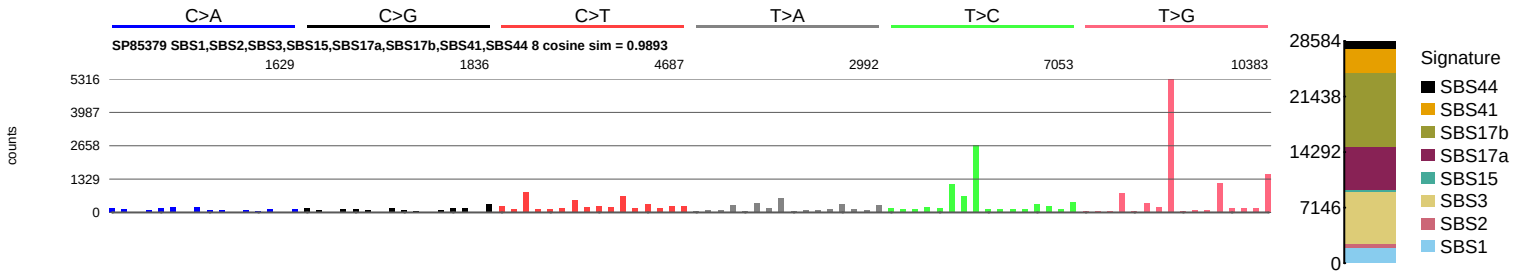
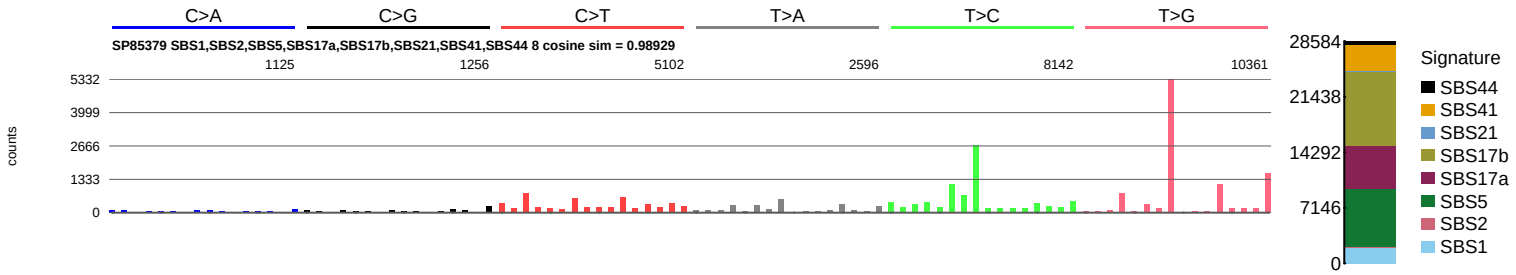
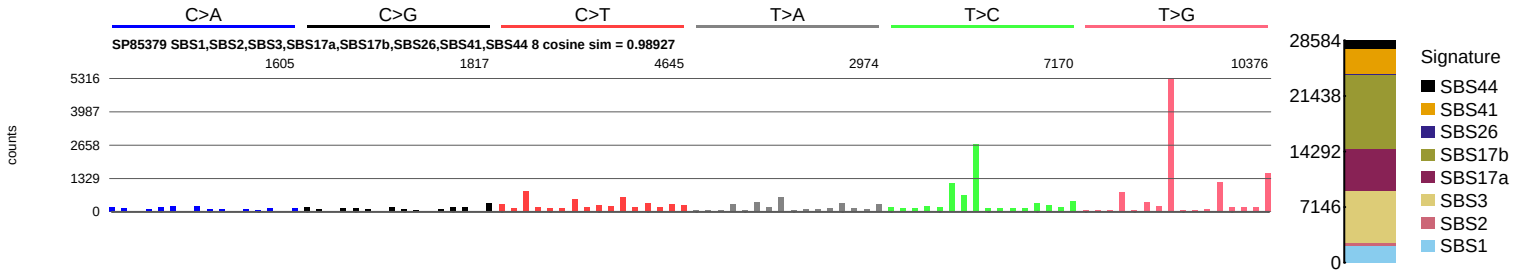
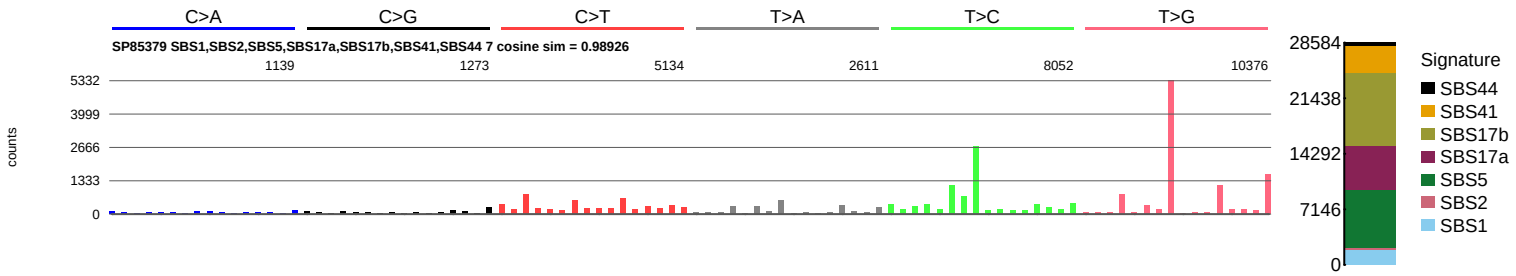
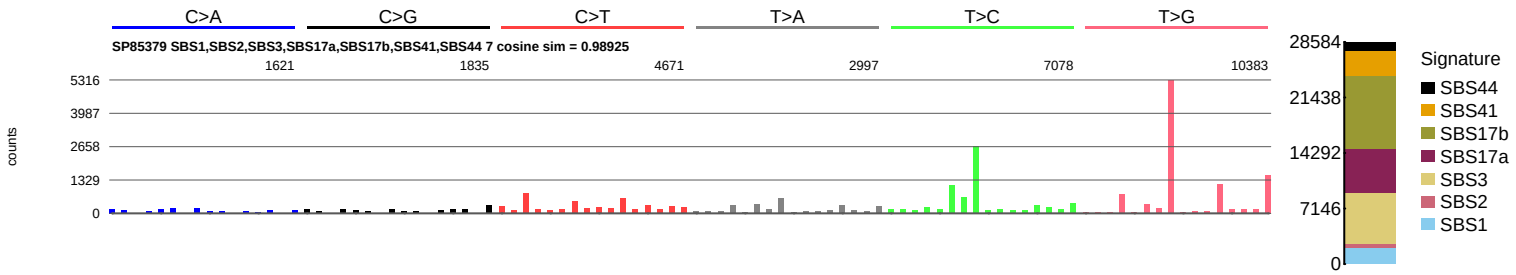
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98931
1601
1814
4657
2966
7169
10379
5316
3987
2658
1329
0
counts
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.9893
1629
1836
4687
2992
7053
10383
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98929
1125
1256
5102
2596
8142
10361
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98927
1605
1817
4645
2974
7170
10376
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98926
1139
1273
5134
2611
8052
10376
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98925
1621
1835
4671
2997
7078
10383
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
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SBS17b
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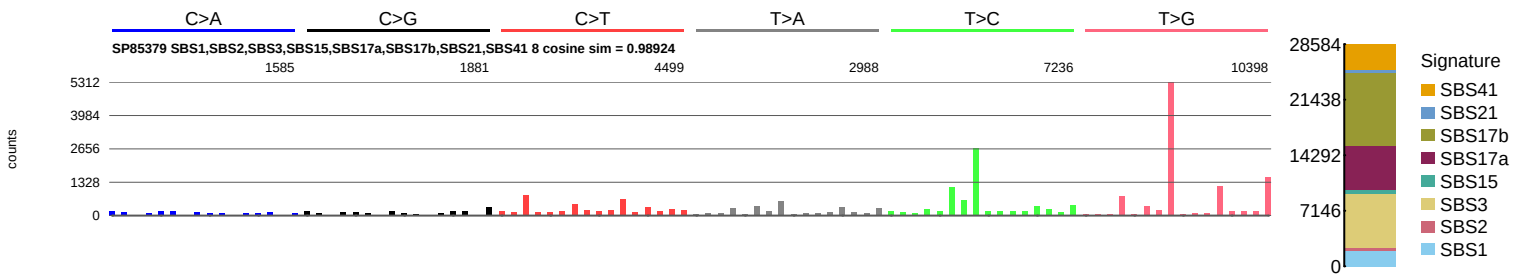
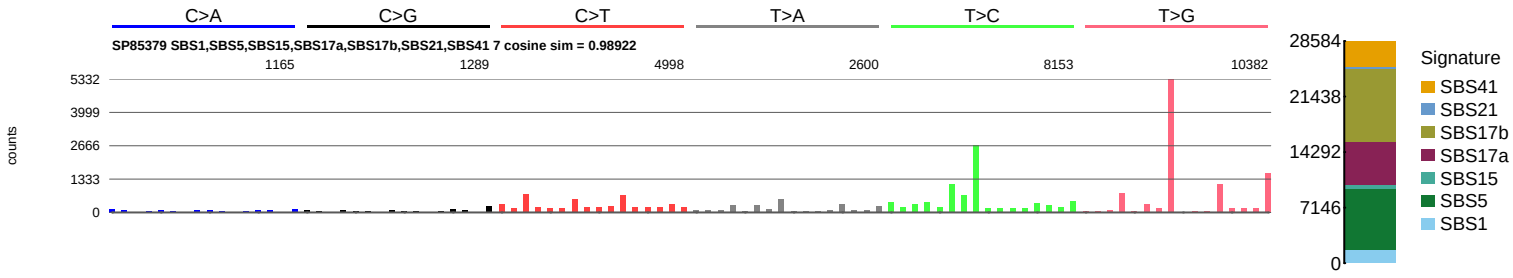
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98924
1585
1881
4499
2988
7236
10398
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98922
1165
1289
4998
2600
8153
10382
counts
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3999
2666
1333
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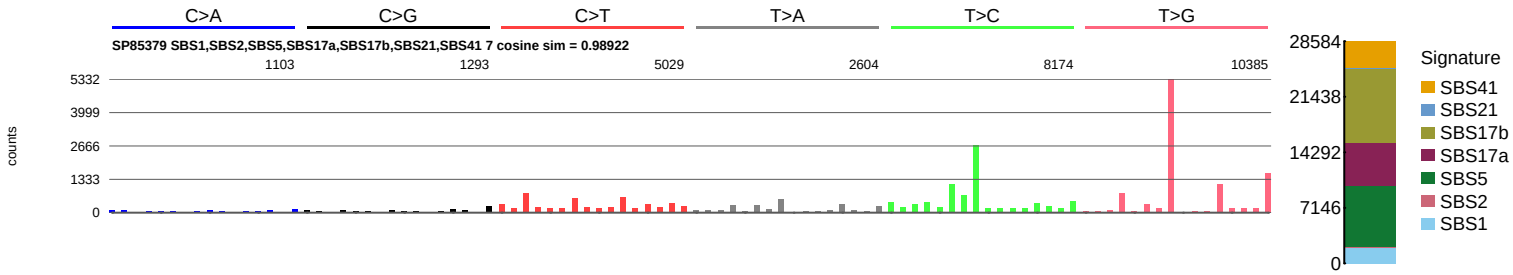
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1103
1293
5029
2604
8174
10385
counts
5332
3999
2666
1333
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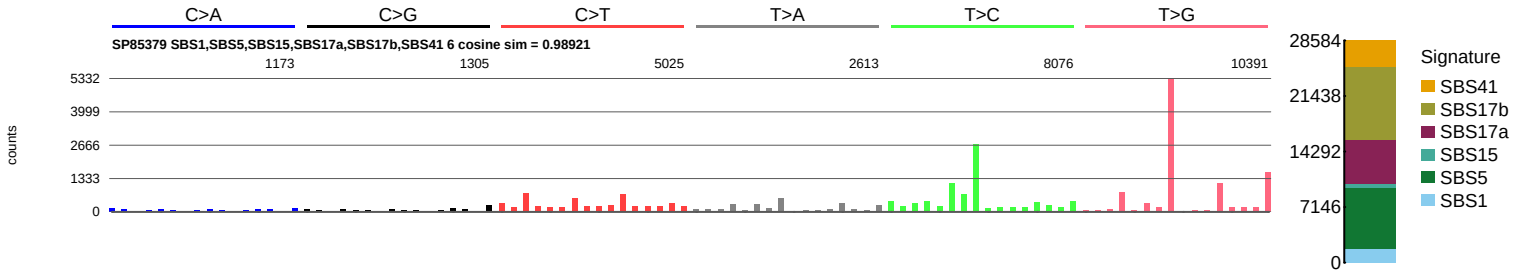
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SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98921
1173
1305
5025
2613
8076
10391
counts
5332
3999
2666
1333
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14292
7146
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SBS17a
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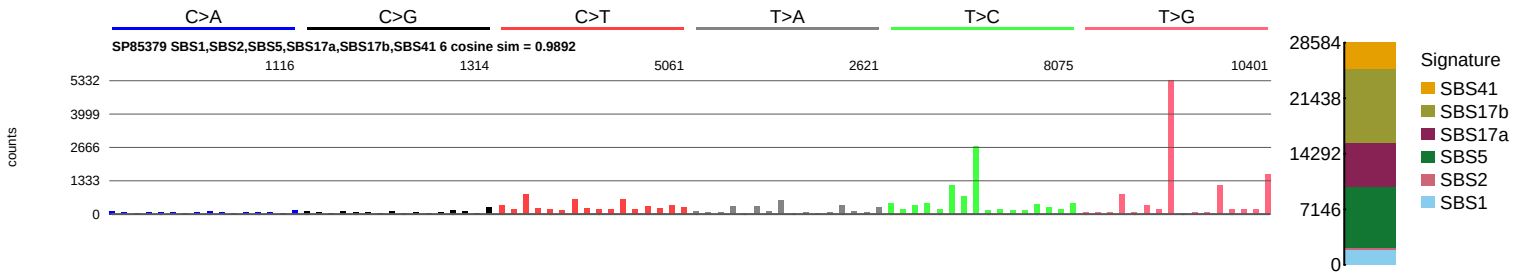
SBS5
SBS1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SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9892
1116
1314
5061
2621
8075
10401
counts
5332
3999
2666
1333
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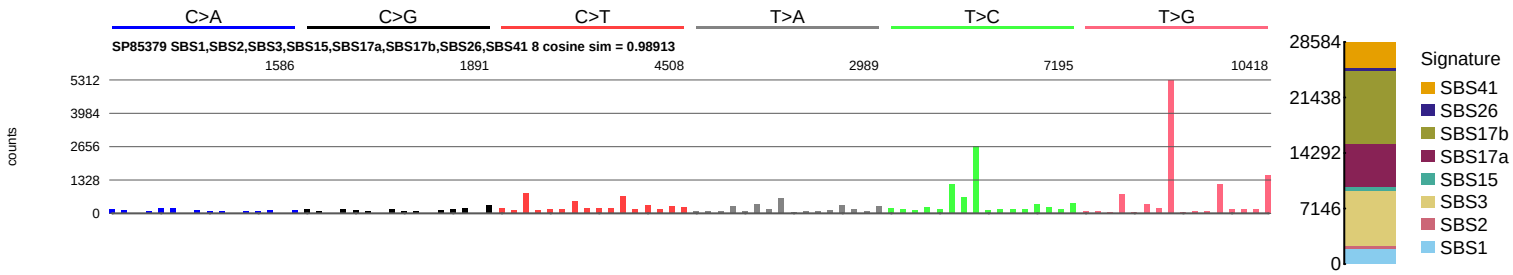
SBS17a
SBS5
SBS2
SBS1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SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98913
1586
1891
4508
2989
7195
10418
counts
5312
3984
2656
1328
0
28584
21438
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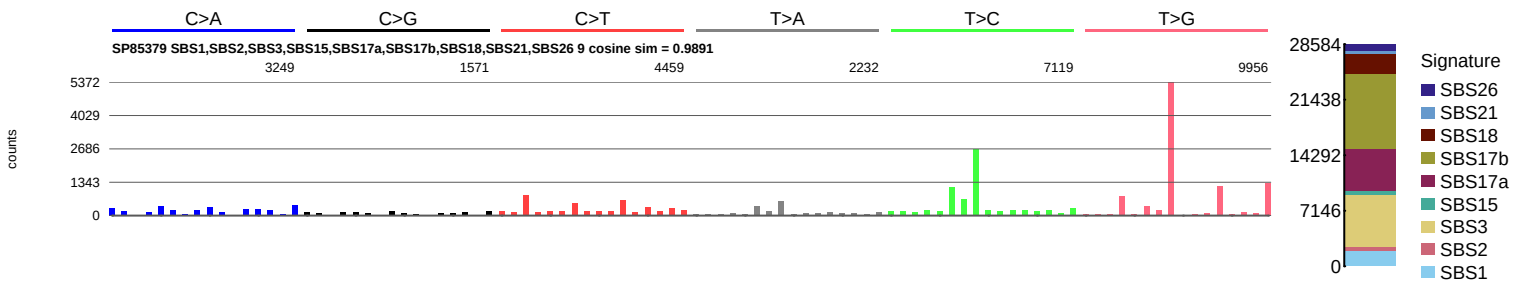
SBS2
SBS1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SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.9891
3249
1571
4459
2232
7119
9956
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
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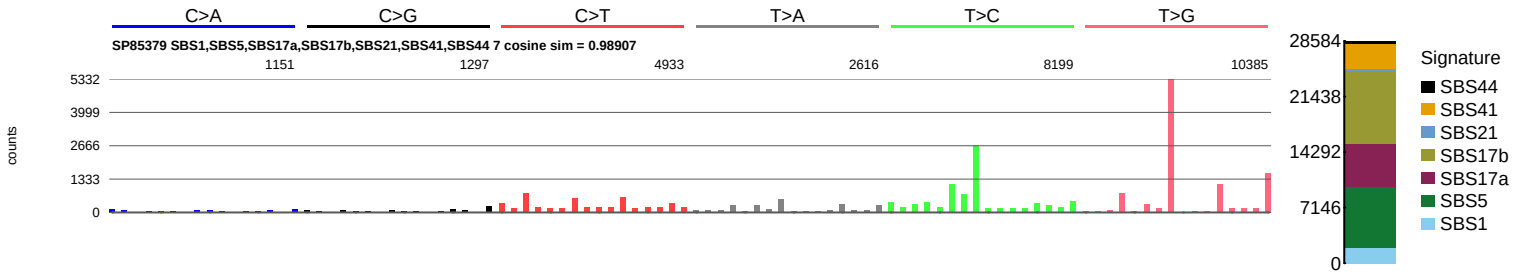
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98907
1151
1297
4933
2616
8199
10385
counts
5332
3999
2666
1333
0
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14292
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SBS17b
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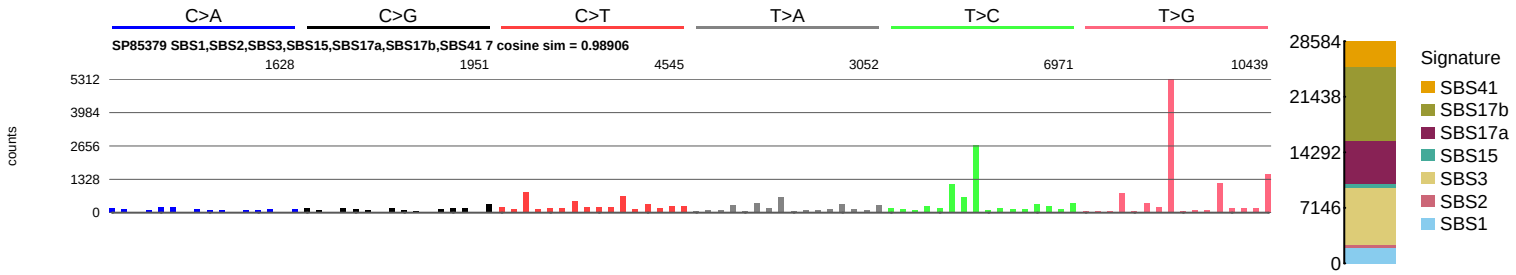
SBS5
SBS1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SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98906
1628
1951
4545
3052
6971
10439
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
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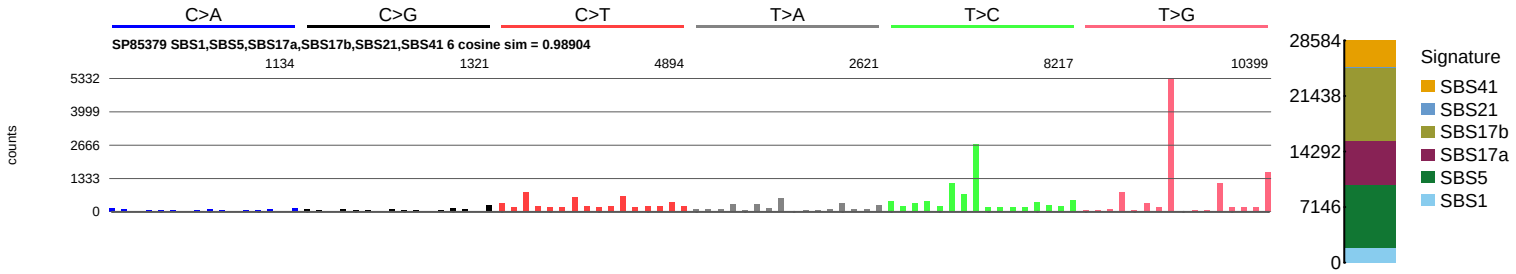
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98904
1134
1321
4894
2621
8217
10399
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
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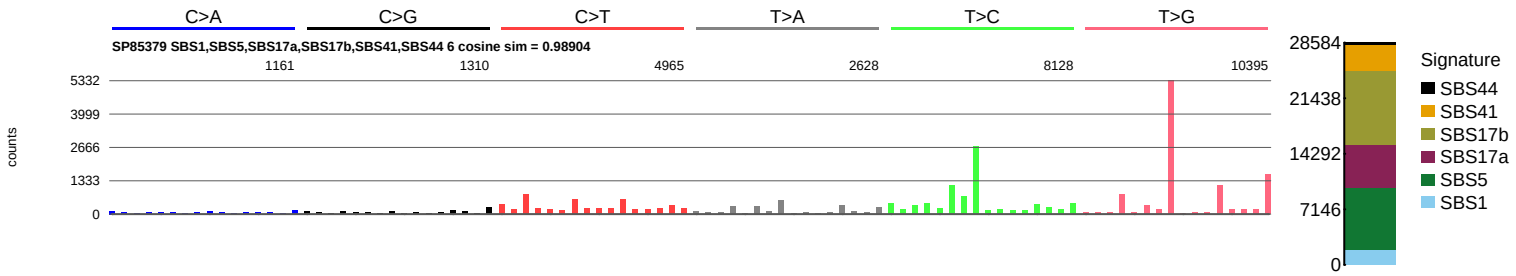
SBS21
SBS17b
SBS17a
SBS5
SBS1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C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98904
1161
1310
4965
2628
8128
10395
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
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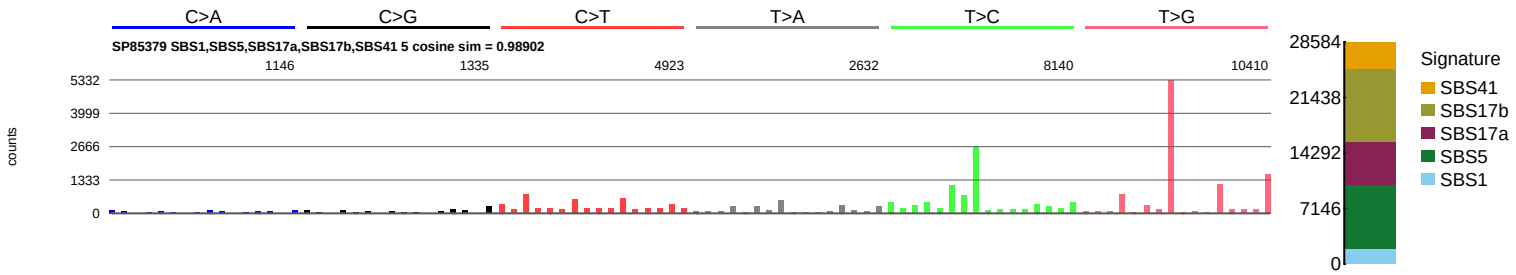
SBS17b
SBS17a
SBS5
SBS1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T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98902
1146
1335
4923
2632
8140
10410
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
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Signature
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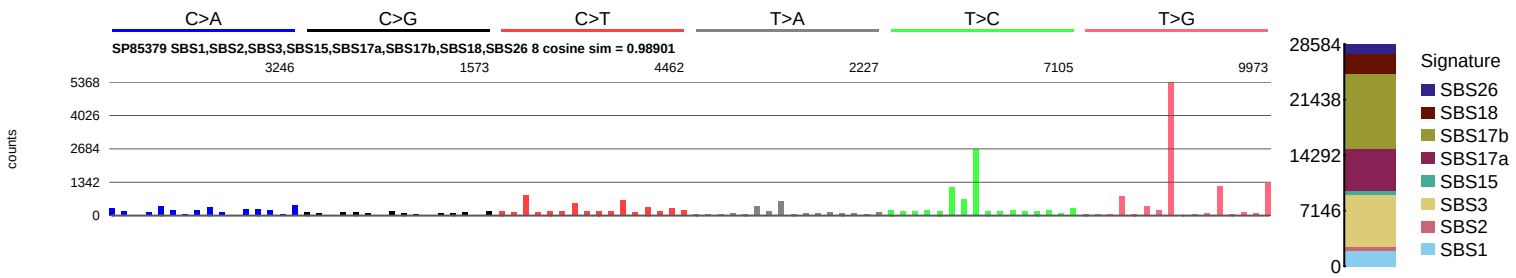
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98901
3246
1573
4462
2227
7105
9973
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
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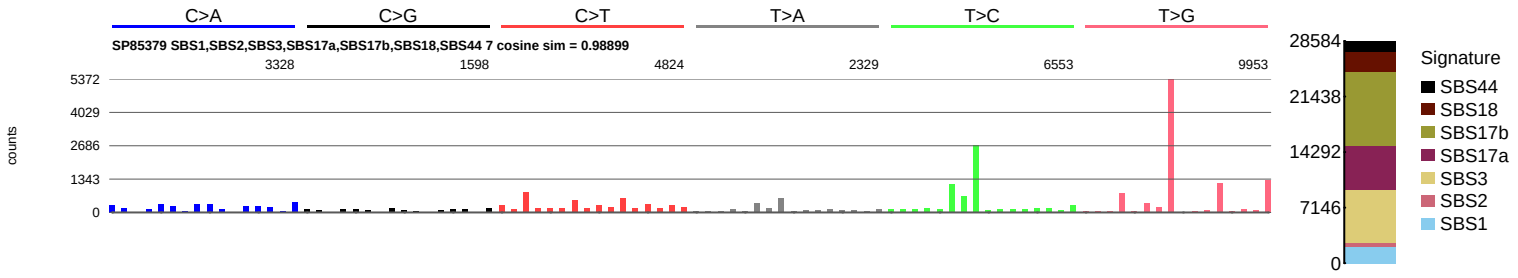
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98899
3328
1598
4824
2329
6553
9953
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
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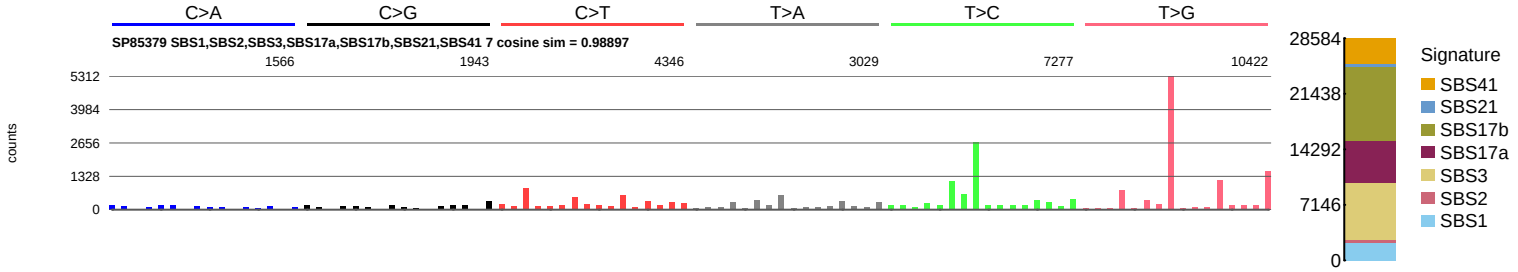
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98897
1566
1943
4346
3029
7277
10422
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
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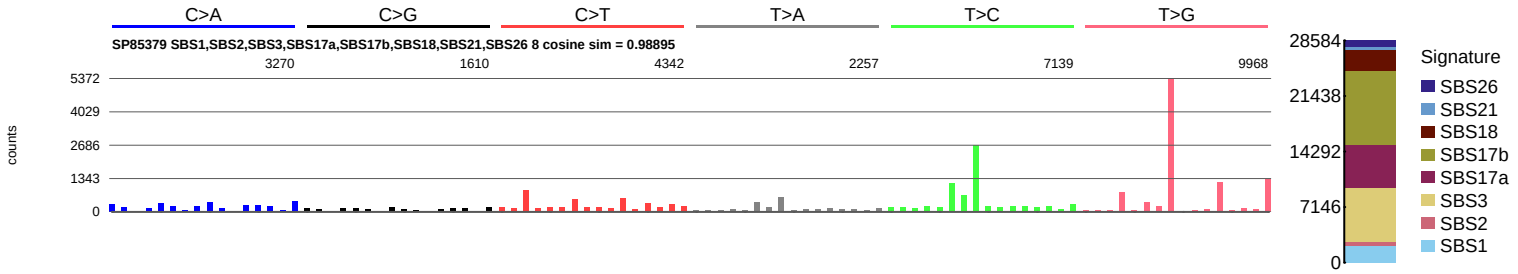
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98895
3270
1610
4342
2257
7139
9968
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
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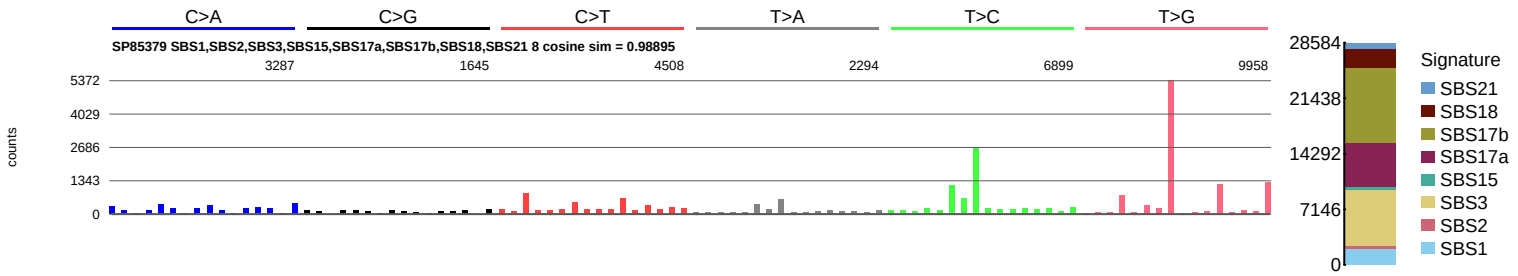
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98895
3287
1645
4508
2294
6899
9958
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
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Signature
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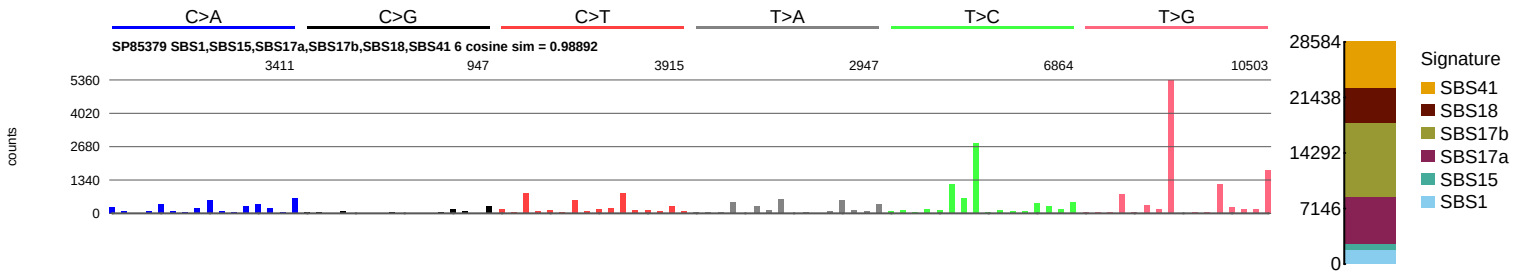
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98892
3411
947
3915
2947
6864
10503
counts
5360
4020
2680
1340
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21438
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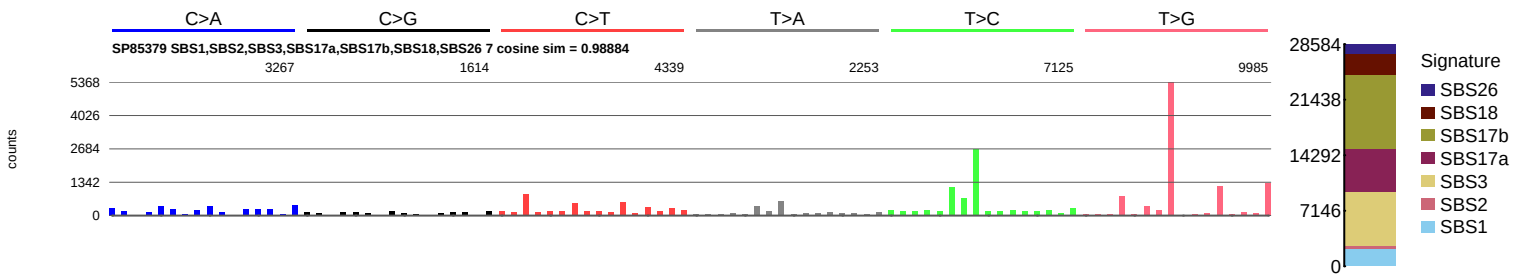
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98884
3267
1614
4339
2253
7125
9985
counts
5368
4026
2684
1342
0
28584
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7146
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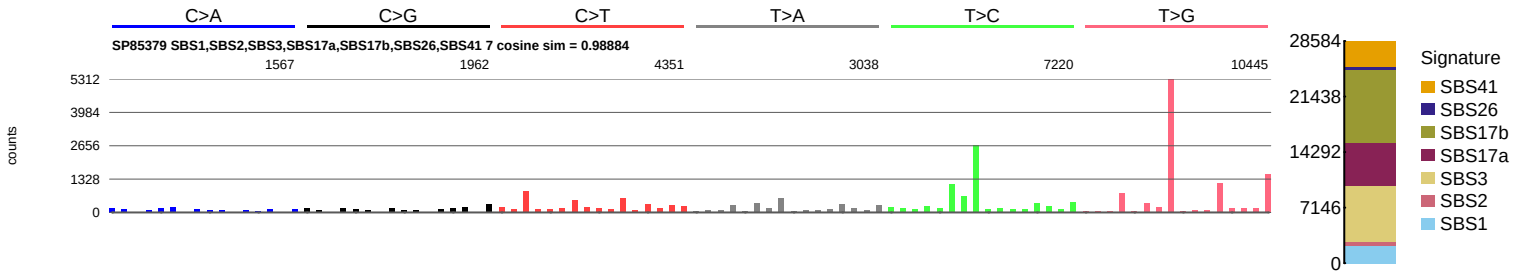
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98884
1567
1962
4351
3038
7220
10445
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
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SBS26
SBS17b
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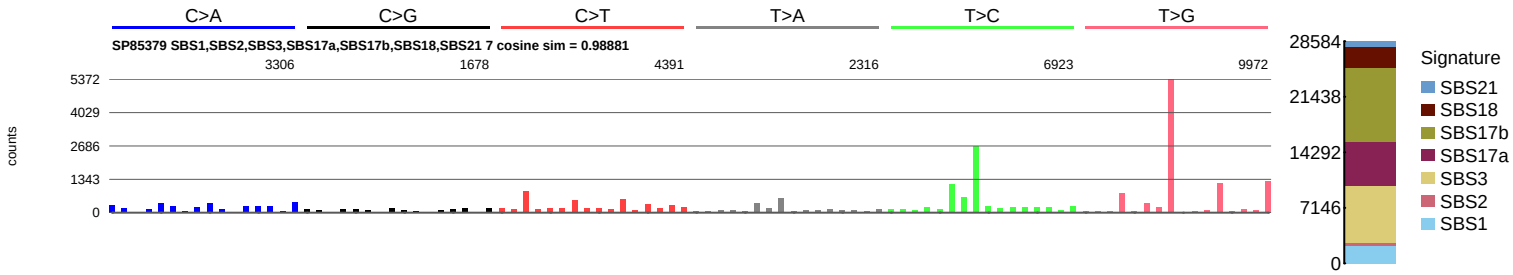
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98881
3306
1678
4391
2316
6923
9972
counts
5372
4029
2686
1343
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28584
21438
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7146
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SBS17b
SBS17a
SBS3
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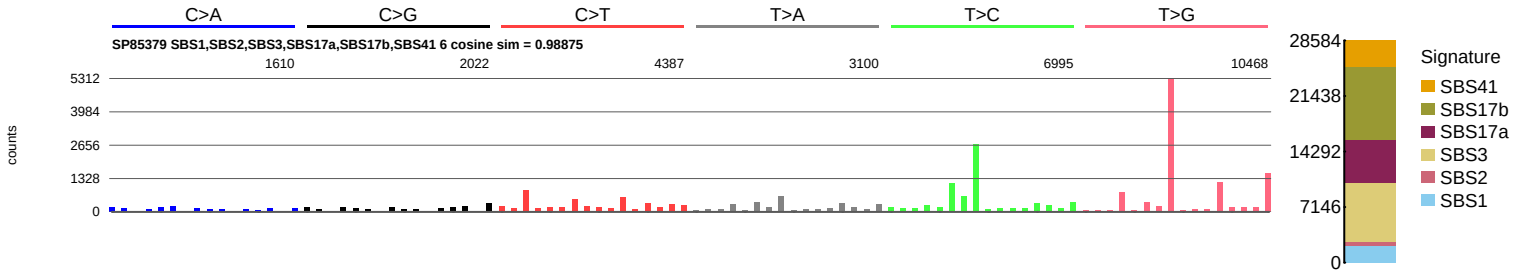
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98875
1610
2022
4387
3100
6995
10468
counts
5312
3984
2656
1328
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28584
21438
14292
7146
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SBS17b
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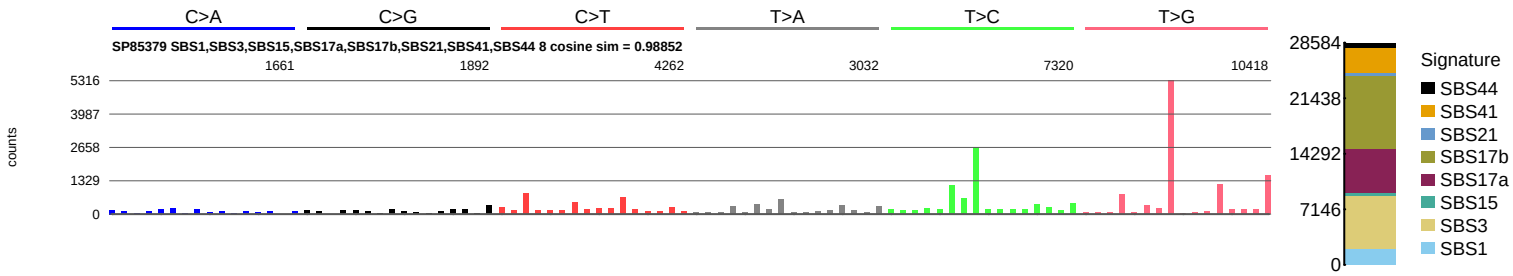
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98852
1661
1892
4262
3032
7320
10418
counts
5316
3987
2658
1329
0
28584
21438
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SBS17b
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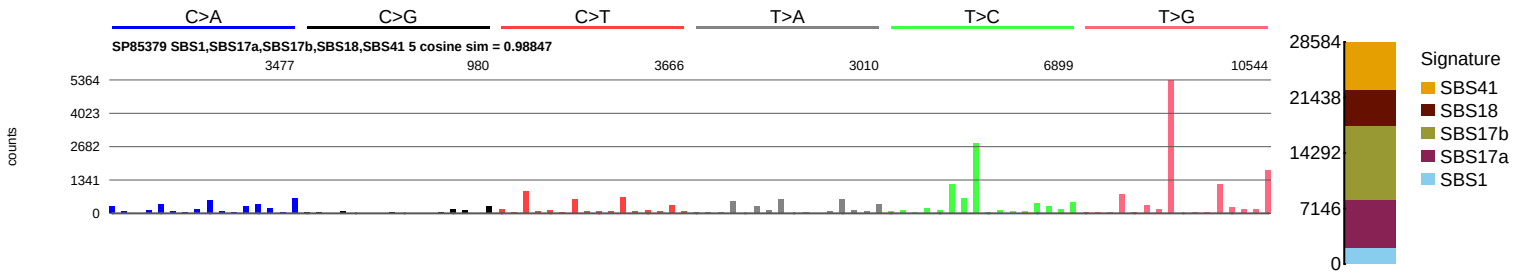
SBS3
SBS1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C>G
C>T
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T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98847
3477
980
3666
3010
6899
10544
counts
5364
4023
2682
1341
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21438
14292
7146
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SBS18
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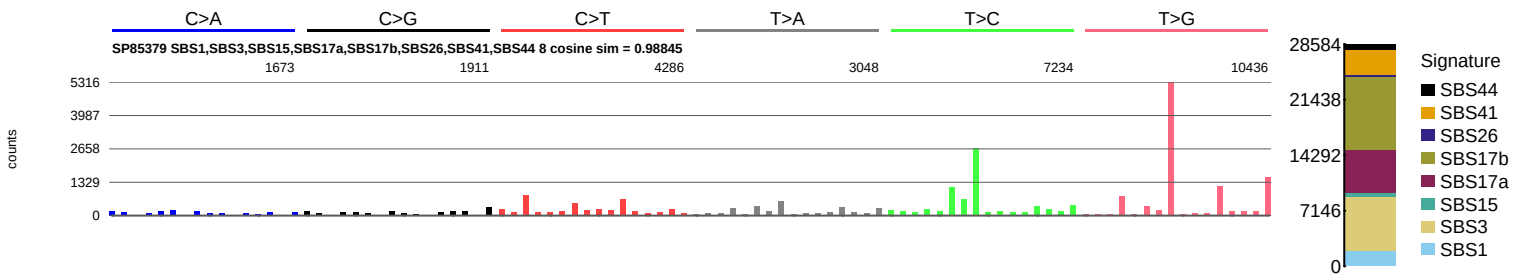
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98845
1673
1911
4286
3048
7234
10436
counts
5316
3987
2658
1329
0
28584
21438
14292
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SBS3
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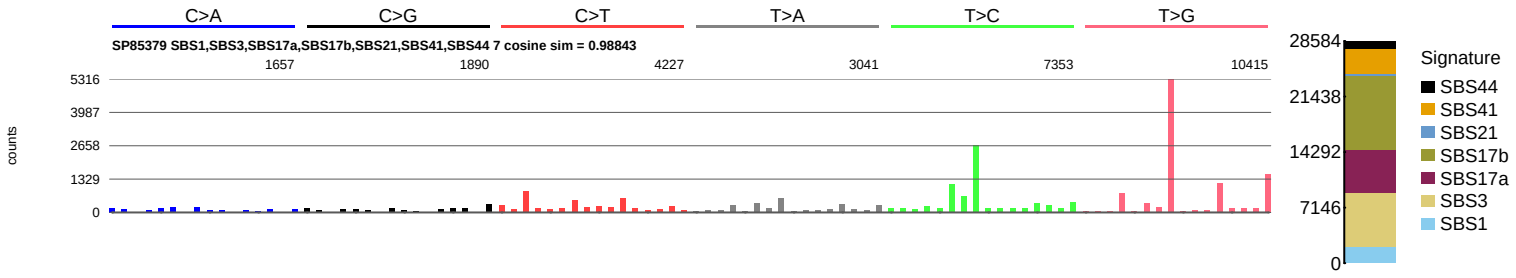
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C>G
C>T
T>A
T>C
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SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98843
1657
1890
4227
3041
7353
10415
counts
5316
3987
2658
1329
0
28584
21438
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7146
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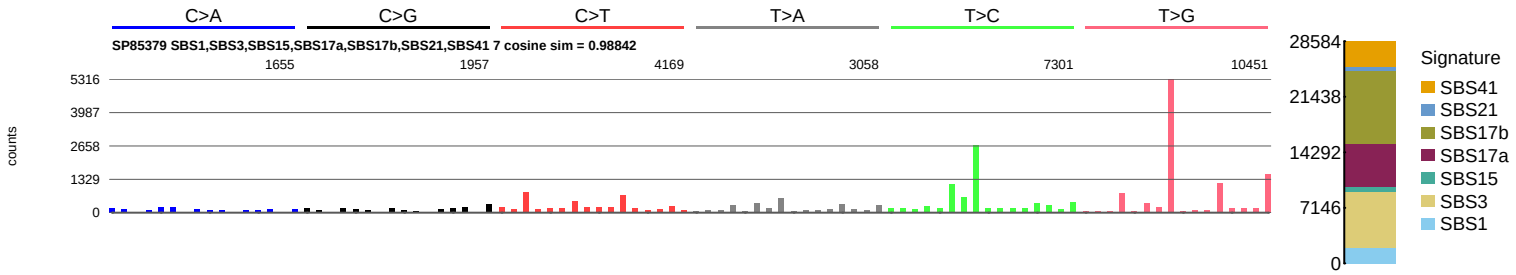
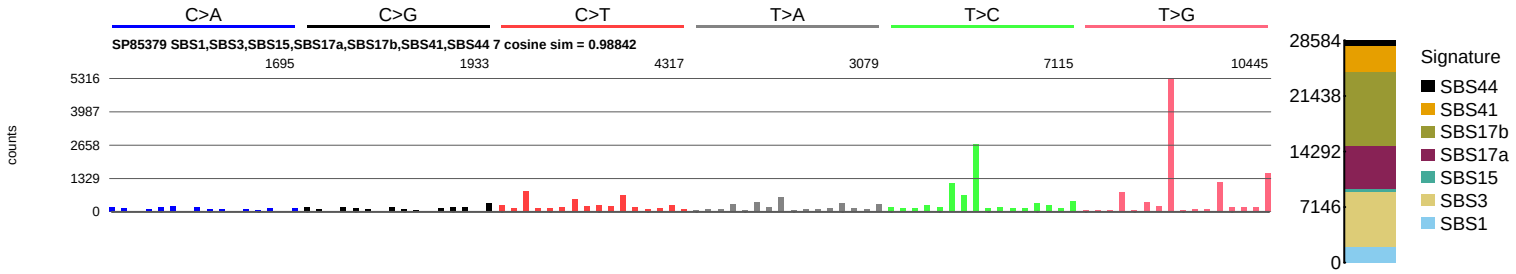
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98842
1655
1957
4169
3058
7301
10451
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1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SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98842
1695
1933
4317
3079
7115
10445
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
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Signature
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SBS41
SBS17b
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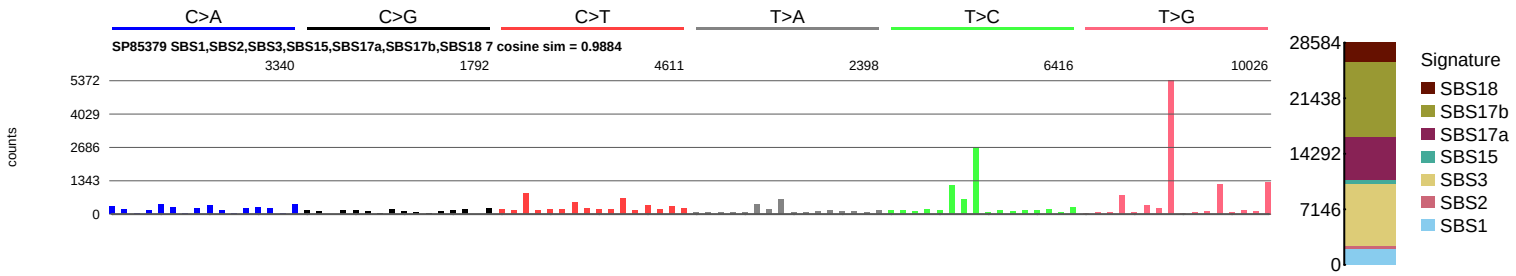
SBS3
SBS1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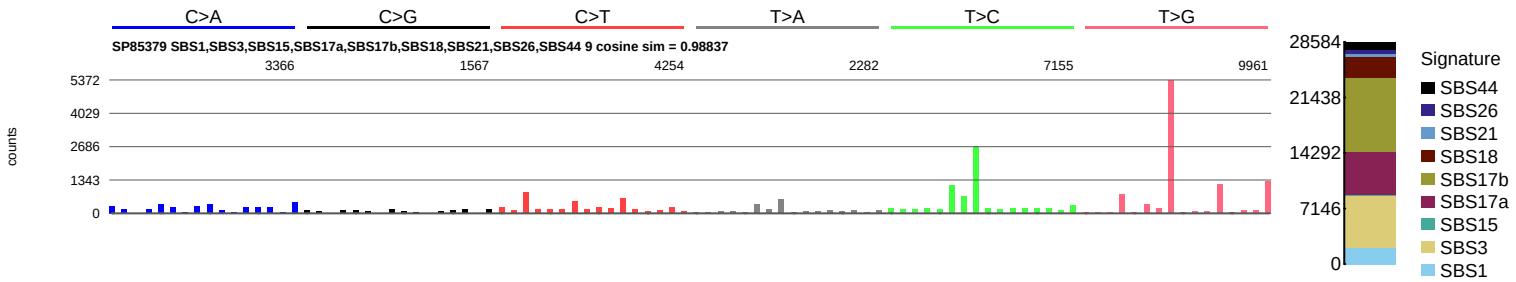
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.9884
3340
1792
4611
2398
6416
10026
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98837
3366
1567
4254
2282
7155
9961
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
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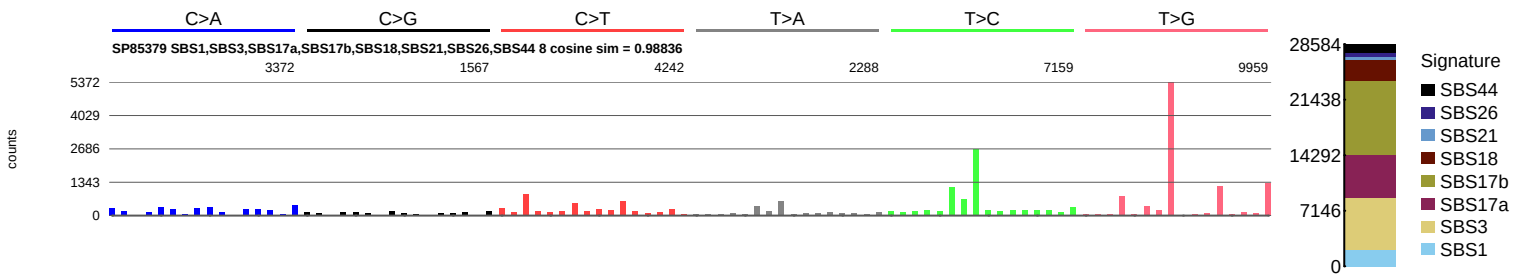
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98836
3372
1567
4242
2288
7159
9959
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
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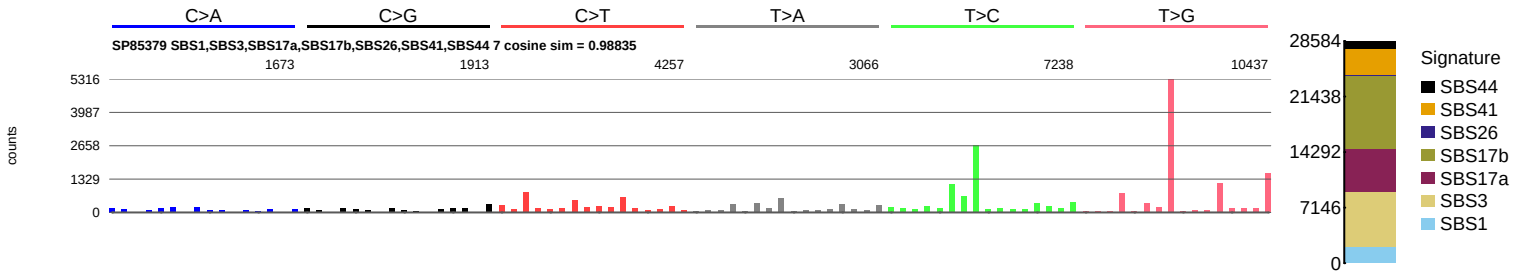
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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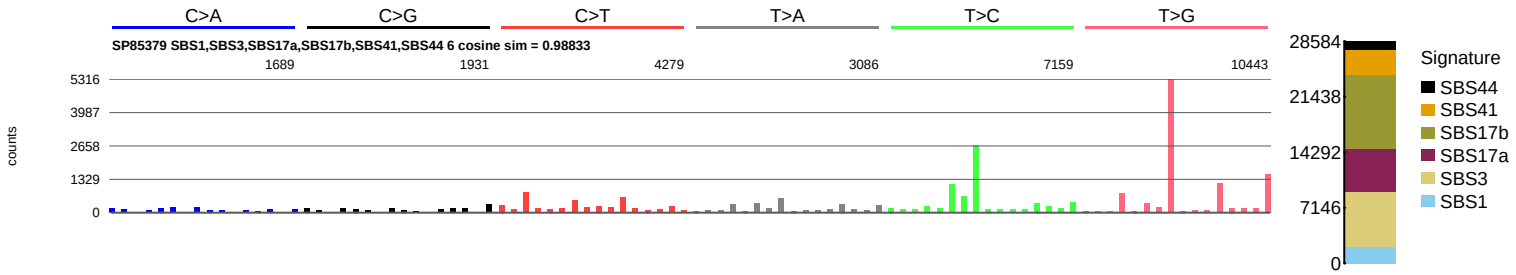
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98835
1673
1913
4257
3066
7238
10437
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1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C>G
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T>C
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SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98833
1689
1931
4279
3086
7159
10443
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
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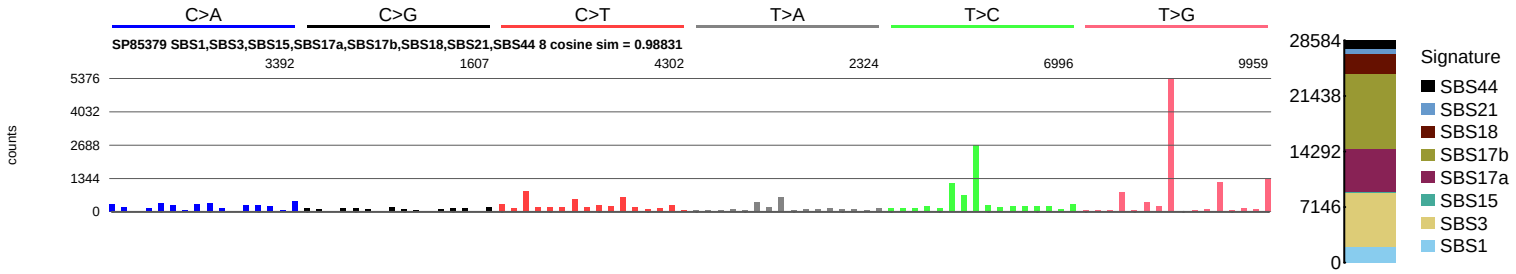
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS1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T>C
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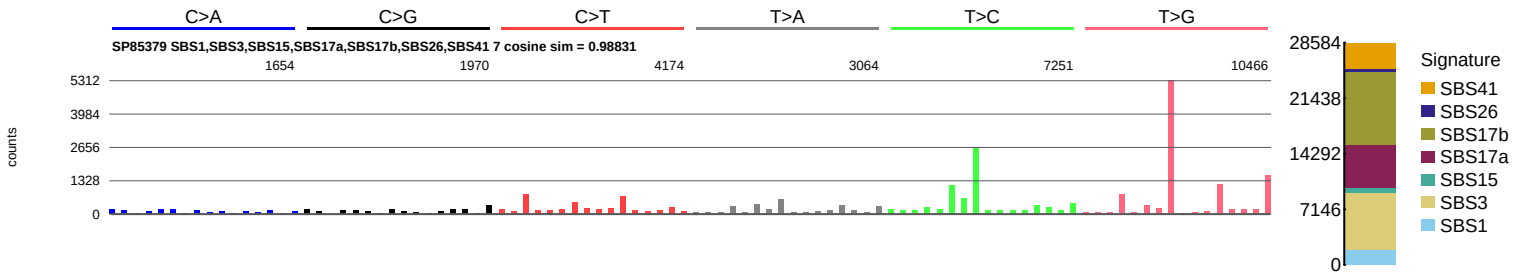
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98831
3392
1607
4302
2324
6996
9959
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
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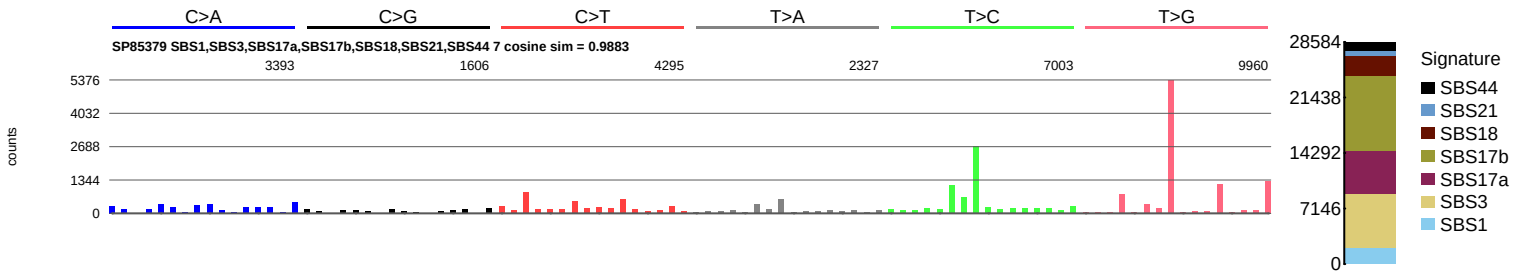
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98831
1654
1970
4174
3064
7251
10466
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
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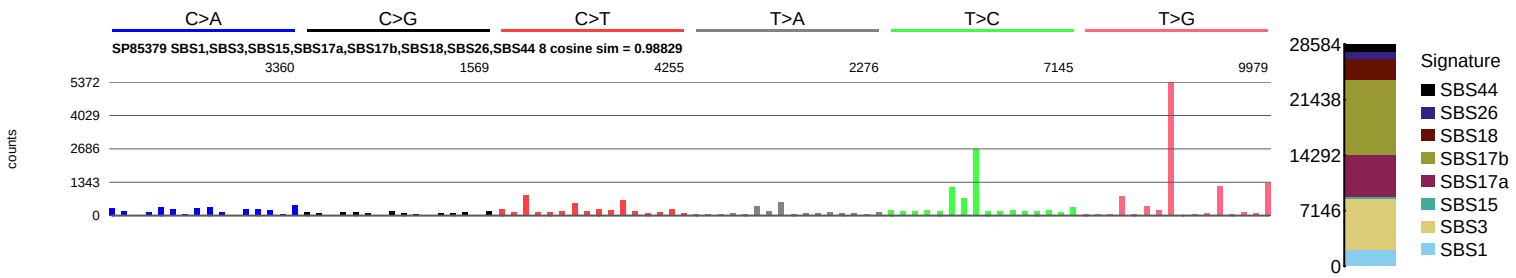
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.9883
3393
1606
4295
2327
7003
9960
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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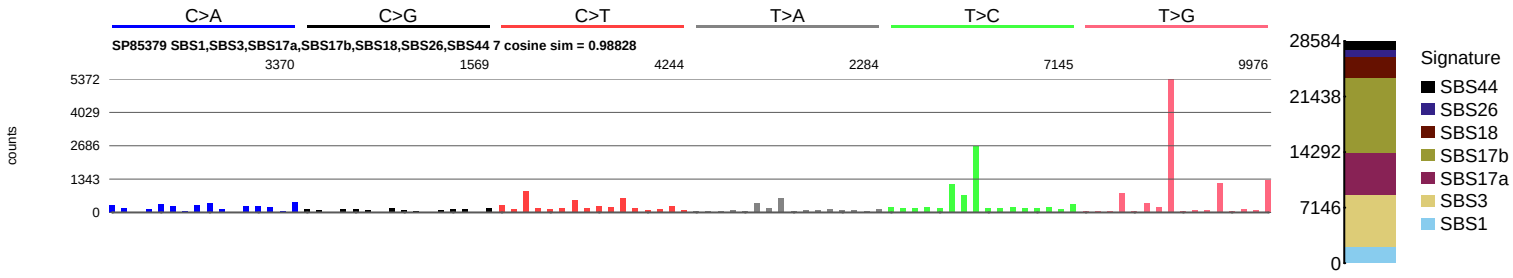
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98829
3360
1569
4255
2276
7145
9979
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98828
3370
1569
4244
2284
7145
9976
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
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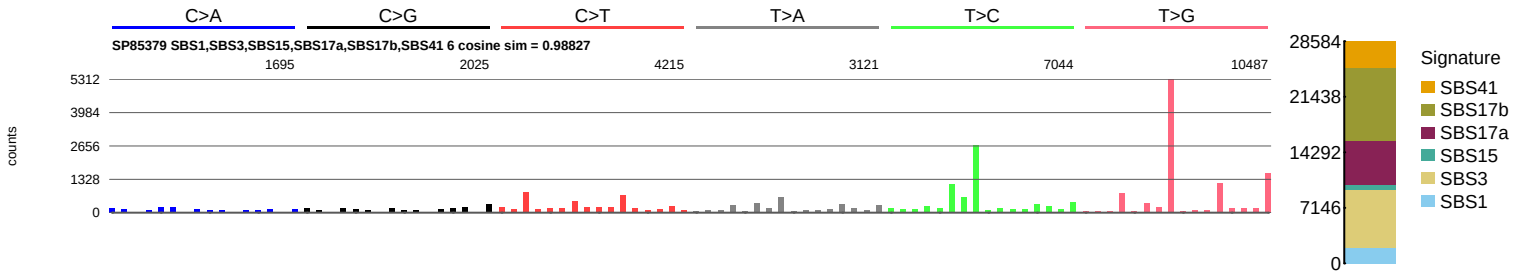
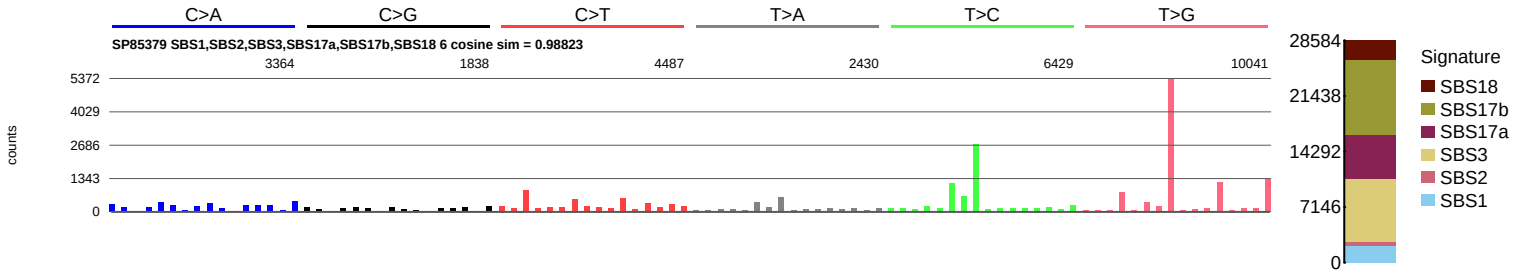
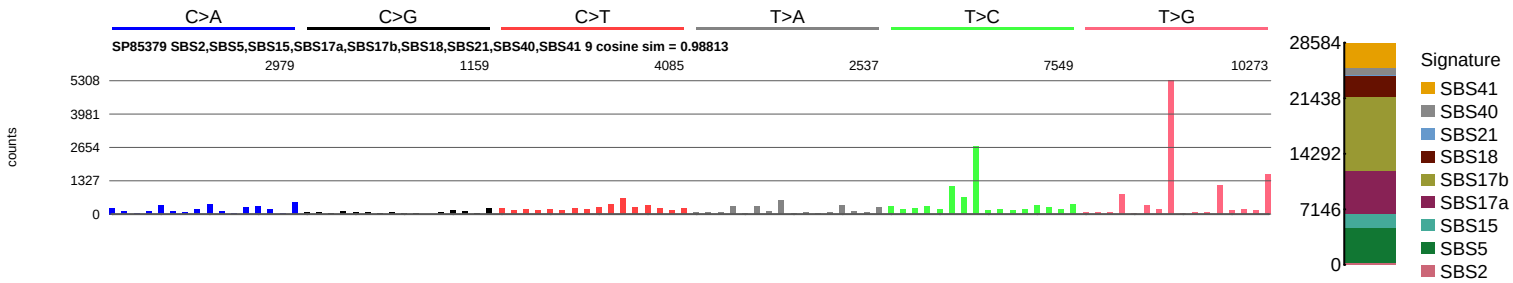
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98827
1695
2025
4215
3121
7044
10487
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98823
3364
1838
4487
2430
6429
10041
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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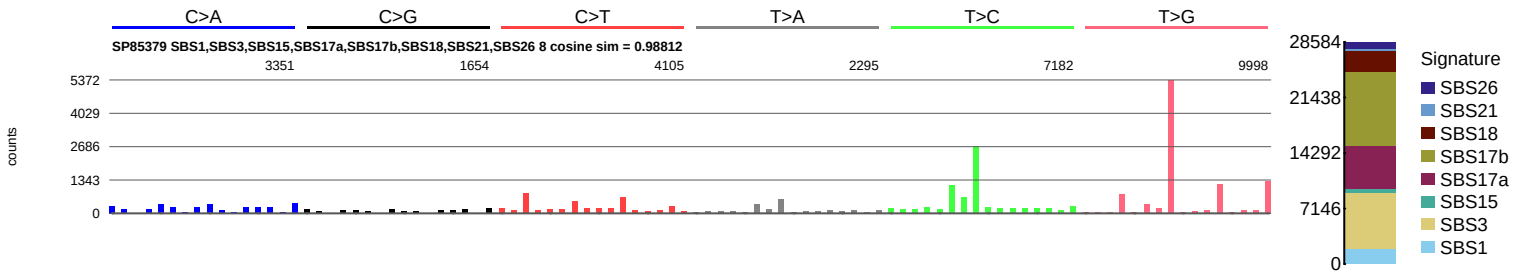
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98813
2979
1159
4085
2537
7549
10273
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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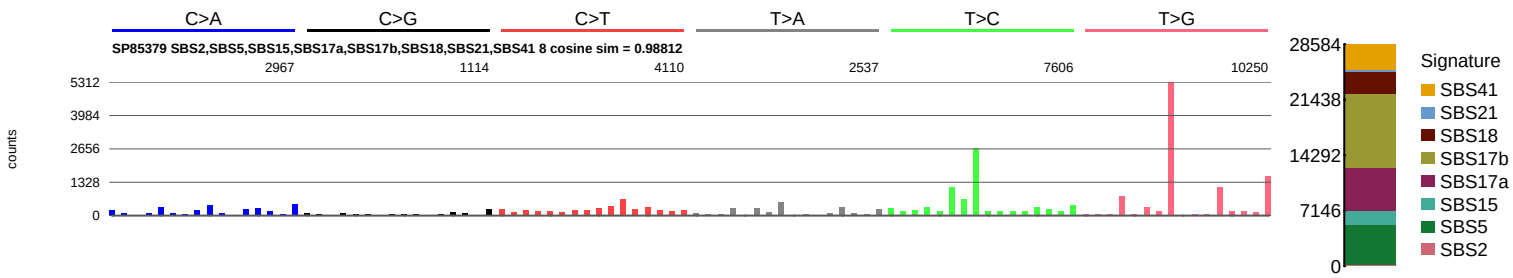
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98812
3351
1654
4105
2295
7182
9998
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98812
2967
1114
4110
2537
7606
10250
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
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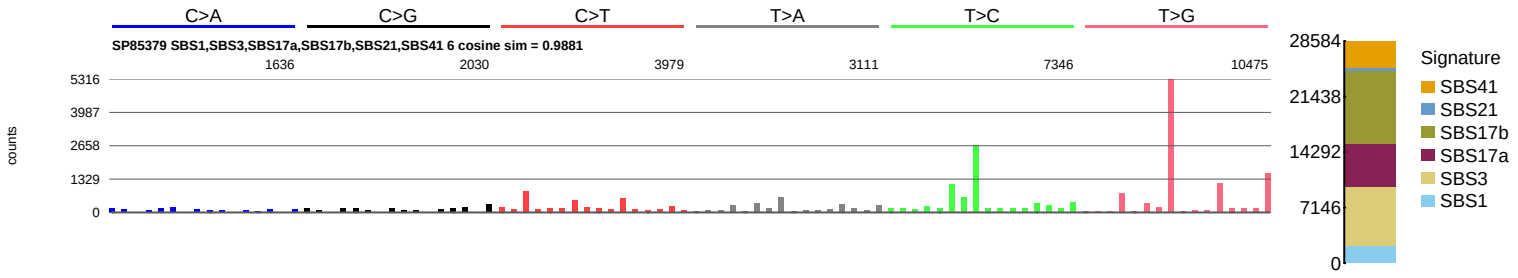
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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C>G
C>T
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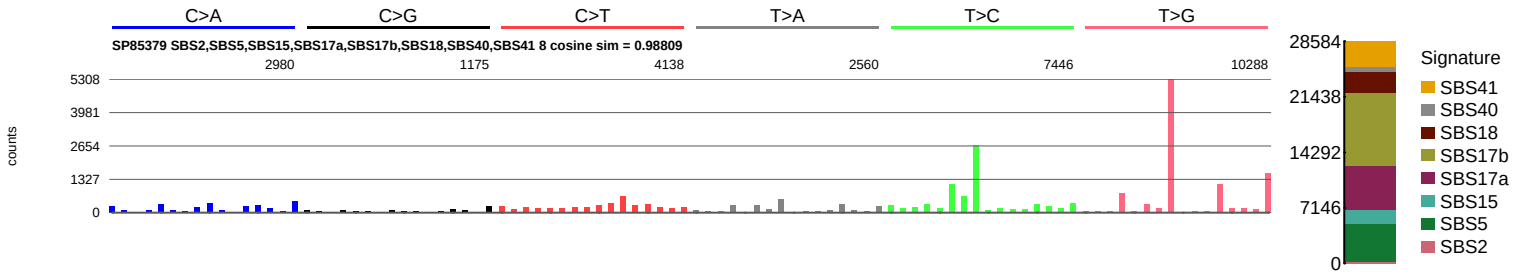
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.9881
1636
2030
3979
3111
7346
10475
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1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T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98809
2980
1175
4138
2560
7446
10288
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
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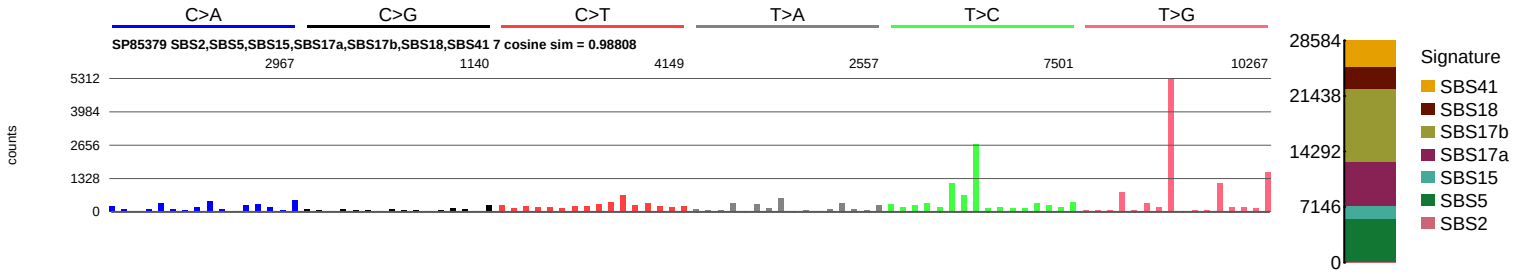
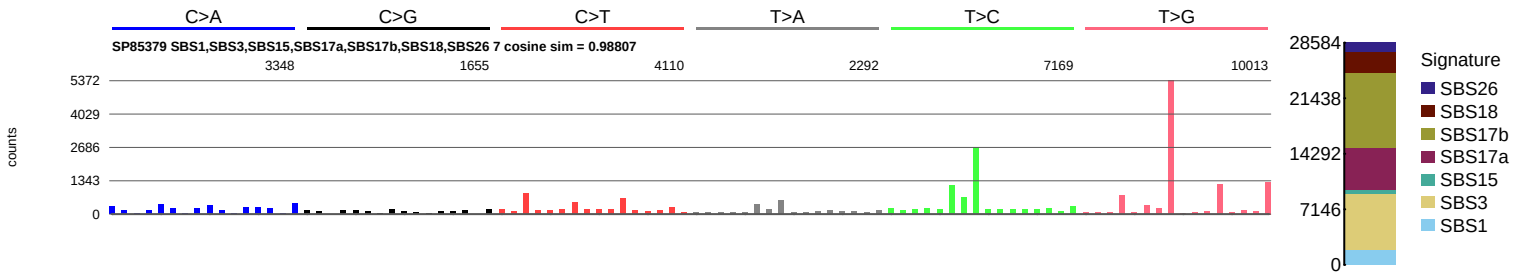
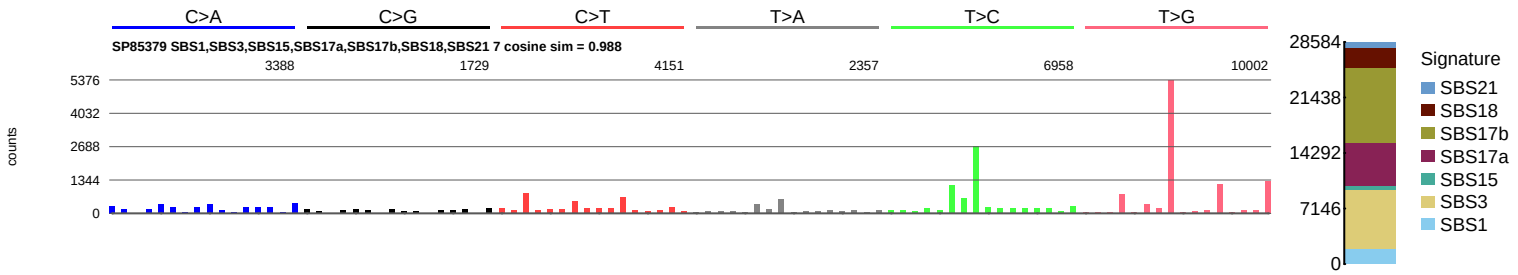
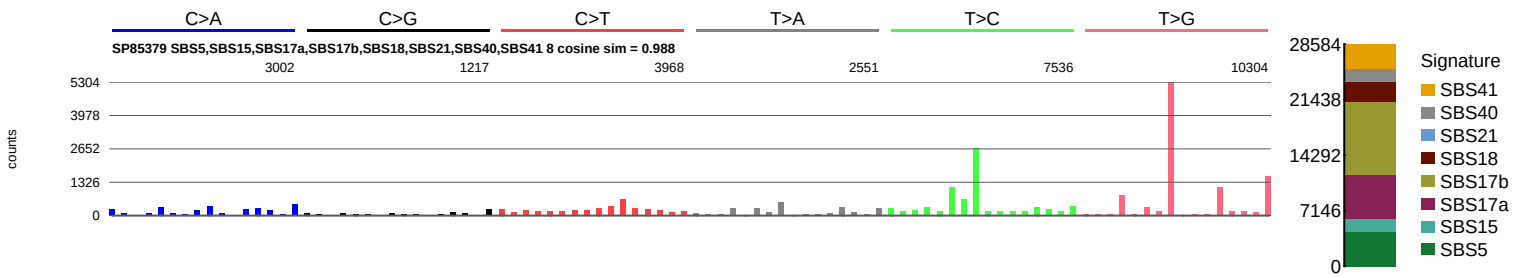
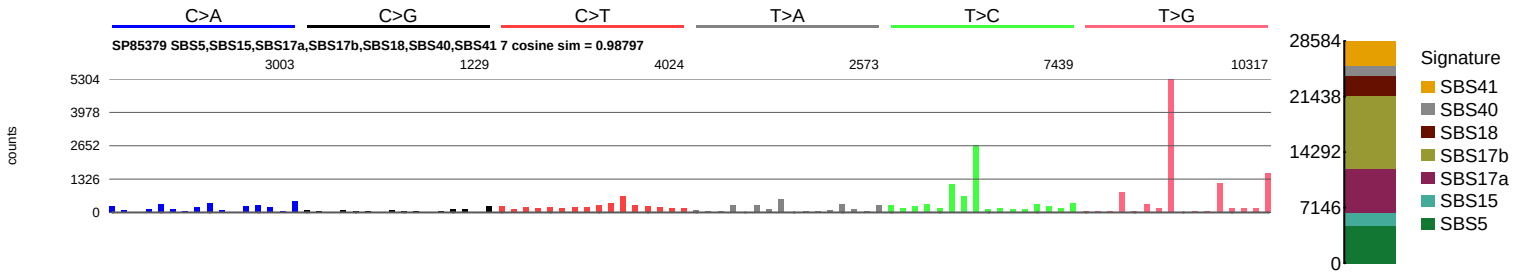
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98808
2967
1140
4149
2557
7501
10267
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98807
3348
1655
4110
2292
7169
10013
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.988
3388
1729
4151
2357
6958
10002
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.988
3002
1217
3968
2551
7536
10304
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98797
3003
1229
4024
2573
7439
10317
counts
5304
3978
2652
1326
0
28584
21438
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SBS40
SBS18
SBS17b
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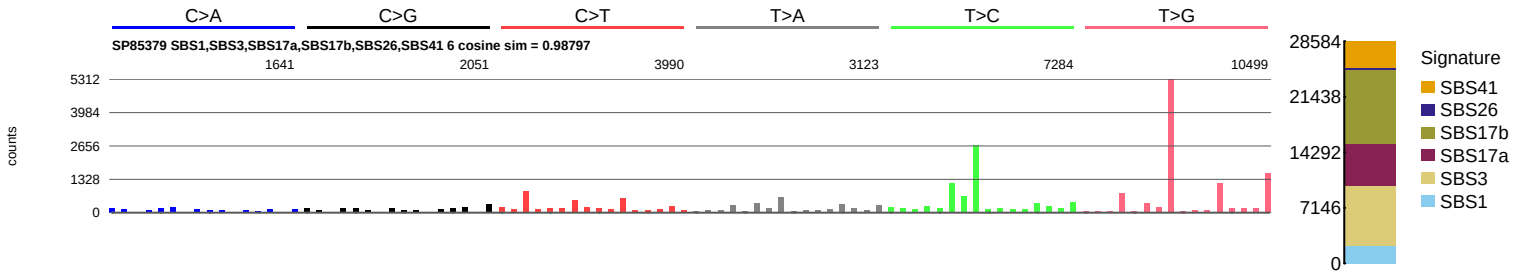
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98797
1641
2051
3990
3123
7284
10499
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
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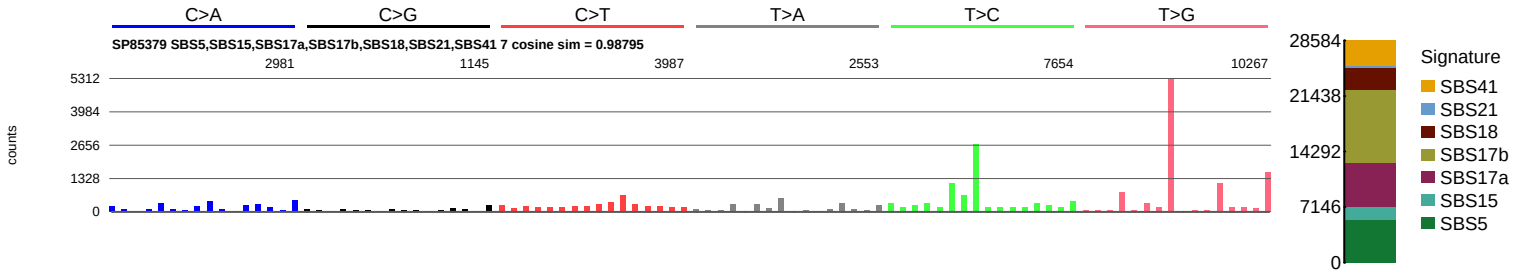
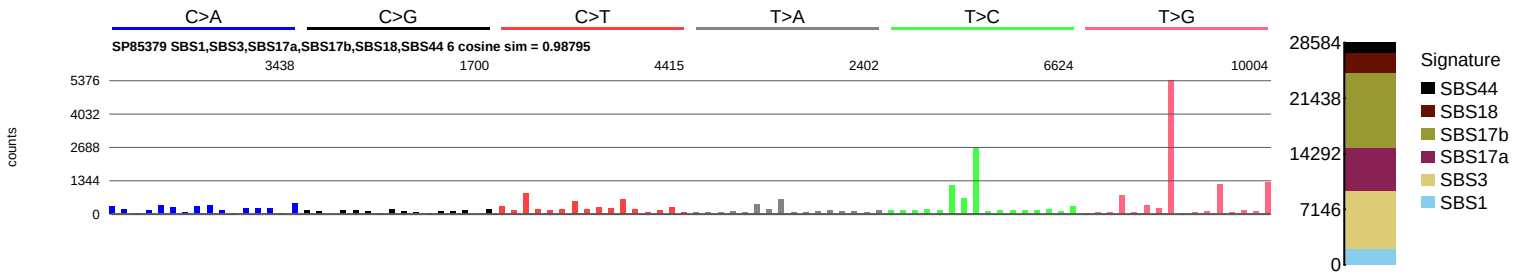
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98795
2981
1145
3987
2553
7654
10267
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5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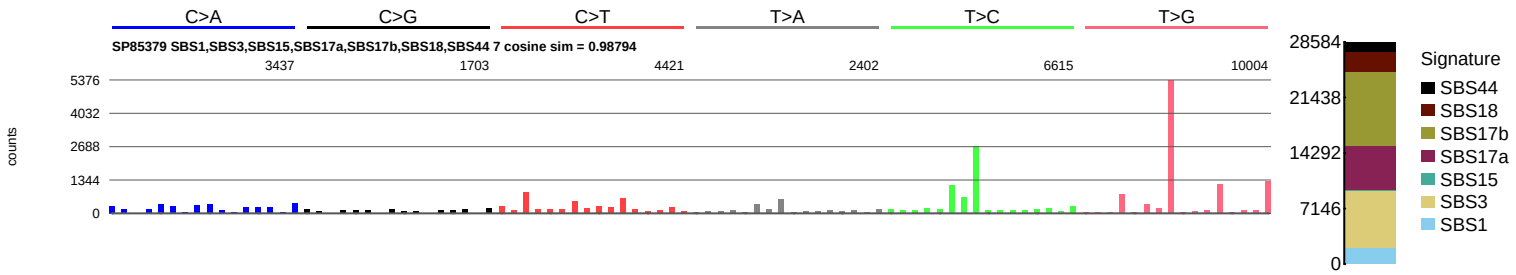
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98795
3438
1700
4415
2402
6624
10004
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS1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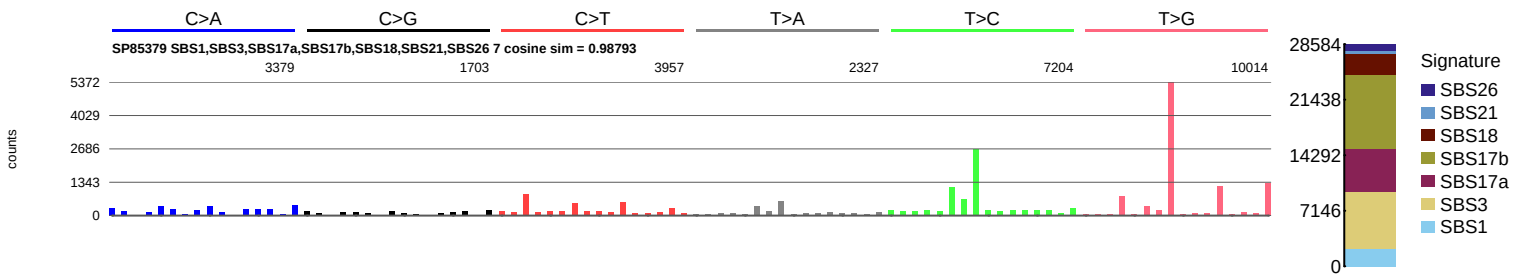
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98794
3437
1703
4421
2402
6615
10004
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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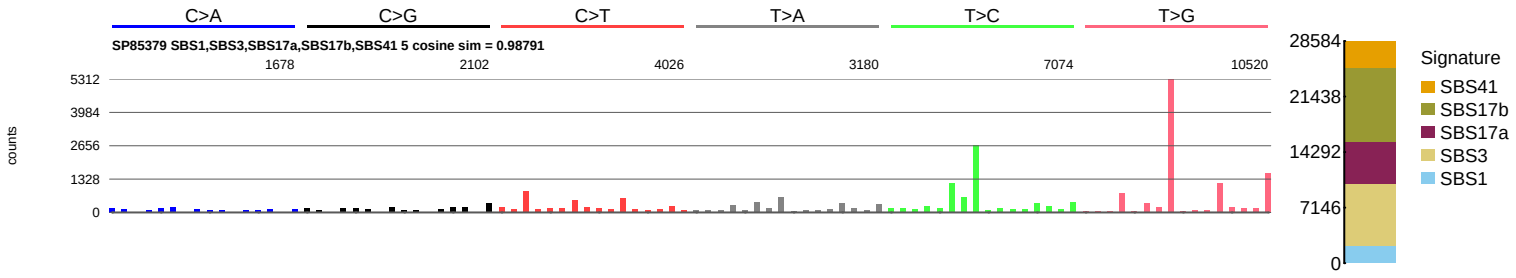
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98793
3379
1703
3957
2327
7204
10014
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
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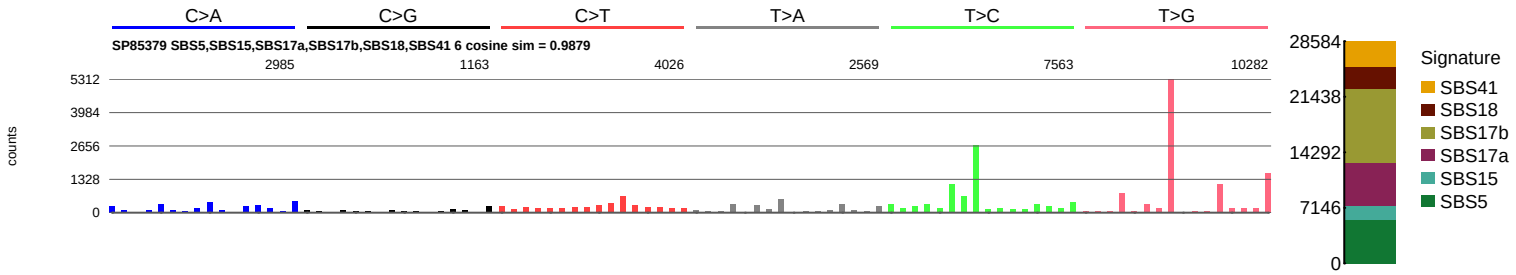
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98791
1678
2102
4026
3180
7074
10520
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS1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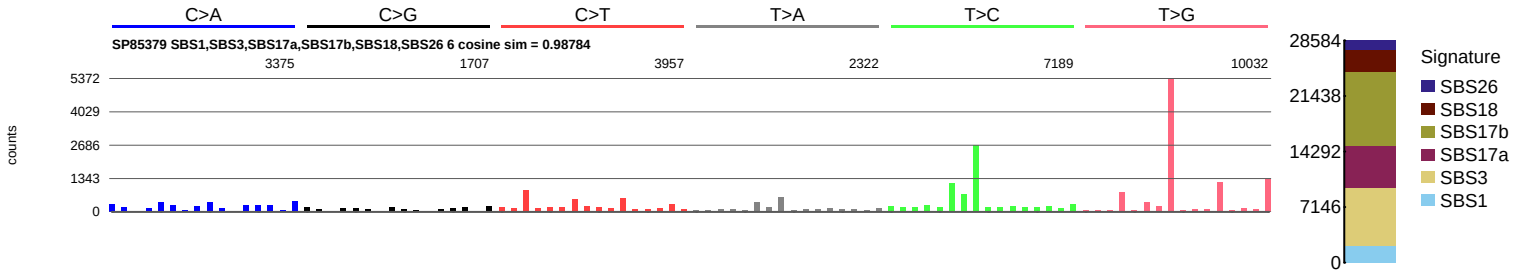
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.9879
2985
1163
4026
2569
7563
10282
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5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C>G
C>T
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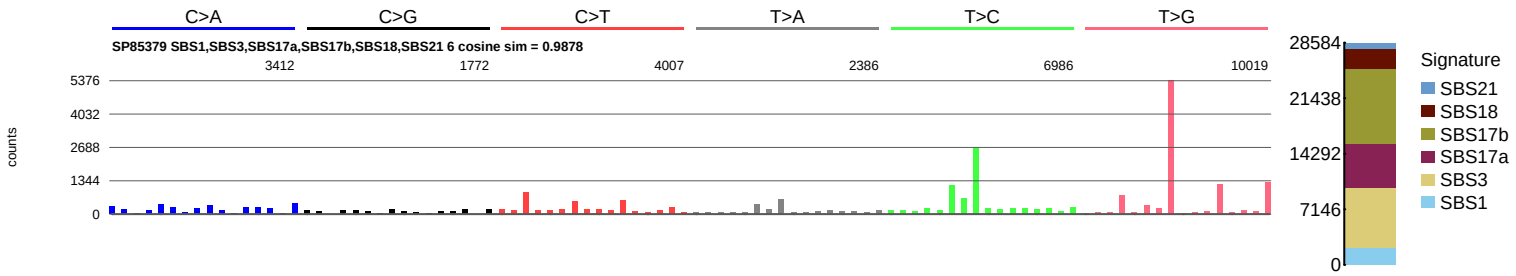
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98784
3375
1707
3957
2322
7189
10032
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>T
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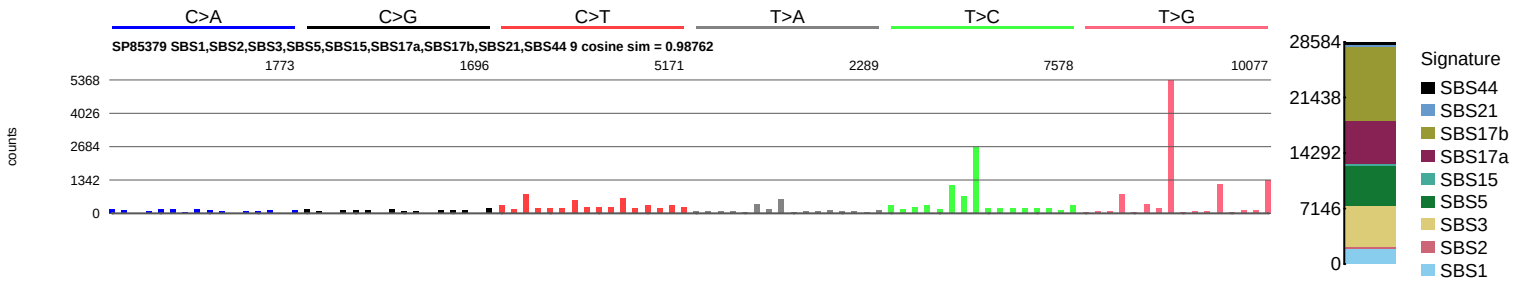
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.9878
3412
1772
4007
2386
6986
10019
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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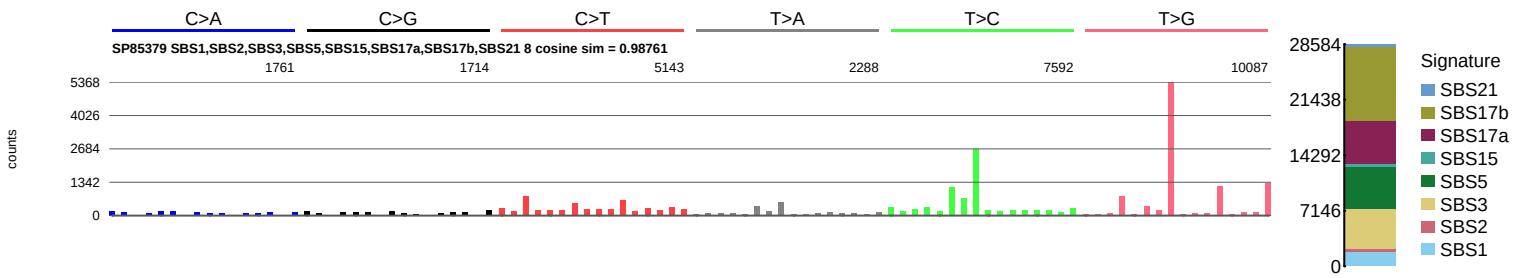
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98762
1773
1696
5171
2289
7578
10077
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98761
1761
1714
5143
2288
7592
10087
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
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Signature
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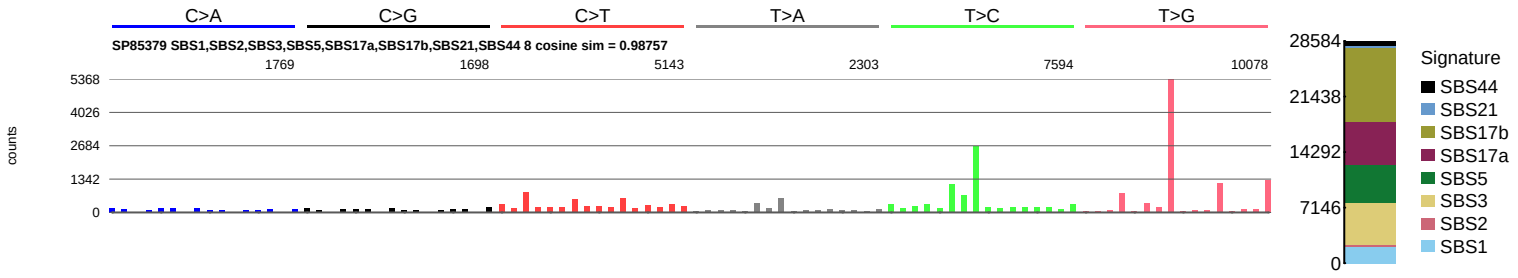
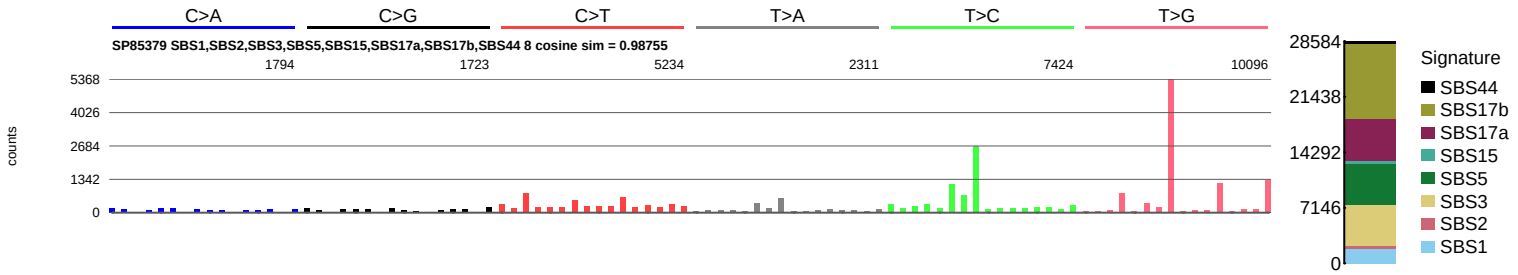
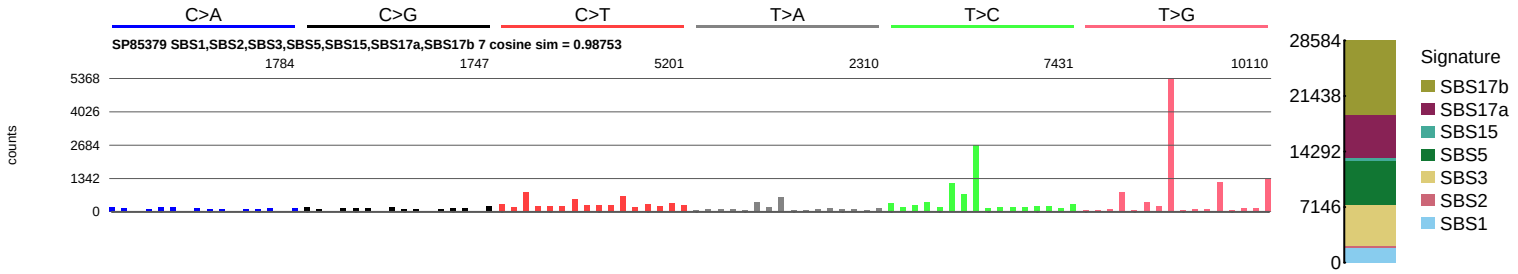
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98757
1769
1698
5143
2303
7594
10078
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98755
1794
1723
5234
2311
7424
10096
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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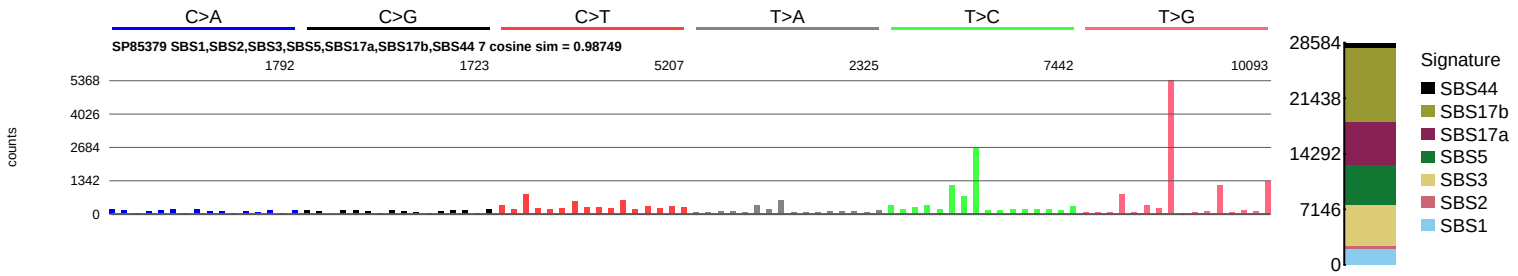
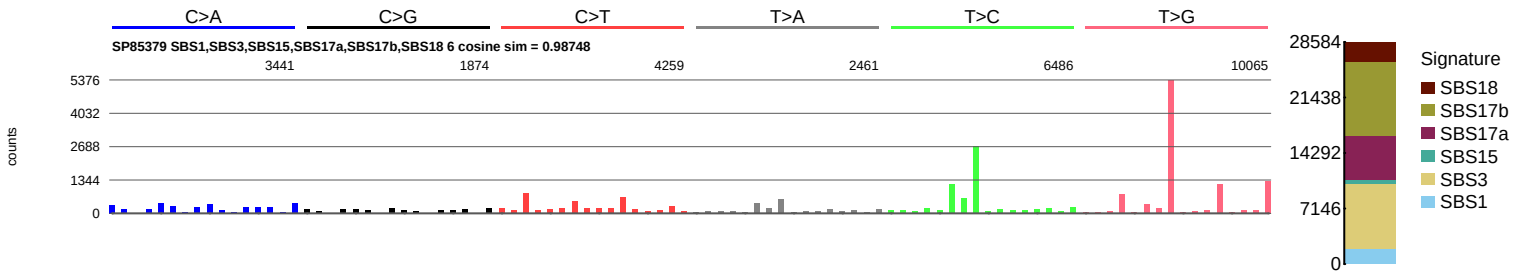
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98753
1784
1747
5201
2310
7431
10110
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98749
1792
1723
5207
2325
7442
10093
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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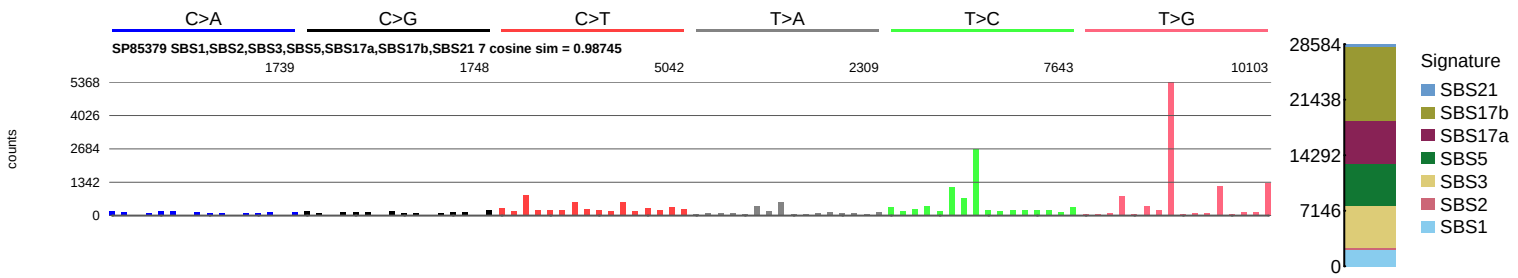
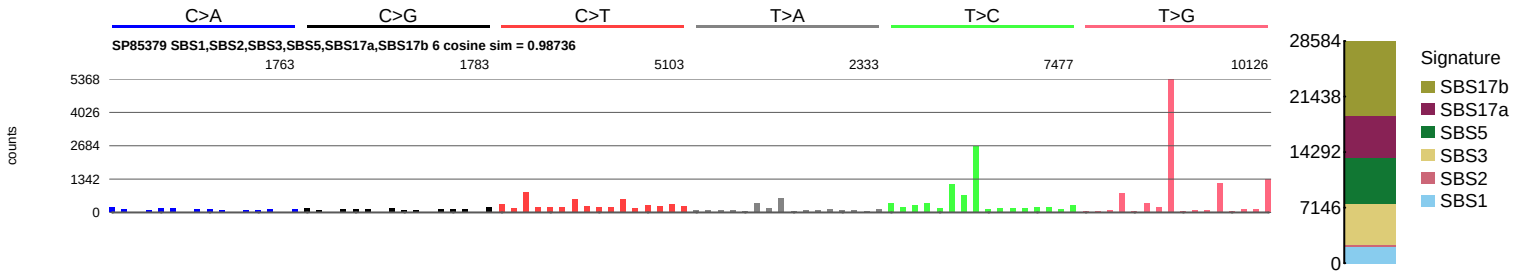
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98748
3441
1874
4259
2461
6486
10065
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98745
1739
1748
5042
2309
7643
10103
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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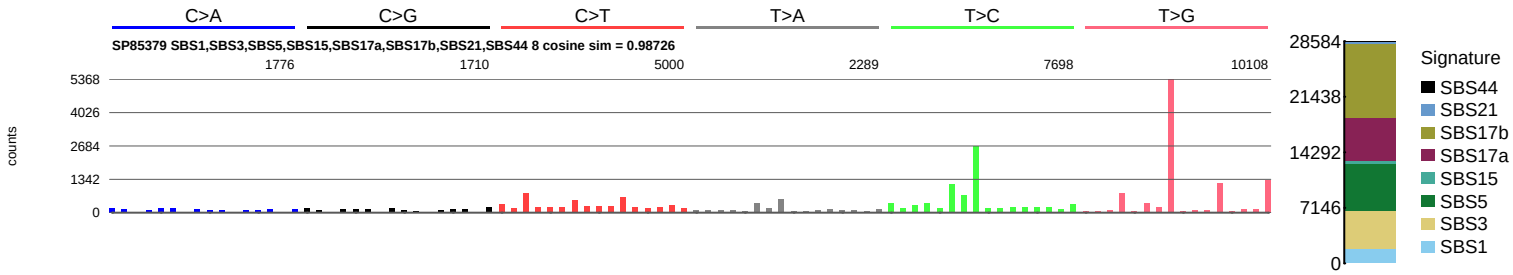
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b 6 cosine sim = 0.98736
1763
1783
5103
2333
7477
10126
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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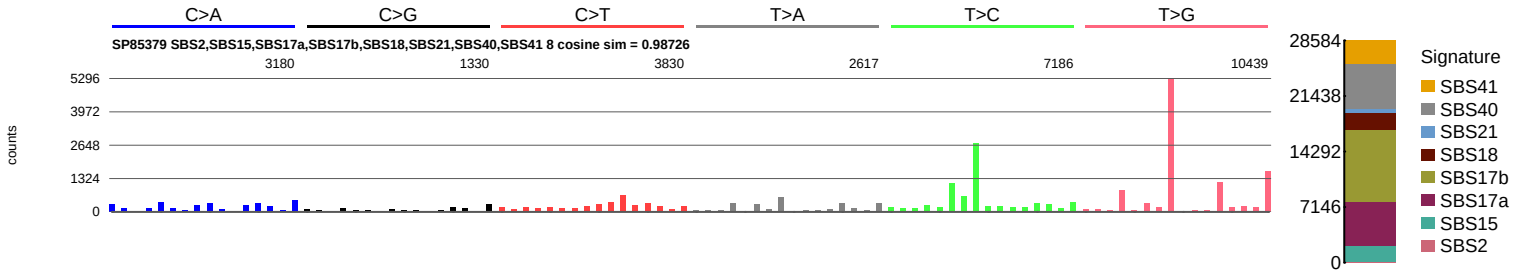
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98726
1776
1710
5000
2289
7698
10108
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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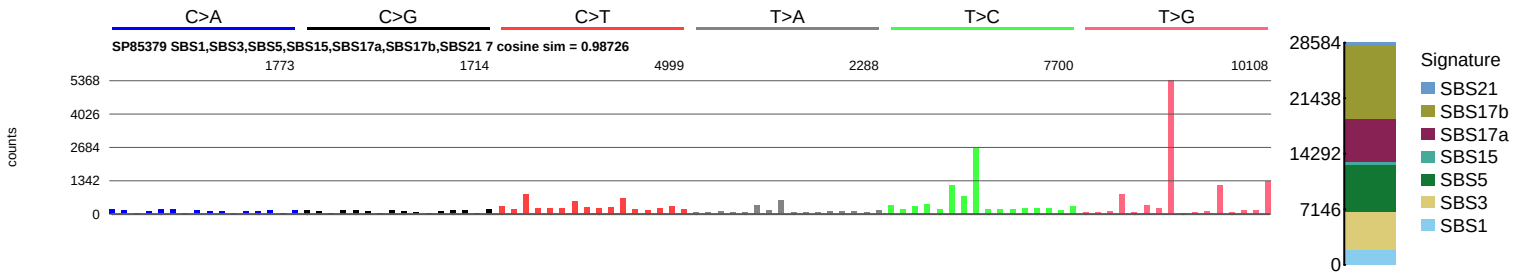
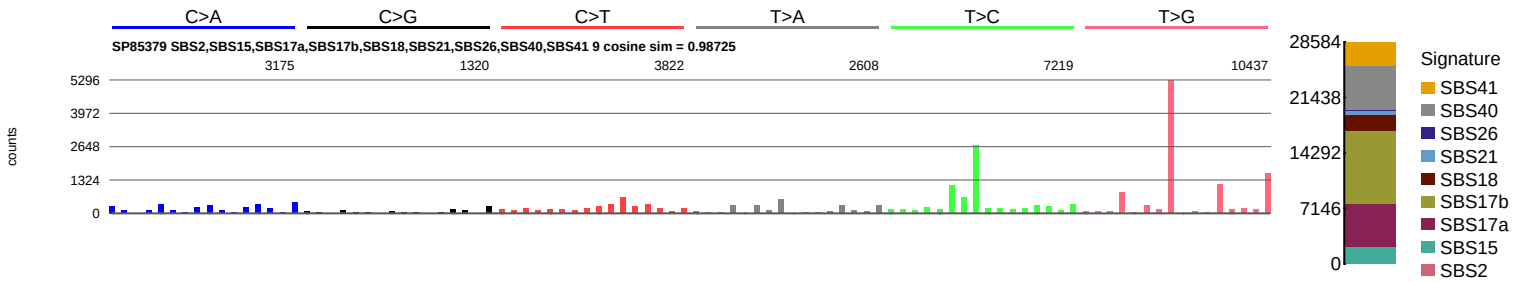
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98726
3180
1330
3830
2617
7186
10439
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98726
1773
1714
4999
2288
7700
10108
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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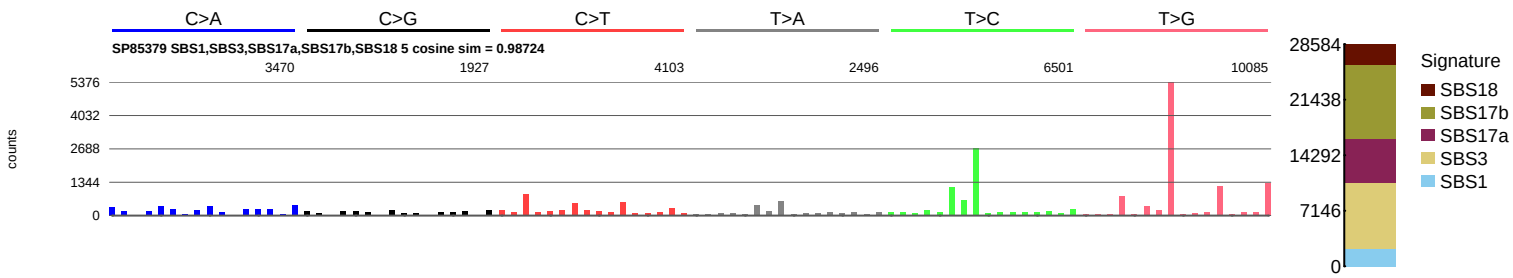
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98725
3175
1320
3822
2608
7219
10437
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2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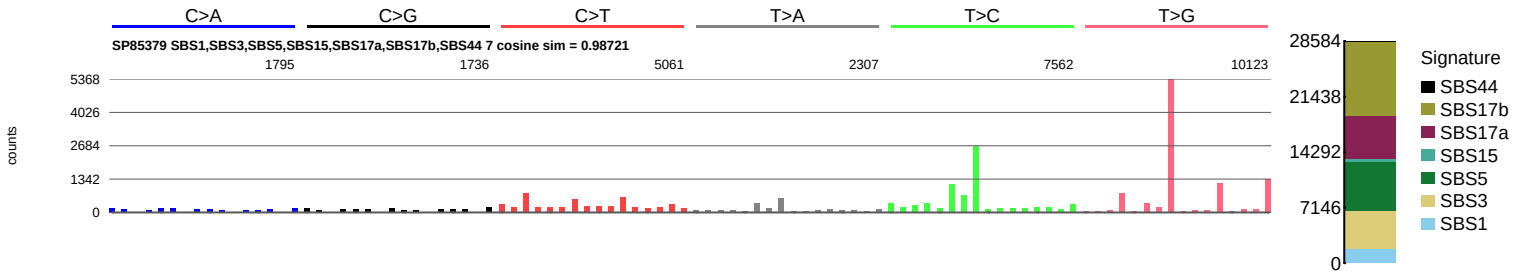
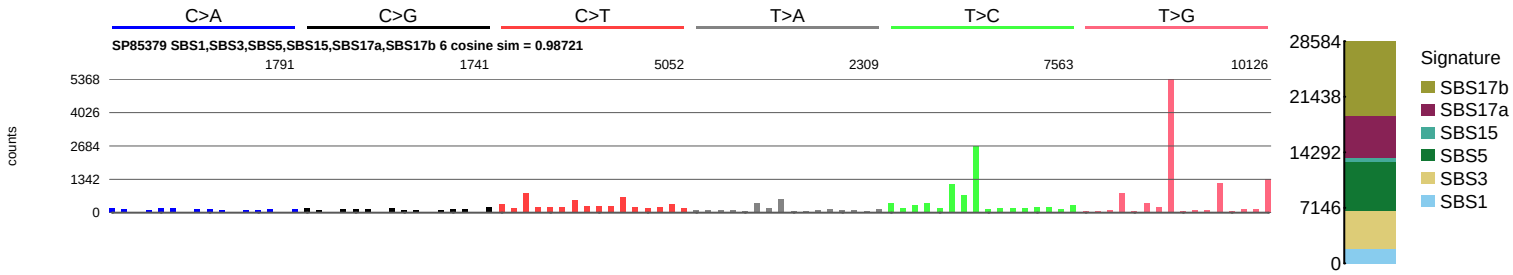
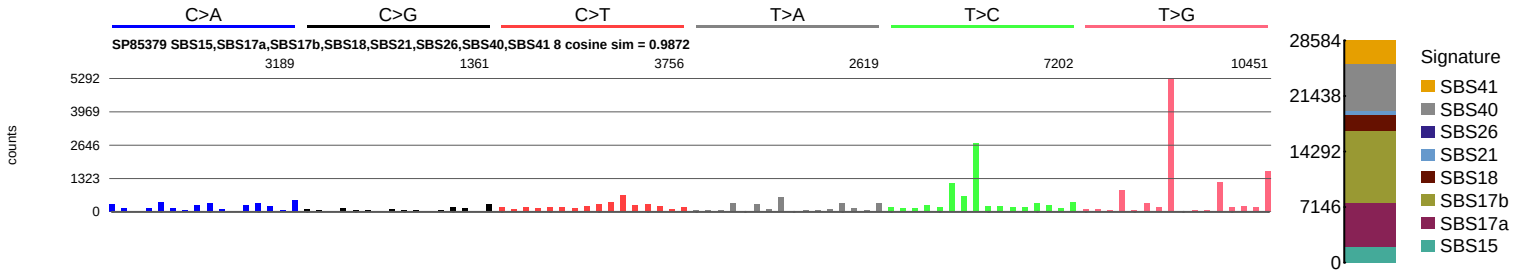
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98724
3470
1927
4103
2496
6501
10085
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98721
1795
1736
5061
2307
7562
10123
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98721
1791
1741
5052
2309
7563
10126
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.9872
3189
1361
3756
2619
7202
10451
counts
5292
3969
2646
1323
0
28584
21438
14292
7146
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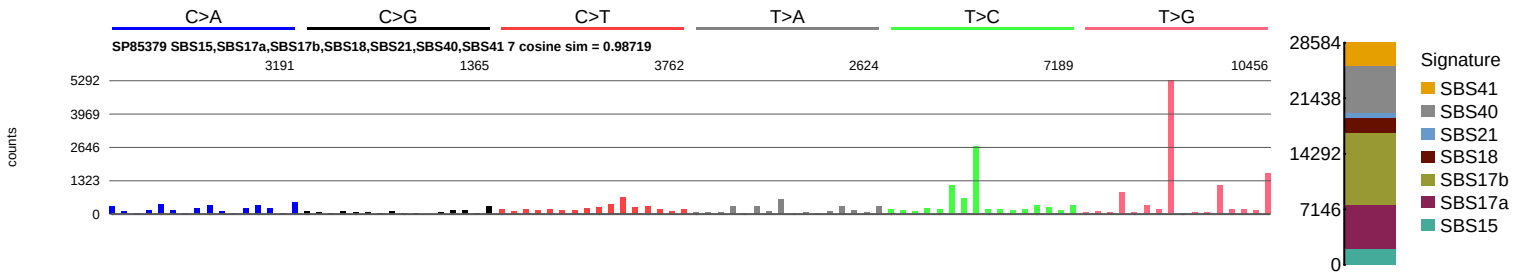
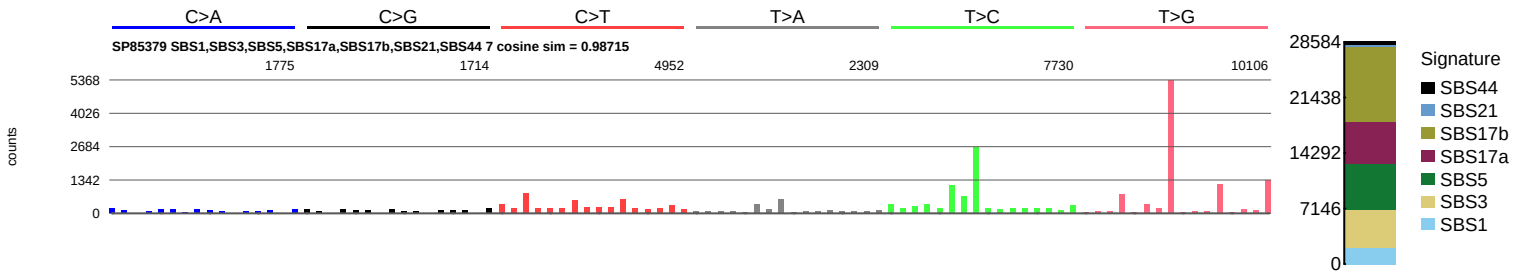
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98719
3191
1365
3762
2624
7189
10456
counts
5292
3969
2646
1323
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
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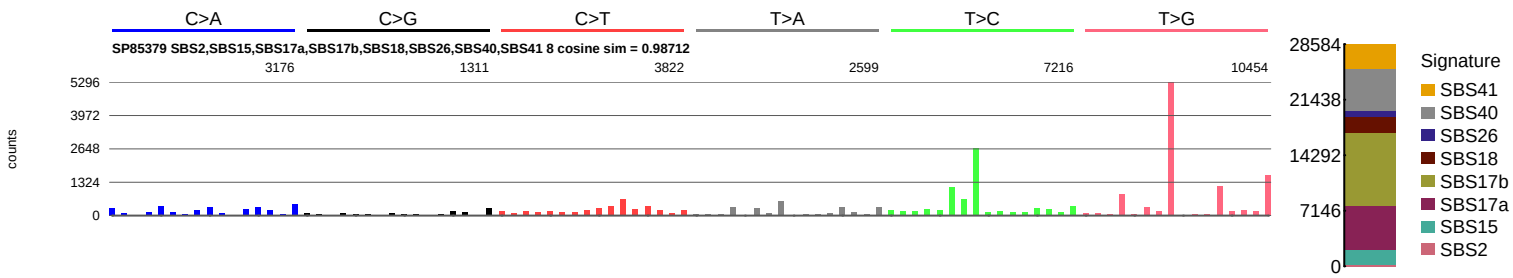
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98715
1775
1714
4952
2309
7730
10106
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1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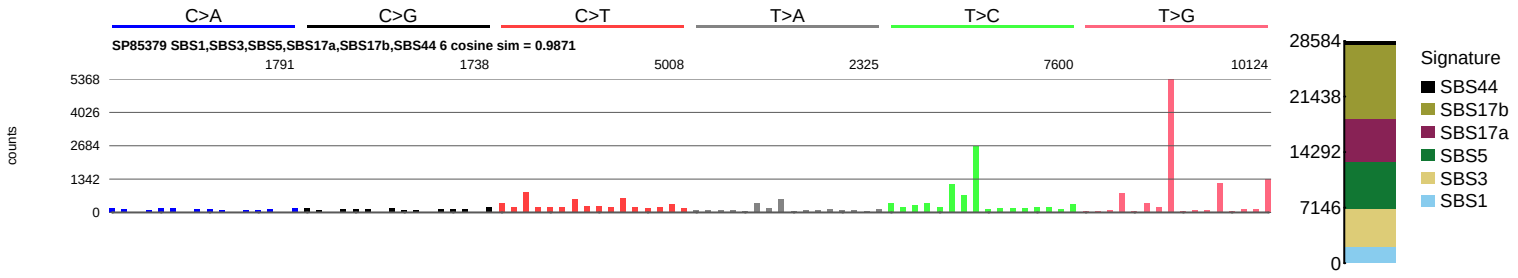
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98712
3176
1311
3822
2599
7216
10454
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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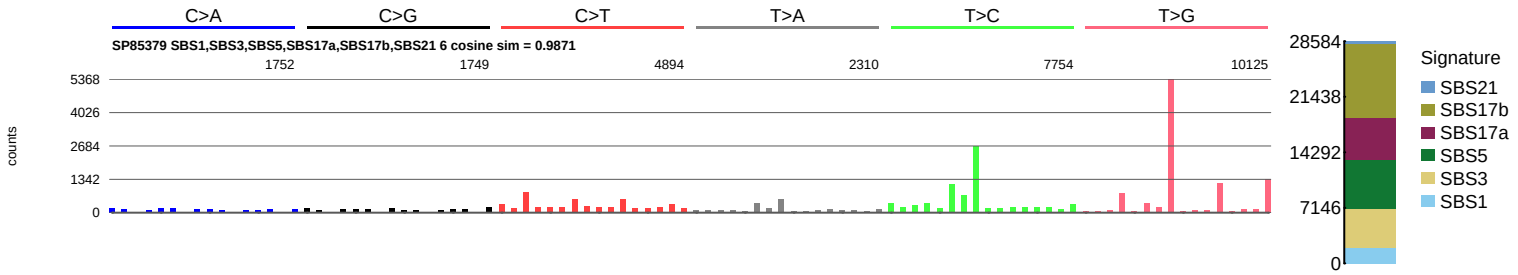
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.9871
1791
1738
5008
2325
7600
10124
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
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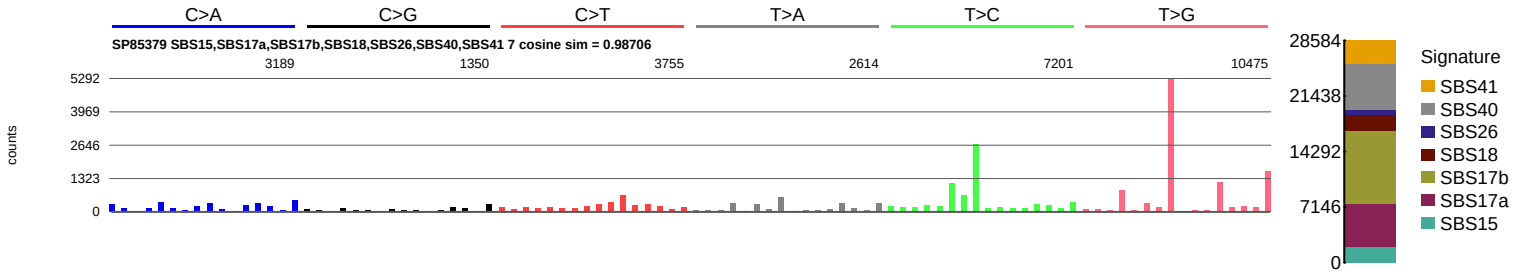
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.9871
1752
1749
4894
2310
7754
10125
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
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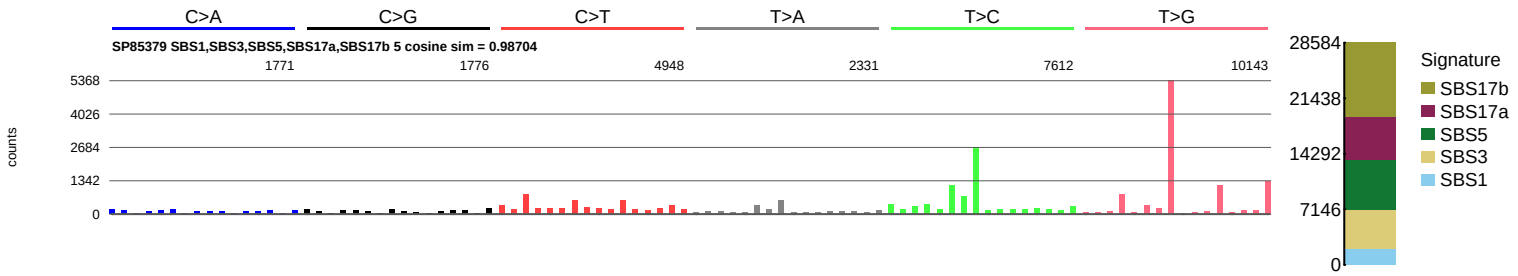
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98706
3189
1350
3755
2614
7201
10475
counts
5292
3969
2646
1323
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15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C>G
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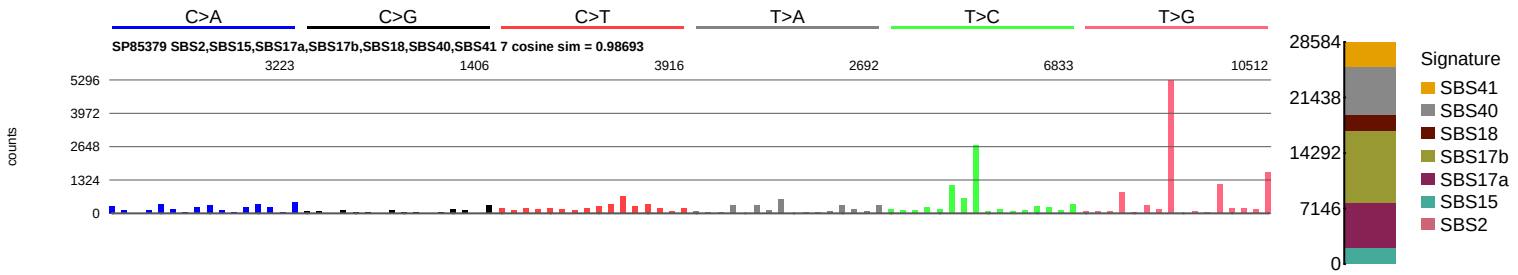
T>C
T>G
SP85379 SBS1,SBS3,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98704
1771
1776
4948
2331
7612
10143
counts
5368
4026
2684
1342
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS1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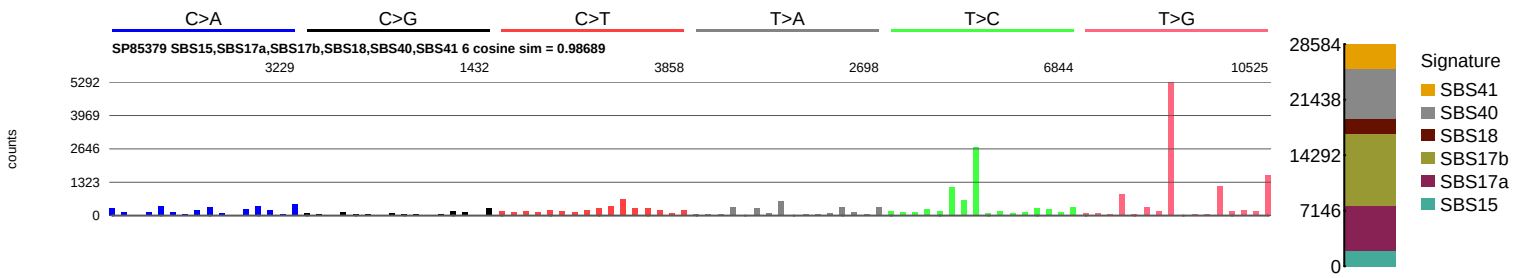
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98693
3223
1406
3916
2692
6833
10512
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2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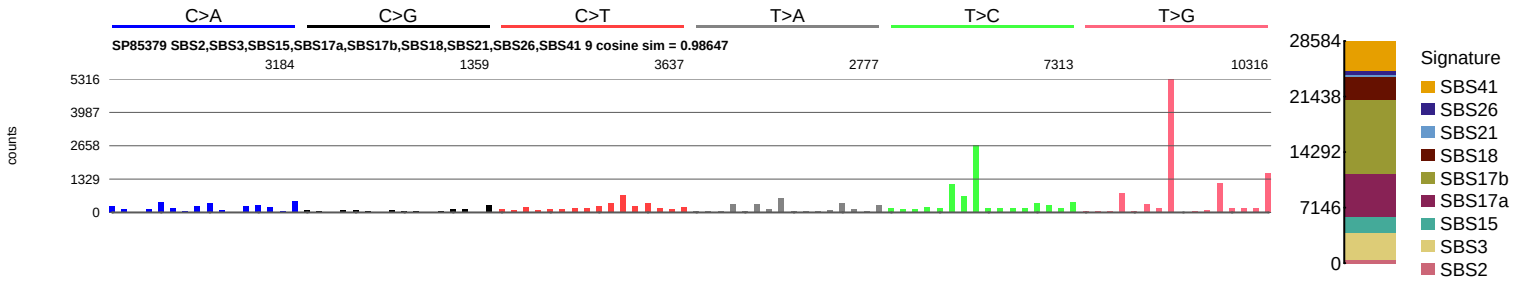
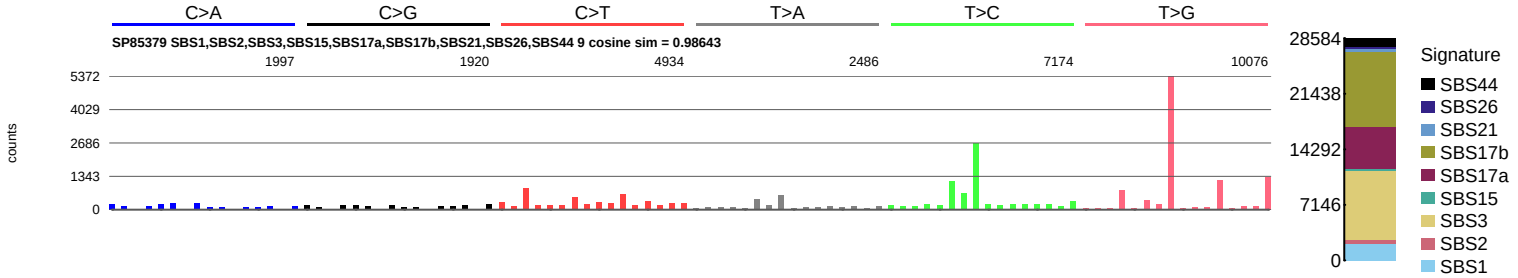
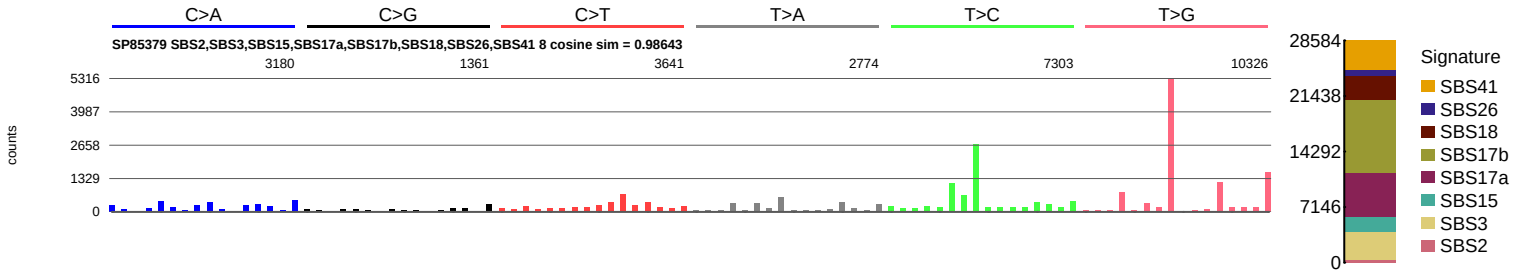
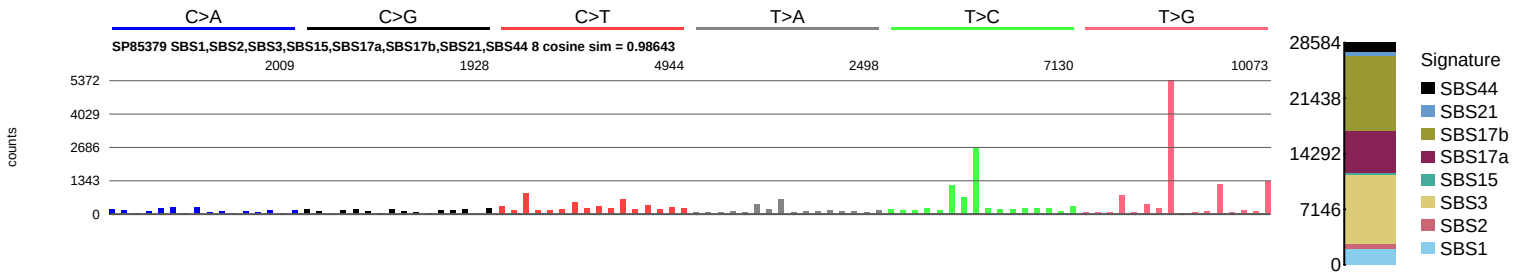
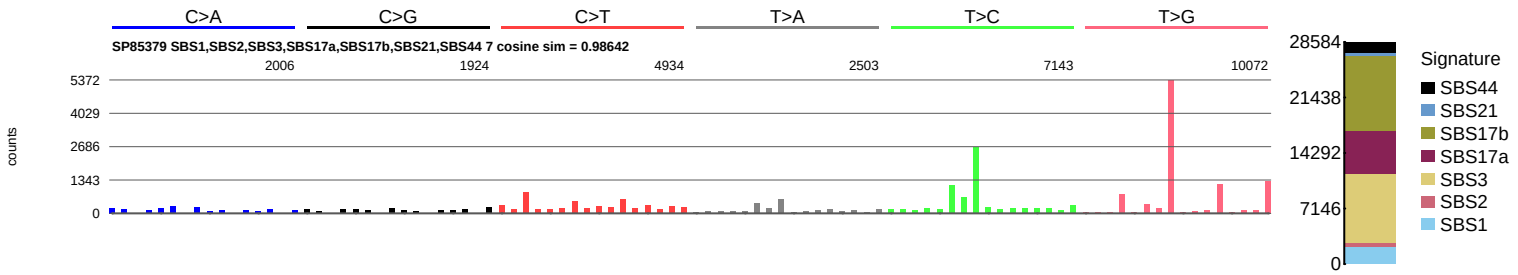
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98689
3229
1432
3858
2698
6844
10525
counts
5292
3969
2646
1323
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15

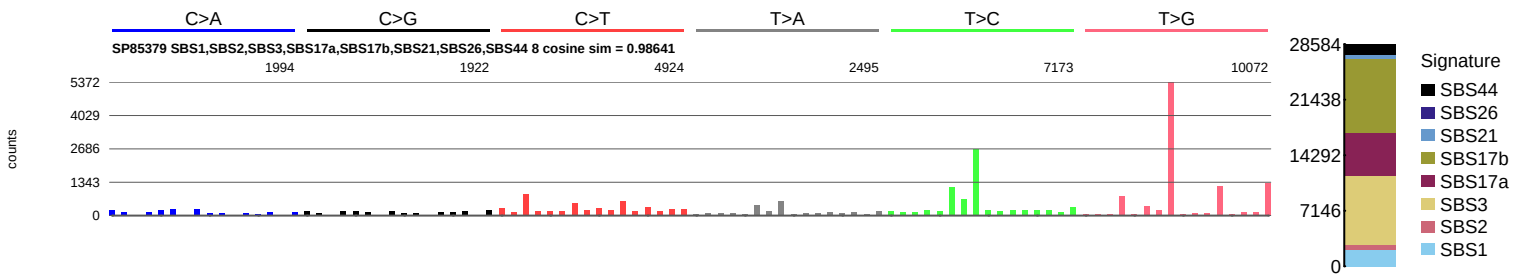

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98647
3184
1359
3637
2777
7313
10316
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

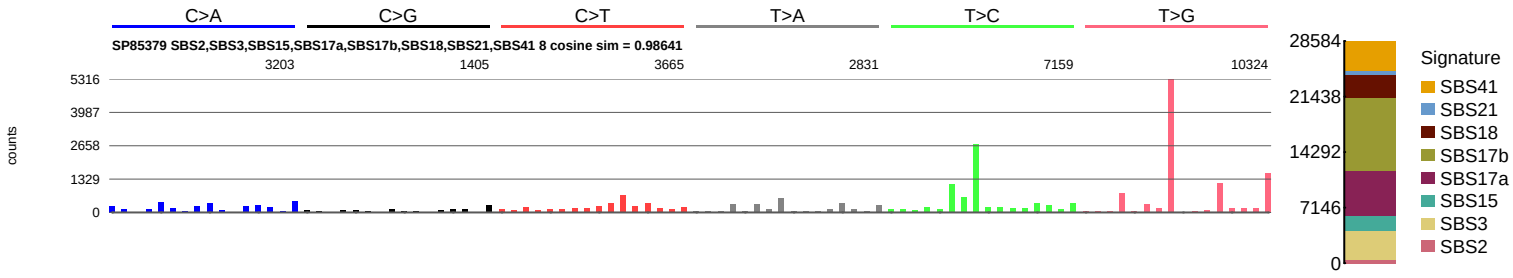

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98643
1997
1920
4934
2486
7174
10076
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

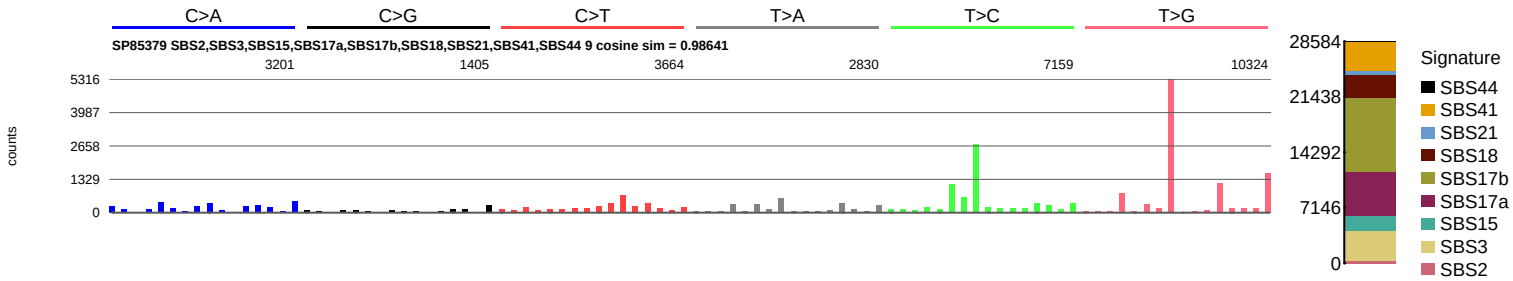

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98643
3180
1361
3641
2774
7303
10326
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

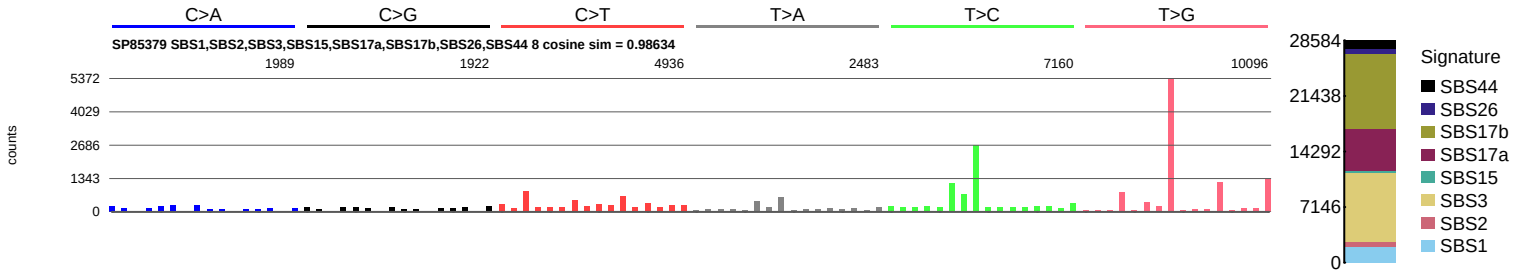

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98643
2009
1928
4944
2498
7130
10073
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

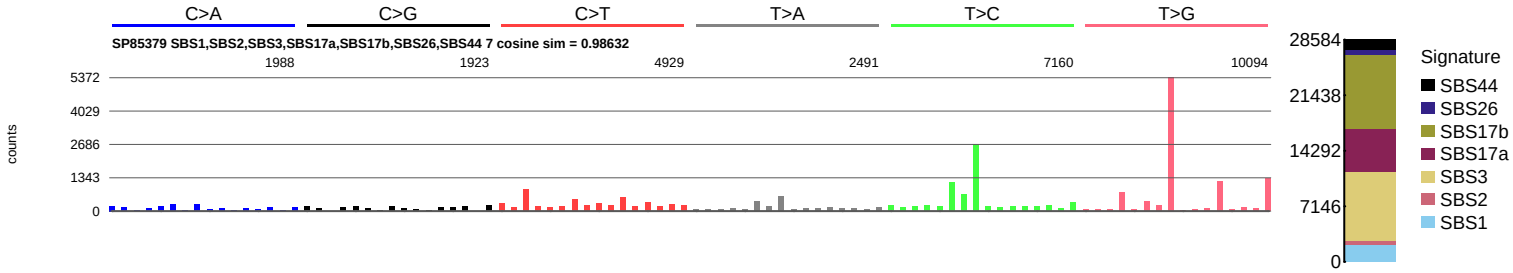

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98642
2006
1924
4934
2503
7143
10072
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

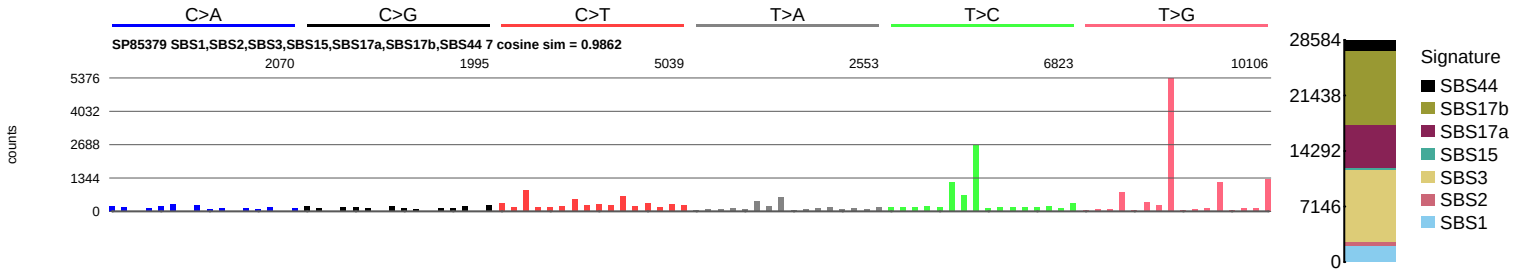

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98641
1994
1922
4924
2495
7173
10072
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

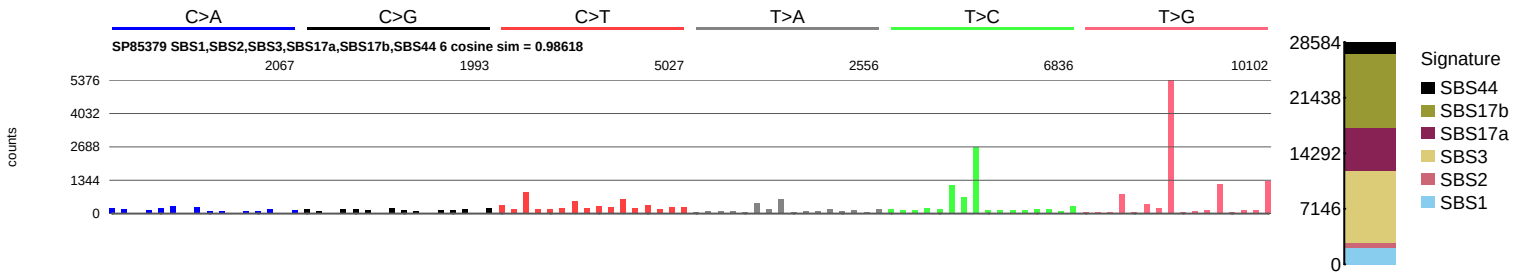

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98641
3203
1405
3665
2831
7159
10324
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

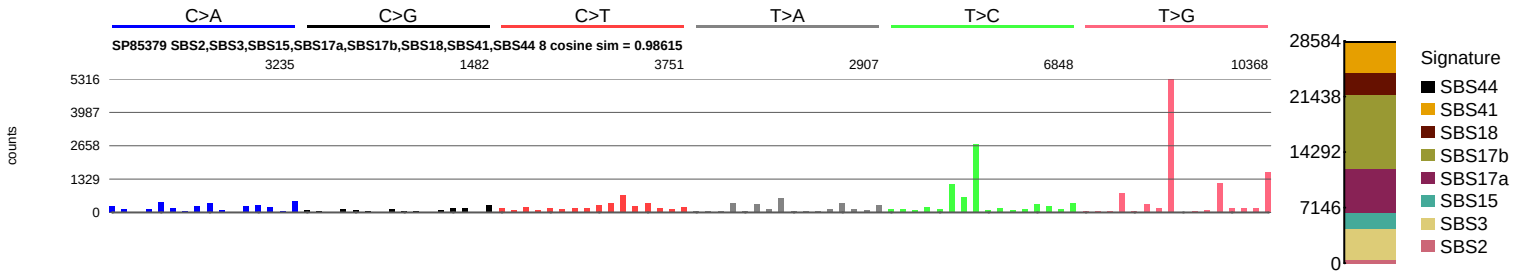

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98641
3201
1405
3664
2830
7159
10324
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

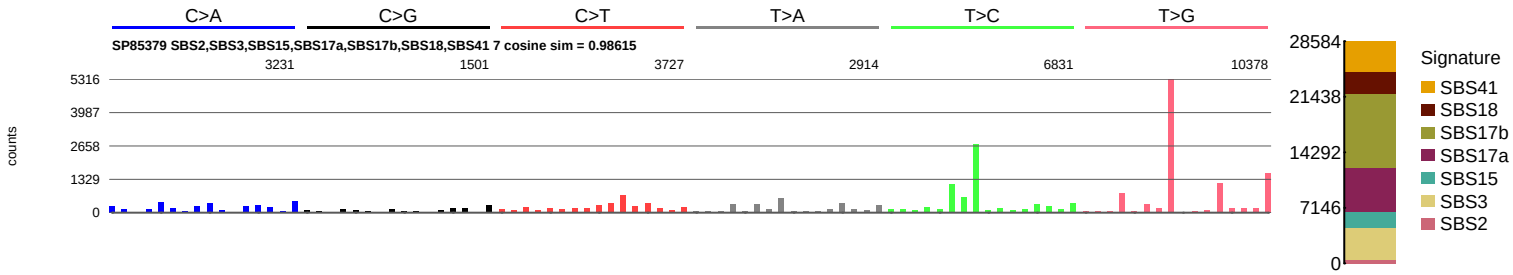

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98634
1989
1922
4936
2483
7160
10096
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

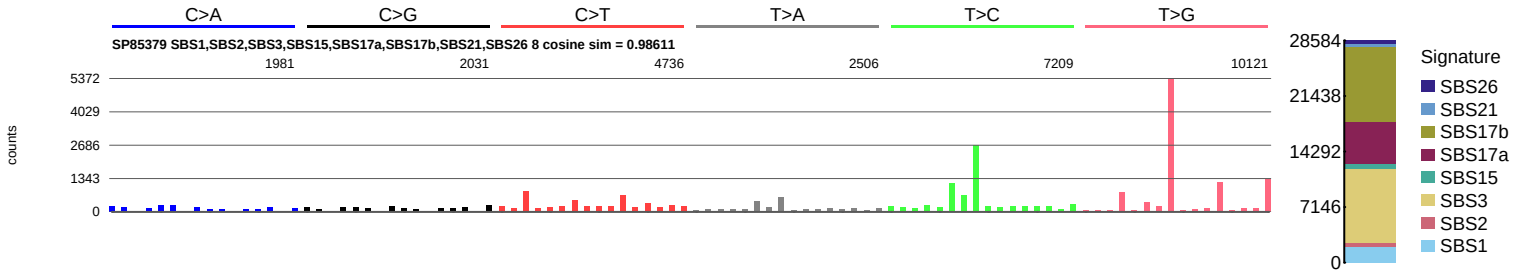

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98632
1988
1923
4929
2491
7160
10094
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

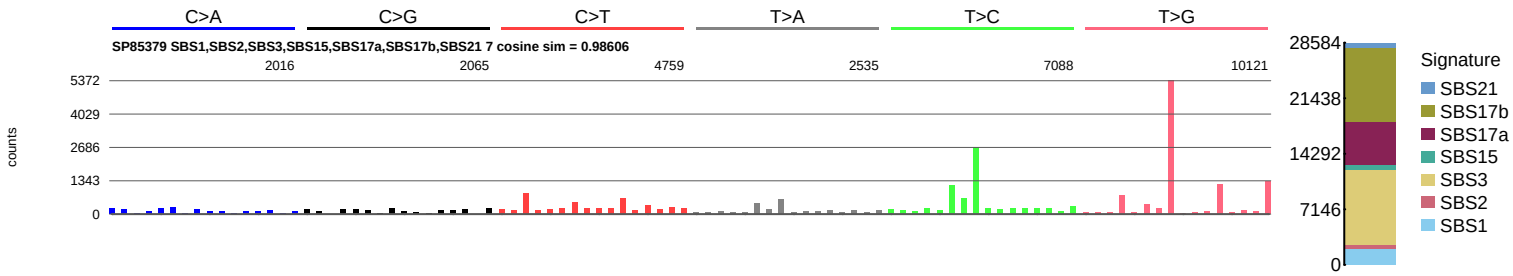

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.9862
2070
1995
5039
2553
6823
10106
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

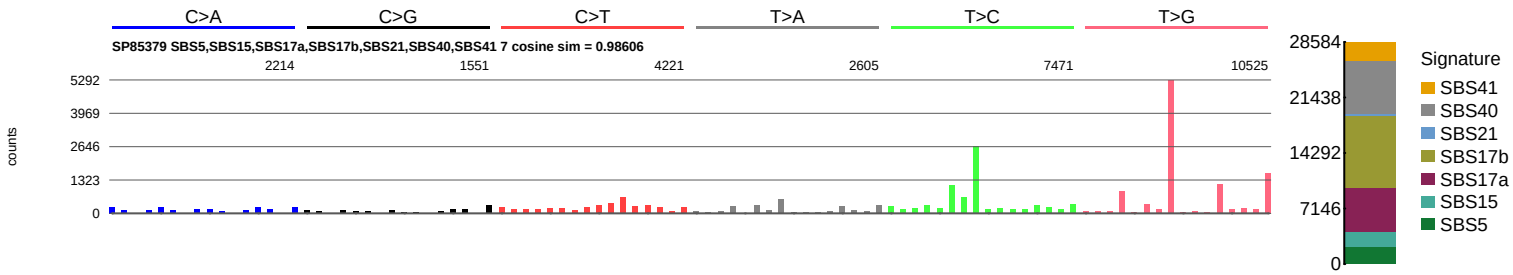

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98618
2067
1993
5027
2556
6836
10102
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS2
SBS1

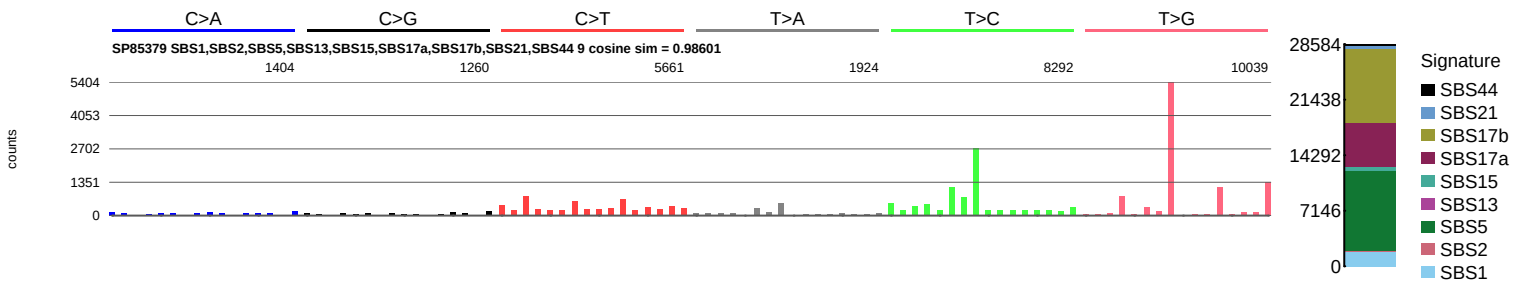

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98615
3235
1482
3751
2907
6848
10368
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98615
3231
1501
3727
2914
6831
10378
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98611
1981
2031
4736
2506
7209
10121
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98606
2016
2065
4759
2535
7088
10121
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98606
2214
1551
4221
2605
7471
10525
counts
5292
3969
2646
1323
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5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SP85379 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98601
1404
1260
5661
1924
8292
10039
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
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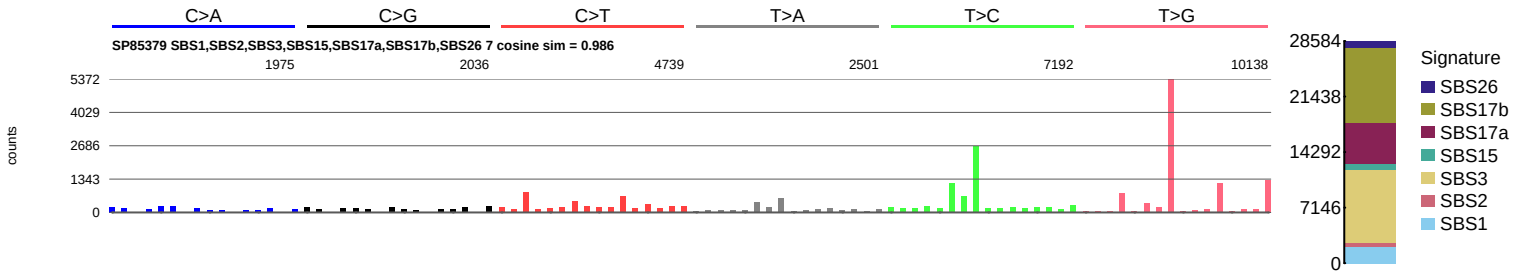
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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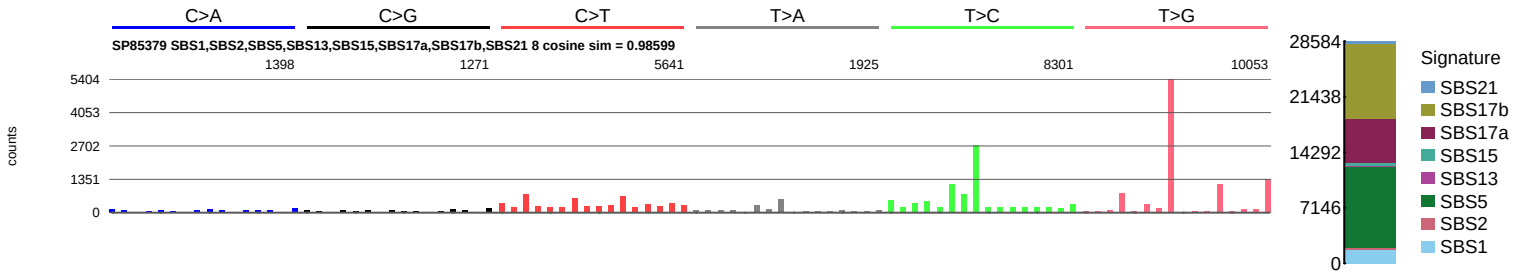
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.986
1975
2036
4739
2501
7192
10138
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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SP85379 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98599
1398
1271
5641
1925
8301
10053
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
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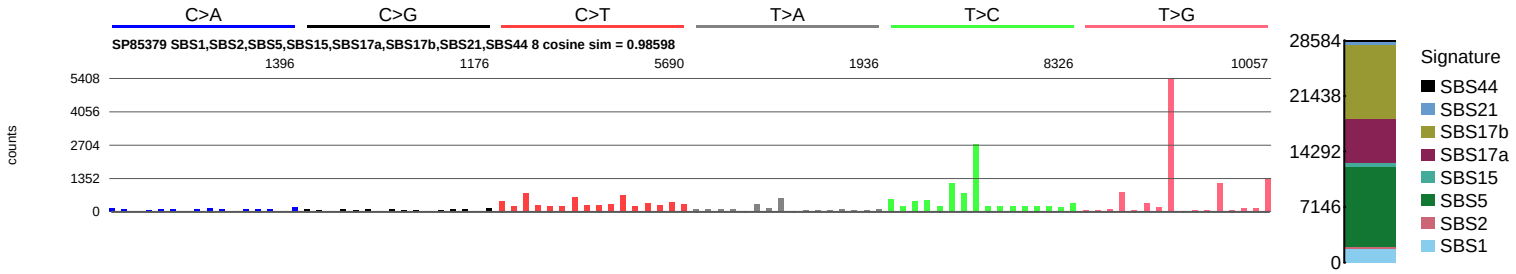
SBS2
SBS1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SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98598
1396
1176
5690
1936
8326
10057
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
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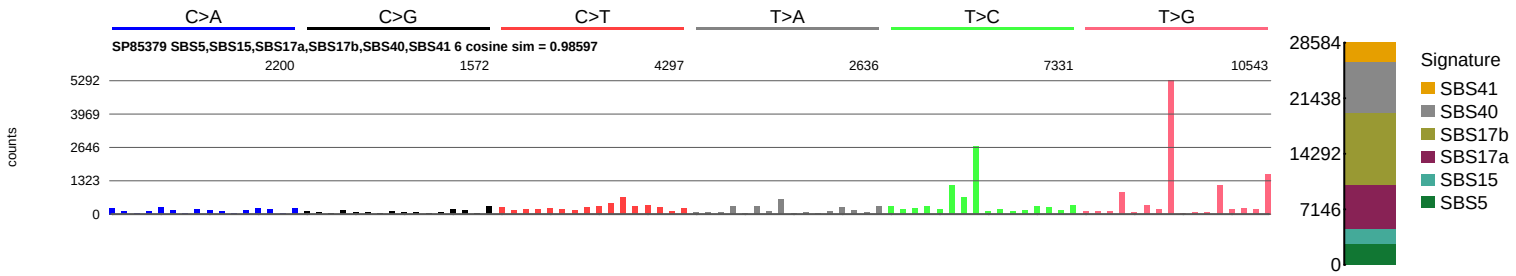
SBS2
SBS1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T>C
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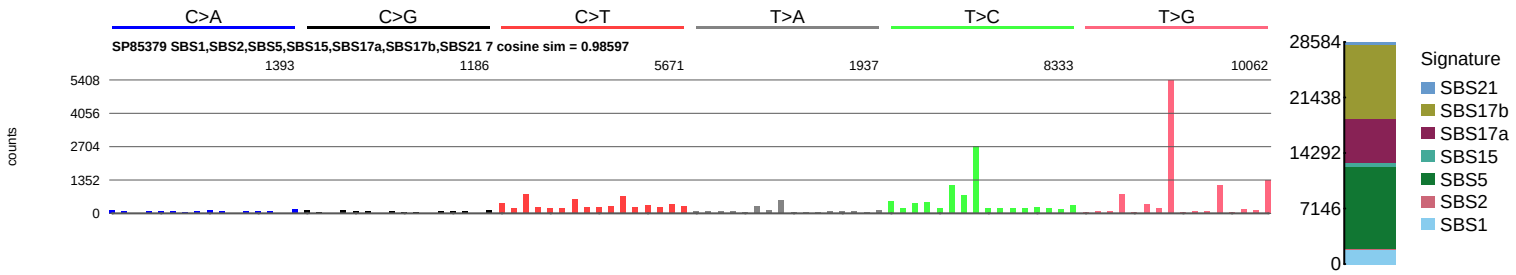
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98597
2200
1572
4297
2636
7331
10543
counts
5292
3969
2646
1323
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98597
1393
1186
5671
1937
8333
10062
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
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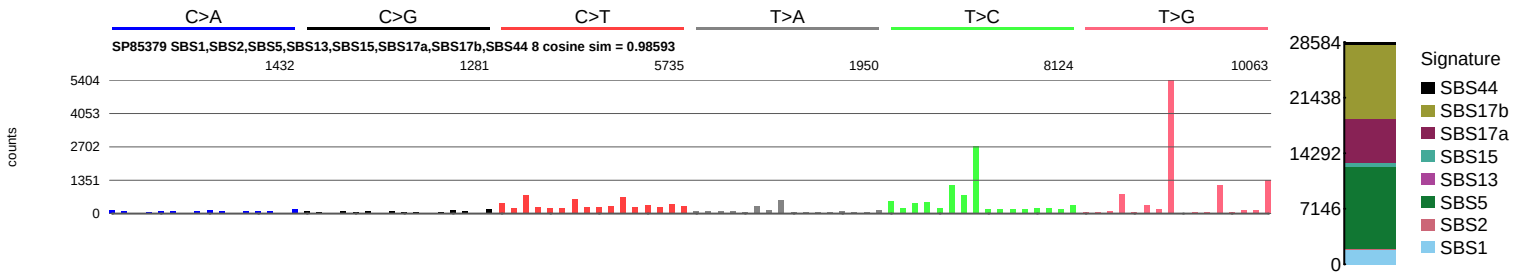
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98593
1432
1281
5735
1950
8124
10063
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
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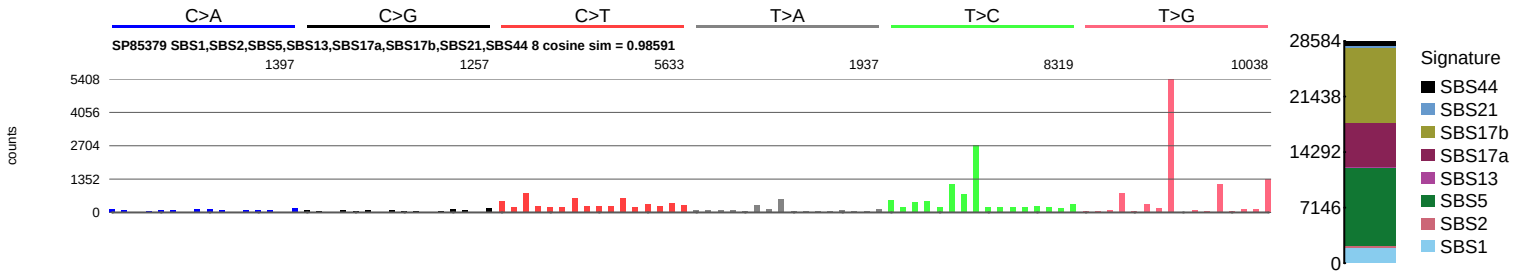
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98591
1397
1257
5633
1937
8319
10038
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
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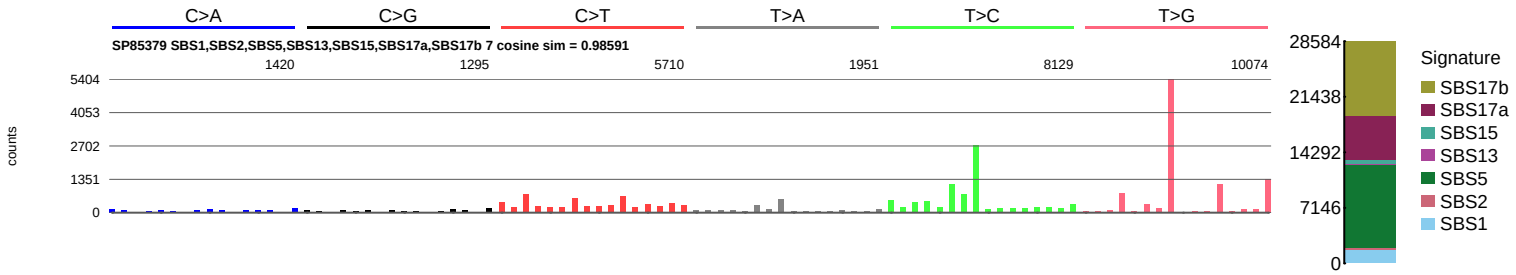
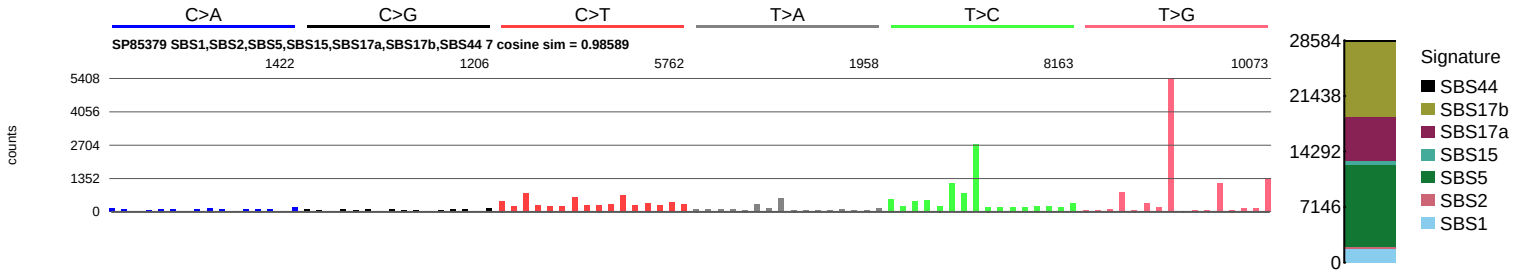
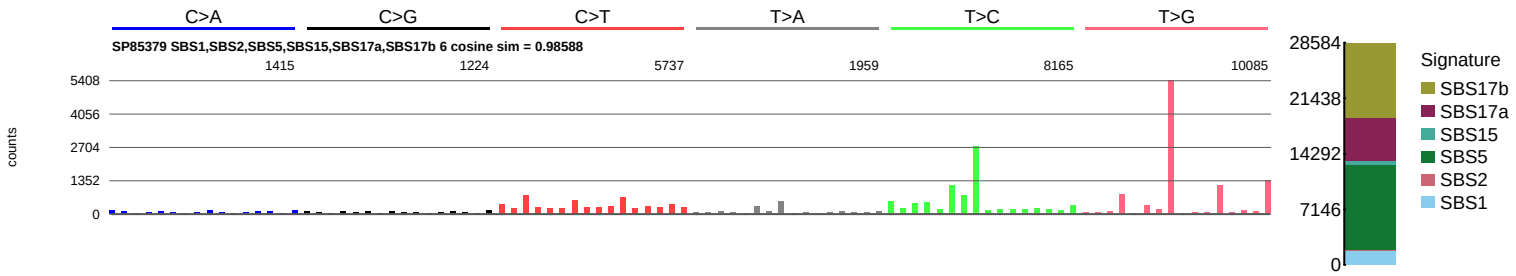
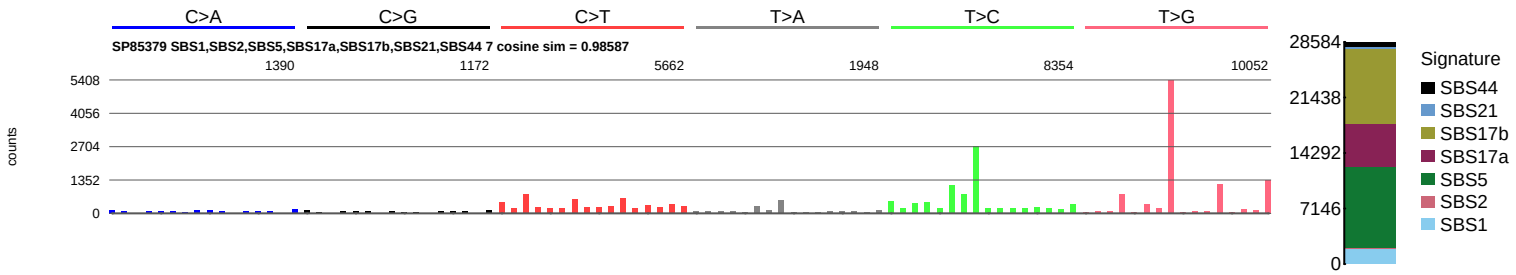
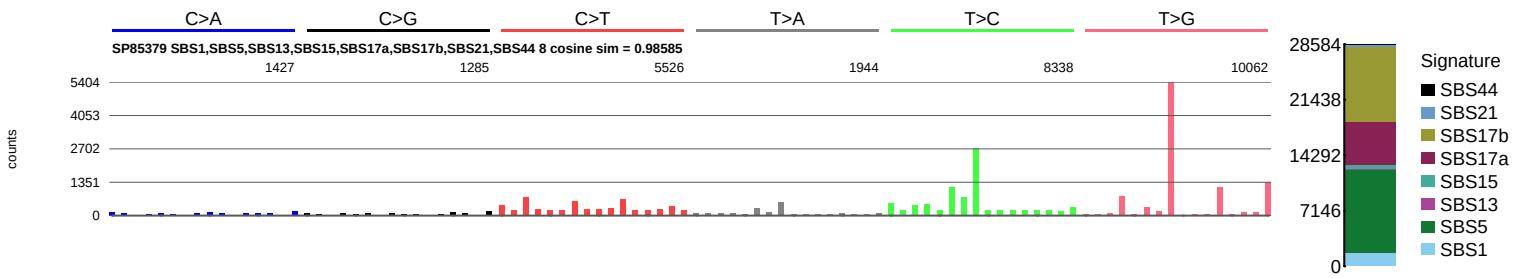
SBS2
SBS1

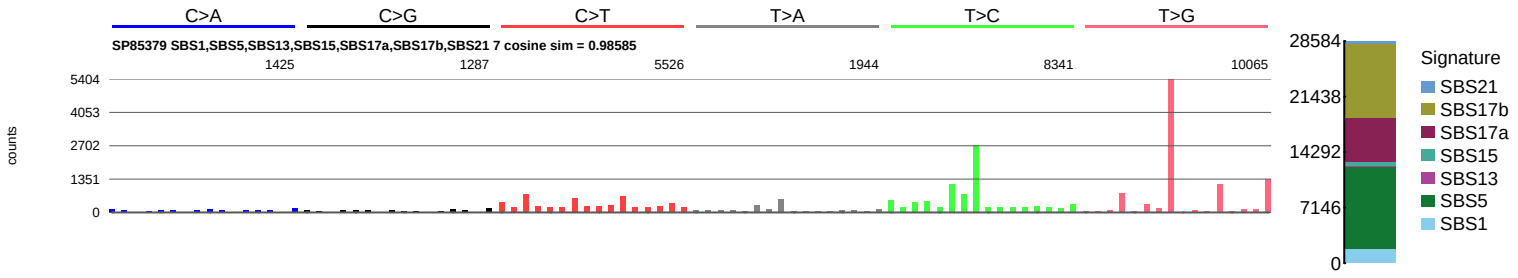

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98591
1420
1295
5710
1951
8129
10074
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

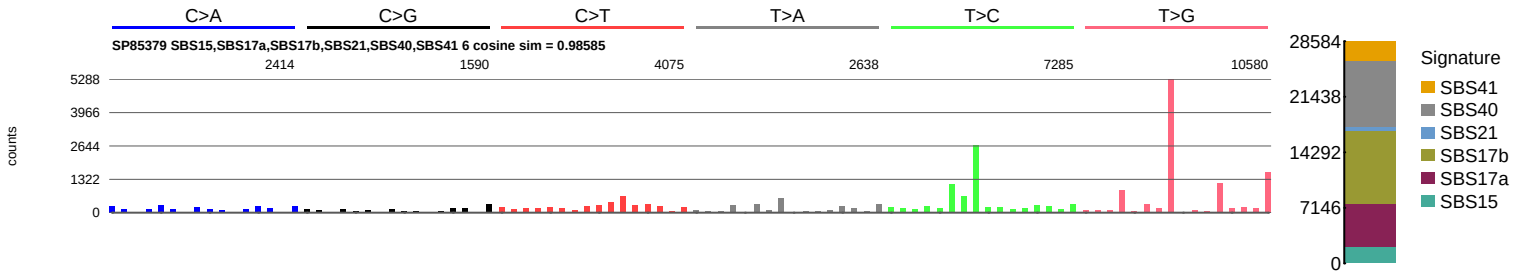

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98589
1422
1206
5762
1958
8163
10073
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

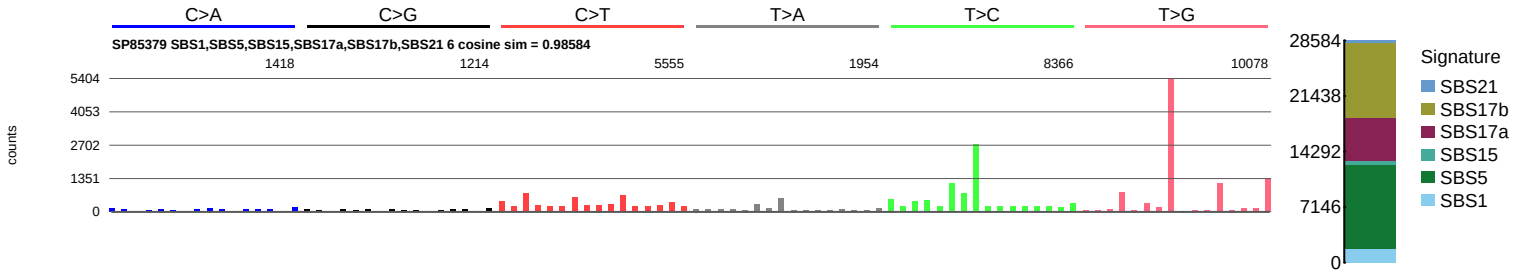

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98588
1415
1224
5737
1959
8165
10085
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98587
1390
1172
5662
1948
8354
10052
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98585
1427
1285
5526
1944
8338
10062
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98585
1425
1287
5526
1944
8341
10065
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98585
2414
1590
4075
2638
7285
10580
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98584
1418
1214
5555
1954
8366
10078
counts
5404
4053
2702
1351
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28584
21438
14292
7146
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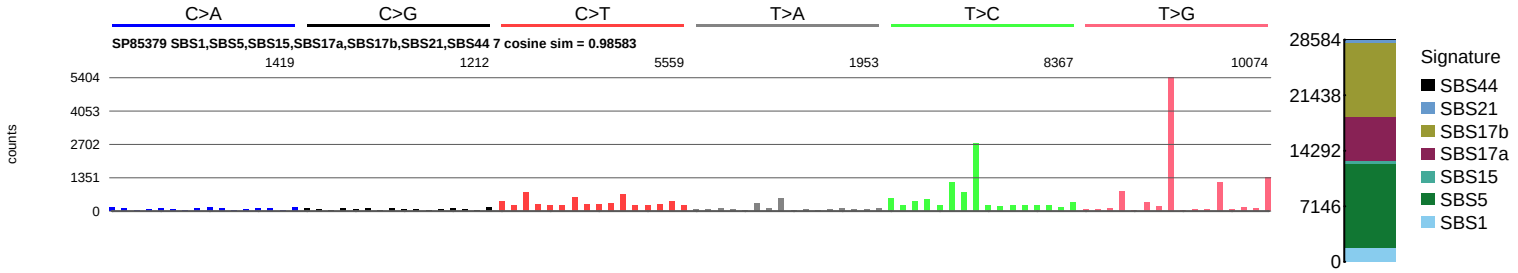
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98583
1419
1212
5559
1953
8367
10074
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
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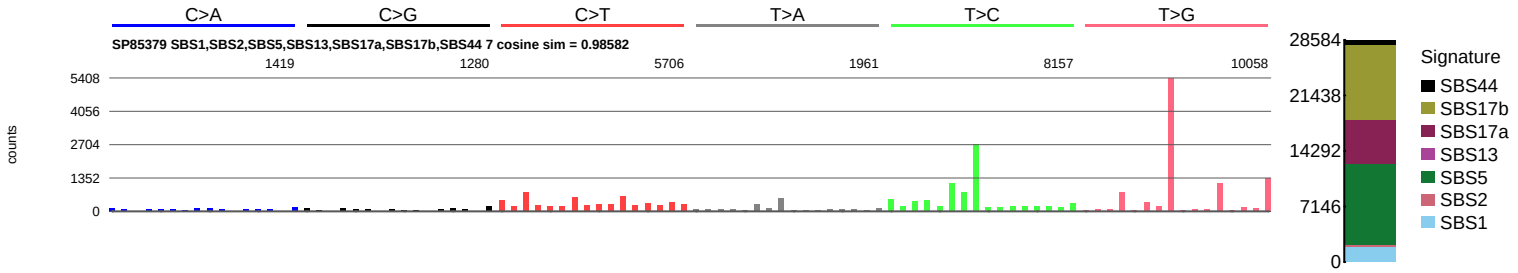
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98582
1419
1280
5706
1961
8157
10058
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
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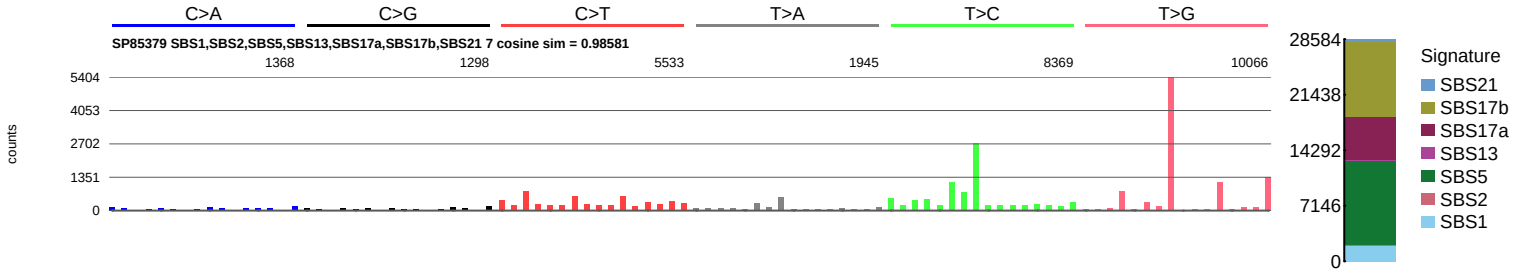
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98581
1368
1298
5533
1945
8369
10066
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
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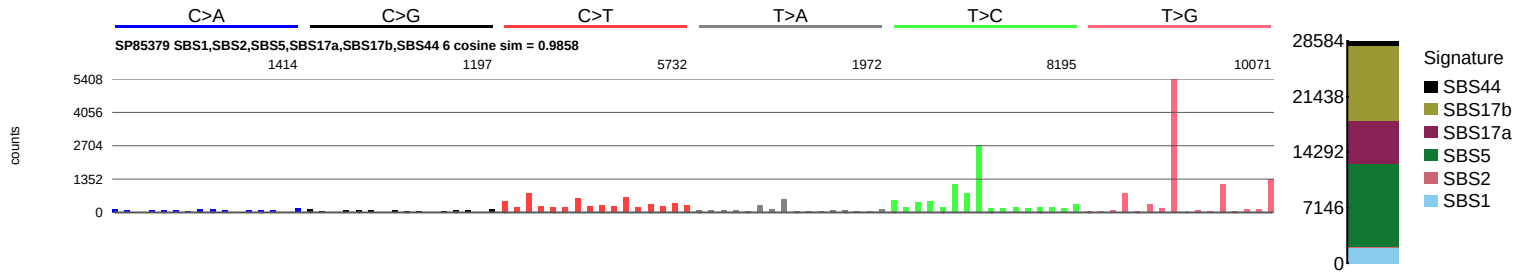
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.9858
1414
1197
5732
1972
8195
10071
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
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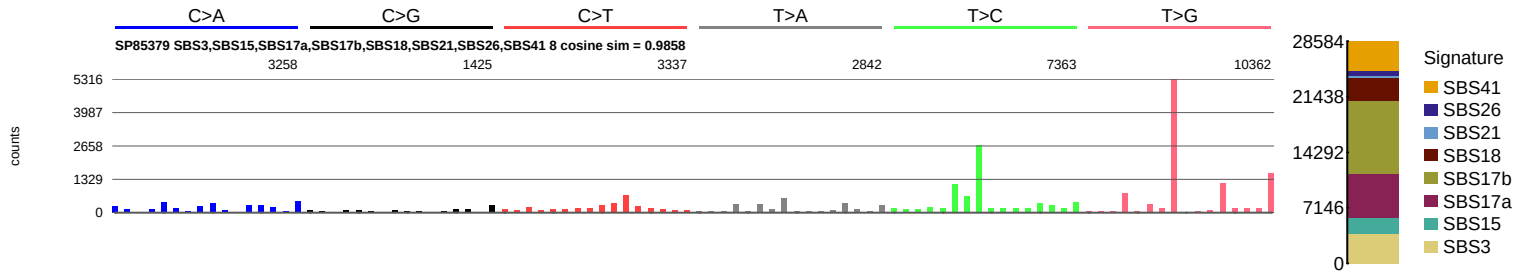
SBS44
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.9858
3258
1425
3337
2842
7363
10362
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
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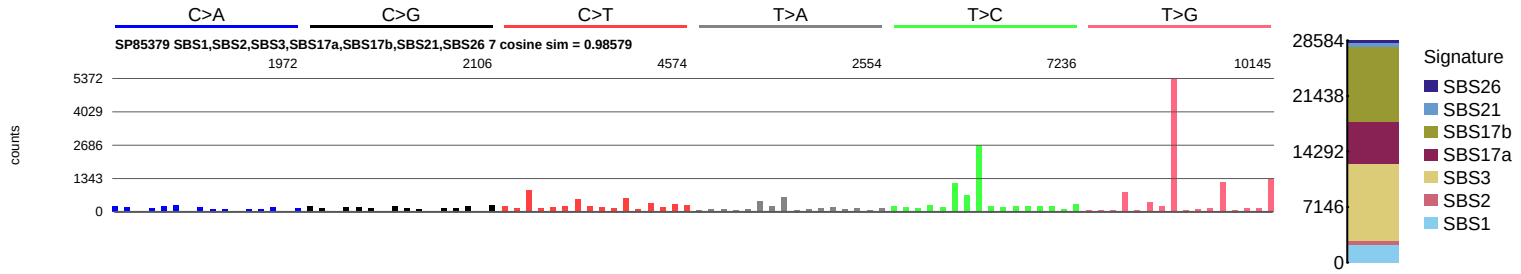
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98579
1972
2106
4574
2554
7236
10145
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
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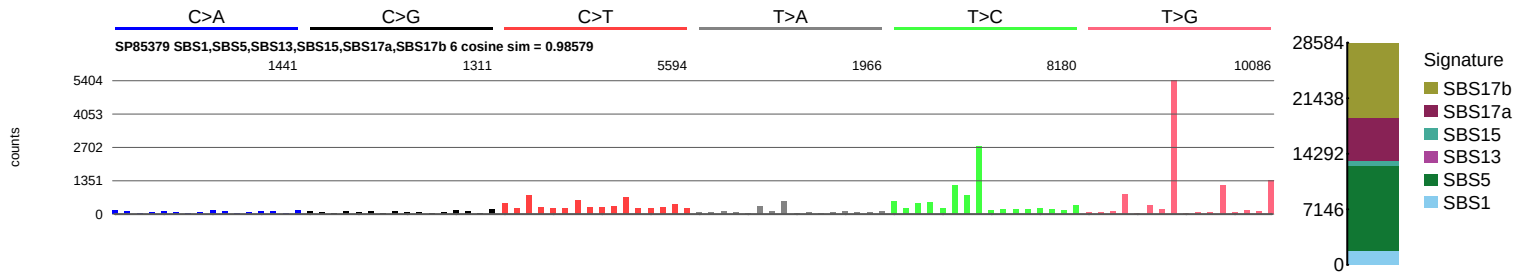
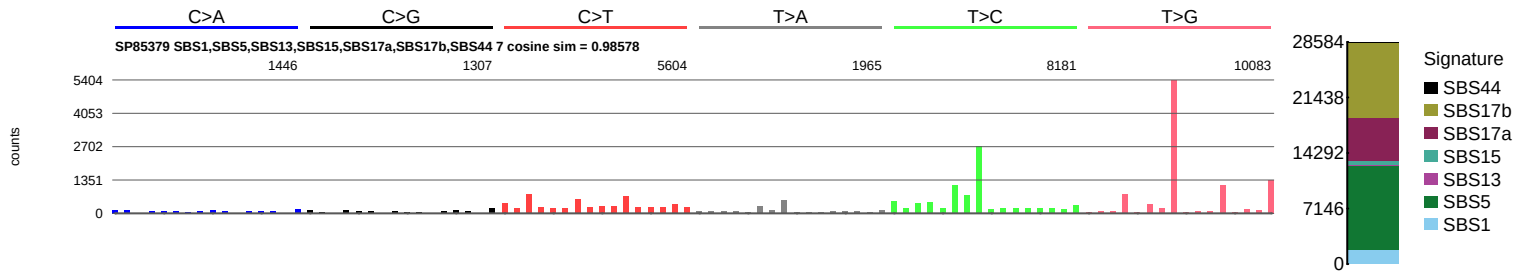
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98579
1441
1311
5594
1966
8180
10086
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98578
1446
1307
5604
1965
8181
10083
counts
5404
4053
2702
1351
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28584
21438
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7146
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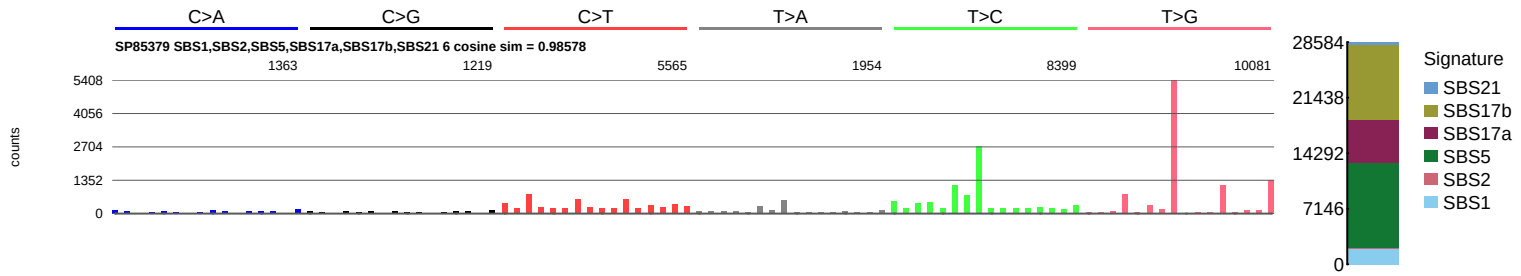
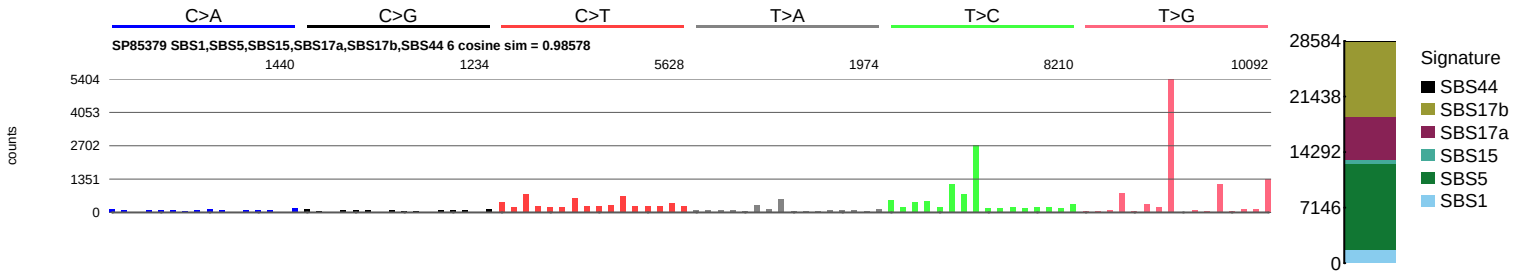
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98578
1363
1219
5565
1954
8399
10081
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
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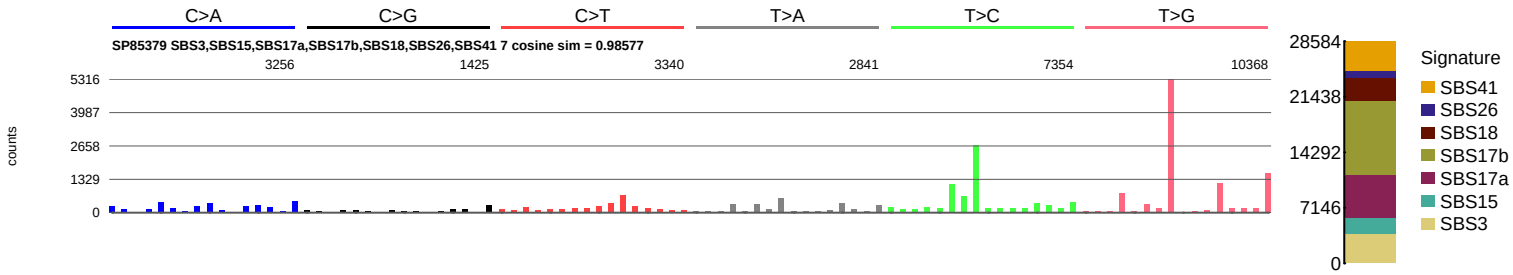
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98578
1440
1234
5628
1974
8210
10092
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS5
SBS1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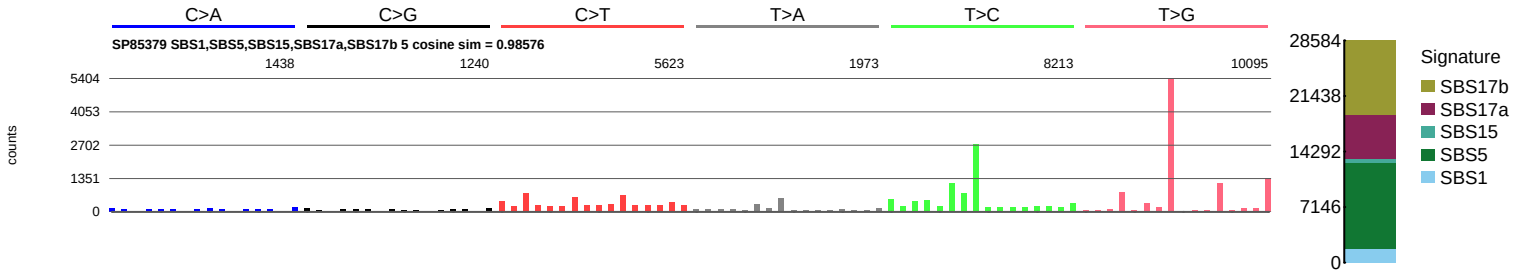
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98577
3256
1425
3340
2841
7354
10368
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3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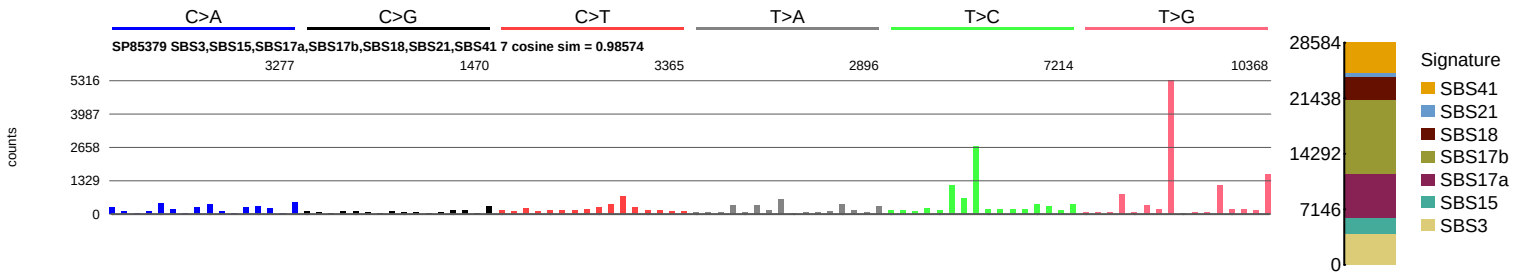
SP85379 SBS1,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98576
1438
1240
5623
1973
8213
10095
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
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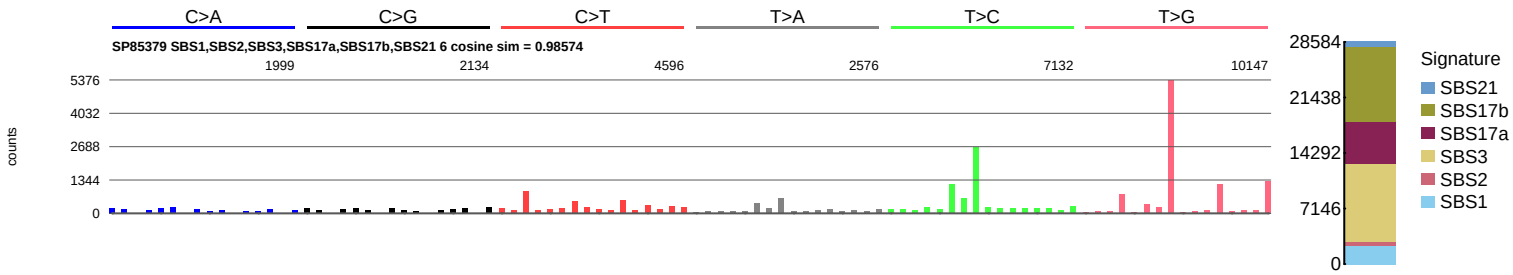
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98574
3277
1470
3365
2896
7214
10368
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
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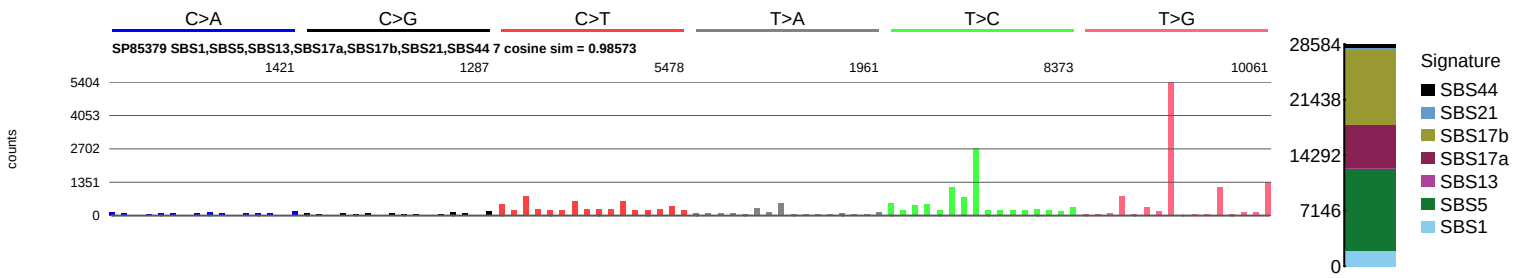
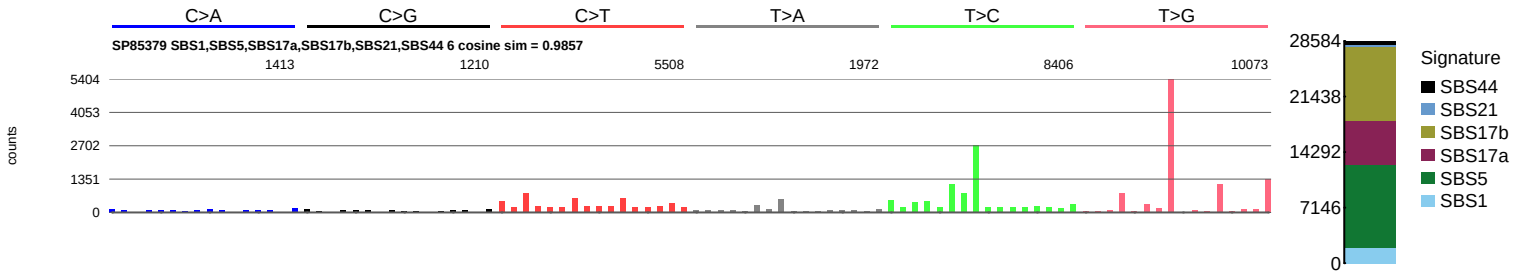
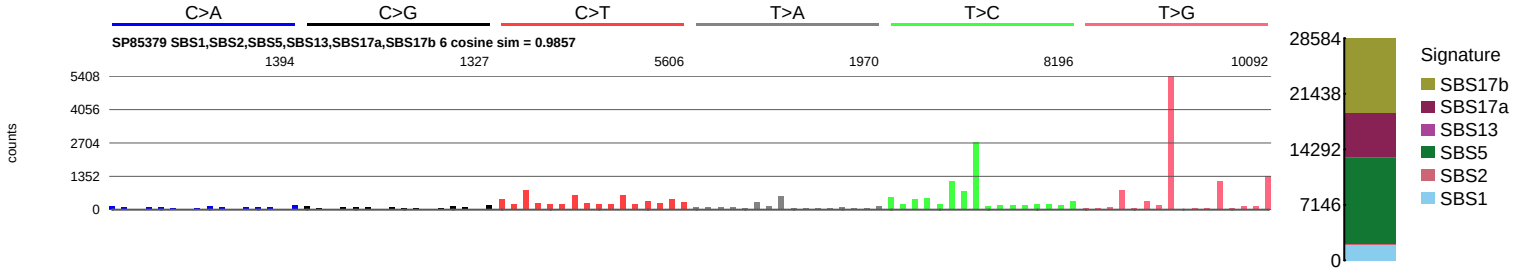
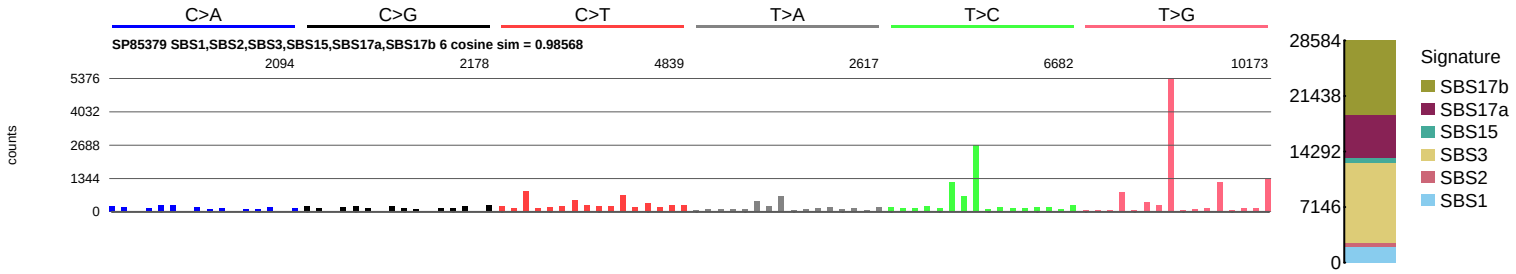
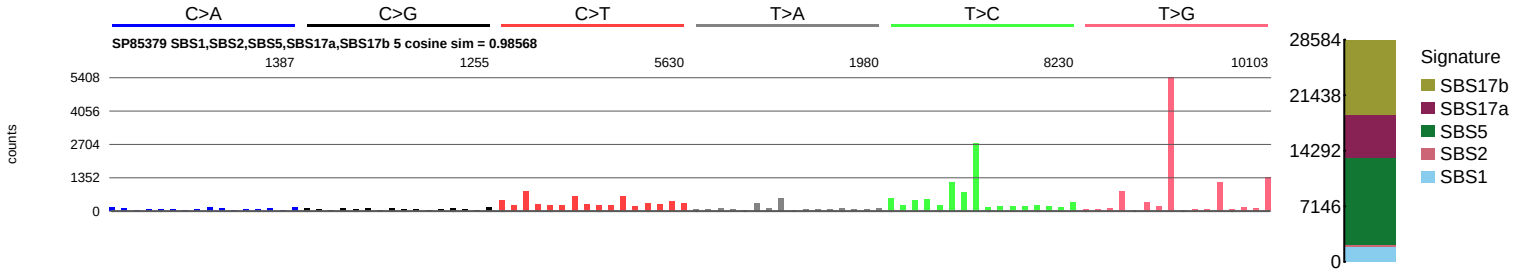
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98574
1999
2134
4596
2576
7132
10147
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

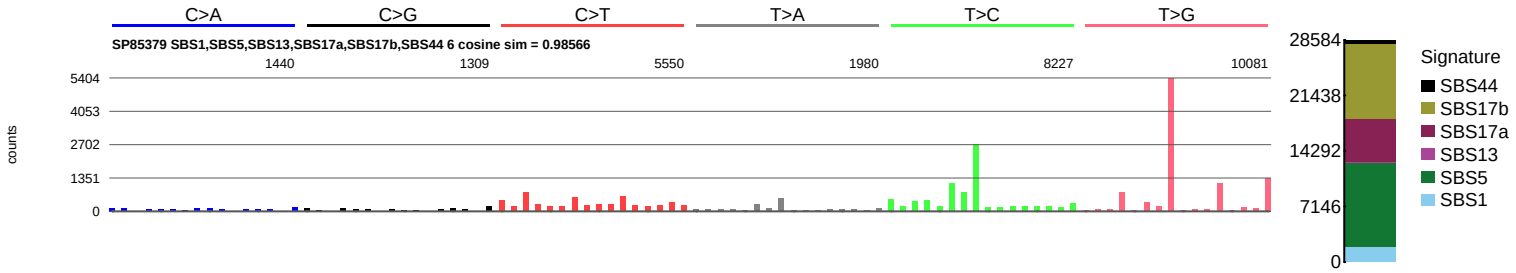

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98573
1421
1287
5478
1961
8373
10061
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

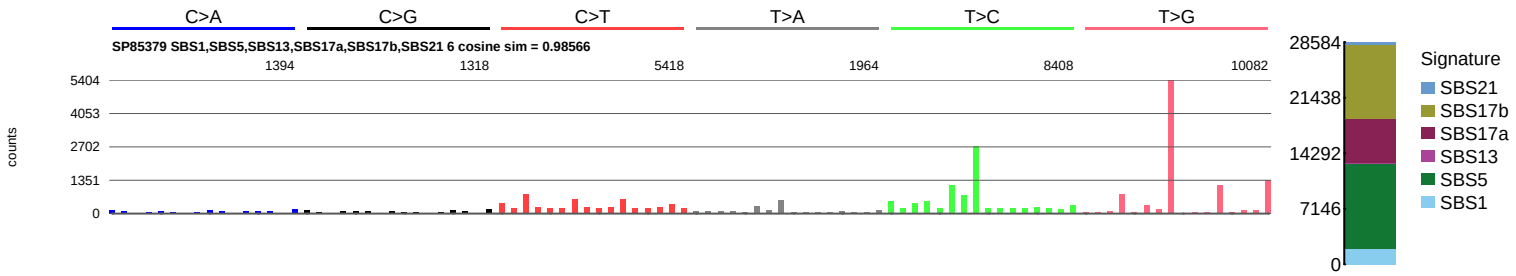

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.9857
1413
1210
5508
1972
8406
10073
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS1

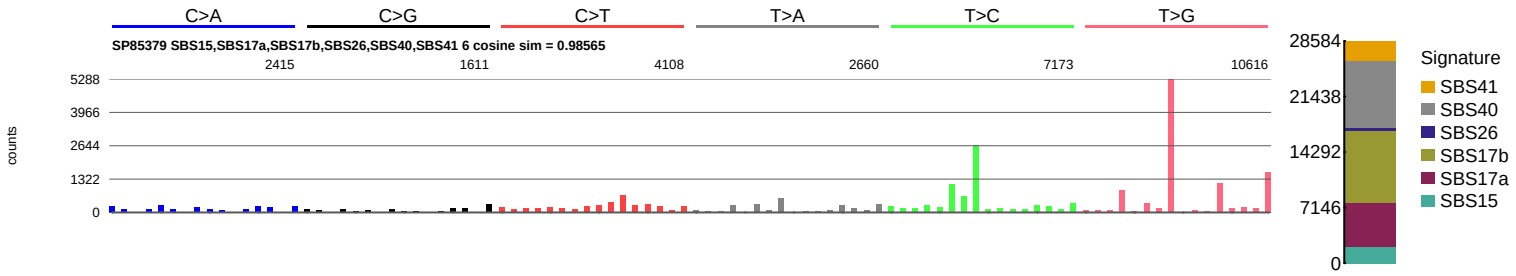

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.9857
1394
1327
5606
1970
8196
10092
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

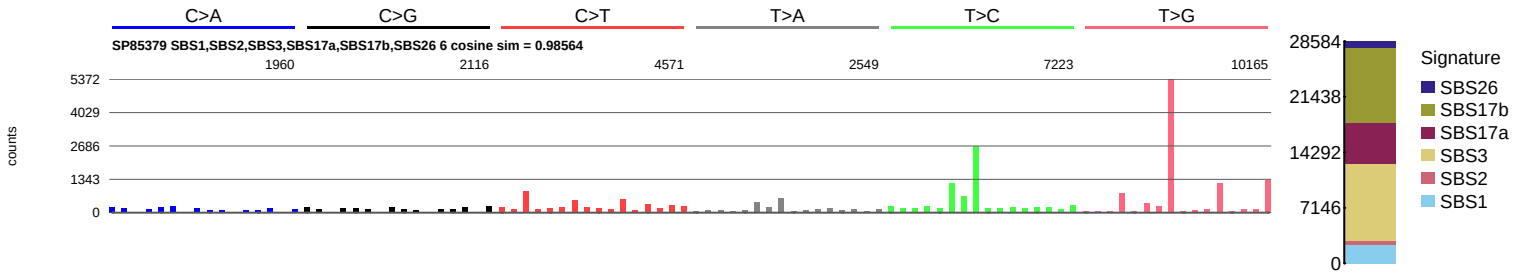

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98568
2094
2178
4839
2617
6682
10173
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

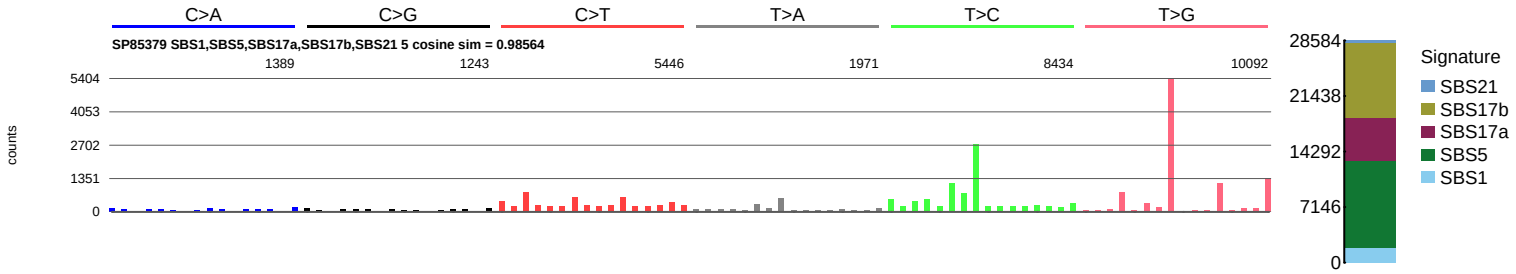

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98568
1387
1255
5630
1980
8230
10103
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS5
SBS2
SBS1

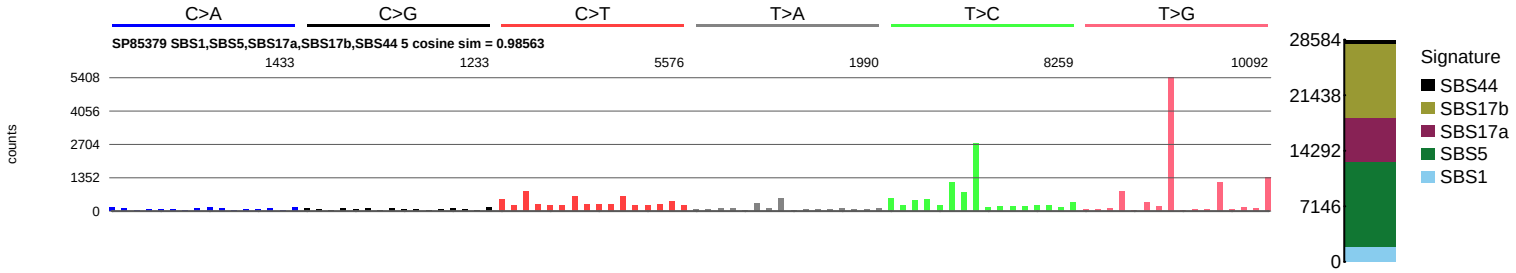

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98566
1440
1309
5550
1980
8227
10081
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS1

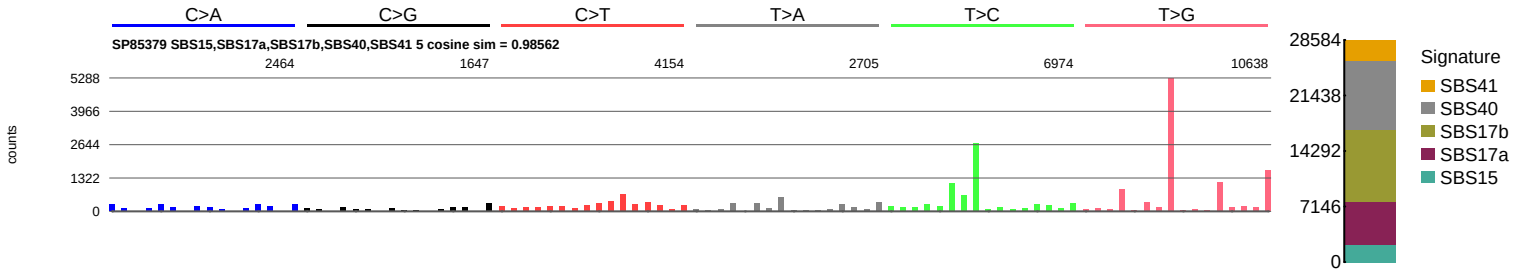

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98566
1394
1318
5418
1964
8408
10082
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98565
2415
1611
4108
2660
7173
10616
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98564
1960
2116
4571
2549
7223
10165
counts
5372
4029
2686
1343
0
28584
21438
14292
7146
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98564
1389
1243
5446
1971
8434
10092
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98563
1433
1233
5576
1990
8259
10092
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS1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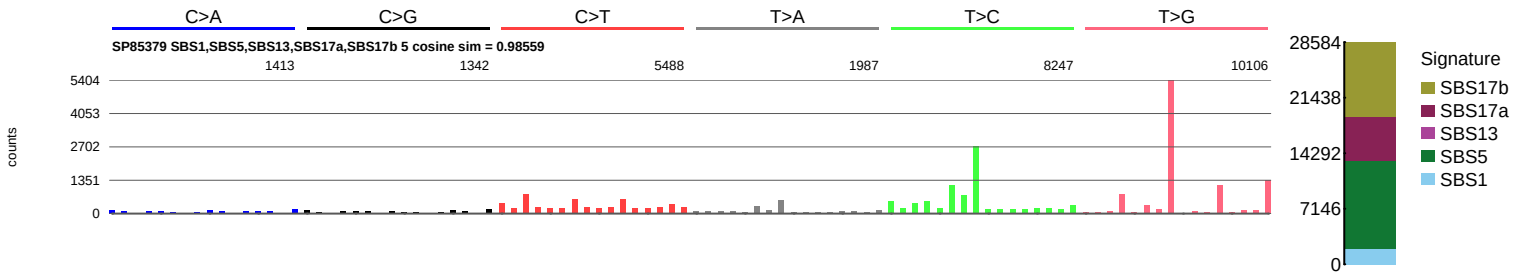
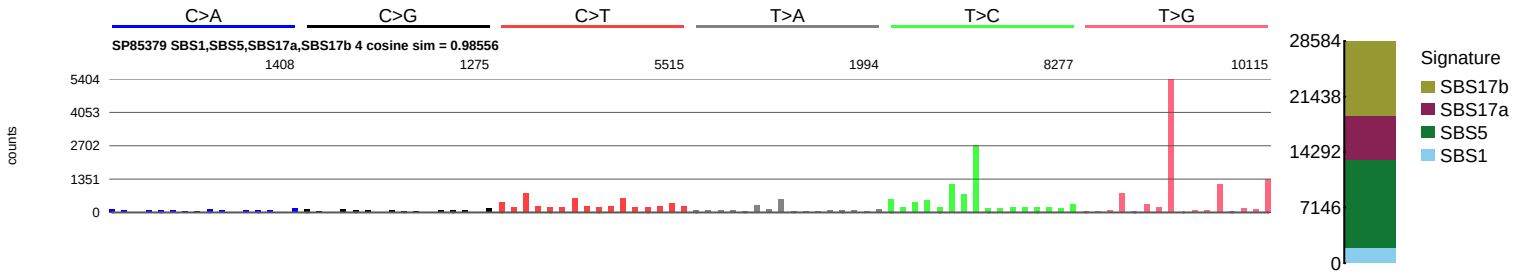
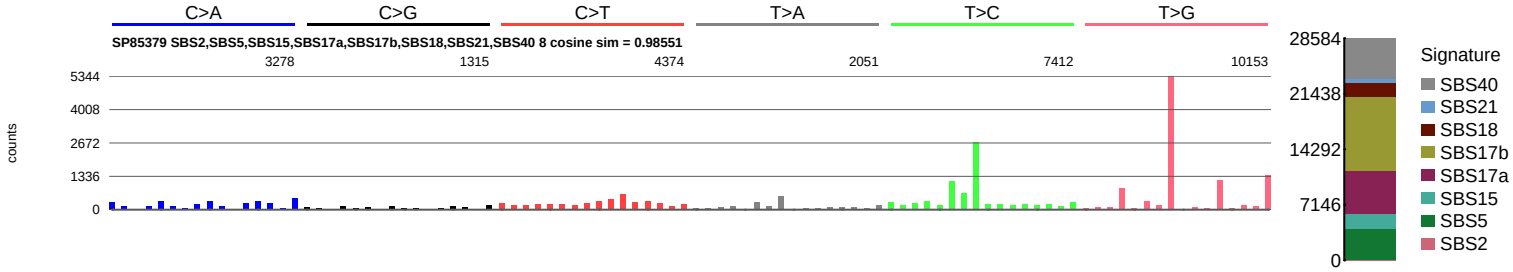
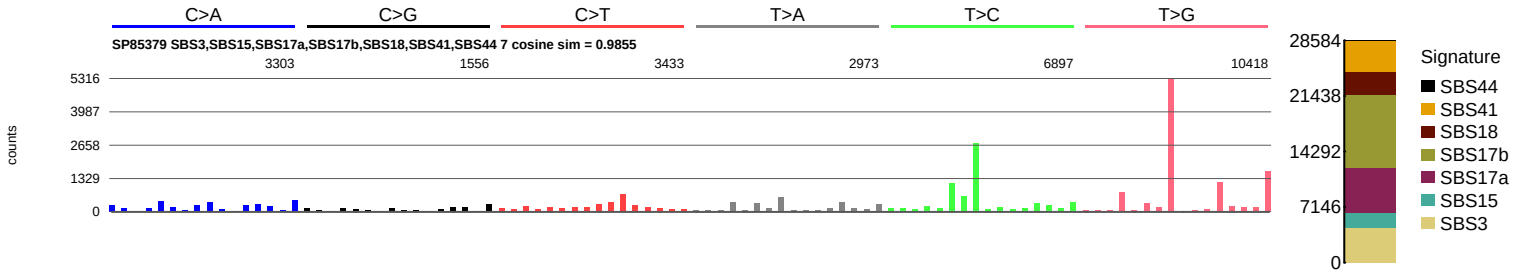
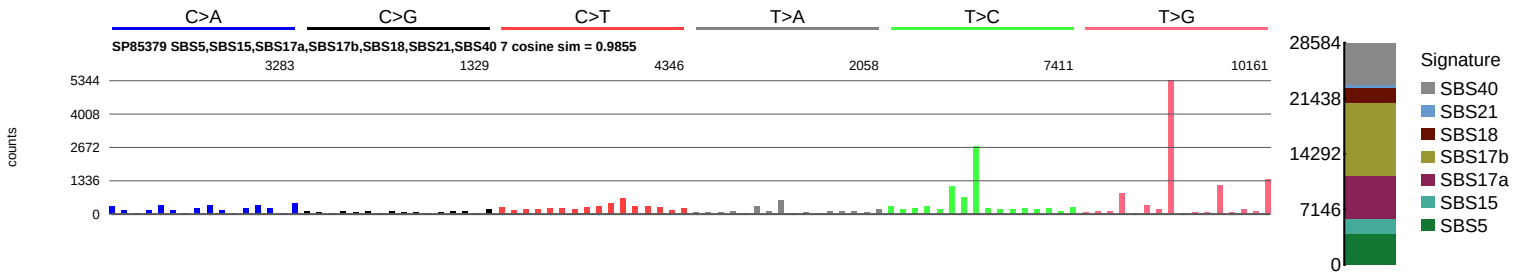
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98562
2464
1647
4154
2705
6974
10638
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15

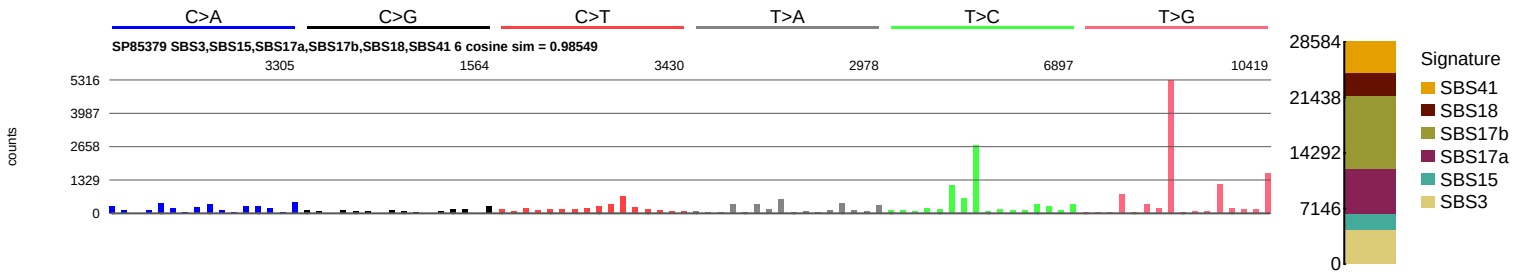

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98559
1413
1342
5488
1987
8247
10106
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS1

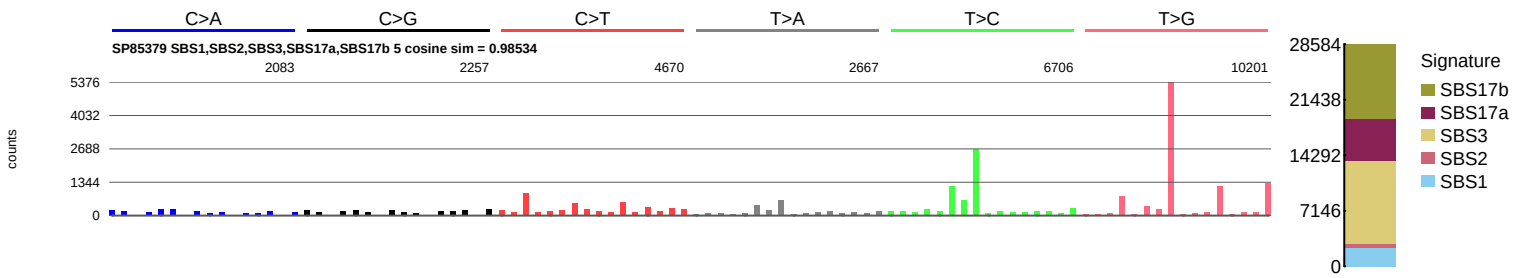

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS5,SBS17a,SBS17b 4 cosine sim = 0.98556
1408
1275
5515
1994
8277
10115
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS5
SBS1

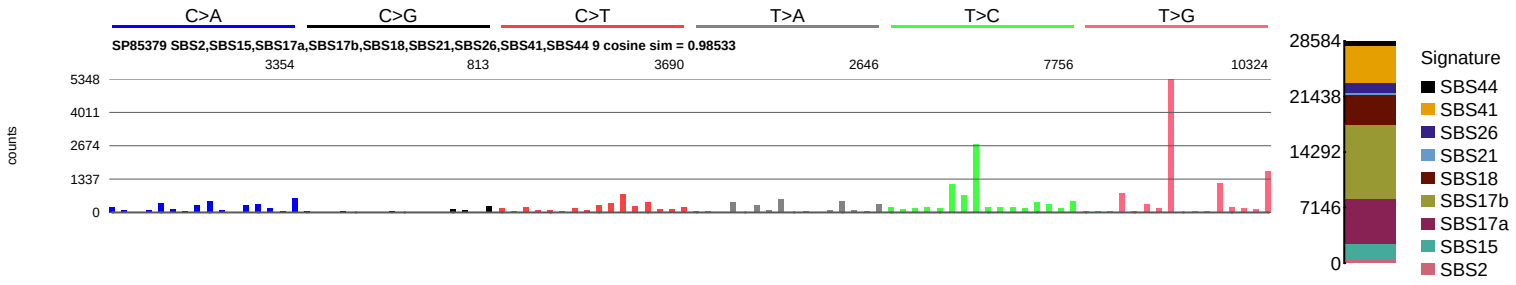

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98551
3278
1315
4374
2051
7412
10153
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.9855
3303
1556
3433
2973
6897
10418
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.9855
3283
1329
4346
2058
7411
10161
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98549
3305
1564
3430
2978
6897
10419
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS3,SBS17a,SBS17b 5 cosine sim = 0.98534
2083
2257
4670
2667
6706
10201
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98533
3354
813
3690
2646
7756
10324
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
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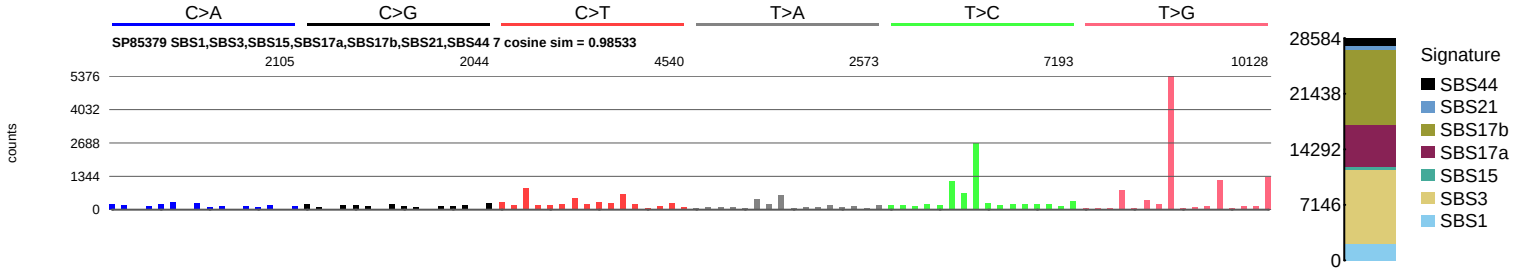
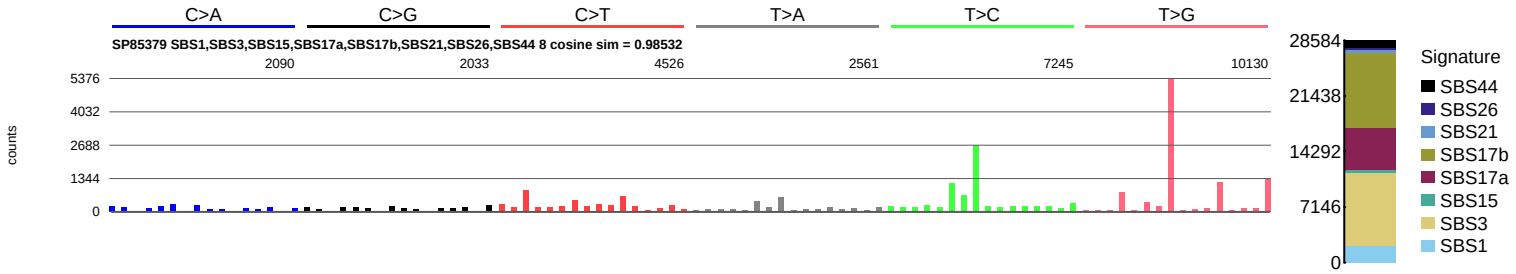
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98533
2105
2044
4540
2573
7193
10128
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1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SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98532
2090
2033
4526
2561
7245
10130
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS17b
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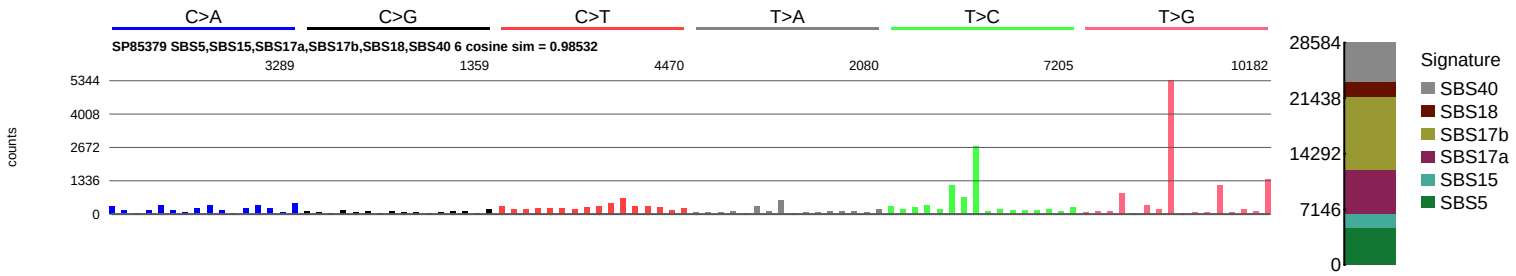
SBS15
SBS3
SBS1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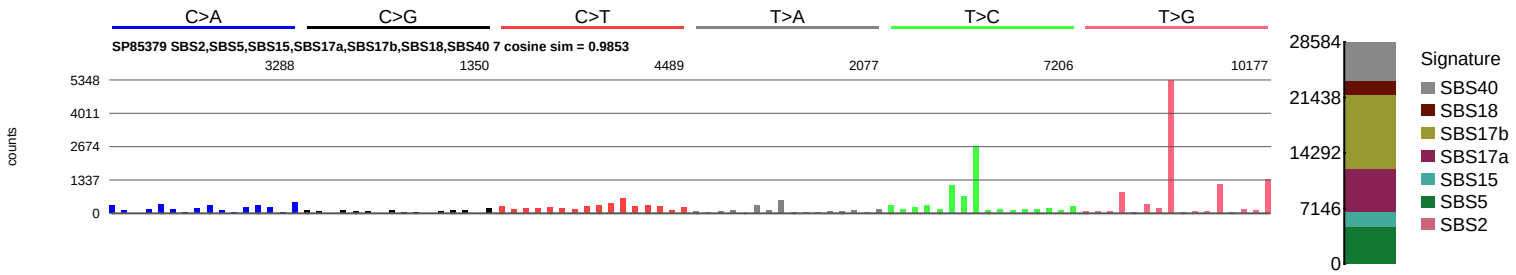
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98532
3289
1359
4470
2080
7205
10182
counts
5344
4008
2672
1336
0
28584
21438
14292
7146
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5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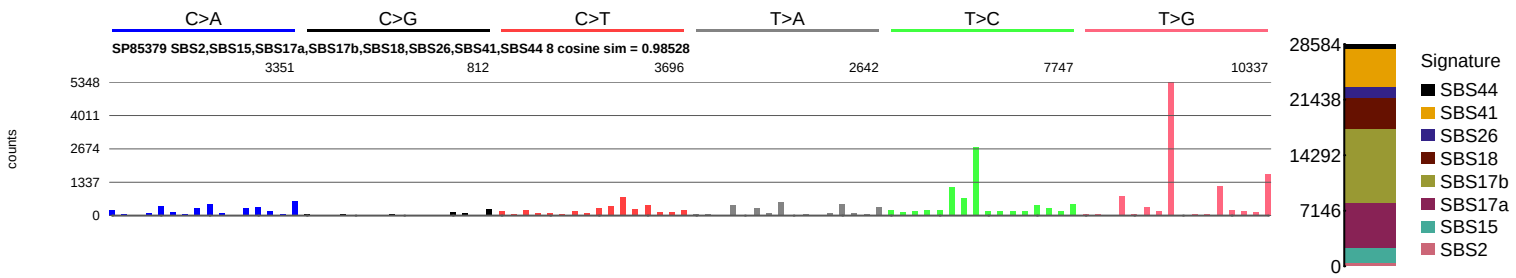
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.9853
3288
1350
4489
2077
7206
10177
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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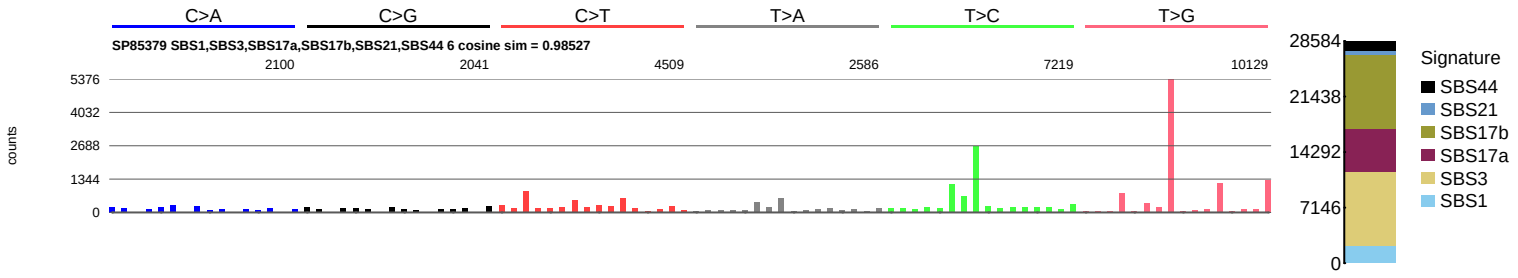
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98528
3351
812
3696
2642
7747
10337
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98527
2100
2041
4509
2586
7219
10129
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
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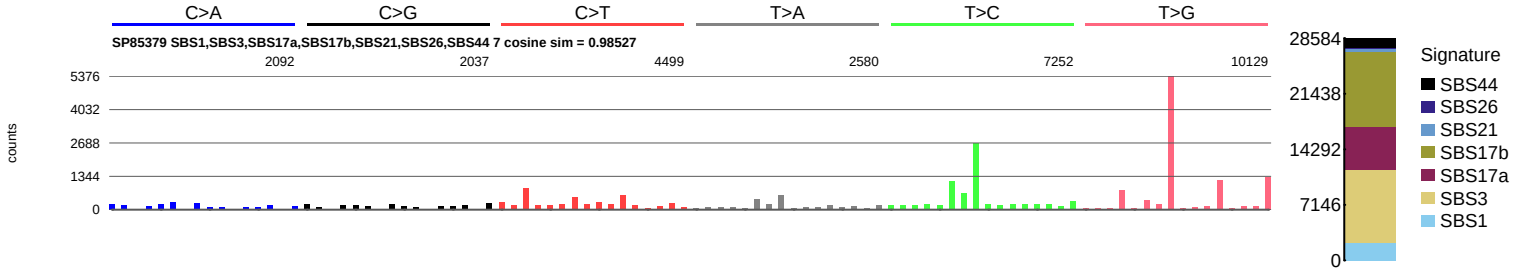
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS1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C>G
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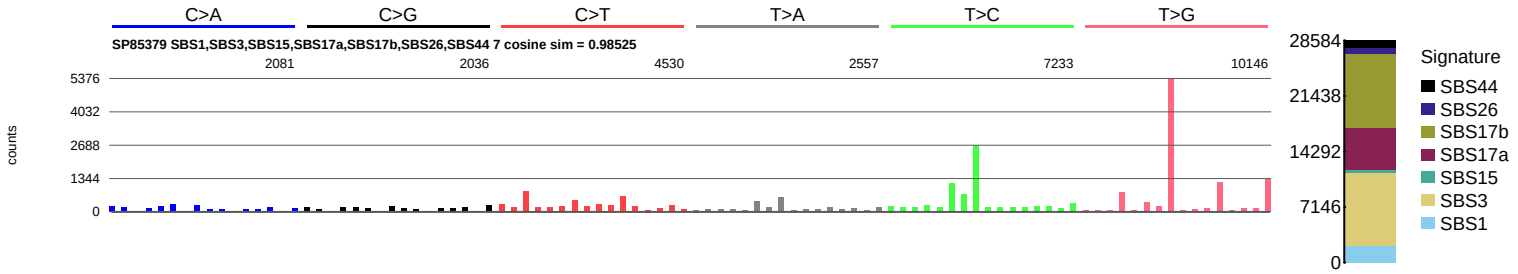
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS44 7 cosine sim = 0.98527
2092
2037
4499
2580
7252
10129
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1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C>G
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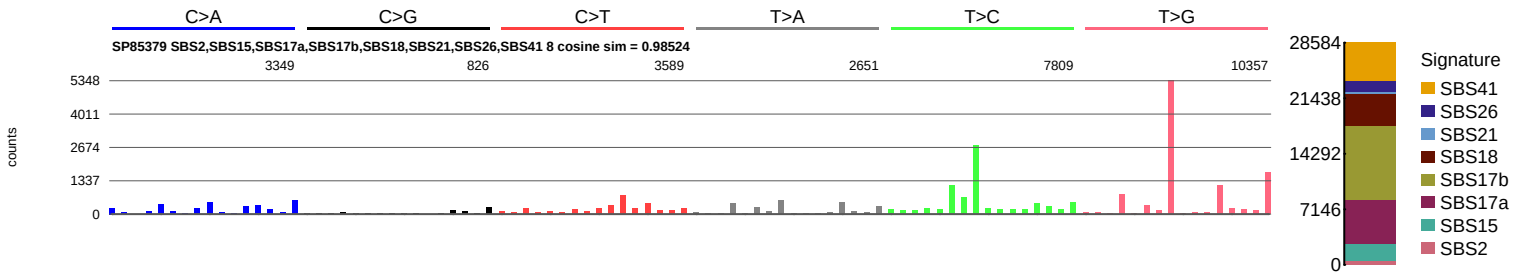
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98525
2081
2036
4530
2557
7233
10146
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
C>T
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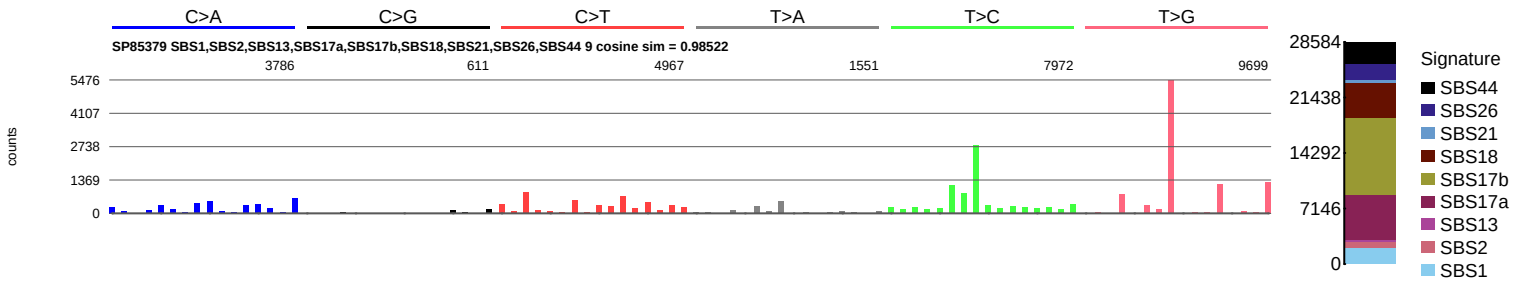
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98524
3349
826
3589
2651
7809
10357
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2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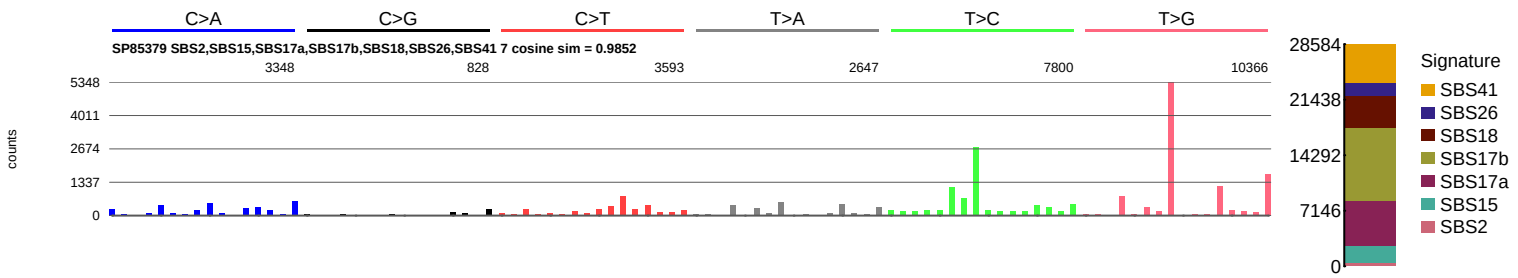
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98522
3786
611
4967
1551
7972
9699
counts
5476
4107
2738
1369
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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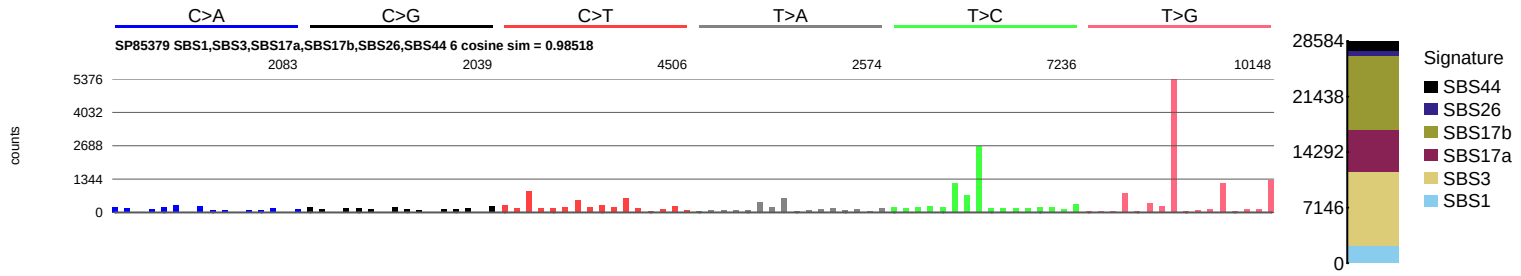
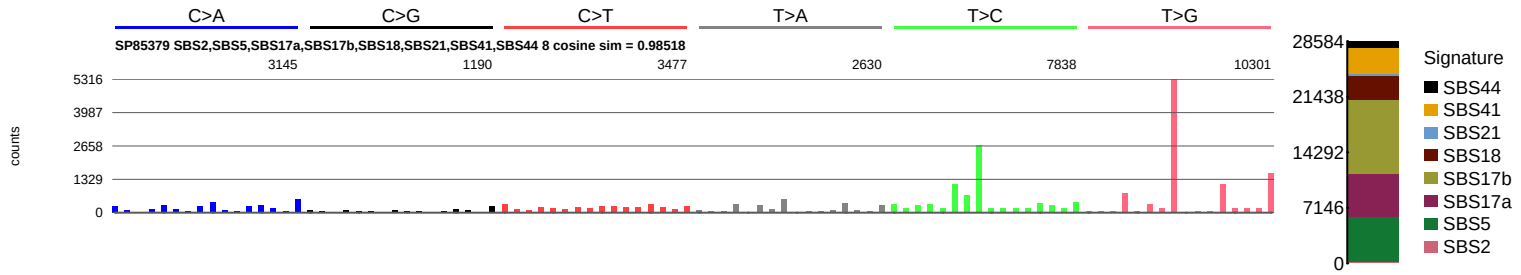
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9852
3348
828
3593
2647
7800
10366
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98518
2083
2039
4506
2574
7236
10148
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98518
3145
1190
3477
2630
7838
10301
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
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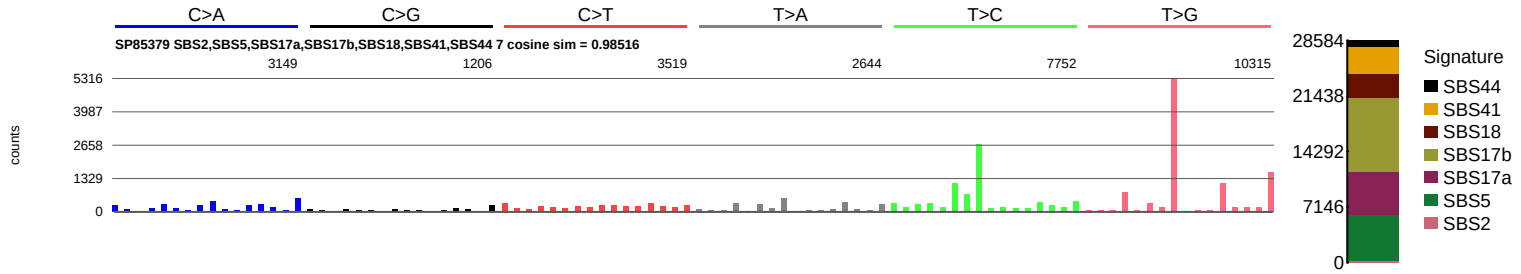
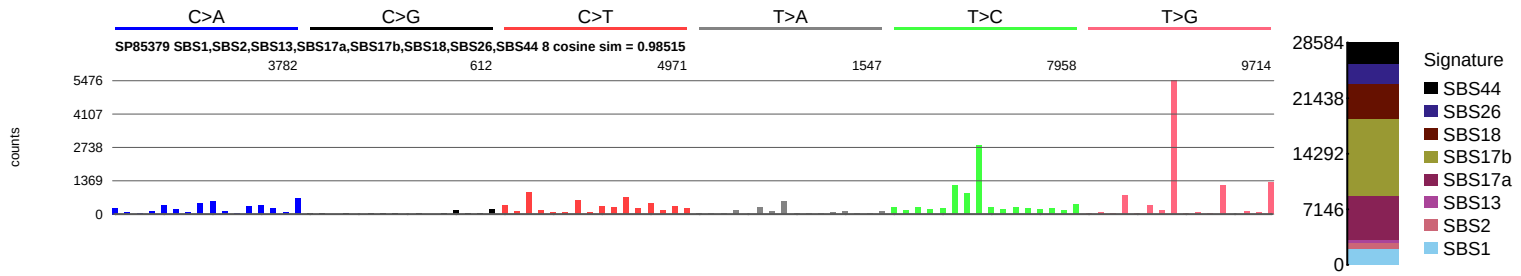
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98516
3149
1206
3519
2644
7752
10315
counts
5316
3987
2658
1329
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2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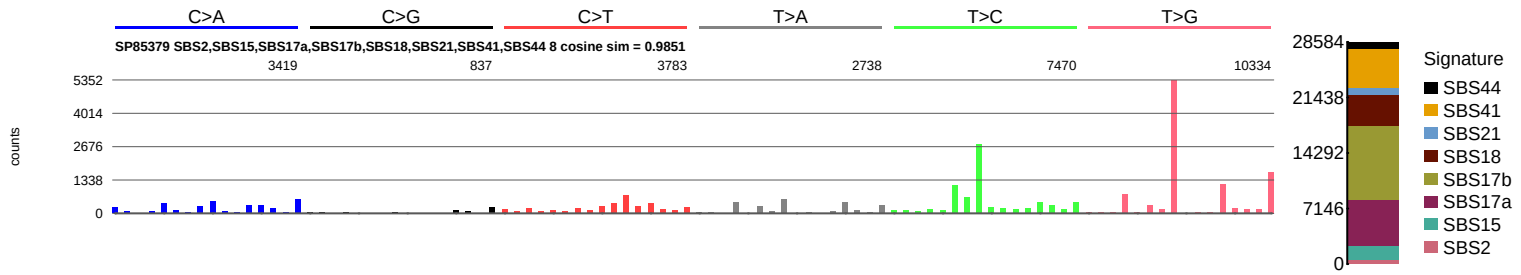
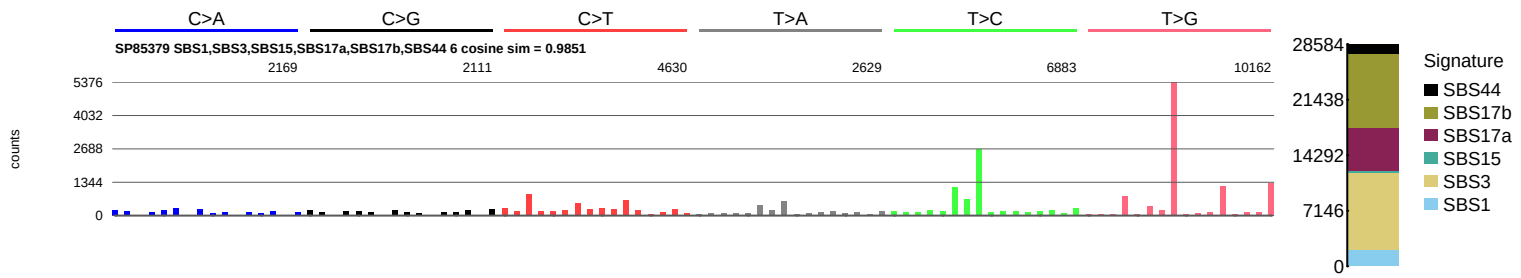
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98515
3782
612
4971
1547
7958
9714
counts
5476
4107
2738
1369
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9851
3419
837
3783
2738
7470
10334
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.9851
2169
2111
4630
2629
6883
10162
counts
5376
4032
2688
1344
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28584
21438
14292
7146
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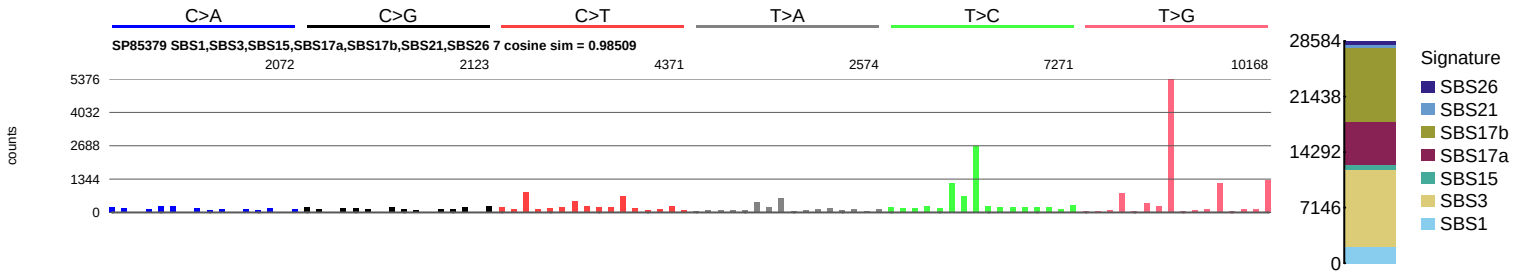
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS1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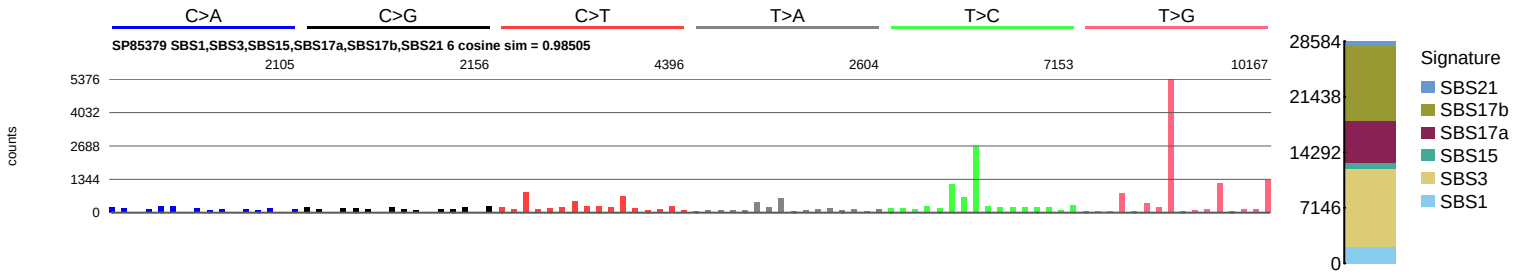
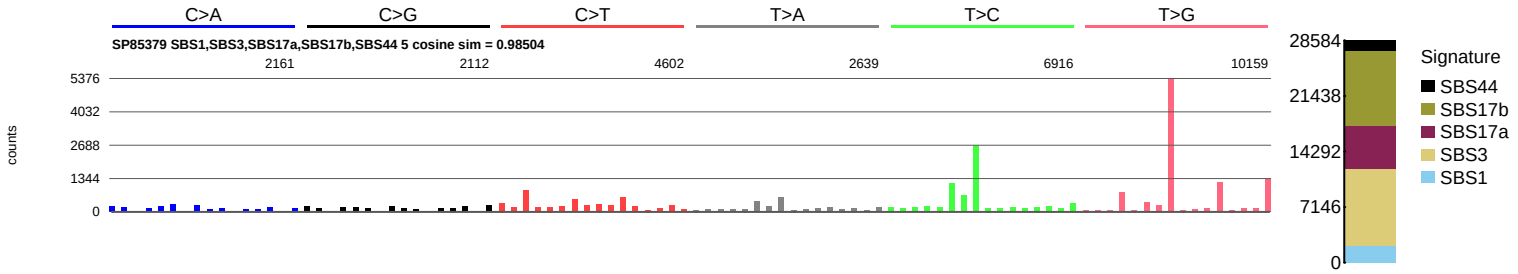
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98509
2072
2123
4371
2574
7271
10168
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98505
2105
2156
4396
2604
7153
10167
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1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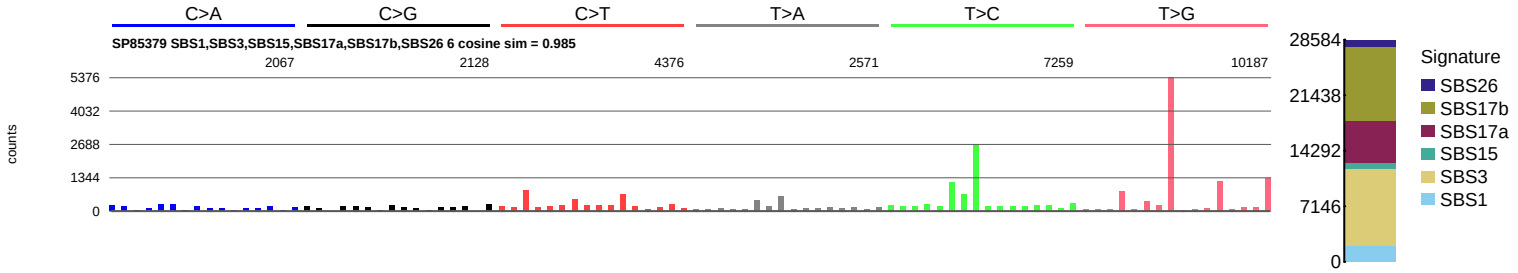
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98504
2161
2112
4602
2639
6916
10159
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.985
2067
2128
4376
2571
7259
10187
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
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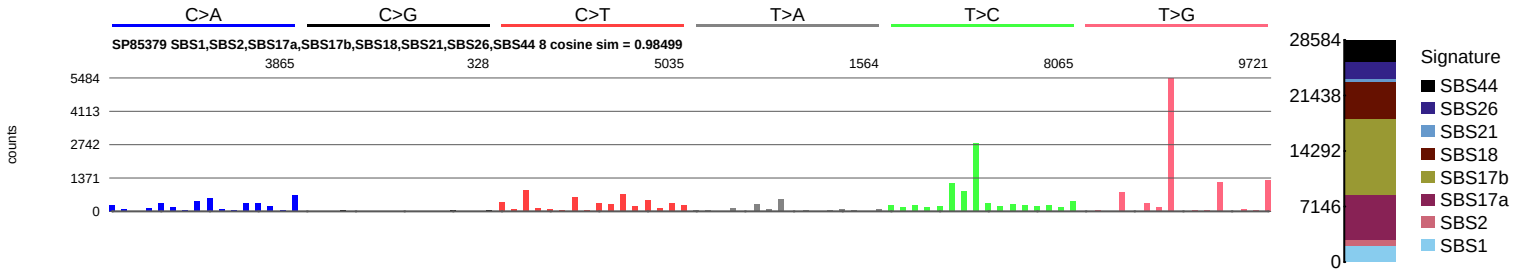
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1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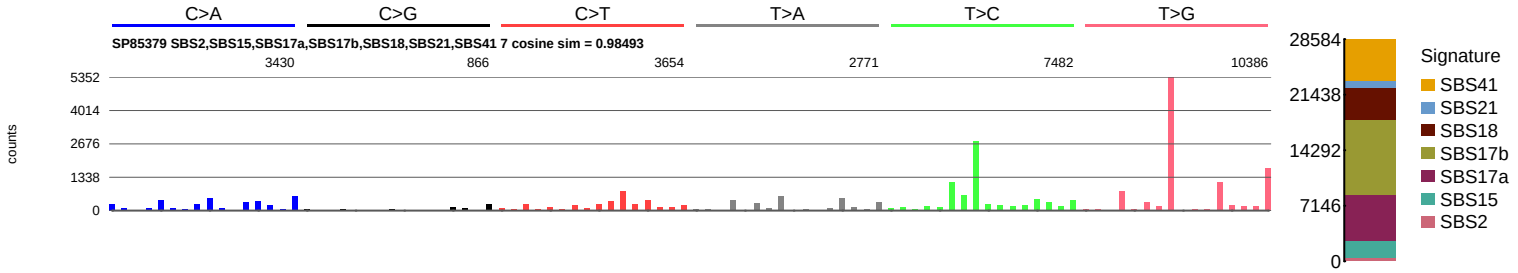
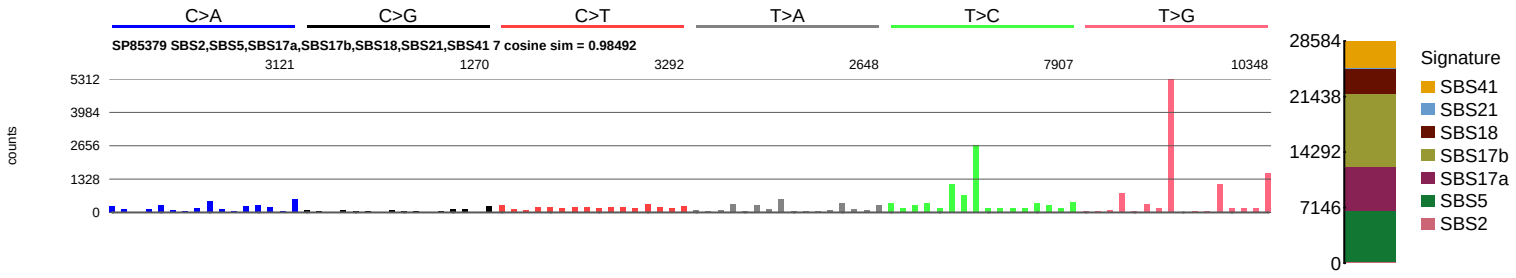
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98499
3865
328
5035
1564
8065
9721
counts
5484
4113
2742
1371
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98493
3430
866
3654
2771
7482
10386
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98492
3121
1270
3292
2648
7907
10348
counts
5312
3984
2656
1328
0
28584
21438
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7146
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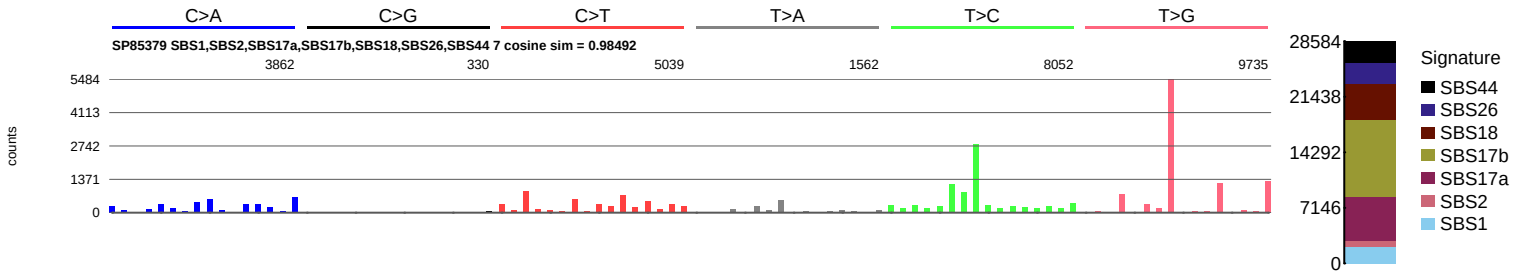
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98492
3862
330
5039
1562
8052
9735
counts
5484
4113
2742
1371
0
28584
21438
14292
7146
0
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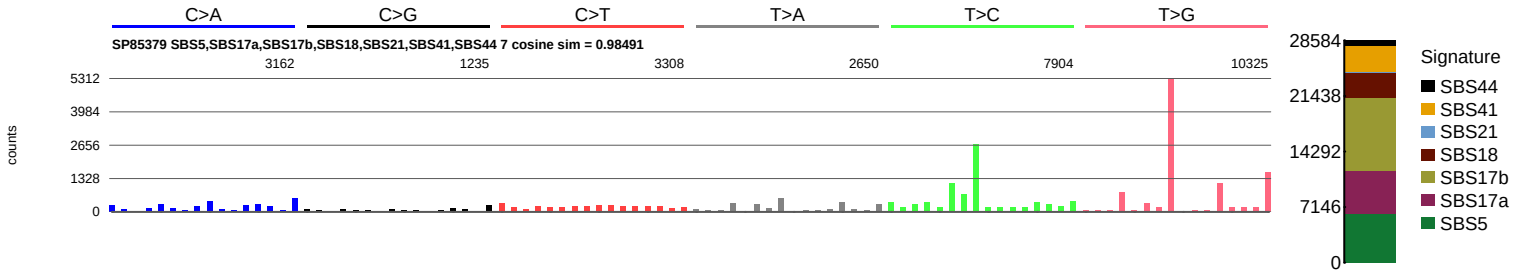
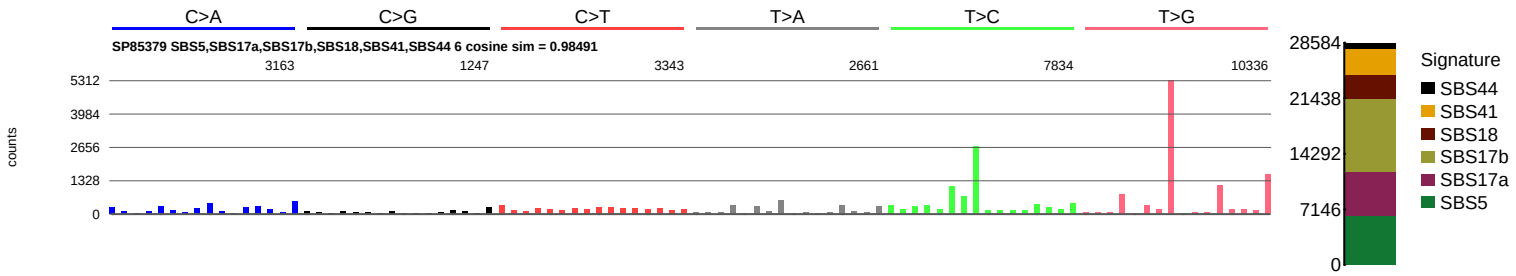
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98491
3162
1235
3308
2650
7904
10325
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
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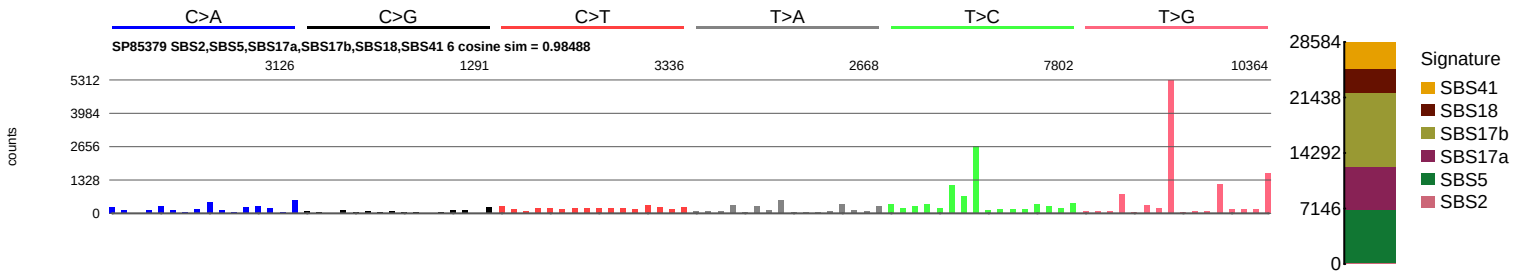
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98491
3163
1247
3343
2661
7834
10336
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5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C>G
C>T
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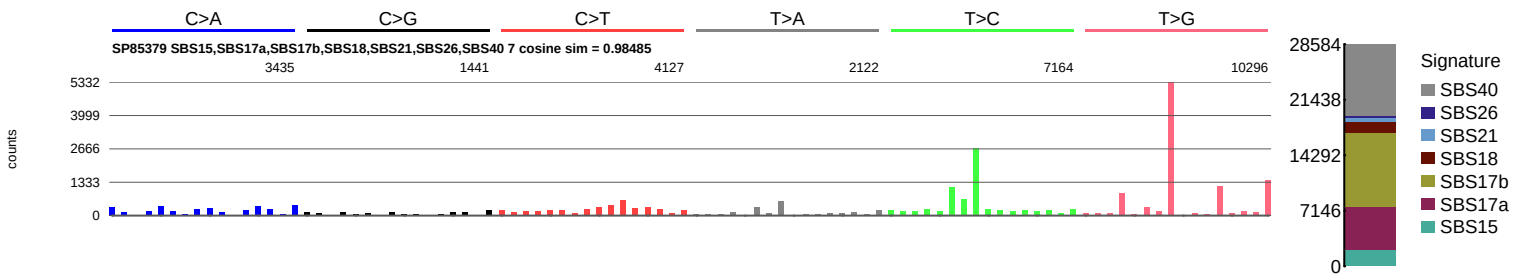
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98488
3126
1291
3336
2668
7802
10364
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
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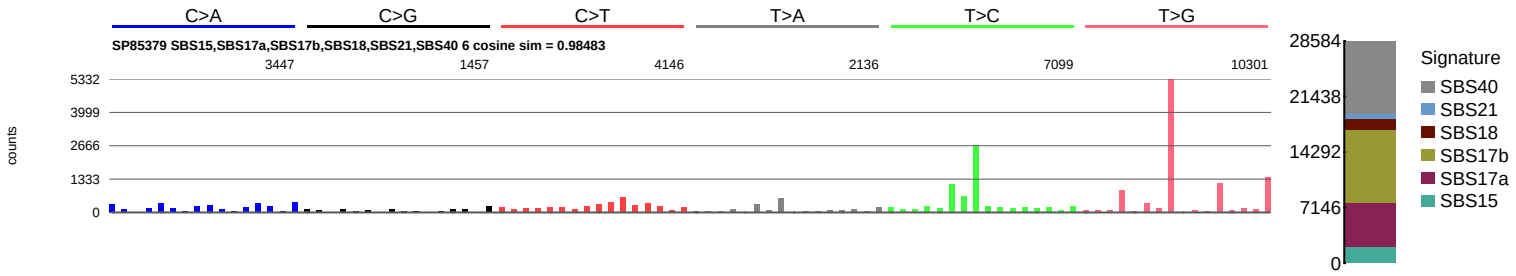
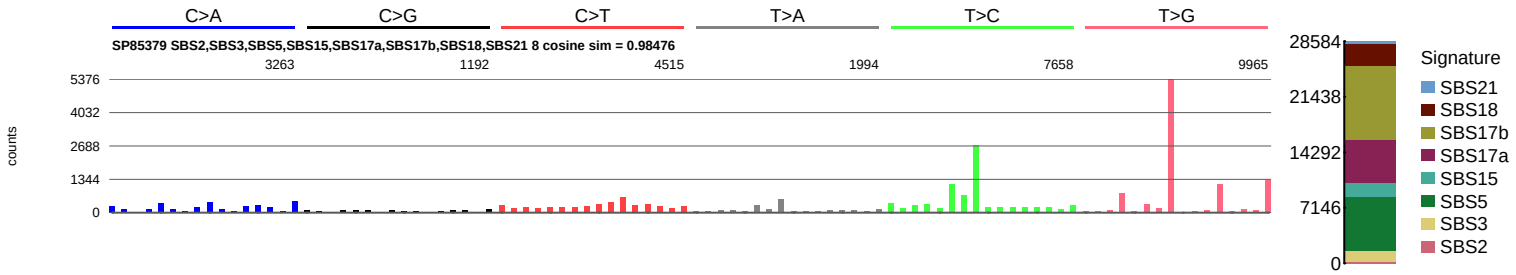
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98485
3435
1441
4127
2122
7164
10296
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98483
3447
1457
4146
2136
7099
10301
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15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C>G
C>T
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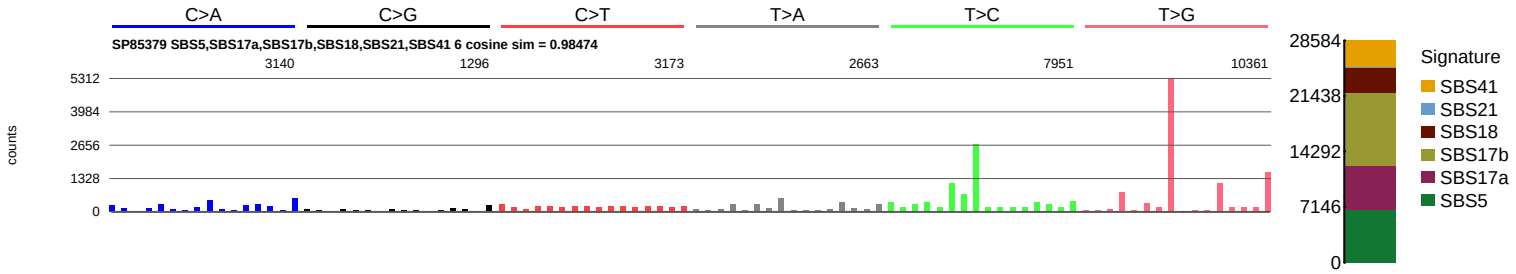
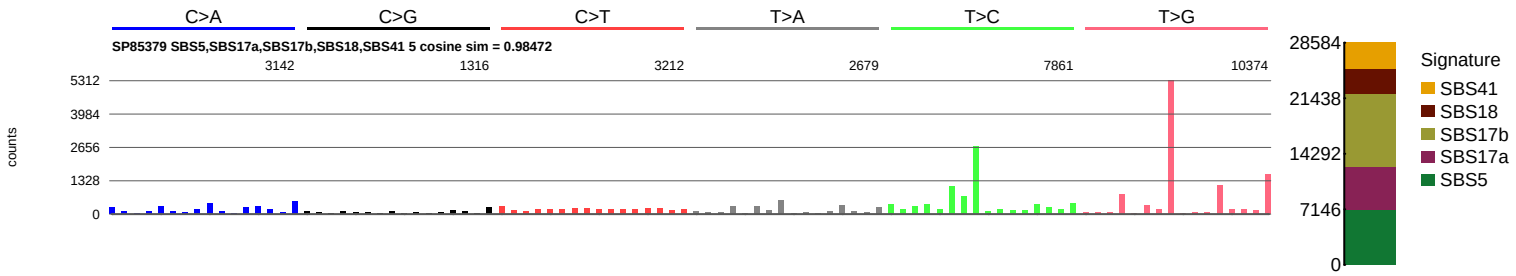
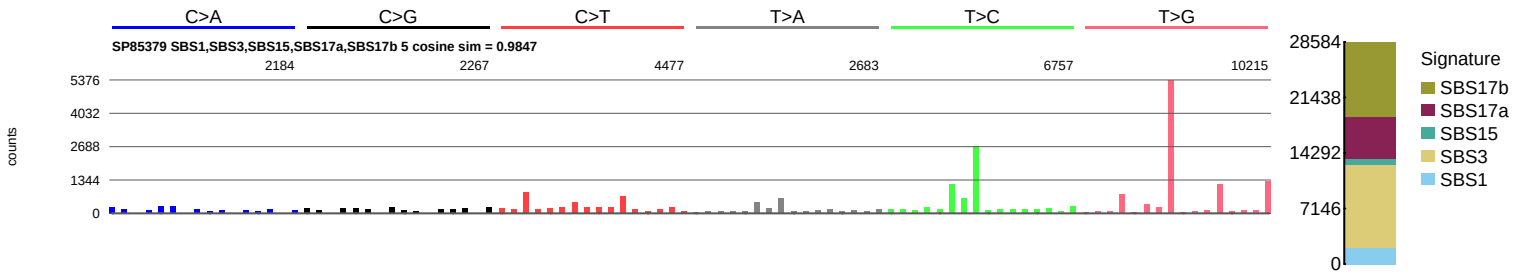
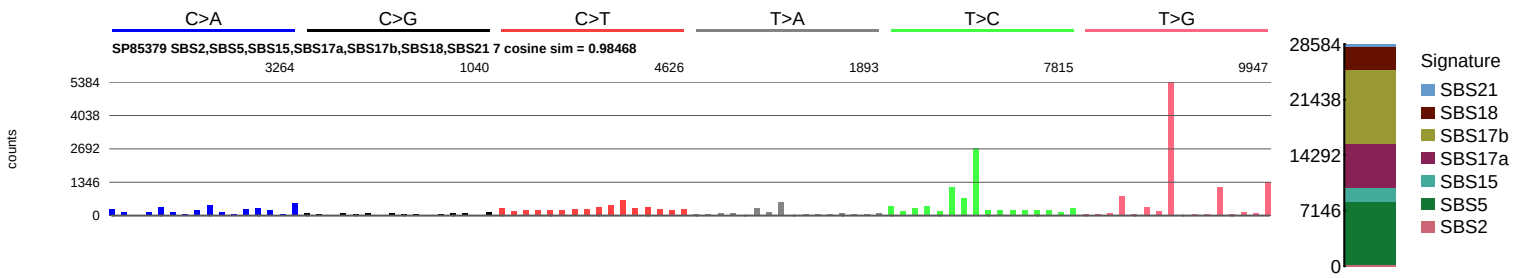
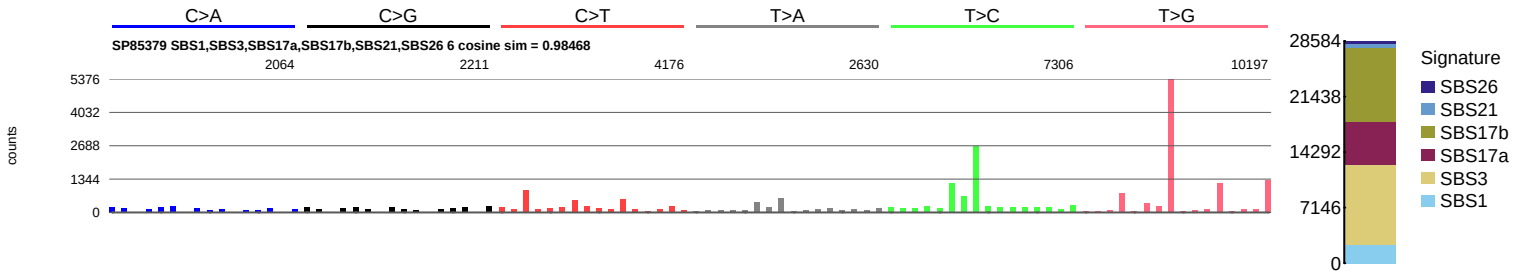
T>G
SP85379 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98476
3263
1192
4515
1994
7658
9965
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

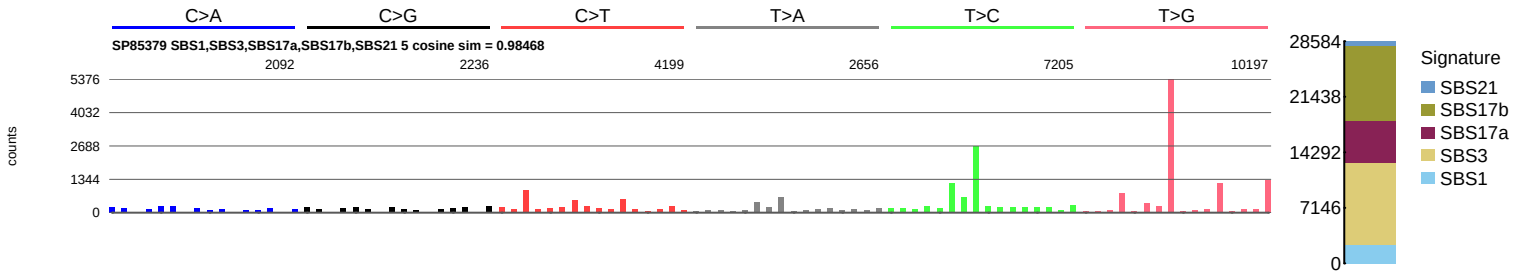

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98474
3140
1296
3173
2663
7951
10361
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5

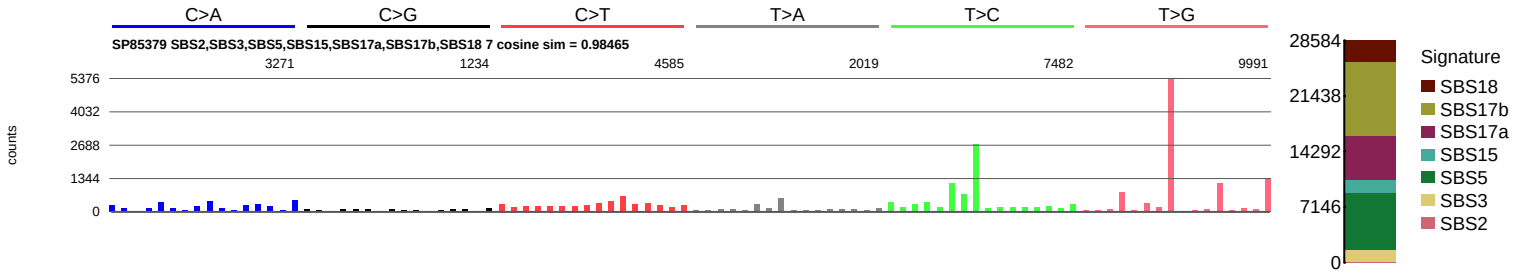

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98472
3142
1316
3212
2679
7861
10374
counts
5312
3984
2656
1328
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5

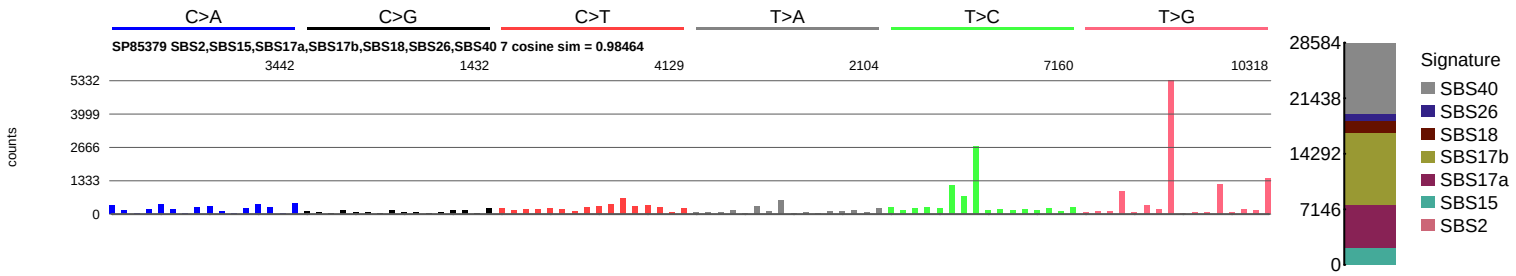

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS15,SBS17a,SBS17b 5 cosine sim = 0.9847
2184
2267
4477
2683
6757
10215
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS1

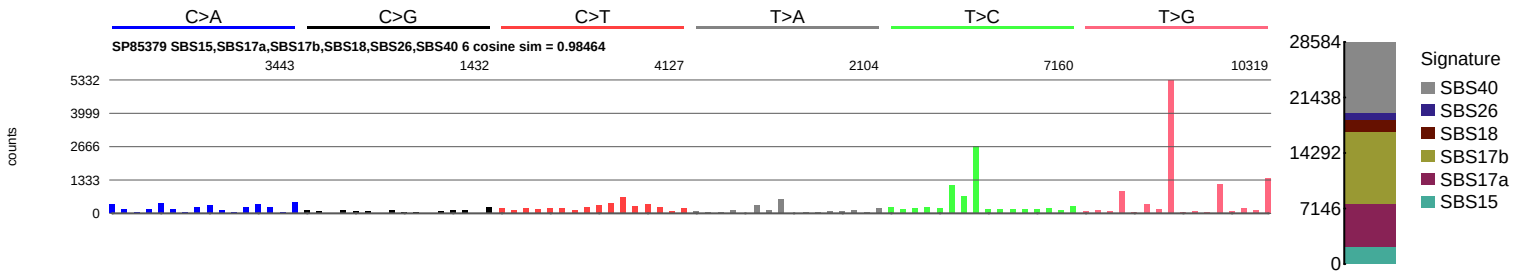

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98468
3264
1040
4626
1893
7815
9947
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

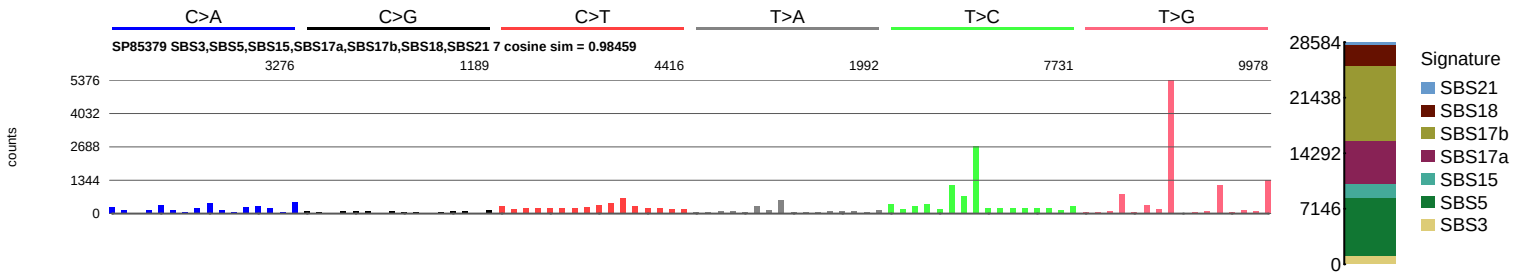

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26 6 cosine sim = 0.98468
2064
2211
4176
2630
7306
10197
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

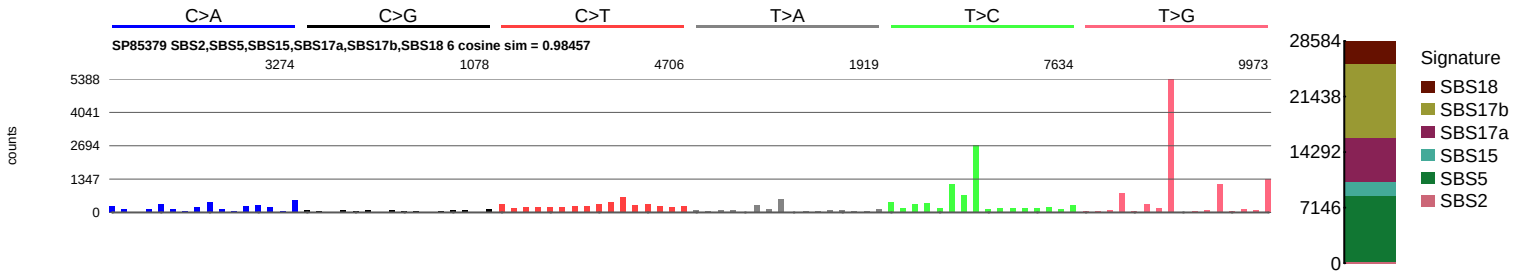

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98468
2092
2236
4199
2656
7205
10197
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98465
3271
1234
4585
2019
7482
9991
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98464
3442
1432
4129
2104
7160
10318
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98464
3443
1432
4127
2104
7160
10319
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98459
3276
1189
4416
1992
7731
9978
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3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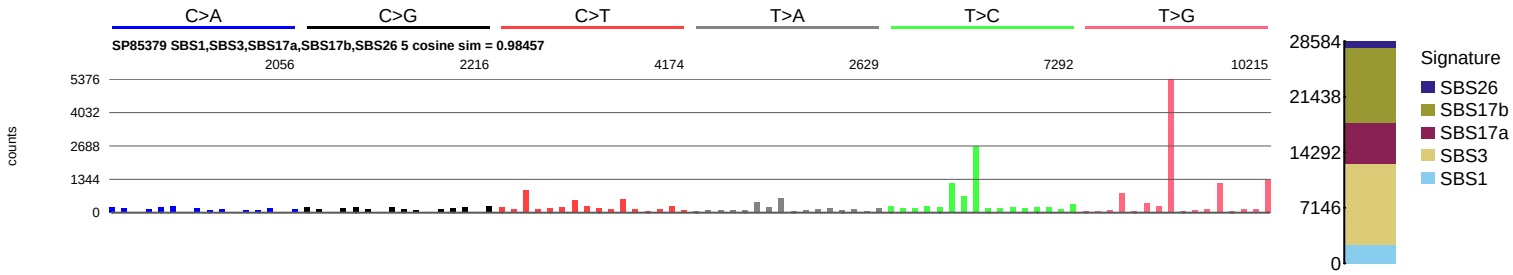
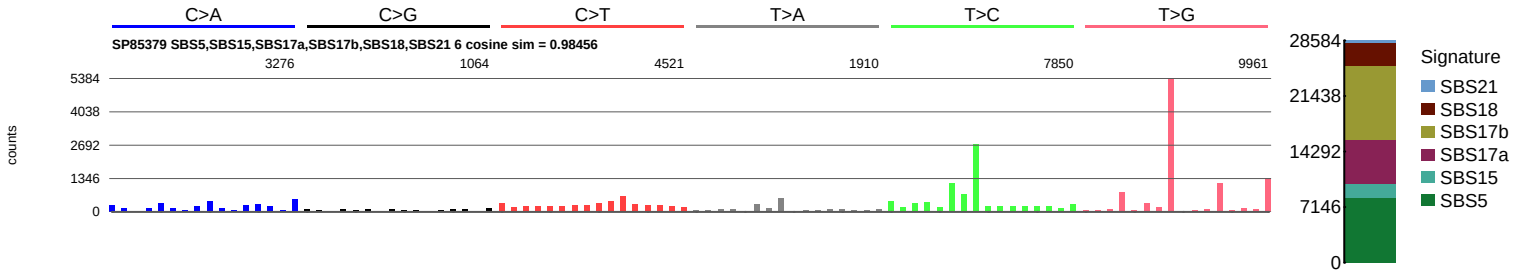
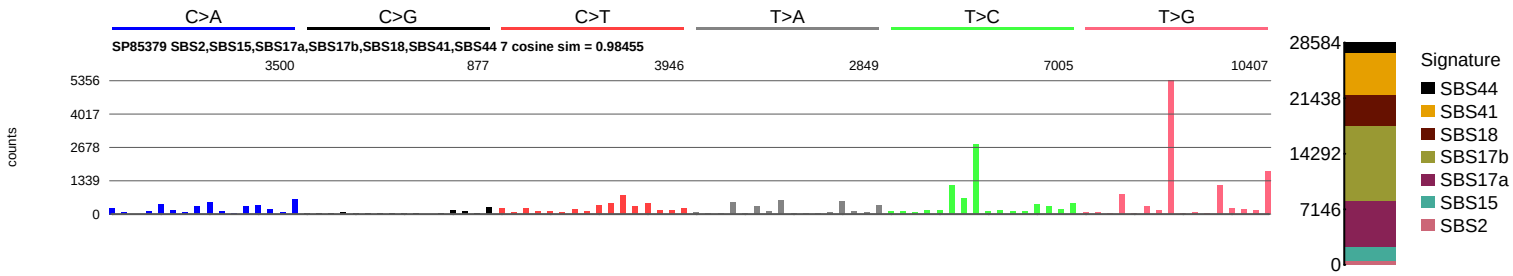
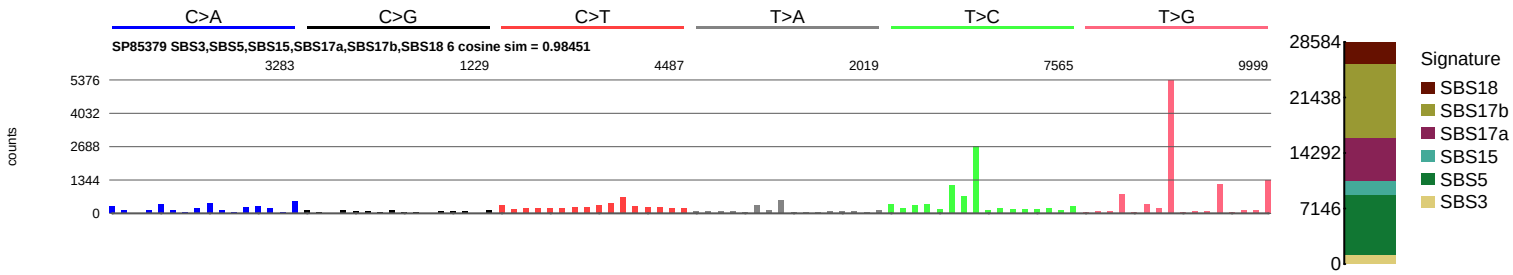
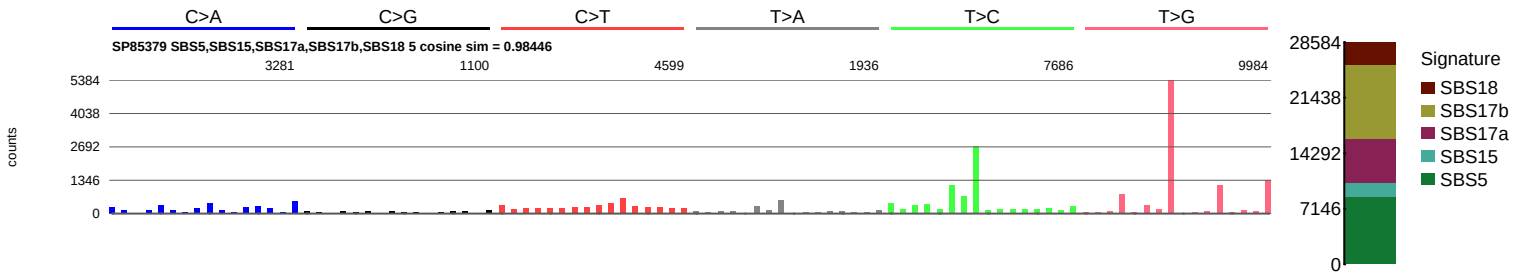
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98457
3274
1078
4706
1919
7634
9973
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

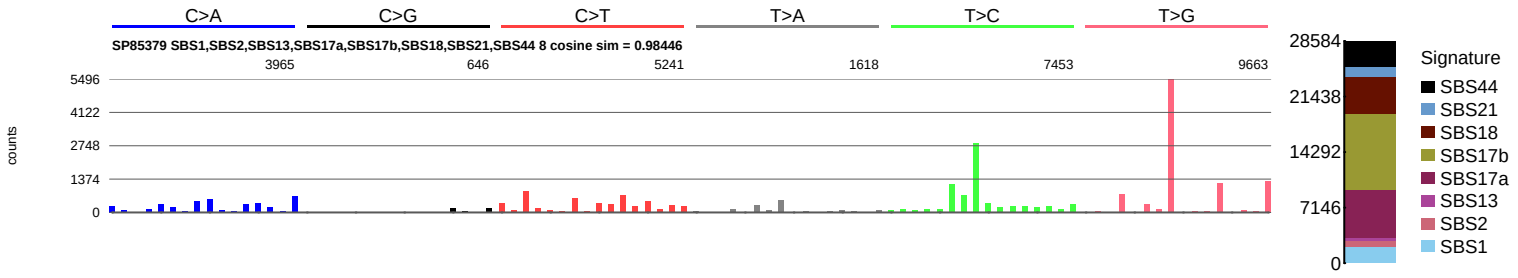

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS3,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98457
2056
2216
4174
2629
7292
10215
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS1

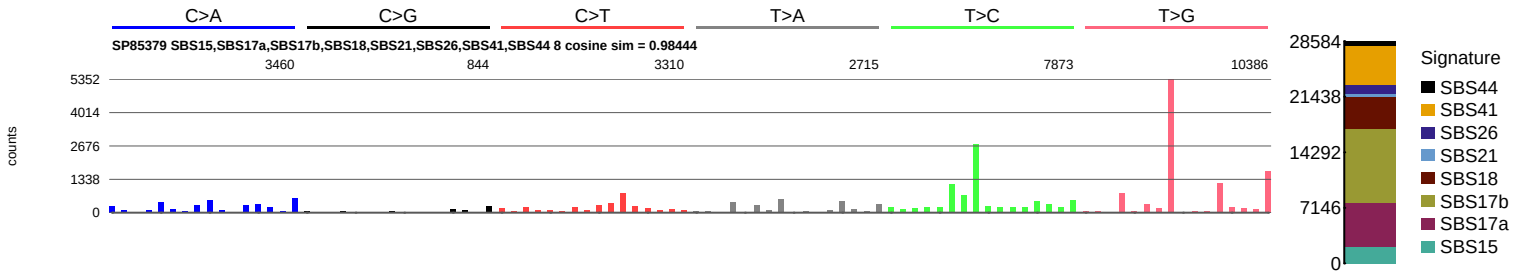

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98456
3276
1064
4521
1910
7850
9961
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98455
3500
877
3946
2849
7005
10407
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98451
3283
1229
4487
2019
7565
9999
counts
5376
4032
2688
1344
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98446
3281
1100
4599
1936
7686
9984
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98446
3965
646
5241
1618
7453
9663
counts
5496
4122
2748
1374
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98444
3460
844
3310
2715
7873
10386
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
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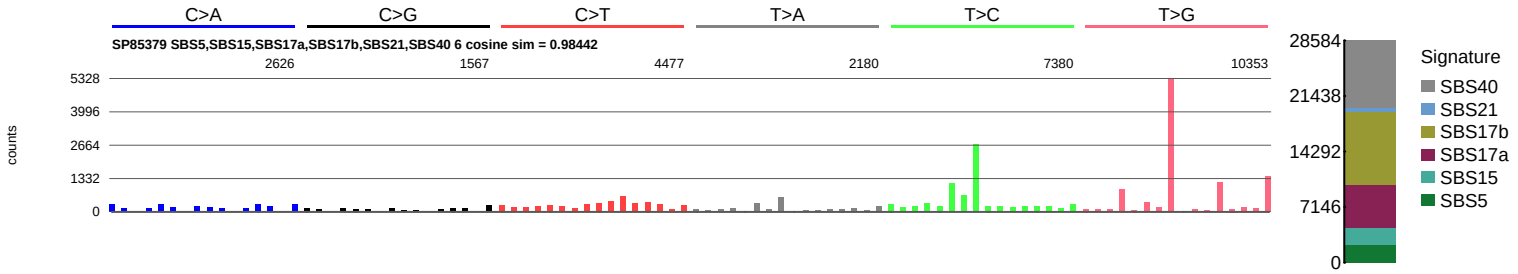
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98442
2626
1567
4477
2180
7380
10353
counts
5328
3996
2664
1332
0
28584
21438
14292
7146
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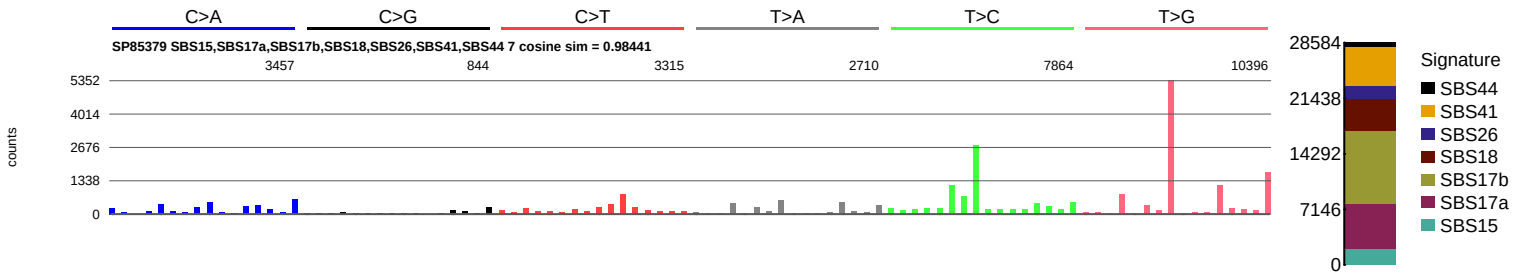
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98441
3457
844
3315
2710
7864
10396
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
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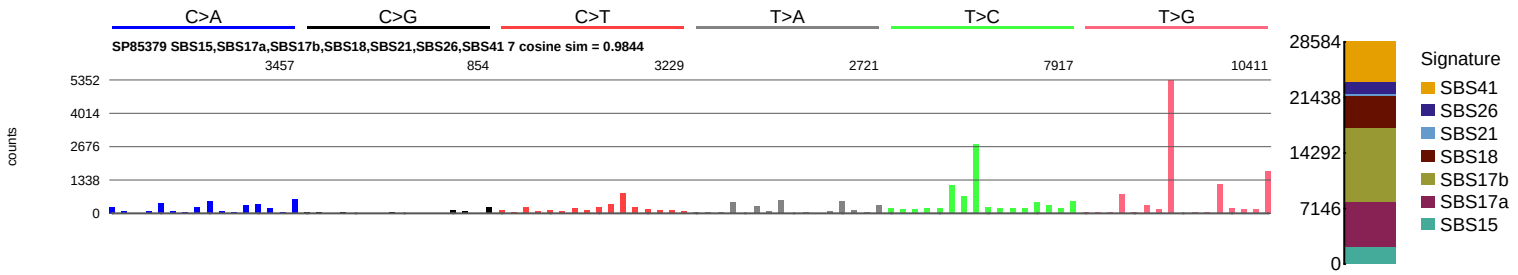
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.9844
3457
854
3229
2721
7917
10411
counts
5352
4014
2676
1338
0
28584
21438
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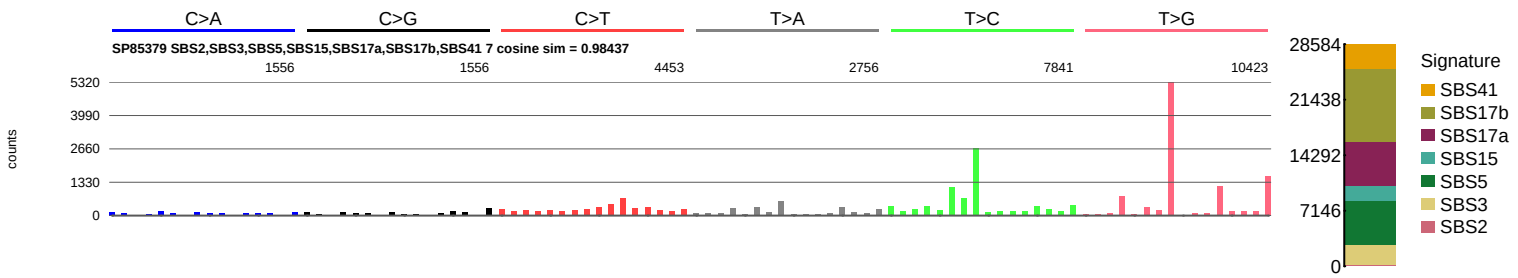
SBS17a
SBS15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SP85379 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98437
1556
1556
4453
2756
7841
10423
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
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SBS17b
SBS17a
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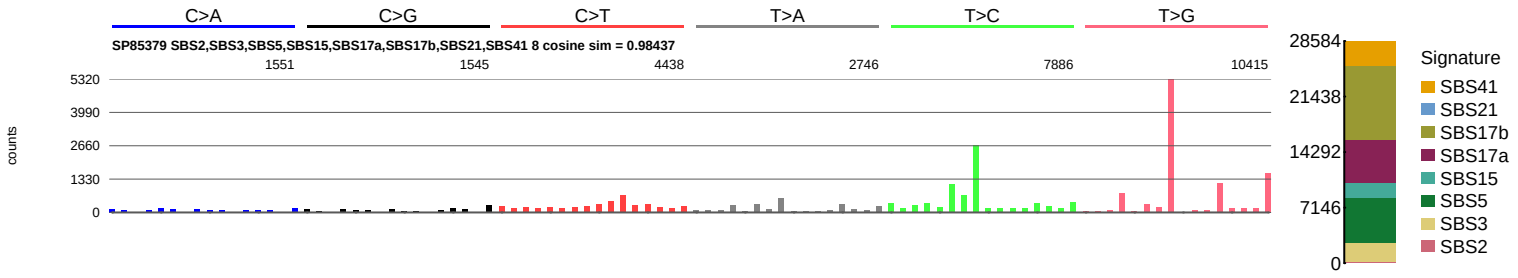
SBS3
SBS2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C>T
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SP85379 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98437
1551
1545
4438
2746
7886
10415
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
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Signature
SBS41
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SBS17b
SBS17a
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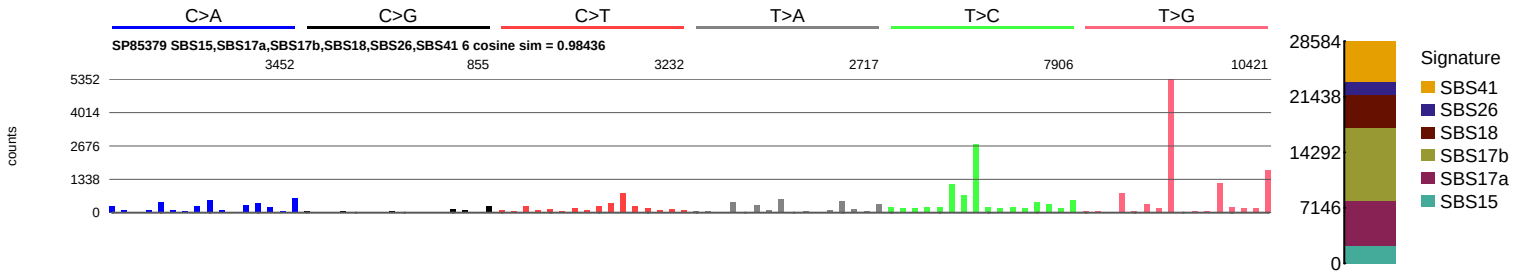
SBS3
SBS2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SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98436
3452
855
3232
2717
7906
10421
counts
5352
4014
2676
1338
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28584
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14292
7146
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SBS41
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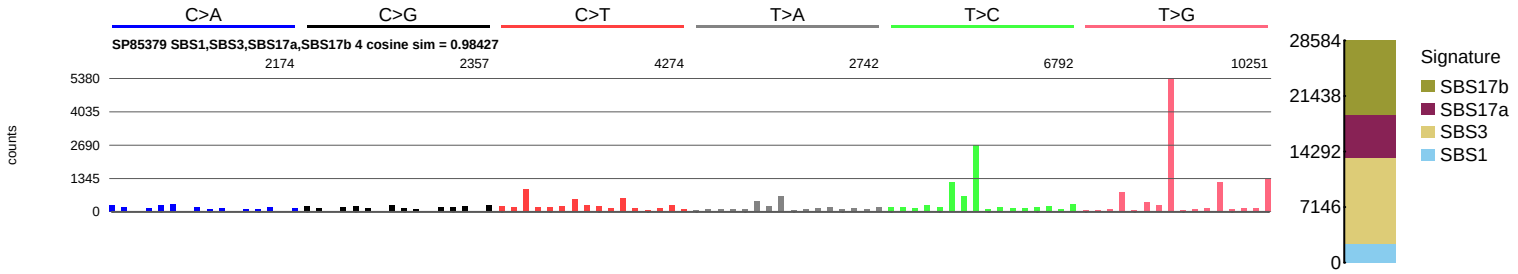
SBS18
SBS17b
SBS17a
SBS15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SP85379 SBS1,SBS3,SBS17a,SBS17b 4 cosine sim = 0.98427
2174
2357
4274
2742
6792
10251
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
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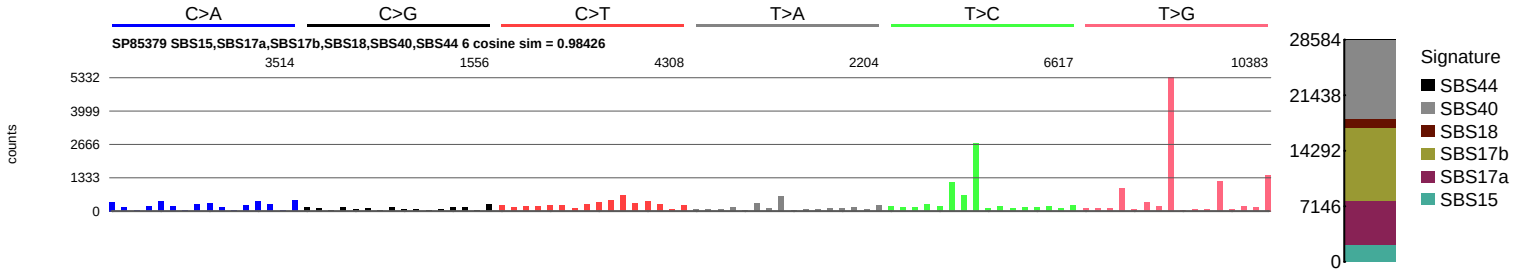
Signature
SBS17b
SBS17a
SBS3
SBS1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C>G
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T>C
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SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98426
3514
1556
4308
2204
6617
10383
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
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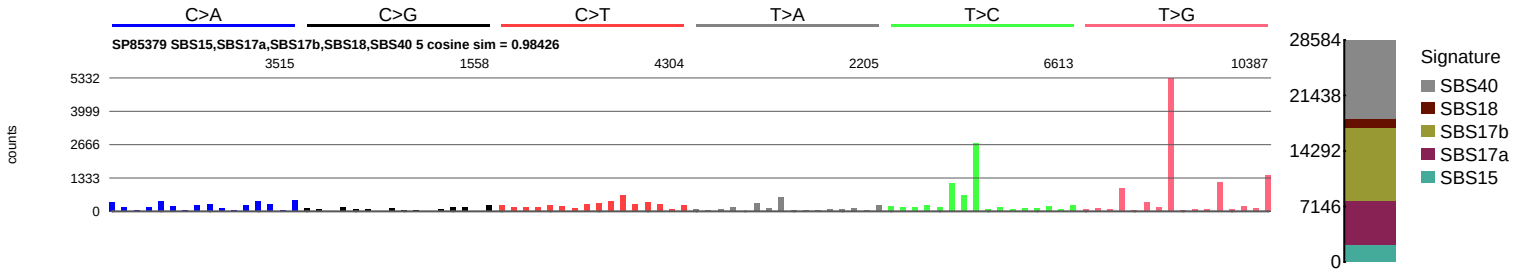
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15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C>G
C>T
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T>C
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SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98426
3515
1558
4304
2205
6613
10387
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
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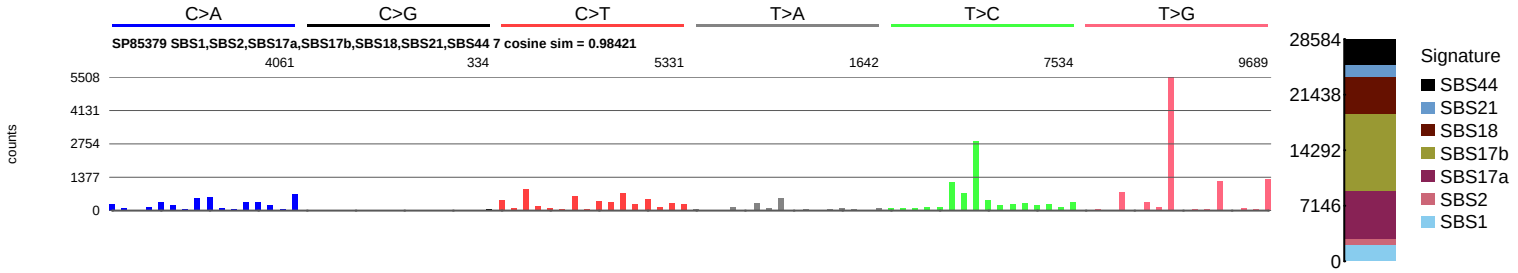
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15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T>C
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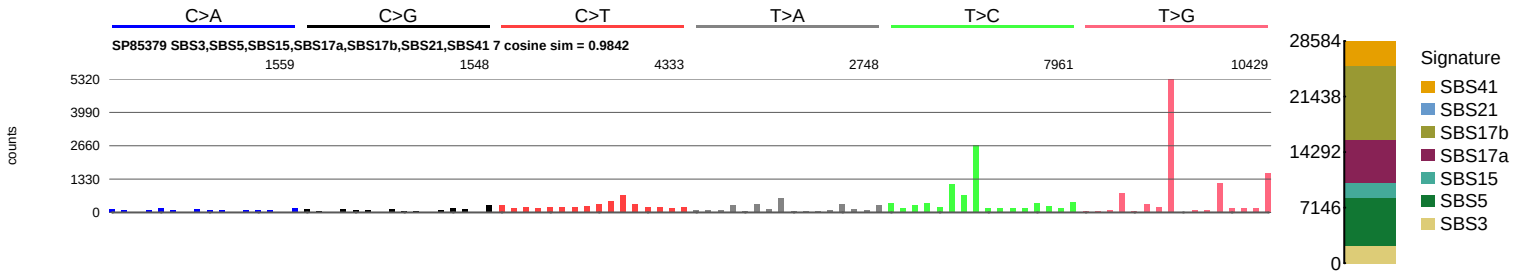
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98421
4061
334
5331
1642
7534
9689
counts
5508
4131
2754
1377
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
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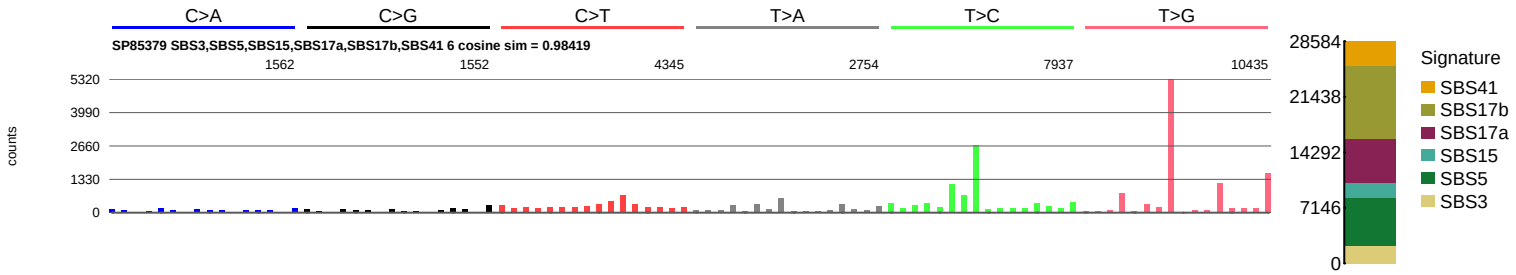
T>C
T>G
SP85379 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9842
1559
1548
4333
2748
7961
10429
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
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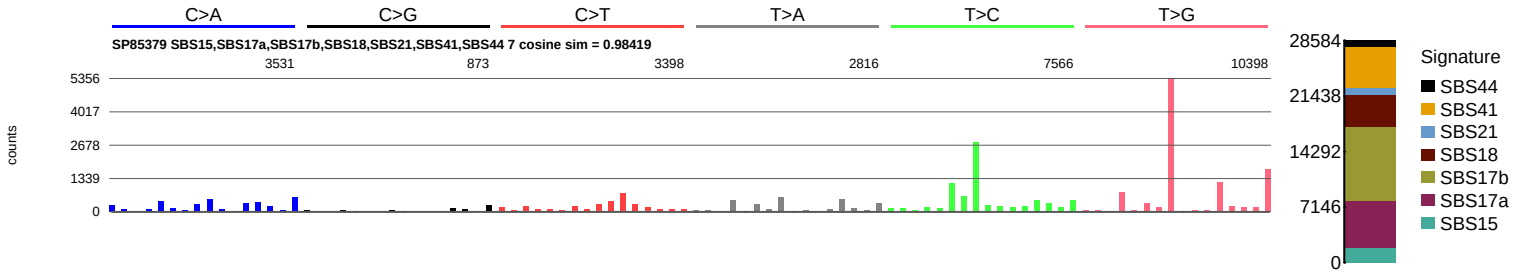
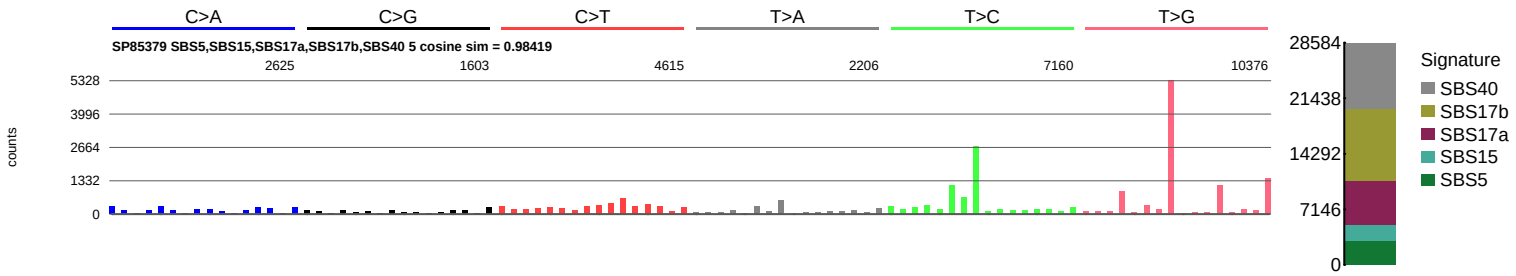
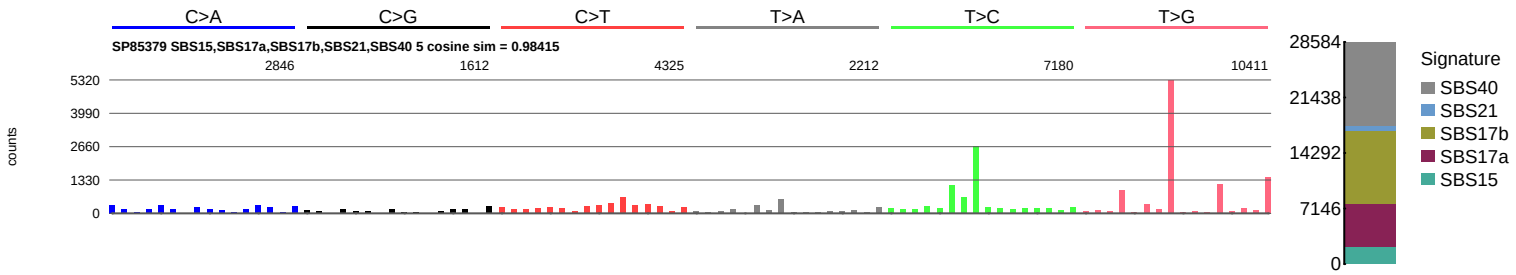
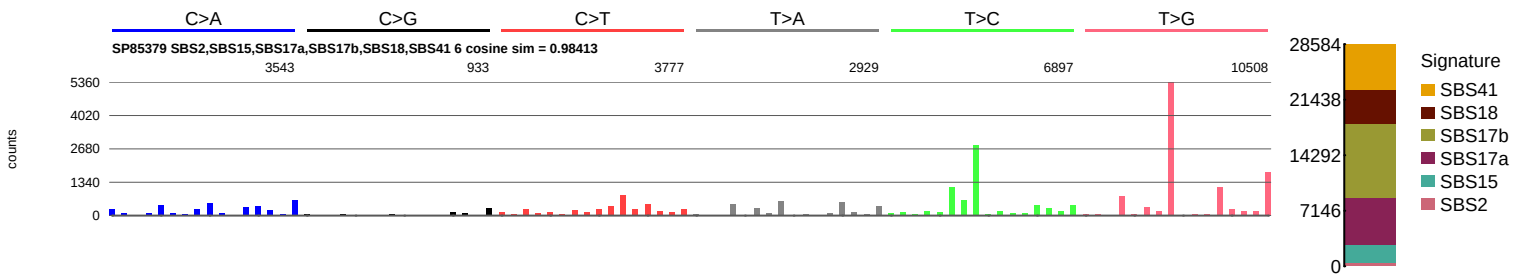
T>A
T>C
T>G
SP85379 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98419
1562
1552
4345
2754
7937
10435
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98419
3531
873
3398
2816
7566
10398
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98419
2625
1603
4615
2206
7160
10376
counts
5328
3996
2664
1332
0
28584
21438
14292
7146
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98415
2846
1612
4325
2212
7180
10411
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98413
3543
933
3777
2929
6897
10508
counts
5360
4020
2680
1340
0
28584
21438
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7146
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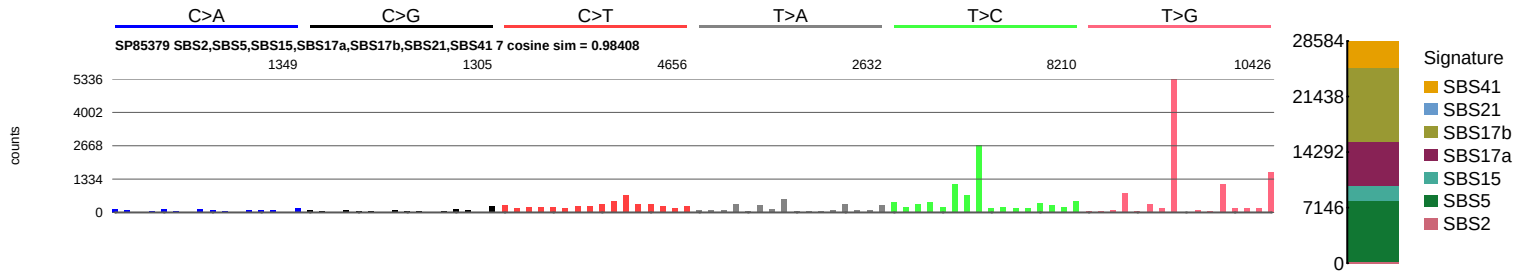
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98408
1349
1305
4656
2632
8210
10426
counts
5336
4002
2668
1334
0
28584
21438
14292
7146
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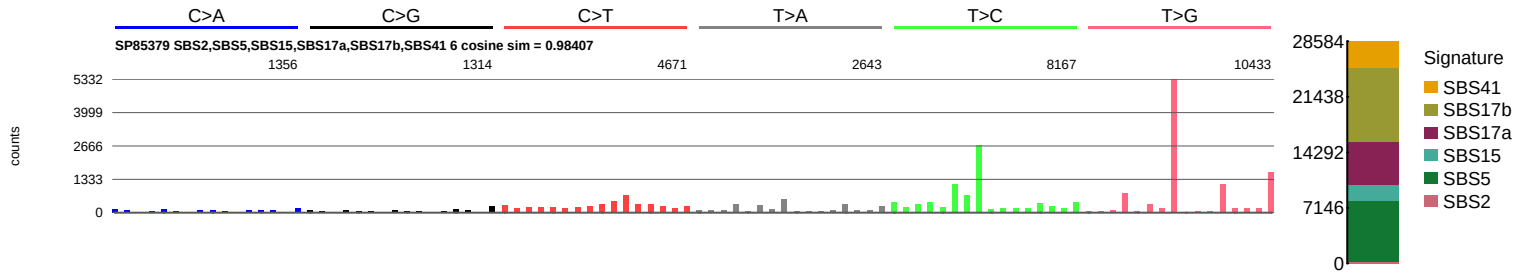
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98407
1356
1314
4671
2643
8167
10433
counts
5332
3999
2666
1333
0
28584
21438
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7146
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SBS41
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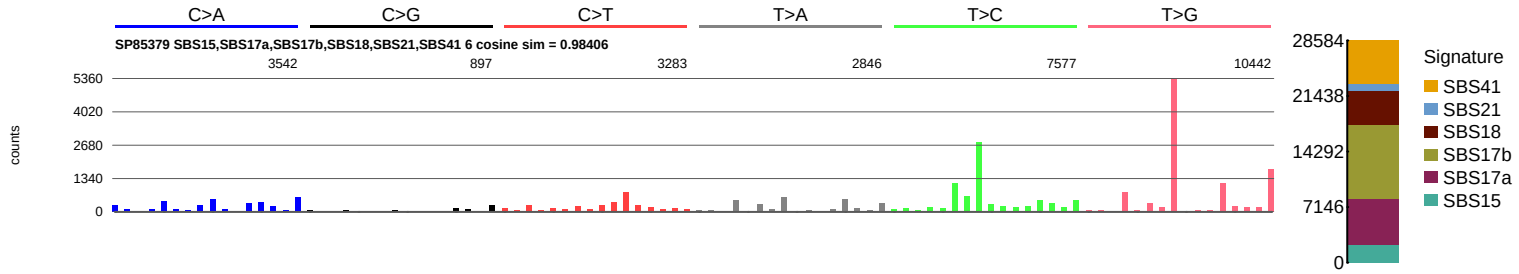
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98406
3542
897
3283
2846
7577
10442
counts
5360
4020
2680
1340
0
28584
21438
14292
7146
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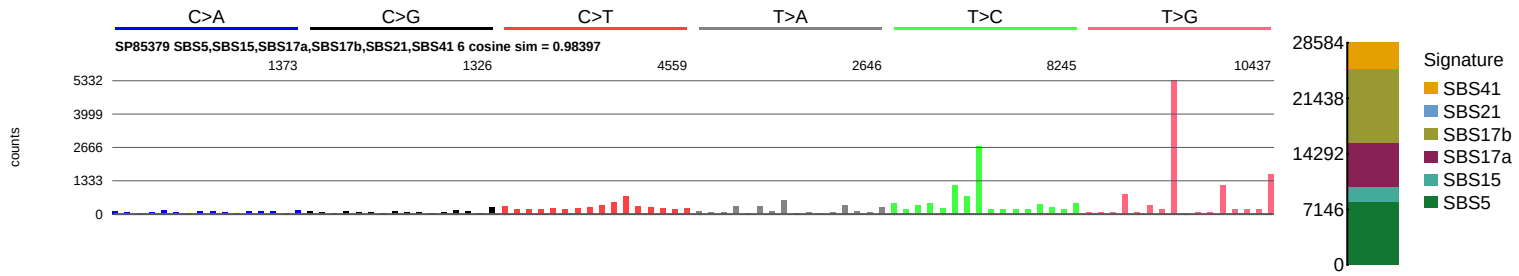
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98397
1373
1326
4559
2646
8245
10437
counts
5332
3999
2666
1333
0
28584
21438
14292
7146
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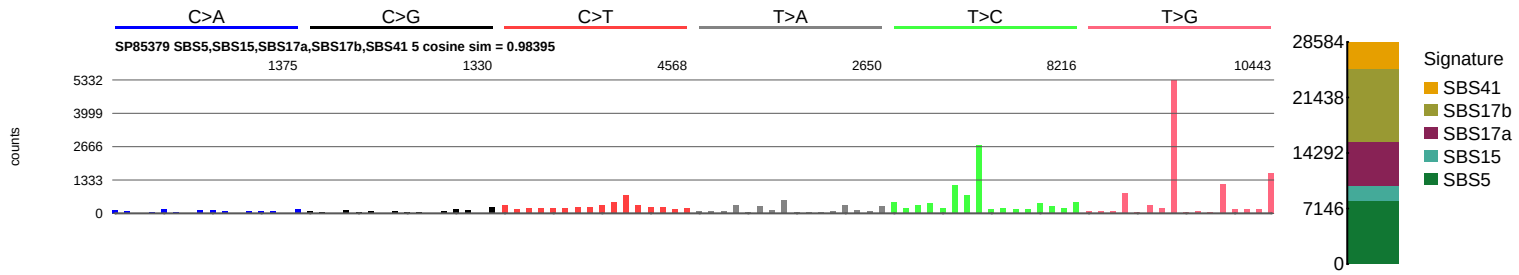
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98395
1375
1330
4568
2650
8216
10443
counts
5332
3999
2666
1333
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28584
21438
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7146
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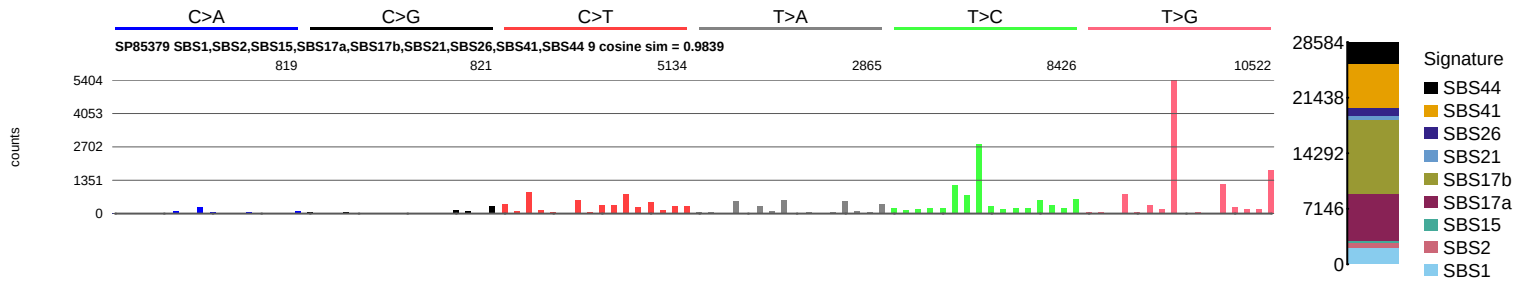
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5


C>A
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C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.9839
819
821
5134
2865
8426
10522
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
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SBS44
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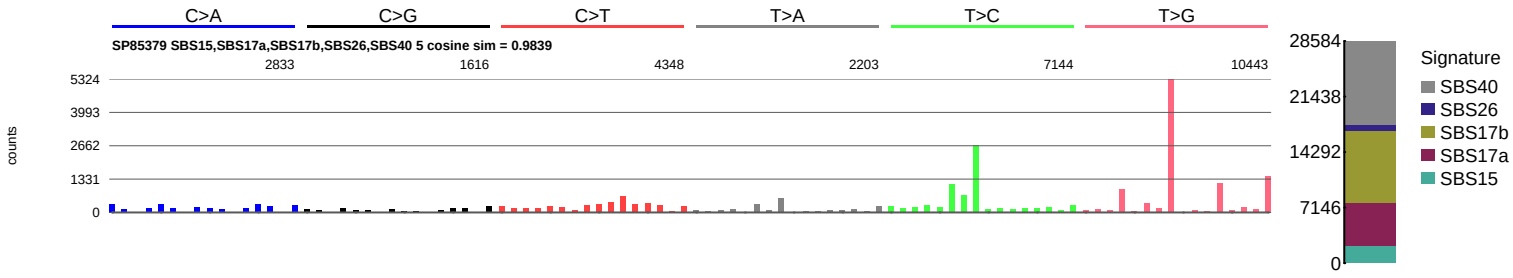
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.9839
2833
1616
4348
2203
7144
10443
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
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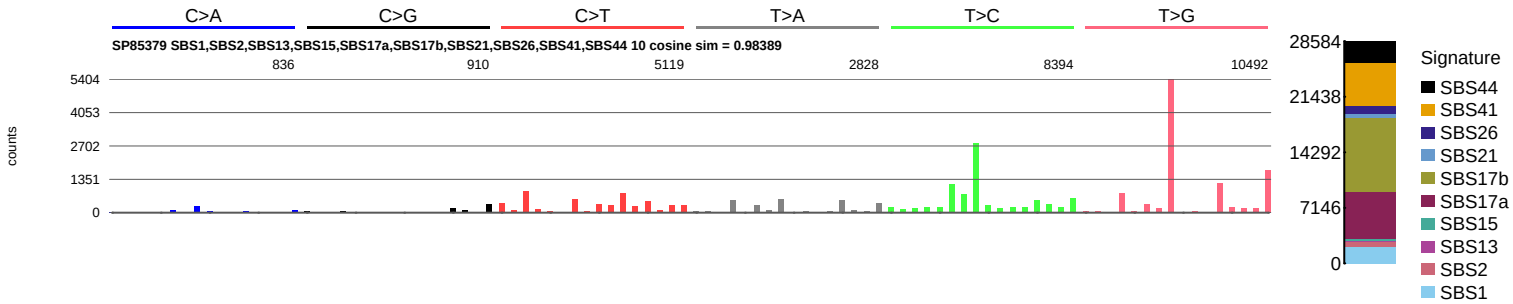
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15


C>A
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C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98389
836
910
5119
2828
8394
10492
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
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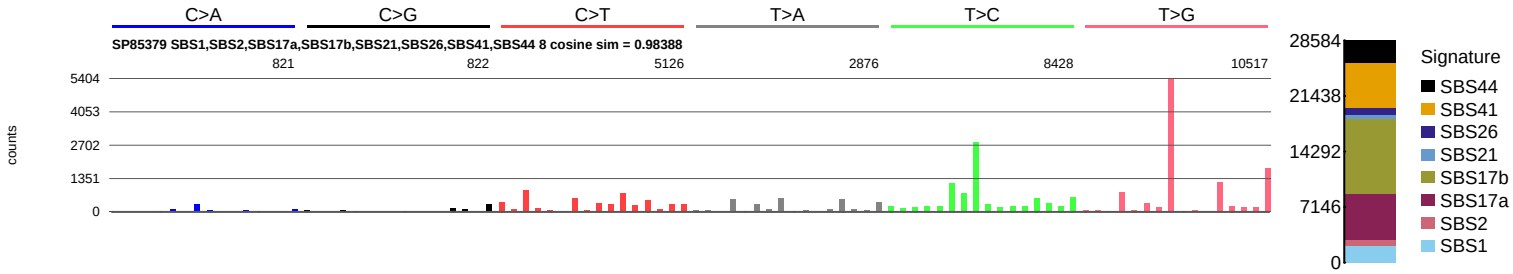
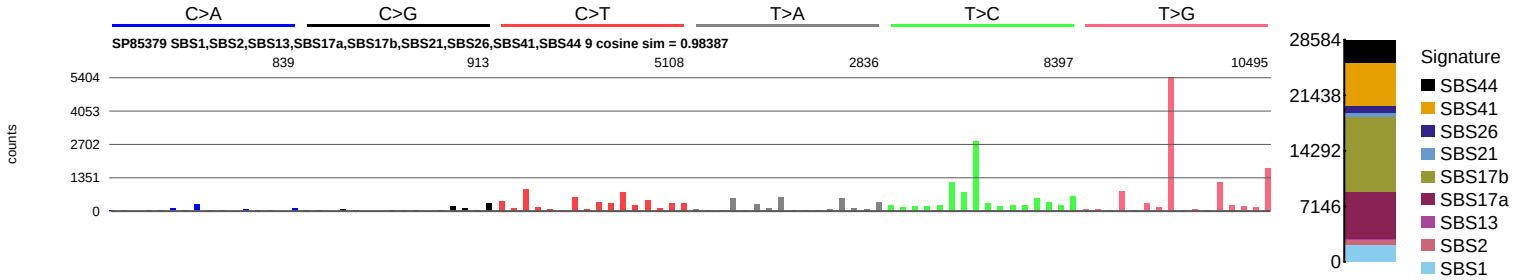
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98388
821
822
5126
2876
8428
10517
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98387
839
913
5108
2836
8397
10495
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
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Signature
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SBS41
SBS26
SBS21
SBS17b
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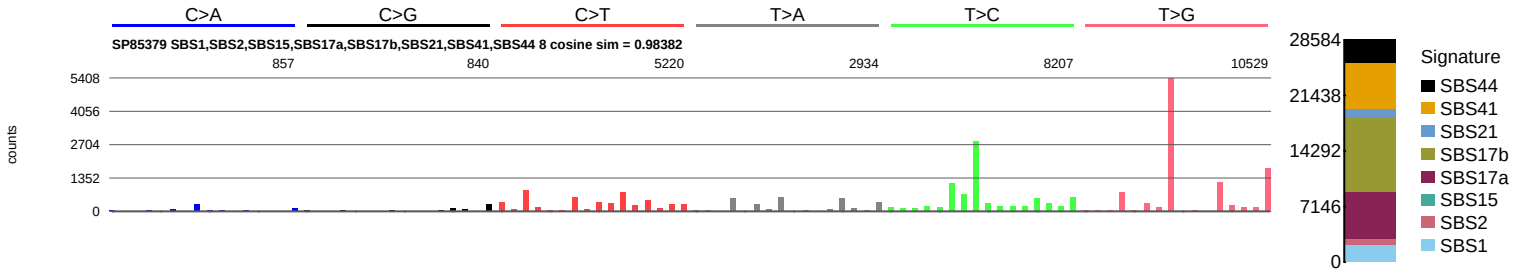
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98382
857
840
5220
2934
8207
10529
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
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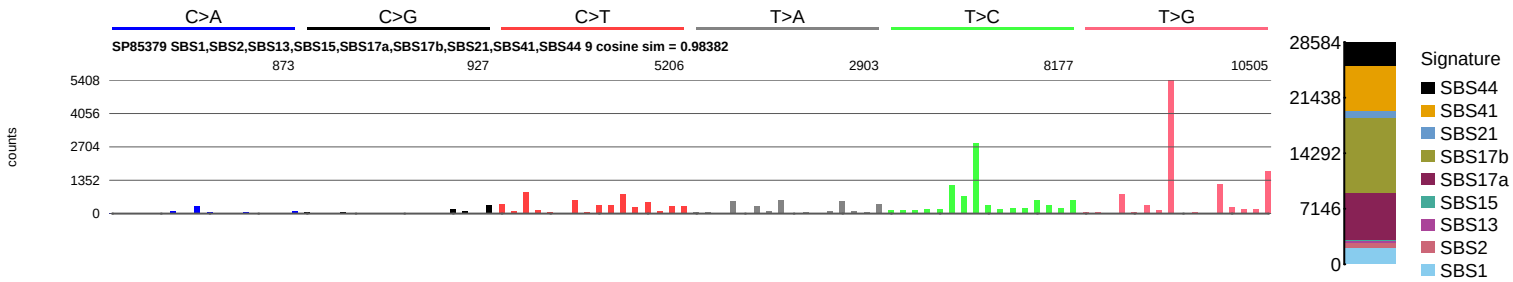
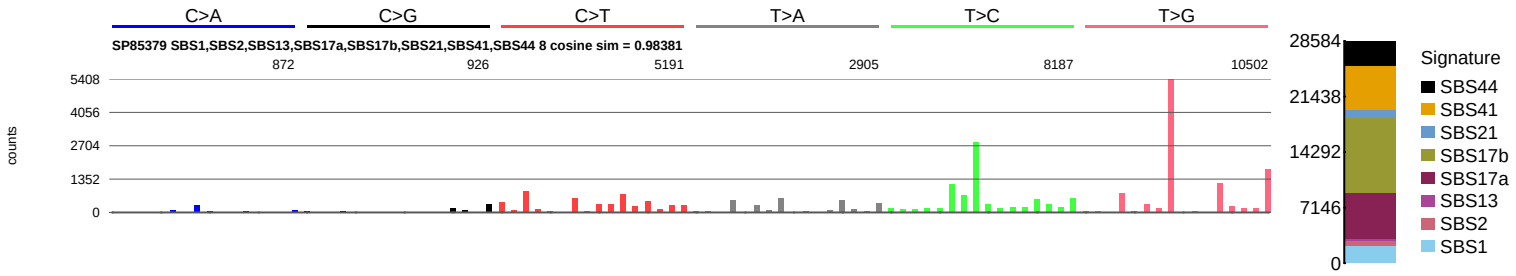
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98382
873
927
5206
2903
8177
10505
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98381
872
926
5191
2905
8187
10502
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
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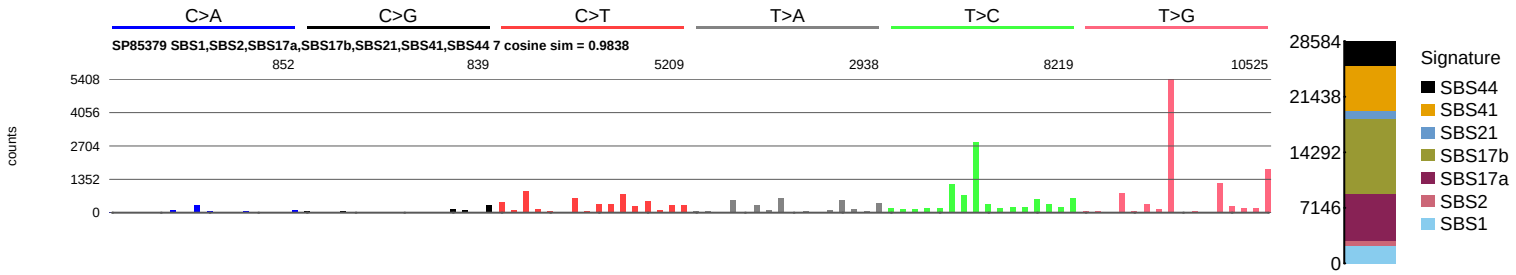
SBS1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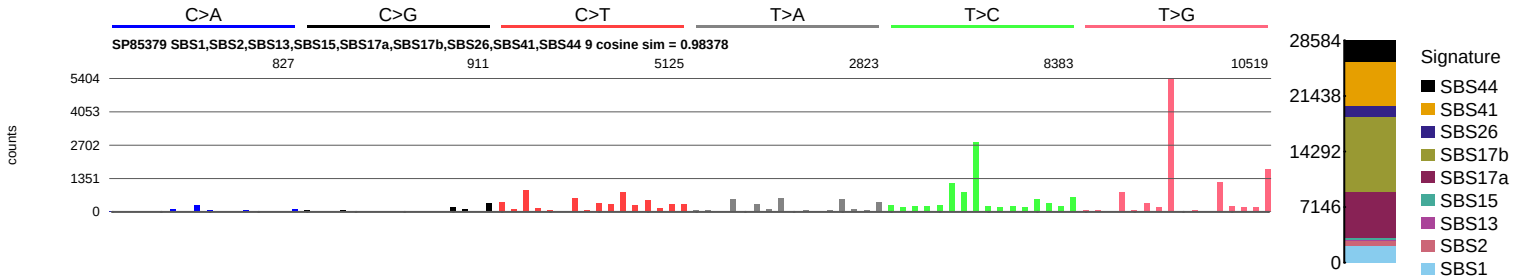
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.9838
852
839
5209
2938
8219
10525
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS2
SBS1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SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98378
827
911
5125
2823
8383
10519
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
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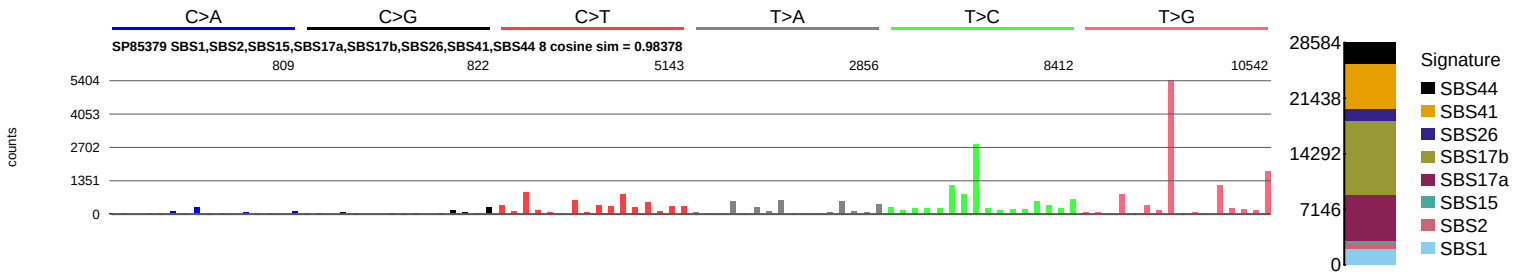
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>T
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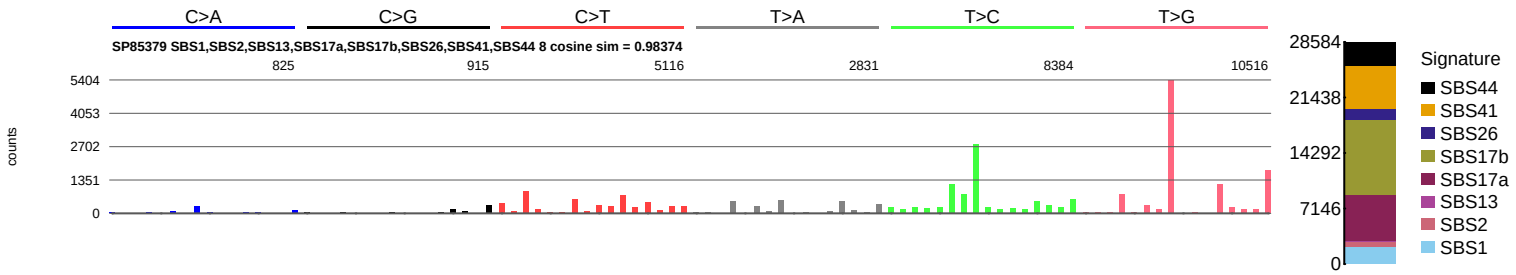
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98378
809
822
5143
2856
8412
10542
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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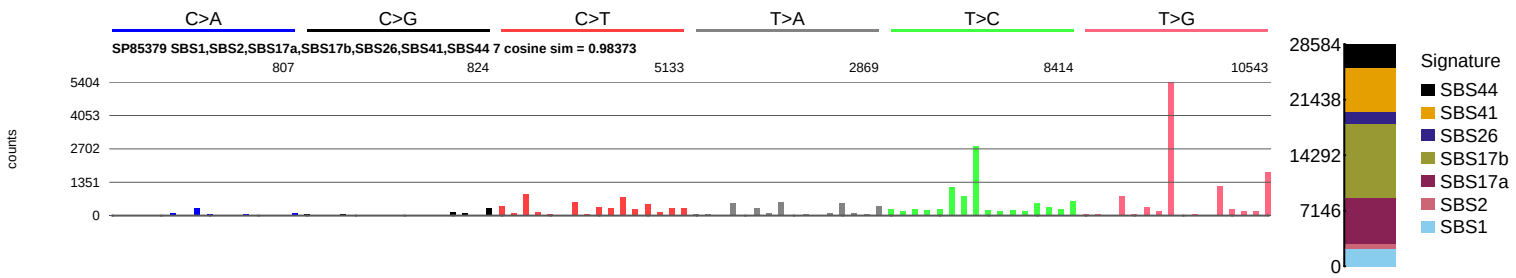
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98374
825
915
5116
2831
8384
10516
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
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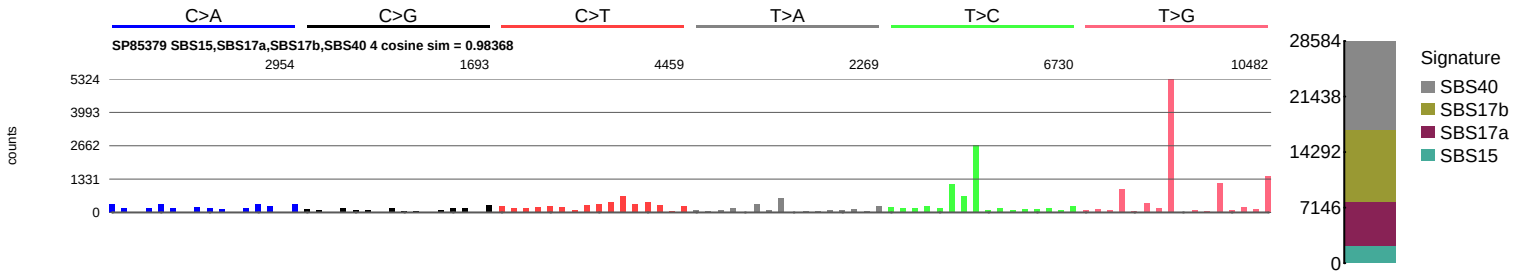
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98373
807
824
5133
2869
8414
10543
counts
5404
4053
2702
1351
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS2
SBS1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T>C
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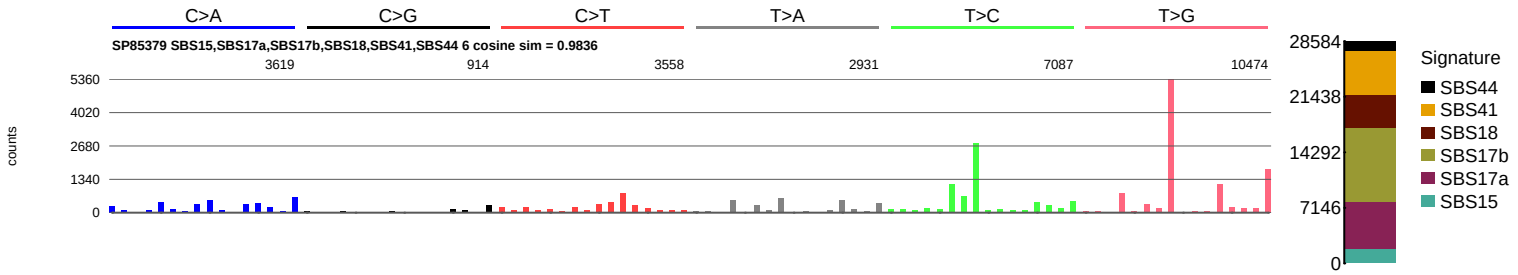
SP85379 SBS15,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98368
2954
1693
4459
2269
6730
10482
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS40
SBS17b
SBS17a
SBS15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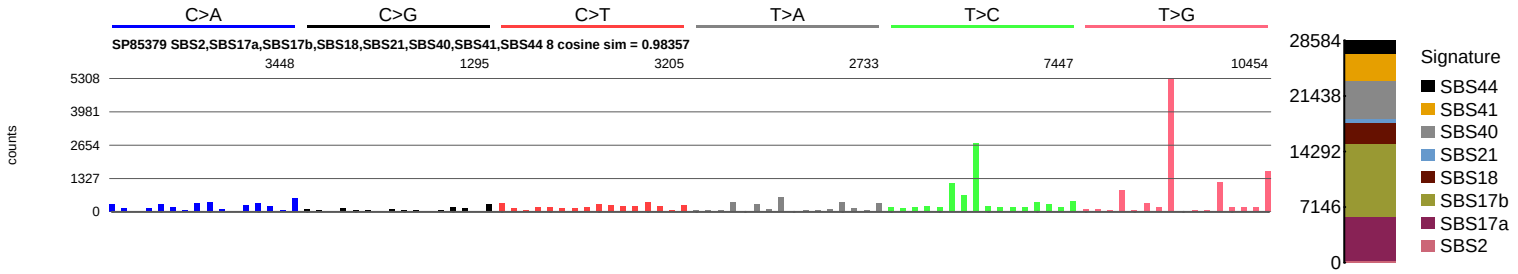
C>G
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.9836
3619
914
3558
2931
7087
10474
counts
5360
4020
2680
1340
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15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C>G
C>T
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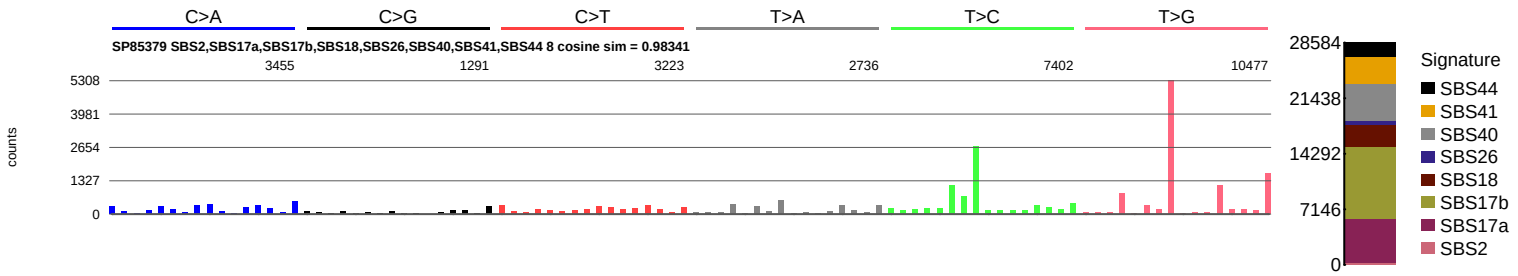
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98357
3448
1295
3205
2733
7447
10454
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2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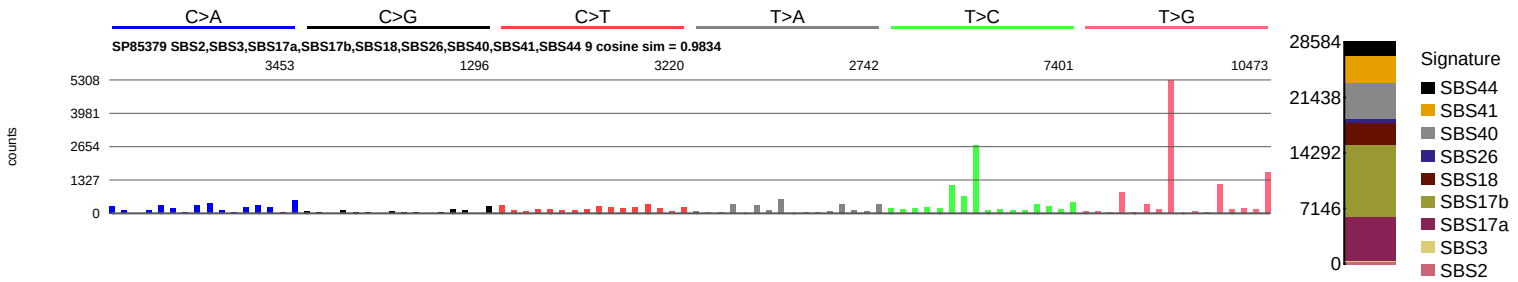
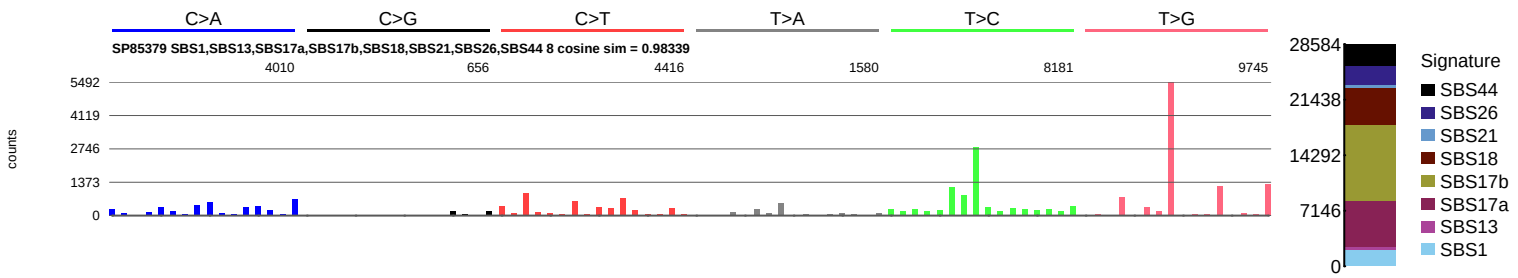
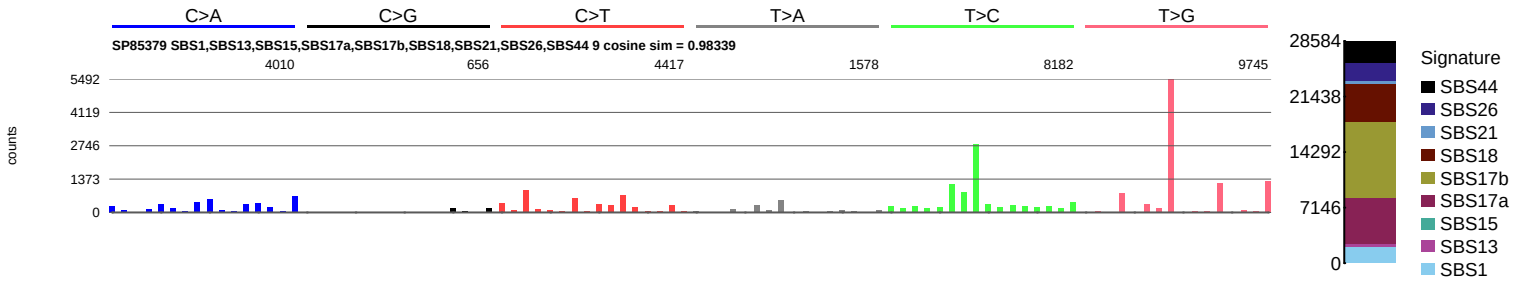
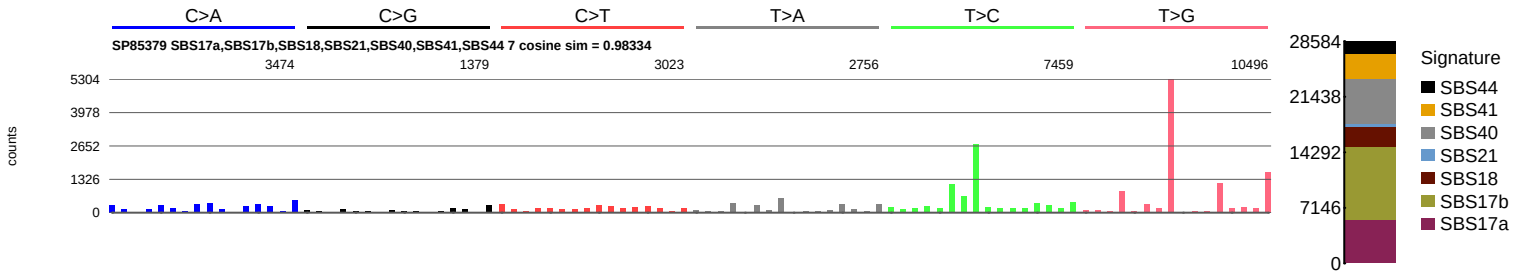
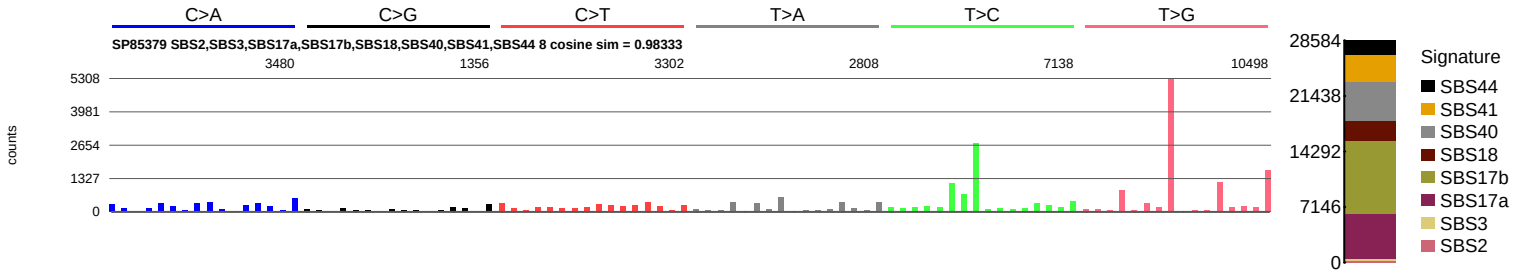
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98341
3455
1291
3223
2736
7402
10477
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9834
3453
1296
3220
2742
7401
10473
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98339
4010
656
4416
1580
8181
9745
counts
5492
4119
2746
1373
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98339
4010
656
4417
1578
8182
9745
counts
5492
4119
2746
1373
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98334
3474
1379
3023
2756
7459
10496
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a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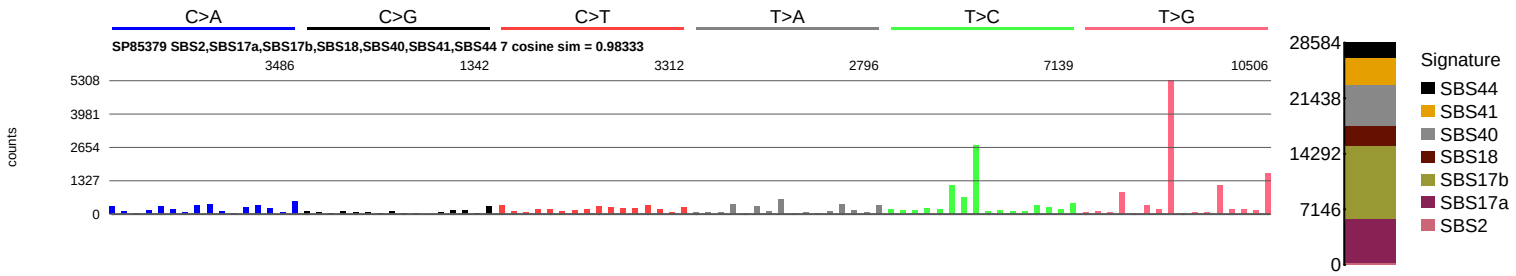
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98333
3480
1356
3302
2808
7138
10498
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98333
3486
1342
3312
2796
7139
10506
counts
5308
3981
2654
1327
0
28584
21438
14292
7146
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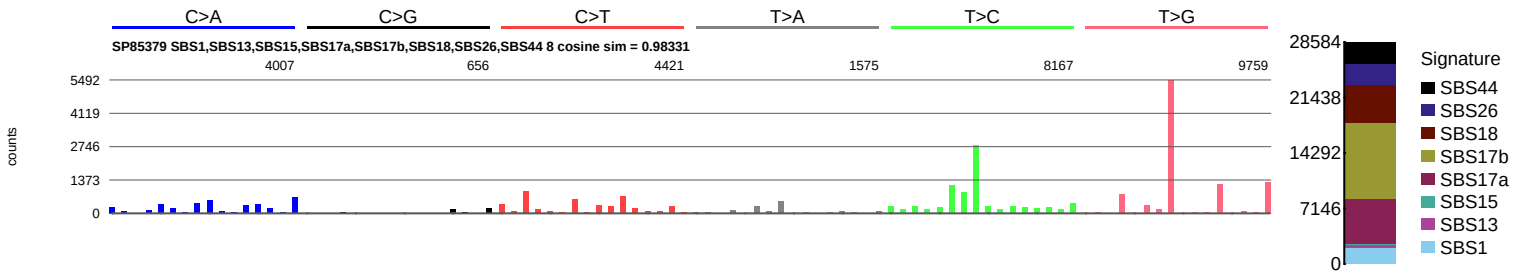
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2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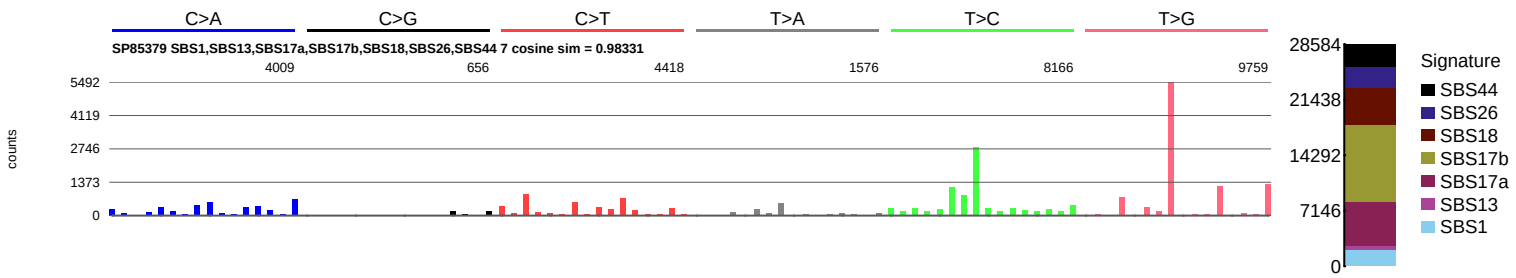
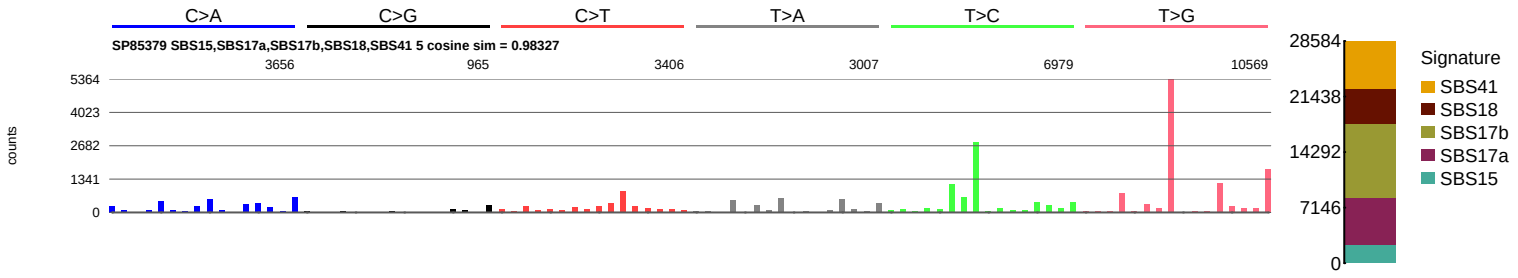
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98331
4007
656
4421
1575
8167
9759
counts
5492
4119
2746
1373
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98331
4009
656
4418
1576
8166
9759
counts
5492
4119
2746
1373
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
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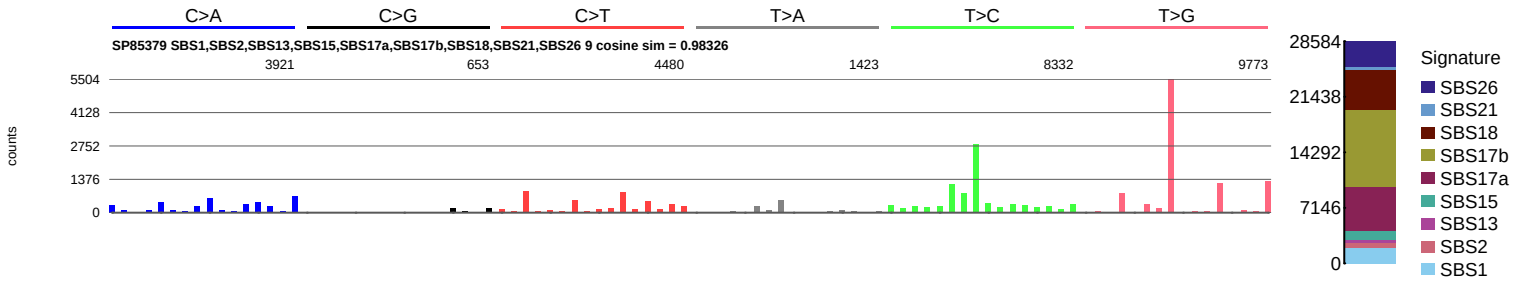
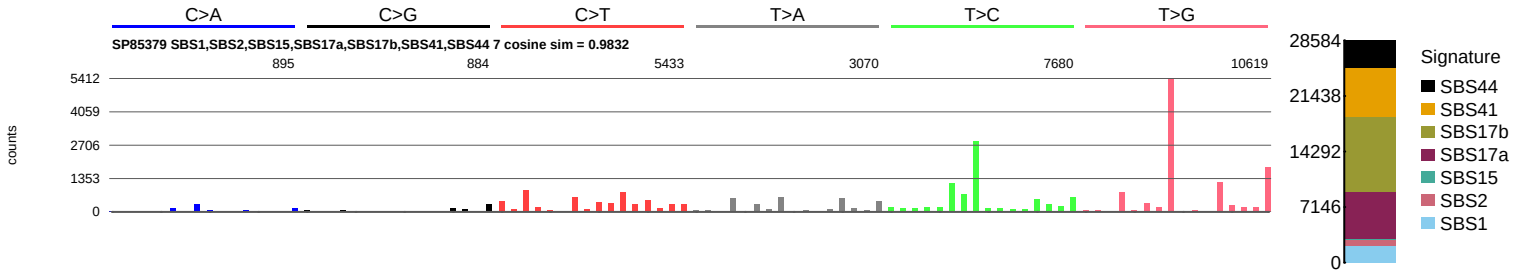
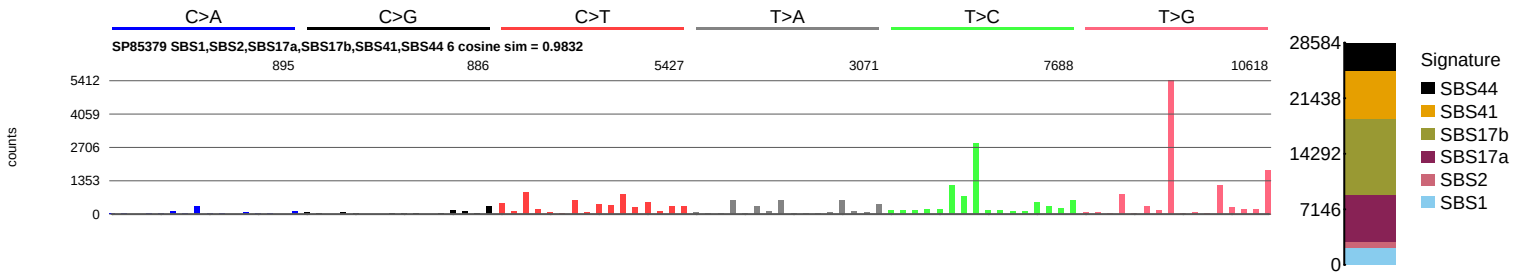
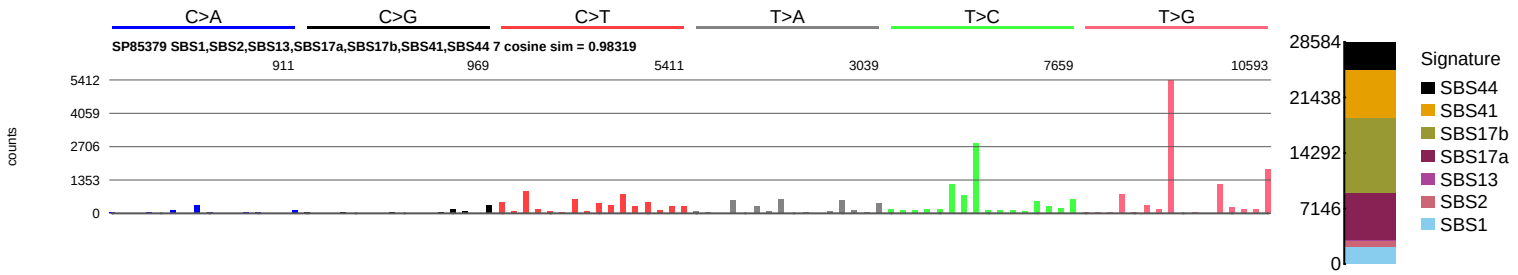
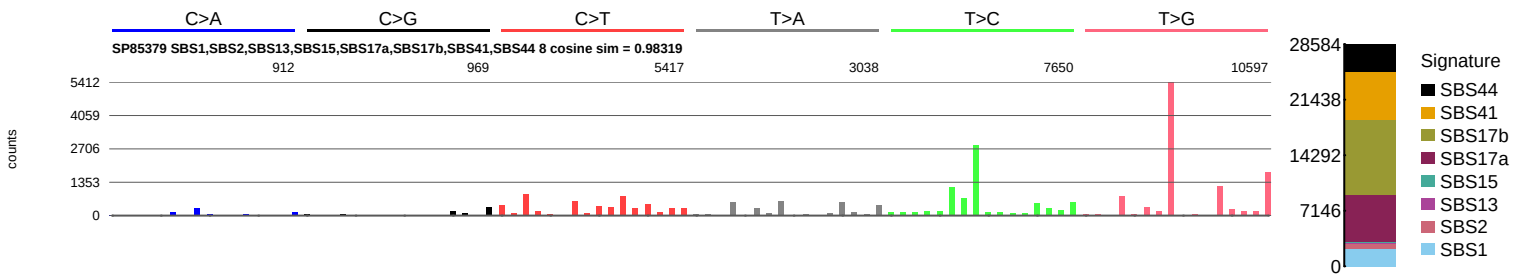
C>T
T>A
T>C
T>G
SP85379 SBS15,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98327
3656
965
3406
3007
6979
10569
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98326
3921
653
4480
1423
8332
9773
counts
5504
4128
2752
1376
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.9832
895
884
5433
3070
7680
10619
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.9832
895
886
5427
3071
7688
10618
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98319
911
969
5411
3039
7659
10593
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98319
912
969
5417
3038
7650
10597
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
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Signature
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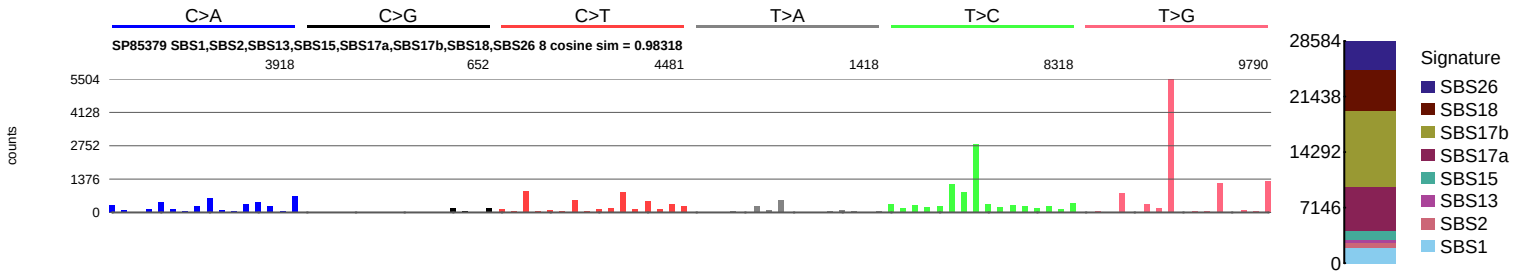
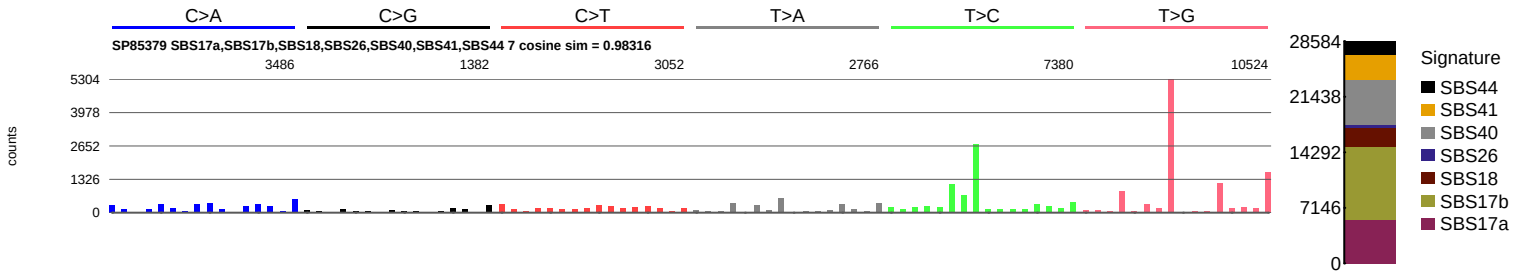
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98318
3918
652
4481
1418
8318
9790
counts
5504
4128
2752
1376
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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SP85379 SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98316
3486
1382
3052
2766
7380
10524
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
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SBS41
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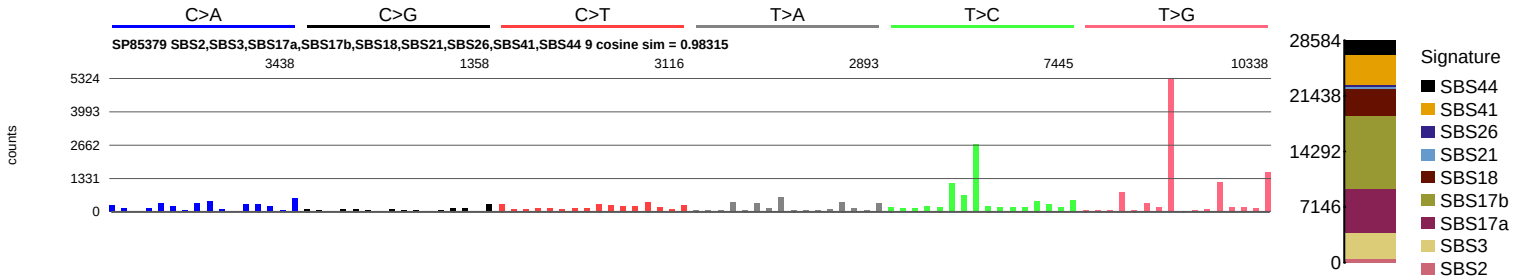
SBS26
SBS18
SBS17b
SBS17a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SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98315
3438
1358
3116
2893
7445
10338
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
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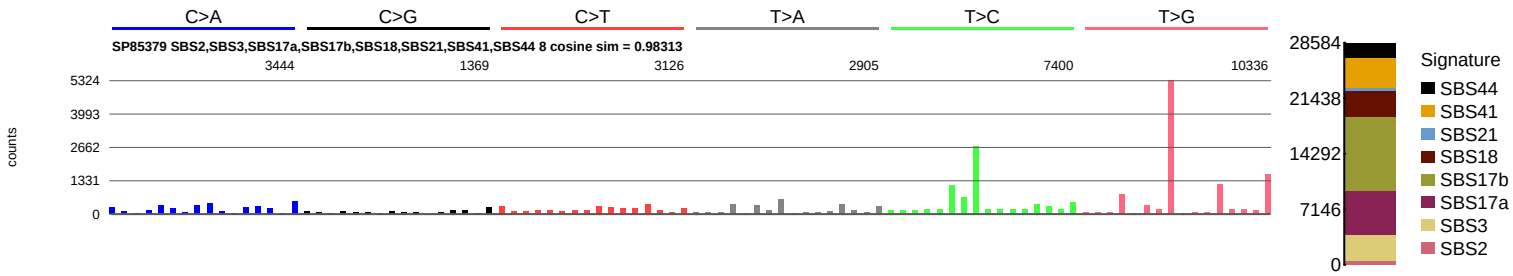
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2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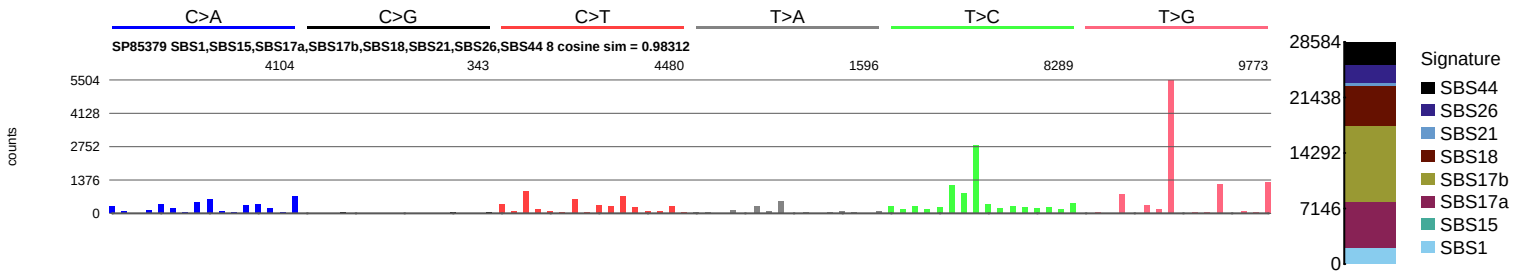
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98313
3444
1369
3126
2905
7400
10336
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2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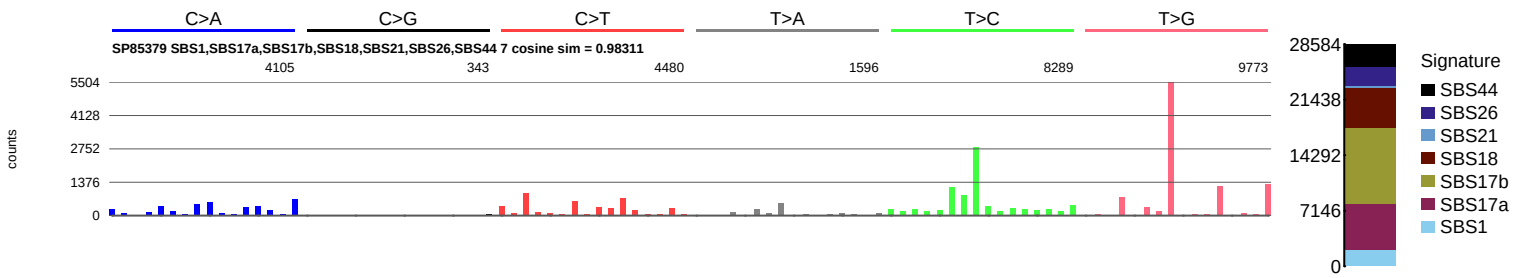
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98312
4104
343
4480
1596
8289
9773
counts
5504
4128
2752
1376
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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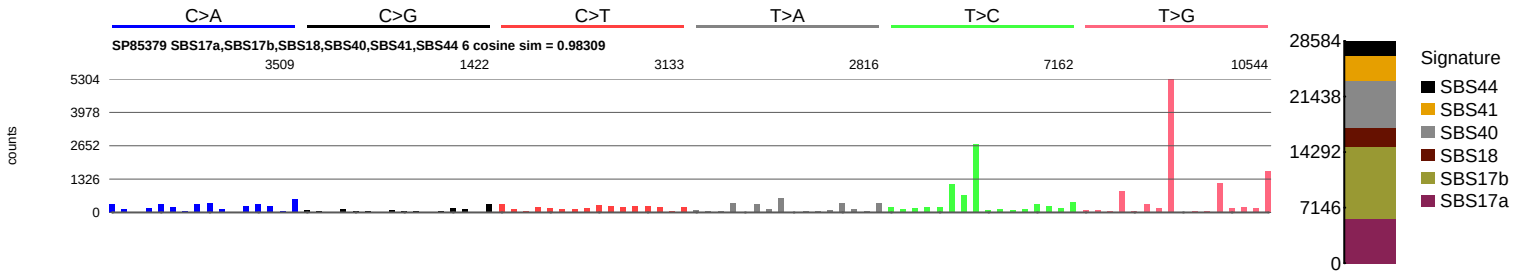
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 7 cosine sim = 0.98311
4105
343
4480
1596
8289
9773
counts
5504
4128
2752
1376
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1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C>T
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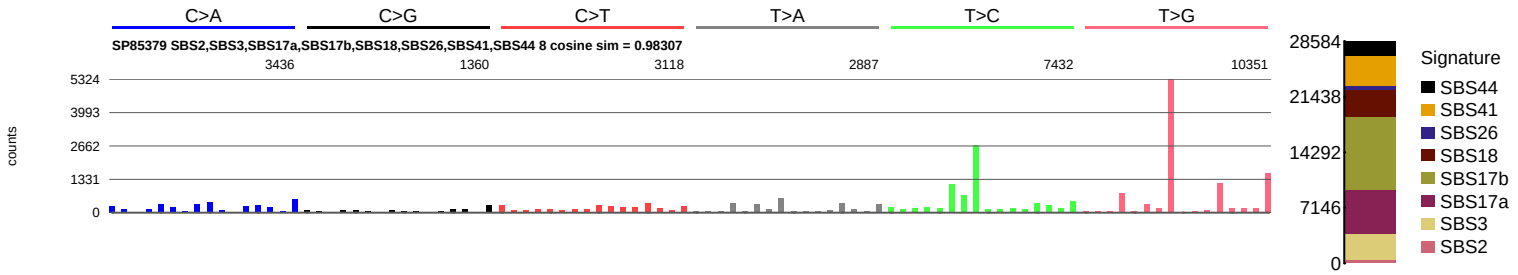
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.98309
3509
1422
3133
2816
7162
10544
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a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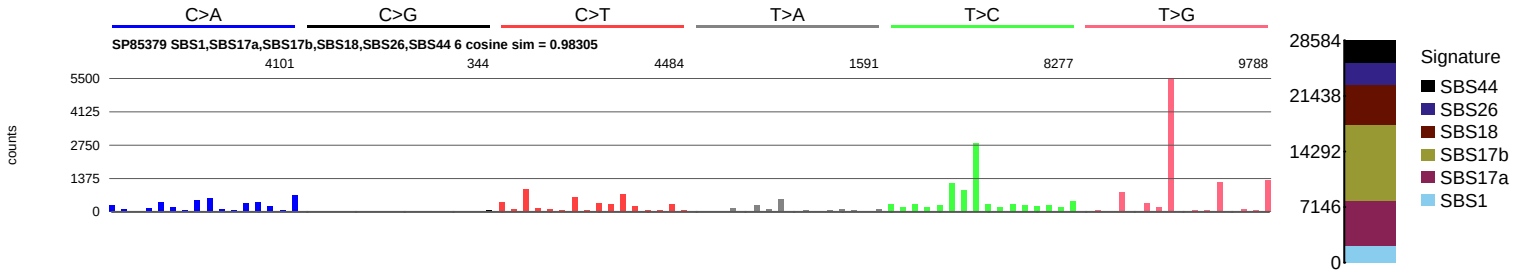
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98307
3436
1360
3118
2887
7432
10351
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2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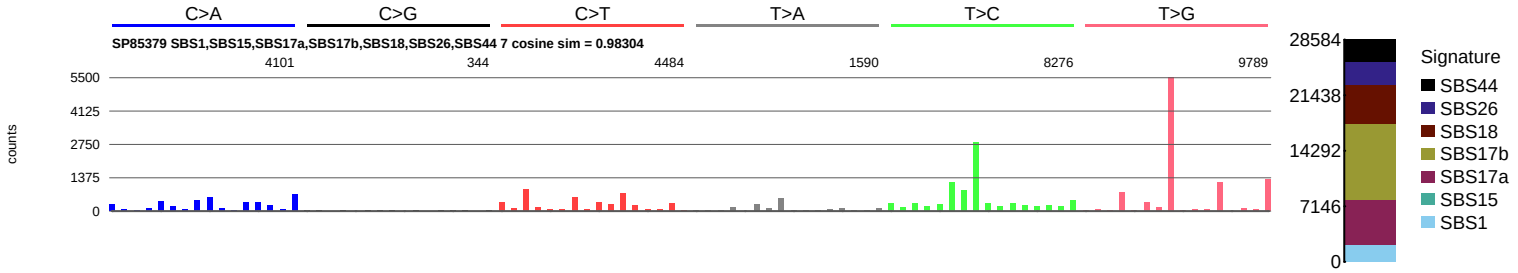
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98305
4101
344
4484
1591
8277
9788
counts
5500
4125
2750
1375
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
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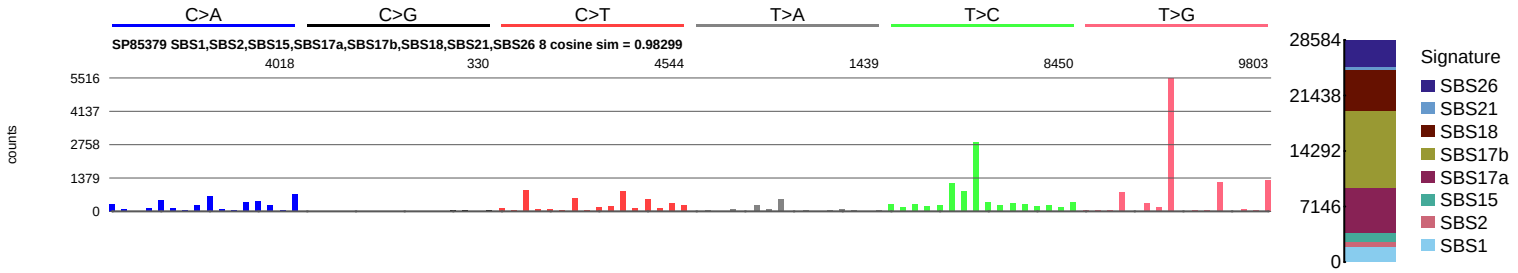
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98304
4101
344
4484
1590
8276
9789
counts
5500
4125
2750
1375
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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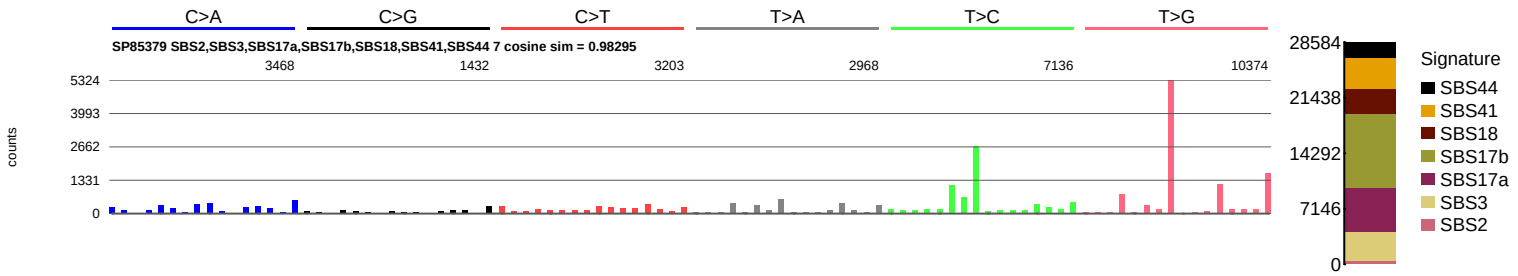
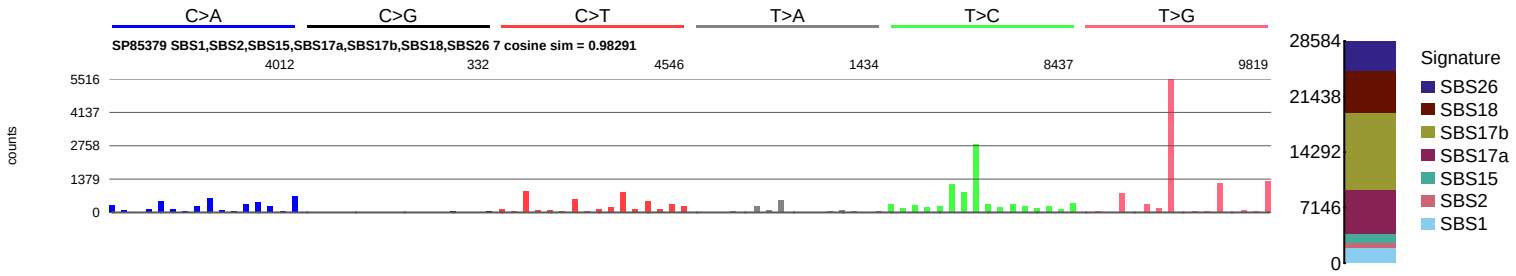
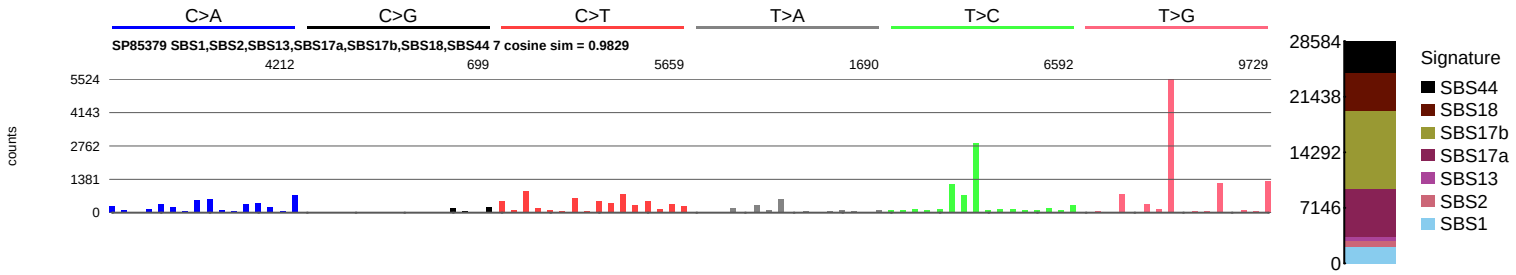
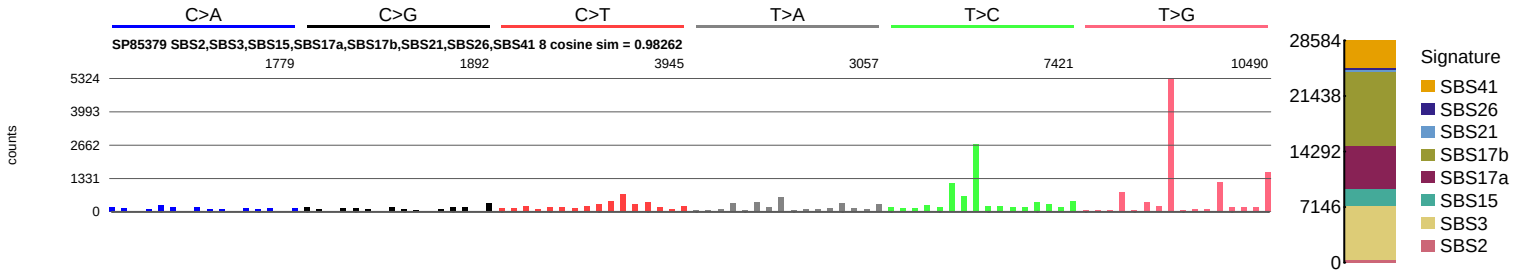
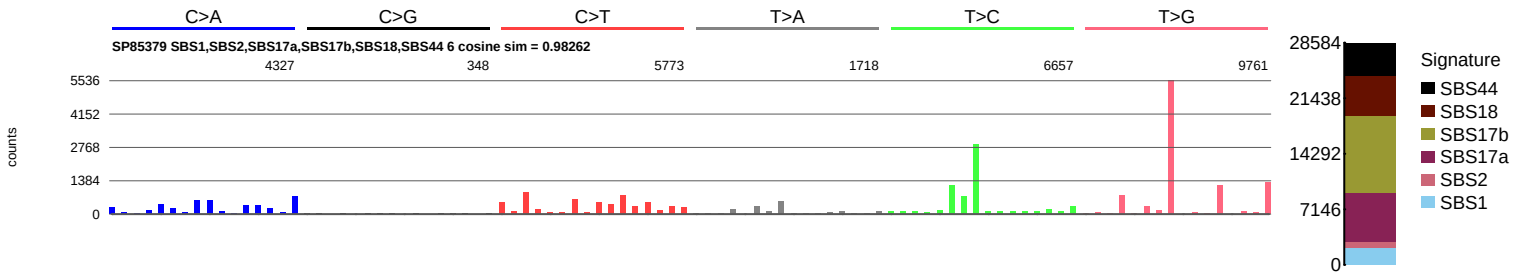
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98299
4018
330
4544
1439
8450
9803
counts
5516
4137
2758
1379
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98295
3468
1432
3203
2968
7136
10374
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98291
4012
332
4546
1434
8437
9819
counts
5516
4137
2758
1379
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9829
4212
699
5659
1690
6592
9729
counts
5524
4143
2762
1381
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98262
1779
1892
3945
3057
7421
10490
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2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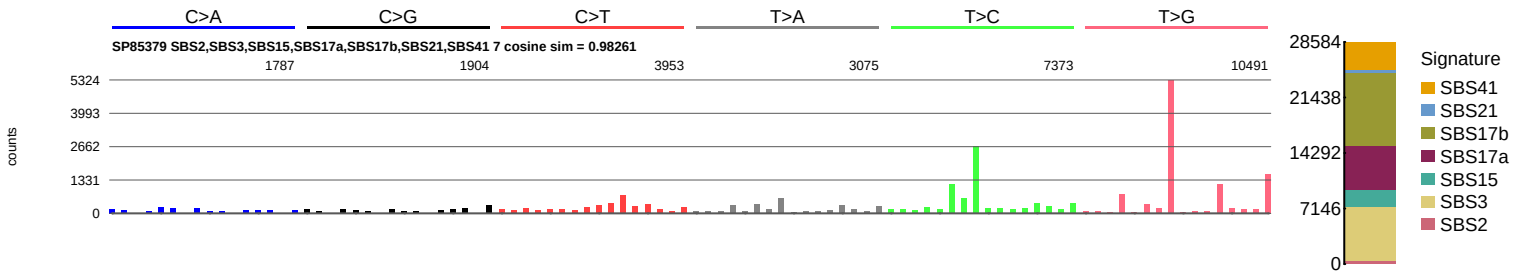
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98262
4327
348
5773
1718
6657
9761
counts
5536
4152
2768
1384
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98261
1787
1904
3953
3075
7373
10491
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
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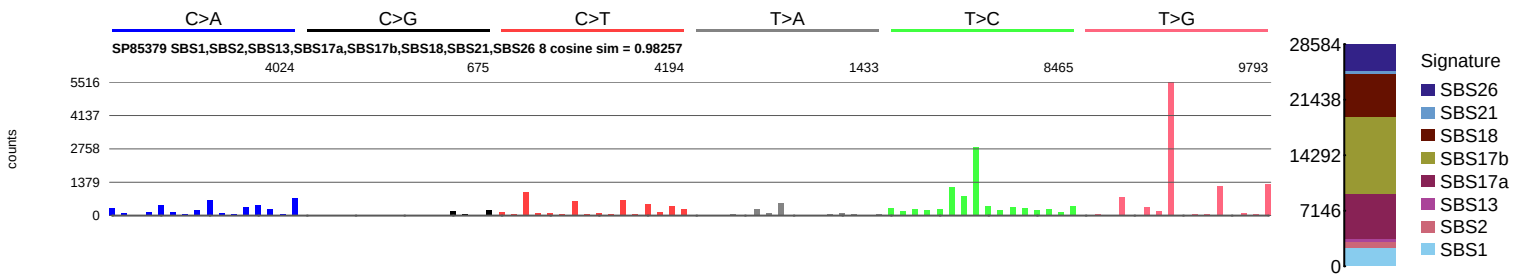
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98257
4024
675
4194
1433
8465
9793
counts
5516
4137
2758
1379
0
28584
21438
14292
7146
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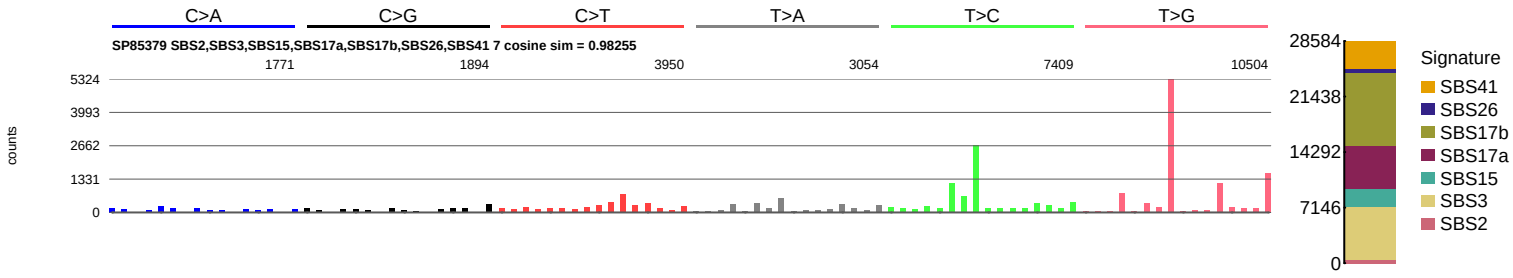
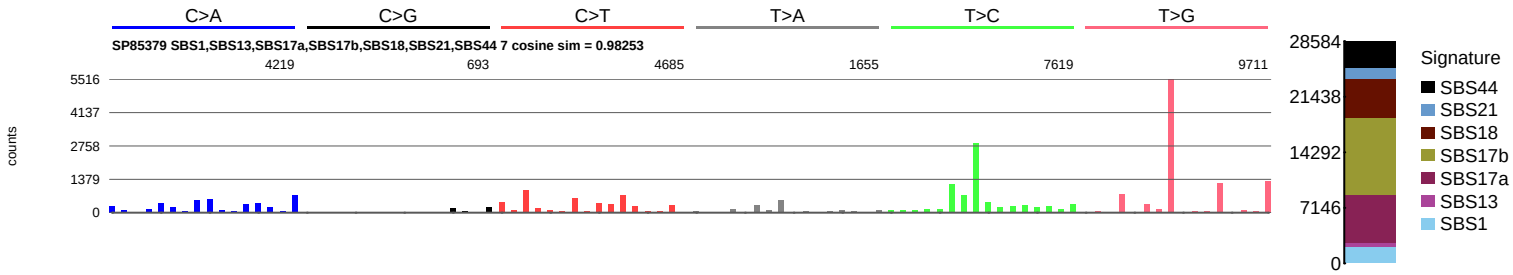
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98255
1771
1894
3950
3054
7409
10504
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98253
4219
693
4685
1655
7619
9711
counts
5516
4137
2758
1379
0
28584
21438
14292
7146
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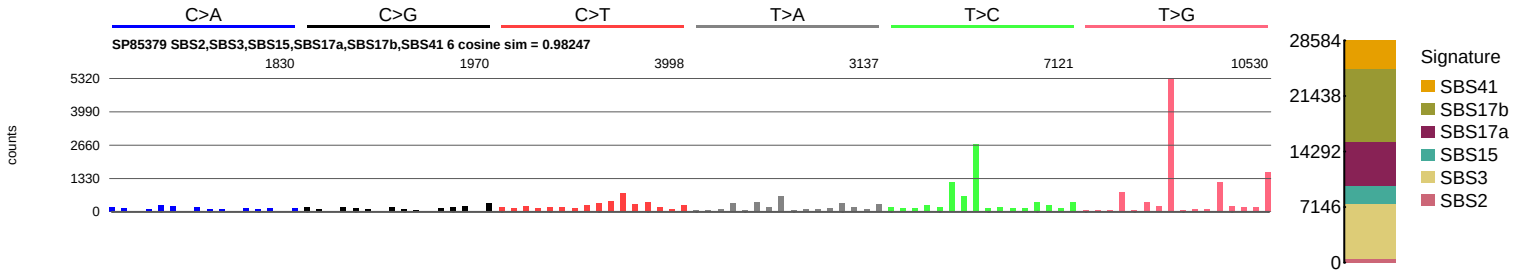
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98247
1830
1970
3998
3137
7121
10530
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
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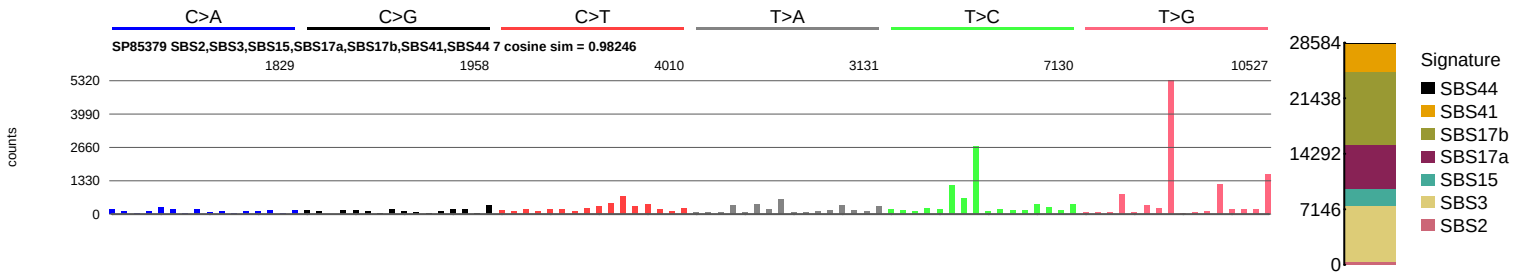
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98246
1829
1958
4010
3131
7130
10527
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
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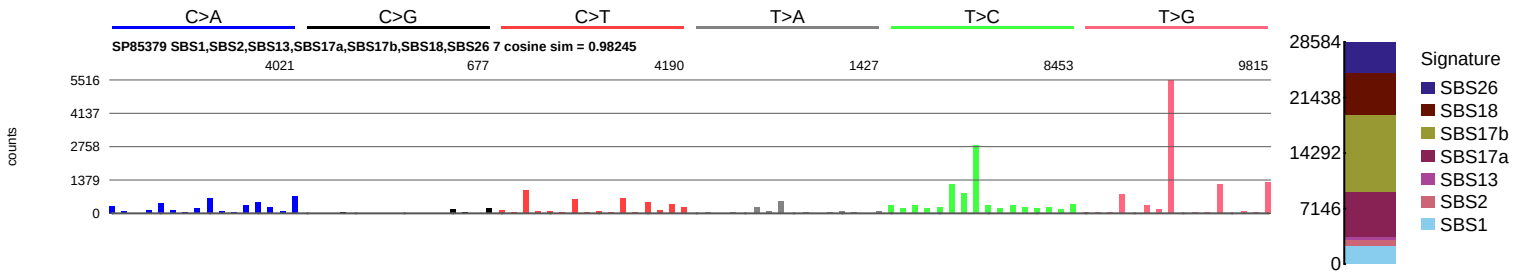
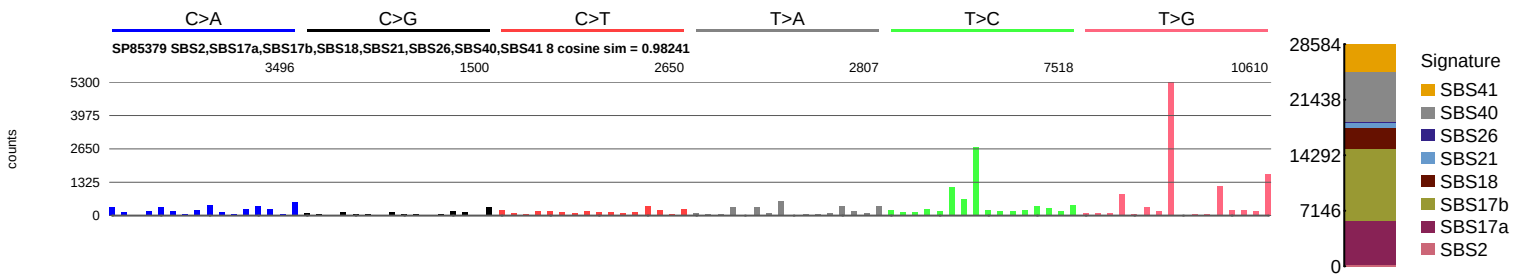
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98245
4021
677
4190
1427
8453
9815
counts
5516
4137
2758
1379
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98241
3496
1500
2650
2807
7518
10610
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
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SBS41
SBS40
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SBS21
SBS18
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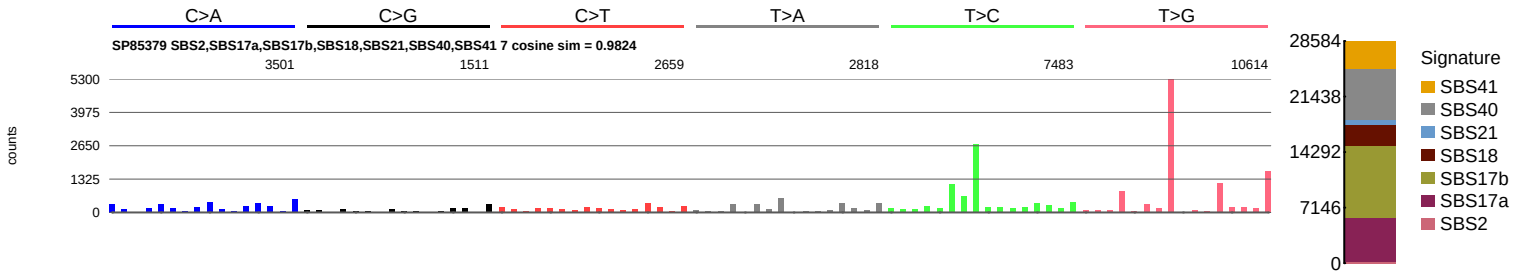
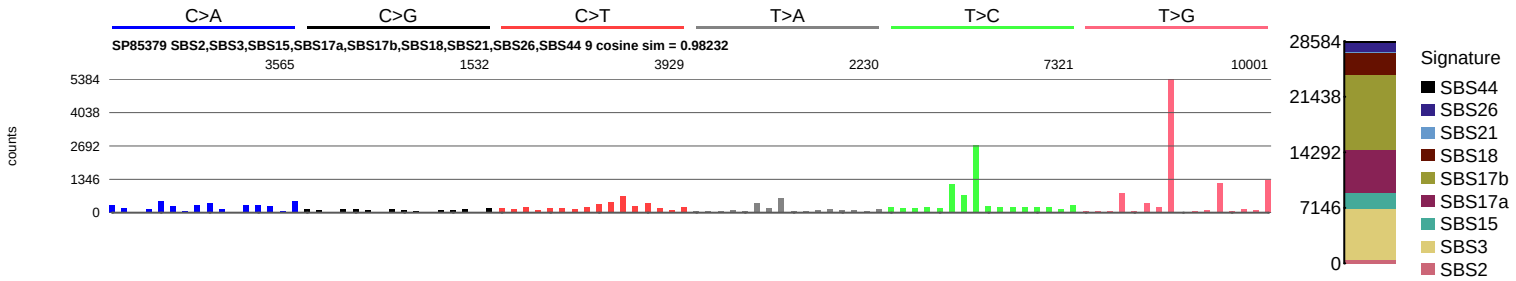
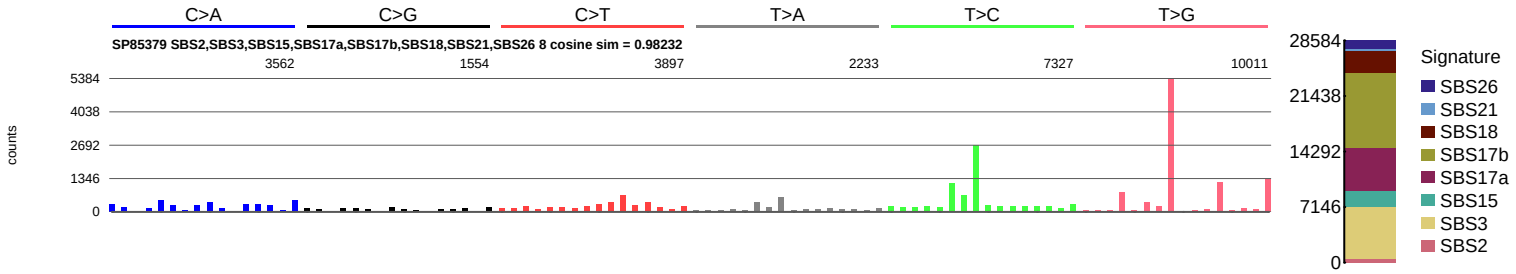
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.9824
3501
1511
2659
2818
7483
10614
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98232
3565
1532
3929
2230
7321
10001
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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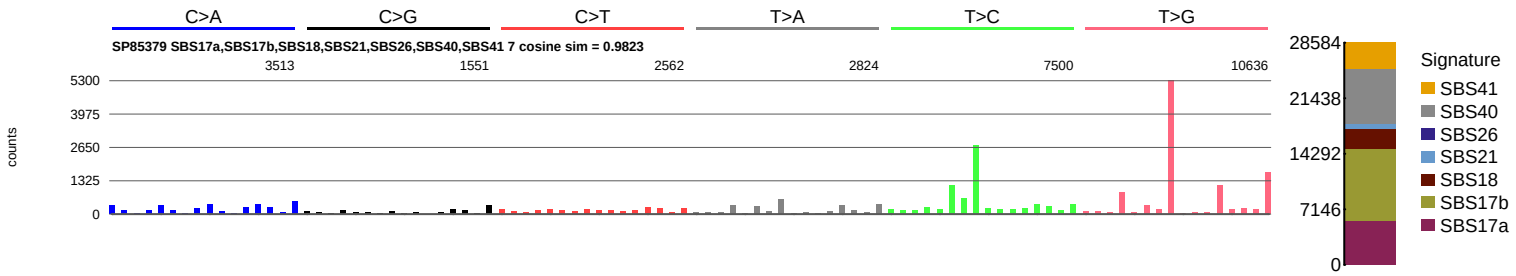
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98232
3562
1554
3897
2233
7327
10011
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
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T>C
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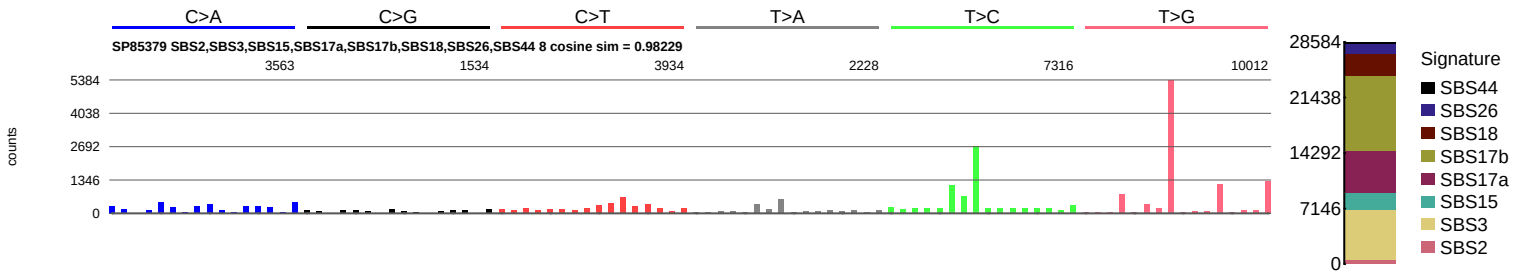
SP85379 SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.9823
3513
1551
2562
2824
7500
10636
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a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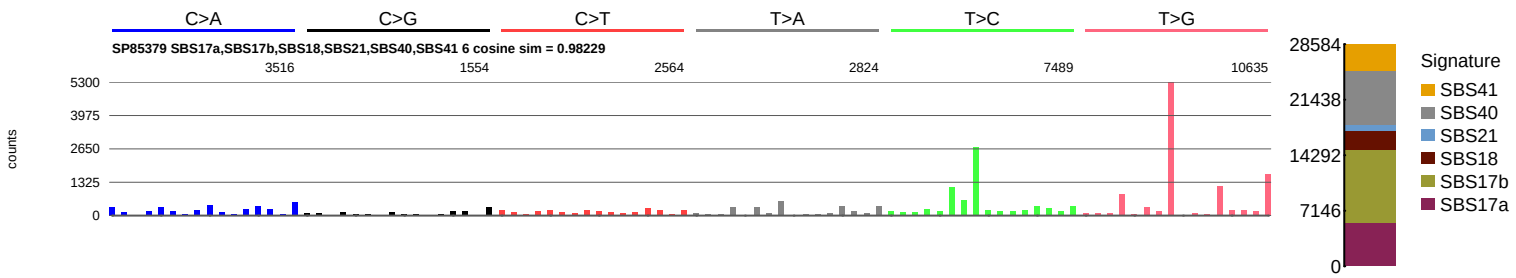
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98229
3563
1534
3934
2228
7316
10012
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
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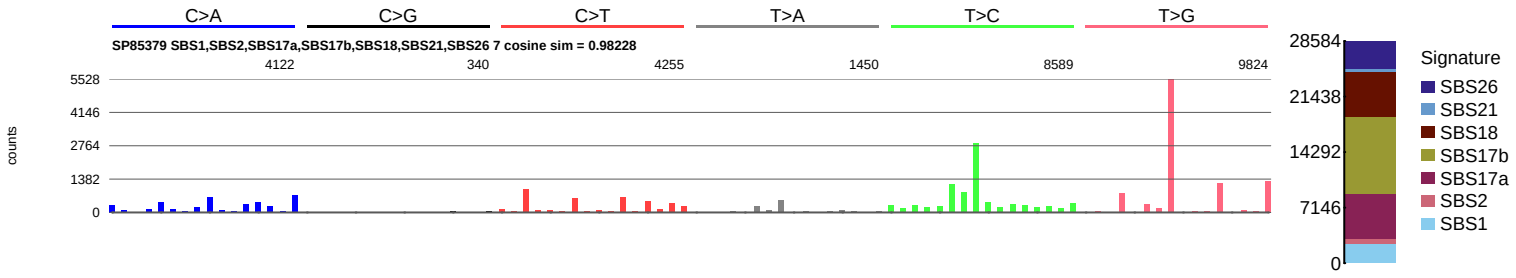
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98229
3516
1554
2564
2824
7489
10635
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
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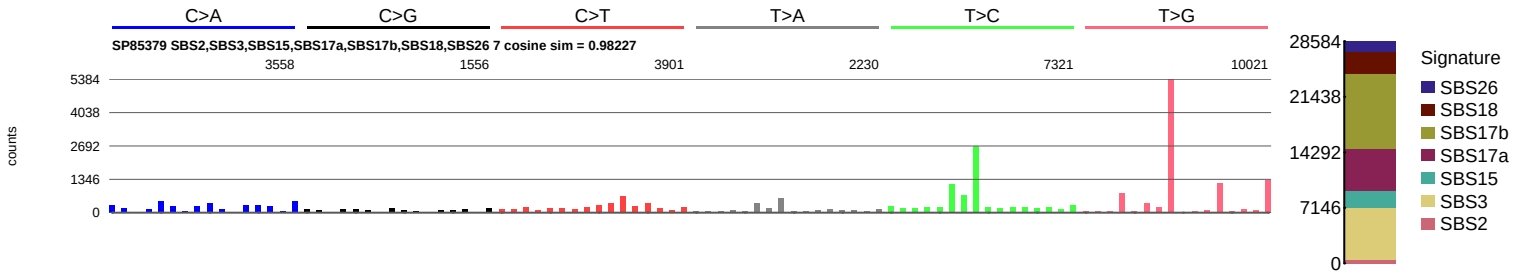
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98228
4122
340
4255
1450
8589
9824
counts
5528
4146
2764
1382
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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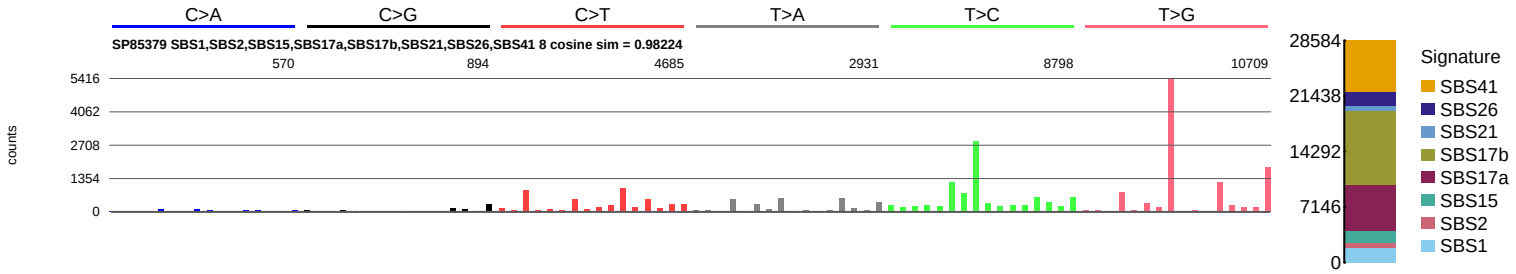
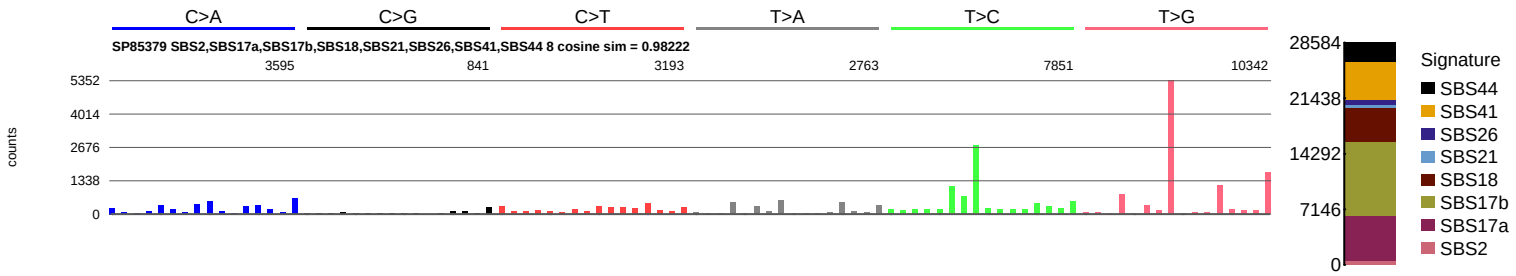
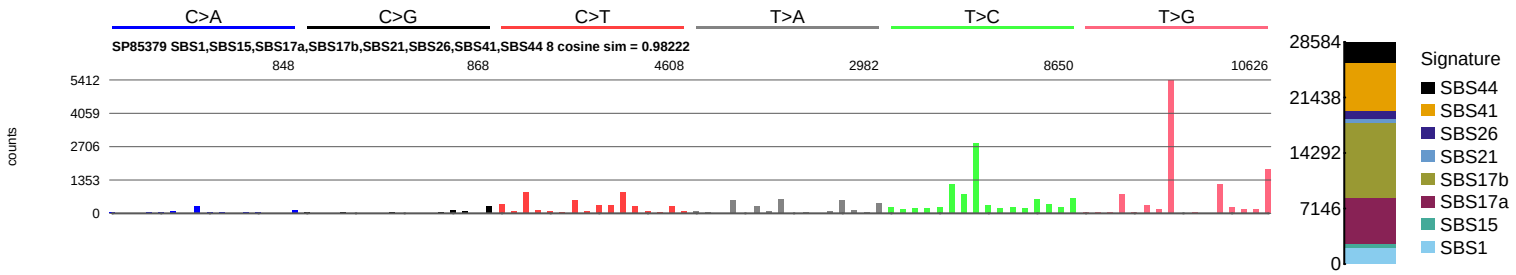
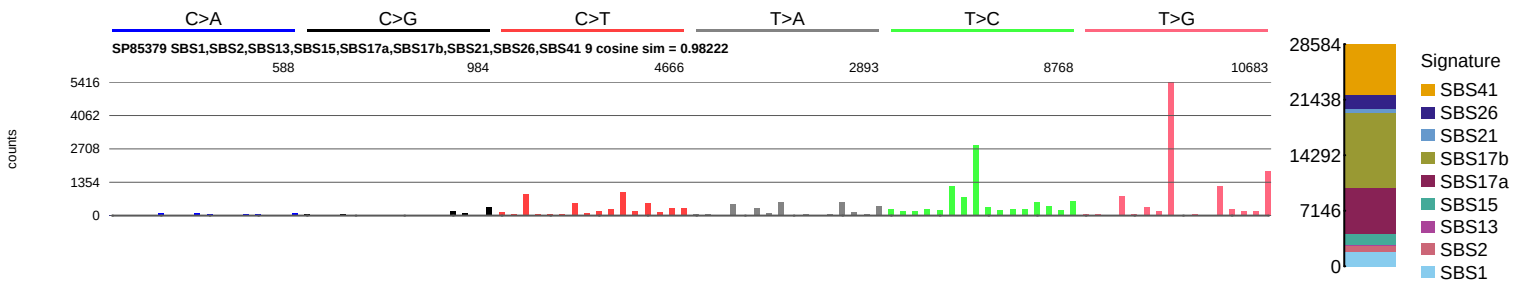
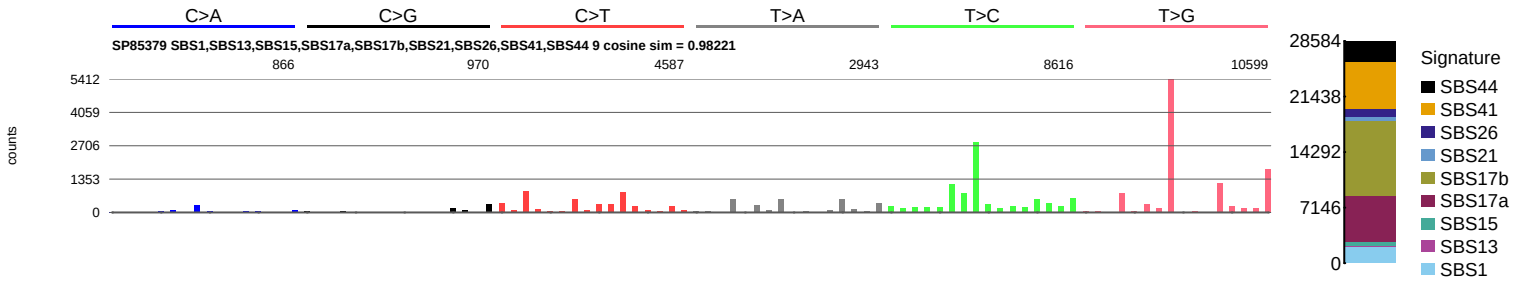
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98227
3558
1556
3901
2230
7321
10021
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

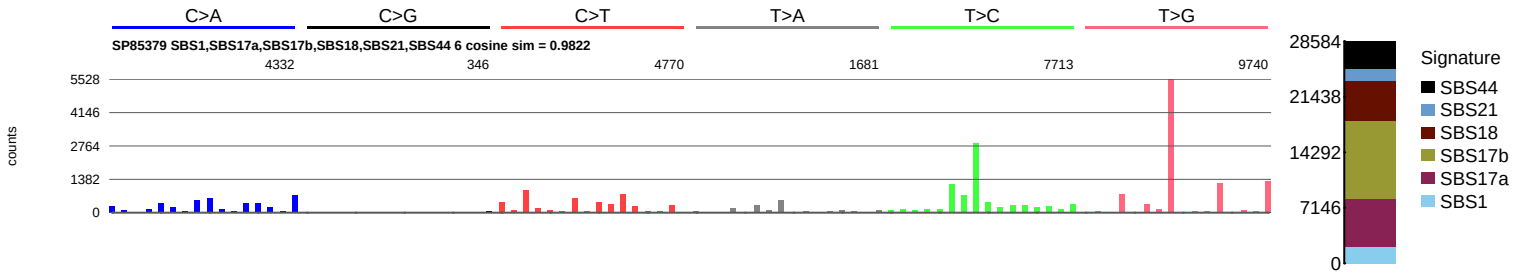

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98224
570
894
4685
2931
8798
10709
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

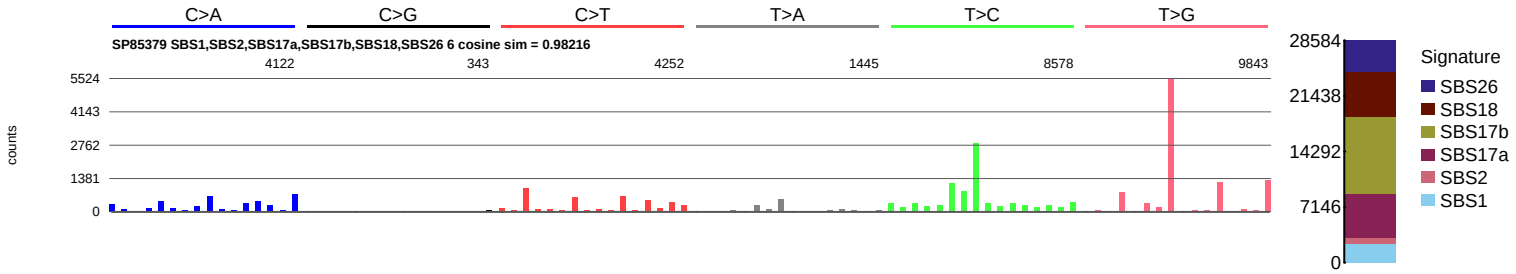

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98222
3595
841
3193
2763
7851
10342
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2

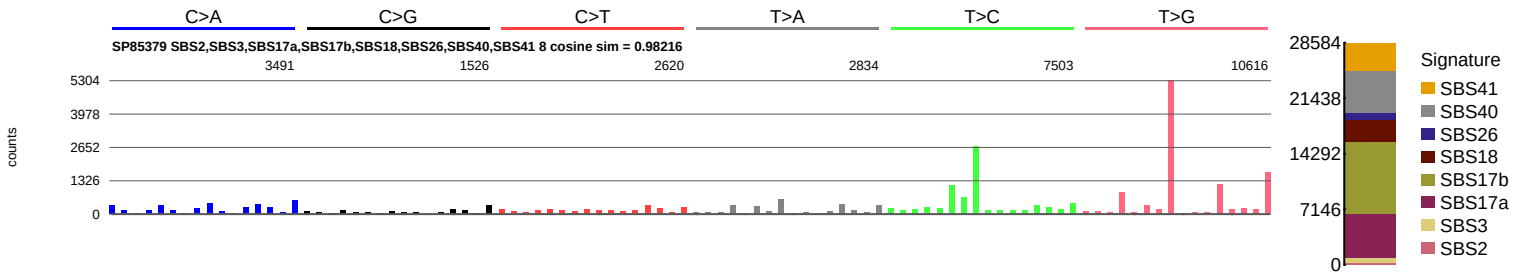

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98222
848
868
4608
2982
8650
10626
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

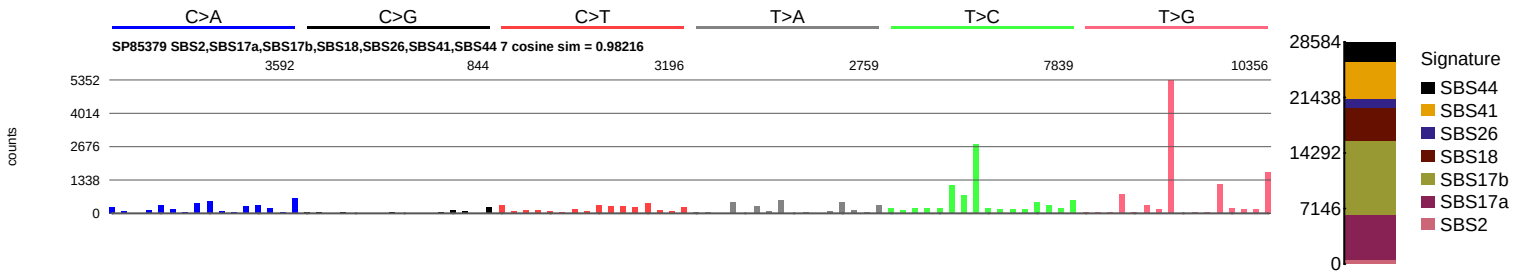

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.98222
588
984
4666
2893
8768
10683
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

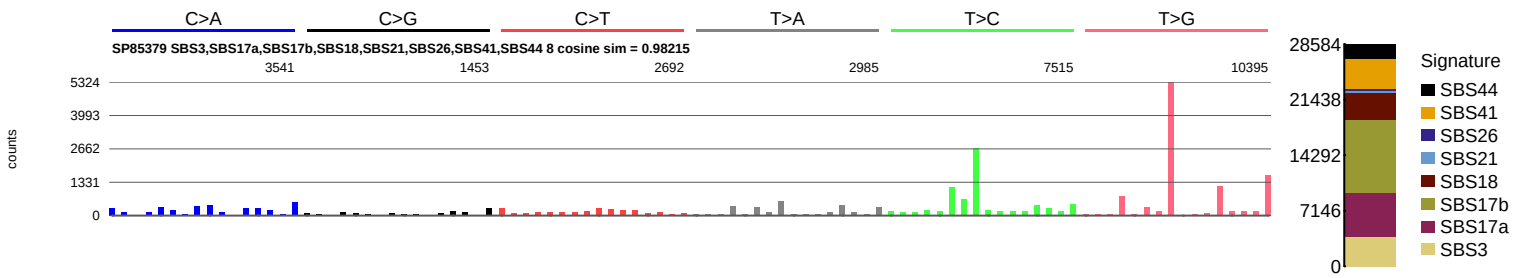

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98221
866
970
4587
2943
8616
10599
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

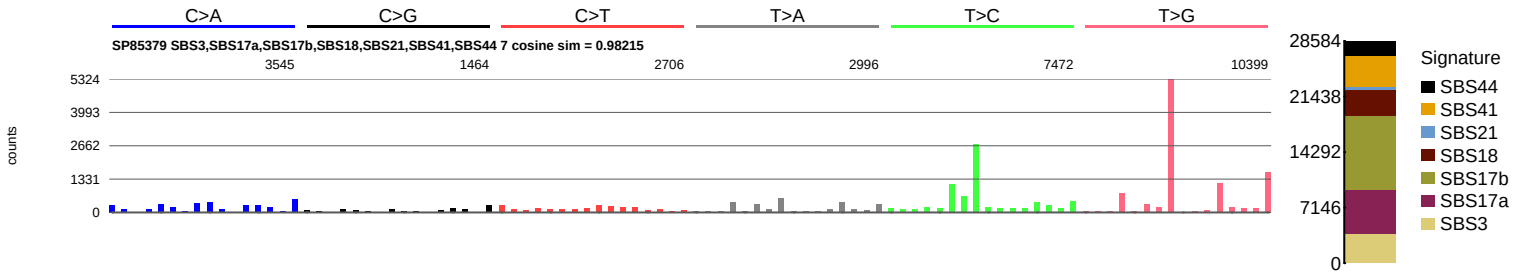

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS44 6 cosine sim = 0.9822
4332
346
4770
1681
7713
9740
counts
5528
4146
2764
1382
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS1

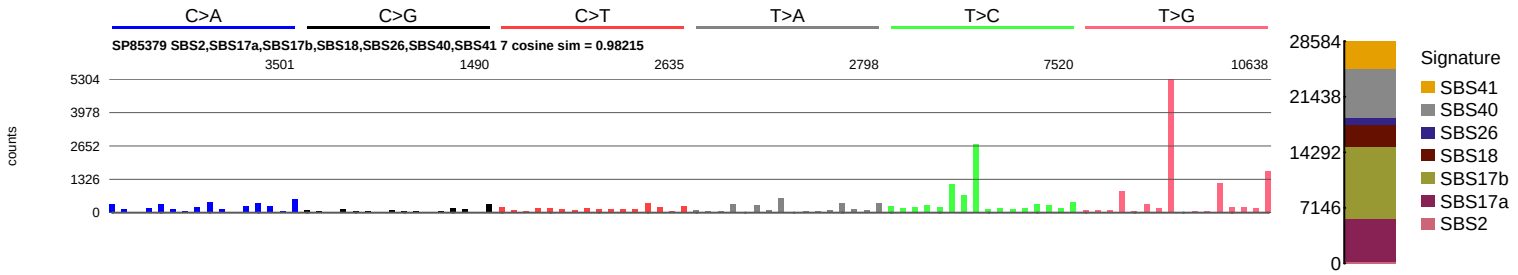

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98216
4122
343
4252
1445
8578
9843
counts
5524
4143
2762
1381
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

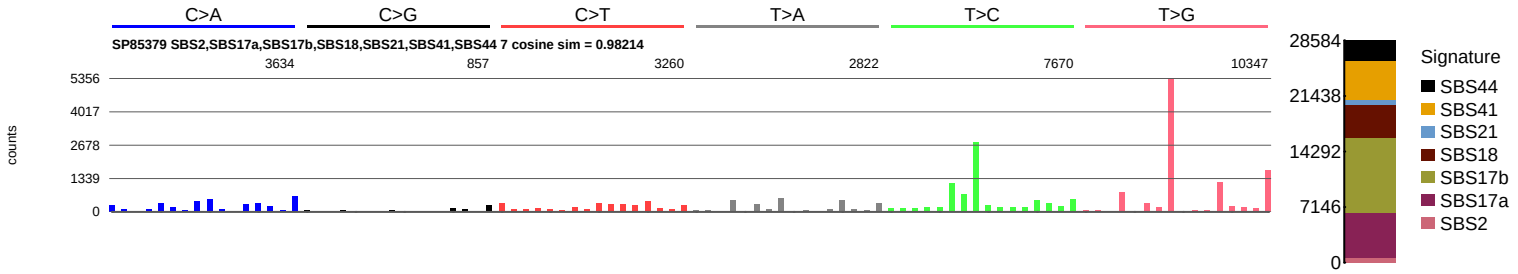

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98216
3491
1526
2620
2834
7503
10616
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

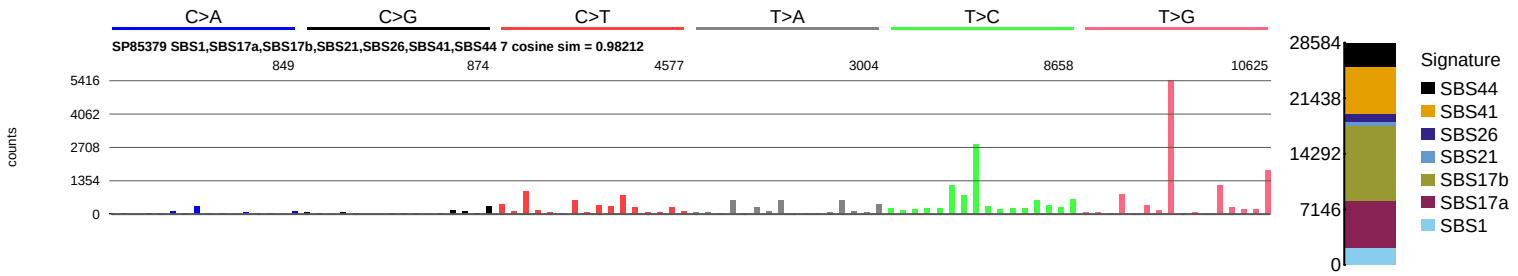

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98216
3592
844
3196
2759
7839
10356
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2

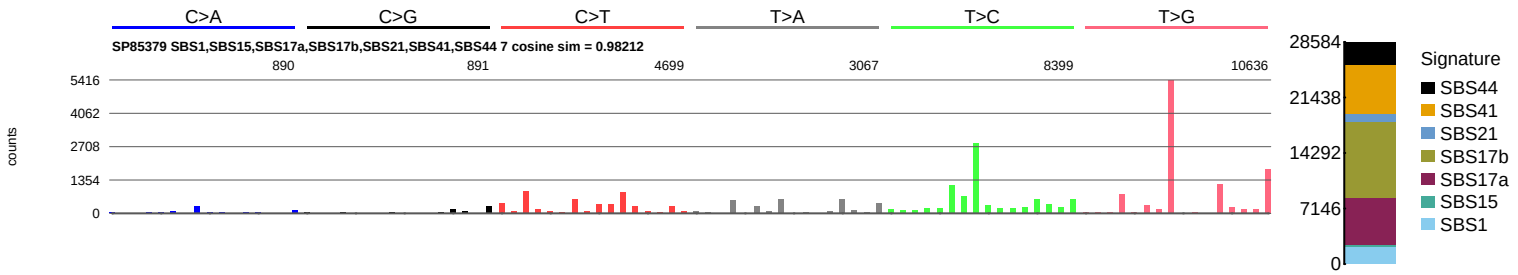

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98215
3541
1453
2692
2985
7515
10395
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3

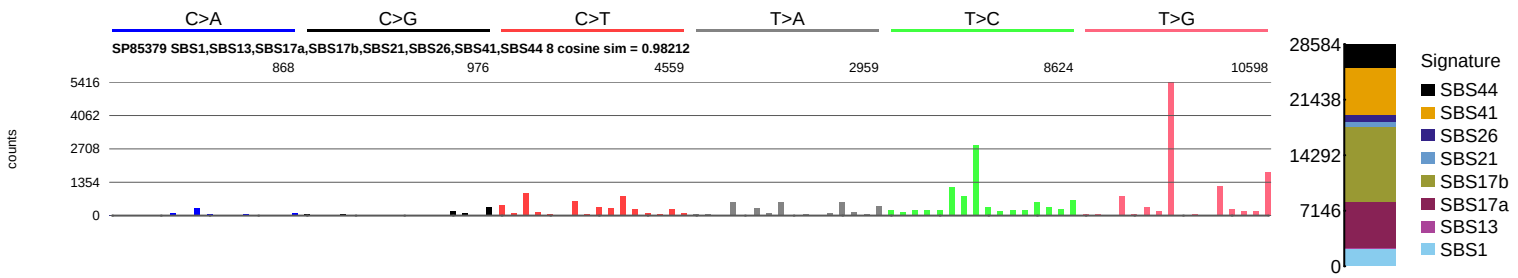

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98215
3545
1464
2706
2996
7472
10399
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3

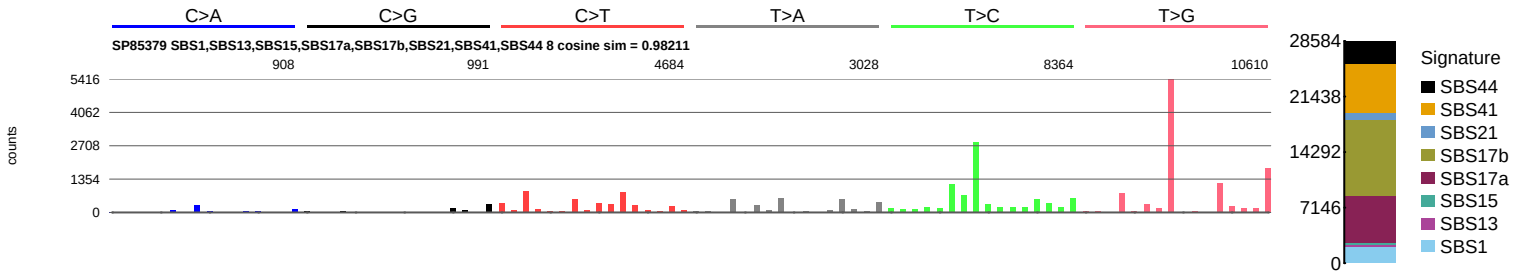

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98215
3501
1490
2635
2798
7520
10638
counts
5304
3978
2652
1326
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2

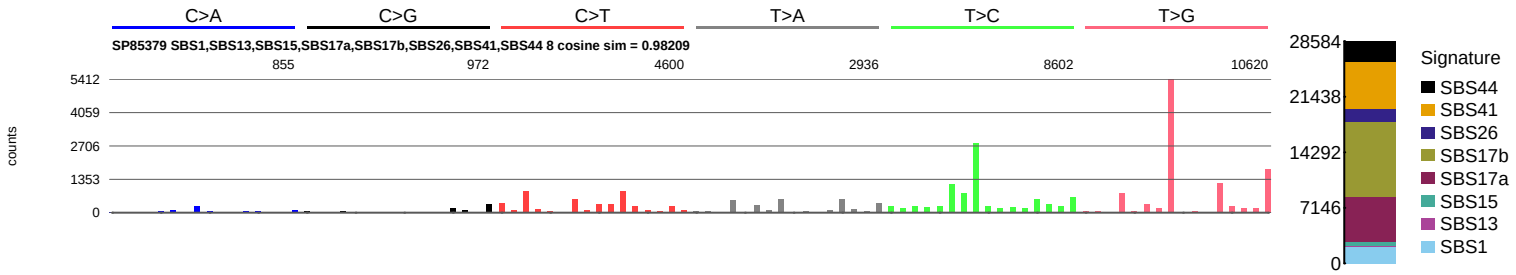

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98214
3634
857
3260
2822
7670
10347
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 7 cosine sim = 0.98212
849
874
4577
3004
8658
10625
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98212
890
891
4699
3067
8399
10636
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98212
868
976
4559
2959
8624
10598
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98211
908
991
4684
3028
8364
10610
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98209
855
972
4600
2936
8602
10620
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
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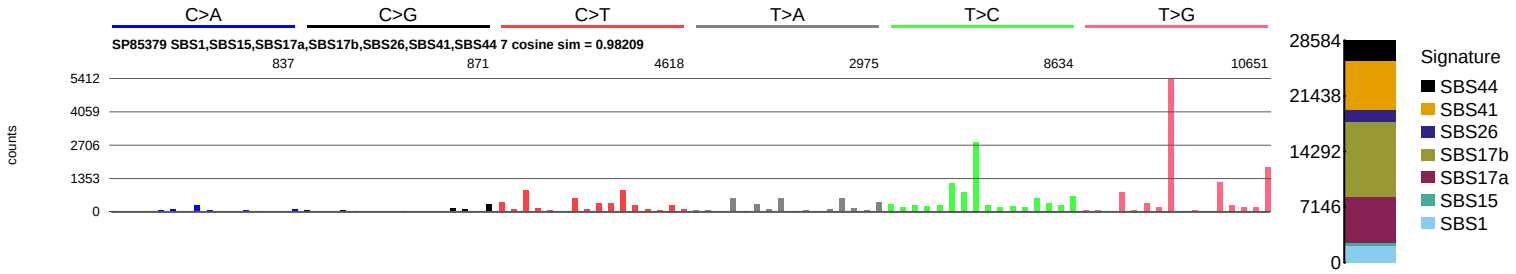
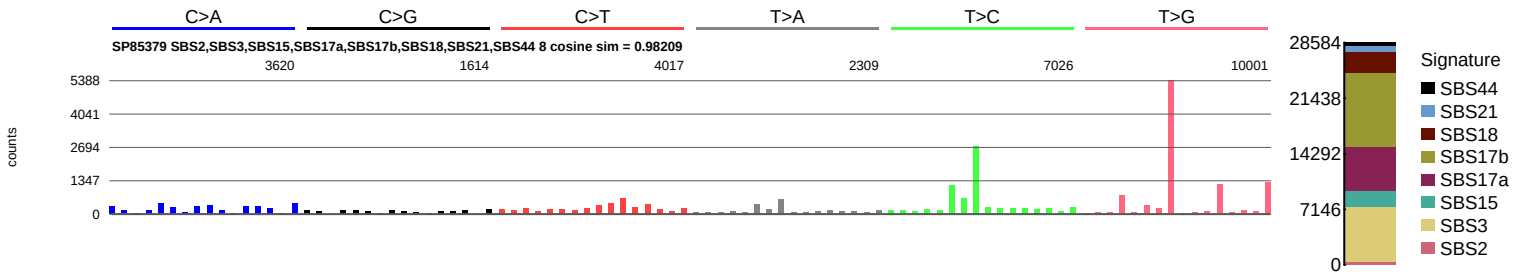
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98209
837
871
4618
2975
8634
10651
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS1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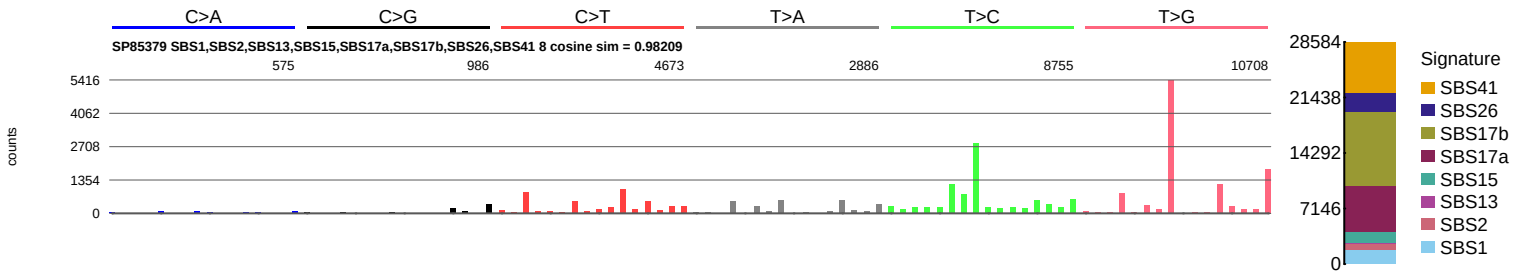
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98209
3620
1614
4017
2309
7026
10001
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
C>T
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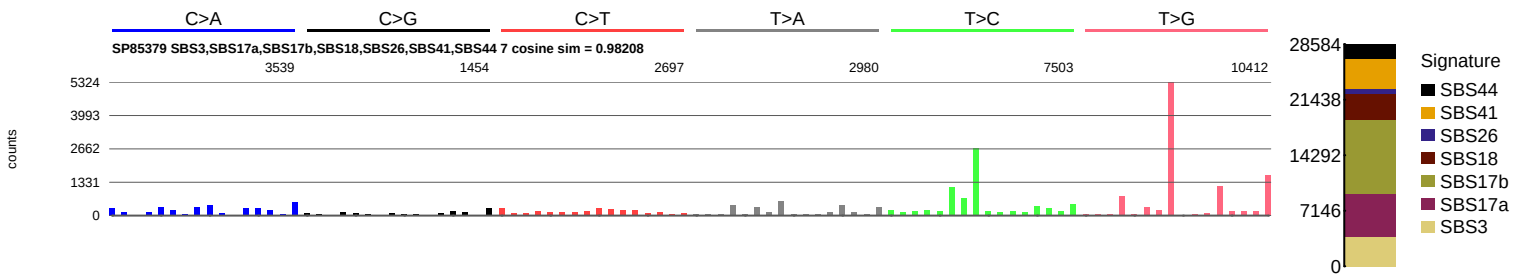
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98209
575
986
4673
2886
8755
10708
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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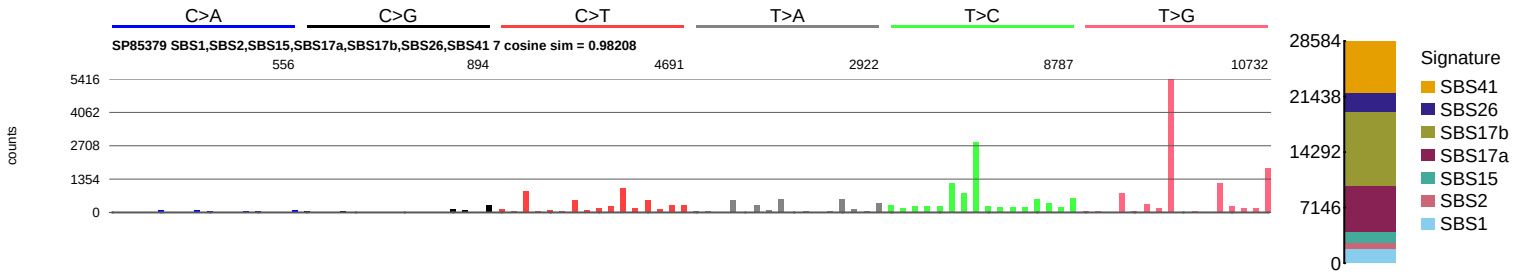
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98208
3539
1454
2697
2980
7503
10412
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
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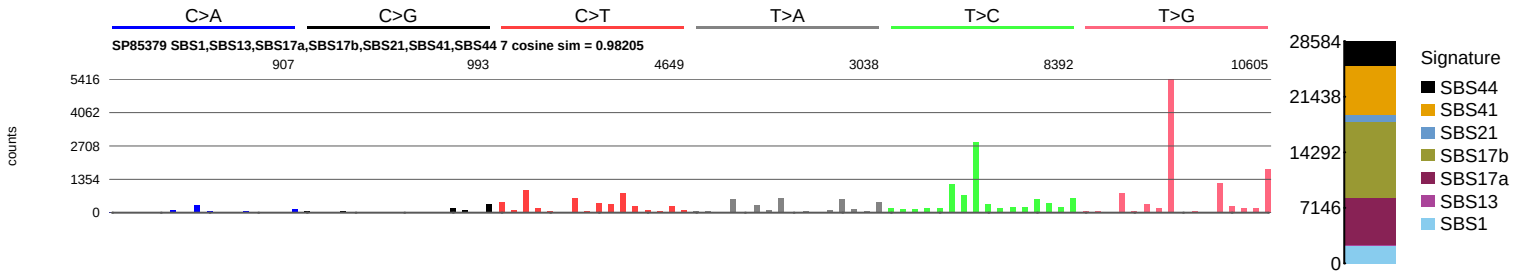
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98208
556
894
4691
2922
8787
10732
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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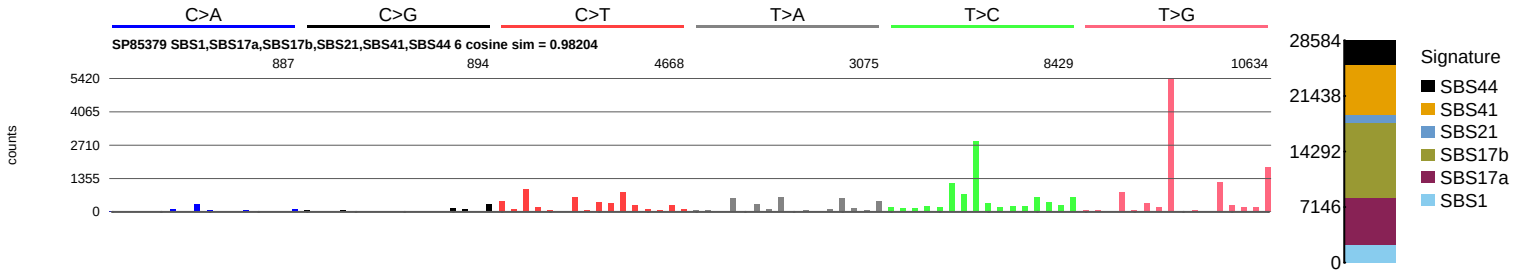
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98205
907
993
4649
3038
8392
10605
counts
5416
4062
2708
1354
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS1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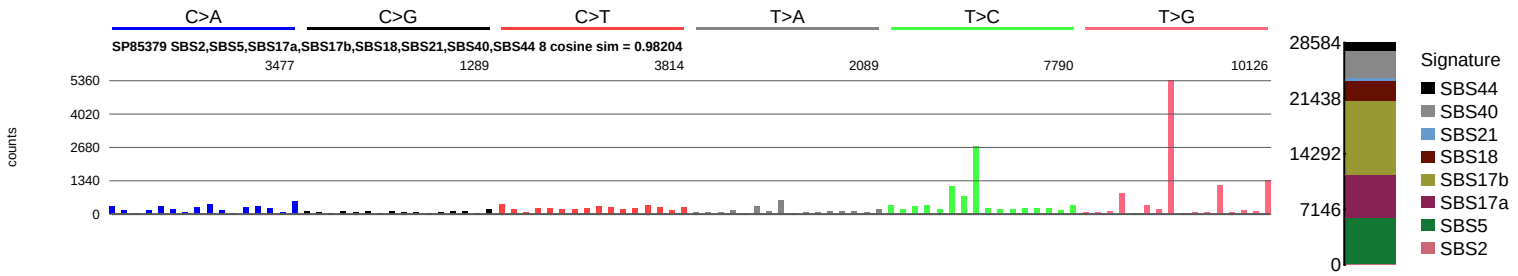
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98204
887
894
4668
3075
8429
10634
counts
5420
4065
2710
1355
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS1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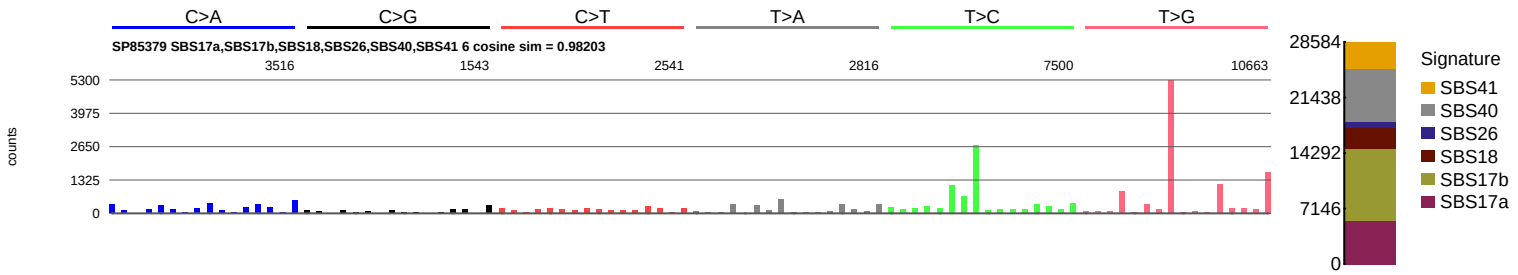
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98204
3477
1289
3814
2089
7790
10126
counts
5360
4020
2680
1340
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>G
C>T
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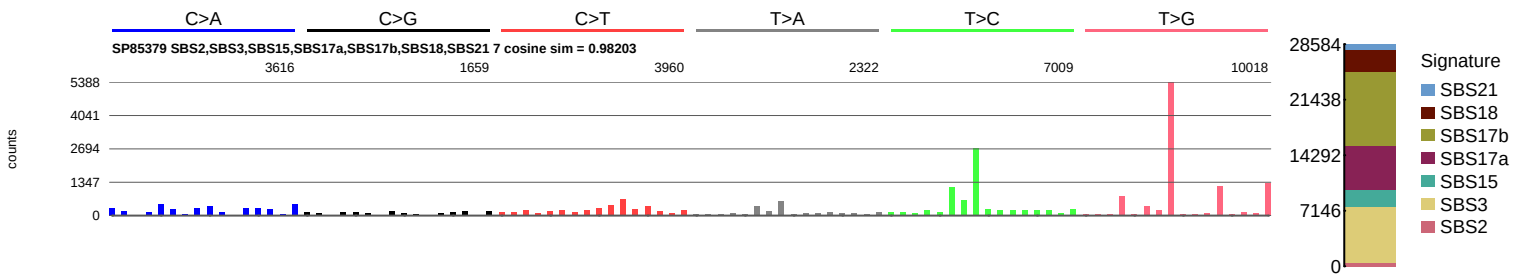
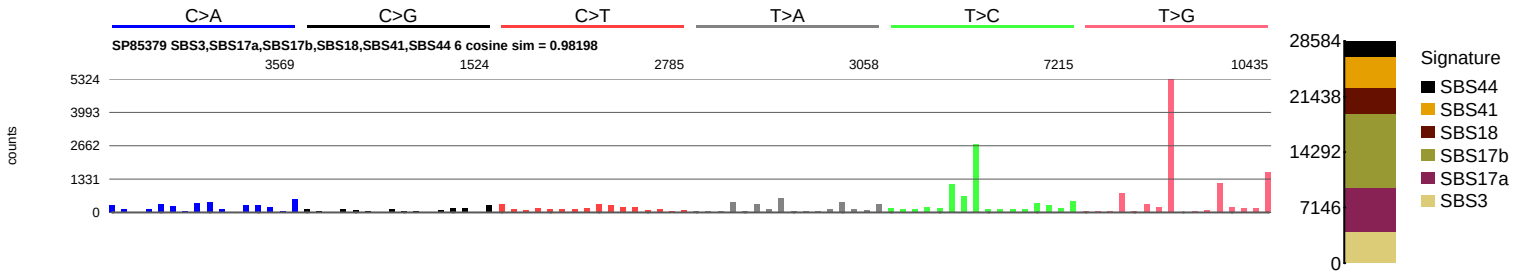
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98203
3516
1543
2541
2816
7500
10663
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98203
3616
1659
3960
2322
7009
10018
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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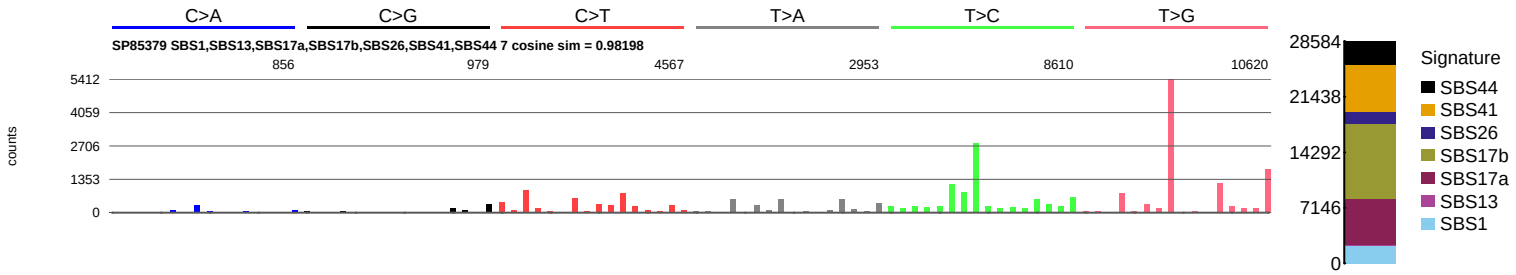
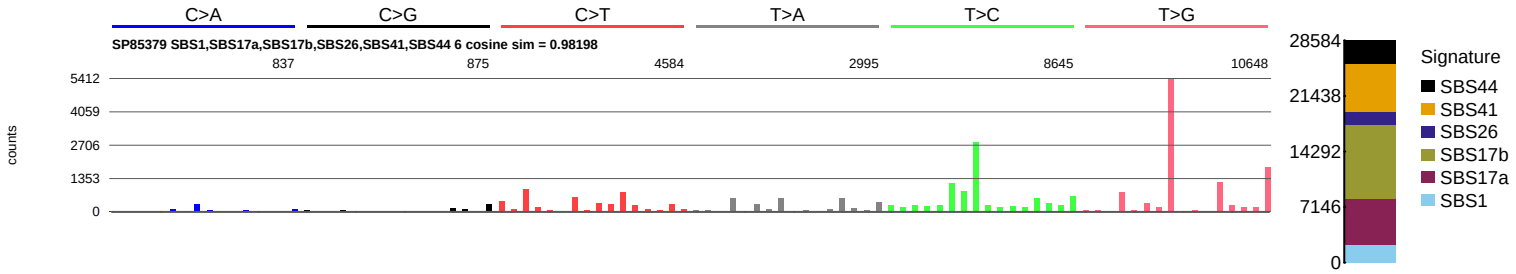
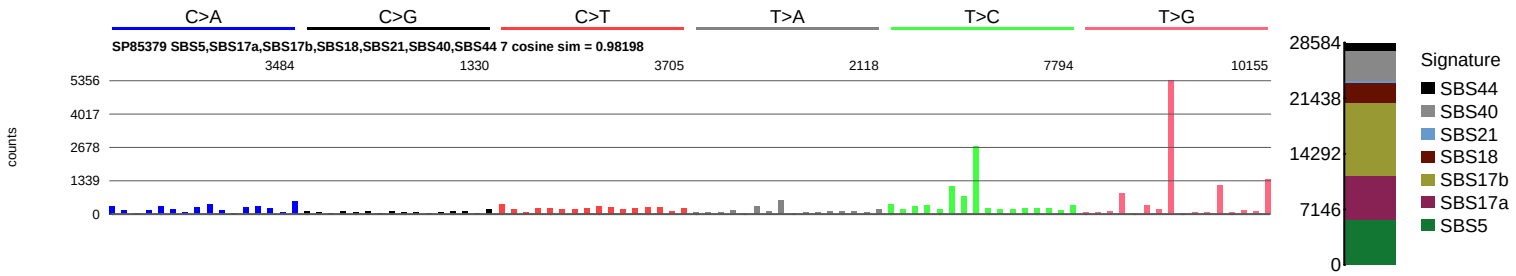
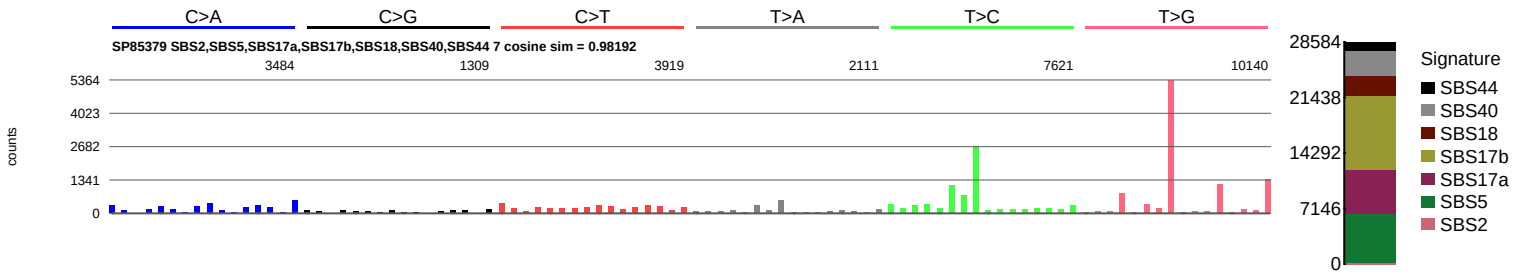
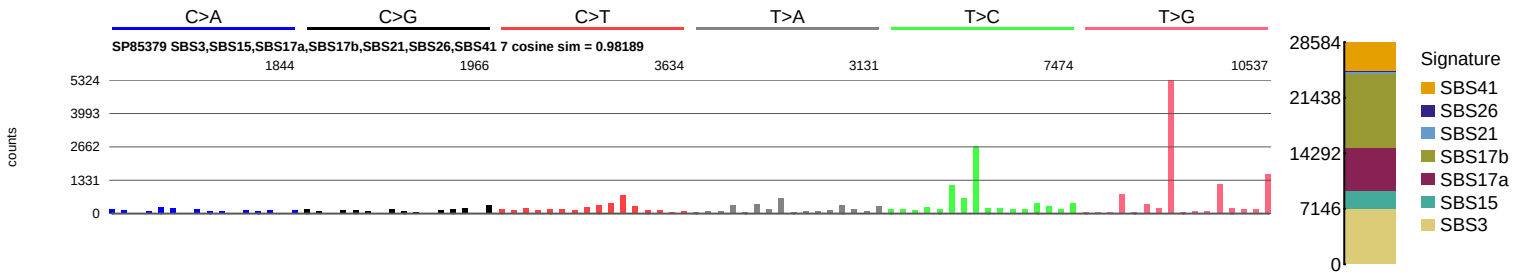
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98198
3569
1524
2785
3058
7215
10435
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3

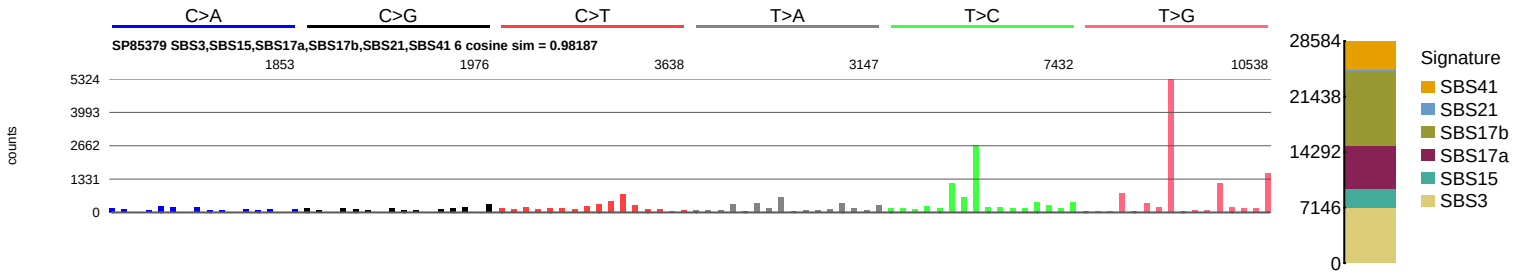

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98198
856
979
4567
2953
8610
10620
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS1

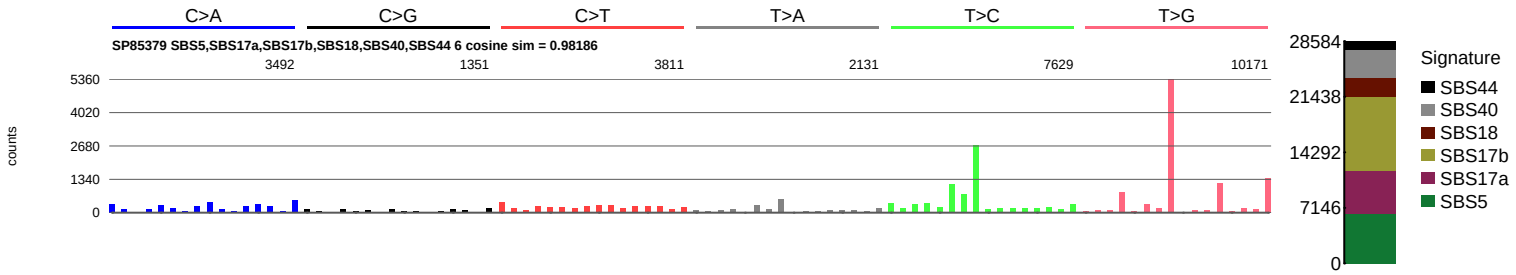

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98198
837
875
4584
2995
8645
10648
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS1

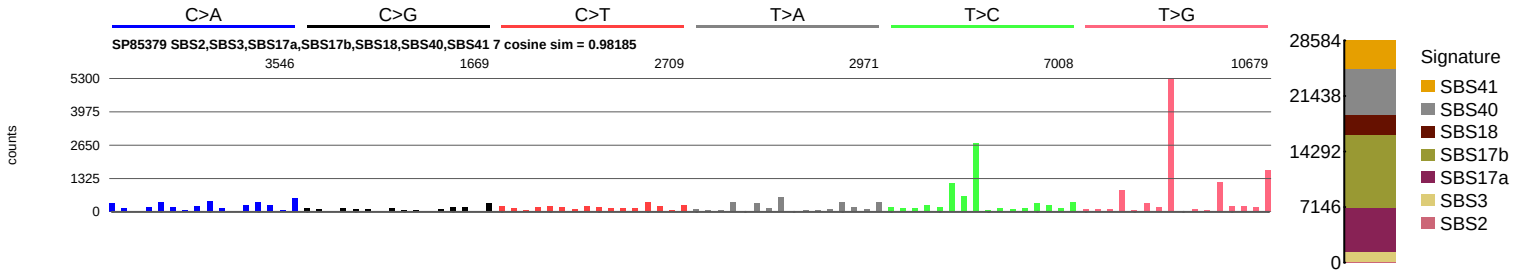

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98198
3484
1330
3705
2118
7794
10155
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5

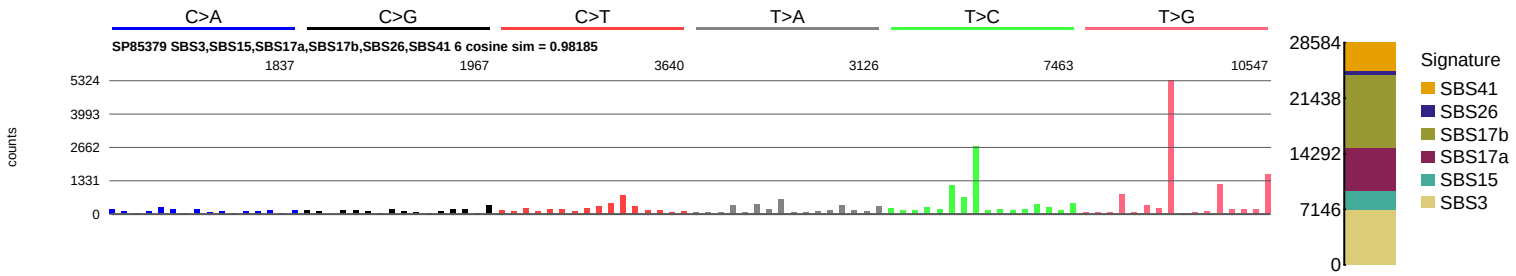

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98192
3484
1309
3919
2111
7621
10140
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2

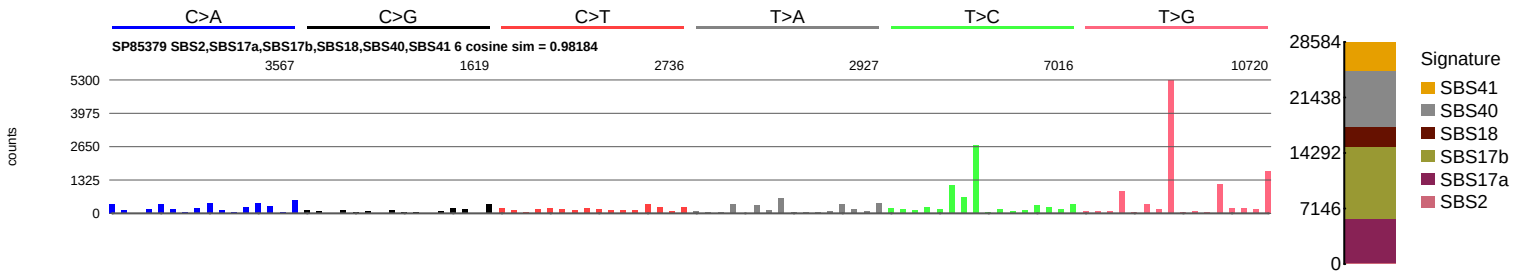

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98189
1844
1966
3634
3131
7474
10537
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3

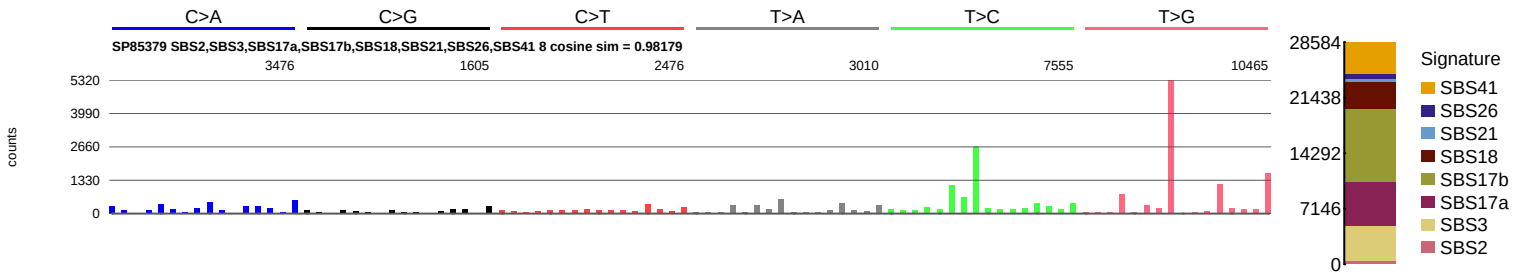

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98187
1853
1976
3638
3147
7432
10538
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3

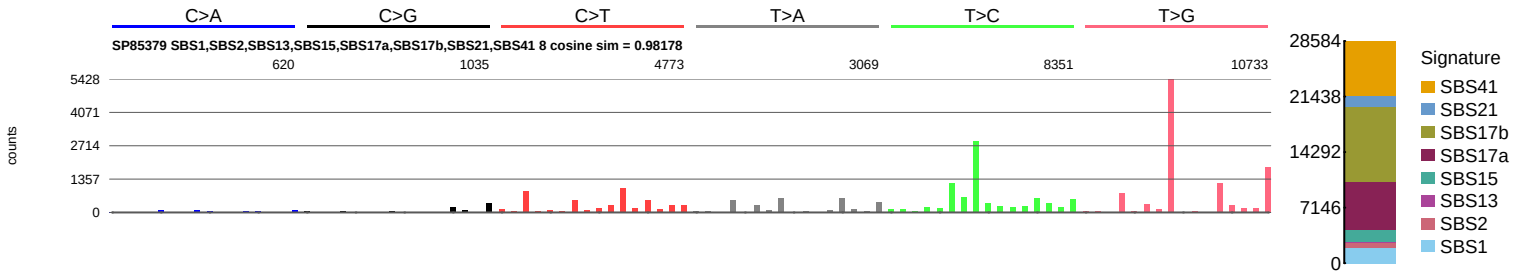

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98186
3492
1351
3811
2131
7629
10171
counts
5360
4020
2680
1340
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5

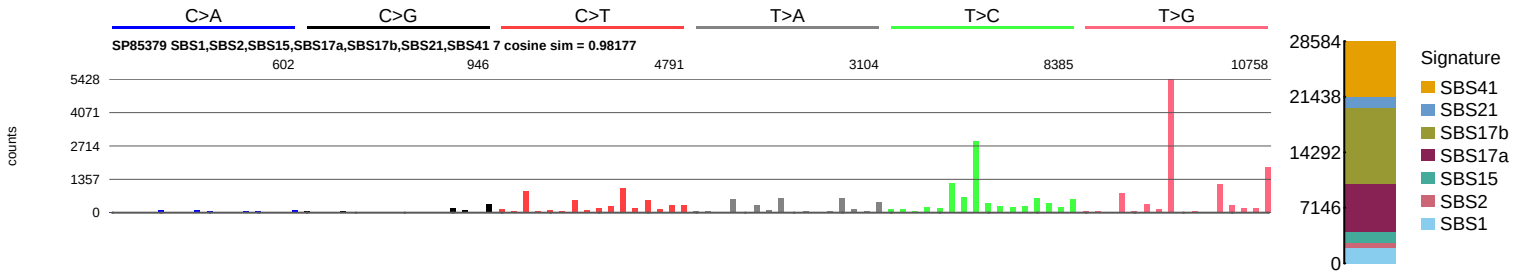

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98185
3546
1669
2709
2971
7008
10679
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98185
1837
1967
3640
3126
7463
10547
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98184
3567
1619
2736
2927
7016
10720
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98179
3476
1605
2476
3010
7555
10465
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98178
620
1035
4773
3069
8351
10733
counts
5428
4071
2714
1357
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98177
602
946
4791
3104
8385
10758
counts
5428
4071
2714
1357
0
28584
21438
14292
7146
0
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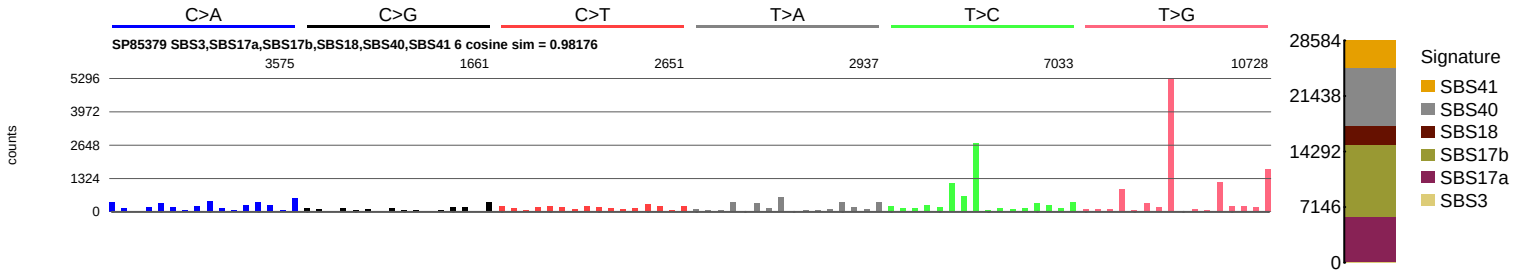
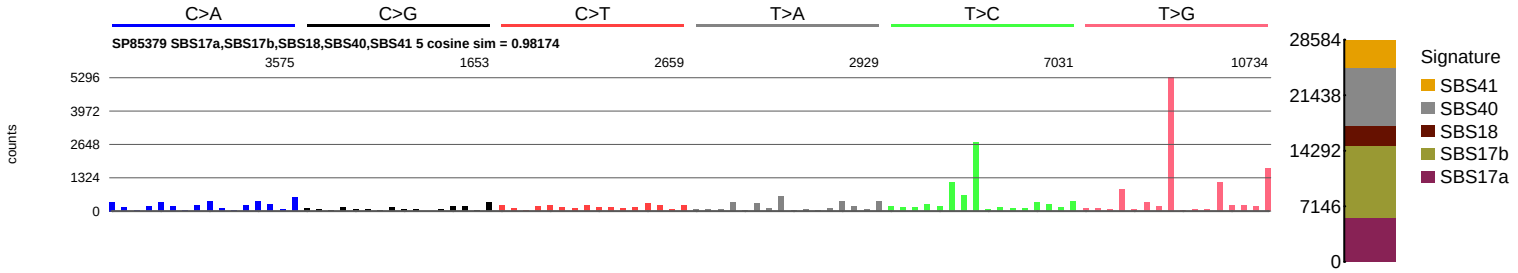
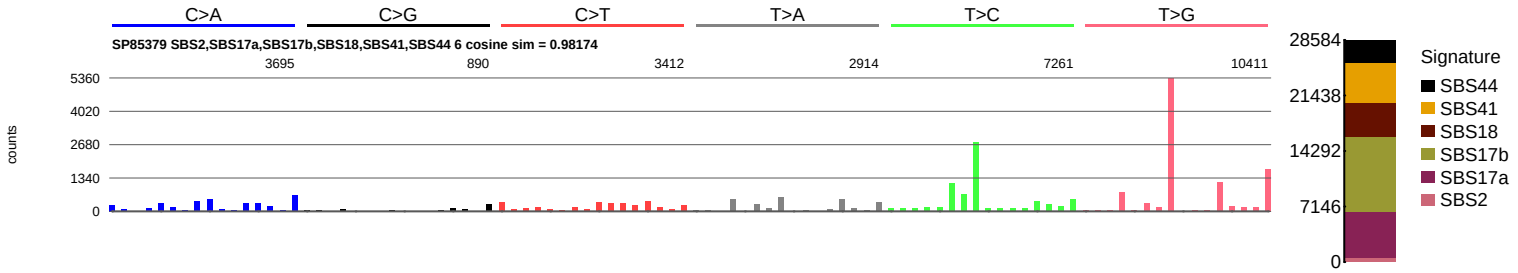
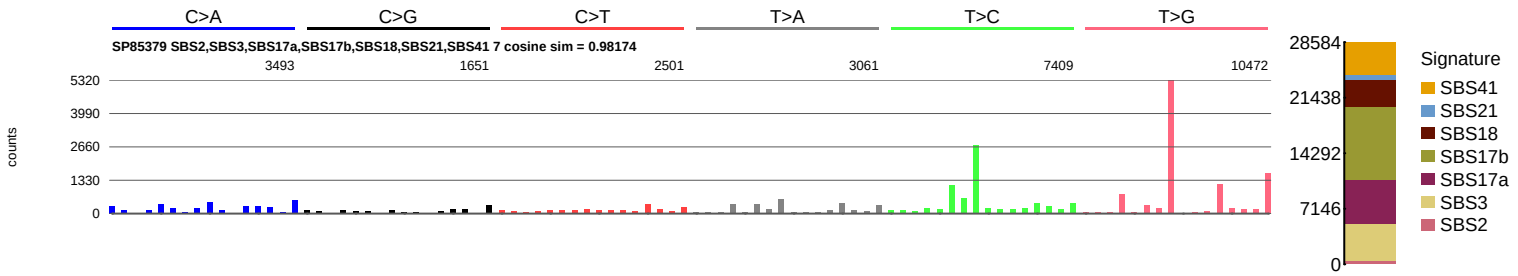
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98176
3575
1661
2651
2937
7033
10728
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS40,SBS41 5 cosine sim = 0.98174
3575
1653
2659
2929
7031
10734
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98174
3695
890
3412
2914
7261
10411
counts
5360
4020
2680
1340
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98174
3493
1651
2501
3061
7409
10472
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
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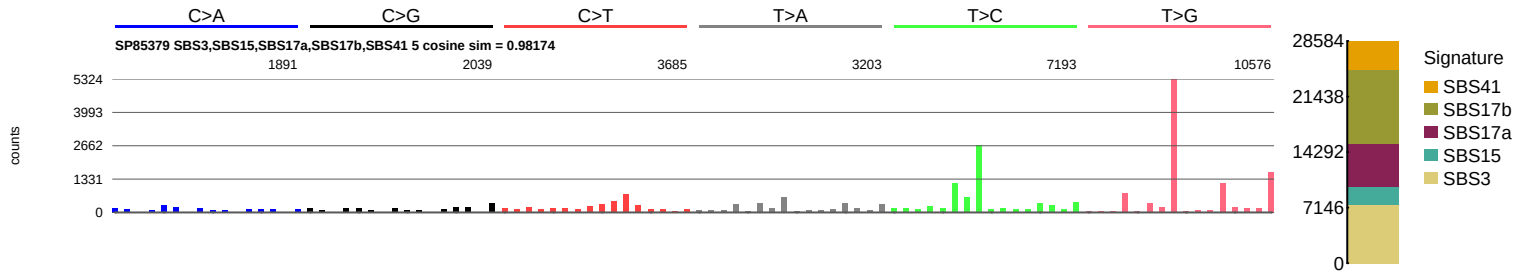
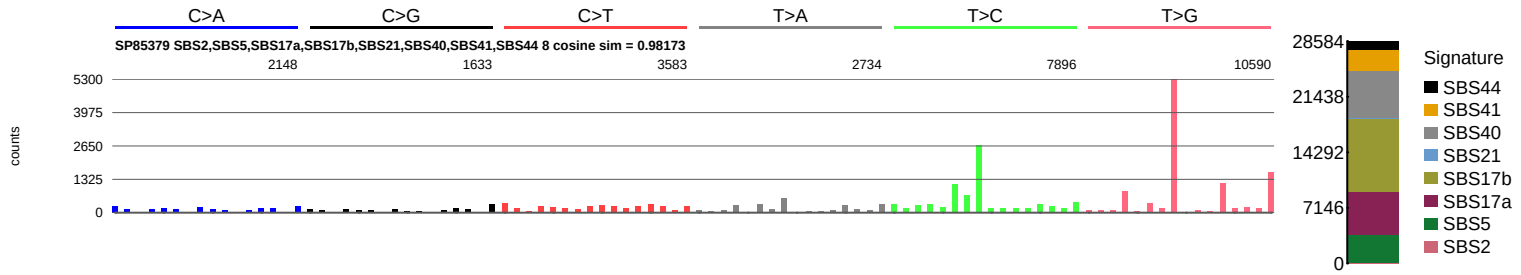
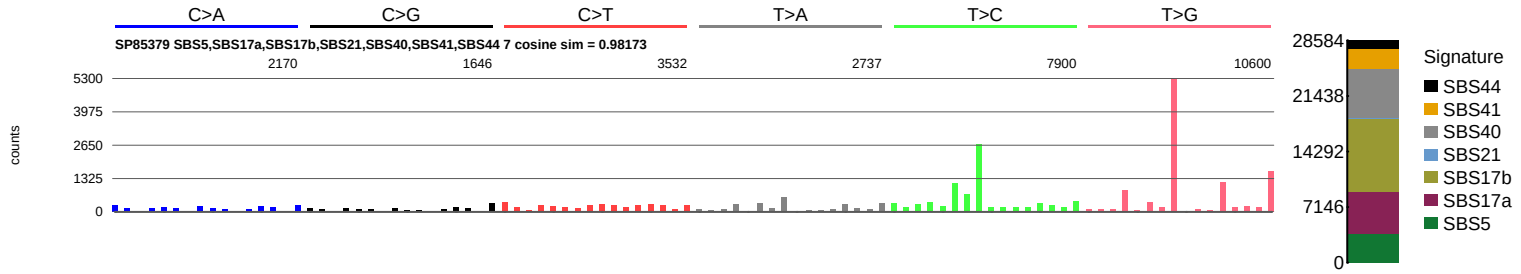
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98174
1891
2039
3685
3203
7193
10576
counts
5324
3993
2662
1331
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98173
2148
1633
3583
2734
7896
10590
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98173
2170
1646
3532
2737
7900
10600
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
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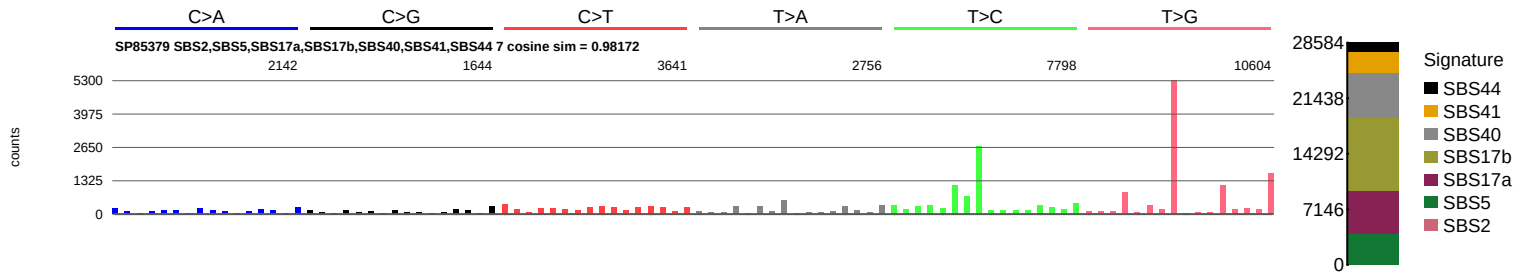
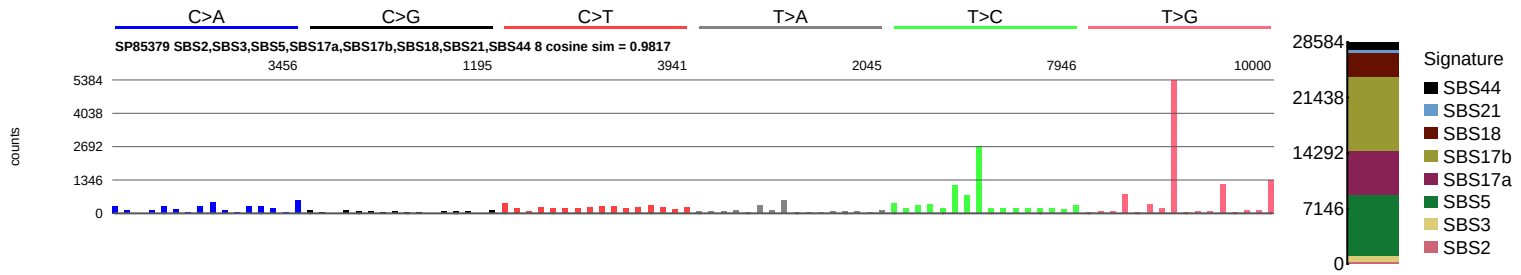
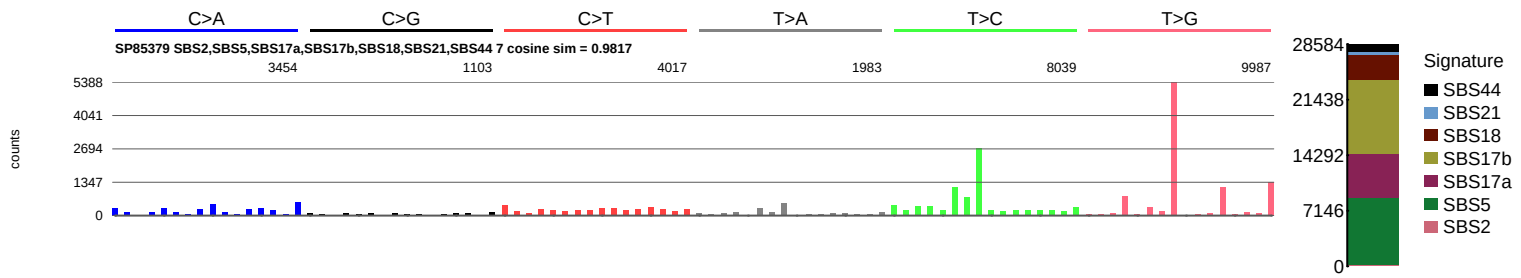
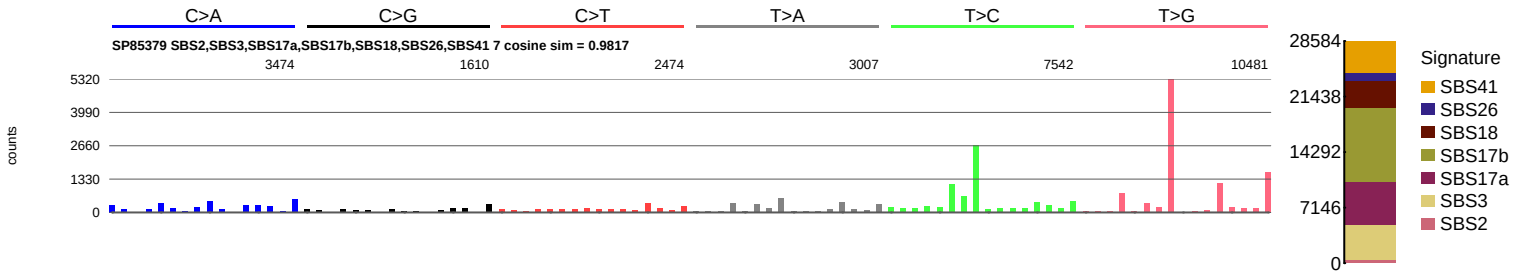
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98172
2142
1644
3641
2756
7798
10604
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.9817
3456
1195
3941
2045
7946
10000
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.9817
3454
1103
4017
1983
8039
9987
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9817
3474
1610
2474
3007
7542
10481
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
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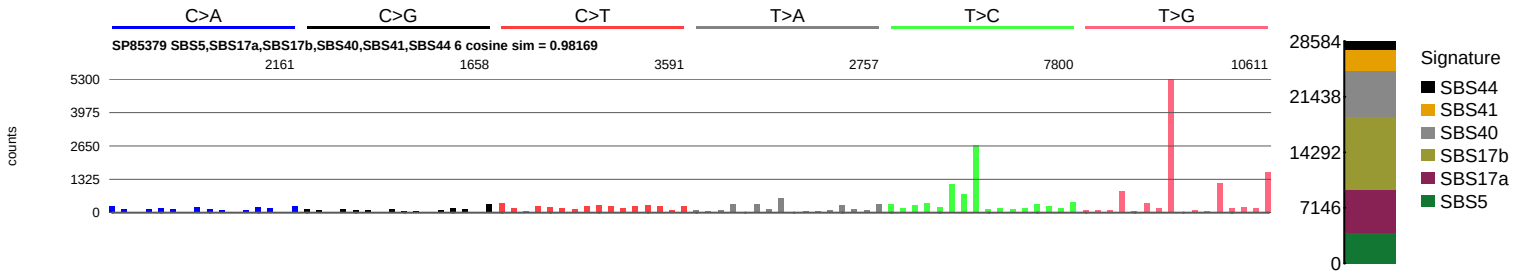
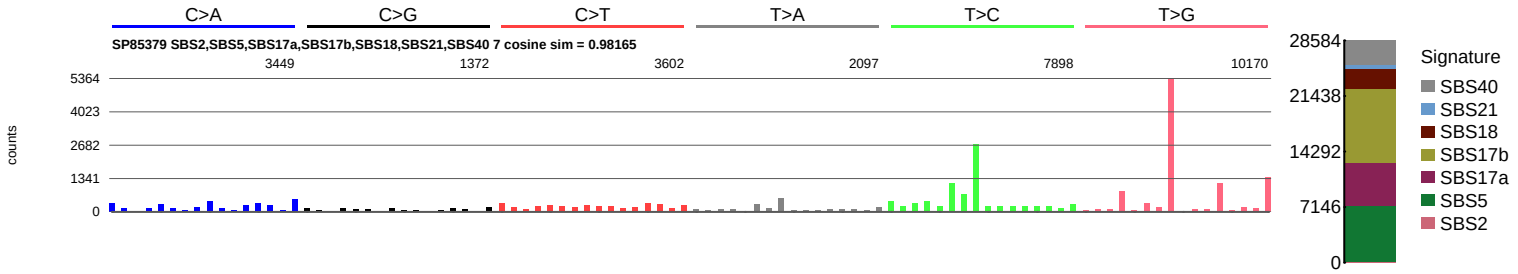
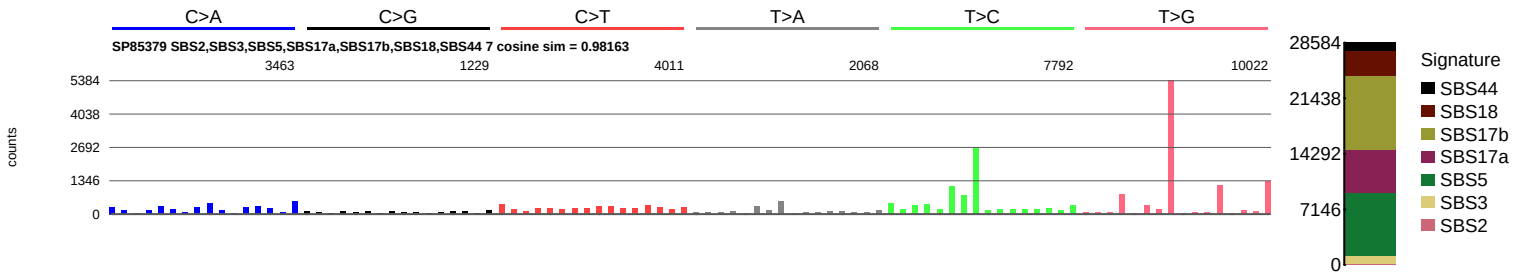
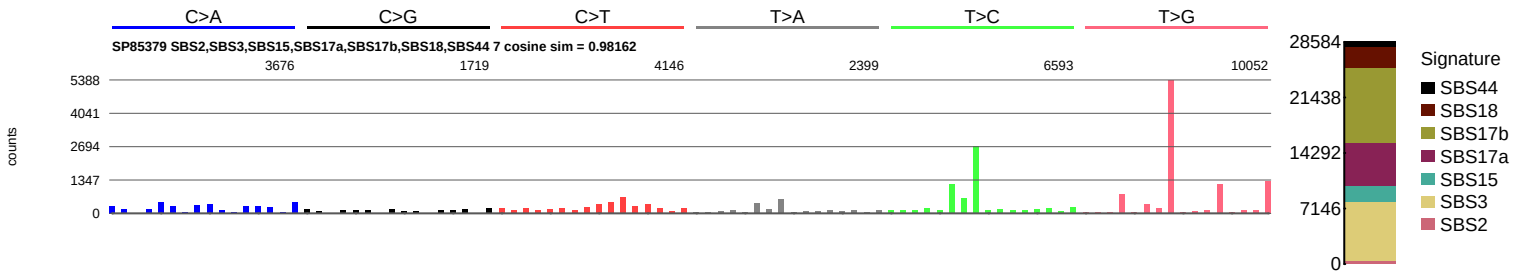
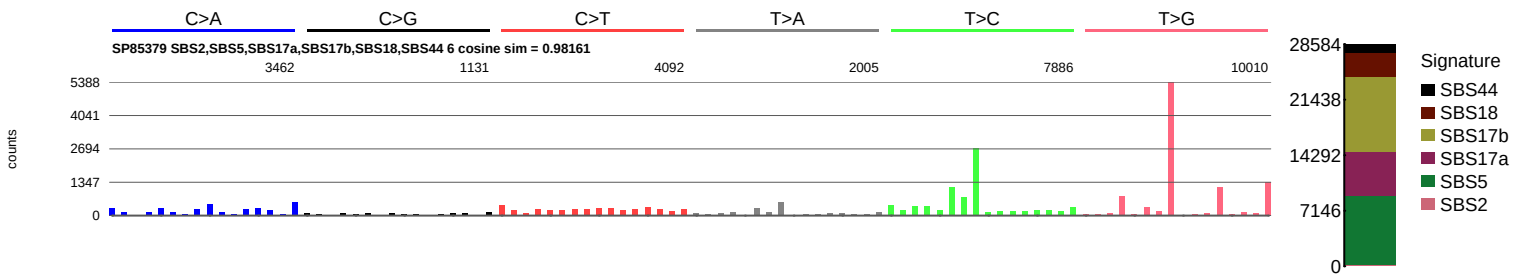
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

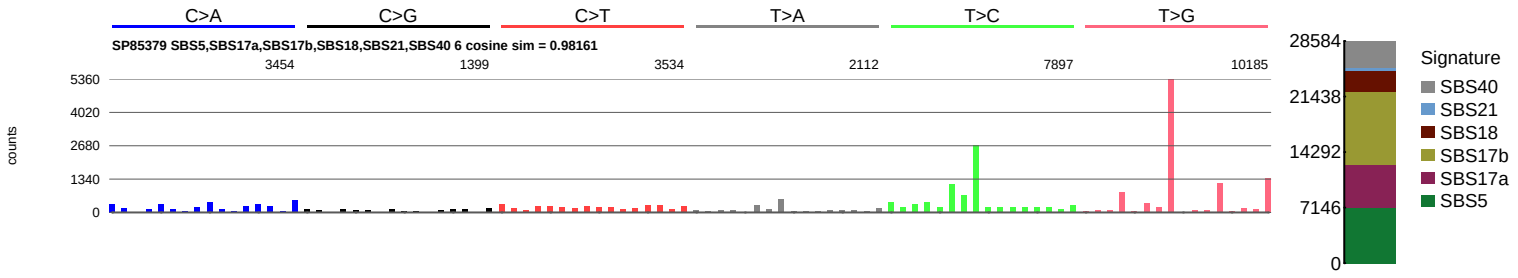

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98169
2161
1658
3591
2757
7800
10611
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5

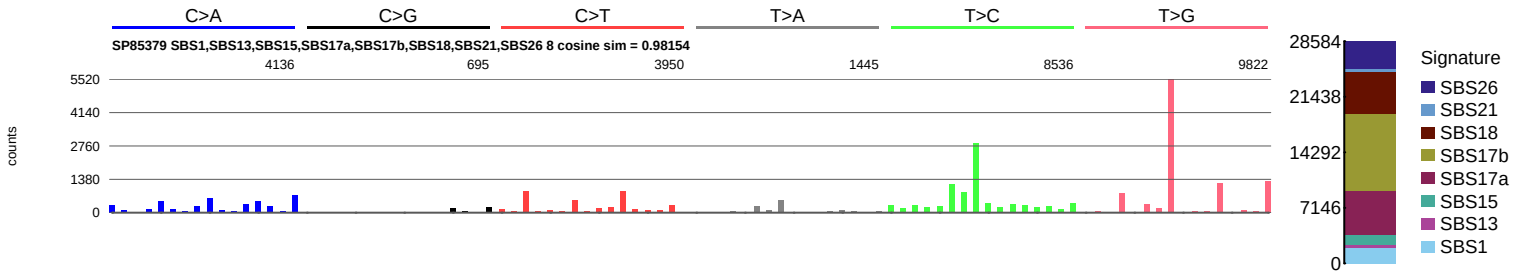

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98165
3449
1372
3602
2097
7898
10170
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

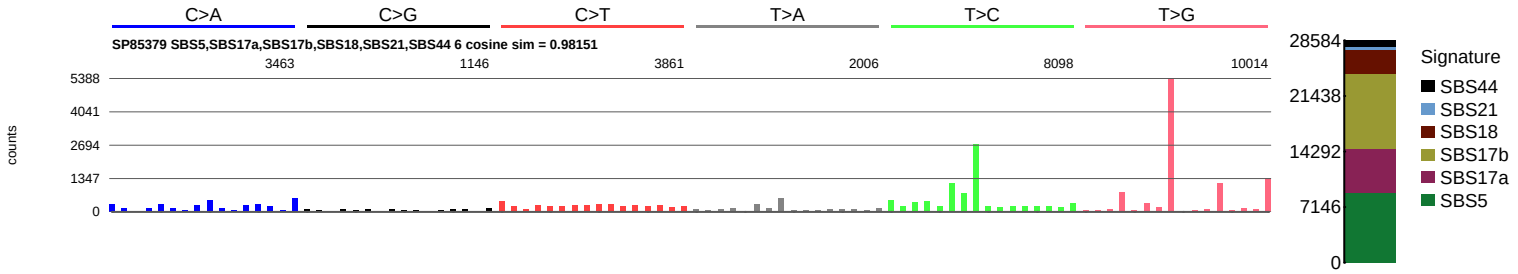

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98163
3463
1229
4011
2068
7792
10022
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

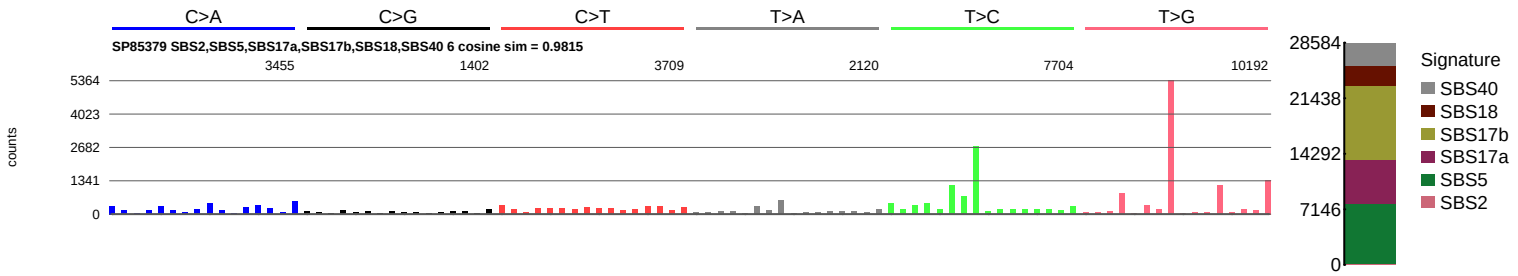

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98162
3676
1719
4146
2399
6593
10052
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

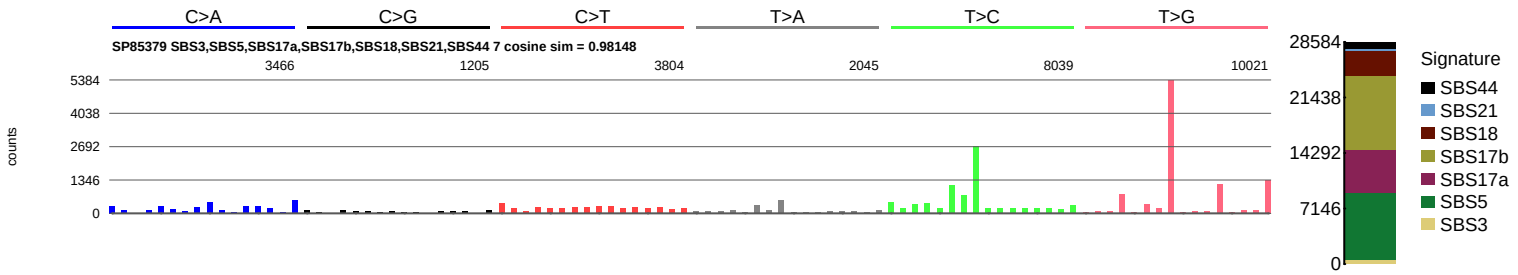

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98161
3462
1131
4092
2005
7886
10010
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2

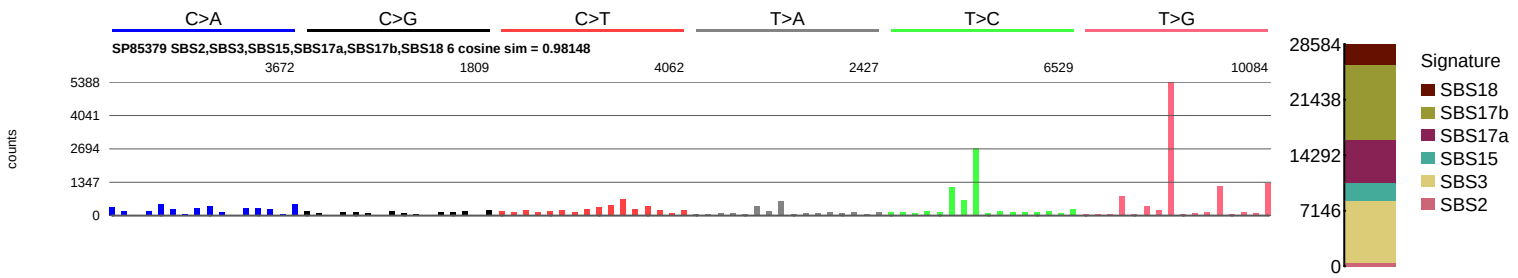

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98161
3454
1399
3534
2112
7897
10185
counts
5360
4020
2680
1340
0
28584
21438
14292
7146
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5

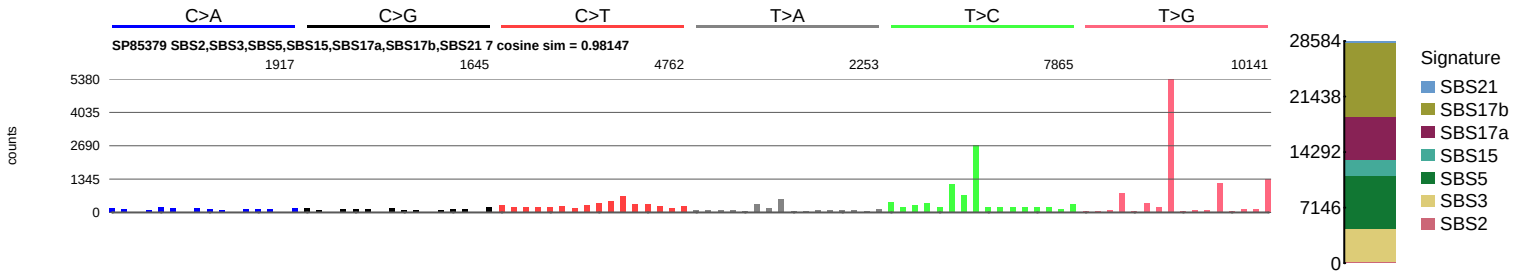

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98154
4136
695
3950
1445
8536
9822
counts
5520
4140
2760
1380
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

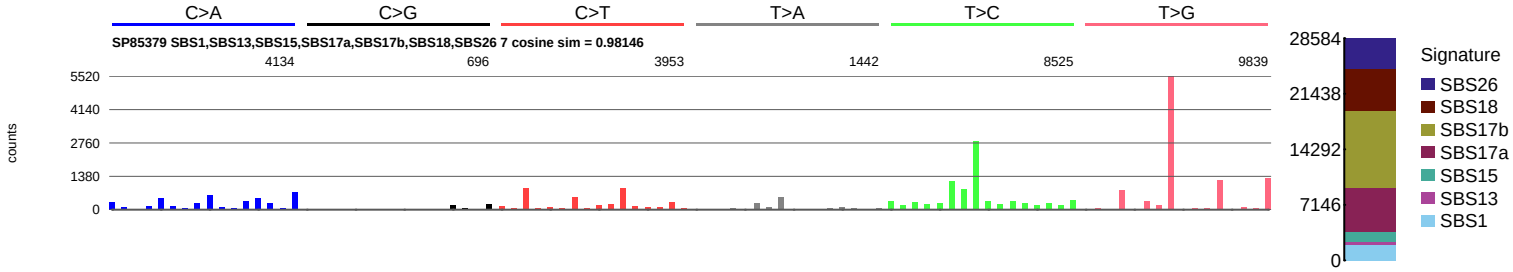

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 6 cosine sim = 0.98151
3463
1146
3861
2006
8098
10014
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5

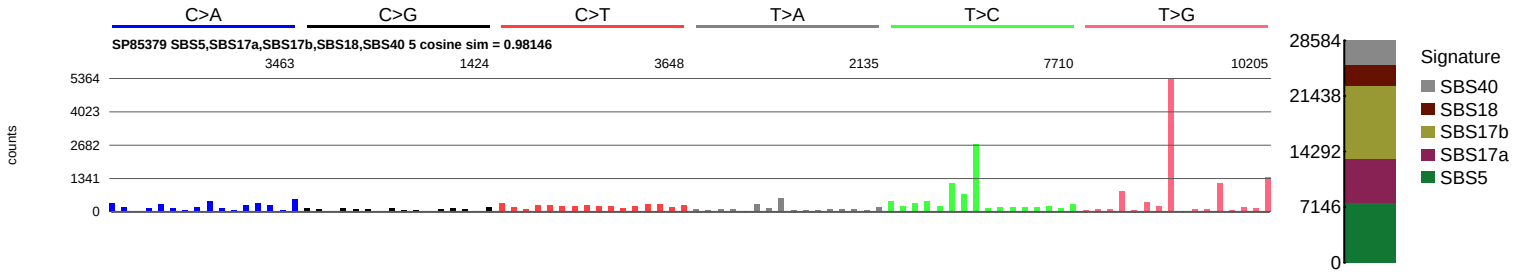

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.9815
3455
1402
3709
2120
7704
10192
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2

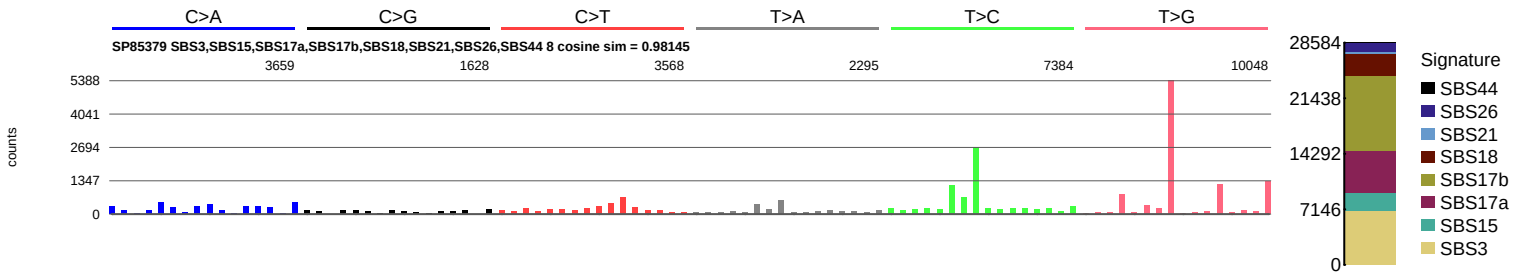

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98148
3466
1205
3804
2045
8039
10021
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3

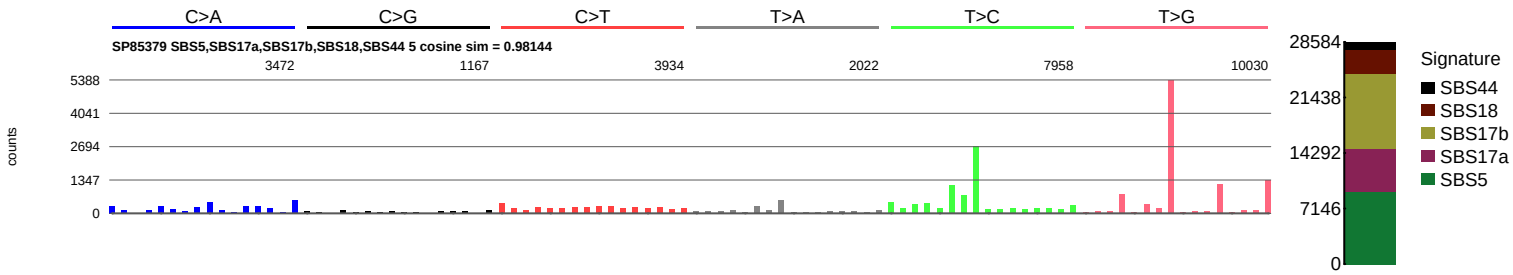

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98148
3672
1809
4062
2427
6529
10084
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

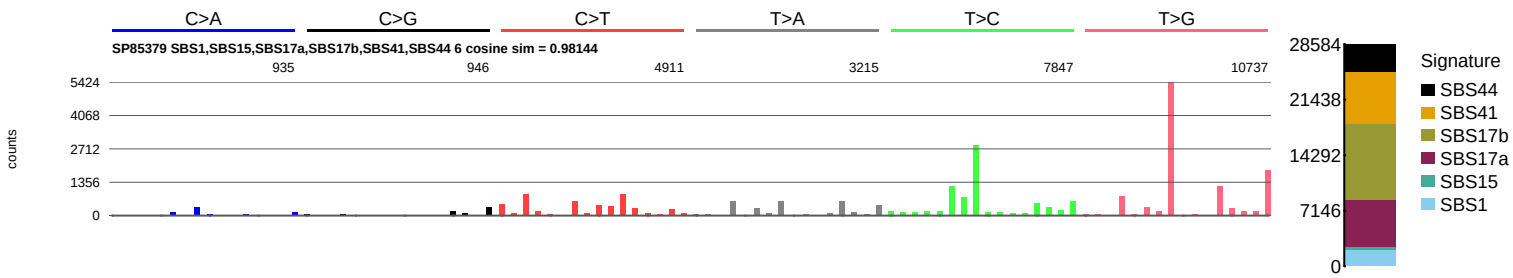

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98147
1917
1645
4762
2253
7865
10141
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

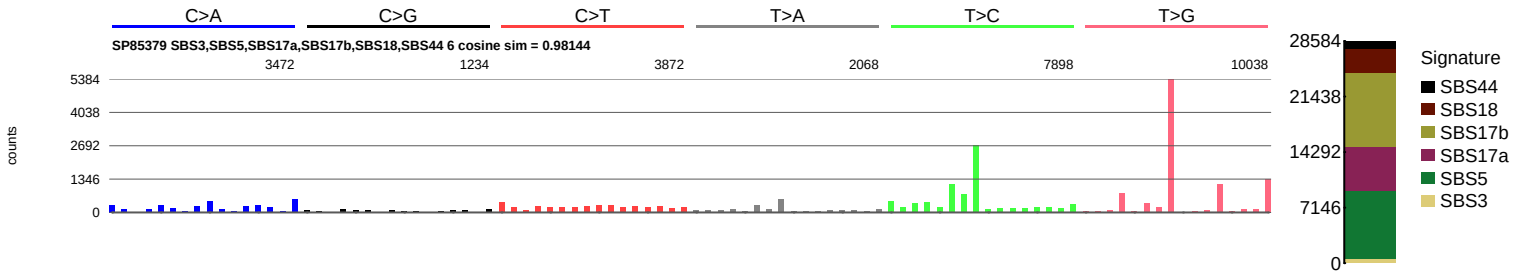

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98146
4134
696
3953
1442
8525
9839
counts
5520
4140
2760
1380
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98146
3463
1424
3648
2135
7710
10205
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98145
3659
1628
3568
2295
7384
10048
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS44 5 cosine sim = 0.98144
3472
1167
3934
2022
7958
10030
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98144
935
946
4911
3215
7847
10737
counts
5424
4068
2712
1356
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS1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C>T
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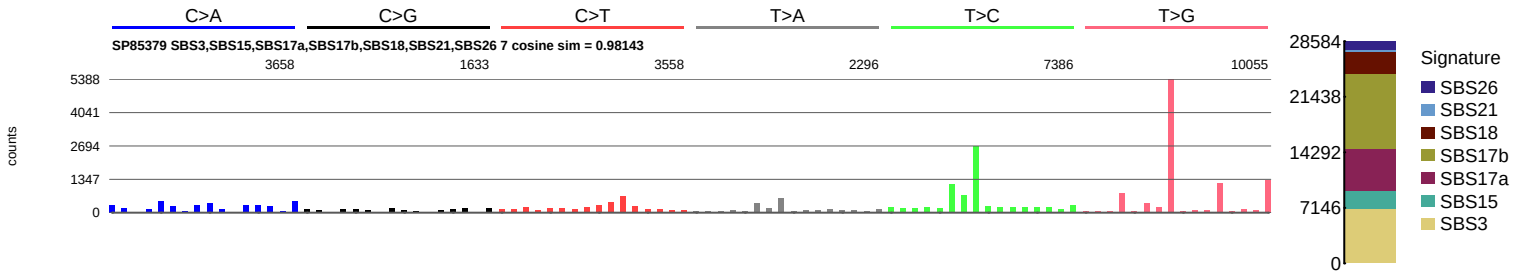
SP85379 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98144
3472
1234
3872
2068
7898
10038
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3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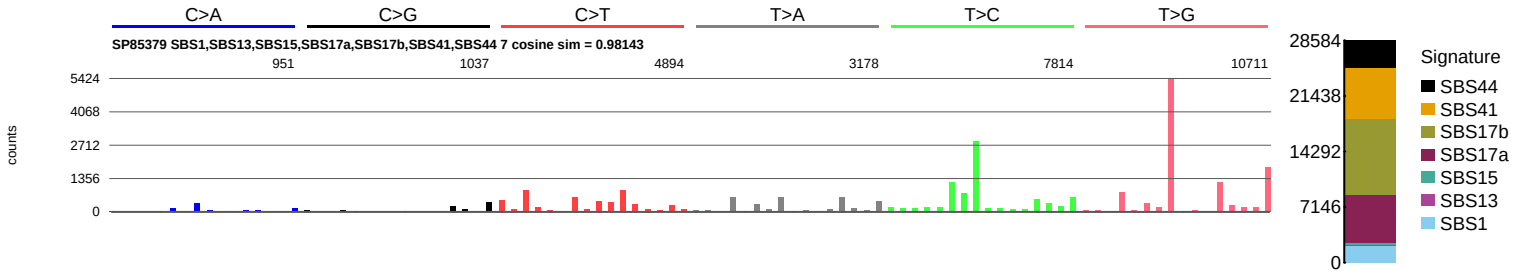
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98143
3658
1633
3558
2296
7386
10055
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3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C>G
C>T
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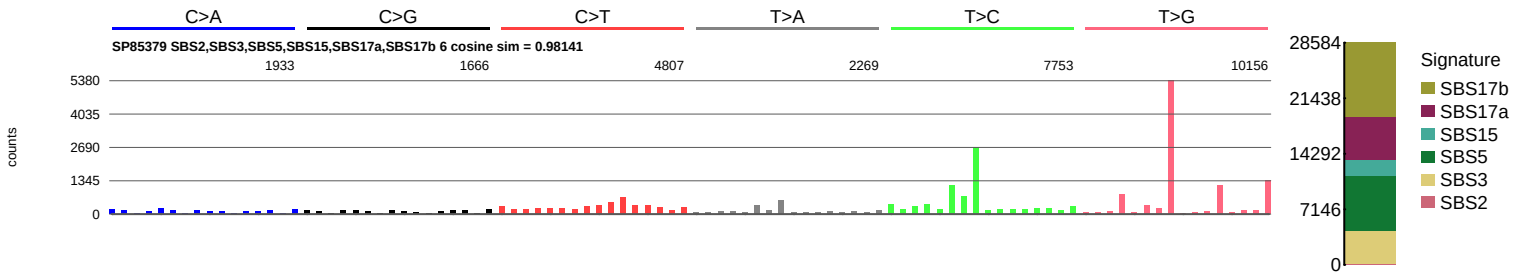
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98143
951
1037
4894
3178
7814
10711
counts
5424
4068
2712
1356
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
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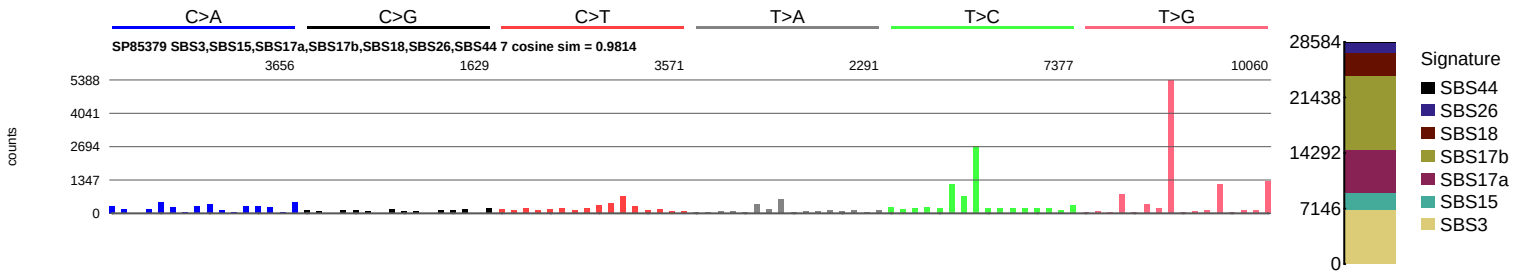
T>G
SP85379 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98141
1933
1666
4807
2269
7753
10156
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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C>G
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SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.9814
3656
1629
3571
2291
7377
10060
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
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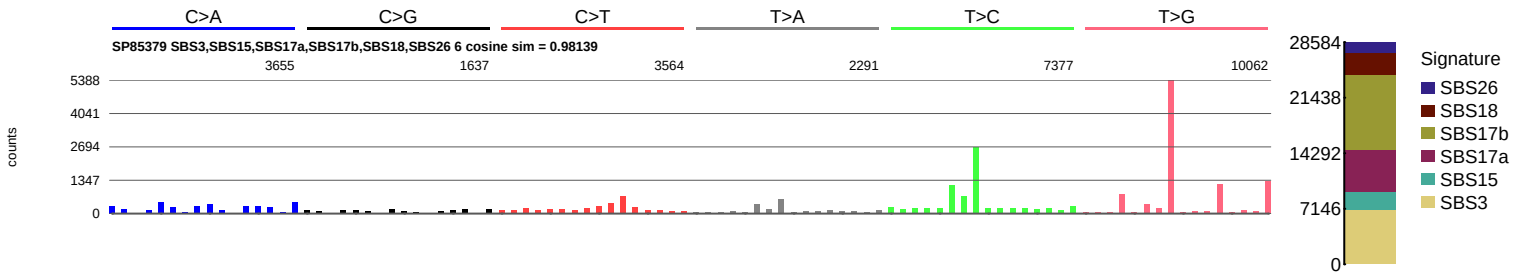
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3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C>G
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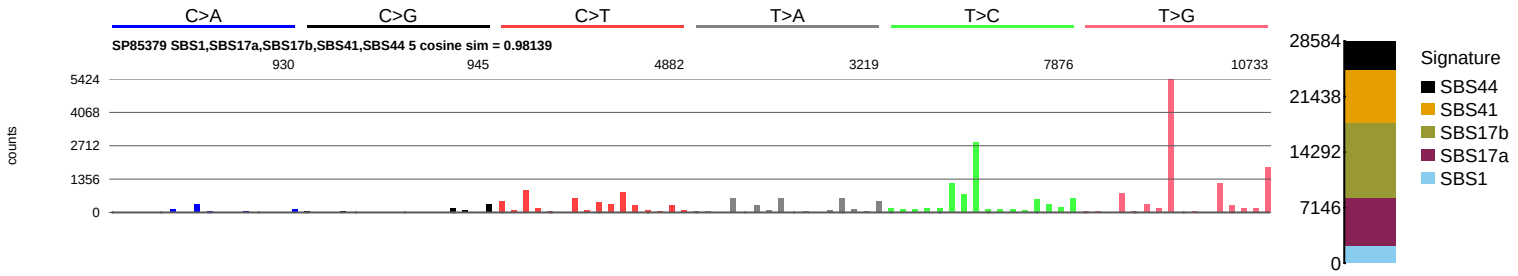
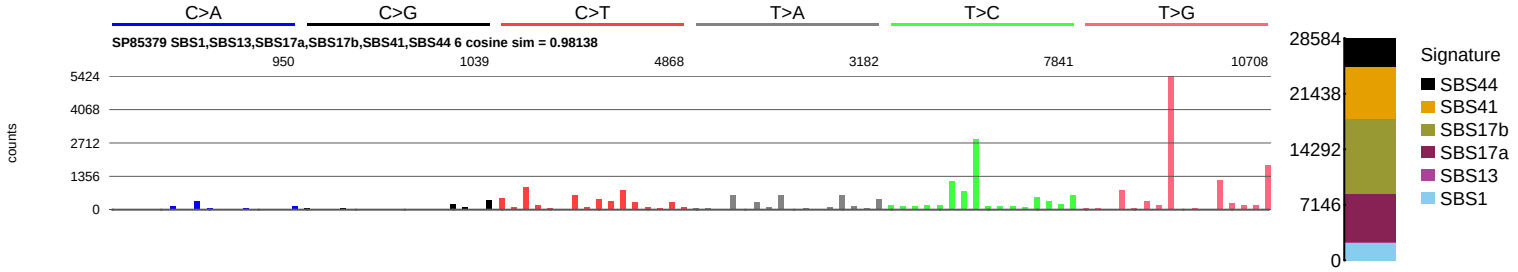
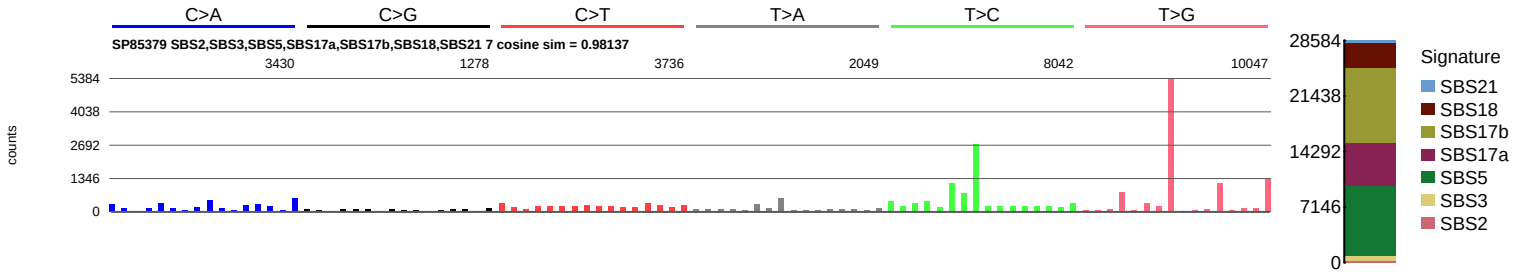
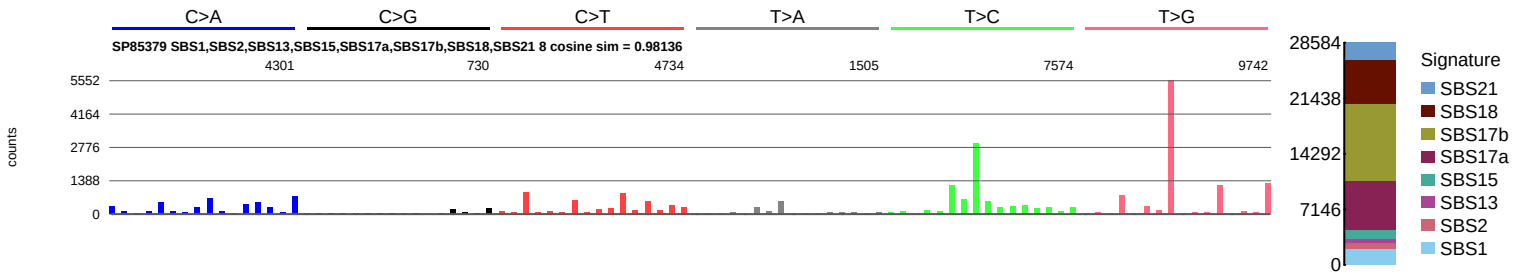
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98139
3655
1637
3564
2291
7377
10062
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98139
930
945
4882
3219
7876
10733
counts
5424
4068
2712
1356
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98138
950
1039
4868
3182
7841
10708
counts
5424
4068
2712
1356
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98137
3430
1278
3736
2049
8042
10047
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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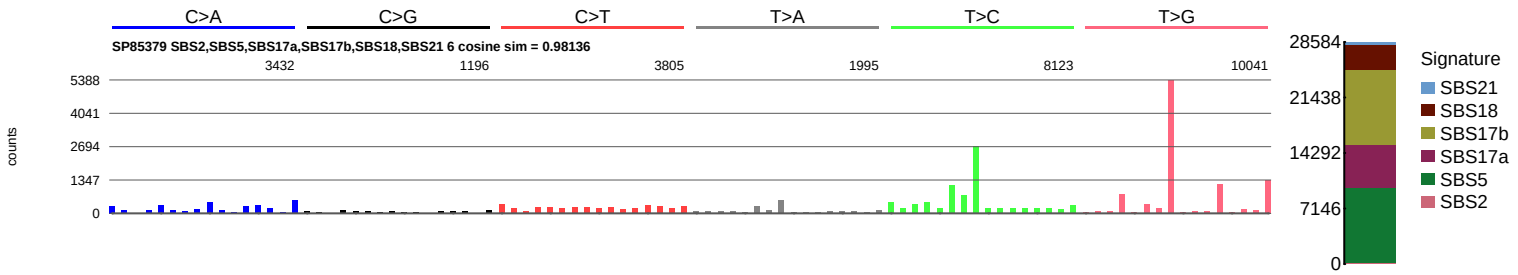
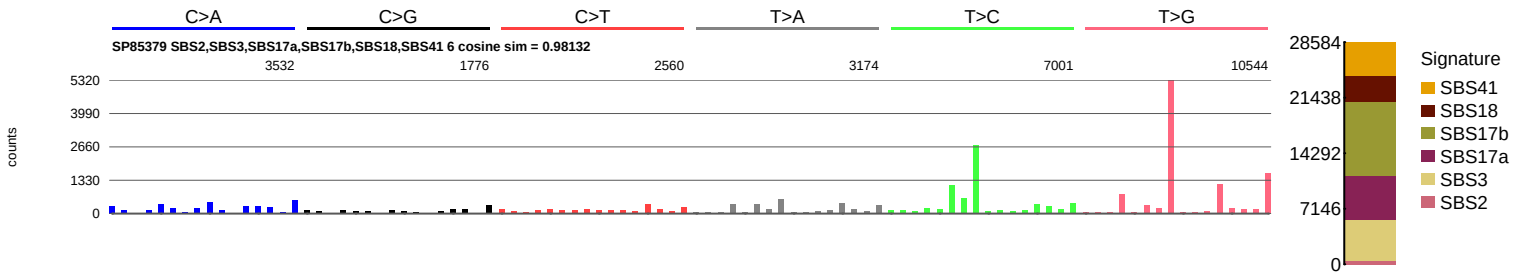
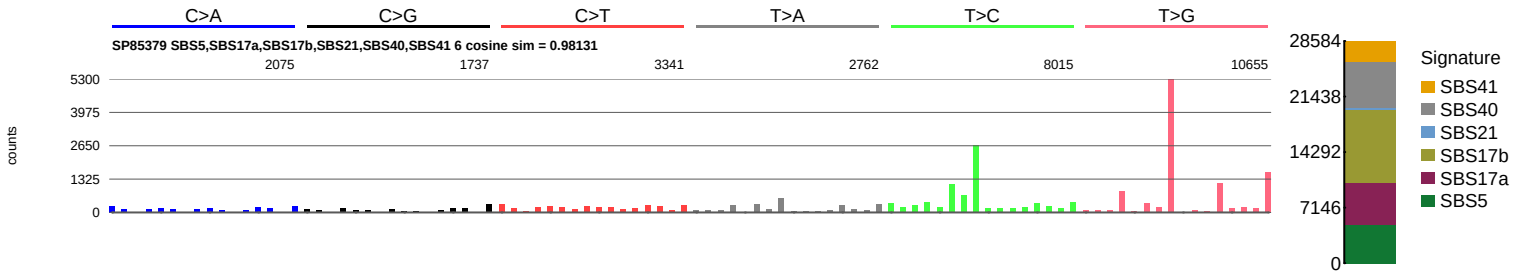
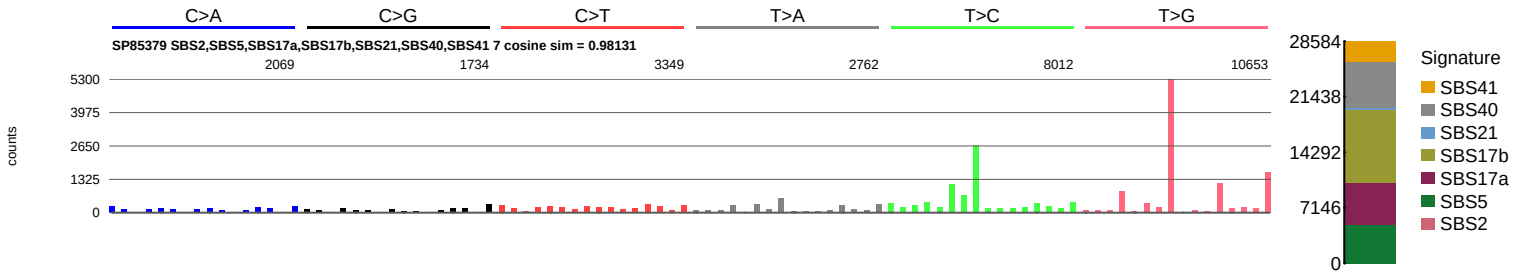
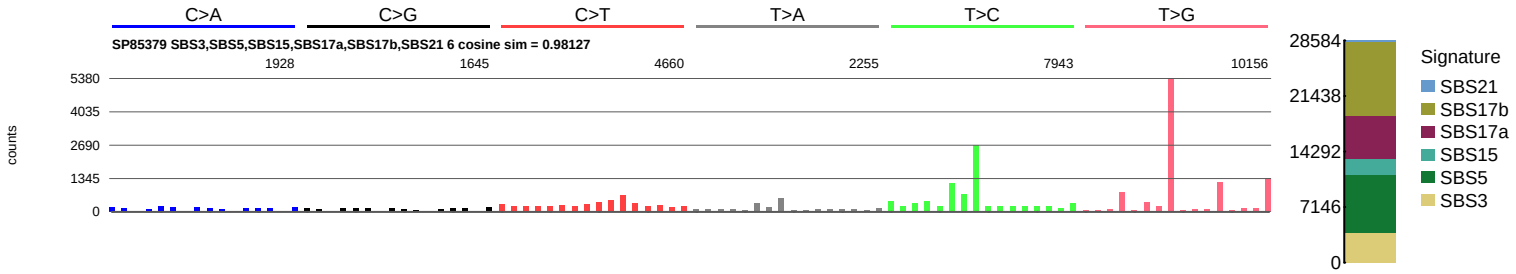
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98136
4301
730
4734
1505
7574
9742
counts
5552
4164
2776
1388
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

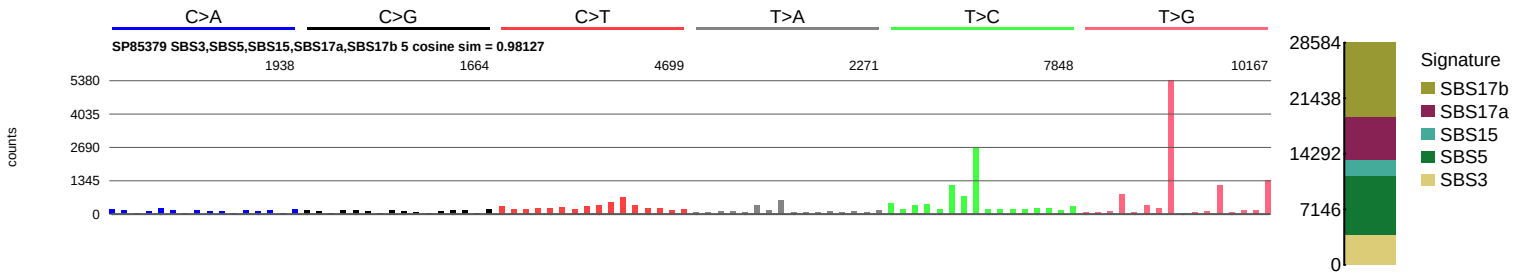

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98136
3432
1196
3805
1995
8123
10041
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

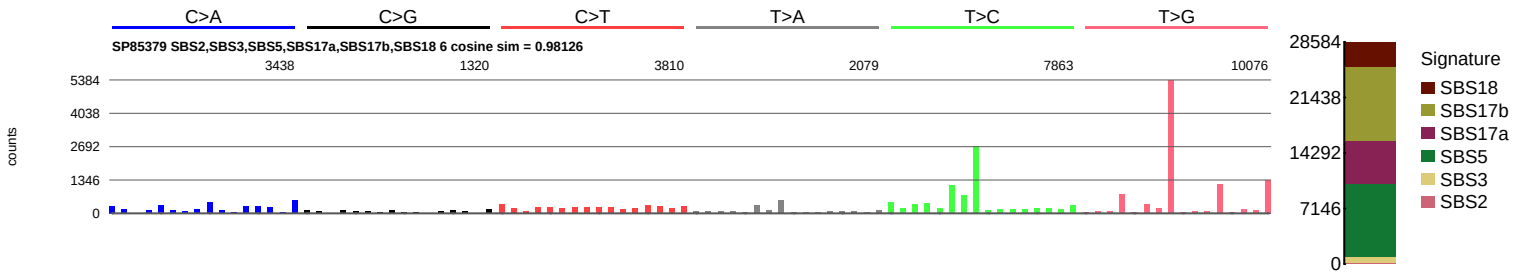

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98132
3532
1776
2560
3174
7001
10544
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2

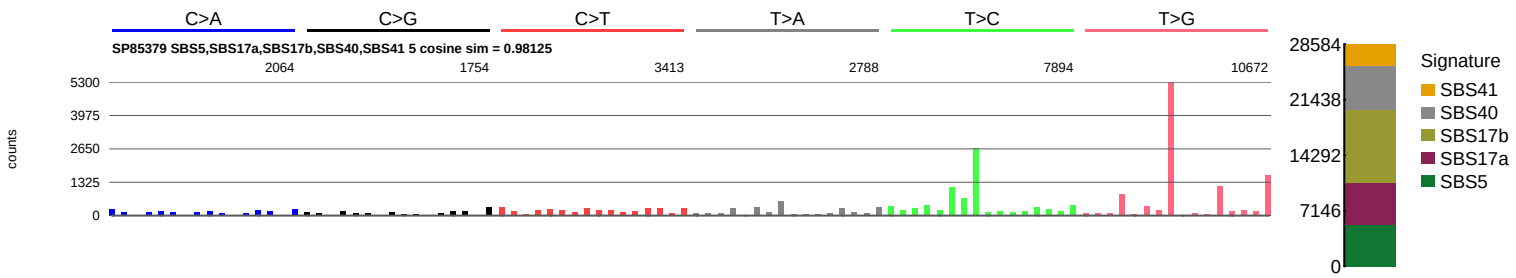

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98131
2075
1737
3341
2762
8015
10655
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5

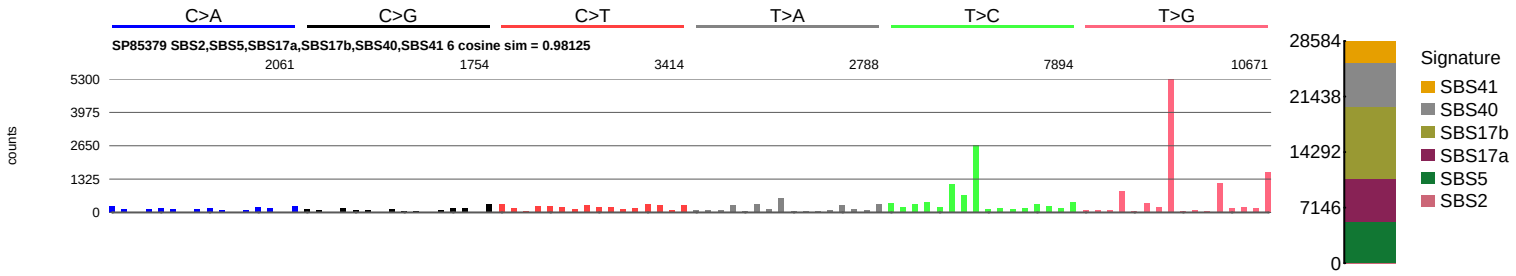

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98131
2069
1734
3349
2762
8012
10653
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98127
1928
1645
4660
2255
7943
10156
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98127
1938
1664
4699
2271
7848
10167
counts
5380
4035
2690
1345
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98126
3438
1320
3810
2079
7863
10076
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98125
2064
1754
3413
2788
7894
10672
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5


C>A
C>G
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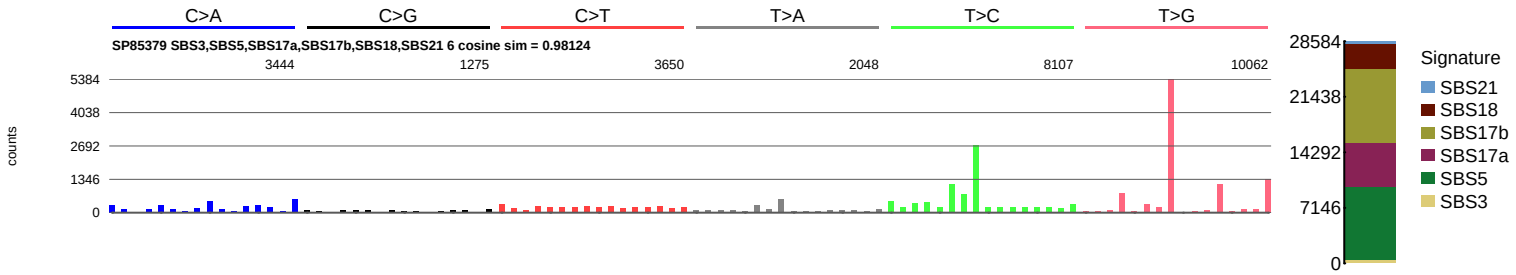
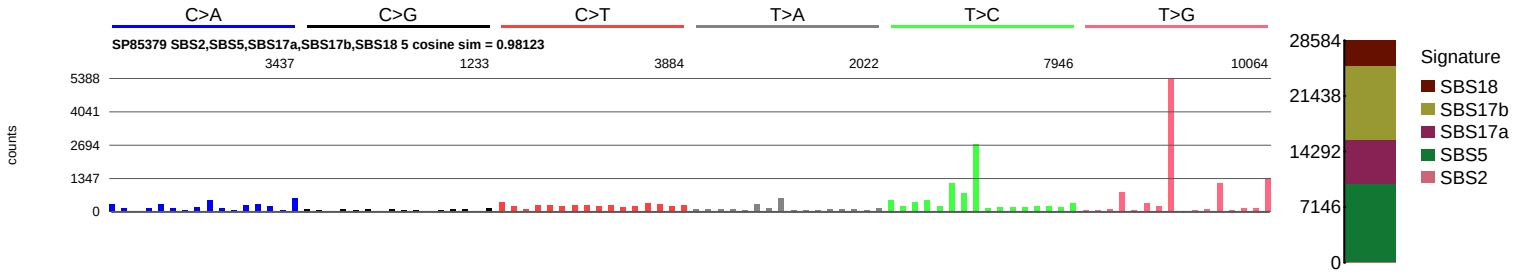
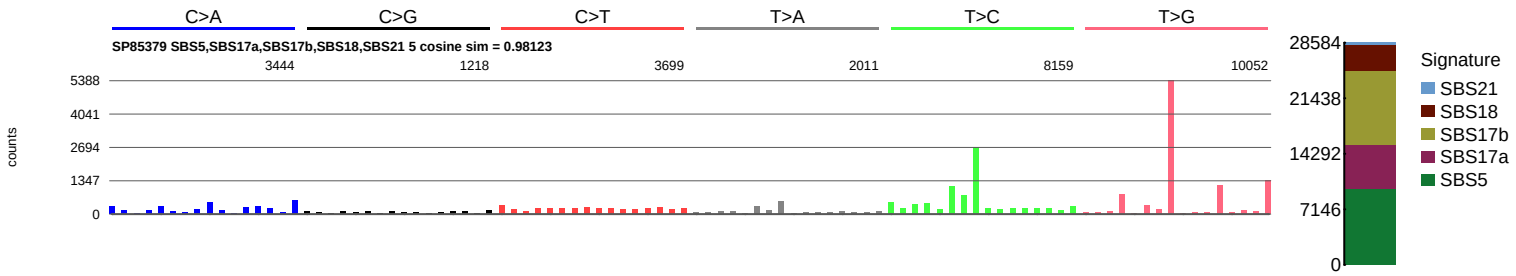
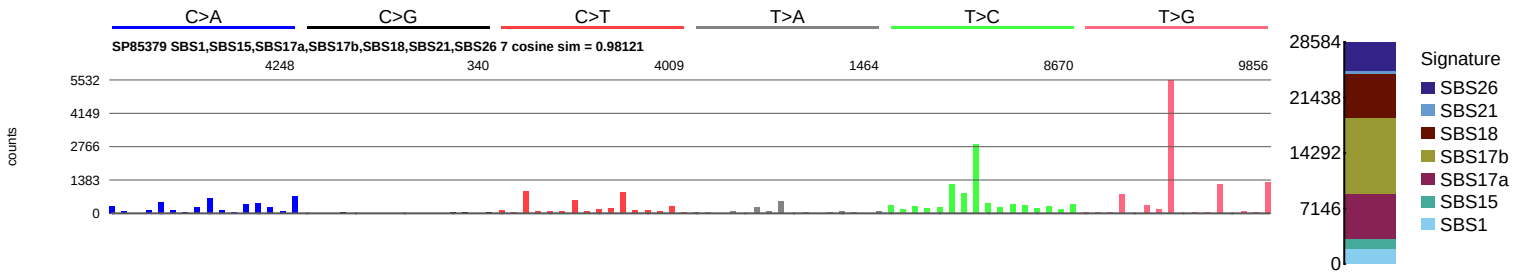
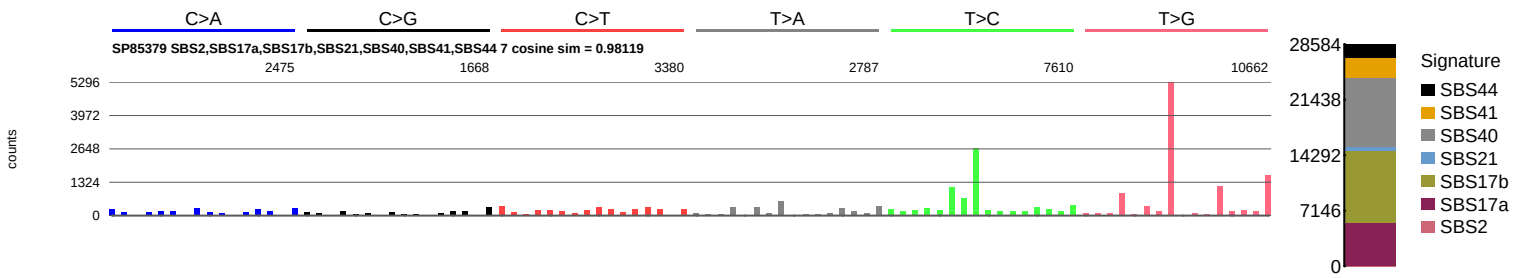
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98125
2061
1754
3414
2788
7894
10671
counts
5300
3975
2650
1325
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98124
3444
1275
3650
2048
8107
10062
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98123
3437
1233
3884
2022
7946
10064
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18,SBS21 5 cosine sim = 0.98123
3444
1218
3699
2011
8159
10052
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98121
4248
340
4009
1464
8670
9856
counts
5532
4149
2766
1383
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98119
2475
1668
3380
2787
7610
10662
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
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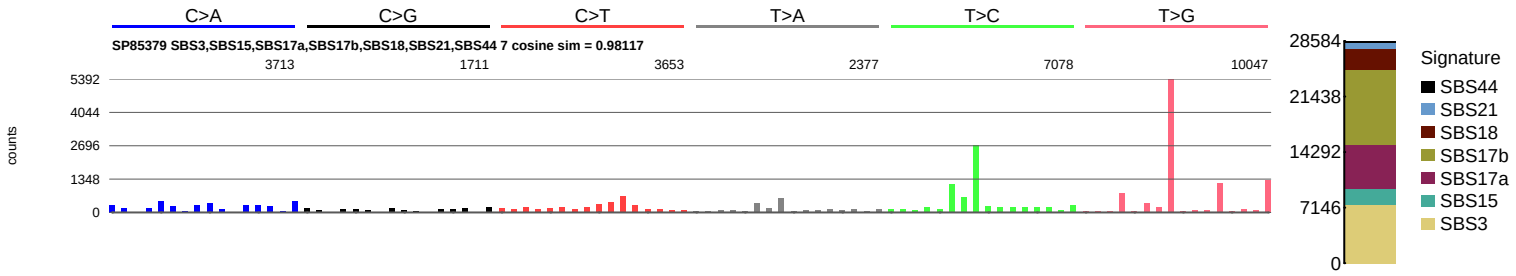
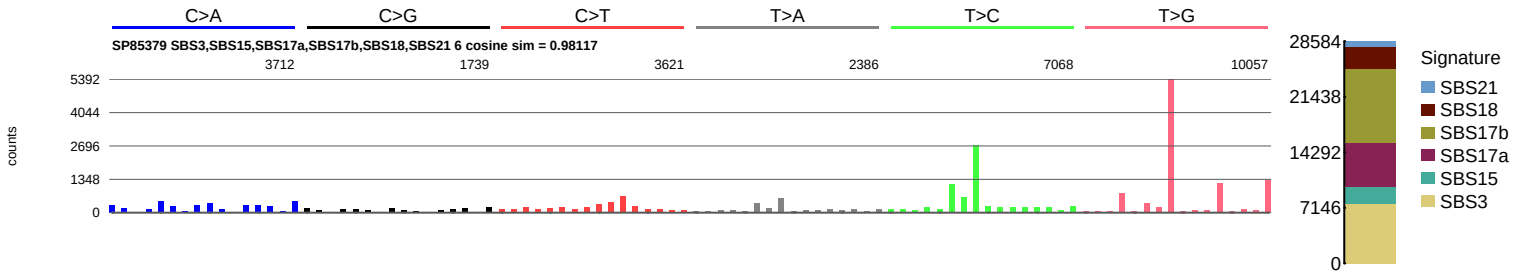
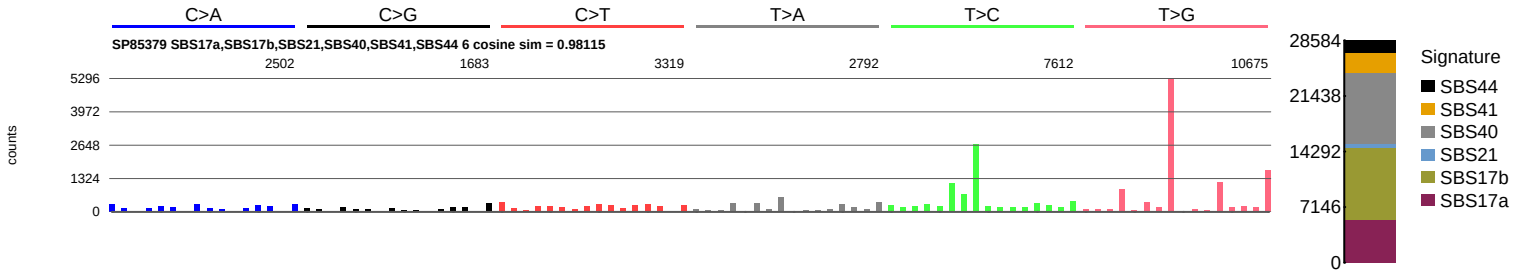
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98117
3713
1711
3653
2377
7078
10047
counts
5392
4044
2696
1348
0
28584
21438
14292
7146
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98117
3712
1739
3621
2386
7068
10057
counts
5392
4044
2696
1348
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
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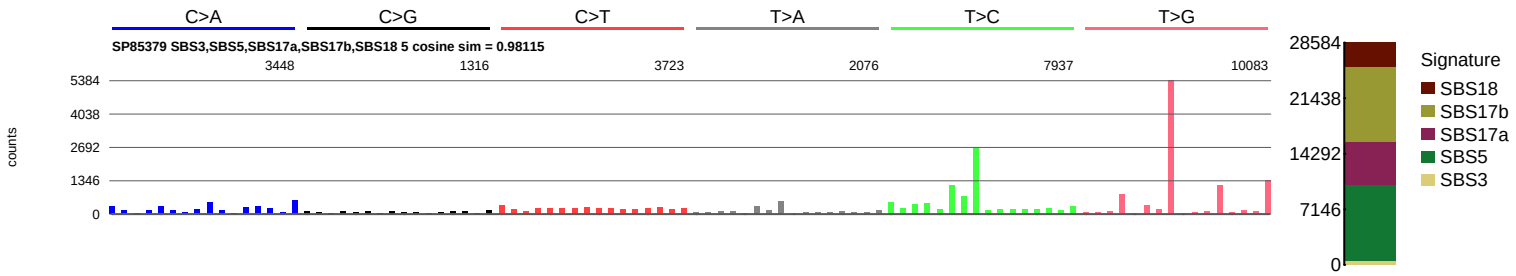
T>C
T>G
SP85379 SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 6 cosine sim = 0.98115
2502
1683
3319
2792
7612
10675
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a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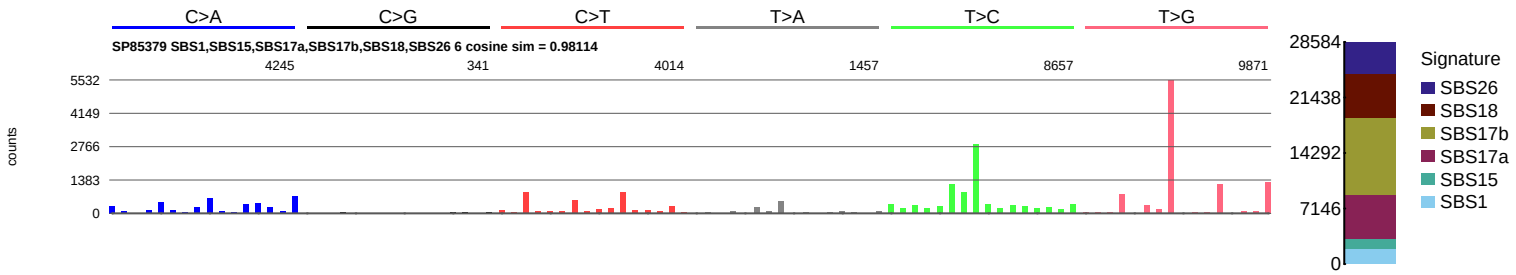
T>G
SP85379 SBS3,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98115
3448
1316
3723
2076
7937
10083
counts
5384
4038
2692
1346
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
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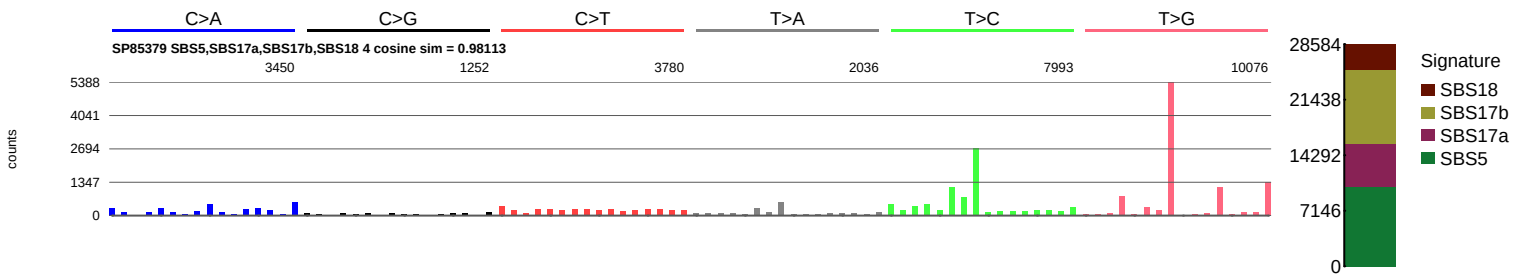
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98114
4245
341
4014
1457
8657
9871
counts
5532
4149
2766
1383
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
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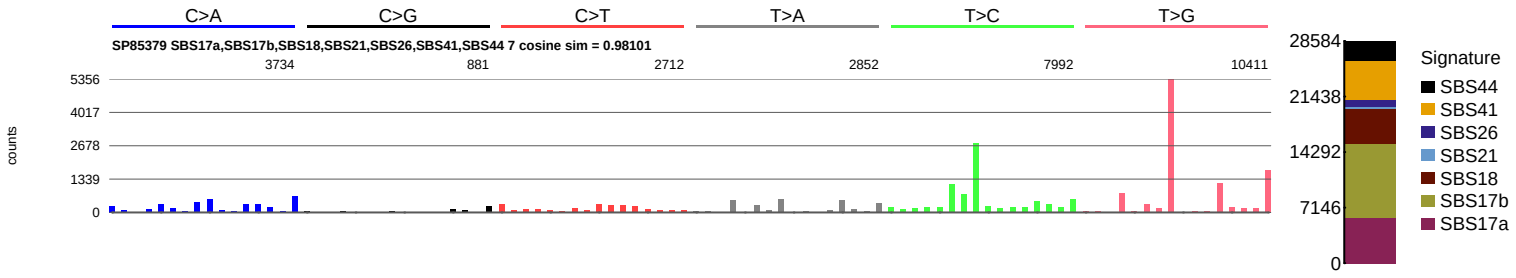
T>C
T>G
SP85379 SBS5,SBS17a,SBS17b,SBS18 4 cosine sim = 0.98113
3450
1252
3780
2036
7993
10076
counts
5388
4041
2694
1347
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
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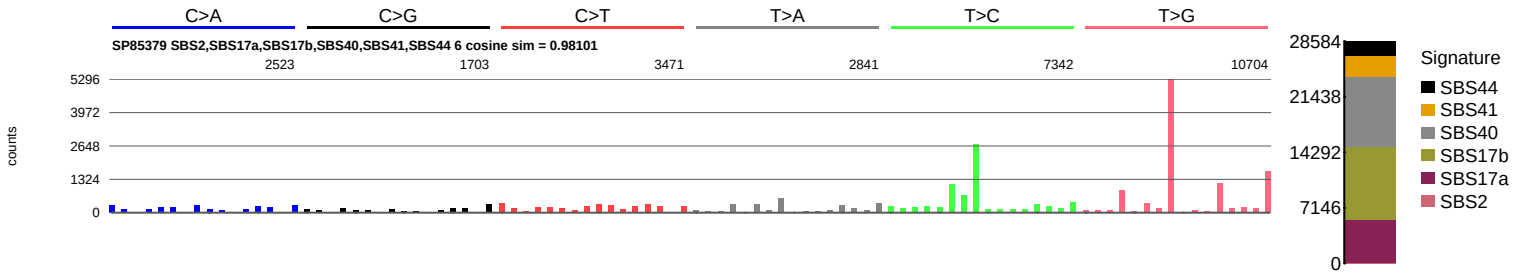
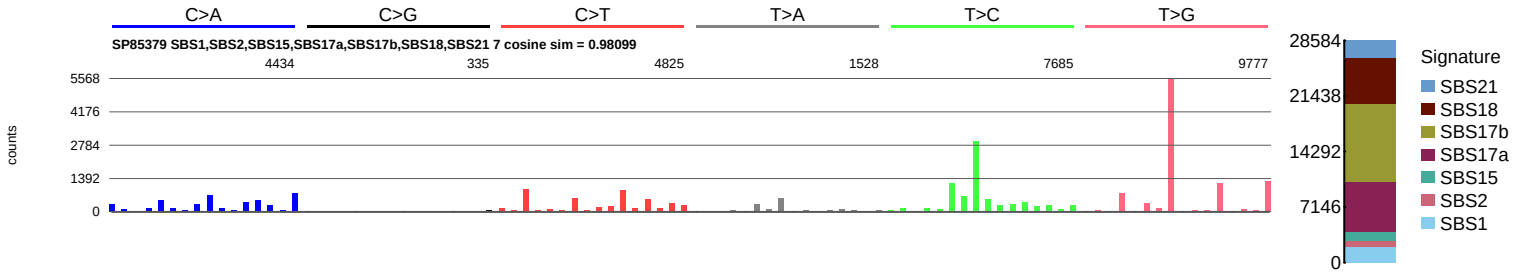
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 7 cosine sim = 0.98101
3734
881
2712
2852
7992
10411
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98101
2523
1703
3471
2841
7342
10704
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
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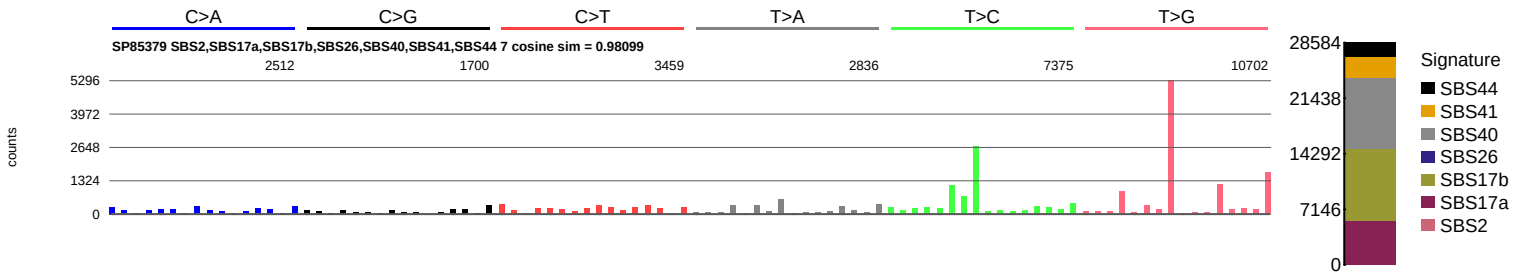
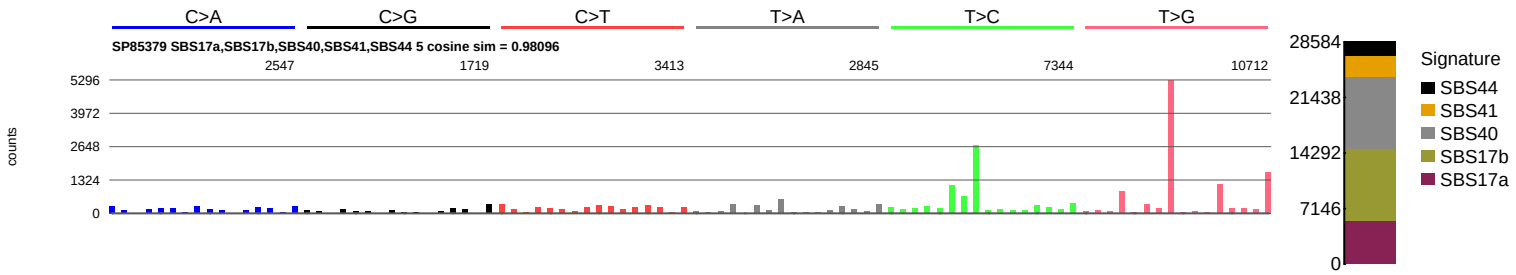
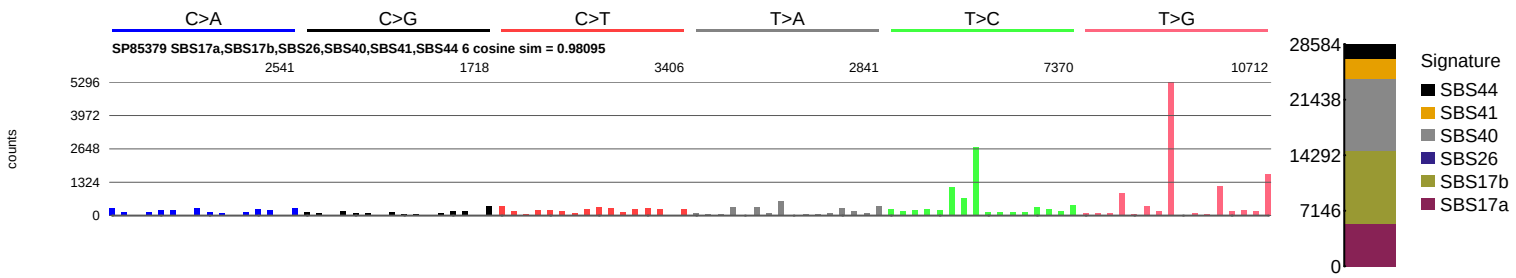
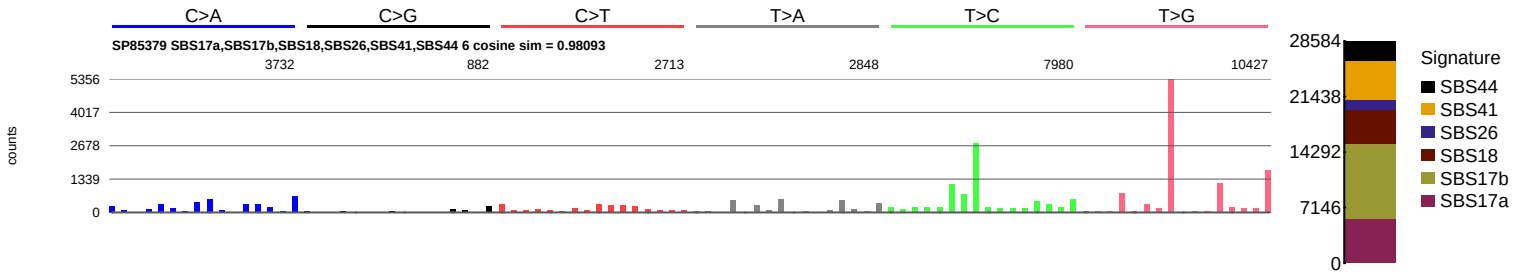
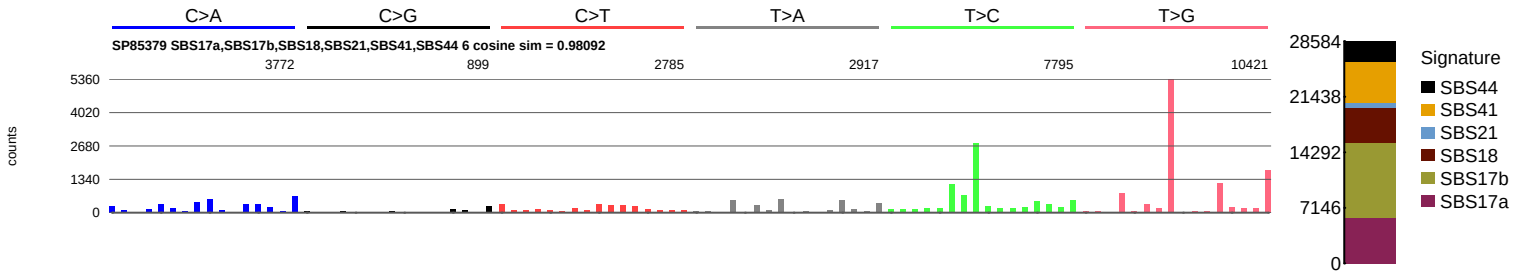
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98099
4434
335
4825
1528
7685
9777
counts
5568
4176
2784
1392
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

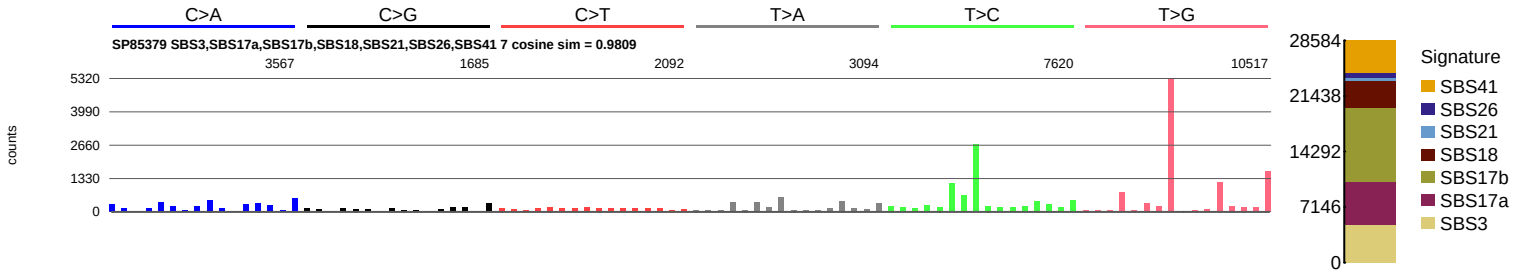

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98099
2512
1700
3459
2836
7375
10702
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2

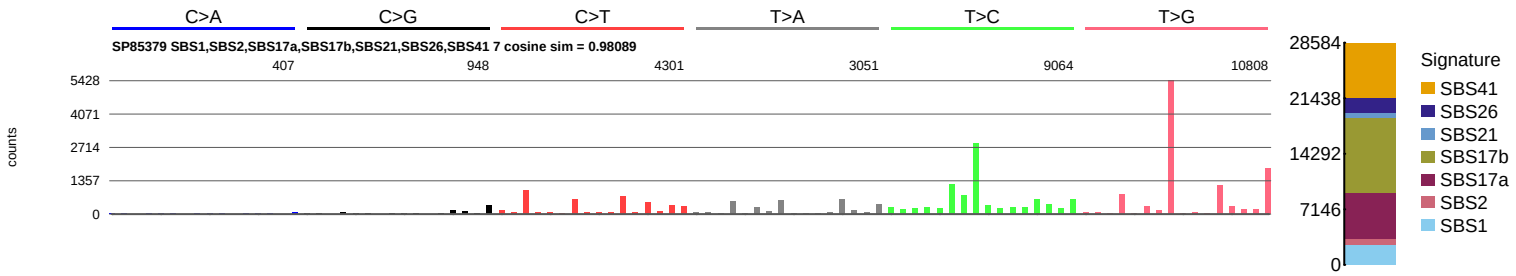

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS40,SBS41,SBS44 5 cosine sim = 0.98096
2547
1719
3413
2845
7344
10712
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a

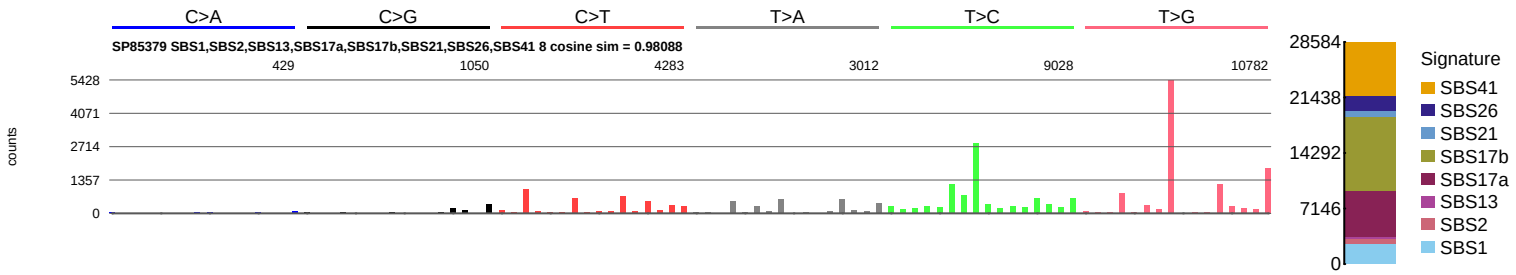

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 6 cosine sim = 0.98095
2541
1718
3406
2841
7370
10712
counts
5296
3972
2648
1324
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a

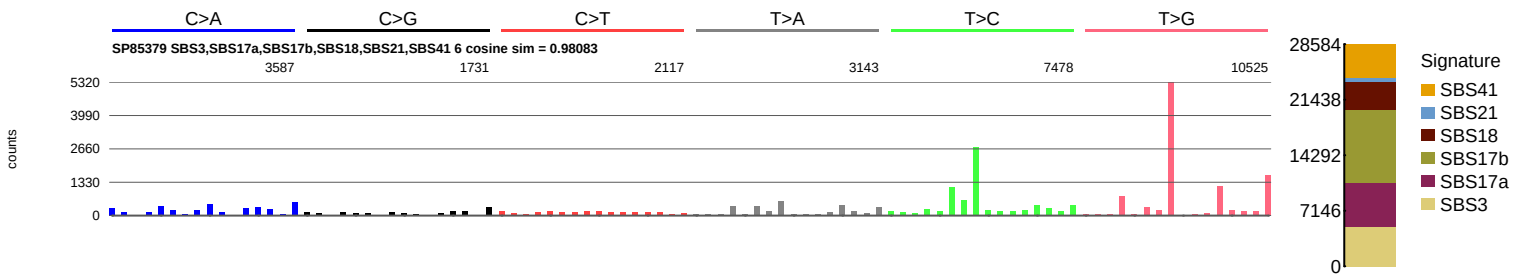

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98093
3732
882
2713
2848
7980
10427
counts
5356
4017
2678
1339
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a

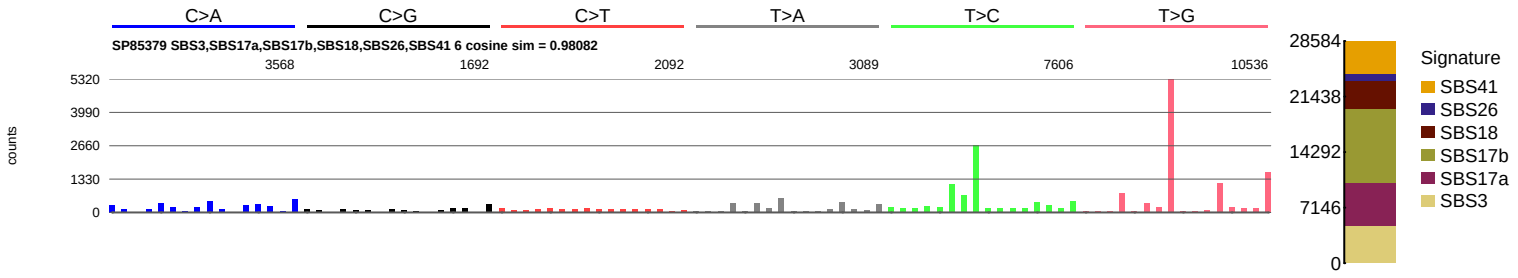

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 6 cosine sim = 0.98092
3772
899
2785
2917
7795
10421
counts
5360
4020
2680
1340
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a

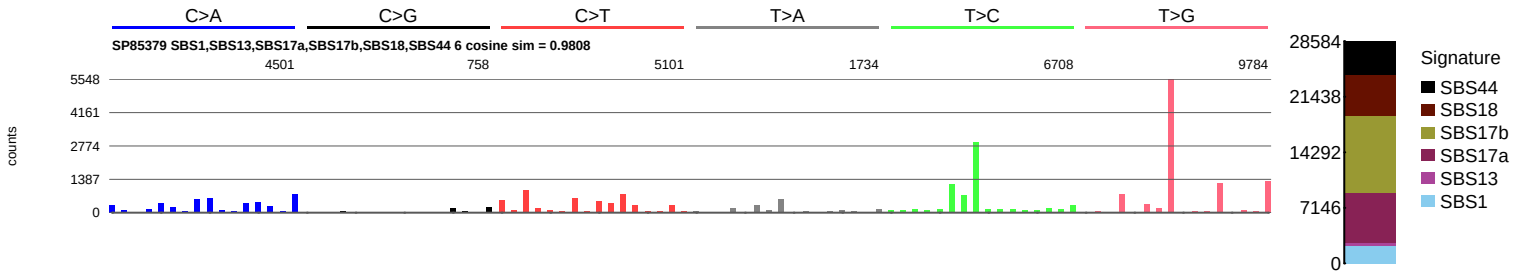

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.9809
3567
1685
2092
3094
7620
10517
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3

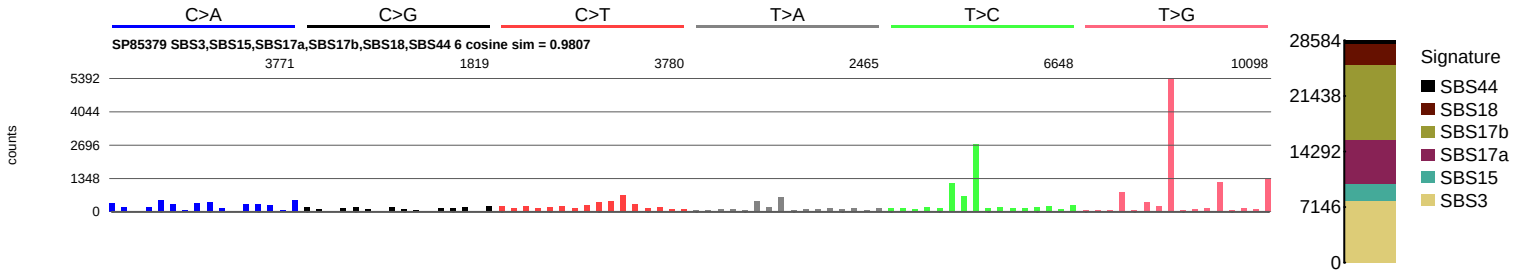

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98089
407
948
4301
3051
9064
10808
counts
5428
4071
2714
1357
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1

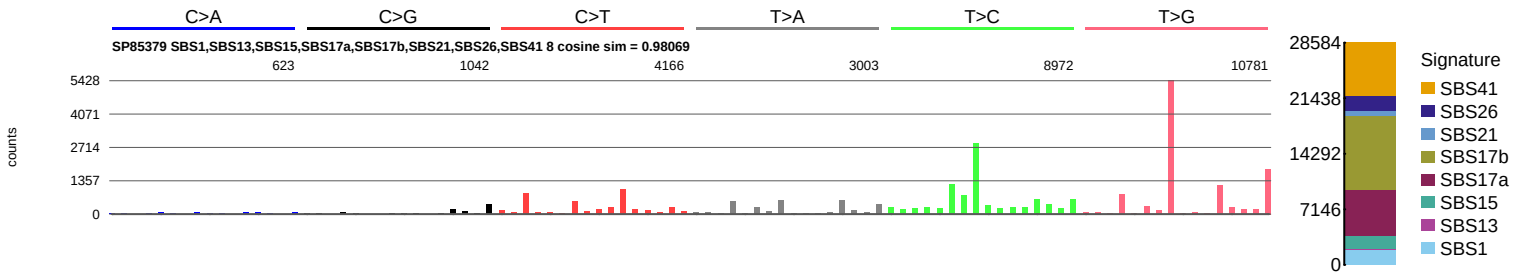

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98088
429
1050
4283
3012
9028
10782
counts
5428
4071
2714
1357
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

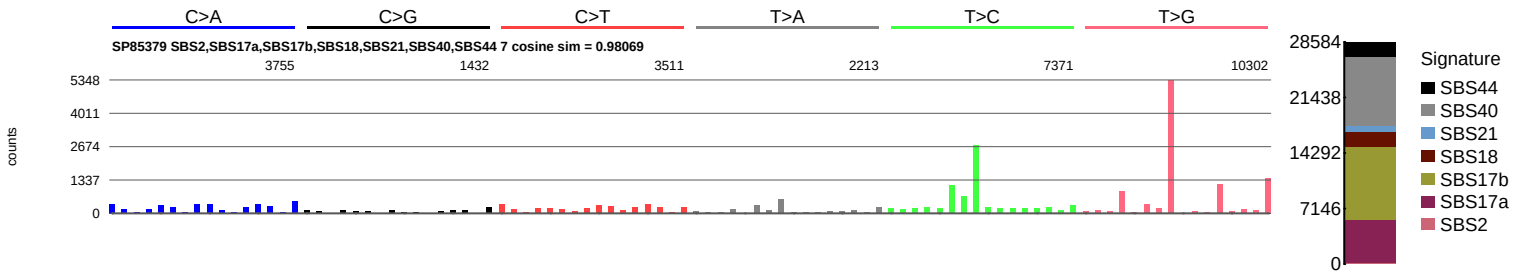

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98083
3587
1731
2117
3143
7478
10525
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3

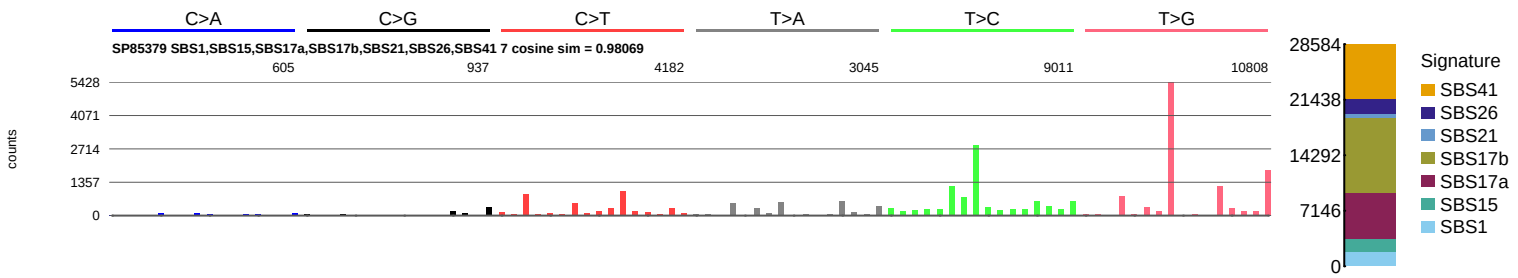

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98082
3568
1692
2092
3089
7606
10536
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3

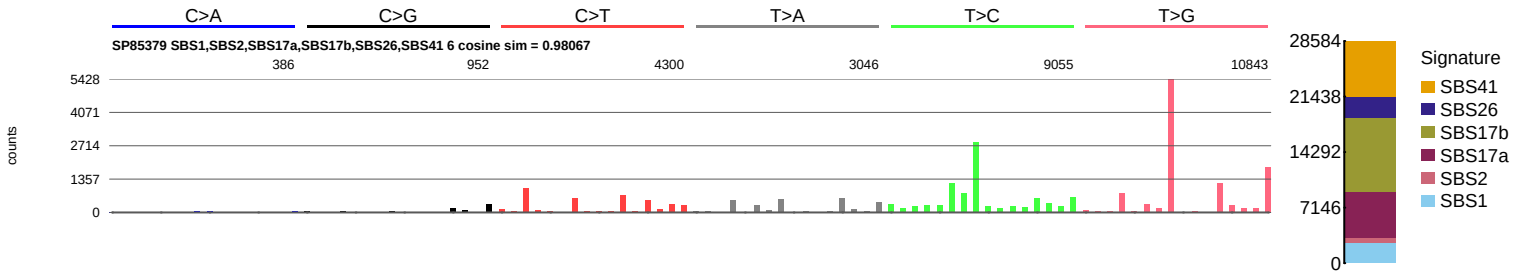

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.9808
4501
758
5101
1734
6708
9784
counts
5548
4161
2774
1387
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS1

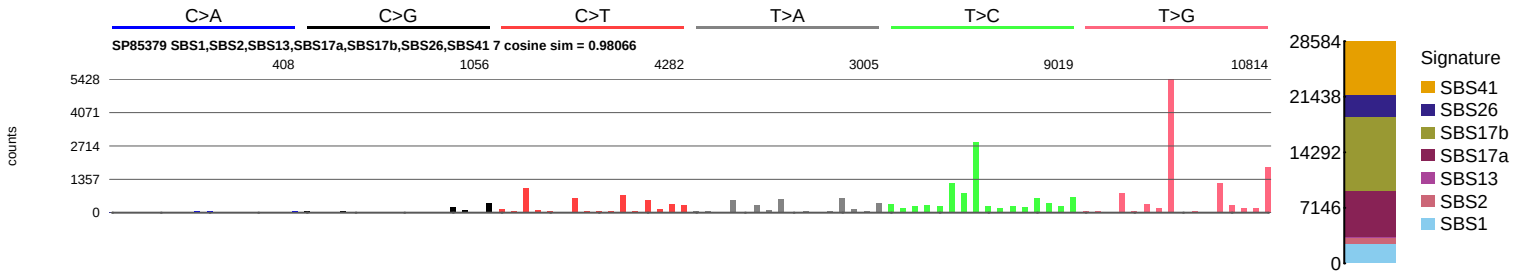

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.9807
3771
1819
3780
2465
6648
10098
counts
5392
4044
2696
1348
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98069
623
1042
4166
3003
8972
10781
counts
5428
4071
2714
1357
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98069
3755
1432
3511
2213
7371
10302
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98069
605
937
4182
3045
9011
10808
counts
5428
4071
2714
1357
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98067
386
952
4300
3046
9055
10843
counts
5428
4071
2714
1357
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98066
408
1056
4282
3005
9019
10814
counts
5428
4071
2714
1357
0
28584
21438
14292
7146
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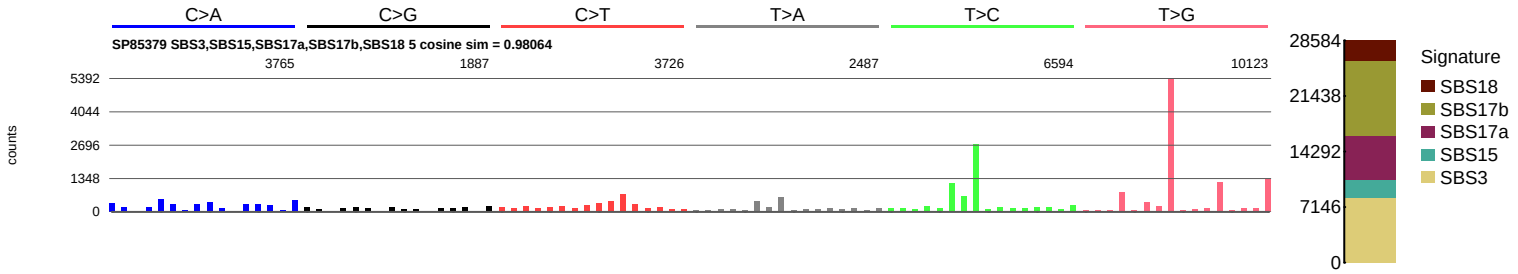
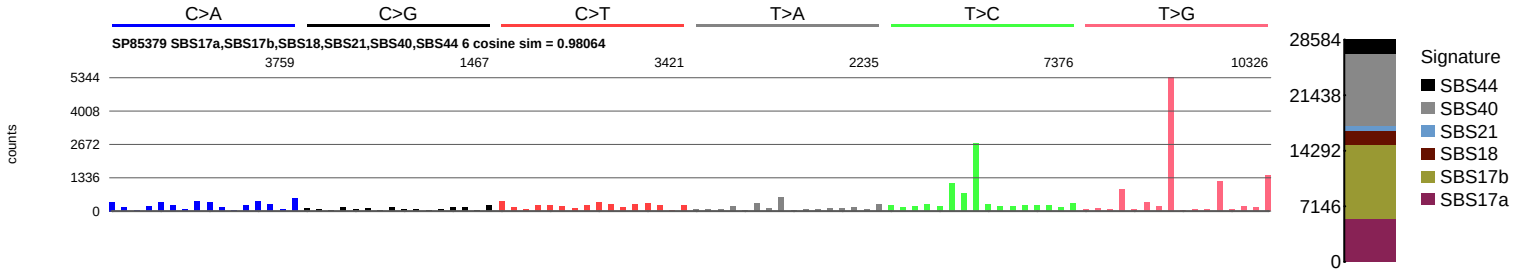
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS15,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98064
3765
1887
3726
2487
6594
10123
counts
5392
4044
2696
1348
0
28584
21438
14292
7146
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.98064
3759
1467
3421
2235
7376
10326
counts
5344
4008
2672
1336
0
28584
21438
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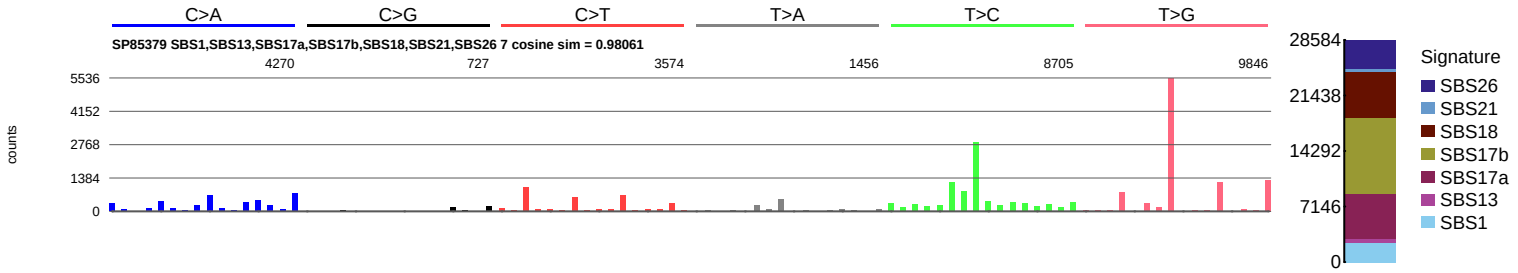
SBS17a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SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98061
4270
727
3574
1456
8705
9846
counts
5536
4152
2768
1384
0
28584
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14292
7146
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SBS21
SBS18
SBS17b
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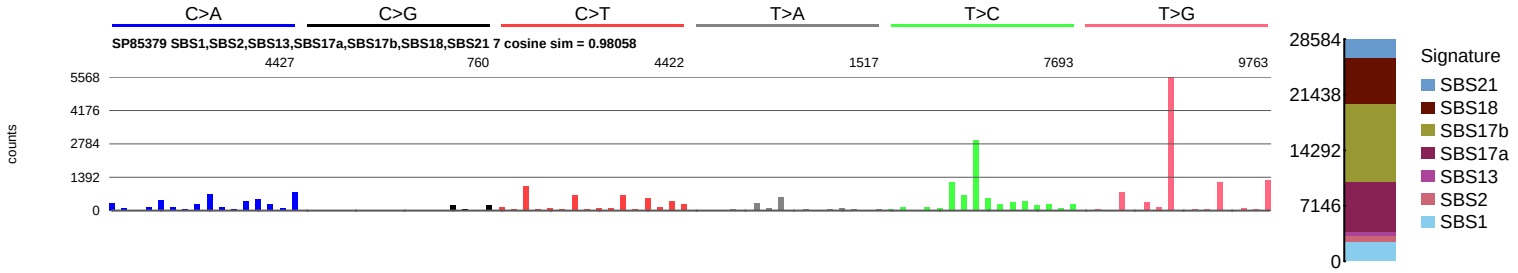
SBS13
SBS1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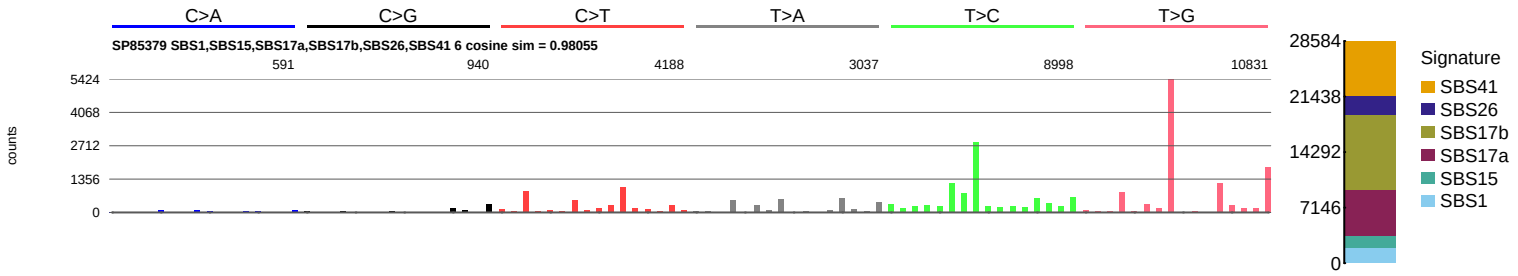
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98058
4427
760
4422
1517
7693
9763
counts
5568
4176
2784
1392
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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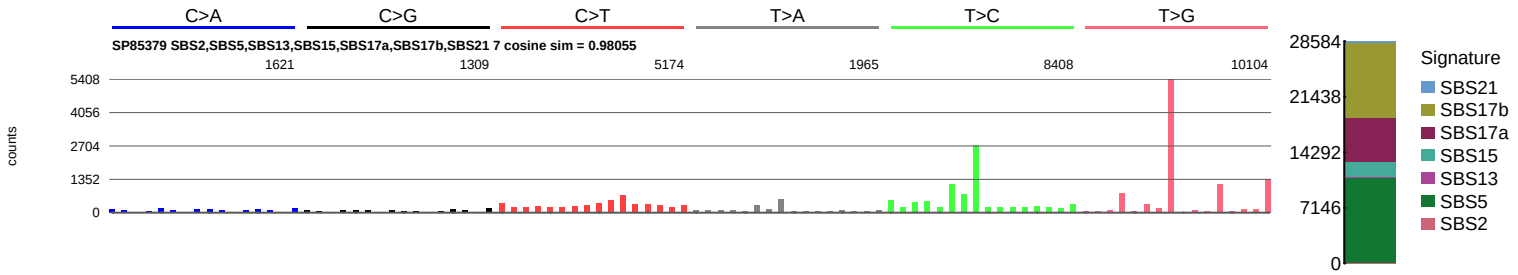
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98055
591
940
4188
3037
8998
10831
counts
5424
4068
2712
1356
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS1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C>G
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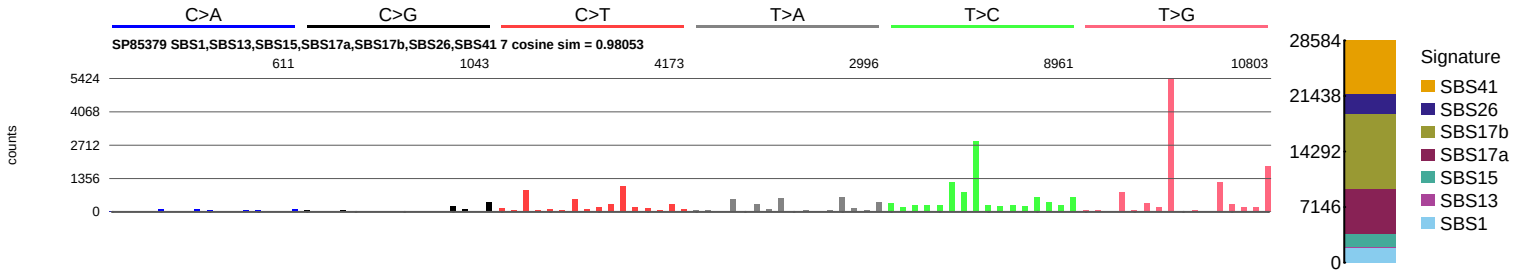
SP85379 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98055
1621
1309
5174
1965
8408
10104
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
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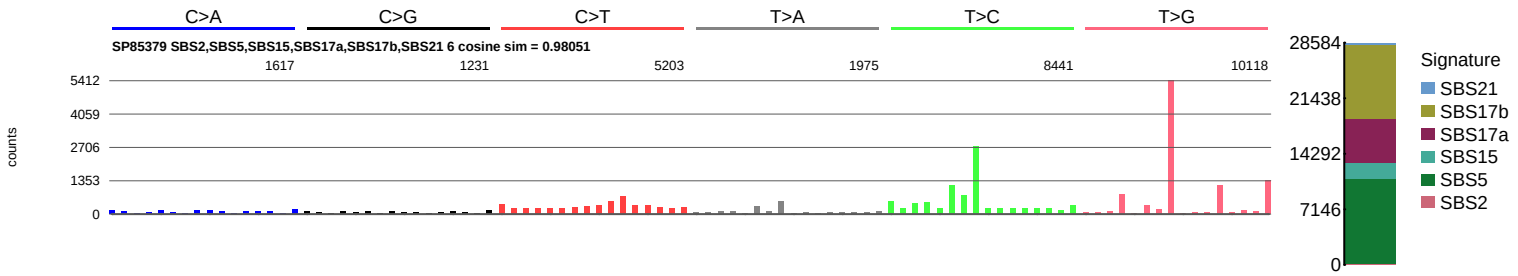
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98053
611
1043
4173
2996
8961
10803
counts
5424
4068
2712
1356
0
28584
21438
14292
7146
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
T>A
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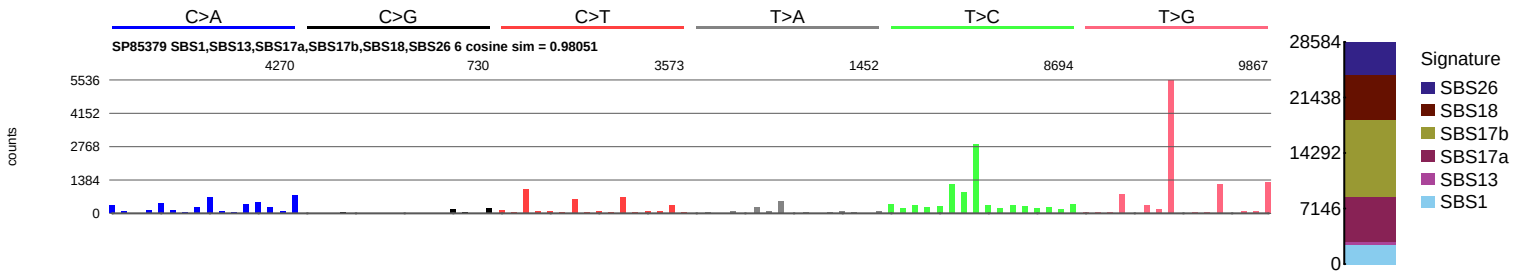
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98051
1617
1231
5203
1975
8441
10118
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2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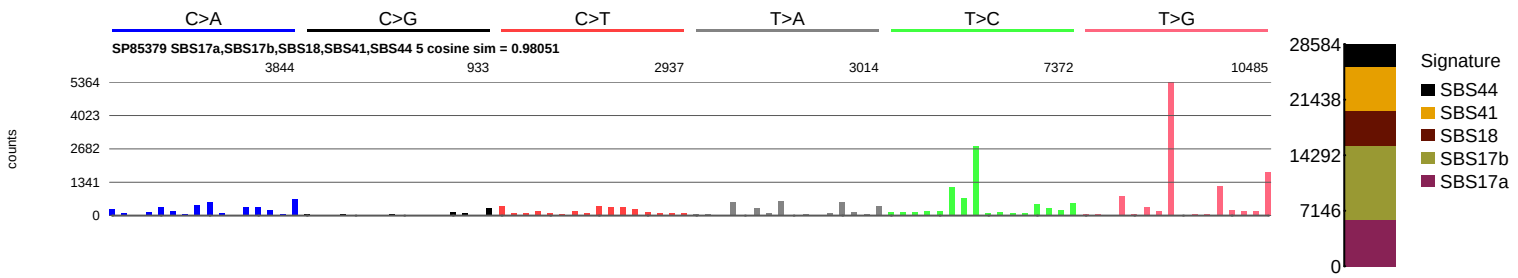
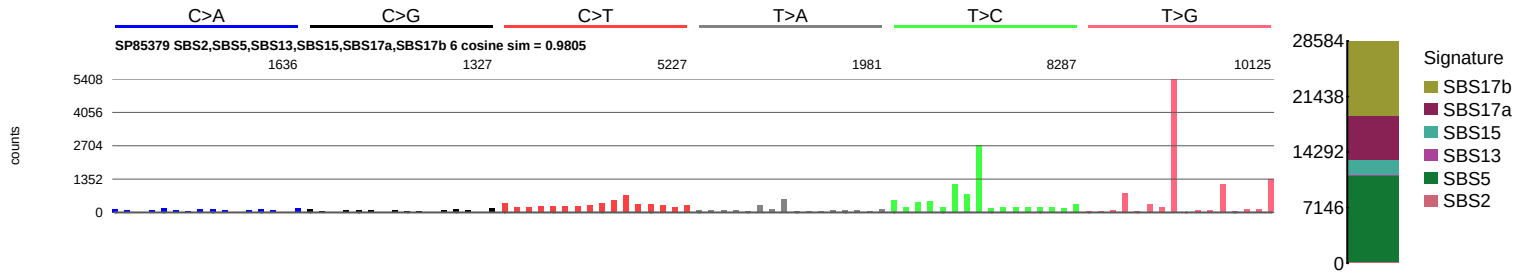
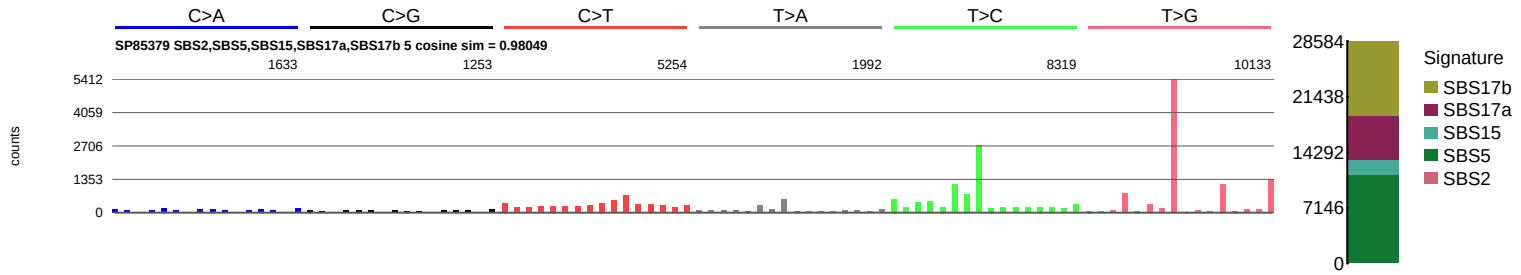
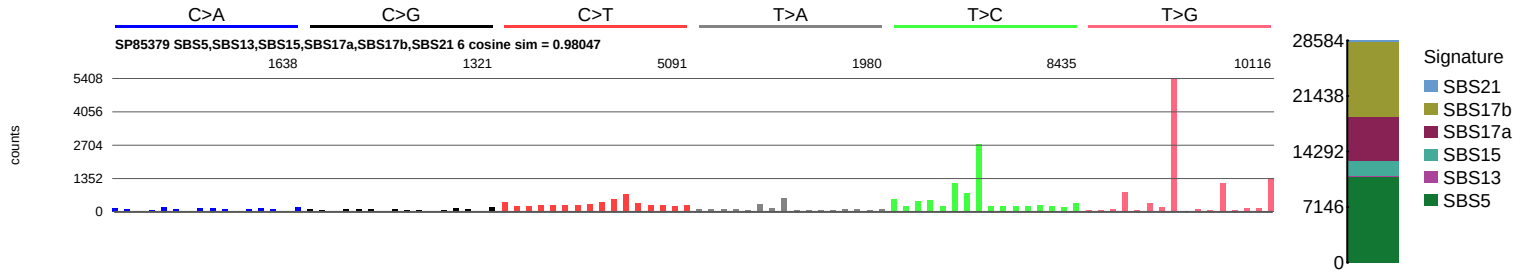
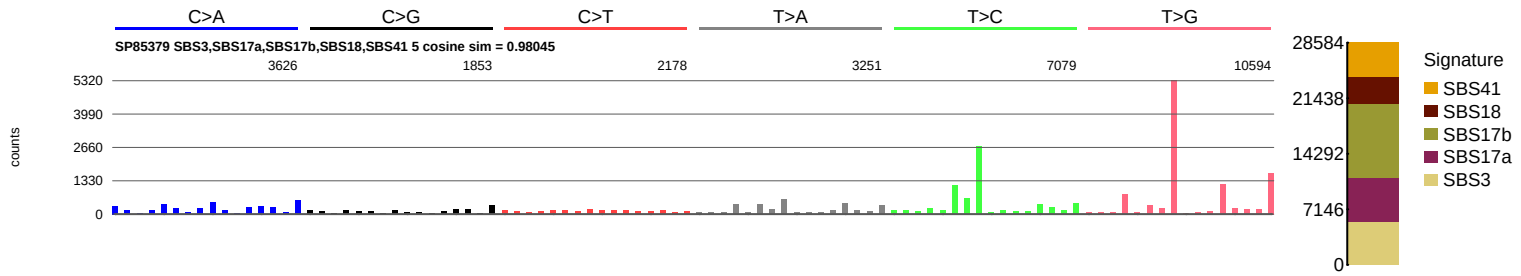
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98051
4270
730
3573
1452
8694
9867
counts
5536
4152
2768
1384
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

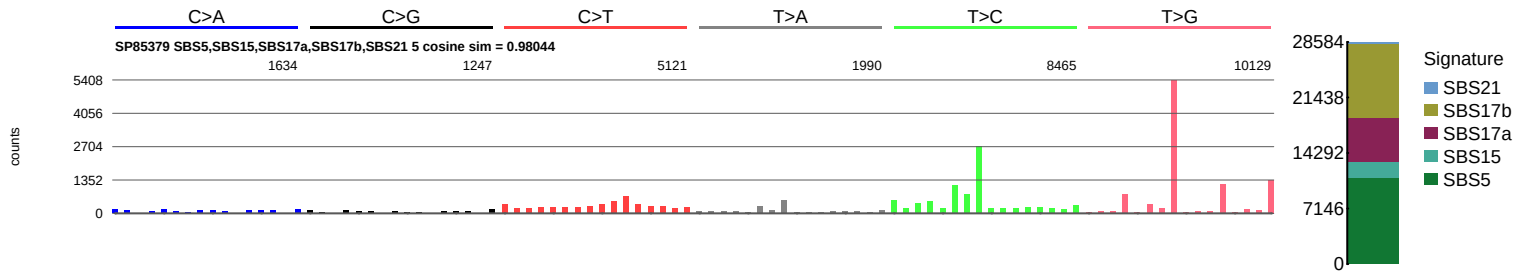

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS41,SBS44 5 cosine sim = 0.98051
3844
933
2937
3014
7372
10485
counts
5364
4023
2682
1341
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.9805
1636
1327
5227
1981
8287
10125
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98049
1633
1253
5254
1992
8319
10133
counts
5412
4059
2706
1353
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98047
1638
1321
5091
1980
8435
10116
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98045
3626
1853
2178
3251
7079
10594
counts
5320
3990
2660
1330
0
28584
21438
14292
7146
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98044
1634
1247
5121
1990
8465
10129
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
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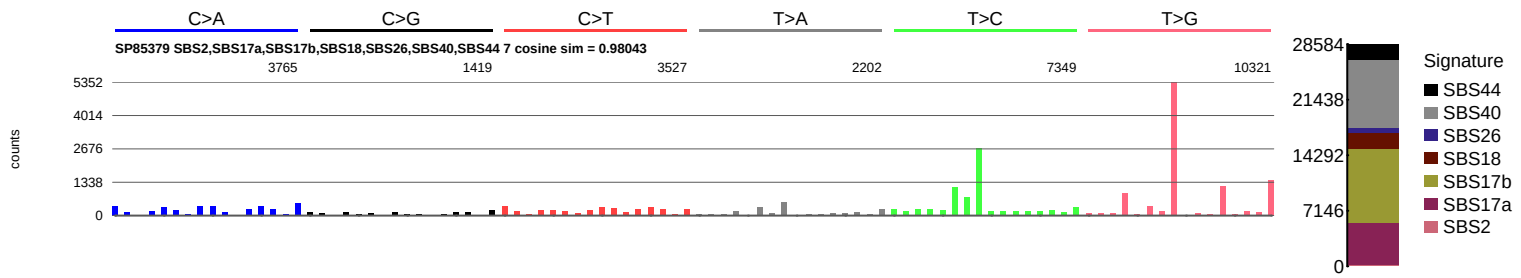
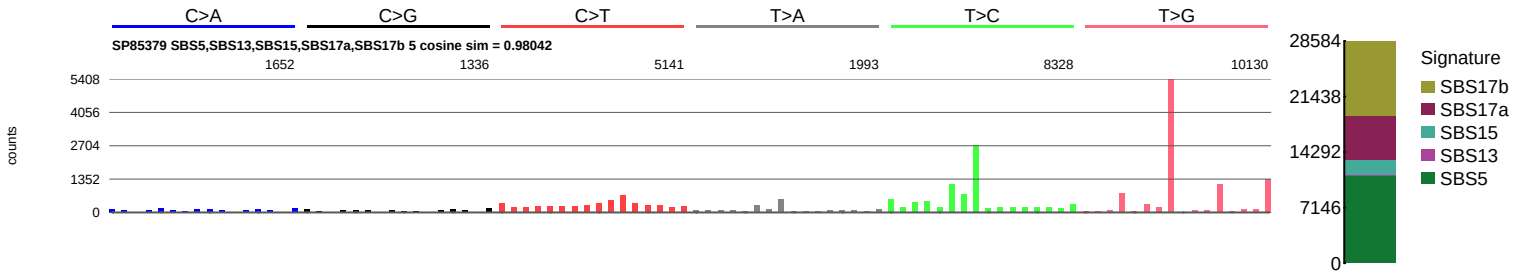
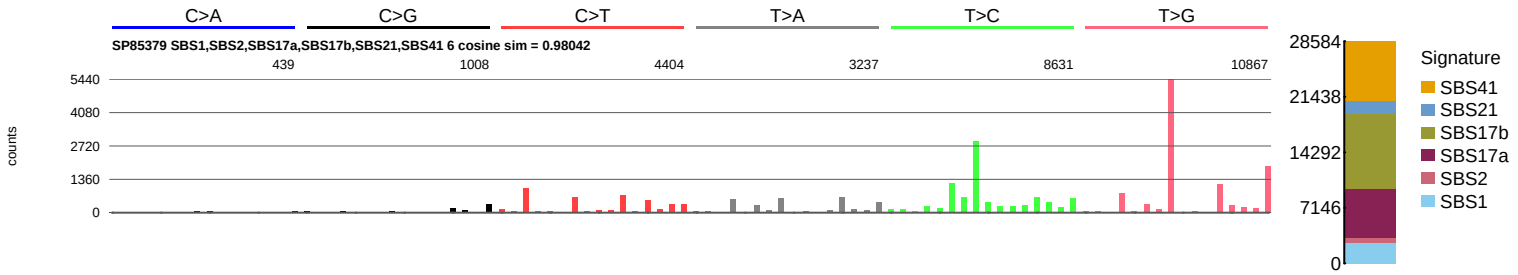
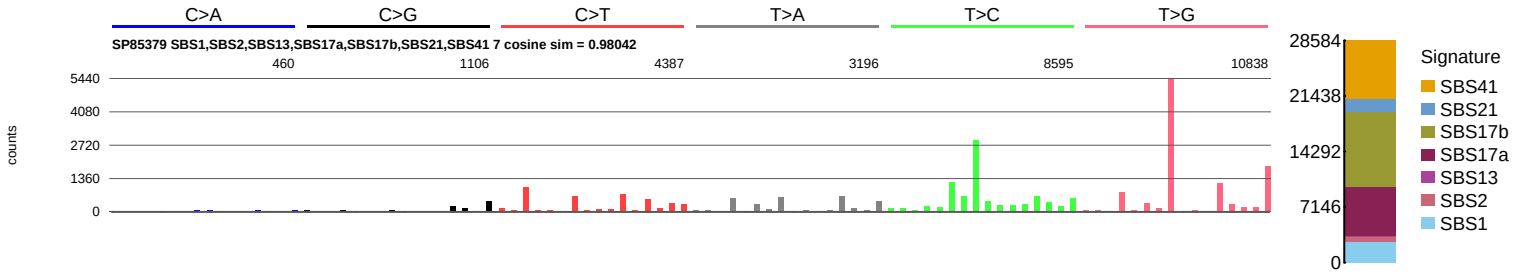
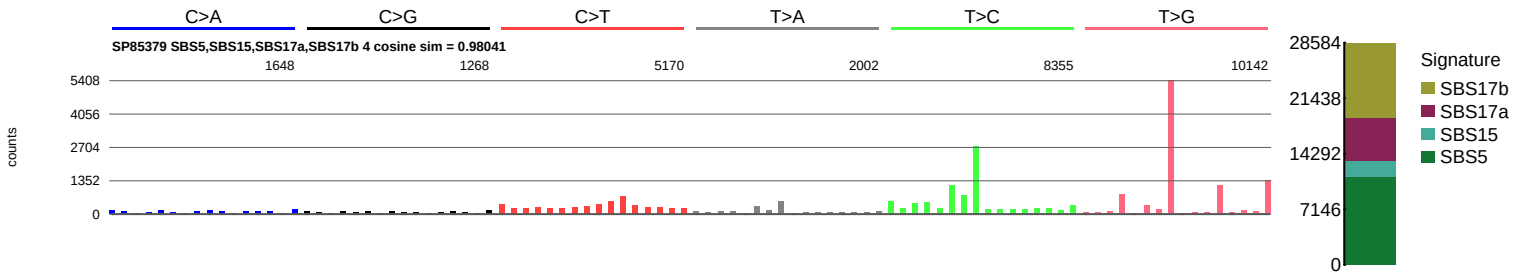
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5

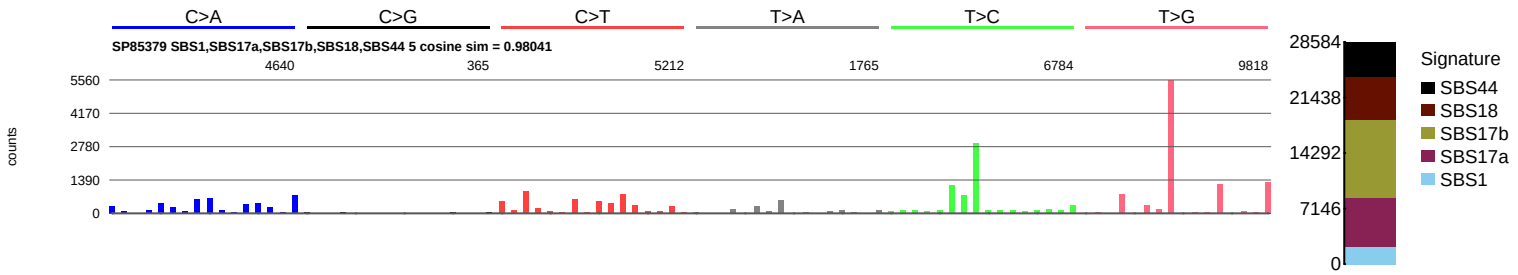

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98043
3765
1419
3527
2202
7349
10321
counts
5352
4014
2676
1338
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2

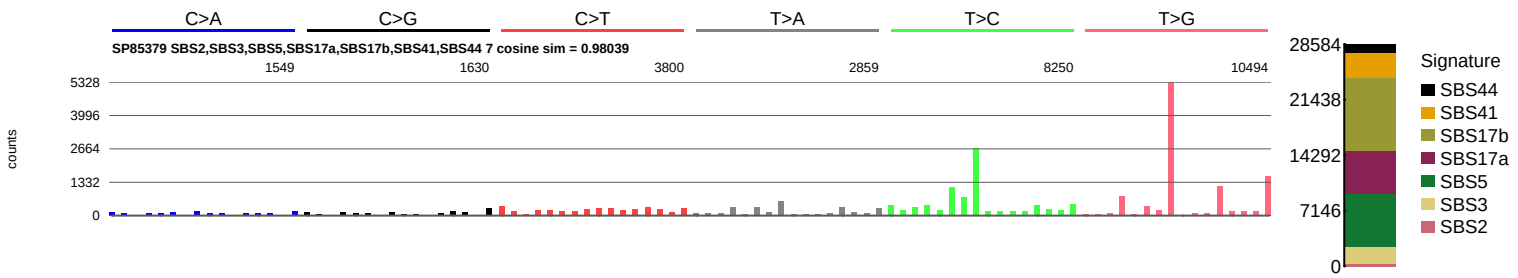

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS13,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98042
1652
1336
5141
1993
8328
10130
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5

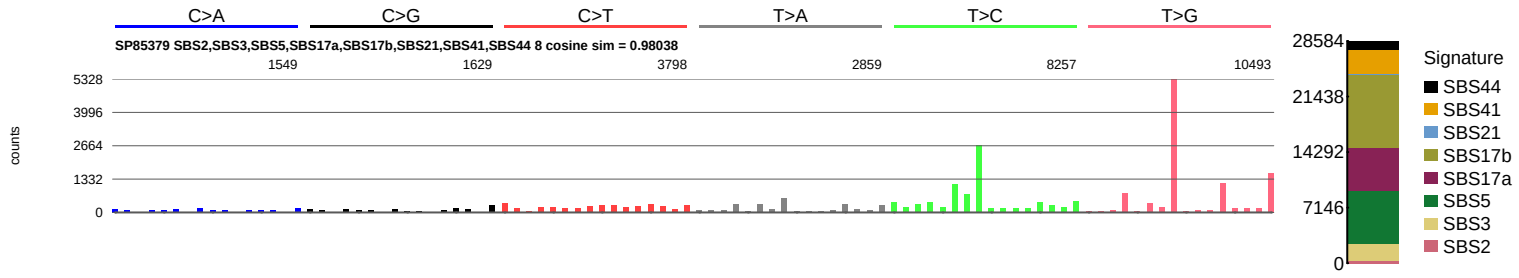

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98042
439
1008
4404
3237
8631
10867
counts
5440
4080
2720
1360
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS2
SBS1

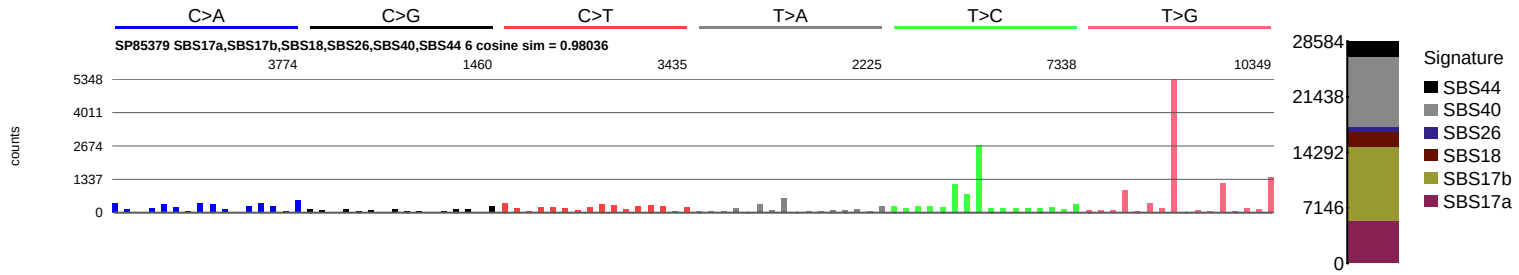

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98042
460
1106
4387
3196
8595
10838
counts
5440
4080
2720
1360
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

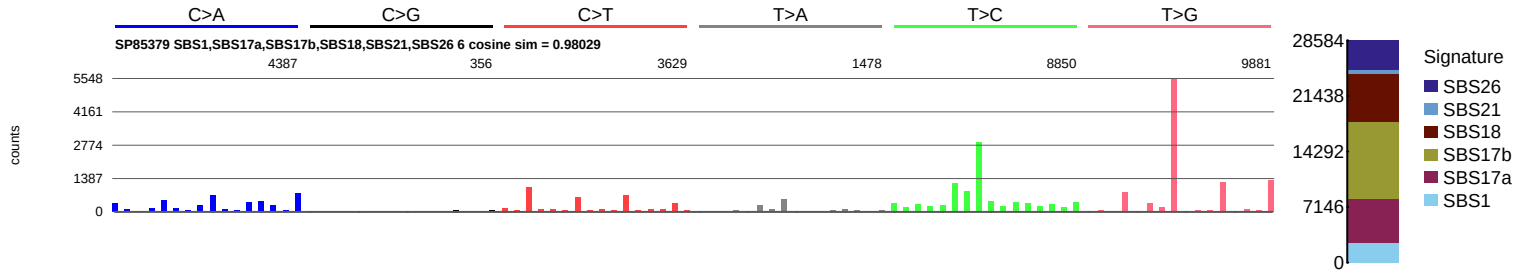

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS5,SBS15,SBS17a,SBS17b 4 cosine sim = 0.98041
1648
1268
5170
2002
8355
10142
counts
5408
4056
2704
1352
0
28584
21438
14292
7146
0
Signature
SBS17b
SBS17a
SBS15
SBS5

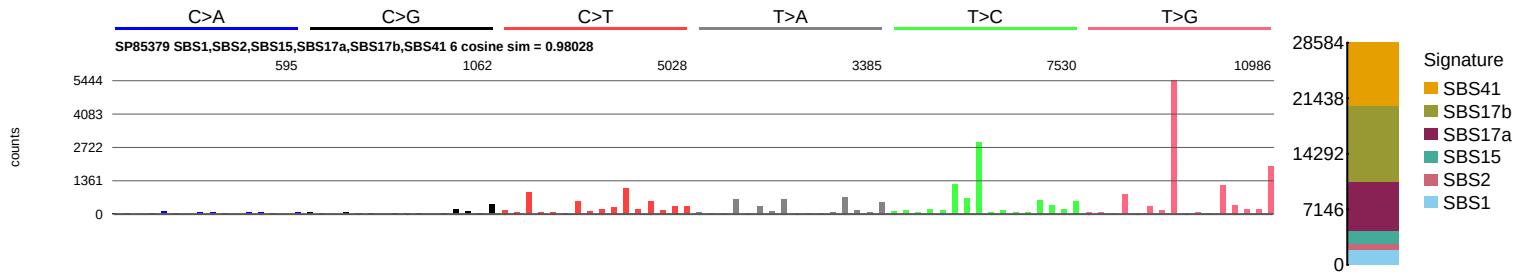

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS44 5 cosine sim = 0.98041
4640
365
5212
1765
6784
9818
counts
5560
4170
2780
1390
0
28584
21438
14292
7146
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS1

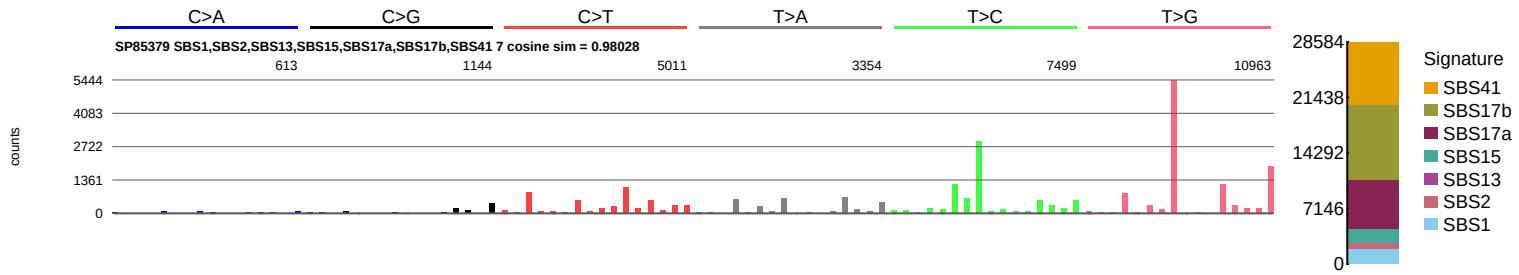

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98039
1549
1630
3800
2859
8250
10494
counts
5328
3996
2664
1332
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2

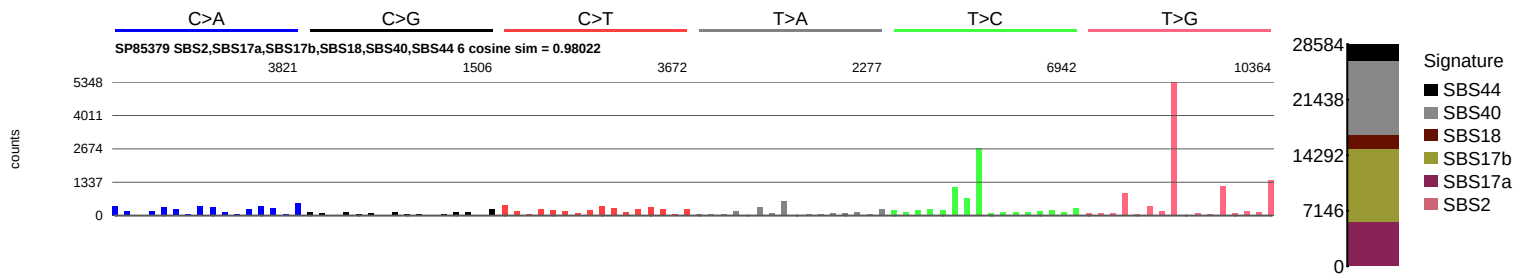

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98038
1549
1629
3798
2859
8257
10493
counts
5328
3996
2664
1332
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2

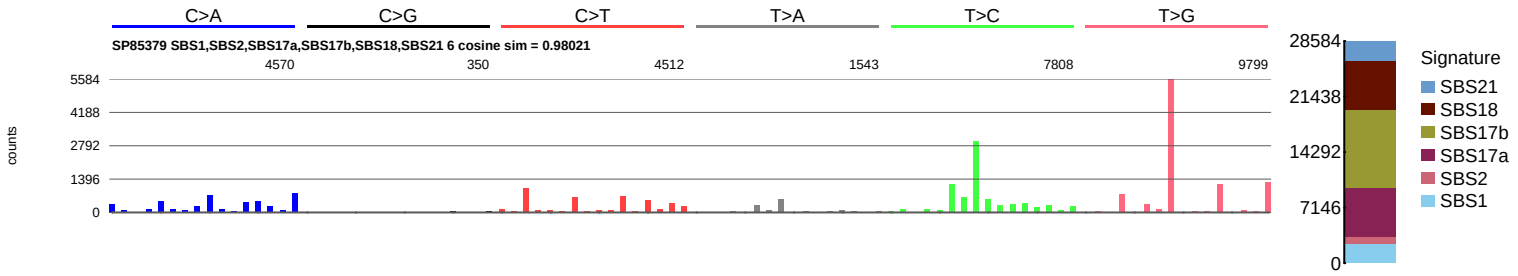

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98036
3774
1460
3435
2225
7338
10349
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a

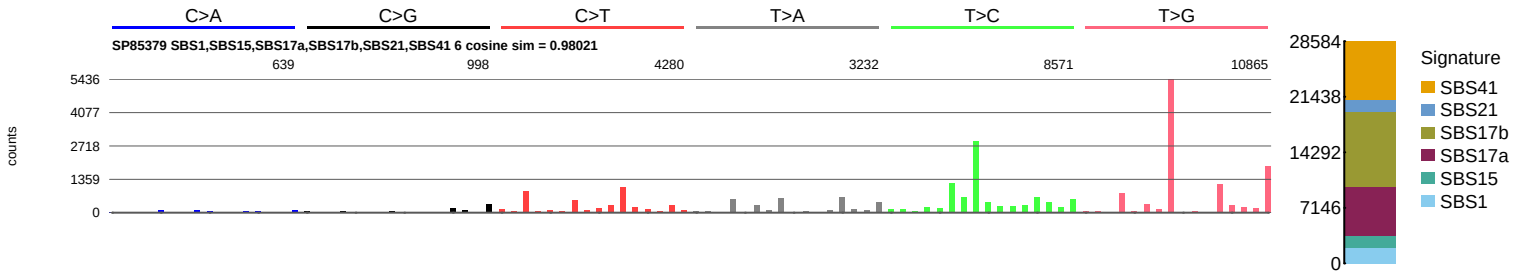

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26 6 cosine sim = 0.98029
4387
356
3629
1478
8850
9881
counts
5548
4161
2774
1387
0
28584
21438
14292
7146
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98028
595
1062
5028
3385
7530
10986
counts
5444
4083
2722
1361
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98028
613
1144
5011
3354
7499
10963
counts
5444
4083
2722
1361
0
28584
21438
14292
7146
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98022
3821
1506
3672
2277
6942
10364
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98021
4570
350
4512
1543
7808
9799
counts
5584
4188
2792
1396
0
28584
21438
14292
7146
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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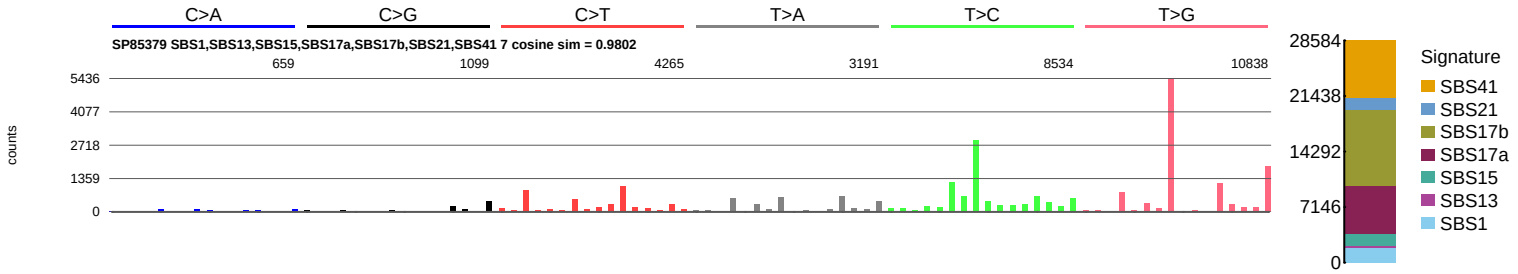
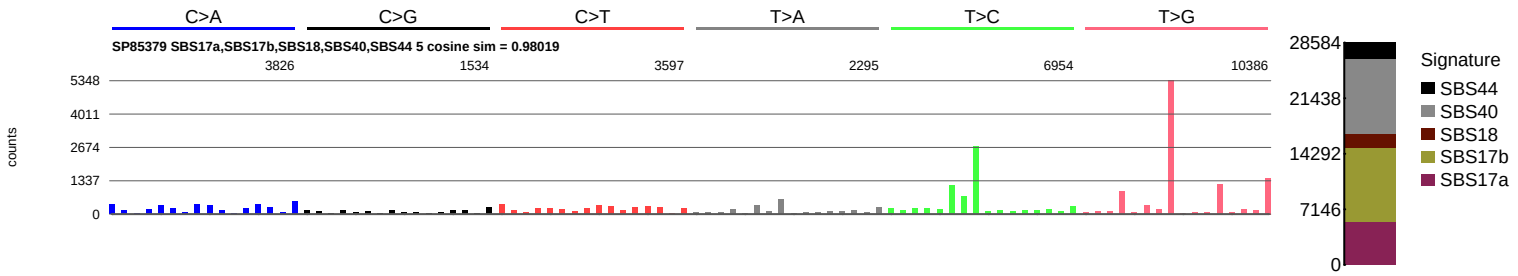
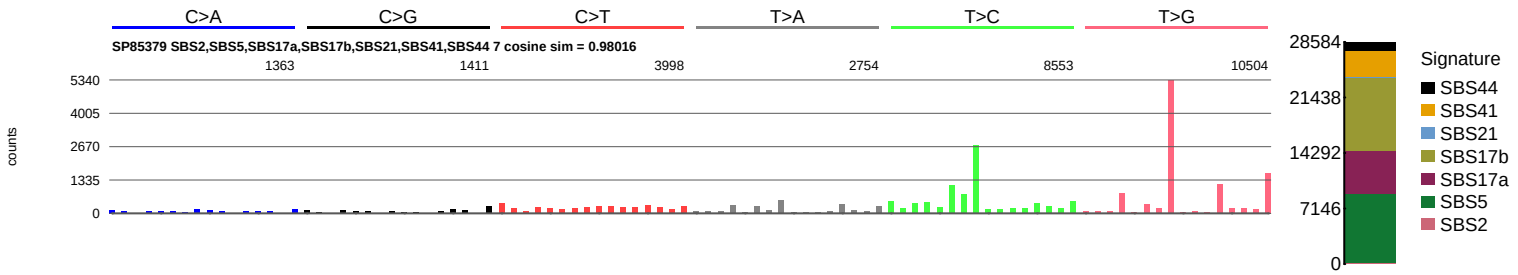
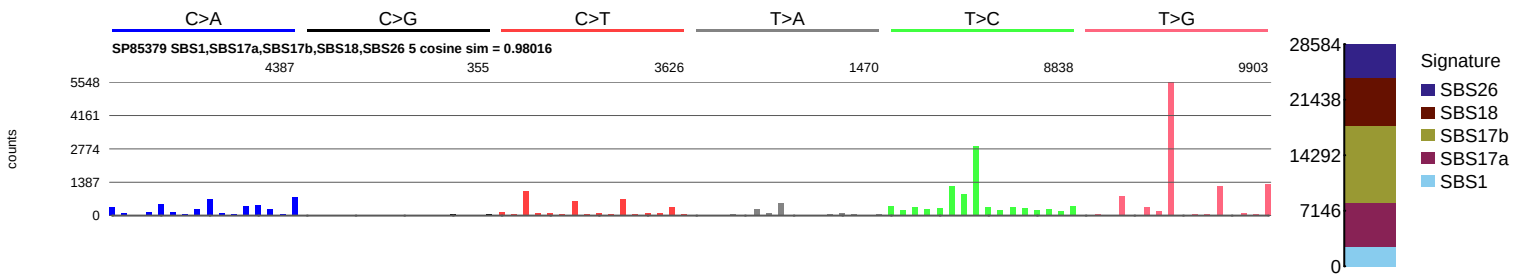
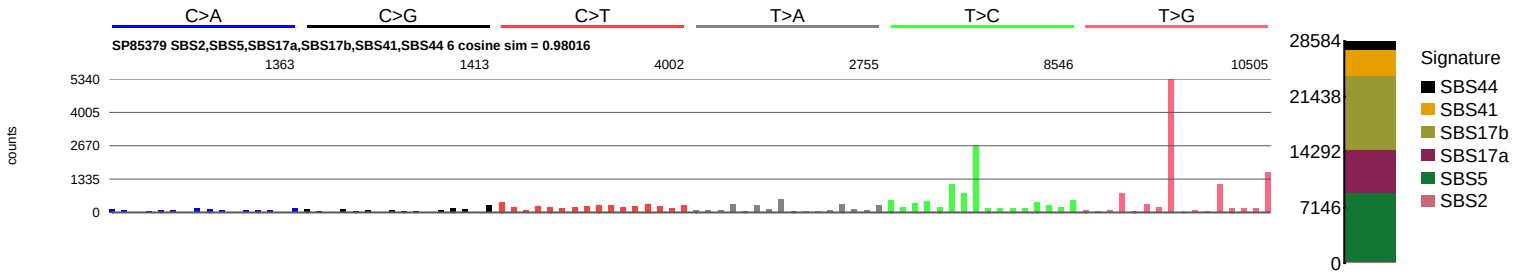
T>C
T>G
SP85379 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98021
639
998
4280
3232
8571
10865
counts
5436
4077
2718
1359
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS1

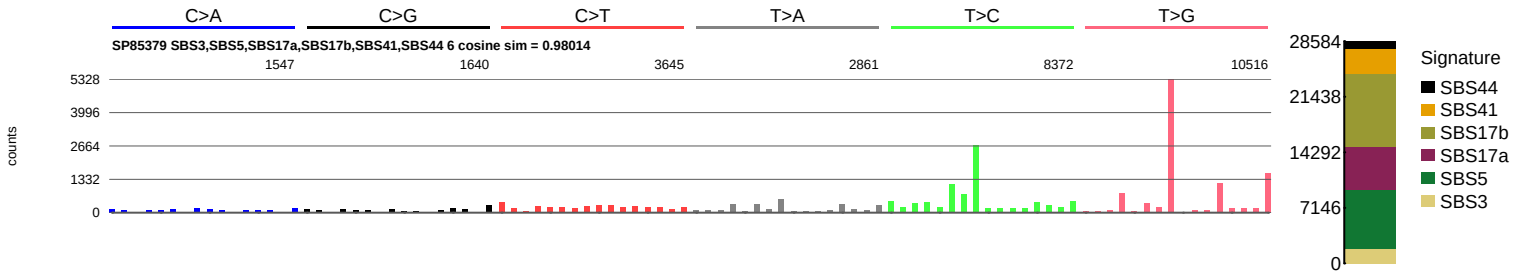

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9802
659
1099
4265
3191
8534
10838
counts
5436
4077
2718
1359
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

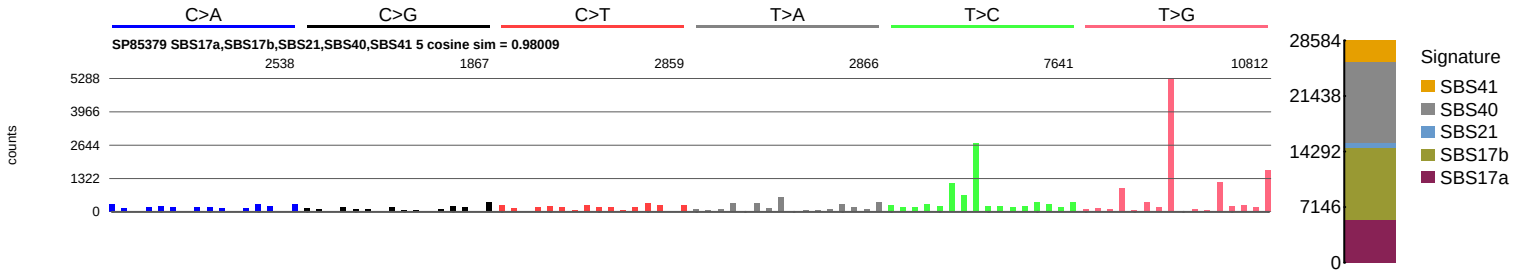

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS18,SBS40,SBS44 5 cosine sim = 0.98019
3826
1534
3597
2295
6954
10386
counts
5348
4011
2674
1337
0
28584
21438
14292
7146
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a

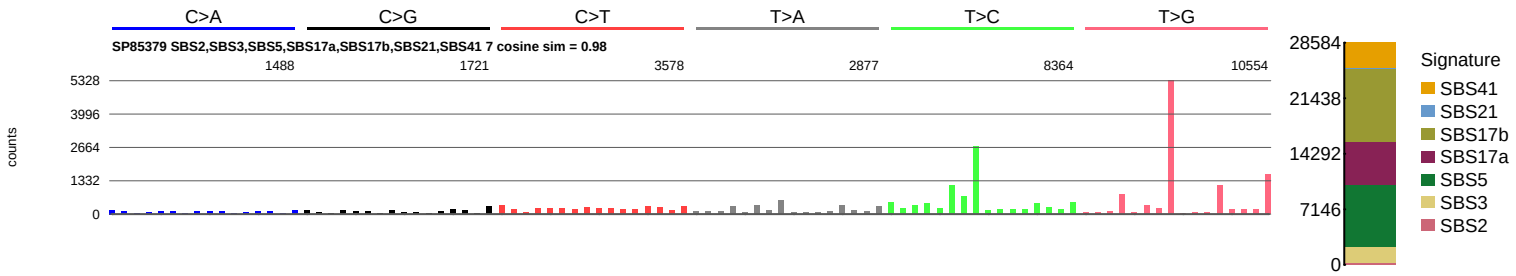

C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98016
1363
1411
3998
2754
8553
10504
counts
5340
4005
2670
1335
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS1,SBS17a,SBS17b,SBS18,SBS26 5 cosine sim = 0.98016
4387
355
3626
1470
8838
9903
counts
5548
4161
2774
1387
0
28584
21438
14292
7146
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS5,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98016
1363
1413
4002
2755
8546
10505
counts
5340
4005
2670
1335
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98014
1547
1640
3645
2861
8372
10516
counts
5328
3996
2664
1332
0
28584
21438
14292
7146
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS17a,SBS17b,SBS21,SBS40,SBS41 5 cosine sim = 0.98009
2538
1867
2859
2866
7641
10812
counts
5288
3966
2644
1322
0
28584
21438
14292
7146
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP85379 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98
1488
1721
3578
2877
8364
10554
counts
5328
3996
2664
1332
0
28584
21438
14292
7146
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2